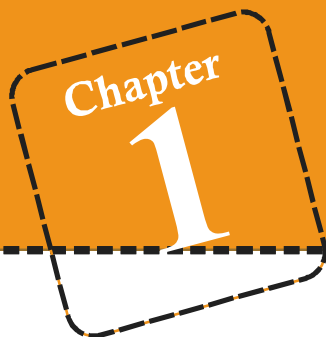


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Integers

INTRODUCTION

We have already learnt about whole numbers, natural numbers and integers in the previous class. In this chapter, we will study more about integers, their properties and operations

INTEGERS

Various Types of Numbers

1. **Natural numbers:** All numbers from 1 to infinite (∞) are known as natural numbers. Thus, 1, 2, 3, ∞ are natural numbers.
2. **Whole numbers:** All natural numbers including zero is known as whole numbers *i.e.*, 0, 1, 2, 3, ∞ are whole numbers $0 + \text{natural numbers} = \text{whole numbers}$.

NOTE: All natural numbers are whole number but zero is the only whole number which is not natural number.

3. **Integers:** All natural numbers, 0 and negative numbers are called integers. Thus..... -5, -4, -3, -2, -1, 0, 1, 2, 3, etc., are all integers.
 - (i) **Positive integers:** All natural numbers are positive integers such as 1, 2, 3, 4,, etc.
 - (ii) **Negative integers:** All negative numbers are negative integers such as ..., -4, -3, -2, -1
 - (iii) Zero is neither negative nor positive integer.

NOTE: 1. Both the positive and negative integer are called directed numbers as they indicate direction. These are also known as signed numbers because of the + or - sign.
2. The sum of any integer and its negative integer is always zero *i.e.*, $a + (-a) = 0$.

ADDITION OF INTEGERS

Following are the rules for addition of integers

Rule 1. If two positive or two negative integers are added, we add their values without considering their signs and put common sign before the sum.

Example: Add:

(i)	$36 + 27$	(ii)	$-36 - 27$
	$+ 36$		$- 36$
	$+ 27$		$- 27$
	$+ 63$		$- 63$

Rule 2. To add a positive and a negative integer, we calculate the difference in their numerical values regardless of their signs and put the sign of greater numerical value integer to the value of difference.

Example: Add:

$$\begin{array}{r} \text{(i)} \quad + 36 - 27 \\ + 36 \\ - 27 \\ + 9 \\ \hline \end{array}$$

$$\begin{array}{r} \text{(ii)} \quad - 36 + 27 \\ - 36 \\ + 27 \\ - 9 \\ \hline \end{array}$$

Properties of Addition of Integers

- Closure property of addition:** The sum of two integers is always an integer. In general, for any two integers a and b , $a + b$ is an integer.
For example: (i) $4 + 3 = 7$, which is an integer
(ii) $4 + (-3) = 1$, which is an integer
Thus, we say that integers are closed under addition.
- Commutative law of addition:** If x and y are any two integers, then, $x + y = y + x$
For example:
(i) $-7 + 8 = 1$ and $8 + (-7) = 1$
 $\therefore -7 + 8 = 8 + (-7)$
(ii) $(-5) + (-8) = -13$ and $(-8) + (-5) = -13$
 $\therefore (-5) + (-8) = (-8) + (-5)$
Thus, we conclude that addition is commutative for integers.
- Associative law of addition:** If x , y and z are any three integers, then
 $(x + y) + z = x + (y + z)$
For example: $\{(-5) + (-6)\} + 7 = -11 + 7 = -4$
 $(-5) + \{(-6) + 7\} = -5 + 1 = -4$
 $\therefore \{(-5) + (-6)\} + 7 = -5 + \{(-6) + 7\}$
- Existence of additive identity:** For any integer x , we have $x + 0 = 0 + x = x$
 0 is called the additive identity for integers.
For example: (i) $0 + 9 = 9 + 0 = 9$ (ii) $(-6) + 0 = 0 + (-6) = -6$
- Existence of additive inverse:** For any integer x , we have $x + (-x) = (-x) + x = 0$
(i) The opposite or additive inverse of x is $(-x)$ and $(-x)$ is x
(ii) The sum of an integer and its opposite is 0 .
For example: $4 + (-4) = 0$ and $(-4) + 4 = 0$
So, additive inverse of 4 is (-4) and (-4) is 4 .

SUBTRACTION OF INTEGERS

For any integers x and y , we define.

- $x - y = x + (\text{additive inverse of } y) = x + (-y)$
- $x - (-y) = x + \{\text{additive inverse of } (-y)\} = x + y$

Properties of Subtraction of Integers

- Closure property for subtraction:** If x and y are any integer, then $x - y$ is always an integer.
For example:
(i) $3 - 5 = -2$, which is an integer
(ii) $(-3) - (-6) = 3$, which is an integer
Thus, we can say that integers are closed under subtraction
- Commutative law of subtraction**
Consider the integers (-6) and (-4) , we have.
 $(-6) - (-4) = -6 + 4 = -2$ and $(-4) - (-6) = -4 + 6 = 2$
 $\therefore (-6) - (-4) \neq (-4) - (-6)$.
Thus, subtraction of an integer is not commutative
- Associative law of subtraction**
Consider the integers 4 , (-5) and (-6) , we have.

$$\{4 - (-5)\} - (-6) = (4 + 5) - (-6) = 9 - (-6) = 9 + 6 = 15$$

$$4 - \{(-5) - (-6)\} = 4 - \{(-5) + 6\} = 4 - 1 = 3$$

$$\text{Since } \{4 - (-5)\} - (-6) \neq 4 - \{(-5) - (-6)\}$$

$$\text{or } a - (b - c) \neq (a - b) - c$$

Thus, subtraction of integers is not associative

MULTIPLICATION OF INTEGERS

Let us now learn how to multiply integers using given rules.

Rule 1: To find the product of two integers with unlike sign, first get their product regardless to their signs, then give minus sign to the product.

For example: (i) $(-40) \times 9 = (-360)$ (ii) $20 \times (-3) = (-60)$

Rule 2: To find the product of two integers with like signs, first find their product regardless to their signs then put plus sign to product.

For example: (i) $3 \times 5 = (+15)$ (ii) $-3 \times -5 = (+15)$

Properties of Multiplication of Integers

- Closure properties for multiplication:** The product of two integers is always an integer.
For example: (i) $3 \times 2 = 6$, which is an integer (ii) $(-3) \times 2 = -6$, which is an integer
Thus, integers are closed under multiplication
- Commutative law of multiplication:** For any two integers x and y , we have $(x \times y) = (y \times x)$
For example: (i) $2 \times (-6) = -12$ and $(-6) \times 2 = -12$
 $\therefore 2 \times (-6) = (-6) \times 2$
(ii) $(-3) \times (-7) = 21$ and $(-7) \times (-3) = 21$
 $\therefore (-3) \times (-7) = (-7) \times (-3)$
Thus, multiplication is commutative for integers.
- Associative law of multiplication:** For any integers x, y and z , we have $(a \times b) \times c = a \times (b \times c)$
Consider the integers 3, (-4) and (-5) , we have
 $\{3 \times (-4)\} \times (-5) = -12 \times (-5) = 60$
and $3 \times \{(-4) \times (-5)\} = 3 \times 20 = 60$
 $\therefore \{3 \times (-4)\} \times (-5) = 3 \times \{(-4) \times (-5)\}$
Thus, the product of three integers is associative
- Distributive law of multiplication over addition:** For any integer x, y and z , we have:
 $x \times (y + z) = (x \times y) + (x \times z)$
Consider the integers 5, (-6) and (-7) , we have $5 \times \{(-6) + (-7)\} = 5 \times (-13) = -65$
and $\{5 \times (-6)\} + \{5 \times (-7)\} = -30 + (-35) = -65$
 $\therefore 5 \times \{(-6) + (-7)\} = \{5 \times (-6)\} + \{5 \times (-7)\}$
- Property of zero:** For every integer, x we have: $(x \times 0) = (0 \times x) = 0$
For example: (i) $5 \times 0 = 0 \times 5 = 0$ (ii) $(-5) \times 0 = 0 \times (-5) = 0$
- Existence of multiplicative identity:** For every integer x we have: $(x \times 1) = (1 \times x) = x$, 1 is known as the multiplicative identity for integers.
For example: (i) $13 \times 1 = 13$ (ii) $(-12) \times 1 = -12$
- Existence of multiplicative inverse:** Multiplicative inverse of a non-zero integer x is $\frac{1}{x}$ as
$$x \left(\frac{1}{x} \right) = \left(\frac{1}{x} \right) \cdot x = 1$$

For example: (i) Multiplicative inverse of $5 = \frac{1}{5}$ (ii) Multiplicative inverse of $-5 = -\frac{1}{5}$

Multiplicative inverse is also known as *reciprocal*.

ILLUSTRATION 1:

Multiply : $(-4) \times (-3) \times (-2)$.

SOLUTION:

$$(-4) \times (-3) \times (-2) = [(-4) \times (-3)] \times (-2) = 12 \times (-2) = (-24).$$

ILLUSTRATION 2:

Find : 28×12

SOLUTION:

$$\begin{aligned} 28 \times 12 &= 28 \times (10 + 2) \\ &= (28 \times 10) + (28 \times 2) = 280 + 56 = 336 \end{aligned}$$

ILLUSTRATION 3 :

Solve : $72 \times (-8) + (-72) \times 2$

SOLUTION:

$$\begin{aligned} 72 \times (-8) + (-72) \times 2 &= 72 \times (-8) + 72 \times (-2) \\ &= 72 \times [(-8) + (-2)] \\ &= 72 \times (-10) = (-720) \end{aligned}$$

DIVISION OF INTEGERS

Following are the rules for division of integers

Rule 1: To find the division of two integers with unlike sign, first get their quotient regardless to their signs, then give minus sign to the product.

For example: (i) $-74 \div 2 = \frac{-74}{2} = -37$ (ii) $96 \div (-3) = \frac{96}{-3} = -32$

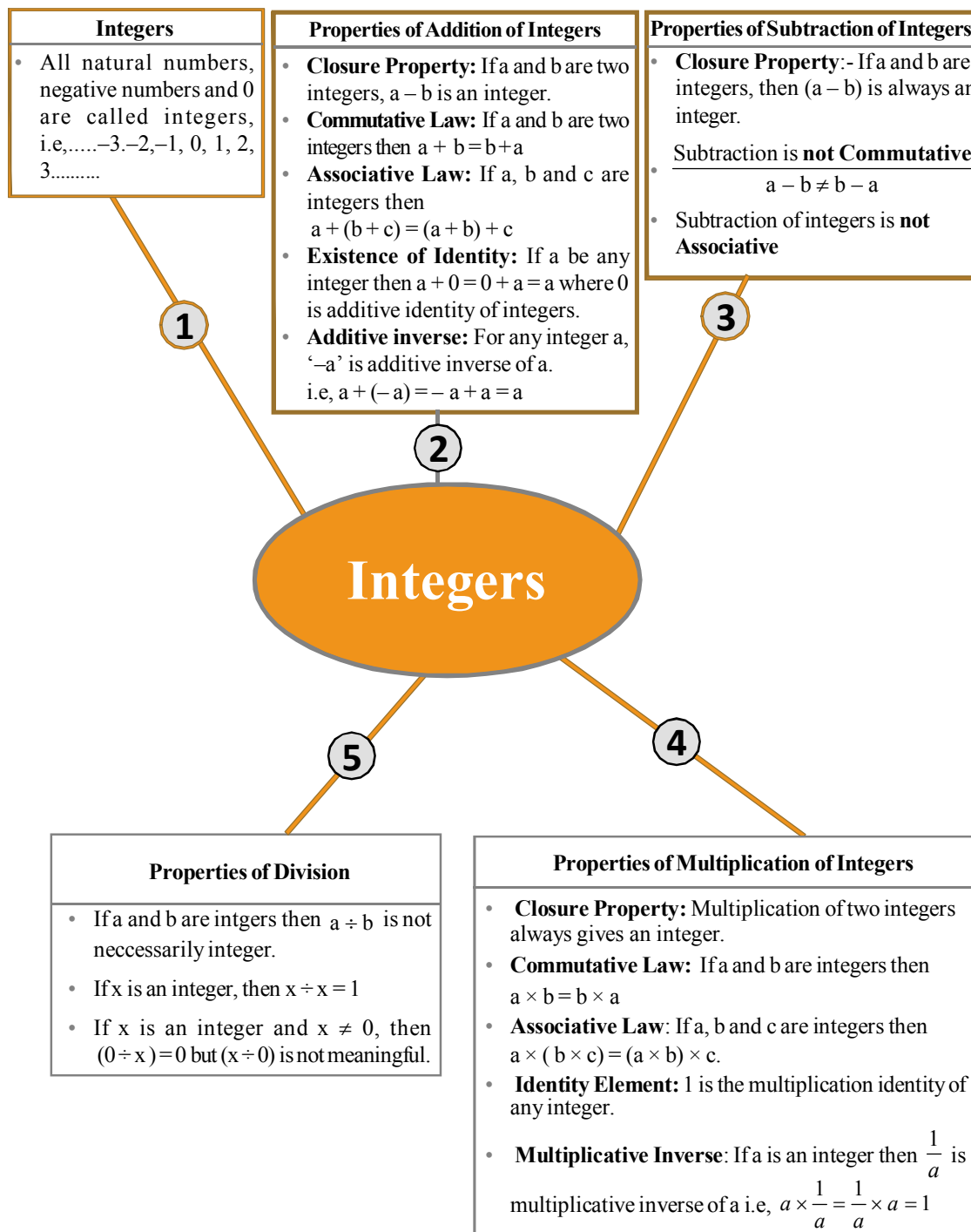
Rule 2: To find the division of two integers with like signs, first find their product regardless to their signs then put plus sign to quotient.

For example: (i) $48 \div 6 = \frac{48}{6} = 8$ (ii) $-155 \div (-5) = \frac{-155}{-5} = 31$

Properties of Division of Integers

- If x and y are integers, then $(x \div y)$ is not necessarily an integer.
For example: (i) 12 and 5 are both integer but $(12 \div 5)$ is not an integer.
(ii) (-12) and 5 are both integer. But $[(-12) \div 5]$ is not an integer.
- If x is an integer and $x \neq 0$, then $x \div x = 1$
For example: (i) $10 \div 10 = 1$ (ii) $(-5) \div (-5) = 1$
- If x is an integer, then $(x \div 1) = x$
For example: (i) $5 \div 1 = 5$ (ii) $(-5) \div 1 = (-5)$
- If x is an integer and $x \neq 0$, then $(0 \div x) = 0$ but $(x \div 0)$ is not meaningful.
For example: (i) $0 \div 9 = 0$ (ii) $6 \div 0 = \text{Meaning less}$
- If x , y and z are integers, then $(x \div y) \div z \neq x \div (y \div z)$ unless $z = 1$.
Thus division on integers is not associative.
For example: Let $x = -6$, $y = 3$, $z = -2$ then,
 $(x \div y) \div z = \{(-6) \div 3\} \div (-2) = (-2) \div (-2) = 1$
 $x \div \{(y) \div (z)\} = (-6) \div \{3 \div (-2)\} = (-6) \div (-1.5) = 4$
 $\therefore (x \div y) \div z \neq x \div \{(y) \div (z)\}$

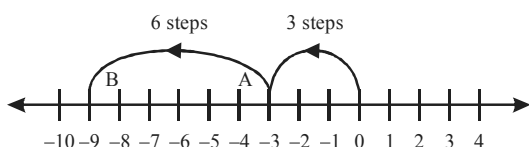
CONCEPT MAP



MISCELLANEOUS SOLVED EXAMPLES

1. Add -3 and -6 on the number line.

Sol. On the number line, we start from 0 and move 3 steps to the left to reach a point A. Now, starting from A, we move 6 steps to the left to reach a point B, as shown below.



Clearly, B represents -9 .

$$\therefore (-3) + (-6) = -9.$$

2. Verify using associative property :

$$4 + \{(-9) + (-3)\} = \{4 + (-9)\} + (-3)$$

Sol. $\{(+4) + (-9)\} + (-3) = (4 - 9) + (-3)$
 $= -5 + (-3) = -8$

$$(+4) + \{(-9) + (-3)\} = (+4) + \{-12\}$$

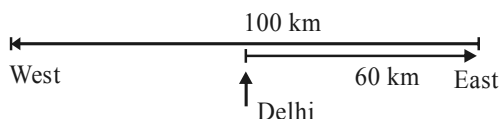
$$= -(12 - 4) = -8$$

$$\therefore \{(+4) + (-9)\} + (-3) = (+4) + \{(-9) + (-3)\}$$

Hence, associative property is verified.

3. A man travelled 60 km to the east of Delhi and then 100 km to the west of it. How far from Delhi was he finally?

Sol.



Take travelling in the east as positive and travelling in the west as negative.

Therefore, distance travelled in east = 60 km
 and distance travelled in west = -100 km

$$\therefore \text{Total distance} = 60 + (-100)$$

$$= -40 \text{ km}$$

$\therefore$ The person is 40 km to the west of Delhi.

4. Find each of the following products:

(i) $(-115) \times 8$ (ii) $9 \times (-3) \times (-6)$

Sol. (i) We have,

$$(-115) \times 8 = -(115 \times 8) = -920$$

(ii) We have,

$$9 \times (-3) \times (-6) = \{9 \times (-3)\} \times (-6)$$

$$= -27 \times (-6) = 162$$

5. Evaluate each of the following products:

(i) $(-3) \times (-6) \times (-9) \times (-12)$

(ii) $(-1) \times (-1) \times (-1) \times \dots 151 \text{ times}$

Sol. (i) Since the number of negative integers in the given product is even. Therefore, their product is positive. Thus, we have

$$(-3) \times (-6) \times (-9) \times (-12)$$

$$= (3 \times 6 \times 9 \times 12)$$

$$= (18 \times 9 \times 12)$$

$$= (162 \times 12) = 1944$$

(ii) The number of negative integers in the given product is odd.

$$\therefore (-1) \times (-1) \times (-1) \times \dots 151 \text{ times}$$

$$= -(1 \times 1 \times 1 \times \dots 151 \text{ times}) = -1$$

6. Calculate and verify using associative law:

(i) $-3 \times (-5 \times 2)$ (ii) $[8 \times (-4)] \times 5$

Sol. (i) $-3 \times (-5 \times 2) = -3 \times (-10) = 30$

$$\text{or } (-3 \times -5) \times 2 = 15 \times 2 = 30$$

(ii) $[8 \times (-4)] \times 5 = (-32) \times 5 = -160$

$$\text{or } 8 \times (-4 \times 5) = 8 \times (-20) = -160$$

Hence, associative property holds.

7. Simplify using distributive law:

(i) $(-18) \times 7 + (-18) \times (-4)$

(ii) $16 \times (-9) + (-8) \times (-9)$

Sol. Using the distributive laws, we get:

(i) $(-18) \times 7 + (-18) \times (-4) = (-18) \times \{7 + (-4)\}$

$$[\because a \times b + a \times c = a \times (b + c)]$$

$$= (-18) \times 3 = -54.$$

$$\begin{aligned} \text{(ii)} \quad 16 \times (-9) + (-8) \times (-9) &= \{16 + (-8)\} \times (-9) \\ &[\because a \times c + b \times c = (a + b) \times c] \\ &= 8 \times (-9) = -72. \end{aligned}$$

8. Calculate the following using suitable property:

(i) $(-7) \times (-5) + (-7) \times 105$

(ii) $(-2) \times 113 \times (-5)$

Sol. (i) $(-7) \times (-5) + (-7) \times 105$
 $= (-7) \times \{(-5) + 105\}$
 [Using the distributive property of multiplication over addition]
 $= (-7) \times 100 = -700$

(ii) $(-2) \times 113 \times (-5)$
 $= 113 \times \{(-2) \times (-5)\}$
 [Using the associative property of multiplication]
 $= 113 \times (+10) = 1130$

9. Divide:

(i) $-20 \div 4$

(ii) $-15 \div (-3)$

(iii) $0 \div (-832)$

(iv) $-56 \div (-1)$

Sol. (i) $-20 \div 4 = -(20 \div 4) = -5.$

(ii) $-15 \div (-3) = +(15 \div 3) = 5$

(iii) $0 \div (-832) = -(0 \div 832) = 0.$

(iv) $-56 \div (-1) = +(56 \div 1) = 56.$

10. Simplify:

$12 - [7 - \{16 - (18 - \overline{6 + 3 - 12})\}]$

Sol. $12 - [7 - \{16 - (18 - \overline{6 + 3 - 12})\}]$
 $= 12 - [7 - \{16 - (18 + 3)\}]$
 $= 12 - [7 - \{16 - 21\}]$
 $= 12 - [7 - \{-5\}]$
 $= 12 - [7 + 5] = 12 - 12 = 0$

1

EXERCISE

Fill in the Blanks :

DIRECTIONS : Complete the following statements with an appropriate word / term to be filled in the blank space(s).

1. $(-a) + b = b +$ additive inverse of _____.
2. $[(-8) + \underline{\hspace{1cm}}] + \underline{\hspace{1cm}} = \underline{\hspace{1cm}} + [(-3) + \underline{\hspace{1cm}}] = -3$
3. If x , y and z are integers then $(x \times \underline{\hspace{1cm}}) \times z = x \times (y \times \underline{\hspace{1cm}})$
4. If we multiply _____ number of negative integers, then the resulting integer is positive.
5. _____ is the multiplicative identity for integers.
6. $-[-\{-73\}] = \underline{\hspace{1cm}}$
7. $-[-\{-0\}] = \underline{\hspace{1cm}}$
8. $(-5) \times \{(2) + \underline{\hspace{1cm}}\} = (-5) \times \underline{\hspace{1cm}} + (-5) \times (-3)$
9. (-3) is _____ more than (-9) .
10. The additive identity element in the set of integers is _____.

True / False :

DIRECTIONS : Read the following statements and write your answer as true or false.

1. Integers do not follow closure property with respect to division.
2. The smallest integer is zero.
3. Zero is larger than every negative integer.
4. The sum of a negative integer and a positive integer is always a positive integer.
5. $42(4 + 2) = (42 \times 4) + (42 \times 2)$ is an example of associative property.
6. $0 \div 5$ is the same as $5 \div 0$
7. 0 is the additive identity element in the set of whole numbers.
8. $(-1) \times (-2) \times (-3) = 1 + 2 + 3$

Match the Columns :

DIRECTIONS : Each question contains statements given in two columns which have to be matched. Statements (A, B, C, D) in column-I have to be matched with statements (p, q, r, s) in column-II.

1. Column-I	Column-II
(A) $(-2) + (-8) \div (-4)$	(p) -13
(B) $3 - (5 - 6 \div 3) - (-2)$	(q) 2
(C) $(-15) + 4 \div (5 - 3)$	(r) 3
(D) $(-3) + (-8) \div (-4)$	(s) 0
$-2 \times (-2)$	

Very Short Answer Questions :

DIRECTIONS : Give answer in one word or one sentence.

1. Find each of the following products:
 - (i) $(-2) \times 36 \times (-5)$
 - (ii) $(-2) \times (-4) \times (-6) \times (-8)$
2. Find the value of:
 - (i) $1487 \times 327 + (-487) \times 327$
 - (ii) $28945 \times 99 - (-28945)$
3. Determine the integer whose product with '-1' is
 - (i) 72
 - (ii) 0
 - (iii) -165
4. Verify the following:
 $(-23) \{(-5) + (+19)\} = (-23) \times (-5) + (-23) \times (+19)$
5. Solve: $-25 + 14 \div (5 - 3)$
6. Simplify: $22 - \frac{1}{4} \{-5 - (-48) \div (-16)\}$
7. Simplify: $36 - [18 - \{14 - (15 - 4 \div 2 \times 2)\}]$
8. Find the quotient in each of the following:
 - (i) $(-15625) \div (-125)$
 - (ii) $30000 \div (-100)$
9. Find the value of
 - (i) $15625 \times (-2) + (-15625) \times 98$
 - (ii) $18946 \times 99 - (-18946)$
10. What will be the sign of the product, if we multiply together 199 negative integers and 101 positive integers?

11. Radha climbs up 5 stairs every minute and then climbs down 2 stairs over the next minute. How many minutes will she take to climb 60 stairs?
12. Which integer is less than every positive integer?

Short Answer Questions :

DIRECTIONS : Give answer in 2-3 sentences.

1. A company manufactures two types of cloth material. If one of them is cotton on which the company earns ₹ 4 on each metre sold. If company sold 500 metres of cotton cloth. What is the profit earned?
2. Aman has ₹ 20 with him. He spent ₹ 8 on Monday, got ₹ 5 as pocket money on Tuesday, gave ₹ 7 as a loan to a friend on Wednesday, ate an ice-cream worth ₹ 10 on Thursday, received a reward of ₹ 5 from his grandfather on Friday. How much does he have on Sunday, if his friend repays the loan on Saturday?
3. Taking $a = -15$ and $b = -23$, show that $a + b = b + a$ but $a - b \neq b - a$
4. Find the value of the following:
 - (i) $\{(-27) + (-73)\} + (-100)$
 - (ii) $\{(-51) + (-83)\} - (-12)$
 - (iii) $\{200 - (-100)\} + (-300)$
5. What must be subtracted from -59 to obtain -21 ?
6. What must be added to 221 to obtain 158 ?
7. Add the following using the number line.
 - (i) $(2) + (-5)$
 - (ii) $(-4) + (-2)$
8. Find the value of the following.
 - (i) $-92 - (-12)$
 - (ii) $0 - (+123)$
 - (iii) $0 - (-150)$
9. Subtract :
 - (i) -887 from -188
 - (ii) 578 from -173
 - (iii) 345 from 282

2

EXERCISE

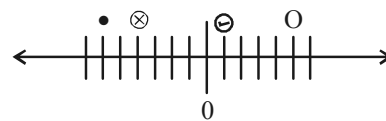
Text-Book Exercise :

1. At Srinagar temperature was -5°C on Monday and then it dropped by 2°C on Tuesday. What was the temperature of Srinagar on Tuesday? On Wednesday, it rose by 4°C . What was the temperature on this day?
2. A plane is flying at the height of 5000 m above the sea level. At a particular point, it is exactly above a submarine floating 1200 m below the sea level. What is the vertical distance between them?
3. Verify $a - (-b) = a + b$ for the following values of a and b .
 - (i) $a = 21, b = 18$
 - (ii) $a = 118, b = 125$
 - (iii) $a = 75, b = 84$
 - (iv) $a = 28, b = 11$
4. Write down a pair of integers whose
 - (i) sum is -3
 - (ii) difference is -5
 - (iii) difference is 2
 - (iv) sum is 0
5. What will be the sign of the product if we multiply together :
 - (i) 8 negative integers and 3 positive integers?
 - (ii) 5 negative integers and 4 positive integers?
 - (iii) (-1) , twelve times?
 - (iv) (-1) , $2m$ times, m is a natural number?
6. Verify $(-30) \times [13 + (-3)] = [(-30) \times 13] + [(-30) \times (-3)]$
7. In a class test containing 15 questions, 4 marks are given for every correct answer and (-2) marks are given for every incorrect answer. Gurpreet attempts all questions but only 9 of her answers are correct.
 - (i) What is her total score?
 - (ii) One of her friends gets only 5 answers correct. What will be her score?

8. Suppose we represent the distance above the ground by a positive integer and that below the ground by a negative integer, then answer the following:
- An elevator descends into a mine shaft at the rate of 5 metre per minute. What will be its position after one hour?
 - If it begins to descend from 15 m above the ground, what will be its position after 45 minutes?
9. Find the product, using suitable properties:
- $8 \times 53 \times (-125)$
 - $15 \times (-25) \times (-4) \times (-10)$
 - $(-41) \times 102$
 - $625 \times (-35) + (-625) \times 65$
10. A certain freezing process requires that room temperature be lowered from 40°C at the rate of 5°C every hour. What will be the room temperature 10 hours after the process begins?
11. A cement company earns a profit of ₹ 8 per bag of white cement sold and a loss of ₹ 5 per bag of grey cement sold.
- The company sells 3,000 bags of white cement and 5,000 bags of grey cement in a month. What is its profit or loss?
 - What is the number of white cement bags it must sell to have neither profit nor loss, if the number of grey bags sold is 6,400 bags.
12. In a test (+ 5) marks are given for every correct answer and (- 2) marks are given for every incorrect answer.
- Radhika answered all the questions and scored 30 marks though she got 10 correct answers.
 - Jay also answered all the questions and scored (- 12) marks though he got 4 correct answers. How many incorrect answers had he attempted?
13. A shopkeeper earns a profit of ₹ 1 by selling one pen and incurs a loss of 40 paise per pencil while selling pencils of her old stock.
- In a particular month she incurs a loss of ₹ 5. In this period, she sold 45 pens. How many pencils did she sell in this period?
 - In the next month she earns neither profit nor loss. If she sold 70 pens, how many pencils did she sell?
14. The temperature at 12 noon was 10°C above zero. If it decreases at the rate of 2°C per hour until midnight, at what time would the temperature be 8°C below zero? What would be the temperature at mid-night?

Exemplar Questions :

- Number of pairs of integers which are smaller than - 3, but their difference is greater than - 3 are.
- If Δ is an operation such that for integers a and b we have $a \Delta b = a \times a + b \times b - a \times b$, then find $(-3) \Delta 2$.
- If $\otimes$, $\ominus$ and $\bullet$ represent some integers on number line, then descending order of these numbers is



- What is the value of $(-43) \times (-99) + 43$?
- Write two integers which are smaller than - 5 but their difference is -5.
 - Write two integers such that one is smaller than -11, and other is greater than -11 but their difference is -11.
- You are at an elevation 380 m above sea level as you start a motor ride. During the ride, your elevation changes by the following metres: 540 m, -268 m, 116 m, -152 m, 490 m, -844 m, 94 m. What is your elevation relative to the sea level at the end of the ride?
- Evaluate the following using distributive property.
 - $53 \times (-9) - (-109) \times 53$
 - $68 \times (-17) + (-68) \times 3$
- If $*$ is an operation such that for integers a and b , we have $a * b = a \times b + (a \times a + b \times b)$ then find
 - $(-3) * (-5)$
 - $(-6) * 2$
- If Δ is an operation such that for integers a and b we have $a \Delta b = a \times b - 2 \times a \times b + b \times b - (-a) \times b + b \times b$ then find $4 \Delta (-3)$.
- Height of a place A is 1800 m above sea level. Another place B is 700 m below sea level. What is the difference between the levels of these two places?
- A green grocer had a profit of ₹ 47 on Monday, a loss of ₹ 12 on Tuesday and loss of ₹ 8 on Wednesday. Find his net profit or loss in 3 days.
- In a test, +3 marks are given for every correct answer and -1 mark are given for every incorrect answer. Sona attempted all the questions and scored +20 marks though she got 10 correct answers.



- (i) How many incorrect answers has she attempted?
- (ii) How many questions were given in the test?

HOTS Questions :

1. In a class test containing 10 questions, 5 marks are awarded for every correct answer and (-2) marks are awarded for every incorrect answer and 0 for each question not attempted.
 - (i) Ravi gets 4 correct and 6 incorrect answers. What is his score?
 - (ii) Heena gets 2 correct and 5 incorrect answers. What is her score?
2. An elevator descends into a mine shaft at the rate of 6 m/min. If the descent starts from 20 m above the ground level, how long will it take to reach -370 m?
3. A person earns ₹125 per day and spends ₹ 31 every day. How much would he earn and spend in 30 days? What is his total savings in 30 days?
4. What is the difference between $0 \div 49$ and $49 \div 0$? Explain.
5. What is the value of $0 - 1 + 2 - 3 + 4 - 5 + 6 - 7 + \dots - 17 + 18 - 19 + 20$.
6. Use $<$ or $>$ to compare $-2 \times (-1) \times 4 \times 2 \times (-3)$ and $(-2) + (-1) + 4 + 2 - (-3)$.
7. Use the given number line to find the sign of the quotient $\frac{a-b}{ab}$.
8. Arrange $-9 \div 3$, $-4 + 6$, $2(-4)$, $1 \mid -7 - 3$, $-5 - 3 \times 2$ in ascending order.
9. Add the additive inverse of $(-12 + 16 - 42 \div 3 \text{ of } 7)$ to the product of -2 , -3 and -4 .

3

EXERCISE

Single Option Correct :

DIRECTIONS : This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONLY ONE is correct.

1. While removing brackets the order in which the brackets are removed is
 - (a) $[], (), \{ \}$
 - (b) $\{ \}, (), []$
 - (c) $(), \{ \}, []$
 - (d) none
2. $38 + 83 = 83 + 38$ is an example of
 - (a) commutative property
 - (b) associative property
 - (c) closure property
 - (d) distributive property
3. The number which is neither positive nor negative is
 - (a) 1
 - (b) 5
 - (c) 0
 - (d) 10
4. Smallest negative number
 - (a) -1
 - (b) -10
 - (c) 0
 - (d) Does not exist
5. $-8 - 2[6 + 4\{7 - (2.8 + 1.2)\}]$
 - (a) -34
 - (b) -44
 - (c) $+42$
 - (d) 34
6. $(-43) - (-12) =$
 - (a) $+55$
 - (b) -21
 - (c) -31
 - (d) $+31$
7. The property represented by $a \times (b + c) = a \times b + a \times c$ is
 - (a) commutative property
 - (b) associative property
 - (c) distributive property
 - (d) none of these
8. If 'a' and 'b' are whole numbers, then $a - b$ is
 - (a) a whole number
 - (b) a natural number
 - (c) cannot be determined
 - (d) none of these
9. Value of $(-1) \times (-1) \times (-1) \dots \dots \dots 11$ times, is
 - (a) $+1$
 - (b) 0
 - (c) -1
 - (d) none

10. Parenthesis type of brackets are denoted by
 (a) $[]$ (b) $\{\}$
 (c) $()$ (d) $][$
11. Closure property is observed w.r.t. _____ operations in integers.
 (a) $+, \div$ (b) $+, \div, \times$
 (c) $+, \times, -$ (d) $+, -, \div$
12. Value of $124 \times 4 - 3 + 118 \div 2$ is
 (a) 552 (b) 496
 (c) 553 (d) -553

More Than One Option Correct :

DIRECTIONS : This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONE or MORE may be correct.

1. Which of the following statements is/are correct about integers?
 (a) For every integer a , we have $a \div 1 = a$.
 (b) For all non-zero integers a and b , $a \times b$ is always greater than either a or b .
 (c) The greater the integer, the lesser is its negative.
 (d) Every integer is a natural number.
2. Which of the following statements are correct?
 (a) Division by zero is not defined.
 (b) -3 is greater than zero.
 (c) Any number multiplied by zero will give zero.
 (d) 0 is an integer.
3. Which of the following statement(s) is/are not correct?
 (a) -5 is less than 0 .
 (b) -3 lies between -4 and 4 .
 (c) multiplicative inverse of -1 is a positive number.
 (d) Product of two negative integers is positive.
4. Which of the following statements is correct?
 (a) 0 is called the additive identity for integers.
 (b) 1 is called the multiplicative identity for integers.
 (c) The additive inverse of 0 is zero itself.
 (d) None of these

5. Addition of integers satisfies which of the following property?
 (a) commutative (b) associative
 (c) closure (d) None of these

Assertion & Reason :

DIRECTIONS : Each of these questions contains an Assertion followed by Reason. Read them carefully and answer the question on the basis of following options. You have to select the one that best describes the two statements.

- (a) If both **Assertion** and **Reason** are **correct** and **Reason** is the **correct** explanation of **Assertion**.
 (b) If both **Assertion** and **Reason** are **correct** but **Reason** is not the **correct** explanation of **Assertion**.
 (c) If **Assertion** is **correct** but **Reason** is **incorrect**.
 (d) If **Assertion** is **incorrect** but **Reason** is **correct**.

1. **Assertion:** $-12 \times 2 = -24$
Reason: Sign of greater number among 2 and 12 is negative, then the result must be negative.
2. **Assertion:** Division by zero is not defined.
Reason: Zero is neither negative nor positive.

Passage Based Questions :

DIRECTIONS : Study the given passage(s) and answer the following questions.

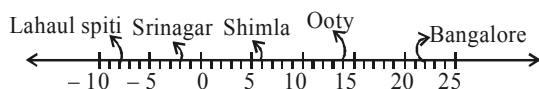
PASSAGE - I

In a class test of 20 questions, 5 marks are given for every correct answer and (-2) marks are given for every incorrect answer. Gurpreet attempts all questions but only 8 of her answers are correct.

1. What was Gurpreet's total score?
 (a) 14 (b) 36
 (c) 16 (d) 28
2. Her friend attempts all questions and get only 7 answers correct. What will be her score?
 (a) 10 (b) 15
 (c) 9 (d) 8
3. If Gurpreet get 10 correct answers, how much more marks she could get?
 (a) 16 (b) 14
 (c) 12 (d) 18

PASSAGE - II

Following number line shows the temperature in degree celsius ($^{\circ}\text{C}$) at different places on a particular day.



4. What is the temperature difference between the hottest and the coldest places among the above?
 - (a) 30°C
 - (b) -30°C
 - (c) 14°C
 - (d) -14°C
5. What is the temperature difference between Lahaul spiti and Srinagar?
 - (a) -6°C
 - (b) 6°C
 - (c) -10°C
 - (d) 10°C
6. The temperature of srinagar and Lahaul Spiti taken together is
 - (a) 10°C
 - (b) -6°C
 - (c) -10°C
 - (d) 6°C

Integer Type Questions :

DIRECTIONS : Answer the following questions. The answer to each of the question is a single digit integer, ranging from 0 to 9.

1. The value of $45 \div \{8 - (-2 \times 5 + 3)\}$ is

2. The deep freezer of a sweet shop should be at -12°C . It is set at -18°C by mistake. What is the difference between these temperature settings.
3. Which is the positive integer closest to zero on the number line?
4. The sum of two integer is -12 . If one of them is -35 , find the sum of digits of other.
5. Find the sum of the smallest positive integer and the greatest negative integer.
6. The additive inverse of -9 is
7. By how much does -3 exceed -5 ?

Multiple Matching Question :

DIRECTIONS : Following question has four statements (A, B, C and D) given in Column-I and five statements (p, q, r, s, t) in Column-II. Any given statement in Column-I can have correct matching with one or more statement(s) given in Column-II.

1. Match the columns.

Column-I	Column-II
(A) $(-1) \times (-1) \times (-1) \times (-1) \dots (30 \text{ times})$	(p) -30
(B) $-3 \times (6 + 4)$	(q) 43
(C) $-5 \times [(-2) \times (-3)]$	(r) Additive inverse of (-1)
(D) $(-10) \times (-5) + (-7)$	(s) $[(-5) \times (-2)] \times (-3)$
	(t) 1

SOLUTIONS

Brief Explanations of Selected Questions

1 EXERCISE

Fill in the Blanks :

- $b +$ additive inverse of a .
- $[(-8) + (-3)] + 8 = (-8) + [(-3) + 8] = -3$
- $(x \times y) \times z = x \times (y \times z)$
- even
- 1
- 73
- 0
- 3, 2
- 6
- 0

True / False :

- | | | |
|----------|----------|----------|
| 1. True | 2. False | 3. True |
| 4. False | 5. False | 6. False |
| 7. True | 8. False | |

Match the Columns :

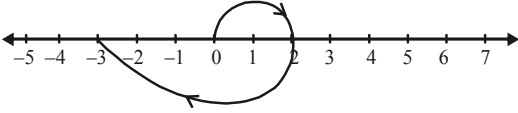
1. (A) $\rightarrow$ (s); (B) $\rightarrow$ (q); (C) $\rightarrow$ (p); (D) $\rightarrow$ (r)

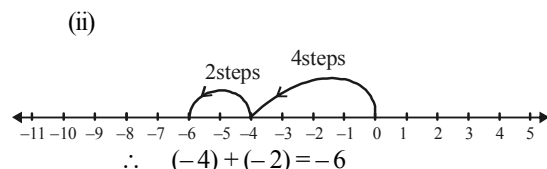
Very Short Answer Questions :

- 360
 - 384
- $1487 \times 327 + (-487) \times 327$
 $= 327 \times (1487 - 487) = 327000$
 - $28945 \times 99 - (-28945)$
 $= 28945 \times 99 + 28945 \times 1$
 $= 28945 \times (99 + 1) = 2894500$
- 72
 - 0
 - 165
- $(-23)\{(-5) + (19)\} = (-23) \times 14 = -322$
 and
 $(-23) \times (-5) + (-23) \times 19 = 115 - 437 = -322$
- $-25 + 14 \div (5 - 3) = -25 + 14 \div 2$
 $= -25 + 7 = (-18)$
- $22 - \frac{1}{4} \{-5 - 3\} = 22 - \frac{1}{4}(-8) = 24$
- $36 - [18 - \{14 - (15 - 4 \div 2 \times 2)\}]$
 $= 36 - [18 - \{14 - (15 - 2 \times 2)\}]$
 $= 36 - [18 - \{14 - 11\}]$
 $= 36 - (18 - 3) = 21$
- $(-15625) \div (-125) = \frac{-15625}{-125} = 125$
 - $30000 \div (-100) = \frac{30000}{-100} = -300$

- $15625 \times (-2) + (-15625) \times 98$
 $= (-15625) \times 2 + (-15625) \times 98$
 $= (-15625) \times (2 + 98)$
 $= (-15625) \times 100 = -1562500$
 - $18946 \times 99 - (-18946)$
 $= 18946 \times 99 + 18946$
 $= 18946 \times (99 + 1)$
 $= 18946 \times 100 = 1894600$
- The number of positive integers will not affect the sign of the product.
 Since 199 is odd and the product of odd number of negative integers is negative, so the product will be negative.
- 39 minutes
- Zero

Short Answer Questions :

- Profit earned on 1 meter cloth = ₹ 8
 Profit earned on 500 metres of cloth
 $= ₹ (500 \times 8) = ₹ 4000$
- Aman has money on Sunday
 $= ₹ (20 - 8 + 5 - 7 - 10 + 5 + 7) = ₹ 12$
- $a + b = (-15) + (-23) = -38$
 $b + a = (-23) + (-15) = -38$
 $\therefore a + b = b + a$
 Now, $a - b = (-15) - (-23) = -15 + 23 = 8$
 $b - a = (-23) - (-15) = -23 + 15 = -8$
 $\therefore a - b \neq b - a$
- $\{(-27) + (-73)\} + (-100)$
 $= (-100) + (-100) = -200$
 - $\{(-51) + (-83)\} - (-12)$
 $= (-134) - (-12) = -122$
 - $\{200 - (-100)\} + (-300)$
 $= 300 + (-300) = 0$
- The required number = $-59 - (-21) = -38$
- The required number = $158 - 221 = -63$
- 
 $\therefore 2 + (-5) = -3$



8. (i) $-92 - (-12) = -92 + 12 = -80$
 (ii) $0 - (+123) = 0 - 123 = -123$
 (iii) $0 - (-150) = 0 + 150 = 150$
9. (i) $-188 - (-887) = -188 + 887 = 699$
 (ii) $-173 - 578 = -751$
 (iii) $282 - 345 = -63$

2 EXERCISE

Text-Book Exercise :

- $-7^{\circ}\text{C}; -3^{\circ}\text{C}$
 - 6200m
 - $(-1) + (-2) = -3$ or $(-5) + 2 = -3$
 - $(-9) - (-4) = -5$ or $(-2) - 3 = -5$
 - $(-7) - (-9) = 2$ or $1 - (-1) = 2$
 - $(-10) + 10 = 0$ or $5 + (-5) = 0$
 - Positive
 - Negative
 - Positive
 - Positive
 - $(-30) \times [13 + (-3)] = (-30) \times 10 = -300$
 $[(-30) \times 13] + [(-30) \times (-3)] = -390 + 90 = -300$
 $\text{So, } (-30) \times [13 + (-3)] = [(-30) \times 13] + [(-30) \times (-3)]$
 - Marks given for one correct answer = 4
 So, marks given for 9 correct answers
 $= 4 \times 9 = 36$

Marks given for one incorrect answer = -2
 So, marks given for 6 (= 15 - 9) incorrect answers = $(-2) \times 6 = -12$
 Therefore, Gurpreet's total score
 $= 36 + (-12) = 24$
 - Marks given for 5 correct answers
 $= 4 \times 5 = 20$

So, marks given for 10 (= 15 - 5) incorrect answers = $(-2) \times 10 = -20$
 Therefore, her friend's total score
 $= 20 + (-20) = 0$
 - Since the elevator is going down, so the distance covered by it will be represented by a negative integer.
 Change in position of the elevator in one minute = -5 m
 Position of the elevator after 60 minutes = $(-5) \times 60 = -300$ m, i.e., 300 m below ground level.
 - Change in position of the elevator in 45 minutes = $(-5) \times 45 = -225$ m, i.e., 225 m below ground level.
 So, the final position of the elevator = $-225 + 15 = -210$ m, i.e., 210 m below ground level.
- 53000
 - 15000
 - 4182
 - 62500
 - -10°C
 - Loss of ₹ 1000
 - 4000 bags
 - Marks given for one correct answer = 5
 So, marks given for 10 correct answers
 $= 5 \times 10 = 50$

Radhika's score = 30
 Marks obtained for incorrect answers
 $= 30 - 50 = -20$

Marks given for one incorrect answer = (-2)
 Therefore, number of incorrect answers
 $= (-20) \div (-2) = 10$
 - Marks given for 4 correct answers
 $= 5 \times 4 = 20$

Jay's score = -12
 Marks obtained for incorrect answers
 $= -12 - 20 = -32$

Therefore, number of incorrect answers
 $= (-32) \div (-2) = 16$
 - Profit earned by selling one pen = ₹ 1
 Profit earned by selling 45 pens = ₹ 45, which we denote by + ₹ 45
 Total loss given = ₹ 5, which we denote by - ₹ 5
 Profit earned + Loss incurred = Total loss
 Therefore, Loss incurred
 $= \text{Total Loss} - \text{Profit earned}$
 $= ₹ (-5 - 45) = ₹ (-50) = -5000$ paise
 Loss incurred by selling one pencil = 40 paise which we write as 40 paise
 So, number of pencils sold
 $= (-5000) \div (-40) = 125$ pencils
 - In the next month there is neither profit nor loss.
 So, Profit earned + Loss incurred = 0
 i.e., Profit earned = - Loss incurred.
 Now, profit earned by selling 70 pens = ₹ 70
 Hence, loss incurred by selling pencils = ₹ 70 which we indicate by
 $- ₹ 70$ or -7,000 paise.
 Total number of pencils sold
 $= (-7000) \div (-40) = 175$ pencils.
 - 9 p.m.; -14°C

Exemplar Questions :

- 5 and – 4 are smaller than – 3 but their difference is $(-4) - (-5) = 1$ which is greater than – 3.

or

– 6 and – 10 are smaller than – 3 but their difference is $(-6) - (-10) = 4$ which is greater than – 3.
- $-3 \Delta 2 = (-3) \times (-3) + 2 \times 2 - (-3) \times 2$
 $= 9 + 4 - (-6) = 13 + 6 = 19.$
- O, $\ominus$, $\otimes$, $\bullet$
- $(-43) \times (-99) + 43 = 43 \times \{(-1)(-99) + 1\}$
 $= 43 \times (99 + 1) = 4300$
- (i) (–17) and (–12) are smaller than –5 and their difference is $(-17) - (-12) = -5$
 (ii) (–12) is smaller than (–11) and (–1) is greater than (–11) but their difference is $(-12) - (-1) = -11$
- Elevation relative to the sea level at the end
 $= 380 + 540 - 268 + 116 - 152 + 490 - 844 + 94$
 $= 356$ m above sea level
- (i) $53 \times (-9) - (-109) \times 53$
 $= 53 \times \{(-9) - (-109)\}$
 $= 53 \times 100 = 5300$
 (ii) $68 \times (-17) + (-68) \times 3$
 $= 68 \times \{(-17) + (-3)\}$
 $= 68 \times (-20) = -1360$
- (i) $(-3) * (-5)$
 $= (-3) \times (-5) + [(-3) \times (-3) + (-5) \times (-5)]$
 $= 15 + (9 + 25) = 49$
 (ii) $(-6) * 2$
 $= (-6) \times 2 + [(-6) \times (-6) + 2 \times 2]$
 $= -12 + (36 + 4) = 28$
- $4 \Delta (-3) = 4 \times (-3) - 2 \times 4 \times (-3) + (-3) \times (-3)$
 $(-4) \times (-3) + (-3) \times (-3)$
 $= (-12) - (-24) + (+108) + 9$
 $= -12 + 24 + 108 + 9 = 129$
- Difference between place A and B
 $= 1800 \text{ m} + 700 \text{ m} = 2500 \text{ m}$
- Let us denote profit by +ve integer and loss by –ve integer. Then
 Net profit/loss $= (+47) + (-12) + (-8) = ₹ 27$
 $\therefore$ He had a net profit of ₹ 27.
- (i) 10 (ii) 30

HOTS Questions :

- (i) Marks given for 1 correct answer = 5
 Marks given for 4 correct answer
 $= 4 \times 5 = 20$

Marks given for 1 incorrect answer = –2

Marks given for 6 incorrect answer
 $= 6 \times (-2) = -12$

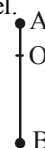
Ravi's total score $= 20 + (-12) = 8$ marks

- Marks given for 2 correct answer
 $= 2 \times 5 = 10$

Marks given for 5 incorrect answer
 $= 5 \times (-2) = -10$

Heena's total score $= 10 + (-10) = 0$

- Let the point O denote the ground level.
 Then, OA = 20 m and OB = –370 m.
 $\therefore AB = 20 - (-370) = 20 + 370$
 $= 390 \text{ m}.$
 Rate of descent = 6 m/min.



Time taken $= \frac{390}{6} \text{ min} = 65 \text{ min} = 1 \text{ hr } 5 \text{ min}.$

- Earning per day = ₹ 125
 Earning in 30 days $= 30 \times 125 = ₹ 3750$
 Expenditure per day = ₹ 31
 Expenditure in 30 days $= 30 \times 31 = ₹ 930$
 His total savings in 30 days
 $= ₹ (3750 - 930) = ₹ 2820$
- $0 \div 49 = 0$ and $49 \div 0$ is meaning less.
- $0 - 1 + 2 - 3 + 4 - 5 + 6 - 7 \dots -17 + 18 - 19 + 20$
 $= 0 + (-1 + 2) + (-3 + 4) + (-5 + 6) + \dots$
 $+ (-17 + 18) + (-19 + 20)$
 $= 0 + 1 + 1 + \dots + 1 + 1 = 10$
- $(-2) \times (-1) \times 4 \times 2 \times (-3)$
 $= -(2 \times 1 \times 4 \times 2 \times 3) = -48$
 and $(-2) + (-1) + 4 + 2 - (-3)$
 $= -3 + 6 + 3 = 6$
 Since $-48 < 6$
 $\therefore -2 \times (-1) \times 4 \times 2 \times (-3) < (-2) + (-1) + 4 + 2 - (-3)$
- Here a is +ve and b is –ve
 $\therefore a - b$ is +ve and ab is –ve
 $\Rightarrow \frac{a-b}{ab}$ is –ve
- $-9 \div 3 = -3$
 $-4 + 6 = 2$
 $2(-4) = -8$
 $1|-7-3| = 1 \times 10 = 10$
 $-5 - 3 \times 2 = -5 - 6 = -11$
 Since $-11 < -8 < -3 < 2 < 10$
 $\therefore -5 - 3 \times 2, 2(-4), -9 \div 3, -4 + 6, 1|-7-3|$
 $(-12 + 16 - 42 \div 3 \text{ of } 7) = (-12 + 16 - 42 \div 21)$
 $= (-12 + 16 - 2) = 2$
 additive inverse of 2 is –2
 Product of –2, –3 and –4
 $= (-2) \times (-3) \times (-4) = -24$
 Now, $-2 + (-24) = -26$

3 EXERCISE

Single Option Correct :

- (c) $() , \{ \} , []$
- (a) Commutative property
- (c)
- (d) Does not exist
- (b) $-8 - 2 [6 + 4 \{ 7 - (4) \}]$
 $= -8 - 2 [6 + 4 \{ 7 - 4 \}]$
 $= -8 - 2 [6 + 4 \{ 3 \}]$
 $= -8 - 2 [6 + 12] = -8 - 2 [18]$
 $= -8 - 36 = -44$
- (c) $(-43) + (+12) = -31$
- (c) Distributive property
- (c) we cannot determine whether $a - b$ is a whole number or not.
- (c) $(-1) \times (-1) \times (-1) \dots \dots \dots 11 \text{ times} = -1$
- (c) Parenthesis ()
- (c) Addition, subtraction, multiplication.
- (a) $[124 \times 4 - 3 + 118 \div 2]$
 $= (124 \times 4) - 3 + 59 = 496 - 3 + 59 = 552$

More Than One Option Correct :

- (a, c)
- (a, c, d)
- (a, b, d)
- (a, b, c)
- (a, b, c) Addition of integers satisfies commutative, associative and closure properties.

Assertion & Reason :

- (c) If **Assertion** is **correct** but **Reason** is **incorrect**.
- (b) If both **Assertion** and **Reason** are correct, but Reason is **not the correct explanation** of Assertion.

Passage Based Questions :

- (c) Score for her correct answers
 $= 8 \times 5 = 40$
 Marks deducted for incorrect answer
 $= 12 \times 2 = 24$
 Total Score $= 40 - 24 = 16$
- (c) Marks for correct answer $= 7 \times 5 = 35$
 Marks deducted for incorrect answer
 $= 13 \times 2 = 26$
 Total score $= 35 - 26 = 9$
- (b) Marks for 10 correct answer
 $= 10 \times 5 = 50$
 Marks deducted for 10 incorrect answers.
 $= 10 \times 2 = 20$
 Total score $= 50 - 20 = 30$
 Difference between marks
 $= 30 - 16 = 14$
- (a) $22 - (-8) = 30^\circ\text{C}$.
- (b) $-2 - (-8) = 6^\circ\text{C}$
- (c) $(-2) + (-8) = -10^\circ\text{C}$

Integer Type Questions :

- (3) $45 \div \{ 8 - (-10 + 3) \} = 45 \div \{ 8 + 7 \}$
 $= 45 \times \frac{1}{15} = 3$
- (6) $-12 - (-18) = 6^\circ\text{C}$
- (1)
- (5) The sum of two integers $= -12$
 One of them $= -35$
 Let the other integer be x .
 $\therefore -35 + x = -12$
 $\Rightarrow x = -12 + 35 = 23$
 Sum of digits of x i.e., 23 is $2 + 3 = 5$
- (0) Smallest positive integer $= 1$
 Greatest negative integer $= -1$
 $\therefore$ Their sum $= 1 + (-1) = 0$
- (9) The additive inverse of -9 is 9.
- (2) Let -3 exceed -5 by x
 $\Rightarrow -3 - x = -5 \Rightarrow -3 + 5 = x \Rightarrow x = 2$

Multiple Matching Question :

- (A) $\rightarrow$ (r, t); (B) $\rightarrow$ (p, s); (C) $\rightarrow$ (p, s), (D) $\rightarrow$ (q)

Chapter

2

Fractions and Decimals

INTRODUCTION

You have learnt fractions and decimals in earlier classes. The study of fractions included different kinds of fractions, comparison and ordering of fractions, representation of the fractions on number line. You shall also learn the concept of decimals in this chapter.

FRACTIONS

The number of the form $\frac{a}{b}$, where $b \neq 0$ and a, b are whole numbers is called a *fraction*.

For fraction $\frac{a}{b}$, a is called *numerator* and b is called *denominator* of the fraction.

TYPES OF FRACTIONS

There are different types of fractions. Let us learn about them.

Proper fraction

A fraction whose numerator is less than its denominator is called a proper fraction. A proper fraction is always less than a whole, e.g. $\frac{5}{7}, \frac{3}{8}, \frac{2}{5}$ etc.

Improper fraction

A fraction whose numerator is greater than or equal to its denominator is called an improper fraction. An improper fraction is always equal to or greater than a whole, e.g. $\frac{4}{3}, \frac{5}{2}, \frac{8}{5}$ etc.

Unit fraction

Fractions with numerator 1 are called unit fractions, e.g. $\frac{1}{4}, \frac{1}{7}, \frac{1}{9}$ etc.

Mixed fraction

A combination of a whole and a proper fraction is called a mixed fraction, e.g. $5\frac{1}{3}, 2\frac{2}{5}, 7\frac{9}{11}$ etc.

EQUIVALENT FRACTIONS

Equivalent fractions are the fractions which represent the same value. They represent the same part of a whole. To find equivalent fractions of any given fraction :

- (i) We may multiply both the numerator and the denominator of the given fraction by the same number.
- (ii) We may divide both the numerator and the denominator of the given fraction by the same number.

ILLUSTRATION : 1

Find two fractions equivalent to $\frac{5}{7}$.

SOLUTION :

$$\frac{5}{7} = \frac{5 \times 2}{7 \times 2} = \frac{10}{14}, \quad \frac{5}{7} = \frac{5 \times 3}{7 \times 3} = \frac{15}{21}$$

LIKE AND UNLIKE FRACTIONS

Fractions with same denominators are called *like fractions*.

For example :

$\frac{5}{15}, \frac{3}{15}, \frac{17}{15}$ and $\frac{31}{15}$ are unlike fractions.

The fractions with different denominators are called *unlike fractions*

For example : $\frac{2}{7}, \frac{9}{11}, \frac{3}{13}$ and $\frac{39}{46}$ are unlike fractions.

COMPARISON OF FRACTIONS

Comparison of Like Fractions

When we compare two fractions having same denominator, then the fraction with greater numerator will be greater.

For example : $\frac{7}{15} > \frac{4}{15}$

Comparison of unlike fractions.

Unlike fractions can be compared by any of the following two methods :

1. By converting them into like fractions :

To compare unlike fractions, we first convert them into like fractions by obtaining their equivalent fraction with a denominator which is LCM of the denominators of both the fractions.

e.g. $\frac{3}{5}$ and $\frac{2}{7}$ can be compared by converting them into like fractions having their denominator equal to LCM of 5 and 7.

L.C.M. of (5, 7) = 35

$$\frac{3 \times 7}{5 \times 7} = \frac{21}{35} \text{ and } \frac{2 \times 5}{7 \times 5} = \frac{10}{35} \quad \text{Now, } \frac{21}{35} > \frac{10}{35} \quad \therefore \frac{3}{5} > \frac{2}{7}$$

2. Cross multiplication method :

To compare any two fractions such as $\frac{a}{b}$ and $\frac{c}{d}$, we cross multiply as shown below.

$$\frac{a}{b} \times \frac{c}{d}$$

Thus, the cross products are ad and bc .

- (a) If $ad > cd$ then $\frac{a}{b} > \frac{c}{d}$ (b) If $ad < cd$ then $\frac{a}{b} < \frac{c}{d}$ (c) If $ad = cb$ then $\frac{a}{b} = \frac{c}{d}$

NOTE: When two fractions have same numerator, but different denominators, then the fraction with smaller denominator is greater. **For example :** $\frac{5}{7} > \frac{5}{9}$

ILLUSTRATION : 2

Compare : $\frac{11}{5}$ and $\frac{13}{3}$

SOLUTION :

$$\frac{11}{5} \text{ and } \frac{13}{3}$$

By Cross multiplication, we see that

$$\frac{11}{5} \times \frac{13}{3}$$

$$\Rightarrow 33 < 65$$

$$\text{So } \frac{11}{5} < \frac{13}{3}$$

- **Addition and subtraction of fractions :** For addition or subtraction, we first find out the LCM of the denominators, then we convert each fraction into an equivalent fraction whose denominator is equal to the LCM. Finally, we add or subtract the like fractions obtained above.

For example : $\frac{3}{5} + \frac{7}{2}$

LCM of 5, 2 = 10

Like fraction $\Rightarrow \frac{3}{5} = \frac{6}{10}$ and $\frac{7}{2} = \frac{35}{10}$

$$\frac{6}{10} + \frac{35}{10} = \frac{41}{10}$$

MULTIPLICATION OF FRACTIONS

Multiplication of a fraction by a whole Number

To multiply a whole number with a proper or an improper fraction, we multiply the whole number with the numerator of the fraction, keeping the denominator same.

To multiply a mixed fraction by a whole number, first convert it into an improper fraction and then multiply.

For example: $2 \times \frac{2}{3} = \frac{2 \times 2}{3} = \frac{4}{3}$

and $1\frac{1}{4} \times 3 = \frac{5}{4} \times 3 = \frac{5 \times 3}{4} = \frac{15}{4}$

Multiplication of a fraction by a fraction

To multiply a fraction by a fraction, multiply their numerators and denominators.

$\therefore$ Product of fractions = $\frac{\text{Product of numerators}}{\text{Product of denominators}}$

For example: $\frac{5}{7} \times \frac{9}{4} = \frac{5 \times 9}{7 \times 4} = \frac{45}{28}$

and $2\frac{3}{4} \times 5\frac{3}{5} = \frac{11}{4} \times \frac{28}{5} = \frac{11 \times \cancel{28}^7}{\cancel{4}_1 \times 5} = \frac{11 \times 7}{1 \times 5} = \frac{77}{5}$

NOTE: When two proper fractions are multiplied, the product is less than each of the fractions.
The value of the product of two improper fractions is more than each of the two fractions.

NOTE: Fraction as an operator 'of', "of" represents multiplication.

ILLUSTRATION : 3

Find 4 of $\frac{1}{3}$

SOLUTION :

$$4 \text{ of } \frac{1}{3} = 4 \times \frac{1}{3} = \frac{4 \times 1}{3} = \frac{4}{3} = 1\frac{1}{3}$$

ILLUSTRATION : 4

Multiply (i) $\frac{5}{6} \times 2\frac{3}{7}$ (ii) $3\frac{4}{7} \times \frac{3}{5}$

SOLUTION :

(i) $\frac{5}{6} \times 2\frac{3}{7} = \frac{5}{6} \times \frac{17}{7} = \frac{5 \times 17}{6 \times 7} = \frac{85}{42}$

(ii) $3\frac{4}{7} \times \frac{3}{5} = \frac{25}{7} \times \frac{3}{\cancel{5}_1} = \frac{5 \times 3}{7 \times 1} = \frac{15}{7}$

RECIPROCAL OF FRACTIONS

Reciprocal of a fraction : When a fraction is inverted its *reciprocal* is obtained. The non zero numbers whose product with each other is 1 are called *reciprocals* of each other.

For example : $\frac{4}{9} \times \frac{9}{4} = 1$

Thus, $\frac{9}{4}$ is the reciprocal of $\frac{4}{9}$ and vice versa

DIVISION OF FRACTIONS

Division of whole number by a fraction

For dividing a whole number by a fraction. First obtain the reciprocal of the fraction and then multiply it with the whole number.

For example : $3 \div \frac{1}{4} = 3 \times \frac{4}{1} = 12$

For dividing a whole number by a mixed fraction, first convert the mixed fraction into improper fraction and then divide

For example: $15 \div 3\frac{1}{2} = 15 \div \frac{7}{2} = 15 \times \frac{2}{7} = \frac{30}{7} = 4\frac{2}{7}$

Division of a fraction by another fraction

To divide two fractions, we change the division sign into multiplication and write reciprocal of divisor i.e, multiply fraction with the reciprocal of the other fraction.

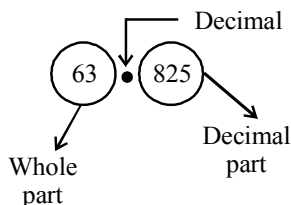
For example : $5\frac{1}{6} \div \frac{9}{2} = \frac{31}{6} \div \frac{9}{2} = \frac{31}{\cancel{6}_3} \times \frac{\cancel{2}^1}{9} = \frac{31}{27} = 1\frac{4}{27}$

DECIMALS

Numbers in form of 1.18, 22.36, 853.4448 are known as *decimal numbers*.

A decimal number have two parts. A whole number part and decimal number part. These two parts are separated by a dot. This dot is known as *decimal point*. The whole number part is in the left of the decimal point and the decimal part to the right of decimal point.

Thus, in 63.825



The number of digits contained in the decimal part shows its decimal places.

For example: 6.438 have three digits in decimal part which is known as three decimal places.

Like and Unlike Decimals

Decimals having the same number of decimal places are called **like decimals**.

For example: 6.238, 38.937, 48.385 are like decimals.

Decimals having different number of decimal places are called **unlike decimals**.

For example: 6.238, 32.73, 65.3984 are unlike decimals.

NOTE: On adding zeros after the last digit of the decimal part of any decimal number does not change the value of the decimal.

For example: 6.28 can be 6.280 or 6.2800 but the value of all three are same.

We use this property to convert the unlike decimal into like decimal.

Comparing decimals:

To compare two given decimals, We have the following steps:

1. Convert the given decimals into like decimals.
2. First compare the whole number part. The decimal with greater whole number part is greater.
3. If the whole-number parts are equal, compare the tenths digits. The decimal with bigger digit in tenths place is greater.
4. If the tenths digits are also equal, compare the hundredths digits, and so on.

ILLUSTRATION : 5

Compare the following.

5.74, 6.03, 0.8, 0.648 and 8.2.

SOLUTION : Converting the given numbers into like decimals, we get them as:

5.740, 6.030, 0.800, 0.648 and 8.200

Clearly $0.648 < 0.800 < 5.740 < 6.030 < 8.200$

ILLUSTRATION : 6

Express the following as mentioned,

(i) 46mm as cm

(ii) 235paise as ₹

(iii) 15mg as g

SOLUTION :

(i) 10mm = 1 cm

So, $46\text{mm} = \frac{46}{10}\text{cm} = 4.6\text{cm}$

(ii) 100paise = ₹1

So, $235\text{paise} = ₹ \frac{235}{100} = ₹2.35 = 2 \text{ rupees } 35 \text{ paise}$

(iii) 1000mg = 1gm

So, $15\text{mg} = \frac{15}{1000}\text{gm} = 0.015\text{gm}$

ADDITION AND SUBTRACTION OF DECIMALS

Addition and subtraction of decimal numbers:

- (i) First convert all the decimal numbers to like decimals

- (ii) Write the given numbers one below the other with decimal points in vertical line.
 (iii) Then add or subtract the numbers and put decimal in result directly under the other decimal points.

ILLUSTRATION 7 :

(i) Add: 0.872, 3.46, 4.309, 3.17 (ii) Subtract: 17.182 from 360.05

SOLUTION :

$ \begin{array}{r} 0.872 \\ 3.460 \\ 4.309 \\ + 3.170 \\ \hline 11.811 \end{array} $	$ \begin{array}{r} 360.050 \\ - 17.182 \\ \hline 342.868 \end{array} $
---	---

MULTIPLICATION OF DECIMAL NUMBER

To multiply a decimal number by a whole number or a decimal number multiply the numbers first as there were no decimal point at all and fix the position by the rule that there are as many decimal places in the product as they are in multiplier and multiplicand put together.

For example :

0.6	×	0.02	=	0.012
1 place		2 places		3 places
7	×	0.003	=	0.021
0 place		3 places		3 places

Multiplication of decimals by 10, 100, 1000:

To multiply a decimal number by 10 or any power of 10 move the decimal point as many places to the right as there are zeros in the multiplier or we can say that multiplying a decimal number by 10^n moves the decimal point n places to the right.

For example: $6.432 \times 10 = 64.32$	Move decimal point 1 place
$38.64 \times 100 = 3864.00$	Move decimal point 2 place
$2.483 \times 1000 = 2483.00$	Move decimal point 3 place

DIVISION OF DECIMAL NUMBERS**Division of decimals by 10, 100, 1000**

While dividing a number by 10, 100 or 1000, the digits of the number and the quotient are same but the decimal point in the quotient shifts to the left by as many places as there are zeros over one.

ILLUSTRATION 8 :

Divide

(i) 2.38 by 10

(ii) 3125.62 by 1000

SOLUTION :

(i) $2.38 \div 10 = 0.238$

(Decimal is shifted to one place to the left)

(ii) $3125.62 \div 1000 = 3.12562$

(Decimal is shifted to three places to the left)

Division of a Decimal Number by a Whole Number

Divide the decimal number, treating it as whole number by the given whole numbers. Put the decimal in quotient so that it has same number of decimal places as the given decimal number.

ILLUSTRATION 9 :

Divide 15.064 by 28

SOLUTION :

$$\begin{array}{r} 0.538 \\ 28 \overline{) 15.064} \\ \underline{-140} \downarrow \\ 106 \downarrow \\ \underline{-84} \downarrow \\ 224 \\ \underline{-224} \\ 0 \end{array}$$

$$\therefore 15.064 \div 28 = 0.538$$

Dividing Decimal Number by multiple of 10

Dividing a whole number or decimal number by 10 or any higher power of 10 (10^n) move the decimal point that number of places (n) to the left.

- For example :**
- (i) $53 \div 10 = 5.3$
 - (ii) $127.5 \div 100 = 1.275$
 - (iii) $6.25 \div 1000 = 0.00625$

Division of a Decimal Number by Another Decimal Number

To divide a decimal number by a decimal number move the decimal point of divisor to the right until it becomes a whole number (i.e., multiply it by 10 or a power of 10). Then move the decimal point of the dividend the same number of places to the right, adding zero if necessary.

ILLUSTRATION 10 :

Divide 49.08 by 0.012

SOLUTION :

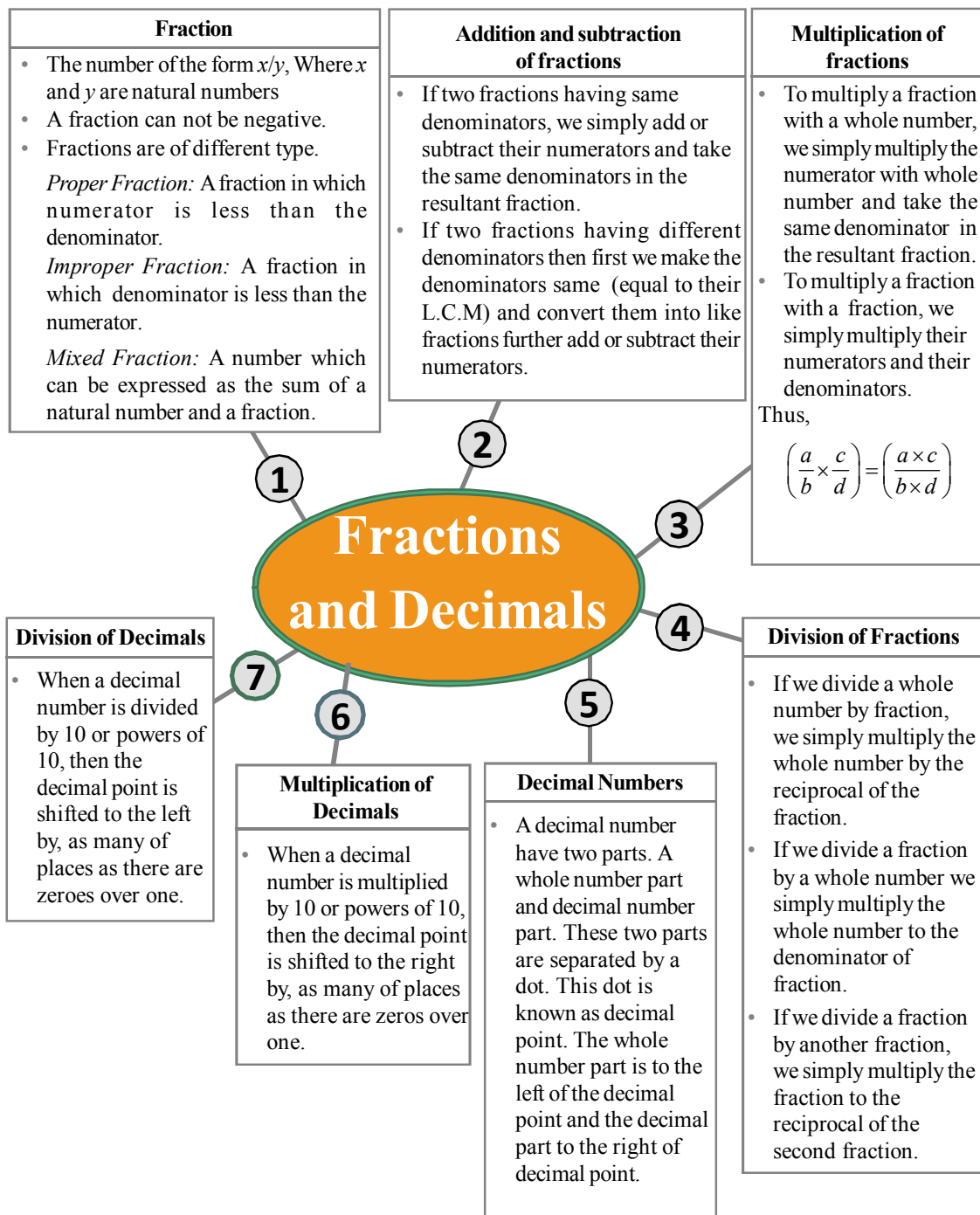
$$\begin{array}{ll} 49.08 \times 1000 = 49080 & \boxed{\text{Moving decimal point}} \\ 0.012 \times 1000 = 12 & \boxed{3 \text{ places to the right}} \end{array}$$

$$\frac{49.08}{0.012} = \frac{49080}{12}$$

$$\begin{array}{r} 4090 \\ 12 \overline{) 49080} \\ \underline{-48} \downarrow \\ 108 \downarrow \\ \underline{-108} \\ 0 \end{array}$$

$$\text{Hence, } 49.08 \div 0.012 = 4090$$

CONCEPT MAP



MISCELLANEOUS

SOLVED EXAMPLES

1. Bongul walked $5\frac{1}{6}$ km to reach the school from home. Lingwan walked $\frac{35}{8}$ km from home to school. Who lives closer to the school and by how many kilometres?

Sol. Bongul walked $5\frac{1}{6}$ km = $\frac{31}{6}$ km

Lingwan walked $\frac{35}{8}$ km

$$\frac{31}{6} \begin{array}{c} \nearrow \searrow \\ \times \end{array} \frac{35}{8}$$

$$248 > 210$$

$$\Rightarrow \frac{31}{6} > \frac{35}{8}$$

$\therefore$ Lingwan lives closer to the school by

$$\frac{31}{6} - \frac{35}{8} = \frac{124 - 105}{24} = \frac{19}{24} \text{ km}$$

2. A bag contains $12\frac{1}{2}$ kg of flour. Two-fifth of it was lost through a hole in the side. How much of it was lost?

Sol. Weight lost = $\frac{2}{5}$ of total weight = $\frac{2}{5}$ of $12\frac{1}{2}$ kg

$$= \frac{2}{5} \text{ of } \frac{25}{2} \text{ kg} = \frac{\cancel{2}}{\cancel{2}} \times \frac{\cancel{25}}{\cancel{2}} \text{ kg} = 5 \text{ kg.}$$

3. A carpenter cuts off from a plank $\frac{5}{12}$ of its length and then $\frac{6}{7}$ of what remains. If the remaining piece is $2\frac{1}{2}$ m long, find the original length of the plank.

Sol. Let the original length of the plank be x m.

$$\text{Length of plank cut off} = \frac{5}{12}x$$

$$\text{Length of plank left} = x - \frac{5}{12}x$$

$$= \left(\frac{12-5}{12}\right)x = \frac{7x}{12}$$

$$\text{Length of plank cut next} = \frac{6}{7} \text{ of } \frac{7x}{12} = \frac{1}{2}x$$

$$\text{Remaining length} = x - \left(\frac{5x}{12} + \frac{1}{2}x\right)$$

$$= \left\{1 - \left(\frac{5}{12} + \frac{1}{2}\right)\right\}x$$

$$= \left(\frac{12-5-6}{12}\right)x = \frac{1}{12}x$$

$$\text{But } \frac{1}{12}x = 2\frac{1}{2}$$

$$\Rightarrow x = \frac{5}{2} \div \frac{1}{12} = 30 \text{ m}$$

Hence, the original length of plank was 30 m.

4. **Simplify:**

$$(i) \quad \frac{3}{19} \text{ of } \left\{ \frac{7}{9} + \left(\frac{3}{4} + \frac{1}{2} - \frac{1}{3} \right) \right\}$$

$$(ii) \quad \left(5\frac{1}{4} - 2\frac{1}{3} \right) + \frac{1}{3} \text{ of } \left(5\frac{1}{2} + 2\frac{1}{5} \right)$$

$$\text{Sol. (i)} \quad \frac{3}{19} \text{ of } \left\{ \frac{7}{9} + \left(\frac{3}{4} + \frac{1}{2} - \frac{1}{3} \right) \right\}$$

$$= \frac{3}{19} \text{ of } \left\{ \frac{7}{9} + \left(\frac{3}{4} + \frac{3-2}{6} \right) \right\}$$

$$\begin{aligned}
 &= \frac{3}{19} \text{ of } \left\{ \frac{7}{9} + \left(\frac{3}{4} \div \frac{1}{6} \right) \right\} \\
 &= \frac{3}{19} \text{ of } \left\{ \frac{7}{9} + \frac{3}{4} \times \frac{6}{1} \right\} \\
 &= \frac{3}{19} \text{ of } \left\{ \frac{7}{9} + \frac{9}{2} \right\} \\
 &= \frac{3}{19} \text{ of } \frac{14+81}{18} = \frac{3}{19} \text{ of } \frac{95}{18} \\
 &= \frac{1}{19} \times \frac{5}{6} = \frac{5}{114}
 \end{aligned}$$

$$\begin{aligned}
 \text{(ii)} \quad &\left(5\frac{1}{4} - 2\frac{1}{3} \right) + \frac{1}{3} \text{ of } \left(5\frac{1}{2} \div 2\frac{1}{5} \right) \\
 &= \left(\frac{21}{4} - \frac{7}{3} \right) + \frac{1}{3} \left(\frac{11}{2} \div \frac{11}{5} \right) \\
 &= \left(\frac{21 \times 3 - 7 \times 4}{12} \right) + \frac{1}{3} \times \left(\frac{11}{2} \times \frac{5}{11} \right) \\
 &= \frac{63-28}{12} + \frac{5}{6} = \frac{35}{12} + \frac{5}{6} \\
 &= \frac{35+5 \times 2}{12} = \frac{45}{12} = \frac{15}{4} = 3\frac{3}{4}
 \end{aligned}$$

5. The cost of $5\frac{1}{4}$ kg of mangoes is ₹ 231. At what rate per kg are the mangoes being sold?

Sol. Cost of $\frac{21}{4}$ kg of mangoes = ₹ 231

$$\Rightarrow \text{Cost of 1 kg of mangoes} = ₹ \left(231 \div \frac{21}{4} \right)$$

$$= ₹ \left(\frac{11}{231} \times \frac{4}{1} \right) = ₹ 44.$$

Hence, the mangoes are being sold at ₹ 44 per kg.

6. Shyama bought 4 kg 350 g potatoes, 3 kg 80 g tomatoes and some onions. If the total weight of the three vegetables is 10 kg 200 g, what is the weight of onions?

Sol. Total weight of all the vegetables
 $= 10 \text{ kg } 200 \text{ g} = 10.200 \text{ kg}.$
 Weight of potatoes = 4 kg 350 g = 4.350 kg.
 Weight of tomatoes = 3 kg 80 g = 3.080 kg.
 Weight of onions
 $= [10.200 - (4.350 + 3.080)] \text{ kg}$
 $= (10.200 - 7.430) \text{ kg} = 2.770 \text{ kg}.$
 Hence, the weight of onions is 2 kg 770 g.

7. Simplify: $36.65 - 15.79 + 85.2 - 57.615$

Sol. We have,
 $36.65 - 15.79 + 85.2 - 57.615$
 Converting the given decimals into like decimals.

$$\begin{aligned}
 &= 36.650 - 15.790 + 85.200 - 57.615 \\
 &= (36.650 + 85.200) - (15.790 + 57.615) \\
 &= 121.850 - 73.405 \\
 &= 48.445
 \end{aligned}$$

8. Each side of a polygon is 2.9 cm in length and its perimeter is 17.4 cm. How many sides does the polygon have?

Sol. Let the number of sides of the polygon be n .
 Length of each side of the polygon = 2.9 cm.
 $\therefore$ perimeter of the polygon = $(2.9 \times n)$ cm.
 But, its perimeter = 17.4 cm (given).
 $\therefore 2.9 \times n = 17.4$

$$\Rightarrow n = \frac{17.4}{2.9} = \frac{174}{29} = 6.$$

Hence, the given polygon has 6 sides.

9. Express in kg using decimals:

- (i) 60 g (ii) 7380 g
 (iii) 6 kg 8 g

Sol. We have:

$$(i) \quad 60 \text{ g} = \frac{60}{1000} \text{ kg} = \frac{6}{100} \text{ kg} = 0.06 \text{ kg}.$$

$$(ii) \quad 7380 \text{ g} = \frac{7380}{1000} \text{ kg} = 7.380 \text{ kg}.$$

$$(iii) \quad 6 \text{ kg } 8 \text{ g} = (6 \times 1000) \text{ g} + 8 \text{ g} = (6008) \text{ g}$$

$$= \frac{6008}{1000} \text{ kg} = 6.008 \text{ kg}.$$

1

EXERCISE

Fill in the Blanks :

DIRECTIONS : Complete the following statements with an appropriate word / term to be filled in the blank space(s).

1. A fraction in which the numerator is greater than the denominator is called _____.
2. The product of two proper fractions is always _____ fraction.
3. $0.2 \times 0.3 =$ _____.
4. Reciprocal of $8\frac{2}{5}$ is _____.
5. $13\frac{1}{2} \div 8 =$ _____.
6. $69\frac{3}{4} \div 7\frac{3}{4} =$ _____.
7. $1.001 \div 14 =$ _____.
8. $204 \div 0.17 =$ _____.
9. $0.47 \times 5.3 =$ _____.
10. $0.023 \times 0.03 =$ _____.
11. $75.57 \div 0.01 =$ _____.

True / False :

DIRECTIONS : Read the following statements and write your answer as true or false.

1. The product of two fractions is always a fraction.
2. The product of 1 and a fraction is the fraction itself.
3. $\frac{5}{2} \times \frac{3}{2} = \frac{3}{2} \times \frac{2}{5}$
4. The sum of any two proper fractions is an improper fraction.
5. The product of a proper and an improper fraction is less than the improper fraction.

6. The product of $6\frac{11}{14} \times 3\frac{1}{2}$ is $23\frac{3}{4}$.
7. $\frac{1}{2}$ of $\frac{6}{7} > \frac{2}{3}$ of $\frac{3}{7}$.
8. 0.35 of a litre is 300 ml.

Match the Columns :

DIRECTIONS : Each question contains statements given in two columns which have to be matched. Statements (A, B, C, D) in column-I have to be matched with statements (p, q, r, s) in column-II.

1.	Column-I	Column-II
(A)	$\frac{22}{15} - \frac{9}{10}$	(p) $\frac{54}{55}$
(B)	$2\frac{1}{9} + 6$	(q) $\frac{15}{4}$
(C)	$\frac{6}{5} \times \frac{9}{11}$	(r) $\frac{17}{30}$
(D)	$\frac{3}{13} \div \frac{4}{65}$	(s) $\frac{73}{9}$

Very Short Answer Questions :

DIRECTIONS : Give answer in one word or one sentence.

1. Find the difference:
 - (i) $\frac{7}{9} - \frac{5}{9}$
 - (ii) $\frac{13}{16} - \frac{7}{12}$
 - (iii) $\frac{11}{15} - \frac{9}{20}$

2. Multiply and reduce to lowest form (if possible):

(i) $\frac{2}{3} \times \frac{5}{4}$ (ii) $\frac{1}{3} \times \frac{15}{8}$

(iii) $\frac{4}{5} \times \frac{12}{7}$

3. Find:

(i) $\frac{1}{5}$ of a rupee (ii) $\frac{5}{8}$ of a day

(iii) $\frac{7}{8}$ of a kilogram (iv) $\frac{4}{5}$ of an hour

4. Write down the reciprocal of:

(i) $\frac{3}{7}$ (ii) $\frac{1}{5}$

(iii) 6 (iv) $3\frac{5}{8}$

5. Simplify:

(i) $\frac{4}{9} \div \frac{2}{3}$ (ii) $3\frac{3}{7} \div \frac{8}{21}$

(iii) $15\frac{3}{7} \div 1\frac{23}{49}$

6. Fill in the blanks:

(i) kg = 1.05 mg

(ii) g = 2.35 mg

(iii) 14.25 km = mm

(iv) m = 37500 cm

7. Multiply.

(i) 0.01×4.76 (ii) 0.1×87

(iii) 0.001×63.3 (iv) 0.01×0.034

8. Divide.

(i) $0.005 \div 0.1$ (ii) $0.073 \div 0.01$

(iii) $0.06 \div 0.001$ (iv) $0.0078 \div 0.001$

9. Add the following decimals.

(i) 5.5, 13.7, 12.8, 0.35 and 1.11

(ii) 3.067, 10.09, 1.287, 71.715 and 4

(iii) 9.09, 90.9, 0.909, 99.09, and 90.099

(iv) 777, 7.7, 7.707, 70.77 and 0.777

(v) 2.367, 15.55, 279.5, 1.32 and 27

10. Subtract:

(i) 37.5 from 48.31

(ii) 2.619 from 31.52

(iii) 761.23 from 1000

(iv) 1.2345 from 7.0091

(v) 0.7894 from 1.023

11. Arrange the following numbers in ascending order.

(i) 3.572; 3.275; 3.725; 3.524

(ii) 7.081; 7.108; 7.801; 7.018

(iii) 21.6; 21.590; 21.71; 21.8

(iv) 4.70; 4.381; 4.8; 4.090

Short Answer Questions :

DIRECTIONS : Give answer in 2-3 sentences.

- A tailor uses $\frac{2}{3}$ metre of cloth to make a blouse for a dancer. If 150 students are taking part in the dance, then how many metres of cloth will the tailor be using?
- In a class of 48 students, $\frac{1}{4}$ of them regularly watch a particular TV programme. How many of the students do not regularly watch the programme?
- The snowfall during three winter months was 30.5 cm, 45.25 cm, and 25.25 cm. How much snow fell altogether in these three months?
- Ashish took 37.75 seconds to finish a race. Anil finished the race 1.8 seconds before Ashish did. How long did it take Anil to run the race?
- Sarita bought $5\frac{3}{4}$ kg potatoes and $3\frac{1}{2}$ kg tomatoes from a vendor. What is the total weight of vegetables bought by her?
- What should be subtracted from $17\frac{3}{4}$ to get $11\frac{2}{3}$?
- Sajal can walk $2\frac{2}{5}$ km in an hour. How much distance will he cover in $3\frac{1}{3}$ hours?
- Multiply 0.089 by 0.76.
- Divide 3.28 by 0.4
- Divide 0.00942 by 0.314

Long Answer Questions :

DIRECTIONS : Give answer in four to five sentences.

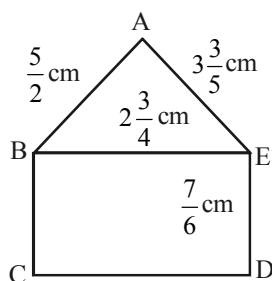
- Simplify: $3\frac{1}{5} + 2\frac{1}{10} - 1\frac{1}{2} - \frac{1}{4}$.
- Milk is sold at ₹ $16\frac{3}{4}$ per litre. Find the cost of $6\frac{2}{5}$ litres of milk.
- Aakash bought vegetables weighing 10 kg. Out of this 3 kg 500 g is onions, 2 kg 75 g is tomatoes and the rest is potatoes. What is the weight of potatoes?
- A car covers a distance of 89.1 km in 2.2 hours. What is the distance covered by it in 1 hour? Also find how much distance is covered in 4.3 hours?
- One apple weighs 74 g and one orange weighs 42 g. Determine the weight of 132 such apples and 46 such oranges in kilograms.
- The product of two numbers is $15\frac{5}{6}$. If one of the numbers is $6\frac{2}{3}$. Find the other.
- The total weight of some bags of cement is 1743 kg. If each bag weighs 49.8 kg. how many bags are there?

2

EXERCISE

Text-Book Exercise :

- Solve:
 - $4 + \frac{7}{8}$
 - $\frac{9}{11} - \frac{4}{15}$
 - $8\frac{1}{2} - 3\frac{5}{8}$
 - $\frac{7}{10} + \frac{2}{5} + \frac{3}{2}$
- Arrange the following in descending order:
 - $\frac{2}{9}, \frac{2}{3}, \frac{8}{21}$
 - $\frac{1}{5}, \frac{3}{7}, \frac{7}{10}$
- Find the perimeters of (i) $\triangle ABE$ (ii) the rectangle BCDE in this figure. Whose perimeter is greater?



- Salil wants to put a picture in a frame. The

picture is $7\frac{3}{5}$ cm wide. To fit in the frame the

picture cannot be more than $7\frac{3}{10}$ cm wide. How much should the picture be trimmed?

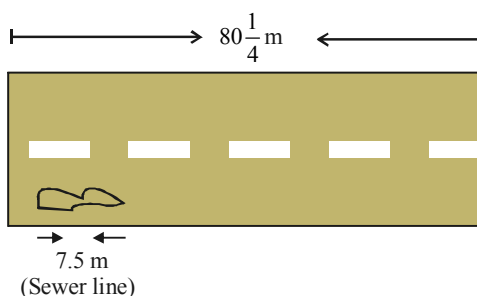
- Michael finished colouring a picture in $\frac{7}{12}$ hour. Vaibhav finished colouring the same picture in $\frac{3}{4}$ hour. Who worked longer? By what fraction was it longer?
- Find:
 - $\frac{2}{3}$ of (a) 18 (b) 27
 - $\frac{4}{5}$ of (a) 20 (b) 35
- Multiply and express as a mixed fraction:
 - $7 \times 2\frac{1}{4}$
 - $4 \times 6\frac{1}{3}$
 - $3\frac{1}{4} \times 6$
 - $3\frac{2}{5} \times 8$

8. Vidya and Pratap went for a picnic. Their mother gave them a water bottle that contained 5 litres of water. Vidya consumed $\frac{2}{5}$ of the water. Pratap consumed the remaining water.
 (i) How much water did Vidya drink?
 (ii) What fraction of the total quantity of water did Pratap drink?
9. Multiply the following fractions:
 (i) $\frac{2}{5} \times 5\frac{1}{4}$ (ii) $6\frac{2}{5} \times \frac{7}{9}$
 (iii) $\frac{3}{2} \times 5\frac{1}{3}$ (iv) $\frac{5}{6} \times 2\frac{3}{7}$
10. Saili plants 4 saplings, in a row, in her garden. The distance between two adjacent saplings is $\frac{3}{4}$ m. Find the distance between the first and the last sapling.
11. Lipika reads a book for $1\frac{3}{4}$ hours everyday. She reads the entire book in 6 days. How many hours in all were required by her to read the book?
12. Find:
 (i) $3\frac{1}{2} \div \frac{8}{3}$ (ii) $\frac{2}{5} \div 1\frac{1}{2}$
 (iii) $3\frac{1}{5} \div 1\frac{2}{3}$ (iv) $2\frac{1}{5} \div 1\frac{1}{5}$
13. Express as rupees using decimals:
 (i) 7 paise (ii) 7 rupees 7 paise
 (iii) 77 rupees 77 paise (iv) 235 paise.
14. Write the following decimal numbers in the expanded form:
 (i) 20.03 (ii) 2.03
 (iii) 200.03 (iv) 2.034
15. Write the place value of 2 in the following decimal numbers:
 (i) 2.56 (ii) 21.37
 (iii) 10.25 (iv) 63.352
16. How much less is 28 km than 42.6 km?
17. Find:
 (i) 153.7×10 (ii) 168.07×10
 (iii) 156.1×100 (iv) 43.07×100
 (v) 0.08×10 (vi) 0.03×1000
18. Find:
 (i) 0.1×51.7 (ii) 0.2×316.8
 (iii) 11.2×0.15 (iv) 1.07×0.02
 (v) 10.05×1.05 (vi) 101.01×0.01
19. Find:
 (i) $0.35 \div 5$ (ii) $651.2 \div 4$
 (iii) $14.49 \div 7$ (iv) $0.80 \div 5$
20. Find:
 (i) $2.7 \div 100$ (ii) $0.78 \div 100$
 (iii) $23.6 \div 100$ (iv) $98.53 \div 100$
21. Find:
 (i) $30.94 \div 0.7$ (ii) $7.75 \div 0.25$
 (iii) $76.5 \div 0.15$ (iv) $2.73 \div 1.3$

Exemplar Questions :

1. If $\frac{2}{3}$ of a number is 10, then what is 1.75 times of that number?
2. In a class of 40 students, $\frac{1}{5}$ of the total number of students like to eat rice only, $\frac{2}{5}$ of the total number of students like to eat chapati only and the remaining students like to eat both. What fraction of the total number of students like to eat both?
3. Renu completed $\frac{2}{3}$ part of her home work in 2 hours. How much part of her home work had she completed in $1\frac{1}{4}$ hours?
4. How many $\frac{1}{16}$ kg boxes of chocolates can be made with $1\frac{1}{2}$ kg chocolates?
5. The largest square that can be drawn in a circle has a side whose length is 0.707 times the diameter of the circle. By this rule, find the length of the side of such a square when the diameter of the circle is
 (i) 14.35 cm (ii) 8.63 cm
6. To find the distance around a circular disc, multiply the diameter of the disc by 3.14. What is the distance around the disc when:
 (i) the diameter is 18.7 cm?
 (ii) the radius is 6.45 cm?

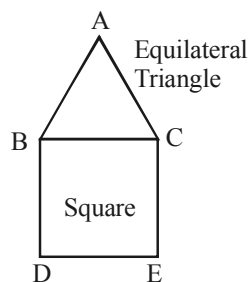
7. Simplify: $\frac{\frac{1}{4} + \frac{1}{5}}{1 - \frac{3}{8} \times \frac{3}{5}}$
8. Divide $\frac{3}{10}$ by $\left(\frac{1}{4} \text{ of } \frac{3}{5}\right)$
9. A flower garden is 22.50 m long. Sheela wants to make a border along one side using bricks that are 0.25 m long. How many bricks will be needed?
10. A public sewer line is being installed along $80\frac{1}{4}$ m of road. The supervisor says that the labourers will be able to complete 7.5 m in one day. How long will the project take to complete?



11. In the morning, a milkman filled $5\frac{1}{2}$ ℓ of milk in his can. He sold to Renu, Kamla and Renuka $\frac{3}{4}$ ℓ each; to Shadma he sold $\frac{7}{8}$ ℓ; and to Jassi he gave $1\frac{1}{2}$ ℓ. How much milk is left in the can?
12. Rama has $6\frac{1}{4}$ kg of cotton wool for making pillows. If one pillow takes $1\frac{1}{4}$ kg, how many pillows can she make?
13. Find the value of :

$$\frac{(0.2 \times 0.14) + (0.5 \times 0.91)}{(0.1 \times 0.2)}$$

14. A square and an equilateral triangle have a side in common. If side of triangle is $\frac{4}{3}$ cm long, find the perimeter of figure formed.



HOTS Questions :

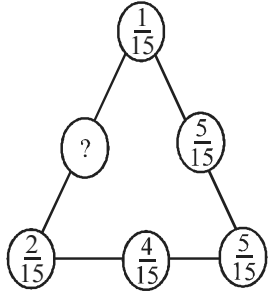
1. A Picture hall has seats for 820 persons. At a recent film show, one usher guessed it was $\frac{3}{4}$ full, another that it was $\frac{2}{3}$ full. The ticket office reported 648 sales. Which usher (first or second) made the better guess?
2. Anuradha can do a piece of work in 6 hours. what part of the work can she do in 1 hour, in 5 hours, in 6 hours?
 It means she can do whole work in 6 hours.
3. What portion of a 'saree' can Rehana paint in 1 hour if it requires 5 hours to paint the whole saree? In $4\frac{3}{5}$ hours?
4. Raj travels 360 km on three fifths of his petrol tank. How far would he travel at the same rate with a full tank of petrol?
5. Rita has bought a carpet of size $4 \text{ m} \times 6\frac{2}{3} \text{ m}$. But her room size is $3\frac{1}{3} \text{ m} \times 5\frac{1}{3} \text{ m}$. What fraction of area should be cut off to fit wall to wall carpet into the room?

3

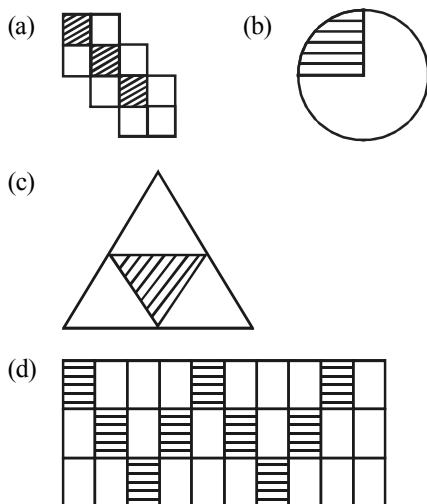
EXERCISE

Single Option Correct :

DIRECTIONS : This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONLY ONE is correct.

- Reciprocal of $2\frac{1}{3}$ is
 (a) $\frac{7}{3}$ (b) $-\frac{7}{3}$
 (c) $-\frac{3}{7}$ (d) $\frac{3}{7}$
- Guru reads $\frac{3}{5}$ of a book. He finds that there are still 80 pages left to be read. Total number of pages in the book are
 (a) 100 (b) 200
 (c) 300 (d) 400
- Ravi had $\frac{5}{6}$ of a cake. He ate $\frac{2}{3}$ of it. What part of the cake did he eat?
 (a) $\frac{5}{9}$ (b) $\frac{10}{12}$
 (c) $\frac{10}{6}$ (d) $\frac{10}{3}$
- The daily consumption of milk of a family is $3\frac{1}{4}$ litres. The quantity of milk consumed by the family during the month of September 2003 is
 (a) 90 litre (b) $100\frac{1}{2}$ litre
 (c) $97\frac{1}{2}$ litre (d) none of these
- $\frac{8}{10} + 0 + \frac{7}{1000} =$ _____
 (a) 0.087 (b) 807
 (c) 0.807 (d) 0.870
- Which number is equal to $\left(\frac{0.1}{0.01} + \frac{0.01}{0.1}\right)$?
 (a) 10.1 (b) 1.10
 (c) 1.01 (d) 10.01
- Compare 12.1280 $\square$ 12.129 (using $>$, $<$, $=$)
 (a) $>$ (b) $<$
 (c) $=$ (d) none
- Divide 125.625 by 0.5.
 (a) 251.25 (b) 2512.5
 (c) 25125 (d) 25.125
- $\frac{2}{3}$ of 4 dozens = _____
 (a) 30 (b) 32
 (c) 28 (d) 48
- The difference between the greatest number and the least number of $\frac{5}{9}, \frac{1}{9}, \frac{11}{9}$ is
 (a) $\frac{2}{9}$ (b) $\frac{4}{9}$
 (c) $\frac{10}{9}$ (d) $\frac{2}{3}$
- What should be placed in the empty space, so that the sum of the fractions on each side of the triangle is same?

 (a) $\frac{7}{15}$ (b) $\frac{9}{15}$
 (c) $\frac{6}{15}$ (d) $\frac{8}{15}$

12. In which figure does the shaded part represents 0.3?



13. Madhavi eats one full bar of chocolate. Then she divides another one into 5 equal parts and eats 3 of those parts. The total number of chocolates that she has eaten is

- (a) $\frac{4}{5}$ (b) $\frac{3}{5}$
(c) $\frac{8}{5}$ (d) $\frac{8}{10}$

14. Which of the following fraction is the largest?

- (a) $\frac{29}{30}$ (b) $\frac{29}{23}$
(c) $\frac{29}{27}$ (d) $\frac{29}{25}$

More Than One Option Correct :

DIRECTIONS : This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONE or MORE may be correct.

1. Fraction for 0.012 is

- (a) $\frac{12}{100}$ (b) $\frac{12}{10}$
(c) $\frac{3}{250}$ (d) $\frac{12}{1000}$

2. On dividing 7 by $\frac{2}{5}$, the result is

- (a) $\frac{14}{2}$ (b) $17\frac{1}{2}$

- (c) $2\frac{4}{5}$ (d) $\frac{35}{2}$

3. $\frac{4}{5}$ of 5 kg apples were used on Monday. The next day $\frac{1}{3}$ of what was left was used. Weight (in kg) of apples left now is

- (a) 0.67 (b) $\frac{1}{14}$
(c) $\frac{2}{3}$ (d) 0.19

4. $\frac{1}{2}$ of $4\frac{2}{7}$ is

- (a) $\frac{15}{7}$ (b) $\frac{30}{7}$
(c) $2\frac{1}{7}$ (d) 4.28

5. The product of 3 and $4\frac{2}{5}$ is:

- (a) $17\frac{2}{5}$ (b) $\frac{24}{5}$
(c) $13\frac{1}{5}$ (d) $\frac{66}{5}$

Assertion & Reason :

DIRECTIONS : Each of these questions contains an Assertion followed by Reason. Read them carefully and answer the question on the basis of following options. You have to select the one that best describes the two statements.

- (a) If both **Assertion** and **Reason** are **correct** and Reason is the **correct explanation** of Assertion.
(b) If both **Assertion** and **Reason** are correct, but Reason is **not the correct explanation** of Assertion.
(c) If **Assertion** is **correct** but **Reason** is **incorrect**.
(d) If **Assertion** is **incorrect** but **Reason** is **correct**.

1. **Assertion:** Proper fraction cannot be written as mixed number.

Reason: In proper fraction denominator is greater than the numerator.

2. **Assertion:** $\frac{2.5}{5} = 0.5$

Reason: Decimal number divided by a natural number always gives either 10 or 100.

Passage Based Questions :

DIRECTIONS : Study the given passage(s) and answer the following questions.

PASSAGE - I

Mrs. Sharma bought $45\frac{1}{4}$ metres of fabric. She used $29\frac{1}{4}$ metres to make 12 curtains. Remaining is used to make four cushion covers.

- How much fabric is used for cushion cover?
(a) 20m (b) 16m
(c) 18m (d) 22m
- How much fabric is needed for one curtains?
(a) $3\frac{7}{16}$ m (b) $2\frac{7}{15}$ m
(c) $2\frac{7}{16}$ m (d) $\frac{7}{16}$ m
- How much fabric is used to make one cushion cover?
(a) $2\frac{2}{3}$ m (b) $3\frac{1}{2}$ m
(c) $3\frac{2}{3}$ m (d) 4m

PASSAGE - II

A group of 20 people went to a restaurant. 9 of them ordered a meal of ₹ 42.20 each and 7 of them ordered a meal of ₹ 47.60 each and rest ordered a meal of ₹ 50 each.

- How much money is spent by 9 people who ordered same meal?
(a) ₹ 369.80 (b) ₹ 379.80
(c) ₹ 309.80 (d) ₹ 389.80
- Total money spent by 20 people is
(a) ₹ 915 (b) ₹ 895
(c) ₹ 913 (d) ₹ 900
- How much more money they would spend if all ordered meal of ₹ 47.60?
(a) ₹ 38 (b) ₹ 40
(c) ₹ 45 (d) ₹ 39

Integer Type Questions :

DIRECTIONS : Answer the following questions. The answer to each of the question is a single digit integer, ranging from 0 to 9.

- How many one-fourths need to be added to $2\frac{1}{4}$ to make 4 ?
- The product of a fractional number and its multiplicative inverse is
- $7.8 = 7 + \frac{k}{10}$. The value of k is
- $\frac{2}{7}$ of a week = _____ days.
- How many one-sevenths make one?
- How many pieces of equal size can be cut from a rope of 30 metres long, each measuring $3\frac{3}{4}$ meters?

Multiple Matching Question :

DIRECTIONS : Following question has four statements (A, B, C and D) given in Column-I and five statements (p, q, r, s, t) in Column-II. Any given statement in Column-I can have correct matching with one or more statement(s) given in Column-II.

- | | |
|--|----------------------|
| 1. Simplify and match: | |
| Column-I | Column-II |
| (A) $4\frac{3}{10} - 1\frac{2}{5} + 8\frac{1}{9}$ | (p) 2.395 |
| (B) $0.25 + 9.81 \times 6.4 + 4\frac{5}{8}$ | (q) - 89.95 |
| (C) $2\frac{3}{8} - 4\frac{7}{9} \times 0.9 + 4.32$ | (r) 67.659 |
| (D) $2\frac{7}{8} + \frac{5}{8} - 0.9345 \times 100$ | (s) 11.01 |
| | (t) $\frac{991}{90}$ |

SOLUTIONS

Brief Explanations of Selected Questions

1 EXERCISE

Fill in the Blanks :

1. improper fraction
2. proper
3. 0.06
4. $\frac{5}{42}$
5. $1\frac{11}{16}$
6. 9
7. 0.0715
8. 1200
9. 2.491
10. 0.00069
11. 7557

True / False :

1. False,
For example : $\frac{4}{7} \times \frac{21}{2} = 6$, which is a natural number
2. True
3. False
4. False
5. True
6. True
7. True
8. False

Match the Columns :

1. (A) $\rightarrow$ (r), (B) $\rightarrow$ (s), (C) $\rightarrow$ (p), (D) $\rightarrow$ (q)

Very Short Answer Questions :

1. (i) $\frac{7}{9} - \frac{5}{9} = \frac{(7-5)}{9} = \frac{2}{9}$.
- (ii) $\frac{13}{16} - \frac{7}{12} = \frac{(39-28)}{48} = \frac{11}{48}$.
- (iii) $\frac{11}{15} - \frac{9}{20} = \frac{(44-27)}{60} = \frac{17}{60}$.
2. (i) $\frac{2}{3} \times \frac{5}{4} = \frac{\overset{1}{\cancel{2}} \times 5}{3 \times \underset{2}{\cancel{4}}} = \frac{1 \times 5}{3 \times 2} = \frac{5}{6}$
- (ii) $\frac{1}{3} \times \frac{15}{8} = \frac{1 \times \overset{5}{\cancel{15}}}{\underset{1}{\cancel{3}} \times 8} = \frac{1 \times 5}{1 \times 8} = \frac{5}{8}$

$$(iii) \frac{4}{5} \times \frac{12}{7} = \frac{4 \times 12}{5 \times 7} = \frac{48}{35} = 1\frac{13}{35}$$

$$3. (i) \frac{1}{5} \text{ of a rupee} = \frac{1}{5} \text{ of 100 paise}$$

$$= \left(\frac{20}{\cancel{100} \times 1} \right) \text{ paise} = 20 \text{ paise.}$$

$$(ii) \frac{5}{8} \text{ of a day} = \frac{5}{8} \text{ of 24 hours}$$

$$= \left(\frac{3}{\cancel{24} \times 5} \right) \text{ hours} = 15 \text{ hours.}$$

$$(iii) \frac{7}{8} \text{ of a kilogram} = \frac{7}{8} \text{ of 1000 g}$$

$$= \left(\frac{125}{\cancel{1000} \times 7} \right) \text{ g} = 875 \text{ g.}$$

$$(iv) \frac{4}{5} \text{ of an hour} = \frac{4}{5} \text{ of 60 min}$$

$$= \left(\frac{12}{\cancel{60} \times 4} \right) \text{ min} = 48 \text{ min.}$$

$$4. (i) \frac{7}{3} \quad (ii) \frac{5}{1} = 5 \quad (iii) \frac{1}{6} \quad (iv) \frac{8}{29}$$

$$5. (i) \frac{4}{9} \div \frac{2}{3} = \frac{4}{9} \times \frac{3}{2} = \frac{\overset{2}{\cancel{4}} \times \overset{1}{\cancel{3}}}{\underset{3}{\cancel{9}} \times \underset{1}{\cancel{2}}} = \frac{2}{3}$$

$$(ii) \quad 3\frac{3}{7} \div \frac{8}{21} = \frac{24}{7} \div \frac{8}{21} = \frac{24}{7} \times \frac{21}{8}$$

$$= \frac{\overset{3}{\cancel{24}} \times \overset{3}{\cancel{21}}}{\underset{1}{\cancel{8}} \times \underset{1}{\cancel{7}}} = 9$$

$$(iii) \quad 15\frac{3}{7} \div 1\frac{23}{49} = \frac{108}{7} \div \frac{72}{49} = \frac{108}{7} \times \frac{49}{72}$$

$$= \frac{\overset{3}{\cancel{108}} \times \overset{7}{\cancel{49}}}{\underset{1}{\cancel{7}} \times \underset{2}{\cancel{72}}} = \frac{21}{2} = 10\frac{1}{2}$$

6. (i) 0.0000015 (ii) 0.00235
 (iii) 14250000 (iv) 375
 7. (i) 0.0476 (ii) 8.7
 (iii) 0.0633 (iv) 0.00034
 8. (i) 0.05 (ii) 7.3
 (iii) 60 (iv) 7.8
 9. (i) 33.46 (ii) 90.159
 (iii) 290.088 (iv) 863.954
 (v) 325.737
 10. (i) 10.81 (ii) 28.901
 (iii) 238.77 (iv) 5.7746
 (v) 0.2336
 11. (i) 3.275, 3.524, 3.572, 3.725
 (ii) 7.018, 7.081, 7.108, 7.801
 (iii) 21.590, 21.6, 21.71, 21.8
 (iv) 4.090, 4.381, 4.70, 4.8

Short Answer Questions :

1. 100m 2. 36 students
 3. 101 cm 4. 35.95 secs
 5. Total weight of vegetables
 $= \left(\frac{23}{4} + \frac{7}{2} \right) \text{ kg} = \frac{(23+14)}{4} \text{ kg} = \frac{37}{4} \text{ kg} = 9\frac{1}{4} \text{ kg}$
 6. Required number
 $= \left(17\frac{3}{4} - 11\frac{2}{3} \right) = \left(\frac{71}{4} - \frac{35}{3} \right)$
 $= \frac{(213-140)}{12} = \frac{73}{12} = 6\frac{1}{12}$
 7. Distance covered by Sajal in 1 hour = $\frac{12}{5}$ km.
 Distance covered by Sajal in $3\frac{1}{3}$ hours

$$= \left(\frac{12}{5} \times \frac{10}{3} \right) \text{ km} = \left(\frac{\overset{4}{\cancel{12}} \times \overset{2}{\cancel{10}}}{\underset{1}{\cancel{5}} \times \underset{1}{\cancel{3}}} \right) \text{ km} = 8 \text{ km}.$$

$$8. \quad 0.089 \times 0.76 = 0.06764.$$

$$9. \quad \frac{3.28}{0.4} = \frac{3.38 \times 10}{0.4 \times 10} = \frac{32.8}{4} = 8.2$$

$$10. \quad \frac{0.00942}{0.314} = \frac{0.00942 \times 1000}{0.314 \times 1000} = \frac{9.42}{314} = 0.03$$

Long Answer Questions :

1. $3\frac{1}{5} + 2\frac{1}{10} - 1\frac{1}{2} - \frac{1}{4} = \frac{16}{5} + \frac{21}{10} - \frac{3}{2} - \frac{1}{4}$
 $= \frac{(64+42-30-5)}{20}$
 $[\because \text{LCM of } 5, 10, 2, 4 = 2 \times 5 \times 2]$
 $= \frac{(106-35)}{20} = \frac{71}{20} = 3\frac{11}{20}$
 2. Cost of 1 litre of milk = ₹ $16\frac{3}{4} = ₹ \frac{67}{4}$.

$$\text{Cost of } 6\frac{2}{5} \text{ litres of milk} = ₹ \left(\frac{67}{4} \times \frac{32}{5} \right)$$

$$= ₹ \left(\frac{67 \times \overset{8}{\cancel{32}}}{\underset{1}{\cancel{4}} \times 5} \right) = ₹ \left(\frac{536}{5} \right) = ₹ 107\frac{1}{5}$$

3. Weight of onions = 3 kg 500 g = 3.500 kg
 Weight of tomatoes = 2 kg 75 g = 2.075 kg
 $\therefore$ Total weight of onions and tomatoes
 $= 5.575 \text{ kg}$
 Weight of potatoes = 10 kg - 5.575 kg
 $= 4.425 \text{ kg}$
 4. Distance covered by the car = 89.1 km.
 Time taken to cover this distance = 2.2 hours.
 So, distance covered by it in 1 hour
 $= \frac{89.1}{2.2} = 40.5 \text{ km}$
 Distance covered by it in 4.3 hours
 $= 4.3 \times 40.5 = 174.15 \text{ km}$

5. Weight of one apple = 74 g
 Weight of 132 apples = $74 \times 132 = 9768$ g
 Weight of one orange = 42 g
 Weight of 46 orange = $42 \times 46 = 1932$ g
 Thus, total weight of 132 apples and 46 oranges

$$= 9768 + 1932 = 11700 \text{ g}$$

$$= \frac{11700}{1000} \text{ kg} = 11.7 \text{ kg}$$

6. Product of two numbers = $15 \frac{5}{6} = \frac{95}{6}$.

$$\text{One of the numbers} = 6 \frac{2}{3} = \frac{20}{3}.$$

$$\text{The other number} = \frac{95}{6} \div \frac{20}{3}$$

$$= \left(\frac{95}{6} \times \frac{3}{20} \right)$$

$$= \frac{19}{4} \times \frac{1}{2} = \frac{19}{8} = 2 \frac{3}{8}.$$

7. Total weight of all the bags = 1743 kg.
 Weight of each bag = 49.8 kg.

$$\text{Number of bags} = \frac{\text{Total weight}}{\text{weight of each bag}}$$

$$= \frac{1743}{49.8} = \frac{17430}{498} = 35.$$

2 EXERCISE

Text-Book Exercise :

1. (i) $4 + \frac{7}{8} = \frac{32+7}{8} = \frac{39}{8} = 4 \frac{7}{8}$
- (ii) $\frac{9}{11} - \frac{4}{15} = \frac{135-44}{165} = \frac{91}{165}$
- (iii) $8 \frac{1}{2} - 3 \frac{5}{8} = \frac{17}{2} - \frac{29}{8}$
- $$= \frac{68-29}{8} = \frac{39}{8} = 4 \frac{7}{8}.$$

$$(iv) \frac{7}{10} + \frac{2}{5} + \frac{3}{2} = \frac{7+4+15}{10}$$

$$= \frac{26}{10} = \frac{13}{5} = 2 \frac{3}{5}$$

$$2. (i) \frac{2}{9} = \frac{2 \times 7}{9 \times 7} = \frac{14}{63}; \frac{2}{3} = \frac{2 \times 21}{3 \times 21} = \frac{42}{63};$$

$$\frac{8}{21} = \frac{8 \times 3}{21 \times 3} = \frac{24}{63}$$

$$[\text{L.C.M.}(3, 9, 21) = 3 \times 3 \times 7 = 63]$$

$$\text{Since, } 42 > 24 > 14, \therefore \frac{2}{3} > \frac{8}{21} > \frac{2}{9}$$

$$(ii) \frac{1}{5} = \frac{1 \times 14}{5 \times 14} = \frac{14}{70}; \frac{3}{7} = \frac{3 \times 10}{7 \times 10} = \frac{30}{70};$$

$$\frac{7}{10} = \frac{7 \times 7}{10 \times 7} = \frac{49}{70}$$

$$[\text{L.C.M.}(5, 7, 10) = 2 \times 5 \times 7 = 70]$$

$$\text{As } 49 > 30 > 14,$$

$$\therefore \frac{7}{10} > \frac{3}{7} > \frac{1}{5}$$

$$3. (i) \text{ Perimeter of } \triangle ABE = \left(\frac{5}{2} + 2 \frac{3}{4} + 3 \frac{3}{5} \right)$$

$$= \left(\frac{5}{2} + \frac{11}{4} + \frac{18}{5} \right) = \frac{50+55+72}{20}$$

$$= \frac{177}{20} = 8 \frac{17}{20} \text{ cm}$$

$$(ii) \text{ Perimeter of rectangle BCDE}$$

$$= 2 \left[\frac{11}{4} + \frac{7}{6} \right] = 2 \left[\frac{33+14}{12} \right]$$

$$= 2 \times \frac{47}{12} = \frac{47}{6} = 7 \frac{5}{6} \text{ cm}$$

$$\text{Perimeter of } \triangle ABE = \frac{177}{20} \text{ cm and}$$

$$\text{Perimeter of rectangle BCDE} = \frac{47}{6} \text{ cm}$$

Changing them to like fractions, we obtain

$$\frac{177}{20} = \frac{177 \times 3}{20 \times 3} = \frac{531}{60};$$

$$\frac{47}{6} = \frac{47 \times 10}{6 \times 10} = \frac{470}{60}$$

As $531 > 470$,

$$\therefore \frac{177}{20} > \frac{47}{6}$$

Thus, perimeter of $\triangle ABE$ is greater than perimeter of rectangle BCDE

4. The picture should be trimmed

$$= \left(7\frac{3}{5} - 7\frac{3}{10} \right) = \left(\frac{38}{5} - \frac{73}{10} \right)$$

$$= \frac{76 - 73}{10} = \frac{3}{10} \text{ cm}$$

5. Time taken by Michael = $\frac{7}{12}$ hour

Time taken by Vaibhav = $\frac{3}{4}$ hour

$$\text{Now, } \frac{3}{4} = \frac{3 \times 3}{4 \times 3} = \frac{9}{12}$$

$$\text{and } \frac{7}{12} = \frac{7 \times 1}{12 \times 1} = \frac{7}{12}$$

Since $9 > 7$, Vaibhav worked longer.

$$\text{Difference} = \frac{9}{12} - \frac{7}{12} = \frac{2}{12} = \frac{1}{6} \text{ hour.}$$

6. (i) (a) $\frac{2}{3} \times 18 = 12$

(b) $\frac{2}{3} \times 27 = 18$

(ii) (a) $\frac{4}{5} \times 20 = 16$

(b) $\frac{4}{5} \times 35 = 28$

7. (i) $7 \times 2\frac{1}{4} = 7 \times \frac{9}{4} = \frac{63}{4} = 15\frac{3}{4}$

(ii) $4 \times 6\frac{1}{3} = 4 \times \frac{19}{3} = \frac{76}{3} = 25\frac{1}{3}$

(iii) $3\frac{1}{4} \times 6 = \frac{13}{4} \times 6 = \frac{39}{2} = 19\frac{1}{2}$

(iv) $3\frac{2}{5} \times 8 = \frac{17}{5} \times 8 = \frac{136}{5} = 27\frac{1}{5}$

8. (i) Water consumed by Vidya = $\frac{2}{5}$ of 5 litres
 $= \frac{2}{5} \times 5 = 2$ litres.

(ii) Fraction of water consumed by Pratap
 $= 1 - \frac{2}{5} = \frac{3}{5}$

9. (i) $\frac{2}{5} \times 5\frac{1}{4} = \frac{2}{5} \times \frac{21}{4} = \frac{21}{10} = 2\frac{1}{10}$

(ii) $6\frac{2}{5} \times \frac{7}{9} = \frac{32}{5} \times \frac{7}{9} = \frac{224}{45} = 4\frac{44}{45}$

(iii) $\frac{3}{2} \times 5\frac{1}{3} = \frac{3}{2} \times \frac{16}{3} = 8$

(iv) $\frac{5}{6} \times 2\frac{3}{7} = \frac{5}{6} \times \frac{17}{7} = \frac{85}{42} = 2\frac{1}{42}$

10. Distance between first and last Sapling

$$= 3 \times \frac{3}{4} = \frac{9}{4} = 2\frac{1}{4} \text{ m.}$$

11. Number of hours, Lipika reads the book per day

$$= 1\frac{3}{4} = \frac{7}{4} \text{ hours}$$

Total number of hours required by her to read

$$\text{the book} = \frac{7}{4} \times 6 = \frac{21}{2} = 10\frac{1}{2}$$

12. (i) $3\frac{1}{2} \div \frac{8}{3} = \frac{7}{2} \times \frac{3}{8} = \frac{21}{16} = 1\frac{5}{16}$

(ii) $\frac{2}{5} \div 1\frac{1}{2} = \frac{2}{5} \times \frac{2}{3} = \frac{4}{15}$

(iii) $3\frac{1}{5} \div 1\frac{2}{3} = \frac{16}{5} \times \frac{3}{5} = \frac{48}{25} = 1\frac{23}{25}$

(iv) $2\frac{1}{5} \div 1\frac{1}{5} = \frac{11}{5} \times \frac{5}{6} = \frac{11}{6} = 1\frac{5}{6}$

13. (i) 7 paise = ₹ $\frac{7}{100}$ = ₹ 0.07
 (ii) 7 rupees 7 paise
 = ₹ 7 + ₹ $\frac{7}{100}$ = ₹ 7.07
 (iii) 77 rupees 77 paise
 = ₹ 77 + ₹ $\frac{77}{100}$ = ₹ 77.77
 (iv) 235 paise = ₹ $\frac{235}{100}$ = ₹ 2.35
14. (i) $20.03 = 2 \times 10 + 0 \times 1 + 0 \times \frac{1}{10} + 3 \times \frac{1}{100}$
 (ii) $2.03 = 2 \times 1 + 0 \times \frac{1}{10} + 3 \times \frac{1}{100}$
 (iii) $200.03 = 2 \times 100 + 0 \times 10 + 0 \times 1 + 0 \times \frac{1}{10} + 3 \times \frac{1}{100}$
 (iv) $2.034 = 2 \times 1 + 0 \times \frac{1}{10} + 3 \times \frac{1}{100} + 4 \times \frac{1}{1000}$
15. (i) 2 ones (ii) 2 tens
 (iii) 2 tenths (iv) 2 thousandths
16. $42.6 - 28.0 = 14.6$ km
 Hence, 28 km is 14.6 km less than 42.6 km.
17. (i) 1537 (ii) 1680.7 (iii) 15610
 (iv) 4307 (v) 0.8 (vi) 30
18. (i) $0.1 \times 51.7 = \frac{1}{10} \times \frac{517}{10} = 5.17$
 (ii) $0.2 \times 316.8 = \frac{2}{10} \times \frac{3168}{10} = 63.36$
 (iii) $11.2 \times 0.15 = \frac{112}{10} \times \frac{15}{100} = 1.68$
 (iv) $1.07 \times 0.02 = \frac{107}{100} \times \frac{2}{100} = 0.0214$
 (v) $10.05 \times 1.05 = \frac{1005}{100} \times \frac{105}{100} = 10.5525$
 (vi) $101.01 \times 0.01 = \frac{10101}{100} \times \frac{1}{100} = 1.0101$

19. (i) $0.35 \div 5 = \frac{35}{100} \times \frac{1}{5} = \frac{7}{100} = 0.07$
 (ii) $651.2 \div 4 = \frac{6512}{10} \times \frac{1}{4} = \frac{1628}{10} = 162.8$
 (iii) $14.49 \div 7 = \frac{1449}{100} \times \frac{1}{7} = \frac{207}{100} = 2.07$
 (iv) $0.80 \div 5 = \frac{80}{100} \div 5 = \frac{80}{100} \times \frac{1}{5} = 0.16$
20. (i) 0.027 (ii) 0.0078
 (iii) 0.236 (iv) 0.9853
21. (i) $30.94 \div 0.7 = \frac{3094}{100} \div \frac{7}{10}$
 = $\frac{3094}{100} \times \frac{10}{7} = \frac{442}{10} = 44.2$
 (ii) $7.75 \div 0.25 = \frac{775}{100} \div \frac{25}{100}$
 = $\frac{775}{100} \times \frac{100}{25} = 31$
 (iii) $76.5 \div 0.15 = \frac{765}{10} \div \frac{15}{100}$
 = $\frac{765}{10} \times \frac{100}{15} = 510$
 (iv) $2.73 \div 1.3 = \frac{273}{100} \div \frac{13}{10} = \frac{273}{100} \times \frac{10}{13}$
 = $\frac{21}{10} = 2.1$

Exemplar Questions :

1. Let the number be x .
 $\therefore \frac{2}{3}$ of $x = 10$
 $\Rightarrow x = \frac{10 \times 3}{2} = 15$
 1.75 times of 15 = $1.75 \times 15 = 26.25$
2. Number of students like rice = $\frac{1}{5}$ of 40 = 8
 Number of students like chapati = $\frac{2}{5}$ of 40 = 16

Number of students like to eat both

$$= 40 - (8 + 16) = 16$$

Fraction of the students like to eat both

$$= \frac{16}{40} = \frac{2}{5}$$

3. Home work done in 2 hours = $\frac{2}{3}$ part

$$\text{Home work done in 1 hour} = \frac{2}{3} \div 2 = \frac{1}{3} \text{ part}$$

$$\text{Home work done in } 1\frac{1}{4} \text{ hours}$$

$$= \frac{1}{3} \times 1\frac{1}{4} = \frac{5}{12} \text{ part}$$

4. Number of boxes

$$= 1\frac{1}{2} \div \frac{1}{16} = \frac{3}{2} \times 16 = 24$$

5. According to the question,

Side of a square = 0.707 times the diameter

$$\begin{aligned} \text{(i) Side of square} &= 0.707 \times 14.35 \text{ cm} \\ &= 10.14545 = 10.15 \text{ cm} \end{aligned}$$

$$\begin{aligned} \text{(ii) Side of square} &= 0.707 \times 8.63 \\ &= 6.10141 = 6.10 \text{ cm} \end{aligned}$$

6. According to the question,

The distance around a circular disc = $3.14 \times$ diameter of disc

$$\begin{aligned} \text{(i) The distance around the disc} &= 3.14 \times 18.7 \\ &= 58.718 = 58.72 \text{ cm (Approx.)} \end{aligned}$$

$$\text{(ii) Radius} = 6.45 \text{ cm}$$

$$\Rightarrow \text{Diameter} = 2 \times 6.45 = 12.9 \text{ cm}$$

$$\begin{aligned} \text{The distance around the disc} &= 3.14 \times 12.9 \\ &= 40.506 = 40.51 \text{ cm (Approx.)} \end{aligned}$$

$$7. \frac{\frac{1}{4} + \frac{1}{5}}{1 - \frac{3}{8} \times \frac{3}{5}} = \frac{\frac{5+4}{20}}{1 - \frac{9}{40}} = \frac{\frac{9}{20}}{\frac{31}{40}} = \frac{9}{20} \times \frac{40}{31} = \frac{18}{31}$$

$$\begin{aligned} 8. \frac{3}{10} \div \left(\frac{1}{4} \text{ of } \frac{3}{5} \right) &= \frac{3}{10} \div \left(\frac{3}{20} \right) \\ &= \frac{3}{10} \times \frac{20}{3} = 2 \end{aligned}$$

9. Number of bricks needed

$$= 22.50 \div 0.25 = 90 \text{ bricks}$$

10. Time taken to complete 7.5 m = 1 day

$$\text{Time taken to complete 1 m} = \frac{1}{7.5} \text{ day}$$

$$\begin{aligned} \text{Time taken to complete } 80\frac{1}{4} \text{ m} &= 80\frac{1}{4} \times \frac{1}{7.5} \\ &= 10.7 \text{ days} \end{aligned}$$

Nearest integer to 10.7 is 11, therefore project will be completed in 11 days.

11. Milk sold to Renu, Kamla and Renuka

$$= \frac{3}{4} + \frac{3}{4} + \frac{3}{4} = \frac{9}{4} \ell$$

$$\begin{aligned} \text{Total milk sold} &= \frac{9}{4} \ell + \frac{7}{8} \ell + 1\frac{1}{2} \ell \\ &= \left(\frac{18+7+12}{8} \right) \ell = \frac{37}{8} \ell \end{aligned}$$

$$\text{Milk left in the can} = \left(5\frac{1}{2} - \frac{37}{8} \right) \ell$$

$$= \frac{11}{2} - \frac{37}{8} = \frac{44-37}{8} = \frac{7}{8} \ell$$

12. Number of pillows = $6\frac{1}{4} \div 1\frac{1}{4}$

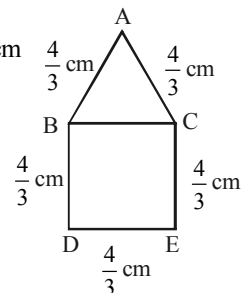
$$= \frac{25}{4} \div \frac{5}{4} = 5 \text{ pillows}$$

13. $\frac{(0.2 \times 0.14) + (0.5 \times 0.91)}{(0.1 \times 0.2)}$

$$= \frac{0.028 + 0.455}{0.02} = \frac{0.483}{0.02} = 24.15$$

14. The perimeter of figure formed

$$\begin{aligned} &= \left(\frac{4}{3} + \frac{4}{3} + \frac{4}{3} + \frac{4}{3} + \frac{4}{3} \right) \text{ cm} \\ &= \frac{20}{3} = 6\frac{2}{3} \text{ cm} \end{aligned}$$



HOTS Questions :

1. Number of seats guessed by first usher

$$= \frac{3}{4} \text{ of } 820 = 615$$

Number of seats guessed by second usher

$$= \frac{2}{3} \text{ of } 820 = 546.666 = 547$$

Number of seats filled = 648

Since 615 is nearer to 648 than 547.

Therefore, first usher made the better guess.

2. Since, Anuradha can do a piece of work in 6 hours.

Therefore, work done in 1 hour = $\frac{1}{6}$ part

Work done in 5 hours = $5 \times \frac{1}{6} = \frac{5}{6}$ part

Work done in 6 hours = $6 \times \frac{1}{6} = 1$ part
= whole work.

It means she can do whole work in 6 hours.

3. Since, Rehana paints whole saree in 5 hours.

$\therefore$ Part of saree painted in 1 hour = $\frac{1}{5}$ part

Part of saree painted in $4\frac{3}{5}$ hours

$$= 4\frac{3}{5} \times \frac{1}{5} = \frac{23}{25} \text{ part}$$

4. Distance covered with $\frac{3}{5}$ part of petrol tank

$$= 360 \text{ km}$$

$\therefore$ Distance covered with full tank of petrol

$$= 360 \div \frac{3}{5} = 600 \text{ km.}$$

5. Area of carpet = $\left(4 \times 6\frac{2}{3}\right) \text{ m}^2 = 4 \times \frac{20}{3} = \frac{80}{3} \text{ m}^2$

Area of floor of room = $\left(3\frac{1}{3} \times 5\frac{1}{3}\right) \text{ m}^2$

$$= \left(\frac{10}{3} \times \frac{16}{3}\right) = \frac{160}{9} \text{ m}^2$$

Area of carpet to be cut off

$$= \left(\frac{80}{3} - \frac{160}{9}\right) \text{ m}^2$$

$$= \left(\frac{240-160}{9}\right) \text{ m}^2 = \frac{80}{9} \text{ m}^2$$

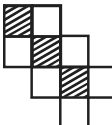
Fraction of carpet area should be cut off

$$= \frac{80}{9} \div \frac{80}{3} = \frac{80}{9} \times \frac{3}{80} = \frac{1}{3}$$

3

EXERCISE

Single Option Correct :

1. (d) $\frac{3}{7}$
2. (b) $\left(1 - \frac{3}{5}\right) \times \text{total pages} = 80$
 $\therefore$ number of pages = $80 \times \frac{5}{2} = 200$
3. (a) $\frac{5}{6} \times \frac{2}{3} = \frac{10}{18} = \frac{5}{9}$
4. (c) September has 30 days
 $30 \times 3\frac{1}{4} = 30 \times \frac{13}{4} = 97\frac{1}{2}$ litre
5. (c) $\frac{8}{10} + 0 + \frac{7}{1000} = 0.807$
6. (a) $\frac{0.1}{0.01} + \frac{0.01}{0.1} = 10 + \frac{1}{10} = 10.1$
7. (b) $12.1280 < 12.129$
8. (a)
9. (b) $\frac{2}{3} \times 4 \text{ dozens} = \frac{2}{3} \times 4 \times 12 = 32$
10. (c) Required difference = $\frac{11}{9} - \frac{1}{9} = \frac{10}{9}$
11. (d) Since $\frac{1}{15} + \frac{8}{15} + \frac{2}{15} = \frac{11}{15}$
12. (a)  $= \frac{3}{10} = 0.3$

13. (c) $1 + \frac{3}{5} = \frac{5+3}{5} = \frac{8}{5}$
14. (b) All fractions are having same numerator. So, the fraction having smallest denominator is the largest.

More Than One Option Correct :

1. (c, d) $0.012 = \frac{12}{1000} = \frac{3}{250}$
2. (b, d) $7 \div \frac{2}{5} = 7 \times \frac{5}{2} = \frac{35}{2} = 17\frac{1}{2}$
3. (a, c) Weight of apple used on Monday
 $= \frac{4}{5}$ of 5 kg = 4 kg
 Weight of apple left = (5 - 4) kg = 1 kg
 Weight of apple used next day
 $= \frac{1}{3}$ of 1 kg = $\frac{1}{3}$ kg
 Weight of apple left = $1 - \frac{1}{3}$
 $= \frac{2}{3}$ kg = 0.67 kg (approx.)
4. (a, c) $\frac{1}{2}$ of $4\frac{2}{7} = \frac{1}{2} \times \frac{30}{7} = \frac{15}{7} = 2\frac{1}{7}$
5. (c, d) $3 \times 4\frac{2}{5} = 3 \times \frac{22}{5} = \frac{66}{5} = 13\frac{1}{5}$

Assertion & Reason :

1. (a)
2. (c) If Assertion is correct but Reason is incorrect.

Passage Based Questions :

1. (b) $45\frac{1}{4} - 29\frac{1}{4} = 16$ m
2. (c) $29\frac{1}{4} \div 12 = \frac{117}{4 \times 12} = \frac{39}{16} = 2\frac{7}{16}$ m
3. (d) $\frac{16\text{m}}{4} = 4$ m
4. (b) $\text{₹}42.20 \times 9 = \text{₹}379.80$
5. (c) $42.20 \times 9 + 47.60 \times 7 + 50 \times 4$
 $= 379.80 + 333.20 + 200 = \text{₹}913$

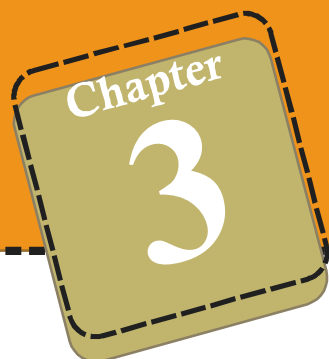
6. (d) Total cost = $47.60 \times 20 = 952$
 Difference = $\text{₹}952 - \text{₹}913 = \text{₹}39$

Integer Type Questions :

1. (7) $4 - 2\frac{1}{4} = 4 - \frac{9}{4} = \frac{7}{4} = 7 \times \frac{1}{4}$
2. (1)
3. (8) $7.8 = 7 + 0.8 = 7 + \frac{8}{10}$
4. (2) $\frac{2}{7}$ of 7 = $\frac{2}{7} \times 7 = 2$
5. (7) $7 \times \frac{1}{7} = 1$
6. (8) $\frac{30}{3\frac{3}{4}} = \frac{30}{15/4} = 30 \times \frac{4}{15} = 8$

Multiple Matching Question :

1. (A) $\rightarrow (s, t); (B) \rightarrow (r); (C) \rightarrow (p); (D) \rightarrow (q)$
- (A) $4\frac{3}{10} - 1\frac{2}{5} + 8\frac{1}{9} = \frac{43}{10} - \frac{7}{5} + \frac{73}{9}$
 $= \frac{387 - 126 + 730}{90} = \frac{991}{90} = 11.01$
- (B) $0.25 + 9.81 \times 6.4 + 4\frac{5}{8}$
 $= 0.25 + 62.784 + \frac{37}{8}$
 $= 63.034 + 4.625 = 67.659$
- (C) $2\frac{3}{8} - 4\frac{7}{9} \times 0.9 + 4.32$
 $= \frac{19}{8} - \frac{43}{9} \times 0.9 + 4.32$
 $= \frac{19}{8} - 4.3 + 4.32$
 $= 2.375 - 4.3 + 4.32$
 $= 6.695 - 4.3 = 2.395$
- (D) $2\frac{7}{8} + \frac{5}{8} - 0.9345 \times 100$
 $= \frac{23}{8} + \frac{5}{8} - 93.45 = \frac{28}{8} - 93.45$
 $= 3.5 - 93.45 = -89.95$



Data Handling

INTRODUCTION

Statistics is a branch of mathematics which deals with collection, organisation, presentation, analysis and interpretation of a collection of numerical figures. In this chapter, you will come across some more kinds of data and graphs.

DATA

A collection of numerical figures giving some particular type of information is called *data*.

For example: The marks obtained by 10 students in a class test (out of 50) are:

24, 43, 49, 38, 36, 36, 31, 40, 42, 15.

Raw data: Data obtained in original form is known as raw data. Data given in above example is raw data.

Collection of Data

Data obtained in original form is called **raw data**. To get better understanding of data, we arrange the data in ascending or descending order. It is known as **array**.

The data which is collected by a person for himself is called **primary data**.

The data which is collected by someone else and used by the investigator for his requirements is called **secondary data**.

ORGANISATION OF DATA

After collection of data, it needs to be recorded and organised in a tabular form. Let us learn it with the help of given illustration.

ILLUSTRATION 1 :

The total number of children in 20 families of a locality are given below:

2, 2, 3, 2, 1, 3, 3, 2, 2, 1, 2, 2, 3, 1, 2, 1, 1, 3, 2, 2

Represent this information in a frequency distribution table.

SOLUTION :

Arranging the data in ascending order we get :

1, 1, 1, 1, 1, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 3, 3, 3, 3, 3.

No. of children	Tally marks	No. of families
1		5
2		10
3		5
	Total	20

MEASURES OF CENTRAL TENDENCY

There are different measures that attempt to describe a whole set of data with a single value that represents the middle or centre of its distribution. These are referred to as **measures of central tendency**.

There are three main measures of central tendency : the mode, the median and the mean.

Arithmetic Mean

The **mean** is the sum of all the values of observations in a given data divided by the number of observations. This is also known as the **arithmetic average**.

$$\text{Mean} = \frac{\text{Sum of the given observations}}{\text{Number of given observations}}$$

ILLUSTRATION 2 :

Find the mean of the numbers

7, 6, 8, 9, 5, 4, 3, 7, 8, 2

SOLUTION :

Sum of the given numbers = $7 + 6 + 8 + 9 + 5 + 4 + 3 + 7 + 8 + 2 = 59$

No. of given observation = 10

$$\text{Mean} = \frac{\text{Sum of observations}}{\text{Number of observations}} = \frac{59}{10} = 5.9$$

Range

Difference between the highest and the lowest value of the data is called the '*range*'.

In the above illustration, Range = maximum value – minimum value = $9 - 2 = 7$

Mode

The **mode** is the most commonly occurring value in a data.

Consider the following data showing the retirement age of 11 people :

54, 55, 54, 56, 54, 57, 60, 58, 57, 58, 60

Let us put the data in frequency distribution table.

Age	Tally bars	Frequency
54		3
55		1
56		1
57		2
58		2
60		2
	Total	11

The most commonly occurring value is 54. Therefore, the mode of this distribution is 54 years.

Median

The median is the middle most observation of the data, when the values are arranged in ascending or descending order.

Case 1: When n is odd:

$$\text{Median} = \left(\frac{n+1}{2} \right) \text{th observation value}$$

Or we can say “when the number of observations is odd, then the middle number is the median”.

Case 2: When n is even

$$\text{Median} = \frac{1}{2} \left\{ \frac{n}{2} \text{th} + \left(\frac{n}{2} + 1 \right) \text{th observation value} \right\}.$$

Or we can say “when the number of observation is even, then the median is the average of the two middle numbers.

ILLUSTRATION 3 :

The weight of 8 students (in kg) are:

32, 38, 40, 30, 41, 35, 43, 30.

Find the median weight.

SOLUTION :

Arranging the weight in ascending order:

30, 30, 32, 35, 38, 40, 41, 43.

Here $n = 8$

$\therefore$ Median weight

$$\begin{aligned} &= \frac{1}{2} \left\{ \frac{n \text{th term value}}{2} + \left(\frac{n}{2} + 1 \right) \text{th term value} \right\} \\ &= \frac{1}{2} \left\{ \frac{8}{2} + \left(\frac{8}{2} + 1 \right) \right\} \\ &= \frac{1}{2} \{ 4 \text{th term value} + 5 \text{th term value} \} \\ &= \frac{1}{2} \times (35 + 38) = \frac{1}{2} \times 73 = 36.5 \text{ kg} \end{aligned}$$

BAR GRAPH

A bar graph is a pictorial representation of data by a number of bars (rectangles) of uniform width created horizontally or vertically with equal spacing between them.

Properties of Bar Graph:

- Bars may be either vertical or horizontal, we often use vertical bars as they give a better look.
- The height (or length) of the bar indicates the value of the data.
- Width of the bar always remains equal but height (or length) changes according to the data.
- Bars are equally spaced.
- Bars may be shaded to make them attractive.
- Bar graphs are useful for visual comparison of the given data.

ILLUSTRATION 4 :

The table below shows the monthly expenditure of a family:

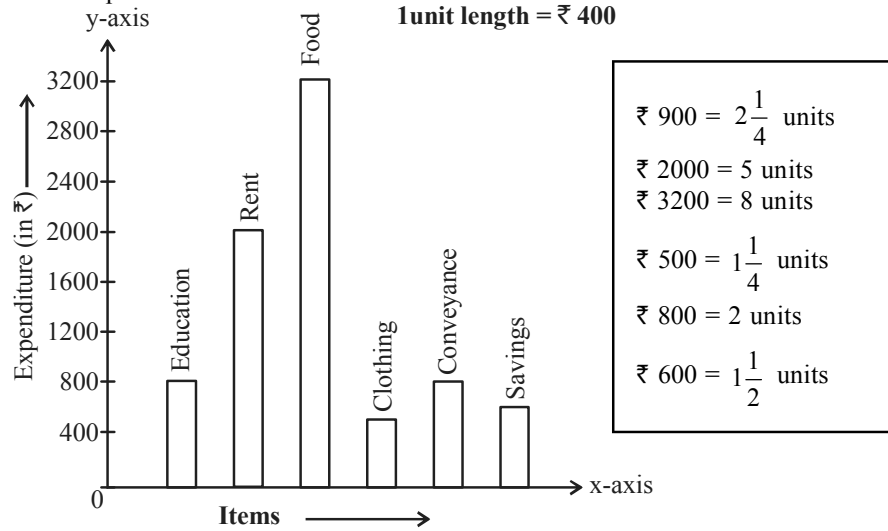
Education	Rent	Food	Clothing	Conveyance	Savings
₹ 900	₹ 2000	₹ 3200	₹ 500	₹ 800	₹ 600

Represent the data in the form of a bar graph.

SOLUTION :

- Step I** : Draw the horizontal and vertical axes. Write 'items' along the horizontal axis and 'expenditure' along the vertical axis.
- Step II** : Choose the widths of the bars leaving equal spaces between them on the horizontal axis.
- Step III** : Choose a convenient scale to represent expenditure on the vertical axis.

Let 1 unit represents ₹ 400.

**Double Bar Graph**

A double bar graph is a bar graph used for displaying two sets of frequency data for each data group. It helps us to compare or represent more than one kind of information, situations, or events instead of just one by using bars.

For example : A mathematics teacher wants to see whether the new technique of teaching she applied after the quarterly test was effective or not. She takes the scores of the 5 weakest children in the quarterly test (out of 25) and in the half yearly test (out of 25):

Students	Ashish	Arun	Kavish	Maya	Rita
Quarterly	10	15	12	20	9
Half yearly	15	18	16	21	15

Since there seems to be a marked improvement in most of the students, the teacher decides that she should continue to use the new technique of teaching.

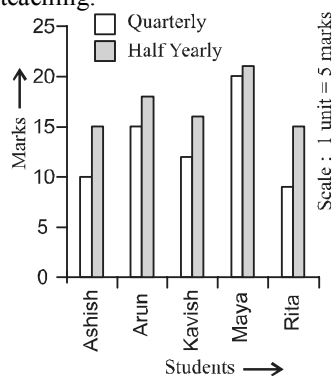
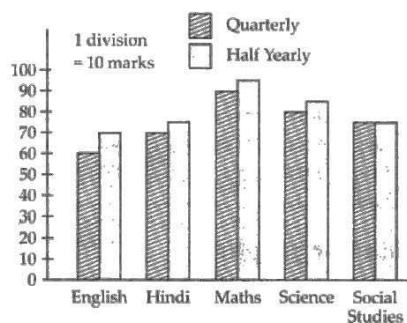


ILLUSTRATION 5 :

The performance of a student in quarterly and half yearly examination is shown through the double bar graph.



In which subject has the student improved his performance the most?

SOLUTION :

Marks scored by a student in English

Quarterly = 60, Half-yearly = 70

∴ Performance improved by $(70 - 60) = 10$ marks

Marks scored by student in Hindi

Quarterly = 70, Half-yearly = 75

∴ Performance improved by $(75 - 70) = 5$ marks

Similarly, performance improved by student in

Maths = $95 - 90 = 5$ marks

Science = $85 - 80 = 5$ marks

Social Studies = $75 - 75 = 0$ marks

Hence, the student improved his performance most in English.

PROBABILITY

Probability is the measure of the likelihood that an event will occur. In other words, the chance of happening of an event is called probability of the event happening. It tells us how likely an event is going to occur. It does not tell us what is going to happen.

For example : When a coin is thrown, it has two possible outcomes, Head or Tail. Which is equally likely to be obtained. Thus, we say that the probability of getting Head or Tail is equal and is $\frac{1}{2}$ for each.

Again for a dice, there are 6 possible outcomes. Thus, probability of getting either of 1, 2, 3, 4, 5 or 6 is equal.

That is, for a dice there are 6 equally likely possible outcomes, we say each of 1, 2, 3, 4, 5, 6 has one-sixth ($\frac{1}{6}$) probability.

ILLUSTRATION 6 :

A box contains pair of socks of two colours (black and white). I have picked out a white socks.

I pick out one more with my eyes closed. What is the probability that it will make a pair?

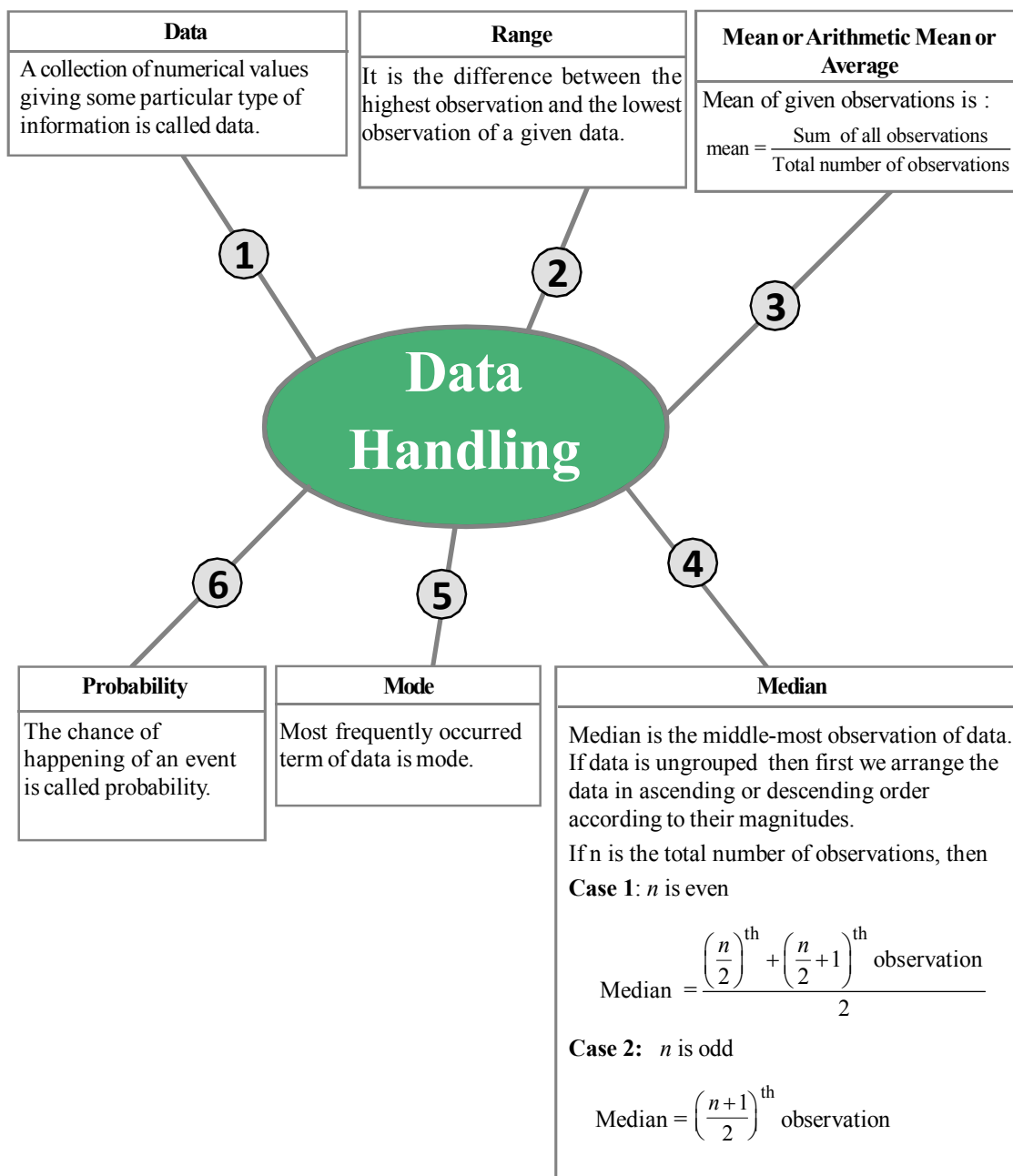
SOLUTION :

It can be observed that while closing the eyes, one can draw either a black socks or a white socks.

$$\text{Probability} = \frac{\text{Number of favourable outcomes}}{\text{Number of possible outcomes}}$$

$$\text{Probability (a pair of white socks will be formed)} = \frac{1}{3}$$

CONCEPT MAP



MISCELLANEOUS

SOLVED EXAMPLES

1. Given below are the heights (in cm) of 16 girls in a class:

154, 150, 152, 154, 154, 150, 148, 152, 152, 152, 154, 150, 152, 154, 152, 152.

Arrange the data in ascending order and prepare the frequency table.

- Sol. Arranging the given data in ascending order, we get.

148, 150, 150, 150, 152, 152, 152, 152, 152, 152, 152, 154, 154, 154, 154, 154.

We may now prepare the frequency table as shown below.

Height (in cm)	Tally Marks	Number of girls (Frequency)
148		1
150		3
152		7
154		5
Total		16

2. Following are the heights of 12 players (measured in centimetres):

144, 138, 154, 140, 141, 145, 149, 139, 148, 146, 147, 149.

- (i) Find the height of the tallest player.
 (ii) Find the height of the shortest player.
 (iii) Find the mean height.

- Sol. (i) The height of the tallest player = 154 cm
 (ii) The height of the shortest player = 138 cm
 (iii) The mean height

$$= \frac{\text{Sum of all observations}}{\text{Number of observations}}$$

$$= \frac{144 + 138 + 154 + 140 + 141 + 145 + 149 + 139 + 148 + 146 + 147 + 149}{12}$$

$$= \frac{1740}{12} = 145$$

3. Find the median of the values: 37, 31, 42, 43, 46, 25, 39, 45, 32

Sol. Arranging the data in ascending order, we have 25, 31, 32, 37, 39, 42, 43, 45, 46

Here, the number of observations $n = 9$ (odd)

$$\therefore \text{Median} = \text{Value of } \left(\frac{9+1}{2}\right)^{\text{th}} \text{ observation}$$

$$\Rightarrow \text{Median} = \text{Value of 5th observation} = 39$$

4. Find the mode for the following set of data.

(i) 1, 2, 4, 1, 3, 4, 5, 3, 1

(ii) 23, 25, 21, 10, 23, 35, 76, 25, 12, 34, 23, 25

- Sol. (i) Here, 1 has the maximum frequency, i.e. 3. Therefore, Mode = 1

(ii) Here, 23 occurs three times and 25 occurs three times as well. Therefore, Mode = 23 and 25

5. Marks obtained by 40 students in an examination are given below:

69, 59, 49, 39, 84, 68, 77, 48, 47, 57, 46, 41, 44, 67, 57, 45, 34, 36, 87, 89, 65, 41, 84, 78, 52, 49, 75, 37, 38, 42, 73, 31, 34, 37, 56, 59, 64, 85, 81, 62

Find the range of the marks.

- Sol. The lowest marks is 31 and the highest marks is 89.
 $\therefore \text{Range} = 89 - 31 = 58$

6. The number of cycles produced in a factory during five consecutive weeks is given below:

Week	Number of cycles produced
First	800
Second	1300
Third	1060
Fourth	920
Fifth	1440

Draw a bar graph representing the above information.

Sol. We can draw the bar graph by following these steps:

Step 1: On a graph paper, draw a horizontal line OX and a vertical line OY, representing the x-axis and y-axis respectively.

Step 2: Along OX, mark the weeks at points taken at equal gaps.

Step 3: Choose the scale: 1 small division = 20 cycles.

Step 4: The heights of the bars are:

Production in the 1st week

$$= \left(\frac{1}{20} \times 800 \right) = 40 \text{ small divisions.}$$

Production in the 2nd week

$$= \left(\frac{1}{20} \times 1300 \right) = 65 \text{ small divisions.}$$

Production in the 3rd week

$$= \left(\frac{1}{20} \times 1060 \right) = 53 \text{ small divisions.}$$

Production in the 4th week

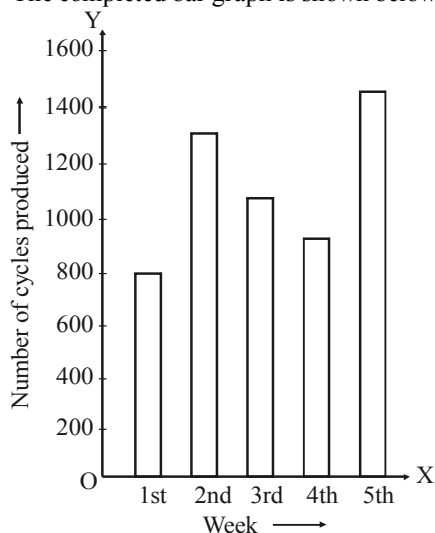
$$= \left(\frac{1}{20} \times 920 \right) = 46 \text{ small divisions.}$$

Production in the 5th week

$$= \left(\frac{1}{20} \times 1440 \right) = 72 \text{ small divisions.}$$

Step 5: Draw bars of equal width and of heights calculated in step 4 at the points marked in step 2.

The completed bar graph is shown below.

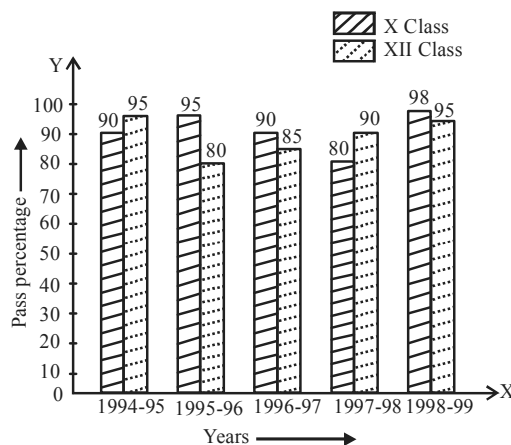


7. The results of pass percentage of Class X and XII in C.B.S.E. examination for 5 years are given in the following table:

Year	X	XII
1994-95	90	95
1995-96	95	80
1996-97	90	85
1997-98	80	90
1998-1999	98	95

Draw bar graphs to represent the data.

Sol.



8. A coin is tossed 100 times in which head is obtained 55 times. On tossing a coin at random, find the probability of getting (i) a head (ii) a tail.

Sol. We have

Total number of trials = 100

Number of times a head comes up = 55

Number of times a tail comes up
= (100 - 55) = 45

∴ Probability of getting a head

$$= \frac{\text{Number of heads}}{\text{Total number of trials}}$$

$$= \frac{55}{100} = 0.55$$

Probability of getting a tail

$$= \frac{\text{Number of tails}}{\text{Total number of trials}} = \frac{45}{100} = 0.45$$

1

EXERCISE

Fill in the Blanks :

DIRECTIONS : Complete the following statements with an appropriate word / term to be filled in the blank space(s).

- The mean of first five prime numbers is _____.
- The probability of getting a number greater than 2 on throwing a die once is _____.
- Mean, mode, median are the measures of _____.
- The mean of a data is defined as _____.
- Median is one of the observations in the data if number of observations is _____.
- The probability of an impossible event is _____.
- The middle most observation in an arranged data is _____.
- The observation that occurs most in a data is called _____.
- When a die is thrown, the probability of getting a number greater than 6 is _____.
- Two collection of data can be compared using _____.
- When a die is thrown there are 6 possible outcomes.
- If a die is thrown, the probability of getting a number less than 1 is 1.
- If a die is thrown, then the probability of getting an even prime number is $\frac{1}{2}$.
- The mean of the data 7, 8, 9, 11, 15, is 12.
- The mode of data is highest value in data.

Match the Columns :

DIRECTIONS : Each question contains statements given in two columns which have to be matched. Statements (A, B, C, D, E) in column-I have to be matched with statements (p, q, r, s, t) in column-II.

True / False :

DIRECTIONS : Read the following statements and write your answer as true or false.

- | 1. | Column-I | Column-II |
|----|---|---------------|
| A. | The data obtained initially is called | (p) mode |
| B. | Arranging numerical figures of a set of data in ascending or descending order is called | (q) frequency |
| C. | The number of times a particular observation occurs is called its | (r) data |
| D. | The number which occurs most frequently in a set of numbers is known as the | (s) an array |
| E. | The collection of a particular type of information in numerical figures is called | (t) raw-data |
- Range of the data is always from the data.
 - The data 12, 13, 14, 15, 16, has every observation as mode.
 - The range of the data 2, - 5, 4, 3, 7, 6 would change if 2 was added to each value in the data.
 - If the extreme observations on both the ends of a data arranged in ascending order are removed, the median gets affected.
 - The measures of central tendency may not lie between the maximum and minimum values of data.

Very Short Answer Questions :

DIRECTIONS : Give answer in one word or one sentence.

- Find the mean of first five natural numbers.
- Find the mean of first seven even natural numbers.
- Find the median of first 15 odd numbers.
- Find the median of first 50 whole numbers.
- The heights of 10 girls were measured in cm and the results are as follows:
135, 150, 139, 128, 151, 132, 146, 149, 143, 141
(i) What is the height of the tallest girl?
(ii) What is the height of the shortest girl?
(iii) What is the range of the data?
(iv) What is the mean height of the girls?
- A die is thrown 100 times and outcomes are noted as given below:

Outcome	1	2	3	4	5	6
Frequency	21	9	14	23	18	15

If a die is thrown at random, find the probability of getting a/an.

- 3
 - 5
 - 4
 - Even number
 - Odd number
 - Number less than 3.
- The marks (out of 100) obtained by a group of students in science test are 85, 76, 90, 84, 39, 48, 56, 95, 81, and 75. Find the
(i) highest and the lowest marks obtained by the students.
(ii) range of marks obtained.
(iii) mean marks obtained by the group.
 - Rainfall as measured in a particular city in 12 days of a rainy season is given below:
2 mm, 5 mm, 9 mm, 3 mm, 14 mm, 0 mm, 9 mm, 3 mm, 5 mm, 5 mm, 2 mm, 3 mm.
Find the average rainfall, median rainfall and modal value of the above data.

Short Answer Questions :

DIRECTIONS : Give answer in 2-3 sentences.

- The pocket-expenses of Rahul during a week are. ₹15.40, ₹18.00, ₹16.50, ₹14.75, ₹12.60 and ₹17.25. Find his mean pocket-expenses per day.

- Given below are the ages of 25 students of class VIII in a school. Prepare a discrete frequency distribution.

15, 16, 16, 14, 17, 17, 16, 15, 15, 16, 16, 17, 15
16, 16, 14, 16, 15, 14, 15, 16, 16, 15, 14, 15

- If the mean of 6, 4, 7, p and 10 is 8, find the value of p .
- The weights of 10 students (in kg) are
40, 52, 34, 47, 31, 35, 48, 41, 44, 38.
Find the median weight.
- The ages in years of 10 teachers in a school are:
32, 41, 28, 54, 35, 26, 23, 33, 38, 40.
(i) What is the age of the oldest teacher and that of the youngest teacher?
(ii) What is the range of the ages of the teachers?
(iii) What is the mean age of these teachers?
- Find the median of the data: 25, 34, 31, 23, 22, 26, 35, 28, 20, 32
- A die is thrown 200 times and the outcomes are noted as shown below:

Outcome	1	2	3	4	5	6
Frequency	21	30	42	38	29	40

When a die is thrown at random, find the probability of getting

- 5
 - 3
 - 4
 - 6
- Given below is the data showing the number of children in 20 families of a locality:
3, 1, 3, 2, 2, 2, 0, 3, 4, 2, 1, 3, 2, 4, 1, 2, 2, 3, 1, 3.
Arrange the data in ascending order and then prepare a frequency table.
 - The weights of new born babies (in kg) in a hospital on a particular day are as follows:
2.3, 2.2, 2.1, 2.7, 2.6, 3.0, 2.5, 2.9, 2.8, 3.1, 2.5, 2.8, 2.7, 2.9, 2.4
(i) Determine the range.
(ii) How many babies were born on that day?
(iii) How many babies weigh below 2.5 kg?
(iv) How many babies weigh more than 2.8 kg?
(v) How many babies weigh 2.8 kg?

Long Answer Questions :

DIRECTIONS : Give answer in four to five sentences.

1. In a mathematics test following marks were obtained by 40 students of class VI. Arrange these marks in a table using tally marks.

8 1 3 7 6 5 5 4 4 2
4 9 5 3 7 1 6 5 2 7
7 3 8 4 2 8 9 5 8 6
7 4 5 6 9 6 4 4 6 6

- Find how many students obtained marks equal to or more than 7?
 - How many students obtained marks below 4?
2. A manufacturing company produces machines. The production details of 5 weeks are given below.

Week	1	2	3	4	5
No. of machines	40	80	60	85	30

Draw a bar graph to show the data.

3. The following data gives the number of students in a class arriving late.

Days	Mon.	Tues.	Wed.	Thurs.	Fri.
Week 1	4	5	2	3	2
Week 2	6	3	4	1	2

Draw a dual bar chart to show the above set of data.

4. For the following data, draw a bar graph.

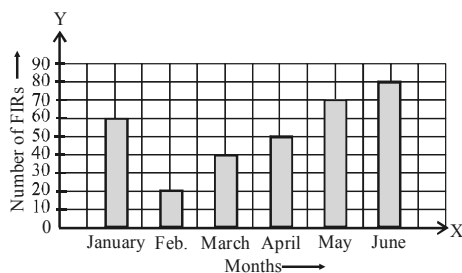
Items	Expenses
Rent	15%
Food	35%
Clothing	10%
Education	20%
Miscellaneous	20%

5. Consider the following data gathered from a survey of a colony:

Favourite Sport	Watching	Participating
Cricket	1240	620
Basket-Ball	470	320
Swimming	510	320
Hockey	420	250
Athletics	250	110

Draw a double bar graph choosing an appropriate scale. What do you infer from the bar graph?

- Which sport is most popular?
 - What is more preferred watching or participating in sports?
6. Observe the bar graph carefully and then answer the following questions:



- What information does the above graph provide?
- How many FIRs were registered from January to June?
- Which month registered the maximum number of FIRs?
- Which month registered the least number of FIRs?

2 EXERCISE

Text-Book Exercise :

1. Following table shows the points of each player scored in four games:

Player	A	B	C
Game 1	14	0	8
Game 2	16	8	11
Game 3	10	6	Did not play
Game 4	10	4	13

Now answer the following questions:

- Find the mean to determine A's average number of points scored per game.
 - To find the mean number of points per game for C, would you divide the total points by 3 or by 4? Why?
 - B played in all the four games. How would you find the mean?
 - Who is the best performer?
2. The rainfall (in mm) in a city on 7 days of a certain week was recorded as follows:

Day	Rainfall
Mon	0.0
Tue	12.2
Wed	2.1
Thurs	0.0
Fri	20.5
Sat	5.5
Sun	1.0

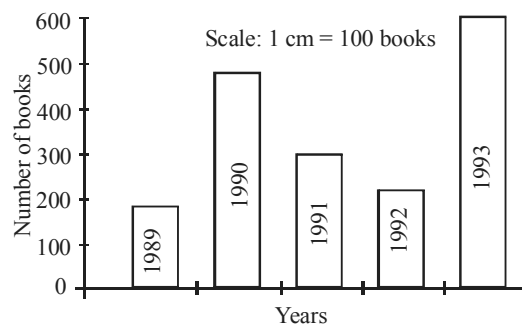
- Find the range of the rainfall in the above data.
 - Find the mean rainfall for the week.
 - On how many days was the rainfall less than the mean rainfall?
3. The runs scored in a cricket match by 11 players are as follows:

6, 15, 120, 50, 100, 80, 10, 15, 8, 10, 15

Find the mean, mode and median of this data. Are the three same?

All three mean, mode and median are not same.

4. Read the bar graph which shows the number of books sold by a bookstore during five consecutive years and answer the following questions:



- About how many books were sold in 1989? 1990? 1992?
 - In which year were about 475 books sold? About 225 books sold?
 - In which years were fewer than 250 books sold?
 - Can you explain how you would estimate the number of books sold in 1989?
5. The performance of a student in 1st term and 2nd term is given. Draw a double bar graph choosing appropriate scale and answer the following:

Subject	1st Term (M.M. 100)	2nd Term (M.M. 100)
English	67	70
Hindi	72	65
Maths	88	95
Science	81	85
S. Science	73	75

- In which subject, has the child improved his performance the most?

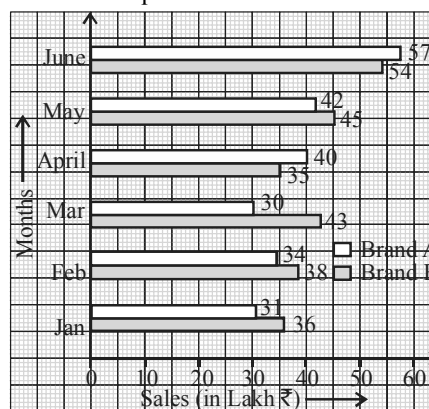
- (ii) In which subject is the improvement the least?
- (iii) Has the performance gone down in any subject?
6. Take the data giving the minimum and the maximum temperature of various cities given in the table.
Plot a double bar graph using the data and answer the following :

Temperatures of Cities as on 20.6.2006		
City	Max.	Min.
Ahmedabad	38°C	29°C
Amritsar	37°C	26°C
Bangalore	28°C	21°C
Chennai	36°C	27°C
Delhi	38°C	28°C
Jaipur	39°C	29°C
Jammu	41°C	26°C
Mumbai	32°C	27°C

- (i) Which city has the largest difference in the minimum and maximum temperature on the given date?
- (ii) Which is the hottest city and which is the coldest city?
- (iii) Name two cities where maximum temperature of one was less than the minimum temperature of the other.
- (iv) Name the city which has the least difference between its minimum and the maximum temperature.
7. Tell whether the following is certain to happen, impossible, can happen but not certain.
- (i) You are older today than yesterday.
- (ii) A tossed coin will land heads up.
- (iii) A die when tossed shall land up with 8 on top.
- (iv) The next traffic light seen will be green.
- (v) Tomorrow will be a cloudy day.
8. There are 6 marbles in a box with numbers from 1 to 6 marked on each of them.
- (i) What is the probability of drawing a marble with number 2?
- (ii) What is the probability of drawing a marble with number 5?

Exemplar Questions :

- The cards bearing letters of the word "MATHEMATICS" are placed in a bag. A card is taken out from the bag without looking into the bag (at random).
 - How many outcomes are possible when a letter is taken out of the bag at random?
 - The probability of getting M is _____.
 - The probability of getting any consonant is _____.
- The mean of 10 observations was calculated as 40. It was detected on rechecking that the value of 45 was wrongly copied as 15. What is the correct mean ?
- The median of observations 11, 12, 14, 18, $x + 2$, 20, 22, 25, 61 arranged in ascending order is 21. Find the value of x .
- Find the median of the given data if the mean is 4.5.
5, 7, 7, 8, x , 5, 4, 3, 1, 2
- What is the probability of the sun setting tomorrow?
- Classify the following events as certain to happen, impossible to happen, may or may not happen:
 - Getting a number less than 1 on throwing a die.
 - Getting head when a coin is tossed.
 - A team winning the match.
 - Christmas will be on 25 December.
 - Today moon will not revolve around the earth.
 - A ball thrown up in the air will fall down after some time.
- Study the double bar graph shown below and answer the questions that follow:



- What information is represented by the above double bar graph?
- In which month sales of Brand A decreased as compared to the previous month?
- What is the difference in sales of both the Brands for the month of June?
- Find the average sales of Brand B for the six months.
- List all months for which the sales of Brand B was less than that of Brand A.
- Find the ratio of sales of Brand A as compared to Brand B for the month of January.

8. Observe the following data:

Government School, Chandpur		
Daily Attendance		Date : 15.4.2009
Class	Total Students	Number of Students Present on that Day
VI	90	81
VII	82	76
VIII	95	91
IX	70	65
X	63	62

- Draw a double bar graph choosing an appropriate scale. What do you infer from the bar graph?
 - Which class has the maximum number of students?
 - In which class, the difference of total students and number of students present is minimum?
 - Find the ratio of number of students present to the total number of students of Class IX.
 - What per cent of Class VI students were absent?
9. The marks out of 100 obtained by Kunal and Soni in the Half Yearly Examination are given below:

Subjects	Kunal	Soni
English	72	81
Hindi	81	89
Maths	92	90
Science	96	82
S. Science	64	75
Sanskrit	85	82

- Draw a double bar graph by choosing appropriate Scale.
- Calculate the total percentage of marks obtained by Soni.
- Calculate the total percentage of marks obtained by Kunal.
- In how many subjects did Soni get more marks than Kunal? Which are those subjects?
- Who got more marks in S.Science and what was the difference of marks?
- In which subject the difference of marks was maximum and by how much?

HOTS Questions :

- If the mean of 4, x and y is 6, then find the mean of x , y and 10.
- The mean of 13 observations is 15. If one of the observations '3' is excluded from the group, what is the new mean?
- The values in data set are 95, 93, 91, 95, 100, 99 and 92. What value can be added to the set so that the mean, median and mode remain the same?
- The mean weight of sample of 10 oranges is 34 g. Later it was shown that the weighing machine had shown the weight of each orange 5g less. Find the correct mean weight of an orange.
- You roll a die. Write the probability of rolling a prime number as a decimal.

3

EXERCISE

Single Option Correct :

DIRECTIONS : This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONLY ONE is correct.

1. In a class test in English, 10 students scored 75 marks, 12 scored 60 marks, 8 scored 40 marks and 3 scored 30 marks. The mode for their scores is
(a) 75 (b) 30
(c) 60 (d) 25
2. The mean of first five prime numbers is
(a) 3 (b) 3.6
(c) 7 (d) 5.6
3. The mean of 20 items of a data is 5; and if each item is multiplied by 3, then the mean will be
(a) 5 (b) 10
(c) 15 (d) 20
4. The mean of $x, x+3, x+6, x+9$ and $x+12$ is
(a) $x+6$ (b) $x+3$
(c) $x+9$ (d) $x+12$
5. The mean of 10 observation is 25. If one observation, namely 25, is deleted, the new mean is
(a) 25 (b) 20
(c) 28 (d) 22
6. The attendance of a class of 45 boys for 10 days is given as 40, 42, 30, 35, 45, 44, 41, 38, 44 and 41 then the mean attendance of the class is
(a) 39 (b) 40
(c) 41 (d) 43
7. Arithmetic mean of 5 scores is 85. Then the sum of the scores is _____
(a) 425 (b) 85
(c) More than 425 (d) less than 400
8. The mean of 6, y, 7, x and 14 is 8. Then
(a) $x+y=13$ (b) $x-y=13$
(c) $2x+3y=13$ (d) $x^2+y^2=15$
9. In which data set are the mean, median, mode and range all the same number?
(a) 1, 2, 3, 3, 2, 1, 2 (b) 1, 2, 3, 1, 2, 3, 1
(c) 1, 3, 3, 3, 2, 3, 1 (d) 2, 2, 1, 2, 3, 2, 3
10. In a class there are 15 boys and 10 girls. If one child is absent, the probability that it is a boy, is
(a) $\frac{1}{5}$ (b) $\frac{2}{5}$
(c) $\frac{3}{5}$ (d) $\frac{4}{5}$
11. The probability of drawing a face card from a standard pack of 52 cards is
(a) $\frac{1}{13}$ (b) $\frac{4}{13}$
(c) $\frac{10}{13}$ (d) $\frac{3}{13}$
12. A letter is chosen at random from the word 'PROBABILITY'. The probability that it is a vowel is
(a) $\frac{6}{11}$ (b) $\frac{3}{11}$
(c) $\frac{7}{11}$ (d) $\frac{4}{11}$
13. If each score is increased by 5 then mean _____
(a) Remains the same
(b) Increases by 5
(c) Decreases by 5
(d) None of these
14. Tally marks are used to find _____
(a) Class intervals (b) Range
(c) Frequency (d) Upper limits

More Than One Option Correct :

DIRECTIONS : This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONE or MORE may be correct.

- Which of the following statement is/are false?
 - The mode is always one of the numbers in a data.
 - The mean is one of the numbers in a data.
 - The median is always one of the numbers in a data.
 - None of these
- Which of the following statement is/are true?
 - The mean of the first five multiples of 2 is 6.
 - The mode of the data set 4, 8, 4, 11, 4, 12, 8 is 4.
 - If there are 19 values in a data set arranged in descending order, then the median is the 5th term.
 - None of these
- The mode of the data 23, 26, 22, 29, 23, 29, 26, 29, 22, 23, is
 - 22
 - 23
 - 29
 - 26
- Which of the following is/are true?
 - The median of any data lies between the minimum and maximum value of observations.
 - The probability of an event which is certain to happen is 0.
 - The range of data : 21, 6, 17, 18, 9, 11, 15 is 15.
 - All of these.
- Which of the following is/are true?
 - The probability of getting an ace out of a deck of cards is greater than 1.
 - Distance between two consecutive bars is always same.
 - All bars in a bar graph have equal width.
 - None of these

Assertion & Reason :

DIRECTIONS : Each of these questions contains an Assertion followed by Reason. Read them carefully

and answer the question on the basis of following options. You have to select the one that best describes the two statements.

- If both **Assertion** and **Reason** are **correct** and Reason is the **correct explanation** of Assertion.
- If both **Assertion** and **Reason** are correct, but Reason is **not the correct explanation** of Assertion.
- If **Assertion** is **correct** but **Reason** is **incorrect**.
- If **Assertion** is **incorrect** but **Reason** is **correct**.

- Assertion:** Median of the data 25, 14, 7, 16, 21 is 16.

Reason: In case, the set of values has even number of values, the median is the mean of two middle values.

- Assertion:** The king, queen and jack of hearts are removed from a deck of 52 playing cards and well shuffled. One card is selected from the remaining cards.

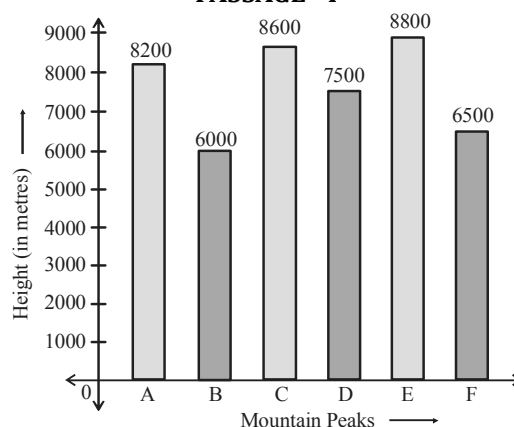
The probability of drawing a '10' of hearts is $\frac{1}{49}$.

Reason: Number of cards remaining after the king, queen and jack of hearts have been removed is 49.

Passage Based Questions :

DIRECTIONS : Study the given passage(s) and answer the following questions.

PASSAGE - I



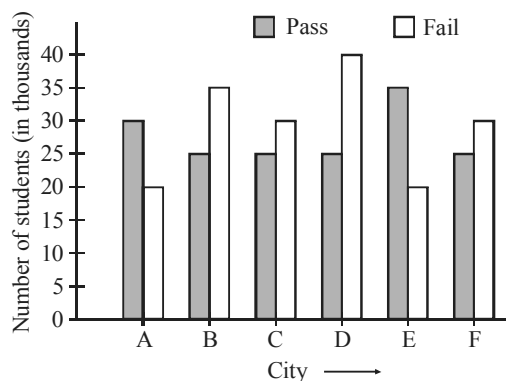
Read the above diagram and answer the following:

- Which is the highest peak?
 - A
 - B
 - C
 - D

2. Write the ratio of the heights of highest peak and the lowest peak.
 - (a) 22 : 15 (b) 15 : 22
 - (c) 20 : 13 (d) 13 : 22
3. Which peak is second highest?
 - (a) B (b) A
 - (c) C (d) E
4. When the heights of the given peaks are written in ascending order, then what is the average of the middle two peaks?
 - (a) 7950m (b) 7560m
 - (c) 7650m (d) 7850m

PASSAGE - II

Study the graph given below and answer the questions that follow.



5. The total number of students failing in the entrance exam from all the cities together, is
 - (a) 175000 (b) 168000
 - (c) 165000 (d) 217500
6. The ratio of the number of students passing in the entrance exam for city B to those passing from city E, is
 - (a) 5 : 7 (b) 1 : 1
 - (c) 5 : 4 (d) 6 : 5
7. The number of students passing in the examination from city A is approximately what percent of the total number of students passing in the examination from all cities together?
 - (a) 15 % (b) 12 %
 - (c) 18 % (d) 21 %

Integer Type Questions :

DIRECTIONS : Answer the following questions. The answer to each of the question is a single digit integer, ranging from 0 to 9.

1. If in a data, 10 numbers arranged in increasing order. If the 7th entry is increased by 4, then the median increased by
2. The mode of the following data is:

Size	1	3	5	7	9
Frequency	6	9	12	3	15
3. For the given data, calculate the mean of its median and mode.
6, 2, 5, 4, 3, 4, 4, 2, 3
4. A bag contains 5 cards written letters A, E, I, O and U on them. What is the chance of drawing out a consonant?

Multiple Matching Question :

DIRECTIONS : Following question has four statements (A, B, C and D) given in Column-I and four statements (p, q, r, s) in Column-II. Any given statement in Column-I can have correct matching with one or more statement(s) given in Column-II.

1. Match the following :

Column-I	Column-II
(A) The mean of the data: 23, 30, 25, 20, 27, is	(p) 18
(B) The median of the data: 52, 37, 65, 23, 43, 49, is	(q) 43
(C) The mode of the data : 15, 18, 23, 25, 27, 18, 12, 25, 34, is	(r) 25
(D) The range of the data : (s) 46 20, 35, 42, 57, 46, 63, is	

SOLUTIONS

Brief Explanations of Selected Questions

1 EXERCISE

Fill in the Blanks :

- 5.6
- $\frac{2}{3}$
- central tendency
- $\frac{\text{Sum of all observations}}{\text{Number of observations}}$
- odd.
- zero.
- median
- mode.
- 0
- double bar graph

True / False :

- False
- True
- False
- False
- False
- True
- False
- False
- False
- False

Match the Columns :

- (A) $\rightarrow$ (t); (B) $\rightarrow$ (s); (C) $\rightarrow$ (q); (D) $\rightarrow$ (p); (E) $\rightarrow$ (r)

Very Short Answer Questions :

- 3
- 8
- 15
- 24.5
- Clearly, the height of the tallest girl is 151 cm
 - The height of the shortest girl is 128 cm.
 - The range = $(151 - 128)$ cm = 23 cm
 - Mean height

$$\frac{128 + 132 + 135 + 139 + 141 + 143 + 146 + 149 + 150 + 151}{10}$$
 cm
 $\Rightarrow$ Mean height = $\frac{1414}{10}$ cm = 141.4 cm
- $\frac{7}{50}$
 - $\frac{9}{50}$
 - $\frac{23}{100}$
 - $\frac{47}{100}$
 - $\frac{53}{100}$
 - $\frac{3}{10}$

- 95,39 marks
 - 56 marks
 - 72.9 marks
- Avg. rainfall = 5 mm, Median = 4 mm, Modal rainfall = 3 mm and 5 mm.

Short Answer Questions :

- Sum of the pocket-expenses
 $= ₹(15.40 + 18.00 + 16.50 + 14.75 + 12.60 + 17.25) = ₹94.50$

$\therefore$ mean pocket-expenses = $₹\left(\frac{94.50}{6}\right)$ per day
 $= ₹15.75$ per day.

2.

Age	Tally Bars	Frequency
14		4
15		8
16		10
17		3
Total		25

- $$8 = \frac{6 + 4 + 7 + p + 10}{5}$$

$$\Rightarrow 40 = 27 + p \Rightarrow p = 13$$
- Arranging the weights in ascending order, 31, 34, 35, 38, 40, 41, 44, 47, 48, 52.

Here, $n = 10$, which is even.

$\therefore$ median weight

$$= \frac{1}{2} \{5\text{th term} + 6\text{th term}\}$$

$$= \frac{1}{2} \{40 + 41\} \text{ kg} = \frac{81}{2} \text{ kg} = 40.5 \text{ kg.}$$
- Let us arrange their ages in ascending order as follows:
23, 26, 28, 32, 33, 35, 38, 40, 41, 54
Age of the oldest teacher is 54 years and

the age of the youngest teacher is 23 years.

- (ii) Range of the ages of teachers
 $= (54 - 23) = 31$ years

- (iii) Mean age of teachers

$$= \frac{23 + 26 + 28 + 32 + 33 + 35 + 38 + 40 + 41 + 54}{10}$$

$$= \frac{350}{10} \text{ years} = 35 \text{ years}$$

6. Arranging the data in ascending order, we get
 20, 22, 23, 25, 26, 28, 31, 32, 34, 35

Here, the number of observation $n = 10$ (even).

$$\text{Median} = \frac{26 + 28}{2} = 27$$

7. (i) $\frac{29}{200}$ (ii) $\frac{21}{100}$
 (iii) $\frac{19}{100}$ (iv) $\frac{1}{5}$

8. Arranging the data in ascending order, we get
 0, 1, 1, 1, 1, 2, 2, 2, 2, 2, 2, 2, 3, 3, 3, 3, 3, 3, 4, 4.
 The frequency table of the given data, is shown below.

Number of children	Tally Bars	Number of families (Frequency)
0		1
1		4
2		7
3		6
4		2
Total		20

9. (i) $(3.1 - 2.1) \text{ kg} = 1.0 \text{ kg}$ (ii) 15
 (iii) 4 (iv) 4 (v) 2

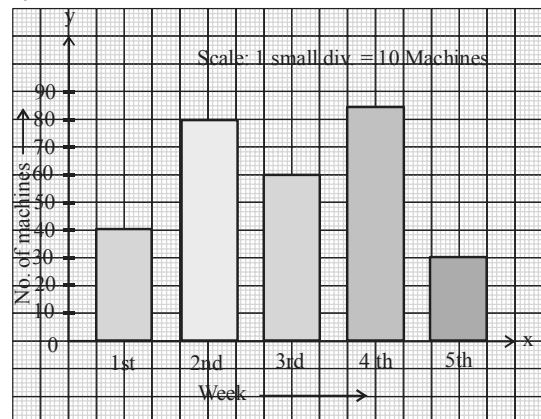
Long Answer Questions :

1.

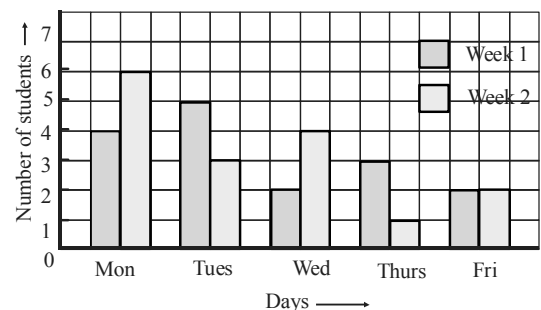
Marks	Tally marks	Frequency
1		2
2		3
3		3
4		7
5		6
6		7
7		5
8		4
9		3
Total		40

(i) 12 (ii) 8

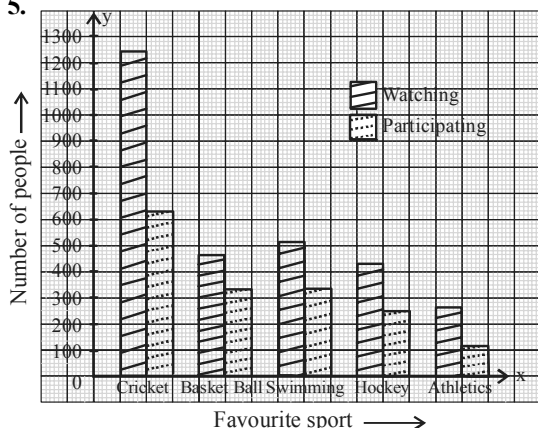
2.



3.



5.



- (i) Cricket (ii) Watching
6. (i) The bar graph shows the number of FIRs registered in different months.
 (ii) Total FIRs = 60 + 20 + 40 + 50 + 70 + 80 = 320.
 (iii) The maximum number of FIRs were registered in June.
 (iv) The least number of FIRs were registered in February.

2

EXERCISE

Text-Book Exercise :

1. (i) A's average number of points

$$= \frac{14+16+10+10}{4} = \frac{50}{4} = 12.5$$
- (ii) To find the mean number of points per game for C, we will divide the total points by 3 because C played 3 games.
 Mean of C's score

$$= \frac{8+11+13}{3} = \frac{32}{3} = 10.6$$
- (iii) Mean of B's Score = $\frac{0+8+6+4}{4} = \frac{18}{4} = 4.5$
- (iv) The average of A is 12.5 which is more than that of B and C. Therefore, A is the best performer.
2. (i) Range = (20.5 - 0.0) mm = 20.5 mm
 (ii) Mean rainfall

$$= \frac{(0.0 + 12.2 + 2.1 + 0.0 + 20.5 + 5.5 + 1.0)}{7}$$

$$= \frac{41.3}{7} = 5.9 \text{ mm}$$

(iii) For 5 days the rainfall was less than the mean rainfall.

3. Arranging the given scores in an ascending order, we have

6, 8, 10, 10, 15, 15, 15, 50, 80, 100, 120

Mean

$$= \frac{6+8+10+10+15+15+15+50+80+100+120}{11}$$

$$= \frac{429}{11} = 39$$

As 15 occurs for the most number of times.

$\therefore$ Mode = 15

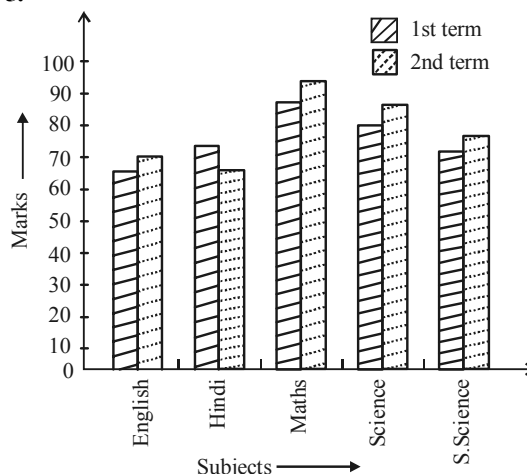
$$\text{Median} = \left(\frac{11+1}{2} \right)^{\text{th}} \text{ term} = 6^{\text{th}} \text{ term} = 15$$

All three mean, mode and median are not same.

4. (i) In 1989, 175 books were sold. In 1990, 475 books were sold. In 1992, 225 books were sold.
 (ii) 475 books were sold in the year 1990 and 225 books were sold in the year 1992.
 (iii) In the year 1989 and 1992, the number of books sold were less than 250.
 (iv) The number of books sold in the year 1989 is about 200.

Scale : 1 cm = 10 Marks

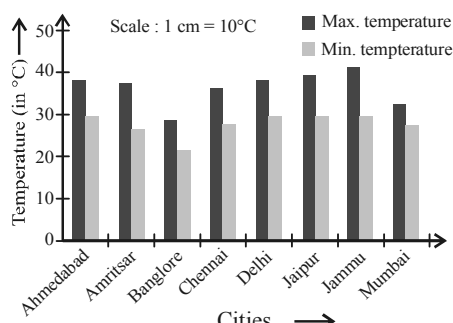
5.



- (i) The child has improved his performance the most in Maths.

- (ii) The improvement was the least in S.Science.
 (iii) The performance in Hindi has gone down.

6.



- (i) Jammu has the largest difference in its minimum and maximum temperatures.
 (ii) Jammu is the hottest city and Bangalore is the coldest city.
 (iii) Bangalore and Jaipur, Bangalore and Ahmedabad
 (iv) Mumbai.
7. (i) Certain
 (ii) Can happen but not certain
 (iii) Impossible
 (iv) Can happen but not certain
 (v) Can happen but not certain
8. (i) Probability of drawing 2 = $\frac{1}{6}$
 (ii) Probability of drawing 5 = $\frac{1}{6}$

Exemplar Questions :

1. (i) There are 11 outcomes namely M, M, A, A, T, T, H, E, I, C, S.
 (ii) Probability of getting 'M' = $\frac{2}{11}$
 (iii) Probability of getting a consonant = $\frac{7}{11}$
2. $40 = \frac{\text{Sum of all observations}}{10}$
 So, sum of all observations = 400
 $\therefore$ Correct sum = Incorrect sum – Incorrect observation + correct observation
 $= 400 - 15 + 45 = 430$
 Correct Mean = $\frac{430}{10} = 43$
3. Median from data = $x + 2$
 or, $21 = x + 2$
 or, $x = 19$

4. Mean = $\frac{5 + 7 + 7 + 8 + x + 5 + 4 + 3 + 1 + 2}{10}$

$$\Rightarrow 4.5 = \frac{42 + x}{10}$$

$$\Rightarrow x = 45 - 42 = 3$$

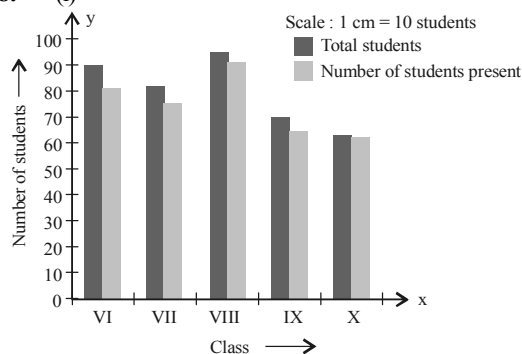
Now, arranging data in ascending order.

1, 2, 3, 3, 4, 5, 5, 7, 7, 8

$$\text{Median} = \frac{5^{\text{th}} \text{ term} + 6^{\text{th}} \text{ term}}{2} = \frac{4 + 5}{2} = 4.5$$

5. The probability of the sun setting tomorrow is 1.
 6. (i) Impossible to happen
 (ii) May or may not happen
 (iii) May or may not happen
 (iv) Certain to happen
 (v) Impossible to happen
 (vi) Certain to happen
7. (i) Comparison of sales of brand A and brand B during the month of January to June.
 (ii) March
 (iii) 3 Lakhs
 (iv) 41.8 Lakhs (approx.)
 (v) April, June
 (vi) 31 : 36

8. (i)



The double bar graph shows the total number of students and those of present on 15.4.2009.

- (ii) Class VIII
 (iii) In class X, the difference of total students and number of students present is $63 - 62 = 1$, which is minimum.
 (iv) Required ratio = $65 : 70 = 13 : 14$
 (v) No. of students absent = $90 - 81 = 9$
 Percent of absent students

$$= \left(\frac{9}{90} \times 100 \right) \% = 10\%$$

9. (ii) 83.2%
 (iii) 81.7%
 (iv) Three subjects (English, Hindi and S.Sc.)
 (v) Soni, 11 marks
 (vi) In Science 14 marks

HOTS Questions :

1. Given, $\frac{4+x+y}{3} = 6$
 $\Rightarrow x+y = 6 \times 3 - 4 = 14$
 $\therefore \frac{x+y+10}{3} = \frac{14+10}{3} = 8$
2. Sum of 13 observations = $15 \times 13 = 195$
 If the observation 3 is excluded, the new sum
 $= 195 - 3 = 192$
 $\therefore$ New Mean = $\frac{192}{12} = 16$
3. Mean = $\frac{95+93+91+95+100+99+92}{7} = 95$
 If we can add 95, the mean, mode and median remains same.
4. $34 \text{ g} + 5 \text{ g} = 39 \text{ g}$
5. Probability of rolling prime = $\frac{3}{6} = \frac{1}{2} = 0.5$

3

EXERCISE

Single Option Correct :

1. (c) Clearly, the maximum students (12) scored 60 marks. Hence, the Mode is 60.
2. (d) Mean = $\frac{2+3+5+7+11}{5} = \frac{28}{5} = 5.6$
3. (c) Then sum of 20 items = $20 \times 5 = 100$
 Multiplying each item by 3 and adding the result, we get
 $3 \times 20 \times 5 = 300$
 Hence, the new mean = $\frac{300}{20} = 15$.
4. (a)
 Mean = $\frac{x+(x+3)+(x+6)+(x+9)+(x+12)}{5}$

$$= \frac{5x+30}{5} = x+6$$

5. (a) The sum of 10 observations is 250.
 $\therefore$ new mean = $\frac{250-25}{9} = \frac{225}{9} = 25$.
6. (b) 7. (a)
8. (a) $\frac{6+y+7+x+14}{5} = 8$,
 or $27+x+y=40$ or $x+y=13$.
9. (a) Mean = $\frac{1+2+3+3+2+1+2}{7} = \frac{14}{7} = 2$
 Mode = 2 (2 occurs most)
 Data in ascending order: 1, 1, 2, 2, 2, 3, 3
 Median = 4th observation = 2
10. (c) Probability of boy being absent
 $= \frac{15}{15+10} = \frac{15}{25} = \frac{3}{5}$
11. (d) Total face cards = 12
 $\therefore P(\text{face card}) = \frac{12}{52} = \frac{3}{13}$
12. (d) Required probability = $\frac{4}{11}$
13. (b) Increases by 5
14. (c)

More Than One Option Correct :

1. (b, c) Only statement (a) is true.
 2. (a, b) 3. (b, c) 4. (a, c)
 5. (b, c)

Assertion & Reason :

1. (b) **Assertion** and **Reason**, both are correct but **reason** is not the correct explanation for **Assertion**.
2. (a) **Assertion** and **Reason**, both are correct and **Reason** is correct explanation of **Assertion**.

Passage Based Questions :

1. (b) E is the highest peak.
 2. (a) Height of highest peak = 8800 m
 Height of lowest peak = 6000 m
 Ratio = $8800 : 6000 = 22 : 15$

3. (c) Second highest peak is C.
 4. (d) The ascending order of heights is,
 $6000\text{ m} < 6500\text{ m} < 7500\text{ m} < 8200\text{ m} < 8600\text{ m} < 8800\text{ m}.$

The two middle peaks are
 7500 m $8200\text{ m}.$

$$\text{Average} = \frac{7500 + 8200}{2}$$

$$= \frac{15700}{2} = 7850\text{ m}.$$

5. (a) Total number of students failing in the exam = $20 + 35 + 30 + 40 + 20 + 30 = 175$ (in thousands) = 175000

6. (a) Required ratio = $\frac{25000}{35000} = \frac{5}{7} = 5 : 7$

7. (c) Required % = $\frac{30 \times 100}{165} = \frac{1000}{55}$
 $= \frac{200}{11} = 18\%(\text{approx.})$

Integer Type Questions :

1. (0) The median would be average of the 5th and 6th observations. Since they are unaffected by increase in 7th entry, the median will be unchanged.
 2. (9) From the table, it is clear that the size corresponding to the maximum frequency (15) is 9. Hence, the mode is 9.
 3. (4) Arranging data in ascending order, we have 2, 2, 3, 3, 4, 4, 4, 5, 6

$$\therefore \text{Median} = \left(\frac{9+1}{2} \right)^{\text{th}} \text{ term} = 5^{\text{th}} \text{ term} = 4$$

$$\text{Mode} = 4$$

$$\text{Mean of median and mode} = \frac{4+4}{2} = 4$$

4. (0) Probability of drawing a consonant

$$= \frac{0}{5} = 0$$

Multiple Matching Question :

1. (A) $\rightarrow$ (r); (B) $\rightarrow$ (s); (C) $\rightarrow$ (p, r); (D) $\rightarrow$ (q)

Chapter

4

Simple Equations

INTRODUCTION

An equation is a condition of equality between two mathematical expressions. It is a condition on a variable. In this chapter, we shall discuss the meaning of the solution of an equation and the methods to obtain it. We shall also discuss the formulation and solution of some practical problems.

EQUATIONS

A statement of equality which contains an unknown quantity is called an **equation**.

For example : $x + 4 = 11$

An equation consists of:

- a left-hand-side expression (abbreviated LHS),
- a right-hand side expression (abbreviated RHS),
- an equal sign between them.

The equal sign indicates that the value of the expression to the right side of equality sign should be equal to the value of the expression to the left side of equality sign.

The following is the general pattern.

$$\boxed{\text{Left Hand Side}} = \boxed{\text{Right Hand Side}}$$

NOTE: An equation remains the same, when the expressions on the left and on the right are interchanged.

ILLUSTRATION 1 :

Write the following statements in the form of equations :

- A number decreased by 10 is 40.
- Nine added to thrice a number gives 45.
- If d is divided by -4 , the result is 3.
- q multiplied by r is 36.

SOLUTION :

- Let the number be x .
Number decreased by 10 = $x - 10$
According to the question,
 $x - 10 = 40$

- Let the number be x
Thrice a number = $3x$
According to question
 $3x + 9 = 45$

- Required quotient = $\frac{d}{(-4)}$

According to question

$$-\frac{d}{4} = 3$$

- Product of q and $r = qr$
According to question $qr = 36$

SOLUTION OR ROOT OF THE EQUATION

A number which satisfies an equation is called the **solution** or **root** of the equation.

For example : In $5 + x = 7$, 2 is the solution or root of the equation.

If we want to check it, then substitute the value, $x = 2$ in LHS.

$$\therefore 5 + x = 7$$

$$\Rightarrow 5 + 2 = 7$$

$$\Rightarrow 7 = 7$$

Thus, LHS = RHS. Hence $x = 2$ is a solution of the given equation.

SOLVING AN EQUATION

Here we have two methods to solve an equation.

1. Balancing Method

Since an equation resembles a balance, therefore to find the solution, if a change is made in one side of an equation, the same change will have to be made in the other side of the equation. The following four facts help us in solving equations.

1. The same number can be added to both sides of an equation.
2. The same number can be subtracted from both sides of an equation.
3. Both sides of an equation can be multiplied by the same number.
4. Both sides of an equation can be divided by the same number.

ILLUSTRATION 2 :

Solve the equation : (i) $7 = x - 2$ (ii) $\frac{z}{4} = 3$

SOLUTION :

$$(i) \quad 7 = x - 2$$

Adding 2 to both sides,

$$7 + 2 = x - 2 + 2$$

$$\Rightarrow 9 = x$$

$$(ii) \quad \frac{z}{4} = 3$$

Multiplying both sides by 4,

$$\frac{z}{4} \times 4 = 3 \times 4$$

$$\Rightarrow z = 12$$

2. Transposition Method

Any term of an equation may be taken from one side to the other with a change in its sign. This does not effect the equality of equation. This process is called **transposition**.

Working Rules for transposition Method

- Step 1** : Obtain the linear equation and identify the variables and constants.
- Step 2** : Simplify the LHS and RHS to their simplest form by removing brackets.
- Step 3** : Transpose all terms containing the variable on LHS and constant terms on RHS.
- Step 4** : Simplify LHS and RHS to the simplest form so that each side contains just one term.
Also, make the coefficient of the variable 1.

ILLUSTRATION 3 :Solve $5x - 6 = 4x - 2$ **SOLUTION:**

$$5x - 6 = 4x - 2$$

$$\Rightarrow 5x - 4x = -2 + 6$$

[Transposing $4x$ to LHS and -6 to RHS]

$$\Rightarrow x = 4$$

Check:Substituting $x = 4$ in the given equation, we get

$$\text{LHS} = 5 \times 4 - 6 = 20 - 6 = 14, \text{ and}$$

$$\text{RHS} = 4 \times 4 - 2 = 16 - 2 = 14$$

$$\therefore \text{LHS} = \text{RHS}$$

ILLUSTRATION 4 :Solve : $2y + \frac{11}{4} = \frac{1}{3}y + 2$ **SOLUTION:**

$$2y + \frac{11}{4} = \frac{1}{3}y + 2 \quad \Rightarrow 2y - \frac{1}{3}y = 2 - \frac{11}{4}$$

$$\Rightarrow \frac{6y - y}{3} = \frac{8 - 11}{4} \quad \Rightarrow \frac{5y}{3} = -\frac{3}{4}$$

Multiplying both sides by $\frac{3}{5}$, we get

$$\frac{5}{3}y \times \frac{3}{5} = -\frac{3}{4} \times \frac{3}{5} \quad \Rightarrow y = -\frac{9}{20}$$

APPLICATION OF SIMPLE EQUATIONS

In this section, we shall study the formulation and solution of some practical problems.

We often refer to these problems as word problems. These word problems are first translated in the form of an equation containing variables and constants and then solved.

ILLUSTRATION 5 :When you add 3 to one-third of z , you get 30. Find the value of z .**SOLUTION:**

$$\frac{z}{3} + 3 = 30 \Rightarrow \frac{z}{3} = 30 - 3 \Rightarrow \frac{z}{3} = 27 \Rightarrow z = 27 \times 3 = 81$$

ILLUSTRATION 6 :

A number is multiplied by 3 and 7 is taken away from the product to get the answer 17. What is the number?

SOLUTION:Let the number be x .3 times the number = $3x$ Taking 7 from it = $3x - 7$

$$\therefore 3x - 7 = 17$$

$$\Rightarrow 3x = 17 + 7 = 24$$

$$\Rightarrow x = 8$$

ILLUSTRATION 7 :

The sum of ages of father and his son is 75 years. If the age of the son is 25 years, find the age of the father.

SOLUTION:Let the age of the father be x years.

The age of son = 25 years.

Sum of the two ages = $(x + 25)$ years.

But the sum of the ages is given as 75 years.

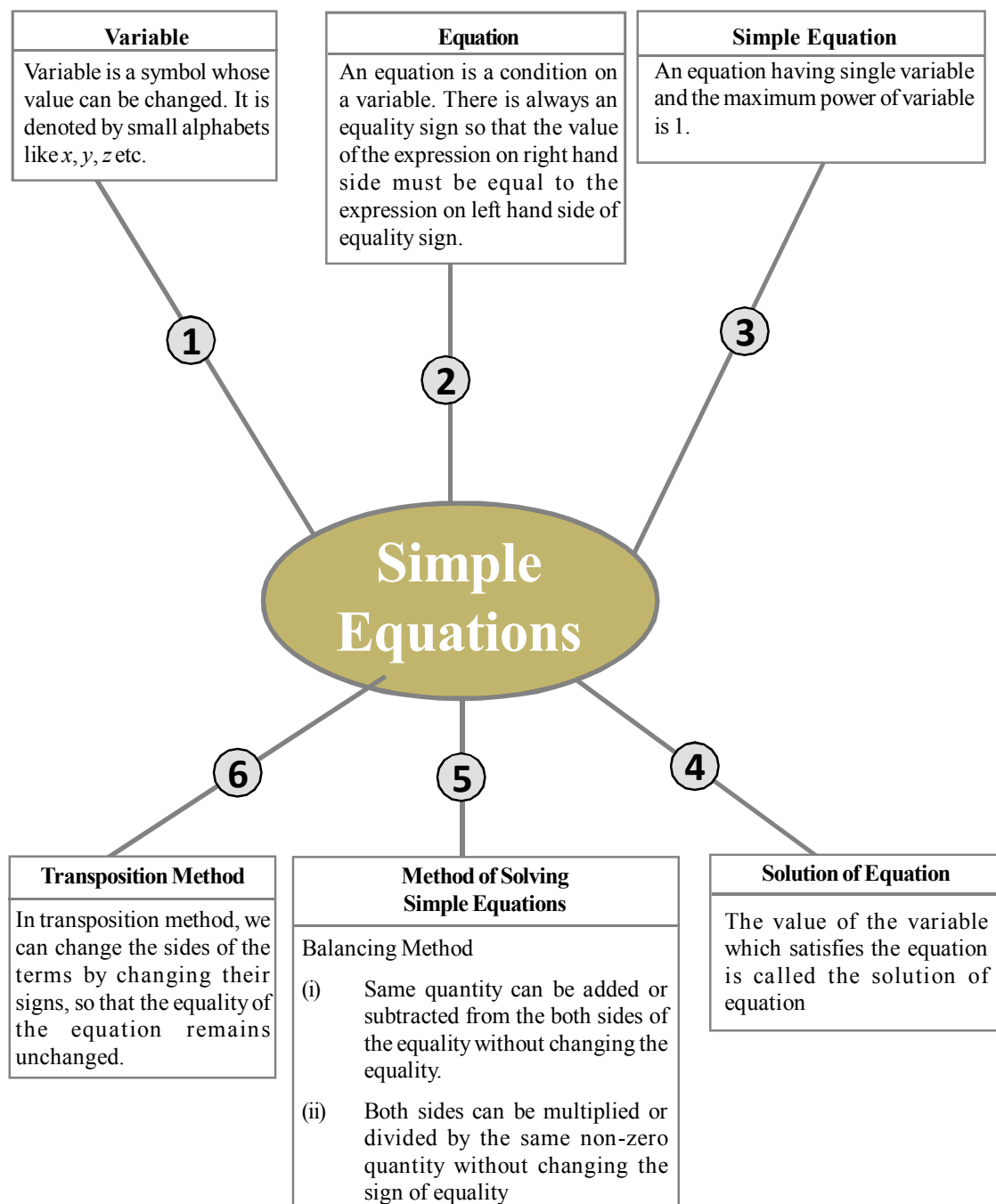
$$\therefore x + 25 = 75$$

$$\Rightarrow x = 75 - 25$$

$$\Rightarrow x = 50$$

Thus, the age of the father is 50 years.

CONCEPT MAP



MISCELLANEOUS

SOLVED EXAMPLES

1. The length of a rectangular plot exceeds its breadth by 5 metres. If the perimeter of the plot is 142 metres, find the dimensions of the plot.

Sol. Let the breadth of the plot be x metres.
Then, the length = $(x + 5)$ metres.

$$\begin{aligned}\therefore \text{perimeter} &= 2(\text{length} + \text{breadth}) \\ &= 2(x + 5 + x) \text{ metres} \\ &= (4x + 10) \text{ metres.}\end{aligned}$$

$$\therefore 4x + 10 = 142$$

$$\Rightarrow 4x = 132 \quad \Rightarrow x = \frac{132}{4} = 33.$$

Thus, breadth = 33 metres.

And, length = $(33 + 5)$ metres = 38 metres.

2. One number is four less than three times another. If their sum is increased by five, the result is twenty-five. Find the numbers.

Sol. Let the other number be x .

First number = 4 less than three times of x
 $= 3x - 4$

Then, the sum of the two numbers
 $= 3x - 4 + x = 4x - 4$

It is also given that if the sum of the numbers increased by five, the result is twenty-five.

$$\text{So, } 4x - 4 + 5 = 25$$

$$\text{or } 4x + 1 = 25$$

$$\text{or } 4x = 25 - 1 = 24$$

$$\text{or } x = 6$$

$$\begin{aligned}\text{Now, the first number} &= 3x - 4 \\ &= 3 \times 6 - 4 = 14\end{aligned}$$

and the second number = $x = 6$

Hence, the numbers are 14 and 6.

3. Solve the equation $5x - 3x = 10 - 2$

$$\begin{aligned}\text{Sol. } 5x - 3x &= 10 - 2 \\ \Rightarrow 5x - 3x &= 8 \quad \Rightarrow 2x = 8\end{aligned}$$

$$\Rightarrow x = \frac{8}{2} \quad [\text{Dividing both sides by 2}]$$

$$\Rightarrow x = 4$$

4. Solve and check the equation:

$$\frac{5x-4}{8} = \frac{x-3}{5} = \frac{x+6}{4}$$

Sol. We have,

$$\frac{5x-4}{8} - \frac{x-3}{5} = \frac{x+6}{4}$$

Multiplying both sides by 40, the LCM of 8, 5 and 4, we have

$$\begin{aligned}5(5x-4) - 8(x-3) &= 10(x+6) \\ \Rightarrow 25x - 20 - 8x + 24 &= 10x + 60\end{aligned}$$

$$\Rightarrow 17x + 4 = 10x + 60$$

$$\Rightarrow 17x - 10x = 60 - 4$$

[Transposing $10x$ to LHS and 4 to RHS]

$$\Rightarrow 7x = 56 \quad \Rightarrow x = \frac{1}{7} \times 56 = 8$$

Thus, $x = 8$ is a solution of the given equation.

5. The sum of three consecutive integers is 10 more than twice the smallest of the integers. Find the integers.

Sol. Let the smallest integer be x .

Then the three integers are $x, x + 1, x + 2$.

Their sum $x + x + 1 + x + 2 = 3x + 3$.

Ten more than twice the smallest integer
 $= 2x + 10$

$$\therefore 3x + 3 = 2x + 10$$

$$\Rightarrow 3x - 2x = 10 - 3 = 7$$

$$\therefore x = 7$$

$\therefore$ The three integers are 7, 8, and 9.

6. Three less than half the age of father is the daughter's age. If the daughter is 22 years old, find the father's age.

Sol. Let the father's age be x years.

$\therefore$ According to the question,

$$\Rightarrow \frac{x}{2} - 3 = 22 \quad \Rightarrow \frac{x}{2} = 22 + 3$$

$$\Rightarrow \frac{x}{2} = 25 \quad \Rightarrow x = 2 \times 25 = 50$$

$\therefore$ Father's age is 50 years.

7. Verify that $y = 9$ is the solution of the equation $\frac{y}{3} + 5 = 8$.

Sol. Putting $y = 9$

$$\text{L.H.S.} = \frac{9}{3} + 5 = 3 + 5 = 8 \text{ and R.H.S.} = 8.$$

Thus, for $y = 9$, we have L.H.S. = R.H.S.

Hence, $y = 9$ is the solution of the given equation.

8. Solve: $0.3x + 0.4 = 0.28x + 1.16$

Sol. We have,

$$0.3x + 0.4 = 0.28x + 1.16$$

$$0.3x - 0.28x = 1.16 - 0.4$$

[Transposing $0.28x$ to L.H.S. and 0.4 to R.H.S.]

$$\Rightarrow (0.3 - 0.28)x = 1.16 - 0.4$$

$$\Rightarrow 0.02x = 0.76 \quad \Rightarrow x = \frac{0.76}{0.02} = 38$$

Thus, $x = 38$ is the solution of the given equation.

1

EXERCISE

Fill in the Blanks :

DIRECTIONS : Complete the following statements with an appropriate word / term to be filled in the blank space(s).

- If $5m + 18 = 8$, then $m =$ _____.
- If $x = 5$, then $\frac{x}{2} + \frac{1}{2} =$ _____.
- The root of the equation $0.8x + 9 = 17$, is ____.
- If $\frac{x}{5} - 9 = 1$ then $x =$ _____.
- If $x = 10$, then $0.2x + 0.2 =$ _____.
- If 10 less than a number is 65, then the number is _____.
- If a number is increased by 20, it becomes 45. Then the number is _____.
- If 84 exceeds another number by 12, then the other number is _____.
- If $x - \frac{7}{8} = \frac{7}{8}$, then $x =$ _____.
- Any value of the variable which makes both sides of an equation equal, is known as a _____ of the equation.
- The root of the equation $y - 13 = 9$ is _____.

True / False :

DIRECTIONS : Read the following statements and write your answer as true or false.

- The root of the equation $\frac{2x+3}{4} = x + 8$, is 7.
- 5 is the solution of the equation $3x + 2 = 17$.
- $4x - 5 = 7$ does not have an integer as its solution.
- $\frac{3}{2}$ is the solution of the equation $8x - 5 = 7$.

- If $4x - 7 = 11$, then $x = 4$.
- $5a$ means 5 is added to a .
- Variables are denoted usually by the numbers 1, 2, 3, 4, 5, 6.
- If we subtract same value from both the sides of an equation, then the value from L.H.S does not match with the value of R.H.S.
- We can write, one fifth of a number minus 4 gives us 3, in the form of an equation as $\frac{1}{5}x - 4 = 3$.
- Unknown quantity is known as variable.

Match the Columns :

DIRECTIONS : Each question contains statements given in two columns which have to be matched. Statements (A, B, C, D) in column-I have to be matched with statements (p, q, r, s) in column-II.

- Match the columns:

Column-I	Column-II
(A) The root of the equation $\frac{4x}{5} + 9 = 2x - 3$ is	(p) 64
(B) If 96 is divided into two parts such that one part is twice the other, then the greater part is	(q) 24
(C) The root of the equation $\frac{7x}{2} + 2 = 14 + 3x$ is	(r) 50
(D) One-fifth of a number is 5 more than one-tenth of the number. The number is	(s) 10

Very Short Answer Questions :

DIRECTIONS : Give answer in one word or one sentence.

- Write the following statements in equation form.
 - The sum of 5 and x is 12
 - x decreased by 7 is 13.
 - Six less than two times y is 32
 - If 8 is taken away from the number n , the result is -4.
 - q multiplied by 7 gives -35.
 - Eight decreased by 3 times a number equals 2.
 - The product of 7 and the number x is 42.
 - Quotient of m divided by 5 is 6.
- Solve : $3x - 5 = 0$
- Solve : $8x - 3 = 9 - 2x$
- Verify by substitution that:
 - $x = 3$ is the root of $5 + 3x = 14$
 - $x = 2$ is the root of $3x - 2 = 8x - 12$
 - $x = 4$ is the root of $\frac{3x}{2} = 6$
 - $y = 2$ is the root of $y - 3 = 2y - 5$
- Solve for x : $12x - 7 = 7x - 13$
- Solve for m : $1.2m + 2.6 = 5$
- If $\frac{x}{2} + \frac{x}{3} = 5$, then find x
- If $1.7y + 2.3y = 2$, then find y .
- Solve : $5(2 - 3x) - 17(2x - 5) = 16$
- Solve : $3(x + 2) - 2(x - 1) = 7$
- Solve : $14 = \frac{7x}{10} - 8$

Short Answer Questions :

DIRECTIONS : Give answer in 2-3 sentences.

- Four-fifths of a number is greater than three-fourths of the number by 4. Find the number.
- The sum of two consecutive even numbers is 86. Find the numbers.
- If one-fourth, half and one-third of a number are added to the number itself, then the result is equal 25. Find the number.

- Divide ₹ 98 into two parts such that one part is 6 times to the other part.
- A number is multiplied by 5 and 25 is subtracted from the product. The result is equal to four times the number itself. Find the number.
- Thirty years ago, the age of a man was three-fifth of his present age. Find his present age (in years).
- Solve the following equations:
 - $5(6x + 1) = 3(-8 + x) + 2$
 - $4 - 4(x - 5) = 2(2 - x) - 6$
 - $10y - 0.2(7 - y) = 19$
 - $\frac{12 - x}{5} = x$
- The length of a rectangle is three times its width. If the perimeter of the rectangle is 96 metres, find the length and breadth of the rectangle.
- There are only 10 paise coins in a purse. If the total value of the money in the purse is ₹ 25, then find the number of coins in the purse.
- If 20% of 60% of a number is 144, then find the number.

Long Answer Questions :

DIRECTIONS : Give answer in four to five sentences.

- Solve: $3(x + 3) - 2(x - 1) = 5(x - 5)$ and check the result.
- Solve: $\frac{12}{7}(x - 5) = 24 + 8x$
- Solve: $\frac{5x - 2}{3} + \frac{4x + 3}{2} = \frac{3x + 19}{2}$
- Solve: $\frac{9m + 4}{5} - \frac{27m + 1}{8} - \frac{1}{2} = 0$
- If $\frac{0.3(3x - 4)}{5} + \frac{0.4x + 3.6}{2} = 3.5x$, then find x .
- Solve: $\frac{y - 8}{3} = \frac{7 - 4y}{7}$
- Solve : $x - \left(2x - \frac{3x - 4}{7}\right) = \frac{4x - 27}{3} - 3$

2

EXERCISE

Text-Book Exercise :

- Check whether the value given in the brackets is a solution to the given equation or not:
 - $n + 5 = 19$ ($n = 1$)
 - $7n + 5 = 19$ ($n = -2$)
 - $7n + 5 = 19$ ($n = 2$)
 - $4p - 3 = 13$ ($p = 0$)
- Solve the following equations by trial and error method:
 - $5p + 2 = 17$
 - $3m - 14 = 4$
- Write the following equations in the statement form:
 - $p + 4 = 15$
 - $m - 7 = 3$
 - $2m = 7$
 - $\frac{m}{5} = 3$
 - $\frac{3m}{5} = 6$
 - $3p + 4 = 25$
 - $4p - 2 = 18$
 - $\frac{p}{2} + 2 = 8$
- Set up an equation in the following cases:
 - The teacher tells the class that the highest marks obtained by a student in her class is twice the lowest marks plus 7. The highest score is 87. (Take the lowest score to be l .)
 - In an isosceles triangle, the vertex angle is twice either base angle. (Let the base angle be b in degrees. Remember that the sum of angles of a triangle is 180° .)
- Solve the following equation:
 - $-4 = 5(p - 2)$
 - $16 = 4 + 3(t + 2)$
 - $4 + 5(p - 1) = 34$
 - $0 = 16 + 4(m - 6)$
- Construct 3 equations starting with $x = 2$.
 - Construct 3 equations starting with $x = -2$.
- Solve the following:
 - In an isosceles triangle, the base angle are

equal. The vertex angle is 40° . what are the base angles of the triangle? (Remember, the sum of three angles of a triangle is 180°).

- Sachin scored twice as many runs as rahul. Together, their runs fell two short of a double century. How many runs did each one score?

- Solve the following:

- Laxmi's father is 49 years old. He is 4 years older than three times Laxmi's age. What is Laxmi's age?
- People of Sundargram planted trees in the village garden. Some of the trees were fruit trees. The number of non-fruit trees were two more than three times the number of fruit trees. what was the number of fruit trees planted if the number of non-fruit trees planted was 77?

Exemplar Questions :

- The age of Sohan Lal is four times that of his son Amit. If the difference of their ages is 27 years. find the age of Amit.
- The interest received by Karim is ₹30 more than that of Ramesh. If the total interest received by them is ₹70, find the interest received by Ramesh.
- 1 subtracted from one-third of a number gives 1. Find the number.
- Each of the 2 equal sides of an isosceles triangle is twice as large as the third side. If the perimeter of the triangle is 30 cm. Find the length of each side of the triangle.
- The sum of two consecutive multiples of 2 is 18. Find the numbers.
- In a class of 60 students, the number of girls is one third the number of boys. Find the number of girls and boys in the class.
- Thrice a number decreased by 5 exceeds twice the number by 1. Find the number.

8. A girl is 28 years younger than her father. The sum of their ages is 50 years. Find the ages of the girl and her father.
9. In a certain examination, a total of 3768 students secured first division in the years 2006 and 2007. The number of first division in 2007 exceeded those in 2006 by 34. How many students got first division in 2006?
10. If one side of a square is represented by $18x - 20$ and the adjacent side is represented by $42 - 13x$, find the length of the side of the square.

HOTS Questions :

1. My younger sister's age today is 3 times, what it will be 3 years from now minus 3 times what her age was 3 years ago. Find her present age.
2. In a bag, the number of one rupee coins is three times the number of two rupees coins. If the worth of the coins is ₹120, find the number of 1 rupee coins.
3. A number is as much greater than 27 as it is less than 73. Find the number.
4. A man travelled two fifth of his journey by train, one-third by bus, one-fourth by car and the remaining 3km on foot. What is the length of his total journey?

3

EXERCISE

Single Option Correct :

DIRECTIONS : This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONLY ONE is correct.

1. What will be the equation form for the following statement? The sum of 3 times x and 11 is 32.
 - (a) $x + 3 \times 11 = 32$
 - (b) $\frac{x}{3} + 11 = 32$
 - (c) $3(x + 11) = 32$
 - (d) $3x + 11 = 32$
2. The statement form of the equation $5p = 20$ is
 - (a) 5 when added to p gives 20
 - (b) Five times a number p is 20
 - (c) Twenty times a number p is 20
 - (d) None of the above
3. The equation for the statement: 'half of a number added to 10 is 15' is :
 - (a) $\frac{x}{2} + 10 = 5$
 - (b) $\frac{x}{2} + 10 = 15$
 - (c) $\frac{x}{2} + 15 = 10$
 - (d) $\frac{x}{2} = 10 + 15$
4. In an examination a student was asked to find $\frac{3}{14}$ th of a certain number. By mistake, he found $\frac{3}{4}$ of it. His answer was 150 more than the correct answer. The number given him is :
 - (a) 290
 - (b) 280
 - (c) 240
 - (d) 180
5. 4 is added to a number and the sum is multiplied by 5. If 20 is subtracted from the product and the difference is divided by 8, the result is equal to 10. Find the number.
 - (a) 16
 - (b) 12
 - (c) 8
 - (d) 20

6. A person travelled $\frac{5}{8}$ th of the distance by train, $\frac{1}{4}$ th by bus and the remaining 15 km by boat. The total distance travelled by him was
 (a) 90km (b) 120km
 (c) 150km (d) 180km
7. The total cost of three prizes is ₹ 2550. If the value of second prize is $\frac{3}{4}$ th of the first and the value of 3rd prize is $\frac{1}{2}$ of the second prize, find the value of first prize.
 (a) ₹ 900 (b) ₹ 1500
 (c) ₹ 1200 (d) ₹ 450
8. Value of x in $\frac{x}{4} + \frac{1}{2} = 4$ is
 (a) +28 (b) -28
 (c) +14 (d) -14
9. A number is multiplied by 6 and 12 is added to the product. The result is 84. Then the number is
 (a) -12 (b) 72
 (c) +12 (d) -72
10. In $6(2a - 1) + 8 = 14$, the value of 'a' is
 (a) -1 (b) $3\frac{1}{12}$
 (c) $1\frac{3}{12}$ (d) +1
11. A farmer divides his herd of n cows among his four sons so that first son gets one half the herd, the second son gets one fourth, the third son gets one fifth, and the fourth son gets 7 cows, then n is:
 (a) 180 (b) 140
 (c) 240 (d) 100
12. A shopkeeper sells mangoes in two types of boxes, one small and one large. A large box contains 8 small boxes plus 4 loose mangoes. Which one of the options given below is the correct equation which gives the number of mangoes in each small box? The number of mangoes in a large box is given to be 100.

- (a) $8 + 4m = 100$ (b) $4m - 8 = 100$
 (c) $8m - 4 = 100$ (d) $8m + 4 = 100$

More Than One Option Correct :

DIRECTIONS : This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONE or MORE may be correct.

- 1 is a solution of the equation
 (a) $x + 1 = 0$ (b) $x - 1 = 2$
 (c) $2y + 3 = 1$ (d) $2p + 7 = 5$
- Which of the following equations can be formed using the expression $x = 5$?
 (a) $2x + 3 = 13$ (b) $3x + 2 = 17$
 (c) $x - 5 = 1$ (d) $4x - 9 = 11$
- The solution of which of the following equations is an integer?
 (a) $2x + 6 = 0$ (b) $3x - 5 = 0$
 (c) $5x - 8 = x + 4$ (d) $4x + 7 = x + 2$
- The equation which can be solved in integers is
 (a) $5y - 3 = -18$ (b) $3x - 9 = 0$
 (c) $3z + 8 = 3 + z$ (d) $9y + 8 = 4y - 7$
- Which of the following is allowed in a given equation?
 (a) Adding the same number to both sides of the equation.
 (b) Subtracting the same number from both sides of the equation.
 (c) Multiplying both sides of the equation by the same non-zero number.
 (d) Dividing both sides of the equation by the same number.

Assertion & Reason :

DIRECTIONS : Each of these questions contains an Assertion followed by Reason. Read them carefully and answer the question on the basis of following options. You have to select the one that best describes the two statements.

- If both **Assertion** and **Reason** are **correct** and Reason is the **correct explanation** of Assertion.
- If both **Assertion** and **Reason** are correct, but Reason is **not the correct explanation** of Assertion.

- (c) If **Assertion** is **correct** but **Reason** is **incorrect**.
 (d) If **Assertion** is **incorrect** but **Reason** is **correct**.
 1. **Assertion** : If 5 is added to both sides of the equation $x + a = b$, then the value of the equation doesn't change.

Reason : If we perform the same mathematical operation on both sides of the equation its value doesn't change.

2. **Assertion** : $2x + 3 = 56$ is an equation.
Reason : In an equation, there is always less than sign.
 3. **Assertion** : $4x + 5 = 7$
Reason : The above equation means 4 times a number added to 5 gives 7.

Passage Based Questions :

DIRECTIONS : Study the given passage(s) and answer the following questions.

PASSAGE - I

By transposing a term of an equation, we simply mean changing its sign and carrying it to the other side of the equation.

1. Solve : $3(x-3) = 5(2x+1)$
 (a) 2 (b) $\frac{4}{3}$
 (c) -2 (d) $\frac{3}{2}$
2. Solve : $2x - 3 = \frac{3}{10}(5x - 12)$
 (a) $-\frac{6}{5}$ (b) $\frac{4}{5}$
 (c) $\frac{8}{9}$ (d) $-\frac{16}{9}$
3. Solve for x : $\frac{x+2}{6} - \left[\frac{11-x}{3} - \frac{1}{4} \right] = \frac{3x-4}{12}$
 (a) $\frac{6}{11}$ (b) 10
 (c) 14 (d) 11

PASSAGE - II

The perimeter of a rectangle is twice the sum of its length and breadth.

4. The length of rectangle is 3cm greater than its breadth. The perimeter is 46 cm. The breadth of rectangle is
 (a) 20 cm (b) 23 cm
 (c) 10 cm (d) 13 cm

5. The length of rectangle is 5 cm more than twice its breadth. The perimeter is 52 cm. The length of rectangle is
 (a) 14 cm (b) 19 cm
 (c) 7 cm (d) 26 cm
6. The breadth of rectangle is 6 cm less than its length. The perimeter is 20 cm. The length of rectangle is
 (a) 8 cm (b) 10 cm
 (c) 14 cm (d) 16 cm

Integer Type Questions :

DIRECTIONS : Answer the following questions. The answer to each of the question is a single digit integer, ranging from 0 to 9.

1. If $m = 2$, the value of $\frac{5m}{2} - 4$ is
2. If $3\frac{1}{x} \times 3\frac{3}{4} = 12\frac{1}{2}$, then the value of 'x' is
3. If the total of four consecutive odd numbers is 40, the smallest number is
4. In $\frac{2}{3}p - 2\frac{1}{2} = 3\frac{1}{2}$, the value of p is
5. If $k + 7 = 16$, then the value of $8k - 72$ is

Multiple Matching Question :

DIRECTIONS : Following question has four statements (A, B, C, D) given in Column-I and six statements (p, q, r, s, t, u) in Column-II. Any given statement in Column-I can have correct matching with one or more statement(s) given in Column-II.

1. Match the solutions in column-I with the equations in column-II

Column-I	Column-II
(A) $x = 4$	(p) $3x - 7 = 50$

(B) $x = \frac{21}{16}$	(q) $5x = 35$
-------------------------	---------------

(C) $x = 7$ (r)	$2x + 7 = 45$
-----------------	---------------

(D) $x = 19$	(s) $x + 16 = 20$
--------------	-------------------

(t) $2x - \frac{7}{4} = \frac{7}{8}$

(u) $3x + 7 = 19$

SOLUTIONS

Brief Explanations
of
Selected Questions

1 EXERCISE

Fill in the Blanks :

- $m = -2$
- $\frac{5}{2} + \frac{1}{2} = \frac{6}{2} = 3$
- $x = 10$
- $x = 50$
- $0.2 \times 10 + 0.2 = 2 + 0.2 = 2.2$
- 75 7. 25 8. 72 9. $\frac{7}{4}$
- Solution 11. 22

True / False :

- False 2. True 3. False 4. True
- False 6. False 7. False 8. False
- True 10. True

Match the Columns :

- (A) $\rightarrow$ (s); (B) $\rightarrow$ (p); (C) $\rightarrow$ (q); (D) $\rightarrow$ (r)

$$(A) \quad \frac{4x}{5} + 9 = 2x - 3$$

$$\Rightarrow \frac{4x}{5} - 2x = -3 - 9$$

$$\Rightarrow \frac{-6x}{5} = -12 \quad \Rightarrow x = 10$$

- Let the smaller part be x

$$\therefore 2x + x = 96$$

$$\Rightarrow 3x = 96 \quad \Rightarrow x = 32$$

$$\therefore \text{Larger part} = 2x = 2 \times 32 = 64$$

$$(C) \quad \frac{7x}{2} + 2 = 14 + 3x$$

$$\Rightarrow \frac{7x}{2} - 3x = 14 - 2$$

$$\Rightarrow \frac{x}{2} = 12 \quad \Rightarrow x = 24$$

- Let the number be x .

$$\therefore \frac{x}{5} - \frac{x}{10} = 5$$

$$\Rightarrow \frac{2x - x}{10} = 5 \Rightarrow x = 50$$

Very Short Answer Questions :

- $5 + x = 12$
 - $x - 7 = 13$
 - $2y - 6 = 32$
 - $n - 8 = -4$
 - $7q = -35$
 - $8 - 3a = 2$

$$(vii) \quad 7x = 42; \quad (viii) \quad \frac{m}{5} = 6$$

$$2. \quad x = \frac{5}{3} \quad 3. \quad x = \frac{6}{5}$$

$$5. \quad 12x - 7x = -13 + 7$$

$$\Rightarrow 5x = -6 \quad \Rightarrow x = -6/5$$

$$6. \quad 1.2m = 5 - 2.6$$

$$\Rightarrow 1.2m = 2.4 \quad \Rightarrow m = \frac{2.4}{1.2} = 2$$

$$7. \quad \frac{x}{2} + \frac{x}{3} = 5 \quad \Rightarrow \frac{3x + 2x}{6} = 5$$

$$\Rightarrow \frac{5x}{6} = 5 \quad \Rightarrow x = 6$$

$$8. \quad 1.7y + 2.3y = 2$$

$$\Rightarrow 4y = 2 \quad \Rightarrow y = \frac{1}{2}$$

$$9. \quad \frac{79}{49}$$

$$10. \quad -1$$

$$11. \quad \frac{220}{7}$$

Short Answer Questions :

- Let the number be x .

$$\text{Then, } \frac{4}{5}x - \frac{3}{4}x = 4.$$

Multiplying both sides by 20, we get

$$16x - 15x = 80 \quad \Rightarrow x = 80$$

2. Let one of the even numbers be x . Then, the next consecutive even number $= x + 2$.

$$\therefore 2x + 2 = 86$$

$$\Rightarrow 2x = 86 - 2$$

$$\Rightarrow 2x = 84 \Rightarrow x = 42$$

Thus, one of the even number $= 42$

The other consecutive even number
 $= x + 2 = 42 + 2 = 44$.

3. Let the number be x .

$$\therefore \frac{x}{4} + \frac{x}{2} + \frac{x}{3} + x = 25$$

$$\Rightarrow \frac{3x + 6x + 4x}{12} + x = 25$$

$$\Rightarrow \frac{13x + 12x}{12} = 25 \Rightarrow \frac{25x}{12} = 25$$

$$\Rightarrow x = 12$$

4. Let one part be x

$\therefore$ The second part is $6x$

$$x + 6x = 98 \Rightarrow x = 14$$

$\therefore 14$ and 84 are the required two parts.

5. Let the number be x .

$$\therefore 5x - 25 = 4x$$

$$\Rightarrow 5x - 4x = 25 \Rightarrow x = 25$$

6. Let the present age of a man be x years.

$$\therefore x - 30 = \frac{3}{5}x$$

$$5x - 150 = 3x$$

$$5x - 3x = 150 \Rightarrow x = 75 \text{ years}$$

7. (i) $x = -1$ (ii) $x = 13$ (iii) $y = 2$

$$(iv) x = 2$$

8. Let the breadth of the rectangle be x m

$\therefore$ Length of the rectangle $= 3x$ m

Given that, perimeter of the rectangle $= 96$ m

$$\therefore 2 \times (x + 3x) = 96$$

$$\Rightarrow 2 \times 4x = 96$$

$$\Rightarrow 8x = 96 \Rightarrow x = \frac{96}{8} = 12$$

$\therefore$ Breadth of the rectangle $= 12$ m

Length of the rectangle $= 3 \times 12 = 36$ m

9. Let the number of coins $= x$

Value of one coin $= 10$ paise

$\therefore$ Value of x coins $= 10 \times x$ paise

$$= ₹ \frac{10x}{100} = ₹ \frac{x}{10}$$

But, it is given that, this value $= ₹ 25$

$$\therefore \frac{x}{10} = 25$$

$$x = 25 \times 10 = 250$$

10. Let the number be x .

$\therefore 20\%$ of 60% of $x = 144$

$$\Rightarrow 20\% \text{ of } \left[\frac{60}{100} \times x \right] = 144$$

$$\Rightarrow 20\% \text{ of } \frac{3}{5}x = 144$$

$$\Rightarrow \frac{20}{100} \times \frac{3}{5}x = 144$$

$$\Rightarrow \frac{3}{25}x = 144$$

$$\Rightarrow x = 144 \times \frac{25}{3} = 1200$$

Long Answer Questions :

1. $3(x + 3) - 2(x - 1) = 5(x - 5)$

$$\Rightarrow 3x + 9 - 2x + 2 = 5x - 25$$

$$\Rightarrow x + 11 = 5x - 25$$

$$\Rightarrow x - 5x = -25 - 11$$

$$\Rightarrow -4x = -36$$

$$\Rightarrow x = 9$$

Substituting $x = 9$ in the given equation, we get

$$\text{L.H.S} = 3(9 + 3) - 2(9 - 1)$$

$$= 3 \times 12 - 2 \times 8 = 36 - 16 = 20$$

$$\text{and R.H.S} = 5(9 - 5) = 5 \times 4 = 20$$

Thus, for $x = 9$, we have L.H.S $=$ R.H.S

2. We have, $\frac{12}{7}(x - 5) = 24 + 8x$

$$\Rightarrow 12(x - 5) = 7(24 + 8x)$$

$$\Rightarrow 12x - 60 = 168 + 56x$$

$$\Rightarrow 12x - 56x = 168 + 60$$

$$\Rightarrow -44x = 228 \Rightarrow x = -\frac{57}{11}$$

3. $\frac{5x-2}{3} + \frac{4x+3}{2} = \frac{3x+19}{2}$

$$\Rightarrow \frac{2(5x-2) + 3(4x+3)}{6} = \frac{3x+19}{2}$$

$$\Rightarrow \frac{10x-4+12x+9}{6} = \frac{3x+19}{2}$$

$$\Rightarrow 2(22x+5) = 6(3x+19)$$

$$\Rightarrow 44x+10 = 18x+114$$

$$\Rightarrow 44x-18x = 114-10$$

$$\Rightarrow 26x = 104 \Rightarrow x = 4$$

$$4. \quad \frac{9m+4}{5} - \frac{27m+1}{8} - \frac{1}{2} = 0$$

$$\Rightarrow \frac{9m+4}{5} - \frac{27m+1}{8} = \frac{1}{2}$$

$$\Rightarrow \frac{8(9m+4) - 5(27m+1)}{40} = \frac{1}{2}$$

$$\Rightarrow \frac{72m+32-135m-5}{40} = \frac{1}{2}$$

$$\Rightarrow 27-63m = \frac{1}{2} \times 40 \Rightarrow 27-63m = 20$$

$$\Rightarrow 63m = 7 \Rightarrow m = 1/9$$

$$5. \quad \frac{0.3(3x-4)}{5} + \frac{0.4x+3.6}{2} = 3.5x$$

$$\Rightarrow \frac{0.9x-1.2}{5} + \frac{0.4x+3.6}{2} = 3.5x$$

$$\Rightarrow \frac{1.8x-2.4+2.0x+18}{10} = 3.5x$$

$$\Rightarrow 3.8x+15.6 = 35x$$

$$\Rightarrow 15.6 = 31.2x$$

$$\Rightarrow x = \frac{15.6}{31.2} \Rightarrow x = \frac{1}{2} = 0.5$$

$$6. \quad \text{We have, } \frac{y-8}{3} = \frac{7-4y}{7}$$

Multiplying both sides by the L.C.M. of 3 and 7 that is 21, we get

$$\Rightarrow 7(y-8) = 3(7-4y)$$

$$\Rightarrow 7y-56 = 21-12y$$

$$\Rightarrow 7y+12y = 21+56$$

$$\Rightarrow 19y = 77 \Rightarrow y = \frac{77}{19}$$

$$7. \quad x - \left(2x - \frac{3x-4}{7} \right) = \frac{4x-27}{3} - 3$$

$$\Rightarrow x - 2x + \frac{3x-4}{7} = \frac{4x-27}{3} - 3$$

$$\Rightarrow -x + \frac{3x-4}{7} = \frac{4x-27}{3} - 3$$

$$\Rightarrow \frac{-7x+3x-4}{7} = \frac{4x-27-9}{3}$$

$$\Rightarrow \frac{-4x-4}{7} = \frac{4x-36}{3}$$

$$\Rightarrow 3(-4x-4) = 7(4x-36)$$

$$\Rightarrow -12x-12 = 28x-252$$

$$\Rightarrow -12x-28x = -252+12$$

$$\Rightarrow -40x = -240 \Rightarrow x = 6$$

2 EXERCISE

Text-Book Exercise :

- Putting $n = 1$
L.H.S. = $n + 5 = 1 + 5 = 6$
and R.H.S. = 19
As L.H.S. $\neq$ R.H.S.
Therefore, $n = 1$ is not a solution of the given equation, $n + 5 = 19$.
 - Putting $n = -2$
L.H.S. = $7n + 5 = 7 \times (-2) + 5$
 $= -14 + 5 = -9$ and R.H.S. = 19
As L.H.S. $\neq$ R.H.S.
Therefore, $n = -2$ is not a solution of the given equation, $7n + 5 = 19$.
 - Putting $n = 2$
L.H.S. = $7n + 5 = 7 \times (2) + 5$
 $= 14 + 5 = 19 = \text{R.H.S.}$
As L.H.S. = R.H.S.,
Therefore, $n = 2$ is a solution of the given equation, $7n + 5 = 19$.
 - Putting $p = 0$
L.H.S. = $4p - 3 = (4 \times 0) - 3 = -3$
and R.H.S. = 13
As L.H.S. $\neq$ R.H.S.,
Therefore, $p = 0$ is not a solution of the given equation, $4p - 3 = 13$.
- $5p + 2 = 17$
Putting $p = 1$
L.H.S. = $(5 \times 1) + 2 = 7 \neq \text{R.H.S.}$
Putting $p = 2$
L.H.S. = $(5 \times 2) + 2 = 10 + 2$
 $= 12 \neq \text{R.H.S.}$

Putting $p = 3$

$$\text{L.H.S.} = (5 \times 3) + 2 = 17 = \text{R.H.S.}$$

Hence, $p = 3$ is a solution of the given equation.

(ii) $3m - 14 = 4$

Putting $m = 4$,

$$\text{L.H.S.} = (3 \times 4) - 14 = -2 \neq \text{R.H.S.}$$

Putting $m = 5$,

$$\text{L.H.S.} = (3 \times 5) - 14 = 1 \neq \text{R.H.S.}$$

Putting $m = 6$,

$$\begin{aligned} \text{L.H.S.} &= (3 \times 6) - 14 = 18 - 14 \\ &= 4 = \text{R.H.S.} \end{aligned}$$

Hence, $m = 6$ is a solution of the given equation.

3. (i) The sum of p and 4 is 15.
 (ii) 7 subtracted from m is 3.
 (iii) Twice of a number m is 7.
 (iv) One-fifth of m is 3.
 (v) Three-fifth of m is 6.
 (vi) Three times of a number p , when added to 4, gives 25.
 (vii) when 2 is subtracted from four times of a number p , it gives 18.
 (viii) When 2 is added to half of a number p , it gives 8.

4. (i) Let the lowest marks be l .
 Highest marks = $2 \times$ Lowest marks + 7
 $\therefore 87 = 2 \times l + 7$
 $\Rightarrow 2l + 7 = 87$
 (ii) An isosceles triangle has two of its angles of equal measure.
 Let base angle be b .
 Vertex angle = $2 \times$ Base angle = $2b$
 Sum of all angles of a triangle = 180°
 $\therefore b + b + 2b = 180^\circ \Rightarrow 4b = 180^\circ$

5. (i) $-4 = 5(p - 2)$
 $\Rightarrow -\frac{4}{5} = p - 2 \Rightarrow -\frac{4}{5} + 2 = p$
 $\Rightarrow \frac{-4+10}{5} = p \Rightarrow p = \frac{6}{5}$

- (ii) $16 = 4 + 3(t + 2)$
 $\Rightarrow 16 - 4 = 3(t + 2)$
 $\Rightarrow 12 = 3(t + 2)$
 $\Rightarrow \frac{12}{3} = t + 2 \Rightarrow 4 = t + 2$
 $\Rightarrow 4 - 2 = t \Rightarrow t = 2$

(iii) $4 + 5(p - 1) = 34$
 $\Rightarrow 5(p - 1) = 34 - 4$

$$\Rightarrow p - 1 = \frac{30}{5}$$

$$\Rightarrow p = 6 + 1 \Rightarrow p = 7$$

(iv) $0 = 16 + 4(m - 6)$

$$\Rightarrow 0 = 16 + 4m - 24$$

$$\Rightarrow 0 = -8 + 4m$$

$$\Rightarrow 4m = 8 \Rightarrow m = 2$$

6. (i) $x = 2$

Multiplying both sides by 3,

$$3x = 6 \quad \dots(i)$$

Subtracting 5 from both sides,

$$3x - 5 = 6 - 5$$

$$3x - 5 = 1 \quad \dots(ii)$$

Dividing both sides by 7,

$$\frac{3x}{7} - \frac{5}{7} = \frac{1}{7} \quad \dots(iii)$$

(ii) $x = -2$

Subtracting 3 from both sides,

$$x - 3 = -2 - 3$$

$$\Rightarrow x - 3 = -5 \quad \dots(i)$$

Multiplying both sides by 7,

$$7 \times (x - 3) = -5 \times 7$$

$$7x - 21 = -35 \quad \dots(ii)$$

Subtracting 12 from both sides,

$$7x - 21 - 12 = -35 - 12$$

$$\Rightarrow 7x - 33 = -47 \quad \dots(iii)$$

7. (i) Let the base angles be b .

Vertex angle = 40°

$$\therefore b + b + 40^\circ = 180^\circ$$

$$\Rightarrow 2b + 40^\circ = 180^\circ$$

$$\Rightarrow 2b = 180^\circ - 40^\circ = 140^\circ$$

$$\Rightarrow b = 70^\circ.$$

(ii) Let Rahul's score be x .

Therefore, Sachin's score = $2x$

$$\text{Rahul's score} + \text{Sachin's score} = 200 - 2$$

$$\Rightarrow x + 2x = 198$$

$$\Rightarrow 3x = 198 \Rightarrow x = 66$$

$$\therefore \text{Rahul's score} = 66$$

$$\text{Sachin's score} = 2 \times 66 = 132$$

8. (i) Let Laxmi's age be x years.

Her father's age = $3 \times$ Laxmi's age + 4

$$\Rightarrow 49 = (3 \times x) + 4$$

$$\Rightarrow 3x + 4 = 49 \Rightarrow 3x = 49 - 4$$

$$\Rightarrow 3x = 45 \Rightarrow x = 15$$

- (ii) Let the number of fruit trees be x .
 Number of non-fruit trees
 $= 3 \times \text{Number of fruit trees} + 2$
 $\Rightarrow 77 = (3 \times x) + 2$
 $\Rightarrow 3x + 2 = 77$
 $\Rightarrow 3x = 77 - 2 \Rightarrow 3x = 75$
 $\Rightarrow x = 25$

Exemplar Questions :

- Let the age of Amit be x years.
 $\therefore 4x - x = 27$
 $\Rightarrow 3x = 27 \Rightarrow x = 9$ years
- Let the interest received by Ramesh be ₹ x
 $\therefore (x + 30) + x = 70$
 $\Rightarrow 2x + 30 = 70 \Rightarrow 2x = 40 \Rightarrow x = 20$
- Let the number be x .
 $\therefore \frac{x}{3} - 1 = 1 \Rightarrow \frac{x}{3} = 2 \Rightarrow x = 6$
- Let the length of third side be x cm.
 $\therefore$ The length of each of equal sides $= 2x$ cm.
 $\therefore 2x + 2x + x = 30$
 $\Rightarrow 5x = 30 \Rightarrow x = 6$ cm
 The length of each of equal sides
 $= 2x = 2 \times 6 = 12$ cm
- Let the two consecutive multiples of 2 be $2x$ and $2(x + 1)$.
 $\therefore 2x + 2(x + 1) = 18$
 $\therefore 4x + 2 = 18$
 $\Rightarrow 4x = 16$
 $\Rightarrow x = 4$
 $\therefore$ First number $= 2x = 2 \times 4 = 8$
 Second number $= 2(x + 1) = 2(4 + 1) = 10$
- Let the number of boys be x
 $\therefore x + \frac{x}{3} = 60 \Rightarrow \frac{4x}{3} = 60$
 $\Rightarrow 4x = 180 \Rightarrow x = 45$
 Number of boys are 45
 and Number of girls are $\frac{x}{3} = \frac{45}{3} = 15$
- Let the number be x .
 $\therefore (3x - 5) - 2x = 1$
 $\Rightarrow 3x - 2x - 5 = 1$
 $\Rightarrow x = 1 + 5 \Rightarrow x = 6$
- Let the age of the father be x years.
 $\therefore (x - 28) + x = 50$
 $\Rightarrow 2x - 28 = 50$

$$\Rightarrow 2x = 78 \Rightarrow x = 39$$

$$\therefore \text{Father's age} = 39 \text{ years}$$

$$\text{Girl's age} = x - 28 = 39 - 28 = 11 \text{ years.}$$

- Let the number of student secured first division in 2006 be x .
 $\therefore$ The number of students secured first division in 2007 $= 3768 - x$
 $\therefore (3768 - x) - x = 34$
 $\Rightarrow 3768 - 2x = 34$
 $\Rightarrow 2x = 3768 - 34$
 $\Rightarrow x = \frac{3734}{2} = 1867$
- $18x - 20 = 42 - 13x$
 $\Rightarrow 18x + 13x = 42 + 20$
 $\Rightarrow 31x = 62 \Rightarrow x = 2$
 $\therefore$ The length of side of the square
 $= 18x - 20$
 $= 18 \times 2 - 20 = 16$

HOTS Questions :

- Let younger sister's age be x years.
 $\therefore x = 3(x + 3) - 3(x - 3)$
 $\Rightarrow x = 3x + 9 - 3x + 9$
 $\Rightarrow x = 18$ years.
- Let the number of two rupee coins be x .
 Then number of one rupee coins $= 3x$
 $\therefore 1 \times 3x + 2 \times x = 120$
 $\Rightarrow 3x + 2x = 120$
 $\Rightarrow 5x = 120 \Rightarrow x = 24$
 Hence, there are 24 coins of ₹ 2.
 and number of coins of ₹ 1 $= 3x = 3 \times 24 = 72$
- Let the number be x
 $\therefore x - 27 = 73 - x$
 $\Rightarrow x + x = 73 + 27$
 $\Rightarrow 2x = 100 \Rightarrow x = 50$
- Let the length of total journey be x km.
 $\therefore \frac{2x}{5} + \frac{1}{3}x + \frac{1}{4}x + 3 = x$
 $\Rightarrow \frac{24x + 20x + 15x}{60} + 3 = x$
 $\Rightarrow x - \frac{59}{60}x = 3$
 $\Rightarrow \frac{1}{60}x = 3 \Rightarrow x = 180$ km

3 EXERCISE

Single Option Correct :

1. (d) 2. (b) 3. (b)

4. (b) Let the required number be x .

$$\therefore \frac{3x}{4} - \frac{3x}{14} = 150$$

$$\frac{21x - 6x}{28} = 150$$

$$x = \frac{150 \times 28}{15} = 280$$

5. (a) Let the number be x .

$$\frac{(x+4) \times 5 - 20}{8} = 10$$

$$5x + 20 - 20 = 80 \Rightarrow x = 16$$

6. (b) Let total distance be x km

$$x - \left(\frac{5x}{8} + \frac{x}{4} \right) = 15$$

$$x - \frac{7x}{8} = 15 \Rightarrow x = 120 \text{ km}$$

7. (c) Let first prize = x

$$\text{Second prize} = \frac{3}{4}x$$

$$\text{Third prize} = \frac{1}{2} \times \frac{3x}{4} = \frac{3x}{8}$$

$$\therefore x + \frac{3x}{4} + \frac{3x}{8} = 2550$$

$$\frac{17x}{8} = 2550 \Rightarrow x = ₹ 1200$$

8. (c) $\frac{x}{4} + \frac{1}{2} = 4$

$$x + 2 = 16$$

$$\Rightarrow x = 14$$

9. (c) $x \times 6 + 12 = 84$

$$\Rightarrow 6x = 72 \Rightarrow x = \frac{72}{6} = 12$$

10. (d) $6(2a - 1) + 8 = 14$
 $12a - 6 + 8 = 14$
 $12a = 14 + 6 - 8 = 2$

$$a = \frac{12}{12} = 1$$

11. (b) The first son gets = $\frac{n}{2}$ cows

$$\text{The second son gets} = \frac{n}{4} \text{ cows}$$

$$\text{The third son gets} = \frac{n}{5} \text{ cows}$$

$$\text{And the fourth son gets} = 7 \text{ cows}$$

$$\text{All the sons get} = \left(\frac{n}{2} + \frac{n}{4} + \frac{n}{5} + 7 \right) \text{ cows}$$

$$= \frac{10n + 5n + 4n + 140}{20} \text{ cows}$$

$$= \frac{19n + 140}{20} \text{ cows}$$

$$\text{But, } \frac{19n + 140}{20} = n$$

$$\Rightarrow 19n + 140 = 20n \Rightarrow n = 140 \text{ cows.}$$

12. (d) Let a small box contain m mangoes. A large box contains 4 more than 8 times m , that is, $8m + 4$ mangoes. But this is given to be 100. Thus, $8m + 4 = 100$

More Than One Option Correct :

1. (a, c, d) 2. (a, b, d)

3. (a, c)

$$(a) \quad 2x + 6 = 0 \Rightarrow 2x = -6 \Rightarrow x = -3$$

$$(c) \quad 5x - 8 = x + 4 \Rightarrow 4x = 12 \Rightarrow x = 3$$

4. (a, b, d)

$$(a) \quad 5y - 3 = -18 \Rightarrow 5y = -15 \Rightarrow y = -3$$

$$(b) \quad 3x - 9 = 0 \Rightarrow 3x = 9 \Rightarrow x = 3$$

$$(d) \quad 9y + 8 = 4y - 7 \Rightarrow 5y = -15 \Rightarrow y = -3$$

5. (a, b, c)

Assertion & Reason :

1. (a)

2. (c) $2x + 3 = 56$ is an equation as in an equation there is always an equality sign.

Assertion : correct, Reason : incorrect

3. (a) **Assertion : correct, Reason : correct** and is the correct explanation of assertion.

Passage Based Questions :

1. (c) $3(x-3) = 5(2x+1)$
 $\Rightarrow 3x-9 = 10x+5$
 $\Rightarrow 3x-10x = 9+5$
 $\Rightarrow -7x = 14 \Rightarrow x = -2$

2. (a) $2x-3 = \frac{3}{10}(5x-12)$
 $\Rightarrow 2x-3 = \frac{15x}{10} - \frac{3}{10} \times 12$
 $\Rightarrow 2x-3 = \frac{3x}{2} - \frac{18}{5}$
 $\Rightarrow 2x - \frac{3x}{2} = \frac{-18}{5} + 3$
 $\Rightarrow \frac{4x-3x}{2} = \frac{-18+15}{5}$
 $\Rightarrow \frac{x}{2} = \frac{-3}{5} \Rightarrow x = \frac{-6}{5}$

3. (d) $\frac{x+2}{6} - \left[\frac{11-x}{3} - \frac{1}{4} \right] = \frac{3x-4}{12}$
 $\Rightarrow \frac{x+2}{6} - \left[\frac{4(11-x)-3}{12} \right] = \frac{3x-4}{12}$
 $\Rightarrow \frac{x+2}{6} - \left[\frac{44-4x-3}{12} \right] = \frac{3x-4}{12}$
 $\Rightarrow \frac{x+2}{6} - \left[\frac{41-4x}{12} \right] = \frac{3x-4}{12}$
 $\Rightarrow \frac{2(x+2)-(41-4x)}{12} = \frac{3x-4}{12}$
 $\Rightarrow 2x+4-41+4x = 3x-4$
 $\Rightarrow 6x-37 = 3x-4$
 $\Rightarrow 3x = -4+37$
 $\Rightarrow 3x = 33 \Rightarrow x = 11$

4. (c) Let the breadth of the rectangle be x cm.
 Length = $(x+3)$ cm
 $\therefore 2[(x+3)+x] = 46 \Rightarrow 2x+3 = 23$
 $\Rightarrow x = 10$ cm
5. (b) Let the breadth of the rectangle be x cm.
 Length = $(2x+5)$ cm
 $\therefore 2[(2x+5)+x] = 52 \Rightarrow 3x+5 = 26$
 $\Rightarrow x = 7$ cm
 Length = $2x+5 = 2 \times 7 + 5 = 19$ cm
6. (a) Let the length of the rectangle be x cm.
 Breadth = $(x-6)$ cm
 $\therefore 2[x+(x-6)] = 20$
 $\Rightarrow 2x-6 = 10 \Rightarrow x = 8$ cm

Integer Type Questions :

1. (1)
2. (3) $3\frac{1}{x} \times 3\frac{3}{4} = 12\frac{1}{2}$
 $\frac{3x+1}{x} \times \frac{15}{4} = \frac{25}{2}$
 $10x = 9x+3 \Rightarrow x = 3$
3. (7) Let the smallest number be x .
 $\therefore x+(x+2)+(x+4)+(x+6) = 40$
 $\Rightarrow 4x+12 = 40 \Rightarrow x = 7$
4. (9) $\frac{2}{3}p - \frac{5}{2} = \frac{7}{2}$
 $\frac{2}{3}p = \frac{7}{2} + \frac{5}{2} = \frac{12}{2} = 6$
 $2p = 6 \times 3$
 $p = \frac{6 \times 3}{2} = 9$
5. (0) $k+7 = 16 \Rightarrow k = 9$
 $\therefore 8k-72 = 8 \times 9 - 72 = 0$

Multiple Matching Question :

1. (A) $\rightarrow$ (s, u); (B) $\rightarrow$ (t); (C) $\rightarrow$ (q); (D) $\rightarrow$ (r, p)

Chapter 5

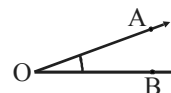
Lines and Angles

INTRODUCTION

In class VI, we have learnt about line, line segment, ray and angles. In this chapter, we shall learn about pair of angles and their properties.

ANGLES

Two rays having the common origin or initial point form an angle.

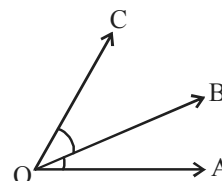


Here, angle AOB is denoted by $\angle AOB$. OA and OB are the arms of the angles and O is the vertex.

PAIRS OF ANGLES

Adjacent Angles

Two angles with same vertex, one common arm and the other arms lying on opposite sides of common arm are called **adjacent angles**. In the figure, $\angle BOC$ and $\angle AOB$ are adjacent angles.

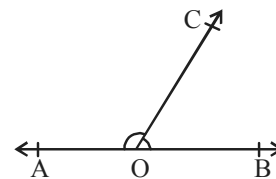


Linear Pair of Angles

Two adjacent angles are said to form linear pair of angles if their non-common arms are two opposite rays, $\angle AOC$ and $\angle BOC$ are two adjacent angles. Their non-common arms OA and OB are two opposite rays.

Linear-pair-angles are always supplementary, i.e. they add up to 180° .

Here, $\angle AOC + \angle COB = 180^\circ$.



NOTE : All adjacent angles do not form a linear pair.

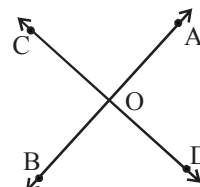
Vertically Opposite Angles

The angles opposite to the common vertex formed by the intersection of two lines having no common arm are known as vertically opposite angles.

If two lines intersect, then the vertically opposite angles are equal.

In the given figure, AB and CD are two intersecting lines and meet at point O. $\angle AOD$ and $\angle COB$ form a pair of vertically opposite angles and similarly $\angle AOC$ and $\angle BOD$ form such a pair. These angles are known as **vertical angles** or **opposite angles**.

For a pair of opposite angles, the following theorem, known as **vertical angle theorem** holds true.



Theorem : If two lines intersect, then the vertically opposite angles are equal.

Proof : Let the two lines $\overleftrightarrow{AB}$ and $\overleftrightarrow{CD}$ intersect at a point O.

Since the ray $\overrightarrow{OA}$ stands on the line $\overleftrightarrow{CD}$, we have :

$$\angle AOC + \angle AOD = 180^\circ \quad [\text{linear pair}]$$

Again, since the ray $\overrightarrow{OD}$ stands on the line $\overleftrightarrow{AB}$, we have:

$$\angle AOD + \angle BOD = 180^\circ \quad [\text{linear pair}]$$

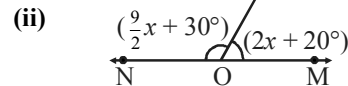
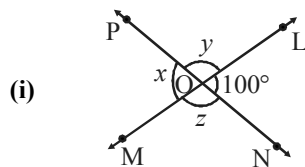
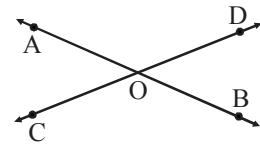
$$\therefore \angle AOC + \angle AOD = \angle AOD + \angle BOD$$

$$\Rightarrow \angle AOC = \angle BOD$$

Similarly, $\angle AOD = \angle BOC$ (Hence Proved)

ILLUSTRATION 1 :

Find the value of x, y, z in the following figures.



SOLUTION :

(i) $\overleftrightarrow{LM}$ and $\overleftrightarrow{PN}$ intersect each other at O.
 $\therefore \angle POM = \angle LON = 100^\circ$

(Vertically Opposite Angles)

$$\therefore x = 100^\circ$$

$$\angle POL + \angle LON = 180^\circ \quad (\text{Linear Pair})$$

$$\Rightarrow y + 100^\circ = 180^\circ$$

$$\Rightarrow y = 180^\circ - 100^\circ = 80^\circ$$

$$\Rightarrow \angle POL = \angle MON = 80^\circ$$

(Vertically Opposite Angles)

$$\therefore z = 80^\circ$$

(ii) $\angle LON + \angle LOM = 180^\circ$
 (Linear Pair)

$$\Rightarrow \frac{9}{2}x + 30^\circ + 2x + 20^\circ = 180^\circ$$

$$\Rightarrow \frac{9}{2}x + 2x + 50 = 180^\circ$$

$$\Rightarrow \frac{9x + 4x}{2} = 180^\circ - 50^\circ$$

$$\Rightarrow \frac{13x}{2} = 130^\circ$$

$$\Rightarrow x = 130^\circ \times \frac{2}{13} = 20^\circ$$

$$\therefore x = 20^\circ$$

Complementary Angles

Two angles are said to be complementary if the sum of their measures is 90° .

For example: If $\angle A = 46^\circ$ and $\angle B = 44^\circ$ then $\angle A$ and $\angle B$ are complementary, since $\angle A + \angle B = 90^\circ$.

Supplementary Angles

Two angles are said to be supplementary if the sum of their measures is 180° . Thus, $\angle A$ and $\angle B$ are supplementary if $\angle A + \angle B = 180^\circ$.

For example: If $\angle A = 60^\circ$ and $\angle B = 120^\circ$, then $\angle A$ and $\angle B$ are supplementary. $\angle A + \angle B = 180^\circ$.

ILLUSTRATION 2 :

Find the supplement of 90° .

SOLUTION :

$$\text{Supplement of } 90^\circ = 180^\circ - 90^\circ = 90^\circ.$$

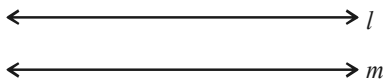
INTERSECTING LINES

Two lines are said to be **intersecting lines** if they have one and only one point in common. This common point is called the **point of intersection**. All intersecting lines form angles at the point of intersection.

PARALLEL LINES AND TRANSVERSAL

Parallel Lines

Two lines or line segments in a plane which do not intersect and the perpendicular distance between them remain constant are called **parallel lines**.



Parallel lines never meet if they are produced towards left or right.

Transversal

A line which intersects two or more lines at distinct points is called a **transversal**.

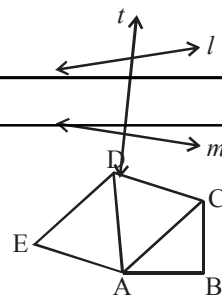
In the adjoining figure, t is a transversal to the lines l and m .

NOTE : The lines which are cut by a transversal may be parallel or non-parallel.

ILLUSTRATION 3 :

Name the pairs of line segments which are parallel in the following figure :

SOLUTION :



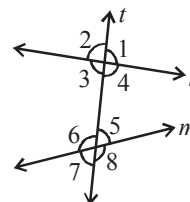
In the given figure, the following pairs of line segments are parallel $AD \parallel BC$; $CD \parallel AE$; $AC \parallel ED$

ANGLES FORMED BY A TRANSVERSAL

l and m are two lines cut by a transversal t . The transversal makes eight angles, which are marked as 1, 2, 3, 4, 5, 6, 7 and 8 in the following figure.

(i) Exterior angles

Exterior angles are the angles that lie outside the lines l and m , cut by a transversal t . In the above figure, $\angle 1$, $\angle 2$, $\angle 7$ and $\angle 8$ are the exterior angles.



(ii) Interior angles

Those angles which lie between the two lines l and m , are interior angles. In other words, interior angles are the angles that are formed by two lines cut by a transversal and are between the two lines.

In the above figure; $\angle 3$, $\angle 4$, $\angle 5$ and $\angle 6$ are the interior angles.

(iii) Corresponding Angles

The pair of angles, one of which is an exterior angle while other is interior angle, that are on the same side of transversal, are corresponding angles. These angles are non-adjacent ($\angle 2$, $\angle 6$); ($\angle 3$, $\angle 7$); ($\angle 1$, $\angle 5$) and ($\angle 4$, $\angle 8$) are corresponding angles.

(iv) Alternate Interior angles

Alternate interior angles are non-adjacent interior angles that lie on opposite sides of transversal.

($\angle 3$, $\angle 5$) and ($\angle 4$, $\angle 6$) are alternate interior angles.

(v) Alternate Exterior Angles

Alternate exterior angles are non-adjacent exterior angles that lie on opposite sides of transversal. ($\angle 2$, $\angle 8$) and ($\angle 1$, $\angle 7$) are alternate exterior angles.

Consecutive Interior Angles

Consecutive Interior Angles are pairs of interior angles that are on the same side of the transversal. These angles are commonly known as Co-interior angles.

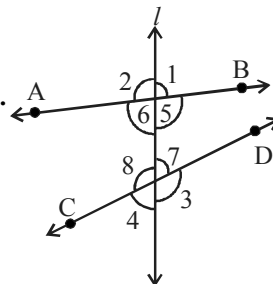
In the above figure, $(\angle 3, \angle 6)$ and $(\angle 4, \angle 5)$ are co-interior angles.

ILLUSTRATION 4 :

In each of the following figures, l is a transversal to the lines AB and CD .

Name the following angles :

- Interior and exterior angles.
- Alternate (interior and exterior) angles.
- Angles corresponding to $\angle 2$, $\angle 8$ and $\angle 3$.



SOLUTION :

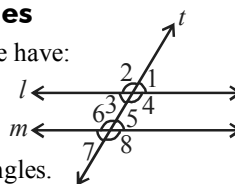
- Exterior angles : $\angle 1, \angle 2, \angle 3, \angle 4$; Interior angles : $\angle 5, \angle 6, \angle 7, \angle 8$
- Alternate interior angles : $(\angle 8, \angle 5), (\angle 7, \angle 6)$. Alternate exterior angles : $(\angle 1, \angle 4), (\angle 2, \angle 3)$
- Angles corresponding to $\angle 2$, $\angle 8$ and $\angle 3$ are $\angle 8, \angle 2, \angle 5$ respectively.

PROPERTIES OF ANGLES FORMED WHEN A TRANSVERSAL CUTS TWO PARALLEL LINES

Angles Formed when a Transversal Cuts Two Parallel Lines

Let two parallel lines l and m be cut by a transversal t , forming angles as shown, we have:

- Corresponding angles:** $(\angle 1, \angle 5), (\angle 2, \angle 6), (\angle 4, \angle 8), (\angle 3, \angle 7)$ are called pairs of corresponding angles. Angles in each pair are equal to each other i.e. $\angle 1 = \angle 5, \angle 2 = \angle 6, \angle 4 = \angle 8$ and $\angle 3 = \angle 7$
- Alternate interior angle:** $(\angle 3, \angle 5), (\angle 4, \angle 6)$, are called alternate interior angles. Angles in each pair are equal to each other. That is, $\angle 3 = \angle 5$ and $\angle 4 = \angle 6$.
- Alternate exterior angle:** $(\angle 2, \angle 8), (\angle 1, \angle 7)$ are called alternate exterior angles which are equal. Angles in each pair are equal to each other. That is, $\angle 2 = \angle 8$ and $\angle 1 = \angle 7$.
- Consecutive interior angles:** $(\angle 3, \angle 6)$ and $(\angle 4, \angle 5)$ are called pairs of interior angles or simply co-interior angles which are supplementary. That is, $\angle 3 + \angle 6 = 180^\circ$ and $\angle 4 + \angle 5 = 180^\circ$.



Test for parallel lines

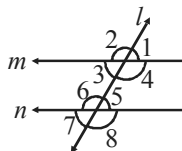
The properties of parallel lines can be used to find whether given lines are parallel or not parallel.

Two lines that are cut by a transversal are parallel if :

- alternate interior angles are equal or
- alternate exterior angles are equal or
- corresponding angles are equal or
- co-interior angles add upto 180° .

ILLUSTRATION 5 :

In fig. if $\angle 2 = 120^\circ$ and $\angle 5 = 60^\circ$, show that $m \parallel n$.



SOLUTION :

We have,

$$\angle 2 = 120^\circ \text{ and } \angle 5 = 60^\circ$$

But, $\angle 2 = \angle 4$ [Vertically opposite angles]

$$\therefore \angle 4 = 120^\circ$$

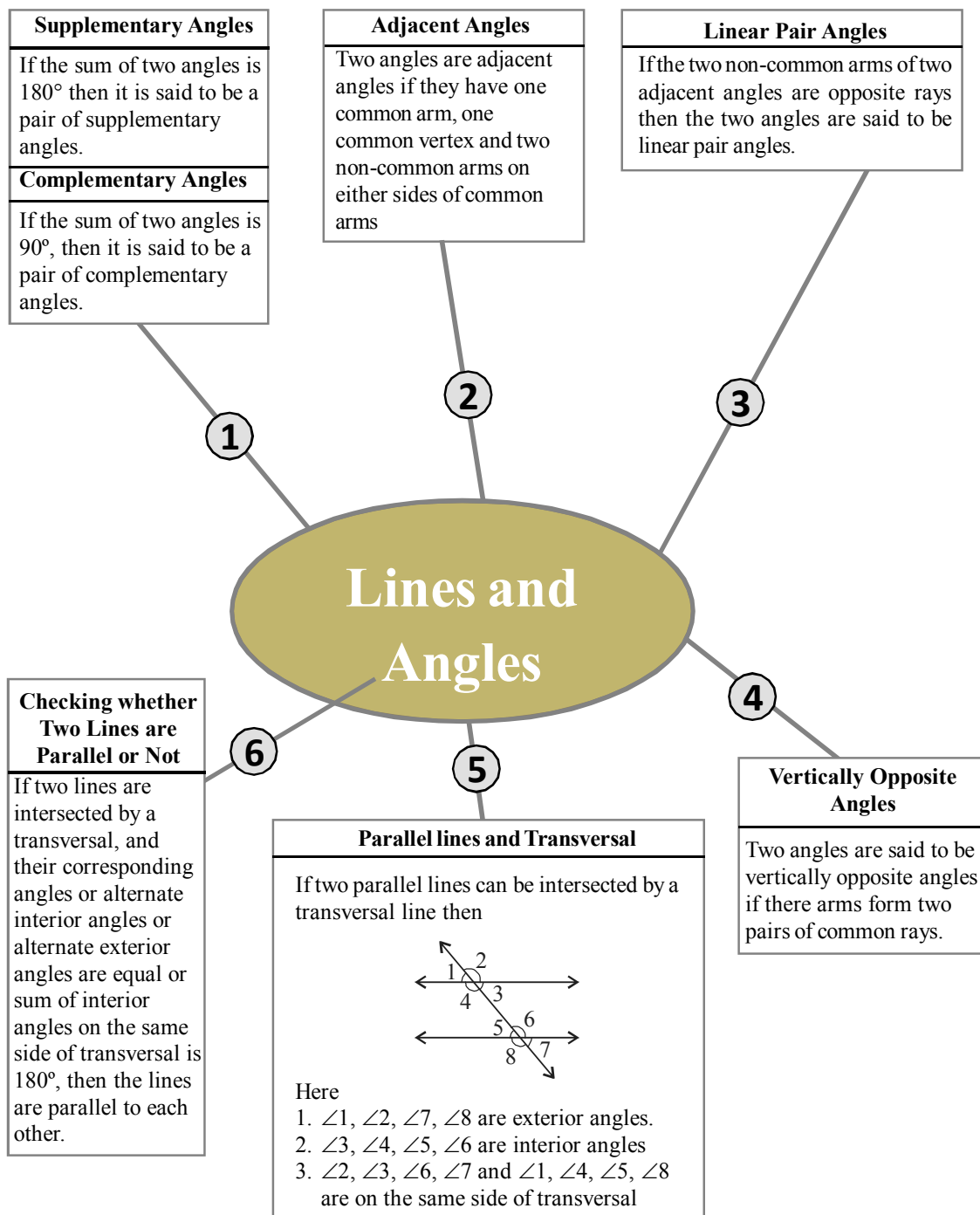
$$\Rightarrow \angle 4 + \angle 5 = 120^\circ + 60^\circ = 180^\circ \text{ [}\because \angle 5 = 60^\circ\text{]}$$

$\Rightarrow \angle 4$ and $\angle 5$ are supplementary angles.

$\Rightarrow$ Consecutive interior angles are supplementary.

$\Rightarrow m \parallel n$.

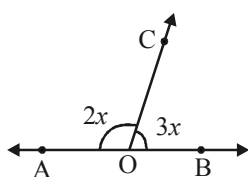
CONCEPT MAP



MISCELLANEOUS

SOLVED EXAMPLES

1. Find the degree measure of each angle in the following linear pair.



Sol. $\angle AOC$ and $\angle BOC$ form a linear pair.

$$\text{Thus, } \angle AOC + \angle BOC = 180^\circ$$

$$\text{or } 2x + 3x = 180^\circ$$

$$\text{or } 5x = 180^\circ$$

$$\text{or } x = 36^\circ$$

$$\text{Now, } 2x = 2 \times 36^\circ = 72^\circ \text{ and } 3x = 3 \times 36^\circ = 108^\circ$$

$$\text{Thus } \angle AOC = 72^\circ \text{ and } \angle BOC = 108^\circ.$$

2. Find the complement of the following angles.

(i) 25° (ii) 37° (iii) 45°

Sol. The sum of the pair of complementary angles is 90° . Therefore

$$\begin{aligned} \text{(i) complementary angle to } 25^\circ \\ = 90^\circ - 25^\circ = 65^\circ \end{aligned}$$

$$\begin{aligned} \text{(ii) complementary angle to } 37^\circ \\ = 90^\circ - 37^\circ = 53^\circ \end{aligned}$$

$$\begin{aligned} \text{(iii) complementary angle to } 45^\circ \\ = 90^\circ - 45^\circ = 45^\circ \end{aligned}$$

3. Find the supplementary angles to the following angles.

(i) 70° (ii) 90° (iii) 85°

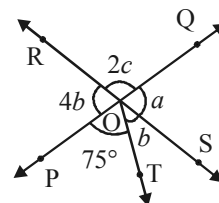
Sol. The sum of supplementary angles is 180° .

$$\begin{aligned} \text{(i) Supplementary angle to } 70^\circ \\ = 180^\circ - 70^\circ = 110^\circ \end{aligned}$$

$$\begin{aligned} \text{(ii) Supplementary angle to } 90^\circ \\ = 180^\circ - 90^\circ = 90^\circ \end{aligned}$$

$$\begin{aligned} \text{(iii) Supplementary angle to } 85^\circ \\ = 180^\circ - 85^\circ = 95^\circ. \end{aligned}$$

4. In figure, two straight lines PQ and RS intersect each other at O. If $\angle POT = 75^\circ$, find the values of a , b and c .



Sol. Since O, R and S lie on the same line.

$$\therefore \angle ROP + \angle POT + \angle TOS = 180^\circ$$

(Linear pair angles)

$$\Rightarrow 4b + 75^\circ + b = 180^\circ$$

$$\Rightarrow 5b + 75^\circ = 180^\circ$$

$$\Rightarrow 5b = 180^\circ - 75^\circ$$

$$\Rightarrow 5b = 105^\circ$$

$$\Rightarrow b = 21^\circ$$

Since PQ and RS intersect at O. Therefore,

$$\angle QOS = \angle POR \text{ [Vertically opposite angles]}$$

$$\Rightarrow a = 4b$$

$$\Rightarrow a = 4 \times 21^\circ = 84^\circ \quad [\because b = 21^\circ]$$

$$\text{Now, } \angle ROQ + \angle QOS = 180^\circ \text{ [Linear pair]}$$

$$\Rightarrow 2c + a = 180^\circ$$

$$\Rightarrow 2c + 84^\circ = 180^\circ \quad [\because a = 84^\circ]$$

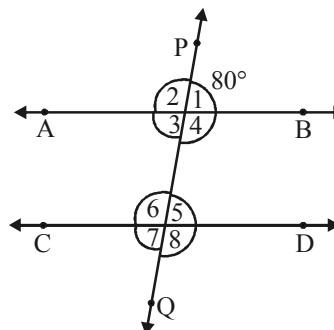
$$\Rightarrow 2c = 180^\circ - 84^\circ$$

$$\Rightarrow 2c = 96^\circ$$

$$\Rightarrow c = 48^\circ$$

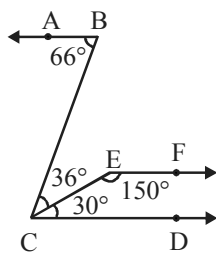
$$\text{Hence, } a = 84^\circ, b = 21^\circ \text{ and } c = 48^\circ.$$

5. In the following figure, if $AB \parallel CD$ and $\angle 1 = 80^\circ$, find the other angles.



Sol. We have $\angle 1 = \angle 3$ (Vertically opposite angles)
 So, $\angle 1 = \angle 3 = 80^\circ$
 Also, $\angle 3 = \angle 5$ (Alternate interior angles)
 So, $\angle 5 = 80^\circ$
 Now, $\angle 4 + \angle 5 = 180^\circ$
 (Angles on the same side of the transversal)
 $\Rightarrow \angle 4 = 180^\circ - \angle 5 = 180^\circ - 80^\circ = 100^\circ$
 $\therefore \angle 2 = \angle 4$ (Vertically opposite angles)
 Or $\angle 2 = 100^\circ$
 Again $\angle 2 = \angle 6$ (Corresponding angles)
 $\therefore \angle 6 = 100^\circ$
 Also, $\angle 3 = \angle 7$ (Corresponding angles)
 $\therefore \angle 7 = 80^\circ$
 and, $\angle 4 = \angle 8$ (Corresponding angles)
 $\therefore \angle 8 = 100^\circ$
 Hence, $\angle 1 = 80^\circ$, $\angle 2 = 100^\circ$, $\angle 3 = 80^\circ$,
 $\angle 4 = 100^\circ$, $\angle 5 = 80^\circ$, $\angle 6 = 100^\circ$, $\angle 7 = 80^\circ$
 $\angle 8 = 100^\circ$.

6. In figure show that $AB \parallel EF$.



Sol. We have,

$$\angle BCD = \angle BCE + \angle ECD$$

$$\Rightarrow \angle BCD = 36^\circ + 30^\circ = 66^\circ$$

$$\therefore \angle ABC = \angle BCD$$

Thus, lines AB and CD are intersected by the line BC such that

$$\angle ABC = \angle BCD$$

i.e., the alternate interior angles are equal.

Therefore,

$$AB \parallel CD \quad \dots (i)$$

$$\text{Now, } \angle ECD + \angle CEF = 30^\circ + 150^\circ = 180^\circ$$

This shows that the sum of the interior angles on the same side of the transversal CE is 180°

$$\therefore EF \parallel CD \quad \dots (ii)$$

From (i) and (ii), we have

$$AB \parallel CD$$

$$\text{And } CD \parallel EF$$

$$\therefore AB \parallel EF$$

1

EXERCISE

Fill in the Blanks :

DIRECTIONS : Complete the following statements with an appropriate word / term to be filled in the blank space(s).

- Two angles are said to be _____ if the sum of their measures is 180° .
- Two angles with common vertex, a common arm and other arms on opposite sides are called _____ angles.
- The sum of adjacent angles formed by a ray standing on a line is _____.
- The sum of all angles around a point is _____.
- When two lines intersect each other then the _____ angles are equal.
- A line which intersects two or more lines at distinct points is called _____.
- When two parallel lines are intersected by a transversal, then the sum of the interior angles on the same side of the transversal is _____.
- Two angles are said to be _____ if the sum of their measures is 90° .
- The supplement of an acute angle is _____ angle.
- The supplement of an obtuse angle is _____ angle.

True / False :

DIRECTIONS : Read the following statements and write your answer as true or false.

- One obtuse angle and one acute angle can make a pair of complementary angles.
- Two right angles are supplementary to each other.
- Each of the angle in a pair of supplementary angles is acute.
- Each of the angle in a pair of complementary angles can be acute.
- Vertically opposite angles form a linear pair.
- An angle of 45° is supplementary angle of 45° angle.
- Each angle in a linear pair is supplement to the other.

- When two parallel lines are intersected by a transversal, then corresponding angles so formed are equal.
- Two lines in a plane which do not intersect each other are called intersecting lines.
- The distance between two parallel lines remains the same everywhere.

Match the Columns :

DIRECTIONS : Each question contains statements given in two columns which have to be matched. Statements (A, B, C, D) in column-I have to be matched with statements (p, q, r, s) in column-II.

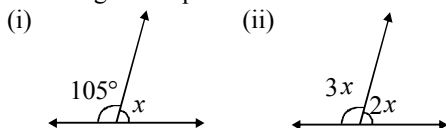
- | 1. | Column-I | Column-II |
|-----|--|--------------------------------|
| (A) | Two angles whose sum is 90° . | (p) Adjacent Angles |
| (B) | Two angles whose sum is 180° . | (q) Complementary angles |
| (C) | Two angles that are formed by two intersecting lines and are not adjacent. | (r) Supplementary angles |
| (D) | Two angles with a common vertex, a common arm and the other arms lying on the opposite sides of the common arm | (s) Vertically opposite angles |

Very Short Answer Questions :

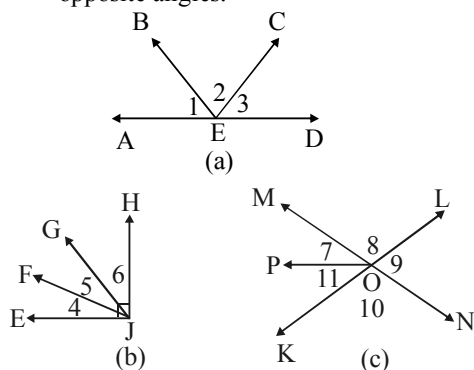
DIRECTIONS : Give answer in one word or one sentence.

- Find the complement of each of the following angles:
(i) 60° (ii) 25° (iii) 72°
- Find the supplement of each of the following angles:
(i) 125° (ii) 64° (iii) 38°

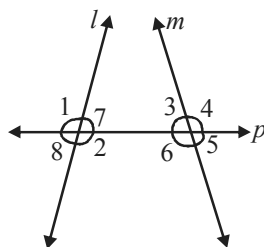
3. Find the degree measure of each angle in the following linear pairs:



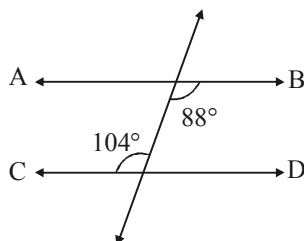
4. (i) In Fig. (a), name two pairs of supplementary angles.
 (ii) In Fig. (b), name two pairs of complementary angles.
 (iii) In Fig. (c), name two pairs of vertically opposite angles.



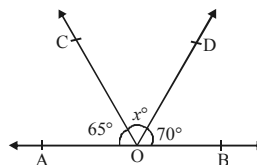
5. Line p is a transversal to lines l and m . Name the following:



- (i) Two pairs of alternate angles.
 (ii) Four pairs of corresponding angles.
 (iii) Two pairs of interior angles on the same side of the transversal.
6. Are the lines AB and CD parallel?

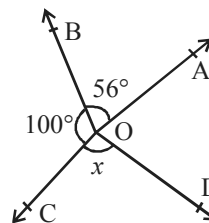


7. In the given figure, AOB is a straight line and the rays OC and OD stand on it.



If $\angle AOC = 65^\circ$, $\angle BOD = 70^\circ$ and $\angle COD = x^\circ$, find the value of x .

8. In the given figure, rays OA, OB, OC and OD are such that $\angle AOB = 56^\circ$, $\angle BOC = 100^\circ$, $\angle COD = x^\circ$ and $\angle DOA = 74^\circ$.

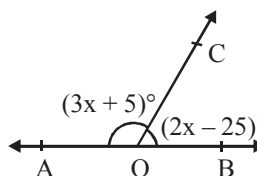


Find the value of x .

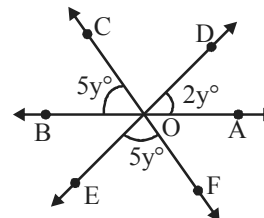
Short Answer Questions :

DIRECTIONS : Give answer in 2-3 sentences.

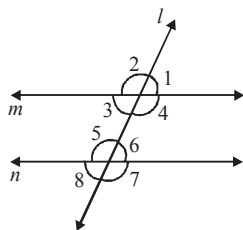
1. In the adjoining figure, what value of x will make AOB a straight line?



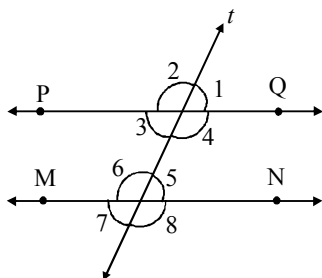
2. Two complementary angles are in the ratio 7 : 8. Find the angles.
 3. An angle is 30° less than two times its supplement. Find the angles.
 4. In figure, determine the value of y .



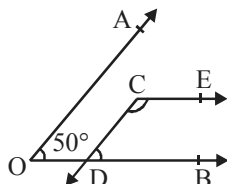
5. In figure, $m \parallel n$ and $\angle 1 = 65^\circ$. Find $\angle 5$ and $\angle 8$.



6. Find the angles in each of the following case.
- The angles are supplementary and the larger is twice the smaller.
 - The angles are complementary and the larger is 20° more than the smaller.
 - The angles are adjacent and form an angle of 120° . The larger is 20° less than three times the smaller.
7. In the figure it is given that $PQ \parallel MN$. If $\angle 1 = (120^\circ - 2x)$ and $\angle 5 = 4x$, find the measures of $\angle 1$ and $\angle 5$.

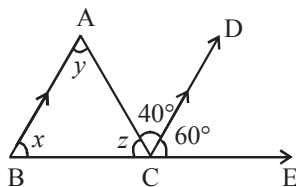


8. In the adjoining figure, it is being given that $AO \parallel CD$, $OB \parallel CE$ and $\angle AOB = 50^\circ$.

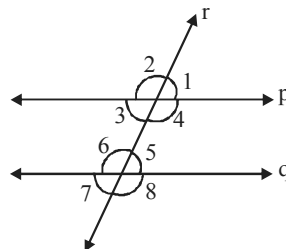


Find the measure of $\angle ECD$.

9. In the given figure, find x , y and z .



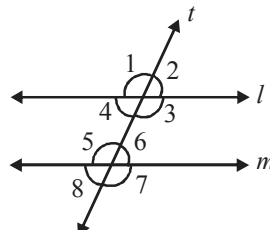
10. In the figure, if $\angle 2 = 120^\circ$ and $\angle 5 = 60^\circ$, show that $p \parallel q$.



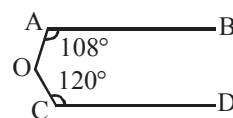
Long Answer Questions :

DIRECTIONS : Give answer in four to five sentences.

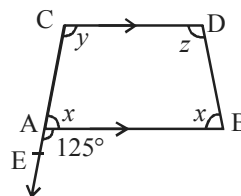
1. In the given figure, $l \parallel m$ and t is a transversal such that $\angle 1 = 135^\circ$. Find the measure of each one of the angles $\angle 2, \angle 3, \angle 4, \angle 5, \angle 6, \angle 7$ and $\angle 8$.



2. In the adjoining figure, it is given that $AB \parallel CD$, $\angle BAO = 108^\circ$ and $\angle OCD = 120^\circ$. Find $\angle AOC$.

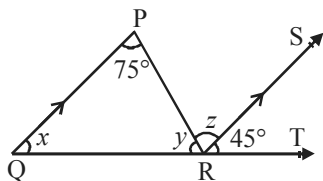


3. In the given figure, $AB \parallel CD$ and CA has been produced to E so that $\angle BAE = 125^\circ$.

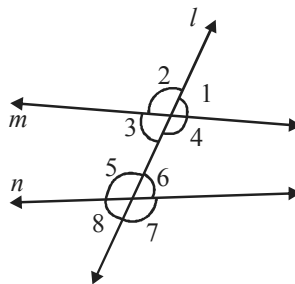


If $\angle BAC = x$, $\angle ABD = x$, $\angle BDC = z$ and $\angle ACD = y$, find the values of x, y, z .

4. Find the value of the angles x , y and z in the following figure.



5. In figure, if $\angle 3 = 61^\circ$ and $\angle 7 = 118^\circ$. Is $m \parallel n$?

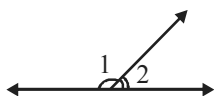


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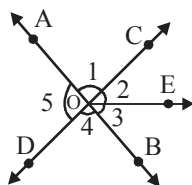
EXERCISE

Text-Book Exercise :

- Identify which of the following pairs of angles are complementary and which are supplementary.
 - $65^\circ, 115^\circ$
 - $63^\circ, 27^\circ$
 - $112^\circ, 68^\circ$
 - $130^\circ, 50^\circ$
- In the given figure, $\angle 1$ and $\angle 2$ are supplementary angles. If $\angle 1$ is decreased, what changes should take place in $\angle 2$ so that both the angles still remain supplementary.

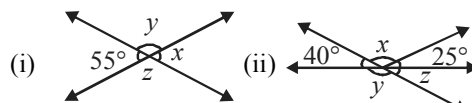


3. In the adjoining figure:

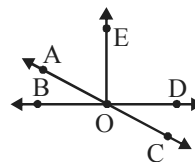


- Is $\angle 1$ adjacent to $\angle 2$?
- Is $\angle AOC$ adjacent to $\angle AOE$?
- Do $\angle COE$ and $\angle EOD$ form a linear pair?
- Are $\angle BOD$ and $\angle DOA$ supplementary?
- Is $\angle 1$ vertically opposite to $\angle 4$?
- What is the vertically opposite angle of $\angle 5$?

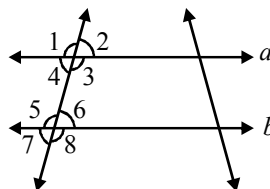
4. Find the value of the angles x , y and z in each of the following:



5. In the adjoining figure, name the following pairs of angles.

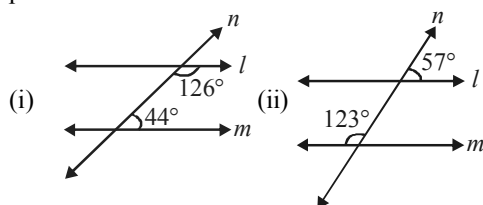


- Obtuse vertically opposite angles
 - Adjacent complementary angles
 - Equal supplementary angles
 - Adjacent angles that do not form a linear pair.
6. State the property that is used in each of the following statements?



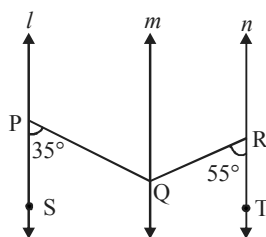
- If $a \parallel b$, then $\angle 1 = \angle 5$.
- If $\angle 4 = \angle 6$, then $a \parallel b$.
- If $\angle 4 + \angle 5 = 180^\circ$, then $a \parallel b$.

7. In the given figures below, decide whether l is parallel to m .

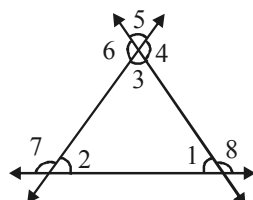


Exemplar Questions :

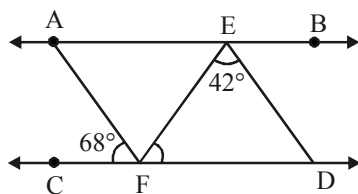
- The sum of two vertically opposite angles is 166° . Find each of the angles.
- In figure, $l \parallel m \parallel n$. $\angle QPS = 35^\circ$ and $\angle QRT = 55^\circ$. Find $\angle PQR$.



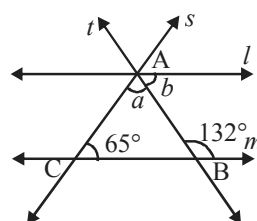
- In figure, write all the pairs of supplementary angles.



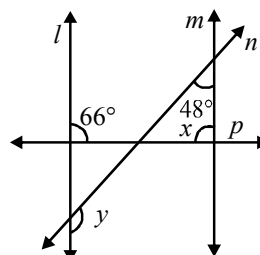
- What is the type of other angle of a linear pair if
 - one of its angles is acute?
 - one of its angles is obtuse?
 - one of its angles is right?
- In figure, $AB \parallel CD$, $AF \parallel ED$, $\angle AFC = 68^\circ$ and $\angle FED = 42^\circ$. Find $\angle EFD$.



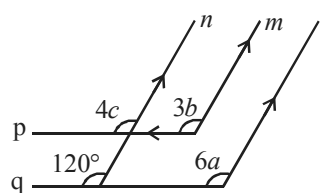
- If a transversal intersects two parallel lines, and the difference of two interior angles on the same side of a transversal is 20° , find the angles.
- Two angles are making a linear pair. If one of them is one-third of the other, find the angles.
- In figure, if $l \parallel m$, find the values of a and b .



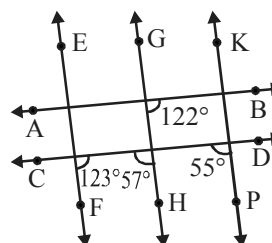
- In figure, two parallel lines l and m are cut by two transversals n and p . Find the values of x and y .



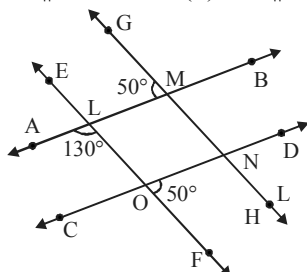
- In figure l , m and n are parallel lines, and the lines p and q are also parallel. Find the values of a , b and c .



- In figure, find out which pair of lines are parallel:

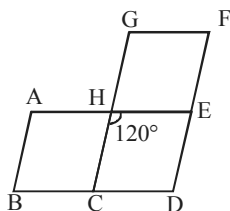


12. In figure, show that
(i) $AB \parallel CD$ (ii) $EF \parallel GH$

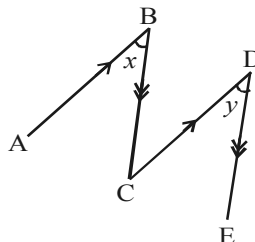


HOTS Questions :

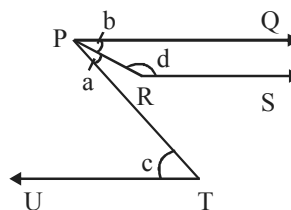
- $\angle 1$ and $\angle 2$ are complementary angles. $\angle 2$ and $\angle 3$ are supplementary angles. If $\angle 1 = 45^\circ$, find the measure of $\angle 3$.
- In figure, $AE \parallel GF \parallel BD$, $AB \parallel CG \parallel DF$ and $\angle CHE = 120^\circ$. Find $\angle ABC$ and $\angle CDE$.



3. In the given figure.
 $AB \parallel CD$ and $BC \parallel DE$.
Show that $\angle x = \angle y$.



4. In figure, PQ, RS and UT are parallel lines.



- If $c = 57^\circ$ and $a = \frac{c}{3}$, find the value of d .
- If $c = 75^\circ$ and $a = \frac{2}{5}c$, find b .

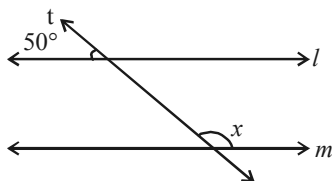
3

EXERCISE

Single Option Correct :

DIRECTIONS : This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONLY ONE is correct.

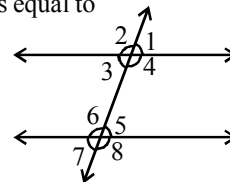
1. In the given figure, it is given that $l \parallel m$, t is a transversal. Then the value of x is



- 130°
- 50°
- 120°
- None of these

2. If $\angle 2 = 2\angle 1$, then $\angle 7$ is equal to

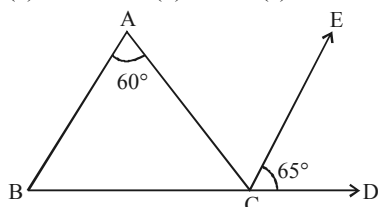
- 50°
- 60°
- 70°
- 80°



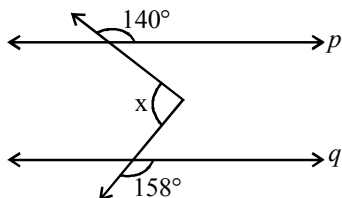
3. Which of the following is $\left(\frac{1}{5}\right)^{\text{th}}$ of its complement?

- 18°
- 15°
- 36°
- 75°

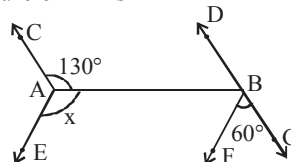
4. Which of the following angles is 20° less than its supplement?
 (a) 100° (b) 35°
 (c) 55° (d) 80°
5. If $l \parallel m$, then x is equal to
 (a) 115°
 (b) 295°
 (c) 195°
 (d) 245°
6. In the given figure, it is given that $\angle A = 60^\circ$, $CE \parallel BA$ and $\angle ECD = 65^\circ$. Then $\angle ACB$ equals to
 (a) 60° (b) 55° (c) 70° (d) 90°



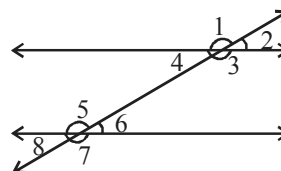
7. Two complementary angles are in the ratio 7 : 8. The angles are
 (a) $30^\circ, 60^\circ$ (b) $40^\circ, 50^\circ$
 (c) $42^\circ, 48^\circ$ (d) $27^\circ, 63^\circ$
8. If $\angle A = (4x + 2)^\circ$, the measure of the complementary of $\angle A$ is equal to
 (a) $(178 + 4x)^\circ$ (b) $(88 - 4x)^\circ$
 (c) $(4x + 88)^\circ$ (d) $(6x - 178)^\circ$
9. If $AB \parallel CD$ and $AC \parallel DE$ then $\angle a + \angle b$ is equal to
 (a) 70°
 (b) 120°
 (c) 100°
 (d) 115°
10. Assume $p \parallel q$ in the figure shown. Then 'x' equals.
 (a) 18° (b) 22°
 (c) 62° (d) 72°



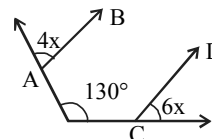
11. In the given figure $AC \parallel BD$ and $AE \parallel BF$. The measure of $\angle x$ is



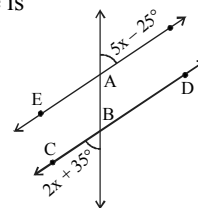
- (a) 130° (b) 110°
 (c) 70° (d) 100°
12. If $\angle 2 = (3x - 10)^\circ$ and $\angle 8 = (5x - 30)^\circ$, then $\angle 2$ and $\angle 8$ are



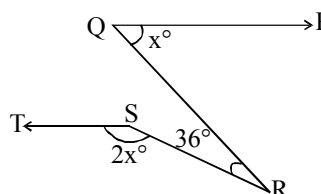
- (a) $100^\circ, 100^\circ$ (b) $20^\circ, 20^\circ$
 (c) $40^\circ, 90^\circ$ (d) $60^\circ, 60^\circ$
13. In the given figure $AB \parallel CD$. Then 'x' is equal to



- (a) 26° (b) 13°
 (c) 23° (d) 46°
14. $\angle ABC$ in the given figure is equal to



- (a) 75° (b) 105°
 (c) 115° (d) 85°
15. $QP \parallel TS$ and $\angle QRS = 36^\circ$, then value of $\angle PQR$ is

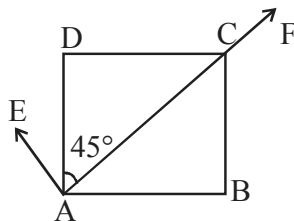


- (a) 48° (b) 52°
 (c) 72° (d) 50°

More Than One Option Correct :

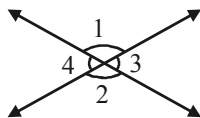
DIRECTIONS : This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONE or MORE may be correct.

1. In the following AD is the bisector of $\angle EAF$

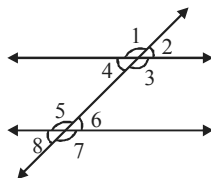


Which one of the following statements is/are correct?

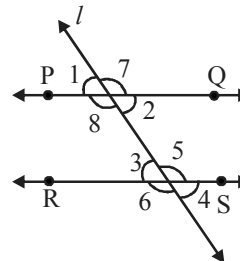
- (a) $\angle EAD$ is an acute angle.
 (b) $\angle BAE$ is an obtuse angle.
 (c) $\angle FAD$ and $\angle DAE$ are complementary to each other.
 (d) $\angle CAD$ and $\angle DAE$ are supplementary to each other.
2. Which of the following is/are true.
- (a) $\angle 1 = \angle 2$
 (b) $\angle 1 + \angle 3 = 180^\circ$
 (c) $\angle 1 = \angle 4$
 (d) $\angle 3 = \angle 4$
3. Which of the following pair of angles are supplementary?
- (a) $48^\circ, 42^\circ$ (b) $90^\circ, 90^\circ$
 (c) $75^\circ, 105^\circ$ (d) $179^\circ, 2^\circ$
4. In figure, pairs of corresponding angles are



- (a) $\angle 1, \angle 5$
 (b) $\angle 3, \angle 6$
 (c) $\angle 3, \angle 5$
 (d) $\angle 3, \angle 7$
5. Angles of a linear pair are
- (a) Complementary (b) Adjacent
 (c) Supplementary (d) None of these



6. In figure $PQ \parallel RS$. Which of the following is true?



- (a) $\angle 1 + \angle 2 = 180^\circ$ (b) $\angle 2 + \angle 5 = 180^\circ$
 (c) $\angle 3 + \angle 8 = 180^\circ$ (d) $\angle 2 + \angle 3 = 180^\circ$

Assertion & Reason :

DIRECTIONS : Each of these questions contains an Assertion followed by Reason. Read them carefully and answer the question on the basis of following options. You have to select the one that best describes the two statements.

- (a) If both **Assertion** and **Reason** are correct and **Reason** is the correct explanation of **Assertion**.
 (b) If both **Assertion** and **Reason** are correct but **Reason** is not the correct explanation of **Assertion**.
 (c) If **Assertion** is correct but **Reason** is incorrect.
 (d) If **Assertion** is incorrect but **Reason** is correct.

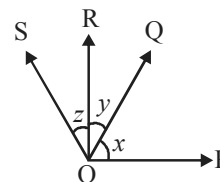
1. **Assertion:** The supplement of 95° is 85° .
Reason: If two angles are supplementary, then their sum is 180° .
2. **Assertion:** The supplement of $\frac{1}{4}$ of 240° is 60° .
Reason: If two angles are complementary then their sum is 90° .

Passage Based Questions :

DIRECTIONS : Study the given passage(s) and answer the following questions.

PASSAGE - I

Observe the following figure.



- $x = 60^\circ, y = 25^\circ$, find $\angle POR$.
(a) 90° (b) 85°
(c) 45° (d) 95°
- $\angle SOQ = 80^\circ, \angle QOR = 25^\circ$, find $\angle SOR$.
(a) 55° (b) 105°
(c) 45° (d) 60°
- If $x = \frac{1}{3}$ of right angle, $y = \frac{2}{3}$ of right angle and $z = \frac{1}{2}$ of right angle, then $\angle POS =$
(a) 60° (b) 45°
(c) 90° (d) 135°

PASSAGE - II

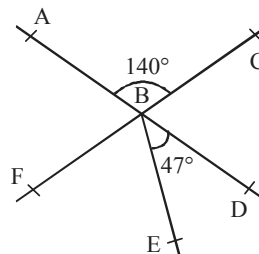
Two angles whose sum is 180° are called supplementary angles.

- Two supplementary angles are in the ratio 5 : 13. Find the angles.
(a) $50^\circ, 130^\circ$ (b) $130^\circ, 60^\circ$
(c) $90^\circ, 90^\circ$ (d) $70^\circ, 110^\circ$
- The angles are supplementary and the larger is 40° less than three times the smaller angle, Find the angles.
(a) $80^\circ, 100^\circ$ (b) $90^\circ, 90^\circ$
(c) $55^\circ, 125^\circ$ (d) $140^\circ, 40^\circ$
- The angles are supplementary and the larger angle is double of the smaller. Find the angles.
(a) $110^\circ, 70^\circ$ (b) $60^\circ, 120^\circ$
(c) $80^\circ, 100^\circ$ (d) $50^\circ, 130^\circ$

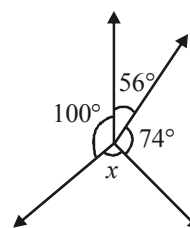
Integer Type Questions :

DIRECTIONS : Answer the following questions. The answer to each of the question is a single digit integer, ranging from 0 to 9.

- If two lines are intersected by a transversal, then the number of pairs of interior angles on the same side of the transversal is
- If the complement of an angle is 89° , then the angle will be
- The angle which makes a linear pair with an angle of 171° is
- In the figure, AD and CF are straight lines. The value of $\angle EBF$ is $(31k^\circ)$. The value of k is



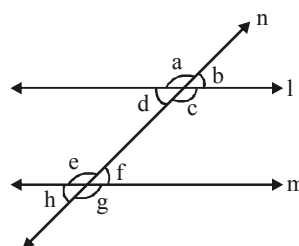
- In the given figure, the value of x is $(26k^\circ)$. Find the value of k.



Multiple Matching Question :

DIRECTIONS : Following question has four statements (A, B, C and D) given in Column-I and five statements (p, q, r, s, t) in Column-II. Any given statement in Column-I can have correct matching with one or more statement(s) given in Column-II.

- Observe the figure and match the columns:



Column-I

- Supplementary angle of $\angle b$
- Vertically opposite angle of $\angle g$
- Adjacent angle of $\angle h$
- Corresponding angle of $\angle d$

Column-II

- $\angle g$
- $\angle h$
- $\angle c$
- $\angle e$
- $\angle a$

SOLUTIONS

Brief Explanations of Selected Questions

1 EXERCISE

Fill in the Blanks :

- | | |
|-------------------------|--------------------------|
| 1. supplementary | 2. adjacent |
| 3. 180° | 4. 360° |
| 5. vertically opposite. | 6. a transversal. |
| 7. 180° | 8. complementary angles. |
| 9. obtuse | 10. acute. |

True / False :

- | | | |
|-----------|----------|-----------|
| 1. False. | 2. True. | 3. False |
| 4. True. | 5. False | 6. False |
| 7. True. | 8. True. | 9. False. |
| 10. True. | | |

Match the Columns :

1. (A) $\rightarrow$ (q); (B) $\rightarrow$ (r); (C) $\rightarrow$ (s); (D) $\rightarrow$ (p)

Very Short Answer Questions :

- $90^\circ - 60^\circ = 30^\circ$.
 - $90^\circ - 25^\circ = 65^\circ$.
 - $90^\circ - 72^\circ = 18^\circ$.
- $180^\circ - 125^\circ = 55^\circ$.
 - $180^\circ - 64^\circ = 116^\circ$.
 - $180^\circ - 38^\circ = 142^\circ$.
- $75^\circ, 105^\circ$ (ii) $72^\circ, 108^\circ$
- $\angle 1$ and $\angle BED$; $\angle 3$ and $\angle AEC$.
 - $\angle 4$ and $\angle FJH$; $\angle 6$ and $\angle EJG$.
 - $\angle 8$ and $\angle 10$; $\angle 9$ and $\angle MOK$.
- $\angle 2$ and $\angle 3$, $\angle 6$ and $\angle 7$
 - $\angle 1$ and $\angle 3$, $\angle 4$ and $\angle 7$, $\angle 8$ and $\angle 6$, $\angle 5$ and $\angle 2$
 - $\angle 2$ and $\angle 6$, $\angle 7$ and $\angle 3$
- The angles shown here are alternate. But they are not equal. Hence, the lines AB and CD are not parallel.
- Since AOB is a straight line
 $\therefore \angle AOC + \angle BOD + x^\circ = 180^\circ$
 $\Rightarrow 65^\circ + 70^\circ + x^\circ = 180^\circ$
 $\Rightarrow x^\circ = 180^\circ - 135^\circ = 45^\circ$

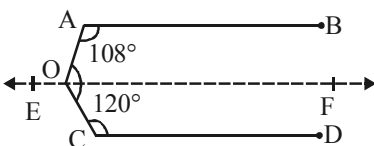
- Sum of all the angles around a point is 360° .
 $\Rightarrow \angle AOB + \angle BOC + \angle COD + \angle AOD = 360^\circ$
 $\Rightarrow 56^\circ + 100^\circ + x^\circ + 74^\circ = 360^\circ$
 $\Rightarrow x^\circ = 360^\circ - 230^\circ = 130^\circ$

Short Answer Questions :

- AOB will be a straight line, if $\angle AOC + \angle BOC = 180^\circ$.
 $\therefore (3x + 5) + (2x - 25) = 180$
 $\Rightarrow 5x - 20 = 180 \Rightarrow 5x = (180 + 20)$
 $\Rightarrow 5x = 200 \Rightarrow x = 40$.
- Let the angles be $7x^\circ$ and $8x^\circ$.
 $\therefore 7x + 8x = 90^\circ$
 $\Rightarrow 15x = 90^\circ \Rightarrow x = 6^\circ$
 $\Rightarrow 7x = 42^\circ$ and $8x = 48^\circ$
- Let one angle be x° . The supplement of this angle is $(180 - x)^\circ$.
 $\therefore x = 2(180 - x) - 30$
 $\Rightarrow x = 360 - 2x - 30$
 $\Rightarrow x + 2x = 330$
 $\Rightarrow 3x = 330 \Rightarrow x = 110^\circ$
 $\Rightarrow \text{Supplement} = 180^\circ - 110^\circ = 70^\circ$
 $\therefore$ Required angles are 110° and 70° .
- $\angle COD = \angle EOF = 5y^\circ$
 $[\because \text{vertically opposite angles.}]$
 Now, OA and OB are opposite rays.
 $\therefore \angle AOD + \angle DOC + \angle COB = 180^\circ$
 $\Rightarrow 2y^\circ + 5y^\circ + 5y^\circ = 180^\circ$
 $\Rightarrow 12y^\circ = 180^\circ \Rightarrow y = 15^\circ$
- $\angle 1 = \angle 3$ [Vertically opposite angles]
 and, $\angle 3 = \angle 8$ [Corresponding angles]
 $\therefore \angle 1 = \angle 8$
 $\Rightarrow \angle 8 = 65^\circ$ [$\because \angle 1 = 65^\circ$ (given)]
 Now, $\angle 5 + \angle 8 = 180^\circ$ [Linear pair]
 $\angle 5 + 65^\circ = 180^\circ$
 $\Rightarrow \angle 5 = 180^\circ - 65^\circ = 115^\circ$
- Let the smaller angle = x .
 $\therefore x + 2x = 180^\circ$
 $\Rightarrow 3x = 180^\circ$
 $\therefore x = 60^\circ, 2x = 120^\circ$

- (ii) Let the smaller angle be x . Then,
 $x + (x + 20^\circ) = 90^\circ$
 $\Rightarrow 2x + 20^\circ = 90^\circ$
 $\Rightarrow 2x = 90^\circ - 20^\circ$
 $\Rightarrow 2x = 70^\circ$
 $\Rightarrow x = 35^\circ, x + 20^\circ = 55^\circ$
- (iii) Let the smaller angle = x . Then,
 Larger angle = $3x - 20^\circ$
 $\therefore x + (3x - 20^\circ) = 120^\circ$
 $\Rightarrow 4x - 20^\circ = 120^\circ$
 $\Rightarrow 4x = 120^\circ + 20^\circ$
 $\Rightarrow 4x = 140^\circ$
 $\therefore x = 35^\circ, 3x - 20^\circ = 85^\circ$
7. $PQ \parallel MN$ and t is the transversal.
 $\therefore 120^\circ - 2x = 4x$
 $\Rightarrow 120^\circ = 4x + 2x$
 $\Rightarrow 6x = 120^\circ \Rightarrow x = 20^\circ$
 $\therefore \angle 1 = 120^\circ - 2x = 120^\circ - 40^\circ = 80^\circ$
 $\angle 5 = \angle 1 = 80^\circ$
8. $AO \parallel CD \Rightarrow \angle AOD = \angle CDB$
 $\Rightarrow \angle CDB = 50^\circ$
 Now, $CE \parallel DB \Rightarrow \angle ECD + \angle CDB = 180^\circ$
 $\Rightarrow \angle ECD = 180^\circ - 50^\circ$
 $\Rightarrow \angle ECD = 130^\circ$
9. Here, $AB \parallel CD$ and AC is transversal.
 $\therefore y = \angle ACD = 40^\circ$
 $AB \parallel CD$ and BE is transversal.
 $\therefore x = \angle DCE = 60^\circ$ [Corresponding angles]
 Now, $60^\circ + 40^\circ + z = 180^\circ$
 $\Rightarrow z = 180^\circ - 100^\circ = 80^\circ$
10. $\angle 1 + \angle 2 = 180^\circ$ (Linear pair)
 $\Rightarrow 120^\circ + \angle 1 = 180^\circ$
 $\therefore \angle 1 = 180^\circ - 120^\circ = 60^\circ$
 Given, $\angle 5 = 60^\circ$
 $\angle 1$ and $\angle 5$ are corresponding angles and
 $\angle 1 = \angle 5$
 $\therefore p \parallel q$.

Long Answer Questions :

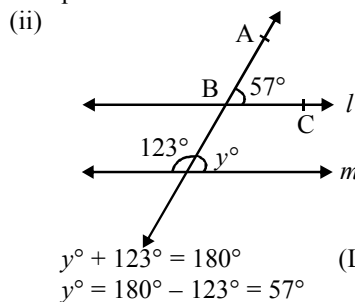
1. It is given that $\angle 1 = 135^\circ$.
 Now, $\angle 3 = \angle 1 = 135^\circ$
 and $\angle 5 = \angle 3 = 135^\circ$ (alternate interior angles)
 Also, $\angle 7 = \angle 5 = 135^\circ$ (vertically opposite $\angle$)
 Now, $\angle 1 + \angle 2 = 180^\circ$ [Linear pair]
 $\Rightarrow 135^\circ + \angle 2 = 180^\circ$
 $\Rightarrow \angle 2 = (180^\circ - 135^\circ) = 45^\circ$
 $\angle 2 = \angle 4 = 45^\circ$
 $\angle 4 = \angle 6 = 45^\circ$
 $\angle 6 = \angle 8 = 45^\circ$
2. Through O , draw $EF \parallel AB$.
- 
- Since $EF \parallel AB$ and AO is a transversal
 $\therefore \angle OAB + \angle AOF = 180^\circ$ [co-interior $\angle$]
 $\Rightarrow 108^\circ + \angle AOF = 180^\circ$
 $\Rightarrow \angle AOF = (180^\circ - 108^\circ) = 72^\circ$.
 Since $EF \parallel CD$ and OC is a transversal
 $\therefore \angle COF + \angle OCD = 180^\circ$ [co-interior $\angle$]
 $\angle COF + 120^\circ = 180^\circ$
 $\angle COF = (180^\circ - 120^\circ) = 60^\circ$.
 $\therefore \angle AOC = \angle AOF + \angle COF$
 $= (72^\circ + 60^\circ) = 132^\circ$.
3. Since $AB \parallel CD$, AC and BD are transversals.
 $x + z = 180^\circ$ (Co-interior angles) (i)
 $y + x = 180^\circ$ (Co-interior angles) (ii)
 $x + \angle EAB = 180^\circ$ (Linear pair)
 $x = 180^\circ - 125^\circ$
 $x = 55^\circ$ (iii)
 From (i) and (iii); $z = 180^\circ - 55^\circ = 125^\circ$
 From (ii) and (iii); $y = 180^\circ - 55^\circ = 125^\circ$
4. QT is a transversal to the parallel lines QP and RS
 $\Rightarrow x = \angle SRT = 45^\circ$ (Corresponding Angles)
 In $\triangle PQR$,
 $x + \angle P + y = 180^\circ$ (Angle Sum Property)
 $\Rightarrow 45^\circ + 75^\circ + y = 180^\circ$
 $\Rightarrow y = 180^\circ - 120^\circ = 60^\circ$
 Also, $y + z + 45^\circ = 180^\circ$
 $\therefore 60^\circ + z + 45^\circ = 180^\circ$
 $105^\circ + z = 180^\circ$
 $\Rightarrow z = 180^\circ - 105^\circ = 75^\circ$
5. Since $\angle 3$ and $\angle 4$ form a linear pair.
 $\therefore \angle 3 + \angle 4 = 180^\circ$
 $\Rightarrow 61^\circ + \angle 4 = 180^\circ$
 $\Rightarrow \angle 4 = 180^\circ - 61^\circ = 119^\circ$
 Now, $\angle 4$ and $\angle 7$ are a pair of corresponding angles such that $\angle 4 = 119^\circ$ and $\angle 7 = 118^\circ$
 i.e., $\angle 4 \neq \angle 7$.
 So, m is not parallel to n .

2 EXERCISE

Text-Book Exercise :

- $65^\circ + 115^\circ = 180^\circ$
 $\therefore$ These angles are supplementary angles.
 - $63^\circ + 27^\circ = 90^\circ$
 $\therefore$ These angles are complementary angles.
 - $112^\circ + 68^\circ = 180^\circ$
 $\therefore$ These angles are supplementary angles.
 - $130^\circ + 50^\circ = 180^\circ$
 $\therefore$ These angles are supplementary angles.
- $\angle 1$ and $\angle 2$ are supplementary angles. If $\angle 1$ is decreased, then $\angle 2$ should be increased by the same measure so that this pair of angles remains supplementary.
- Yes.
 - No. Since they have a common vertex O and a common arm OA. But, their non-common arms, OC and OE, are on the same side of the common arm.
 - Yes.
 - Yes.
 - Yes.
 - $\angle COB$
- $x = 55^\circ$ (Vertically opposite $\angle$ s)
 $x + y = 180^\circ$ ($\therefore$ Linear pair)
 $55^\circ + y = 180^\circ$
 $y = 180^\circ - 55^\circ = 125^\circ$
 $y = z = 125^\circ$ ($\therefore$ Vertically opposite angles)
 - $z = 40^\circ$ ($\therefore$ Vertically opposite angles)
 $y + z = 180^\circ$ ($\therefore$ Linear Pair)
 $y = 180^\circ - 40^\circ = 140^\circ$
 $40^\circ + x + 25^\circ = 180^\circ$
 $65^\circ + x = 180^\circ$
 $x = 180^\circ - 65^\circ = 115^\circ$.
- $\angle BOC$ and $\angle AOD$
 - $\angle AOB$ and $\angle AOE$
 - $\angle EOB$ and $\angle EOD$
 - $\angle AOE$ and $\angle EOD$; $\angle EOD$ and $\angle DOC$; $\angle AOB$ and $\angle AOE$
- Corresponding angles property
 - Alternate interior angles property
 - Interior angles on the same side of transversal are supplementary.

- Sum of the interior angles on the same side of transversal $= 126^\circ + 44^\circ = 170^\circ$. As the sum of interior angles on the same side of transversal is not 180° , therefore, l is not parallel to m .



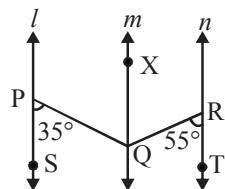
As corresponding angles ($\angle ABC$ and y°) are equal. Hence, $l \parallel m$.

Exemplar Questions :

- Let the measure of one of the vertically opposite angles be x° .

$$\therefore x^\circ + x^\circ = 166^\circ \Rightarrow 2x^\circ = 166^\circ \Rightarrow x^\circ = 83^\circ$$

2.



Since $l \parallel m$ and PQ is a transversal.

$$\therefore \angle PQX = \angle QPS = 35^\circ$$

Since $m \parallel n$ and QR is a transversal.

$$\therefore \angle RQX = \angle QRT = 55^\circ$$

$$\text{Now, } \angle PQR = \angle PQX + \angle RQX \\ = 35^\circ + 55^\circ = 90^\circ$$

- $\angle 7, \angle 2; \angle 1, \angle 8; \angle 5, \angle 6; \angle 6, \angle 3; \angle 3, \angle 4; \angle 4, \angle 5$
- Other angle is obtuse.
 - Other angle is acute.
 - Other angle is also a right angle.
- Since $AF \parallel ED$ and EF is a transversal.
 $\therefore \angle AFE = \angle FED = 42^\circ$

[$\therefore$ alternate interior $\angle$]

$$\text{Now, } \angle AFC + \angle AFE + \angle EFD = 180^\circ$$

[$\therefore$ angles on a straight line]

$$\Rightarrow 68^\circ + 42^\circ + \angle EFD = 180^\circ$$

$$\Rightarrow \angle EFD = 180^\circ - 110^\circ = 70^\circ$$

6. Let the two angles be of measure x° and $x^\circ + 20^\circ$
 $\therefore x^\circ + x^\circ + 20^\circ = 180^\circ$
 [$\because$ Interior angles on the same side of a transversal are supplementary]
 $\Rightarrow 2x^\circ + 20^\circ = 180^\circ$
 $\Rightarrow 2x^\circ = 160^\circ$
 $\Rightarrow x^\circ = 80^\circ$
 The measure of other angle = $x^\circ + 20^\circ$
 $= 80^\circ + 20^\circ = 100^\circ$

7. Let the measure of two angles be x° and $\frac{x^\circ}{3}$

$$\therefore x^\circ + \frac{x^\circ}{3} = 180^\circ \quad [\because \text{linear pair}]$$

$$\Rightarrow \frac{4x^\circ}{3} = 180^\circ \quad \Rightarrow 4x^\circ = 540^\circ$$

$$\Rightarrow x^\circ = \frac{540^\circ}{4} = 135^\circ$$

The measure of other angle

$$= \frac{x^\circ}{3} = \frac{135^\circ}{3} = 45^\circ$$

8. Since $l \parallel m$ and t is a transversal.

$$\therefore \angle b + 132^\circ = 180^\circ$$

$$\Rightarrow \angle b = 180^\circ - 132^\circ = 48^\circ$$

Since $l \parallel m$ and s is a transversal.

$$\therefore (\angle a + \angle b) + 65^\circ = 180^\circ$$

$$\Rightarrow \angle a + 48^\circ + 65^\circ = 180^\circ$$

$$\Rightarrow \angle a = 180^\circ - 113^\circ = 67^\circ$$

9. Since $l \parallel m$ and p is a transversal.

$$\therefore 66^\circ + x = 180^\circ$$

$$\Rightarrow x = 180^\circ - 66^\circ = 114^\circ$$

Since $l \parallel m$ and n is a transversal.

$$\therefore 48^\circ + y = 180^\circ$$

$$\Rightarrow y = 180^\circ - 48^\circ = 132^\circ$$

10. Since $p \parallel q$ and n is a transversal.

$$\therefore 4c = 120^\circ \quad [\because \text{Corresponding angles}]$$

$$\Rightarrow c = \frac{120^\circ}{4} = 30^\circ$$

Since $n \parallel m$ and p is a transversal.

$$\therefore 3b = 4c \quad [\because \text{Corresponding angles}]$$

$$\Rightarrow 3b = 4 \times 30^\circ$$

$$\Rightarrow b = \frac{120^\circ}{3} = 40^\circ$$

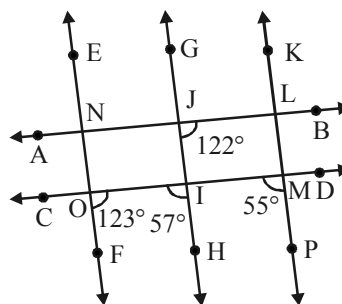
Since $n \parallel l$ and q is a transversal.

$$\therefore 6a = 120^\circ \quad [\because \text{Corresponding angles}]$$

$$\Rightarrow a = \frac{120^\circ}{6} = 20^\circ$$

11. Consider two lines EF and GH and a transversal CD.

Sum of interior angles on the same side of transversal = $123^\circ + 57^\circ = 180^\circ$



Hence, $EF \parallel GH$

Now, consider line GH and KP and a transversal CD.

Since $57^\circ \neq 55^\circ$ [$\because$ Corresponding angles]

$\therefore$ GH is not parallel to KP.

$$\angle NJI + \angle LJI = 180^\circ \quad [\because \text{Linear pair}]$$

$$\Rightarrow \angle NJI + 122^\circ = 180^\circ$$

$$\Rightarrow \angle NJI = 180^\circ - 122^\circ = 58^\circ$$

$$\text{But, } \angle HIO = 57^\circ \quad [\text{Given}]$$

$$\therefore \angle NJI \neq \angle HIO \quad [\text{corresponding angles}]$$

$\Rightarrow$ AB is not parallel to CD.

12. (i) $\angle MLO + \angle ALO = 180^\circ$ [Linear pair]

$$\Rightarrow \angle MLO + 130^\circ = 180^\circ$$

$$\Rightarrow \angle MLO = 180^\circ - 130^\circ = 50^\circ$$

Now $\angle MLO = \angle NOF$ [Corresponding $\angle$]

$$\therefore AB \parallel CD.$$

- (ii) $\angle MLO = \angle GML$ [Alternate interior $\angle$]

$$\therefore EF \parallel GH$$

HOTS Questions :

1. $\angle 1 = 45^\circ$ and $\angle 1 + \angle 2 = 90^\circ$
 [complementary $\angle$]

$$\Rightarrow 45^\circ + \angle 2 = 90^\circ$$

$$\Rightarrow \angle 2 = 90^\circ - 45^\circ = 45^\circ$$

Now, $\angle 2 + \angle 3 = 180^\circ$ [Supplementary $\angle$]

$$\Rightarrow 45^\circ + \angle 3 = 180^\circ$$

$$\Rightarrow \angle 3 = 180^\circ - 45^\circ = 135^\circ$$

2. Since $AE \parallel BD$ and CG is a transversal.

$$\therefore \angle HCD + \angle CHE = 180^\circ$$

$$\Rightarrow \angle HCD + 120^\circ = 180^\circ$$

$$\Rightarrow \angle HCD = 180^\circ - 120^\circ = 60^\circ$$

Since $AB \parallel CG$ and BD is a transversal.

$$\therefore \angle ABC = \angle HCD = 60^\circ$$

Since $CG \parallel DF$ and BD is a transversal.

$$\therefore \angle HCD + \angle CDE = 180^\circ$$

$$\Rightarrow 60^\circ + \angle CDE = 180^\circ$$

$$\Rightarrow \angle CDE = 180^\circ - 60^\circ = 120^\circ$$

3. Since $AB \parallel CD$ and BC is a transversal.

$$\therefore \angle BCD = \angle x \quad \dots\dots (i)$$

Since $BC \parallel DE$ and CD is a transversal.

$$\therefore \angle BCD = \angle y \quad \dots\dots (ii)$$

From (i) and (ii), we have

$$\angle x = \angle y$$

4. (i) Since $PQ \parallel UT$ and PT is a transversal.

$$\therefore \angle QPT = \angle c$$

$$\Rightarrow \angle a + \angle b = 57^\circ [\because \angle c = 57^\circ]$$

$$\Rightarrow \frac{\angle c}{3} + \angle b = 57^\circ$$

$$\Rightarrow \angle b = 57^\circ - \frac{57^\circ}{3} = 57^\circ - 19^\circ = 38^\circ$$

Now, Since $PQ \parallel RS$ and PR is a transversal.

$$\therefore \angle b + \angle d = 180^\circ$$

$$\Rightarrow \angle d = 180^\circ - \angle b$$

$$\Rightarrow 180^\circ - 38^\circ = 142^\circ$$

- (ii) Since $PQ \parallel UT$ and PT is a transversal.

$$\therefore \angle a + \angle b = \angle c$$

$$\Rightarrow \frac{2}{5} \angle c + \angle b = 75^\circ$$

$$\Rightarrow \angle b = 75^\circ - \frac{2}{5} \times 75^\circ$$

$$= 75^\circ - 30^\circ = 45^\circ$$

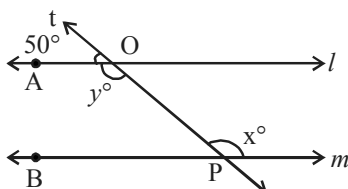
3 EXERCISE

Single Option Correct :

1. (a) Let the transversal t cuts the line l at O and the line m at P .

$$y^\circ + 50^\circ = 180^\circ$$

$$\Rightarrow y^\circ = 130^\circ$$



$\therefore$ Alternate interior angles are equal.

$$\Rightarrow x^\circ = 130^\circ$$

2. (b) $\angle 1 + \angle 2 = 180^\circ$ [linear pair]

$$\angle 1 + 2\angle 1 = 180^\circ$$

$$\angle 1 = \frac{180^\circ}{3} = 60^\circ$$

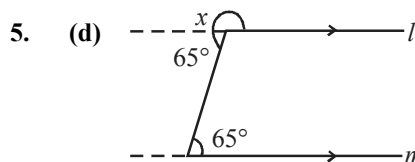
$$\angle 1 = \angle 3 = 60^\circ \text{ (Vertically opposite angles)}$$

$$\angle 3 = \angle 7 \text{ (Corresponding angles)}$$

$$\angle 1 = \angle 7 = 60^\circ$$

3. (b) $75^\circ + 15^\circ = 90^\circ$ and $\frac{1}{5} \times 75^\circ = 15^\circ$

4. (d)



$$x = 180^\circ + 65^\circ = 245^\circ$$

6. (b) $BA \parallel CE$ and AC is transversal

$$\angle A = \angle ACE = 60^\circ$$

$$\text{Now, } \angle ACB + \angle ACE + \angle ECD = 180^\circ$$

$$\angle ACB + 60^\circ + 65^\circ = 180^\circ$$

$$\angle ACB = 180^\circ - 125^\circ = 55^\circ$$

7. (c) Let the angles be $7x$ and $8x$.

$$\therefore 7x + 8x = 90^\circ$$

$$15x = 90^\circ \Rightarrow x = 6^\circ$$

$$7x = 7 \times 6^\circ = 42^\circ, 8x = 8 \times 6^\circ = 48^\circ$$

8. (b) Complementary of $\angle A = 90^\circ - \angle A$

$$= 90^\circ - (4x + 2)^\circ = 90^\circ - 4x^\circ - 2^\circ$$

$$= (88 - 4x)^\circ$$

9. (a) $AB \parallel CD$ and AC is transversal

$$\angle a = \angle ACD \text{ (alternate interior angles)}$$

$$AC \parallel DE \text{ and } CD \text{ is transversal}$$

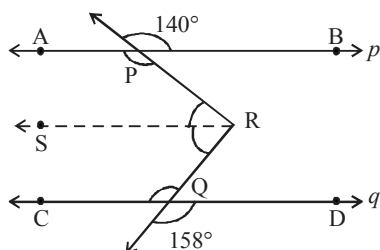
$$\angle b = \angle ACD$$

$$\angle ACD = 360^\circ - 325^\circ = 35^\circ$$

$$\therefore \angle a + \angle b = \angle ACD + \angle ACD = 2\angle ACD$$

$$\angle a + \angle b = 2(35^\circ) = 70^\circ$$

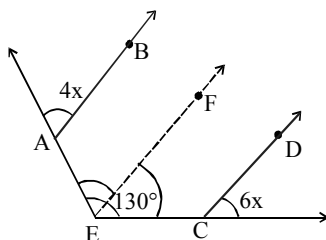
10. (c) Draw a ray RS parallel to p' and q'



$$\begin{aligned}\angle APR &= 140^\circ, \angle CQR = 158^\circ \\ &\text{(Vertically opposite angles)} \\ \angle APR + \angle PRS &= 180^\circ \\ \text{and } \angle CQR + \angle QRS &= 180^\circ \\ \text{adding, } \angle APR + \angle PRS + \angle CQR \\ &\quad + \angle QRS = 360^\circ \\ 140^\circ + \angle PRS + 158^\circ + \angle QRS &= 360^\circ \\ \angle PRS + \angle QRS &= 360^\circ - 298^\circ = 62^\circ \\ \angle x &= 62^\circ\end{aligned}$$

11. (b) $\angle CAB + \angle DBA = 180^\circ$
 $\angle DBA = 180^\circ - \angle CAB$
 $= 180^\circ - 130^\circ = 50^\circ$
 Now, $\angle DBA + \angle ABF + \angle FBG = 180^\circ$
 $\Rightarrow 50^\circ + \angle ABF + 60^\circ = 180^\circ$
 $\Rightarrow \angle ABF = 180^\circ - 110^\circ = 70^\circ$
 Now, $AE \parallel BF \Rightarrow \angle EAB + \angle ABF = 180^\circ$
 $\Rightarrow x + 70^\circ = 180^\circ$
 $\Rightarrow x = 110^\circ$
12. (b) $\angle 2 = \angle 4$ (opposite angles)
 $\angle 4 = \angle 6$ (alternate interior angles)
 $\angle 6 = \angle 8$ (opposite angles)
 $\Rightarrow \angle 2 = \angle 8$
 $3x - 10 = 5x - 30^\circ$
 $2x = 20^\circ \Rightarrow x = 10^\circ$
 $\angle 2 = 3 \times 10^\circ - 10^\circ = 20^\circ$
 $\angle 2 = \angle 8 = 20^\circ$

13. (b)



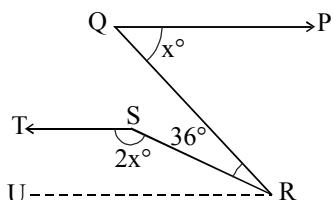
Draw a line $EF \parallel AB$, EF is also parallel

to CD {as $AB \parallel CD$ }
 $4x = \angle AEF$ (corresponding angles)
 $6x = \angle FEC$ (corresponding angles)
 $\therefore 10x = \angle AEF + \angle FEC = 130^\circ$

$$x = \frac{130^\circ}{10} = 13^\circ$$

14. (b) $\angle EAB = 5x - 25^\circ$ (opposite angles)
 $\angle EAB = \angle ABD$ (alternate interior angles)
 $\angle ABD = 2x + 35^\circ$ (opposite angles)
 $\therefore 2x + 35 = 5x - 25$
 $\Rightarrow x = 20^\circ$
 $\angle ABC = 180^\circ - (2x + 35^\circ) = 145^\circ - 2x$
 $\angle ABC = 145^\circ - 2(20^\circ) = 105^\circ$

15. (c) Draw a line UR parallel to QP



$$\begin{aligned}\angle PQR &= \angle QRU = x^\circ \\ \angle QRS + \angle SRU &= x^\circ \\ \angle SRU &= x^\circ - 36^\circ\end{aligned}$$

$$\begin{aligned}\text{Now, } \angle TSR + \angle SRU &= 180^\circ \\ \Rightarrow 2x^\circ + x^\circ - 36^\circ &= 180^\circ \\ \Rightarrow 3x &= 216^\circ \Rightarrow x = \angle PQR = 72^\circ\end{aligned}$$

More Than One Option Correct :

1. (a, b, c) 2. (a, b, d) 3. (b, c)
 4. (a, d) 5. (b, c) 6. (b, c)

Assertion & Reason :

1. (a)
 2. (d) Supplement of $\frac{1}{4}$ of 240°

$$= 180^\circ - \left(\frac{1}{4} \text{ of } 240^\circ \right) = 120^\circ$$

Assertion: incorrect, Reason: correct

Passage Based Questions :

1. (b) $\angle POR = x + y = 60^\circ + 25^\circ = 85^\circ$
 2. (a) $\angle SOQ = y + z = 80^\circ$

$$\Rightarrow z = 80^\circ - y = 80^\circ - 25^\circ = 55^\circ$$

$$\Rightarrow \angle \text{SOR} = z = 55^\circ$$

3. (d) $x = \frac{1}{3}$ of right angle $= \frac{1}{3} \times 90^\circ = 30^\circ$

$$y = \frac{2}{3} \text{ of right angle} = \frac{2}{3} \times 90^\circ = 60^\circ$$

$$z = \frac{1}{2} \text{ of right angle} = \frac{1}{2} \times 90^\circ = 45^\circ$$

$$\begin{aligned} \angle \text{POS} &= x + y + z \\ &= 30^\circ + 60^\circ + 45^\circ = 135^\circ \end{aligned}$$

4. (a) Let these angles be $5x$ and $13x$
 $\therefore 5x + 13x = 180^\circ \Rightarrow 18x = 180^\circ$

$$\Rightarrow x = \frac{180^\circ}{18} = 10^\circ$$

$$\text{The angles are } 5x = 5 \times 10 = 50^\circ$$

$$\text{and } 13x = 13 \times 10 = 130^\circ$$

5. (c) Let the smaller angle be x

$$\text{Larger angle} = (3x - 40^\circ)$$

$$\therefore x + (3x - 40^\circ) = 180^\circ$$

$$\Rightarrow 4x - 40^\circ = 180^\circ$$

$$\Rightarrow 4x = 220^\circ \Rightarrow x = 55^\circ$$

$$\begin{aligned} \text{Larger angle : } (3x - 40^\circ) &= 3 \times 55^\circ - 40^\circ \\ &= 165^\circ - 40^\circ = 125^\circ \end{aligned}$$

6. (b) Let the smaller angle be $= x$

$$\therefore 2x + x = 180^\circ$$

$$3x = 180^\circ \Rightarrow x = 60^\circ$$

$$\text{Smaller angle} = 60^\circ$$

$$\text{Larger angle} = 120^\circ$$

Integer Type Questions :

1. (2)

2. (1) Complementary of $89^\circ = 90^\circ - 89^\circ = 1^\circ$

3. (9) The required angle $= 180^\circ - 171^\circ = 9^\circ$

4. (3) $\angle \text{ABC} = \angle \text{DBF} = 140^\circ$

$$\therefore \angle \text{EBF} = 140^\circ - 47^\circ = 93^\circ$$

$$\Rightarrow 31k^\circ = 93^\circ = 31 \times 3$$

$$\Rightarrow k = 3$$

5. (5) $100^\circ + 56^\circ + 74^\circ + x = 360^\circ$

$$230^\circ + x = 360^\circ$$

$$x = 360^\circ - 230^\circ = 130^\circ = 26 \times 5$$

$$\Rightarrow k = 5$$

Multiple Matching Question :

1. (A) $\rightarrow$ (r, t); (B) $\rightarrow$ (s); (C) $\rightarrow$ (p, s); (D) $\rightarrow$ (q)

Chapter 6

The Triangles and Its Properties

TRIANGLE

A closed figure formed by joining three non-collinear points is called a *triangle*.

Triangle PQR has

- (i) Three vertices P, Q and R.
- (ii) Three sides namely, PQ, QR, and PR
- (iii) Three angles namely, $\angle P$, $\angle Q$ and $\angle R$

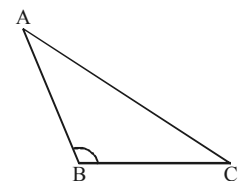
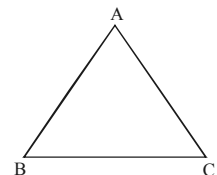
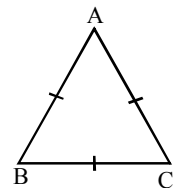
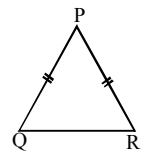
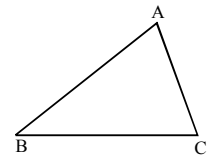
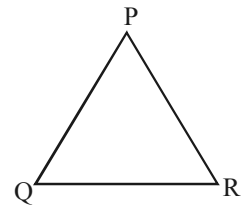
Types of Triangle Based on the Length of its Sides

Triangles can be classified into three categories

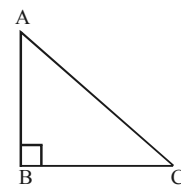
- (a) **Scalene triangle:** A triangle with three unequal sides is called a *scalene triangle*.
In the given figure, $AB \neq BC \neq CA$
- (b) **Isosceles triangle:** A triangle with two equal sides is called an *isosceles triangle*.
In the given figure, $PQ = PR$
- (c) **Equilateral triangle:** A triangle with all its three sides equal is called an *equilateral triangle*.
In the given figure, $AB = BC = CA$

Types of Triangles Based on the Measure of their Angles

- (i) **Acute angled triangle:** A triangle with all three acute angles is called an *acute angled triangle*.
In the given figure, $\angle A$, $\angle B$ and $\angle C$ are all less than 90°
- (ii) **Obtuse angled triangle:** A triangle with one obtuse angle and two acute angles is called an *obtuse angled triangle*. In triangle ABC, $\angle B$ is obtuse angle, $\angle A$ and $\angle C$ are acute angles.



- (iii) **Right angled triangle:** A triangle with one right angle and two acute angles are called a *right angled triangle*.
 $\angle B$ is right angle. $\angle A$ and $\angle C$ are acute angles.

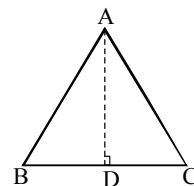


ALTITUDE OF A TRIANGLE

An altitude of a triangle is the perpendicular drawn from a vertex to the opposite side.

In $\triangle ABC$, AD is the altitude from A to BC. i.e, $AD \perp BC$

Altitude gives the height of the triangle.



Altitudes in Different Types of Triangles

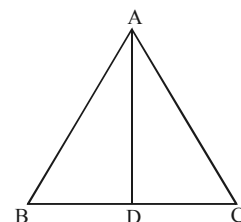
- (i) In scalene triangle, all three altitudes are of different length.
- (ii) In Isosceles triangle, the altitudes to the equal sides are equal in length.
- (iii) In equilateral triangle, all three altitudes are of equal length.

NOTE: There are three altitudes in any triangle and all of them meet at a point called *orthocentre* of the triangle.

MEDIANS OF A TRIANGLE

A median of a triangle is a line segment joining a vertex to the mid-point of the side opposite to that vertex.

In $\triangle ABC$, D is the mid-point of BC,
 so AD is median.



Medians in Different Types of Triangles

- (i) In scalene triangle, all three medians are of different lengths
- (ii) In Isosceles triangle, the medians to the equal sides are equal in length.
- (iii) In equilateral triangle, all three medians are of equal length.

NOTE: There are three medians in any triangle and all of them meet at a point called *centroid* of the triangle.

ANGLE SUM PROPERTY OF A TRIANGLE

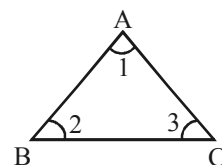
1. **Angle Sum Property of a Triangle:** In any triangle, the sum of angles of a triangle is always 180° .

In $\triangle ABC$,

$$\angle 1 + \angle 2 + \angle 3 = 180^\circ.$$

Facts:

1. A triangle cannot have more than one right angle.
2. A triangle cannot have more than one obtuse angle.
3. In a right angle triangle, the sum of acute angle is 90° .



The Triangles and Its Properties



Justification

To prove : Sum of all angles of $\triangle ABC$ is equal to 180°

That is, $\angle 1 + \angle 2 + \angle 3 = 180^\circ$

Construction : Construct a line $PQ \parallel BC$ and passing through A

Proof : PAQ is a straight line

$\therefore \angle PAQ = 180^\circ$ (Straight angle)

$\Rightarrow \angle 5 + \angle 1 + \angle 4 = 180^\circ \dots(i)$

But, $\angle 5 = \angle 2$
and $\angle 4 = \angle 3$ } $\dots(ii)$

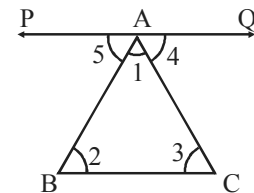
$PQ \parallel BC$ and they are pairs of alternate interior angles

$\therefore$ From (i) and (ii)

$\angle 2 + \angle 1 + \angle 3 = 180^\circ$

That is, $\angle 1 + \angle 2 + \angle 3 = 180^\circ$

$\therefore$ Sum of the angles of $\triangle ABC$ is 180° .



EXTERIOR ANGLE OF A TRIANGLE AND ITS PROPERTY

Exterior Angle

When any side of a triangle is extended, the angle so formed outside the triangle by the extended and adjacent side of the triangle is called an *exterior angle* of that triangle.

Three exterior angles for $\triangle ABC$ are $\angle QAB$, $\angle RBC$ and $\angle PCA$ at the vertices A, B and C, respectively.

Exterior Angle Property

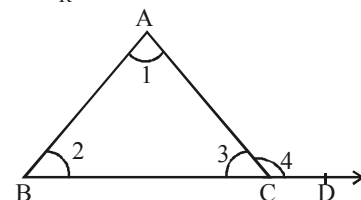
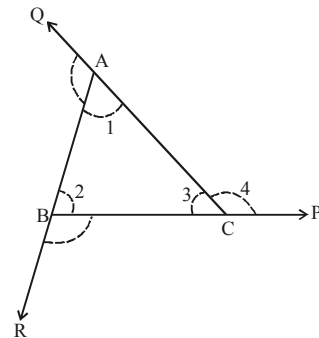
Exterior Angle Property of a Triangle: If any side of a triangle is produced then the exterior angle so formed is equal to the sum of the interior opposite angles.

$\angle ACD = \angle 1 + \angle 2$

or $\angle 4 = \angle 1 + \angle 2$

Here, $\angle 4$ is the exterior angle

and, $\angle 1$ and $\angle 2$ are interior opposite angles



Justification

To prove : In $\triangle ABC$, $\angle 4 = \angle 1 + \angle 2$

Proof : BCD is a straight line

$\therefore \angle BCD = 180^\circ$ (Straight Angle)

$\therefore \angle 3 + \angle 4 = 180^\circ$ (Linear pair)(i)

Also, $\angle 1 + \angle 2 + \angle 3 = 180^\circ$ (Angle Sum Property of a triangle).....(ii)

$\therefore$ From (i) and (ii)

$\angle 3 + \angle 4 = \angle 1 + \angle 2 + \angle 3$

$\Rightarrow \cancel{\angle 3} + \angle 4 = \angle 1 + \angle 2 + \cancel{\angle 3}$

$\Rightarrow \angle 4 = \angle 1 + \angle 2$

Hence proved

TRIANGULAR INEQUALITY

Triangular Inequality: (i) The sum of any two sides of triangle is greater than the third side.

In $\triangle ABC$, according to Triangular Inequality

$$x + y > z$$

$$\text{or } y + z > x$$

$$\text{or } z + x > y$$

(ii) The difference of any two sides of a triangle is less than the third side.

$$\text{i.e., } x - y < z$$

$$y - z < x$$

$$\text{and } z - x < y$$

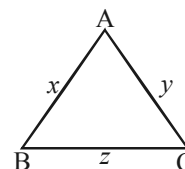


ILLUSTRATION 1 :

In each of the following, there are three positive numbers. State if these numbers could possibly represent the measure of the lengths of the sides of a triangle.

- (i) 4, 5, 3 (ii) 2.5, 1.5, 4

SOLUTION:

These numbers would possibly be the lengths of the sides of a triangle only if the sum of any two of them is greater than the third.

$$(i) \quad 4 + 5 = 9 > 3, \quad 5 + 3 = 8 > 4, \quad 4 + 3 = 7 > 5$$

$\therefore$ Triangle is possible.

$$(ii) \quad 2.5 + 1.5 = 4 \text{ (the third number), i.e., not greater than 4.}$$

$\therefore$ Triangle is not possible.

TWO SPECIAL TRIANGLES : EQUILATERAL AND ISOSCELES

Equilateral Triangle : A triangle in which all three sides are of equal length is called an *equilateral triangle*.

In an equilateral triangle :

- (i) all sides have the same length;
- (ii) each angle is of measure 60° .

Isosceles Triangle : A triangle in which two sides are of equal length is called an *isosceles triangle*.

In an isosceles triangle :

- (i) Two sides have the same length;
- (ii) Angles opposite equal sides are equal.

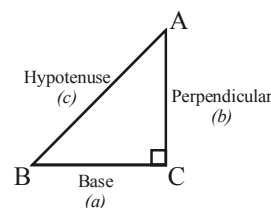
PYTHAGORAS THEOREM

In a right triangle, the square of the hypotenuse equals the sum of the squares of its remaining two sides.

Thus, if $\triangle ABC$ is a right triangle right-angled at C, i.e., AB is the hypotenuse and AC and BC are the other two sides of the right triangle, then $(AB)^2 = (BC)^2 + (CA)^2$

$$\text{i.e., } (\text{Hypotenuse})^2 = (\text{Base})^2 + (\text{Perpendicular})^2$$

$$\text{or } c^2 = a^2 + b^2, \text{ where } a = BC, b = CA \text{ and } c = AB$$



NOTE: Hypotenuse is the side which is opposite to right angle. It is also the longest side in the right triangle.

ILLUSTRATION 2 :

The hypotenuse of a right triangle is 17 cm long. If one of the remaining two sides is 8 cm in length, find the length of other side.

SOLUTION : Let $\triangle ABC$ be right angled at C

Let $AB = 17$ cm $BC = 8$ cm

Then, by Pythagoras' theorem

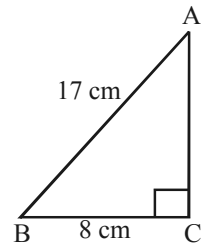
$$AB^2 = BC^2 + AC^2$$

$$AC^2 = AB^2 - BC^2$$

$$= (17)^2 - (8)^2 = 289 - 64 = 225$$

$$\therefore AC \Rightarrow \sqrt{225} = 15 \text{ cm.}$$

Hence the length of the other side is 15 cm.

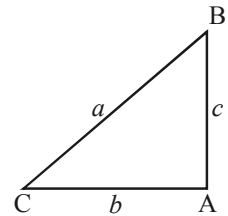


Converse of Pythagoras Theorem

The converse of Pythagoras theorem is also true.

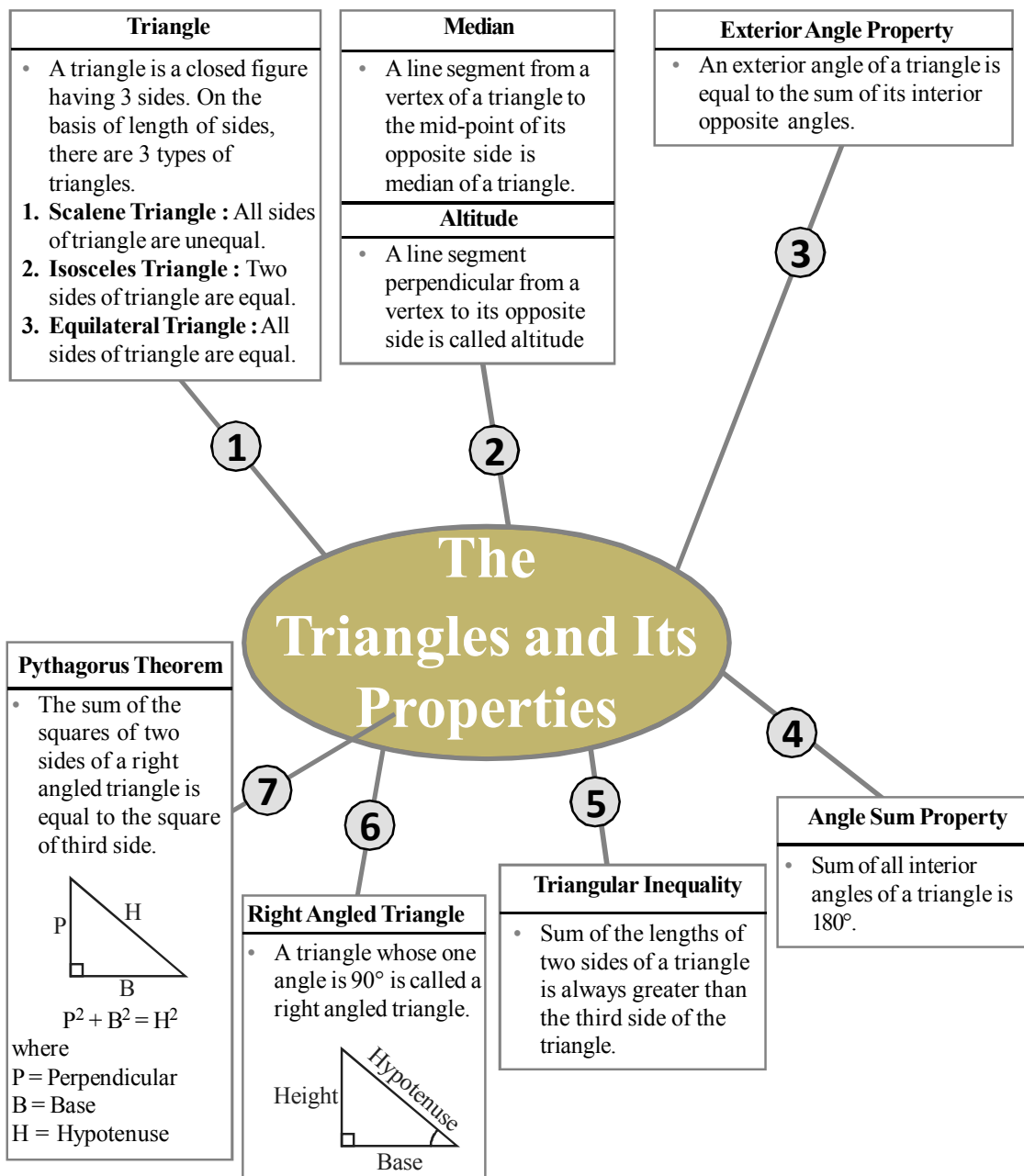
Theorem : If $\triangle ABC$ is right-angled at A, then $a^2 = b^2 + c^2$.

Converse : In $\triangle ABC$, if $a^2 = b^2 + c^2$, then angle A is a right angle.



NOTE: If the Pythagoras property holds, the triangle must be right angled.

CONCEPT MAP



MISCELLANEOUS

SOLVED EXAMPLES

1. Find the angles of a triangle which are in the ratio 3 : 4 : 5.

Sol. Let the measures of the given angles of the triangle be $(3x)^\circ$, $(4x)^\circ$ and $(5x)^\circ$ respectively.

$$\therefore 3x + 4x + 5x = 180^\circ \quad [\because \text{sum of the } \angle \text{ of a triangle is } 180^\circ]$$

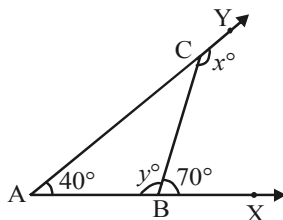
$$\Rightarrow 12x = 180^\circ$$

$$\Rightarrow x = \frac{180^\circ}{12} = 15^\circ.$$

So, the angles measure $(3 \times 15)^\circ$, $(4 \times 15)^\circ$ and $(5 \times 15)^\circ$, i.e., 45° , 60° and 75° .

Hence, the angles of the triangle are 45° , 60° and 75° .

2. In figure, the measures of some of the angles are indicated. Find the values of x° and y° .



Sol. $\angle CBX + \angle CBA = 180^\circ$ (linear pair.)

$$\Rightarrow 70^\circ + y^\circ = 180^\circ$$

$$\Rightarrow y^\circ = 180^\circ - 70^\circ = 110^\circ$$

Again, $\angle BCY$ is exterior of $\triangle ABC$ at C

$$\therefore \angle BCY = \angle CAB + \angle CBA$$

$$\Rightarrow x^\circ = 40^\circ + y^\circ$$

$$\Rightarrow x^\circ = 40^\circ + 110^\circ \quad [\because y^\circ = 110^\circ]$$

$$\Rightarrow x^\circ = 150^\circ$$

Hence, we have $x^\circ = 150^\circ$ and $y^\circ = 110^\circ$.

3. Two sides of a triangle are 6 cm and 8 cm long. What can be the length of its third side?

Sol. Let the length of the third side be x cm.

We know that the sum of any two sides of a triangle is greater than the third.

$$\therefore (6 + 8) > x \Rightarrow x < 14.$$

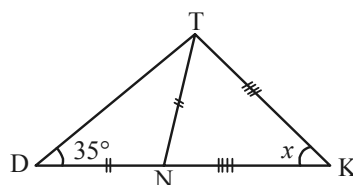
Also, we know that the difference of any two sides of a triangle is less than the third side.

$$\therefore (8 - 6) < x \Rightarrow x > 2.$$

Thus, $2 < x < 14$.

Hence, the length of the third side must be greater than 2 cm and smaller than 14 cm.

4. In the following figure, find the measure of the angle marked x .



Sol. It is given that in $\triangle TND$, $\overline{TN} = \overline{DN}$ and $\angle D = 35^\circ$

$$\therefore \angle TND = 35^\circ \quad [\because \text{base angles are equal}]$$

So, the $\triangle TND$ is an isosceles triangle.

$$\text{Then, } \angle TND = 180^\circ - (\angle TND + \angle D)$$

$$= 180^\circ - (35^\circ + 35^\circ)$$

$$= 180^\circ - 70^\circ$$

$$\therefore \angle TND = 110^\circ$$

$$\text{Also, } \angle TND + \angle KNT = 180^\circ$$

$$\text{or } \angle KNT = 180^\circ - \angle TND$$

$$= (180^\circ - 110^\circ) = 70^\circ$$

In $\triangle TNK$, since $TK = NK$ and $\angle KNT = 70^\circ$

$$\therefore \angle KTN = 70^\circ \quad [\because \text{in } \triangle TNK \text{ base angles are equal}]$$

Now in $\triangle TNK$,

$$\angle KNT + \angle KTN + x = 180^\circ$$

$$\text{or } 70^\circ + 70^\circ + x = 180^\circ$$

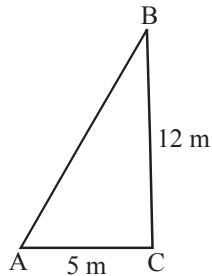
$$\text{i.e., } x = 40^\circ$$

Hence, the value of x , i.e., $\angle K = 40^\circ$.

5. A ladder is placed in such a way that its foot is at a distance of 5 m from a wall and its top reaches a window 12 m above the ground. Determine the length of the ladder.

Sol. Let AB be the ladder and B be the window.

Then, $BC = 12$ m and $AC = 5$ m.



Since $\triangle ABC$ is a right-angled triangle at C.

$$\therefore AB^2 = AC^2 + BC^2$$

$$\Rightarrow AB^2 = 5^2 + 12^2 = 169$$

$$\Rightarrow AB = \sqrt{169} \text{ m} = 13 \text{ m}$$

Hence, the length of the ladder is 13 m.

6. The sides of certain triangles are given below. Determine which of them are right triangles:

(i) $a = 6 \text{ cm}$, $b = 8 \text{ cm}$ and $c = 10 \text{ cm}$

(ii) $a = 5 \text{ cm}$, $b = 8 \text{ cm}$ and $c = 11 \text{ cm}$.

- Sol. (i) Here the larger side is $c = 10 \text{ cm}$.

We have:

$$a^2 + b^2 = 6^2 + 8^2 = 36 + 64 = 100 = c^2.$$

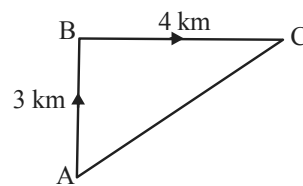
So, the triangle with the given sides is a right triangle.

- (ii) Here, the larger side is $c = 11 \text{ cm}$.

Clearly, $a^2 + b^2 = 25 + 64 = 89 \neq c^2$. So, the triangle with the given sides is not a right triangle.

7. A man drives 3 km towards north and then 4 km towards east. How far is he away from his initial position?

Sol. Let the starting point be A and final point is C as shown in the figure.



In $\triangle ABC$, $\angle B = 90^\circ$

$$\begin{aligned} \therefore (AC)^2 &= (AB)^2 + (BC)^2 \\ &= (3)^2 + (4)^2 \\ &= 9 + 16 = 25 = 5^2 \end{aligned}$$

$$\Rightarrow AC = 5 \text{ km}$$

Hence, man is 5 km away from his initial position.

1

EXERCISE

Fill in the Blanks :

DIRECTIONS : Complete the following statements with an appropriate word / term to be filled in the blank space(s).

- The line segment joining a vertex of a triangle to the mid-point of its opposite side is called its _____.
- In an isosceles triangle, two angles are always _____.
- The longest side of a right angled triangle is called its _____.
- The _____ triangle always has altitude outside itself.
- Measures of each of the angles of an equilateral triangle is _____.
- The sum of an exterior angle of a triangle and its adjacent angle is always _____.
- Line segment drawn from a vertex of a triangle perpendicular to its opposite side is _____.
- A triangle with all its sides unequal is known as _____ triangle.
- In an isosceles triangle, angles opposite to equal sides are _____.
- The sum of the lengths of any two sides of a triangle is always _____ than the length of the third side.

True / False :

DIRECTIONS : Read the following statements and write your answer as true or false.

- Sum of two sides of a triangle is greater than or equal to the third side.
- It is possible to have a right-angled equilateral triangle.
- Sum of any two angles of a triangle is always greater than the third angle.
- It is possible to have a triangle in which each angle is less than 60° .
- The difference between the lengths of any two sides of a triangle is smaller than the length of third side.

- The sum of the measures of three angles of a triangle is greater than 180° .
- If two angles of a triangle are equal, the third angle is also equal to each of the other two angles.
- In a triangle, sum of squares of two sides is equal to the square of the third side.
- It is possible to have a triangle in which two of the angles are obtuse.
- It is possible to have a triangle in which two of the angles are right angles.

Match the Columns :

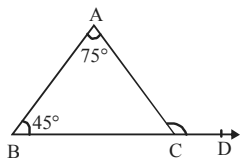
DIRECTIONS : Each question contains statements given in two columns which have to be matched. Statements (A, B, C, D) in column-I have to be matched with statements (p, q, r, s) in column-II.

1.	Column-I	Column-II
(A)	In a ΔPQR , $\angle P = 70^\circ$, $\angle Q = 30^\circ$ The measure of $\angle R$ is	(p) 55°
(B)	In a right angled triangle, one of the acute angle measures 35° . The measure of other acute angle is	(q) 70°
(C)	In an isosceles triangle, each of the equal angle measures 55° . The measure of third angle is	(r) 90°
(D)	In a ΔABC , all three angles are in the ratio $1 : 2 : 3$. The measure of largest angle is	(s) 80°

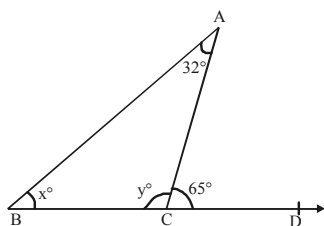
Very Short Answer Questions :

DIRECTIONS : Give answer in one word or one sentence.

- Two angles of a triangle are equal and the third angle measures 70° . Find the measure of each of the unknown angles.
- In the figure given alongside, find the measure of $\angle ACD$.

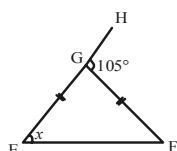
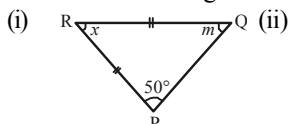


3. In the figure given alongside, find the values of x and y .

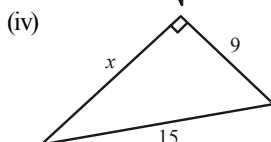
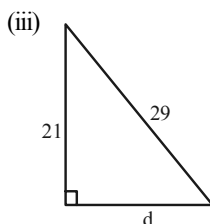
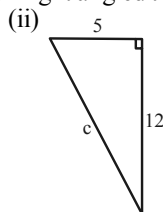
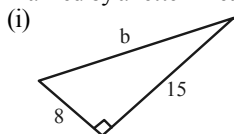


4. Is it possible to draw a triangle, the lengths of whose sides are given below?

- 5 cm, 7 cm, 9 cm
 - 2 cm, 10 cm, 15 cm
 - 3 cm, 4 cm, 5 cm
 - 2 cm, 5 cm, 7 cm
5. The sides of two triangles are given below. Determine which of them is the right triangle :
- $a = 5$ cm, $b = 9$ cm, $c = 11$ cm
 - $a = 15$ cm, $b = 20$ cm, $c = 25$ cm
6. If in $\triangle ABC$, $\angle A = 2x + 15^\circ$, $\angle B = x - 20^\circ$ and $\angle C = 3x - 55^\circ$, find the value of x .
7. Find the angles of the triangle which are in the ratio $2 : 2 : 5$
8. Find the lettered angle in each of the following:



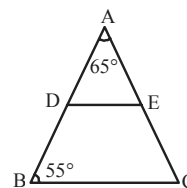
9. Two sides being given, calculate the third side marked by a letter in each right angled triangle.



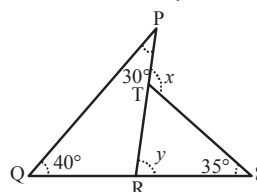
Short Answer Questions :

DIRECTIONS : Give answer in 2-3 sentences.

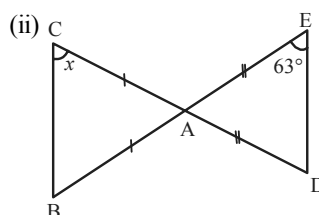
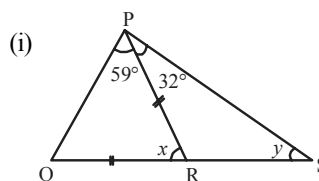
- One side of a triangle is produced and the exterior angle so formed is 120° . If the interior opposite angles be in the ratio $3 : 5$, find the measure of each angle of the triangle.
- In the given figure, $DE \parallel BC$. If $\angle A = 65^\circ$ and $\angle B = 55^\circ$, find



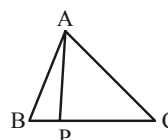
- $\angle ADE$
 - $\angle AED$
 - $\angle C$
3. Find the values of x and y in the figure given.



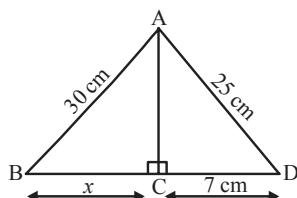
- In $\triangle ABC$, $\angle B$ is three-fourth of $\angle A$ and $\angle C$ is two-third of $\angle B$. Find all the angles.
- Find the value of x in each of the following :



- In the given figure, P is a point on the side BC of $\triangle ABC$. Prove that $(AB + BC + AC) > 2AP$.



7. In the given figure, find x .



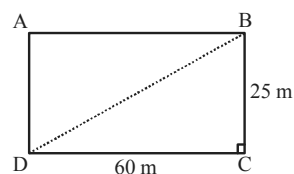
8. Find the perimeter of the rectangle whose length is 40 cm and a diagonal is 41 cm.

Long Answer Questions :

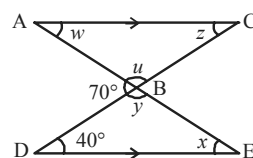
DIRECTIONS : Give answer in four to five sentences.

- In a $\triangle ABC$, if $3\angle A = 4\angle B = 6\angle C$, calculate $\angle A$, $\angle B$ and $\angle C$.
- A tree is broken at a height of 6 m from the ground and its top touches the ground at a distance of 8 m from the base of the tree. Find the original height of the tree.

3. A rectangular field is 60 m long and 25 m wide. How can a person take a shortcut to reach the opposite corner from one corner of the field? What is the length covered?



4. In the figure given below, find the angles x , y , z , u and w .



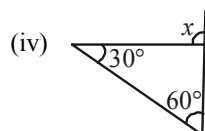
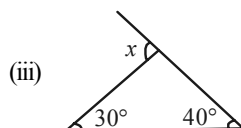
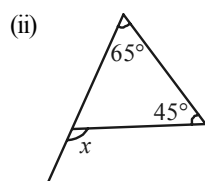
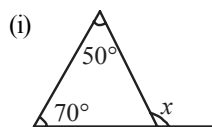
5. Two poles of heights 9 m and 14 m stand upright on a plane ground. If the distance between their feet is 12 m, find the distance between their tops.

2

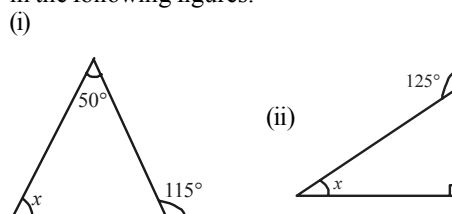
EXERCISE

Text-Book Exercise :

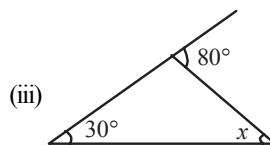
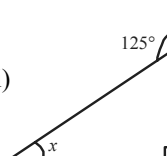
- Verify by drawing a diagram, if the median and altitude of an isosceles triangle can be same.
- Find the value of the unknown exterior angle x in the following diagrams:



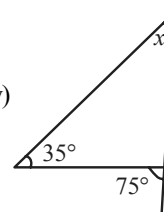
3. Find the value of the unknown interior angle x in the following figures:



(ii)

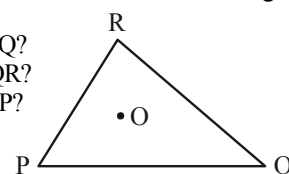


(iv)

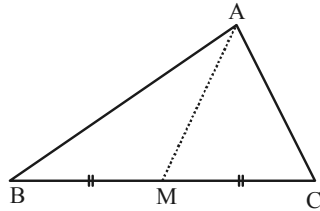


4. Take any point O in the interior of a triangle PQR . Is

- $OP + OQ > PQ$?
- $OQ + OR > QR$?
- $OR + OP > RP$?

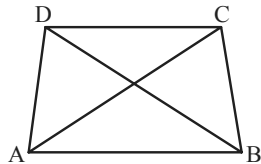


5. AM is median of a triangle ABC.



Is $AB + BC + CA > 2AM$?
(consider the sides of $\triangle ABM$ and $\triangle AMC$.)

6. ABCD is a quadrilateral.

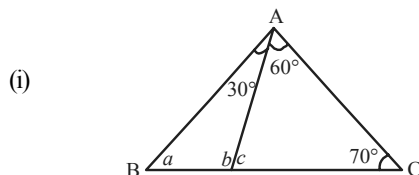


Is $AB + BC + CD + DA > AC + BD$?

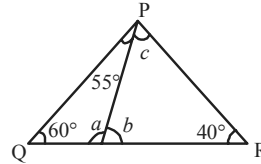
7. ABCD is a quadrilateral.
Is $AB + BC + CD + DA < 2(AC + BD)$?
8. The lengths of two sides of a triangle are 12 cm and 15 cm. Between what two measures should the length of the third side fall?
9. Which of the following can be the sides of a right triangle?
- 2.5 cm, 6.5 cm, 6 cm
 - 2 cm, 2 cm, 5 cm
 - 1.5 cm, 2 cm, 2.5 cm
- In the case of right-angled triangles, identify the right angles.
10. A tree is broken at a height of 5 m from the ground and its top touches the ground at a distance of 12 m from the base of the tree. Find the original height of the tree.
11. The diagonals of a rhombus measure 16 cm and 30 cm. Find its perimeter.

Exemplar Questions :

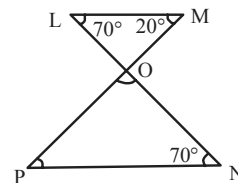
1. In figure (i) and (ii), find the values of a , b and c .



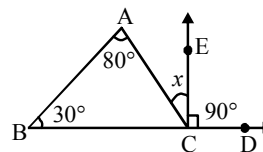
(ii)



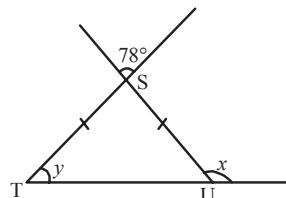
2. Jayanti takes shortest route to her home by walking diagonally across a rectangular park. The park measures 60 metres $\times$ 80 metres. How much shorter is the route across the park than the route around its edges?
3. In figure find the measures of $\angle PON$ and $\angle NPO$.



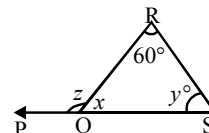
4. Each of the two equal angles of an isosceles triangle is four times the third angle. Find the angles of the triangle.
5. Find the value of x in the figure.



6. In figure, if $ST = SU$, then find the values of x and y .



7. In figure, if y is five times x , find the value of z .

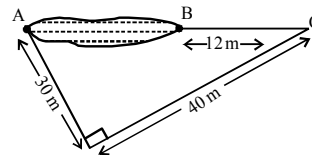


8. The lengths of two sides of an isosceles triangle are 9 cm and 20 cm. What is the perimeter of the triangle? Give reason.
9. The foot of a ladder is 6 m away from its wall and its top reaches a window 8 m above the ground. (i) Find the length of the ladder. (ii) If the ladder is shifted in such a way that its foot is 8 m away from the wall, to what height does its top reach?

HOTS Questions :

1. In a triangle ABC, the measure of $\angle A$ is 40° less than the measure of angle B and 50° less than that of angle C. Find the measure of $\angle A$.
2. If the sides of a triangle are produced in an order, show that the sum of the exterior angles so formed is 360° .
3. The angles of a triangle are arranged in descending order of their magnitudes. If the difference between two consecutive angles is 10° , find the three angles.

4. In $\triangle DEF$, $\angle D = 60^\circ$, $\angle E = 70^\circ$ and the bisectors of $\angle E$ and $\angle F$ meet at O. Find
(i) $\angle F$ (ii) $\angle EOF$.
5. Points A and B are on the opposite edges of a pond as shown in figure. To find the distance between the two points, the surveyor makes a right-angled triangle as shown. Find the distance AB.



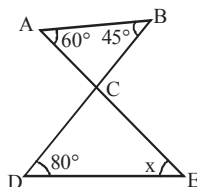
3

EXERCISE

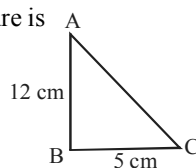
Single Option Correct :

DIRECTIONS : This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONLY ONE is correct.

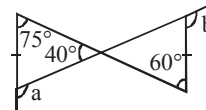
1. If one angle of a triangle is 75° and the other angles are in the ratio 2 : 3. The angles are
(a) $42^\circ, 63^\circ$ (b) $52^\circ, 53^\circ$
(c) $45^\circ, 60^\circ$ (d) $35^\circ, 70^\circ$
2. One of the acute angle of a right angled triangle is 62° . The measure of other acute angle is
(a) 50° (b) 28° (c) 36° (d) 48°
3. Find the value of x in the given figure.
(a) 30° (b) 25° (c) 35° (d) 45°



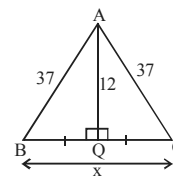
4. Length of AC in given figure is
(a) 12.5 cm
(b) 14 cm
(c) 15 cm
(d) 13 cm



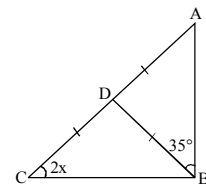
5. In the given figure, $(a + b)$ equals



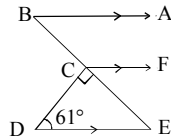
- (a) 235° (b) 215° (c) 195° (d) 225°
6. If one angle of a triangle is equal to the sum of the other two angles, the triangle must be:
(a) scalene (b) right angled
(c) obtuse angles (d) acute angled
7. The value of x in the figure is
(a) 35 (b) 70 (c) 55 (d) 65
8. The perimeter of a rectangle whose one side measure 20 m and the diagonal 29 m is
(a) 82m (b) 78 cm (c) 98 m (d) 88 m
9. In the given triangle, the value of x is



- (a) 35 (b) 70 (c) 55 (d) 65
8. The perimeter of a rectangle whose one side measure 20 m and the diagonal 29 m is
(a) 82m (b) 78 cm (c) 98 m (d) 88 m
9. In the given triangle, the value of x is
(a) 55° (b) 110° (c) 70° (d) 27.5°

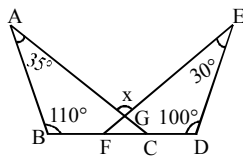


10. In the given figure $BA \parallel CF \parallel DE$. BCE is a straight line and $\angle DCE$ is a right angle. $\angle BCF$ is equal to



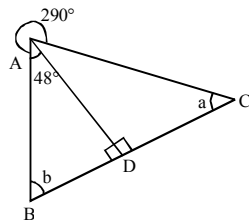
- (a) 150° (b) 115° (c) 151° (d) 105°

11. Value of x in the given diagram is



- (a) 70° (b) 95° (c) 110° (d) 120°

12. In the given figure how much angle a is greater than angle b .



- (a) 26° (b) 28°
(c) 30° (d) 24°

More Than One Option Correct :

DIRECTIONS : This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONE or MORE may be correct.

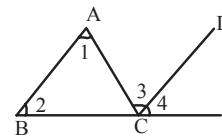
- Two sides of an Isosceles triangle is 5 cm and 6 cm then the third side is
(a) 5 cm (b) 6 cm
(c) 4 cm (d) 7 cm
- You are given lengths of sides of a figure. Which of these can form a triangle?
(a) 3, 7, 15 (b) 6, 4, 9
(c) 8, 7, 12 (d) 5, 5, 10
- In an isosceles triangle, one angle is 70° . The other two angles are
(a) 55° and 55° (b) 70° and 40°
(c) 50° and 60° (d) 80° and 30°
- Which of the following statements is correct?
(a) The sum of any two sides of a triangle is greater than the third side.

- (b) A triangle can have all its angles acute.
(c) A right-angled triangle cannot be equilateral.
(d) Difference of any two sides of a triangle is greater than the third side.

5. Which of the following triplets can be the angles of a triangle?

- (a) $67^\circ, 51^\circ, 62^\circ$ (b) $70^\circ, 83^\circ, 27^\circ$
(c) $90^\circ, 70^\circ, 20^\circ$ (d) $40^\circ, 132^\circ, 18^\circ$

6. In figure, if $AB \parallel CD$, then



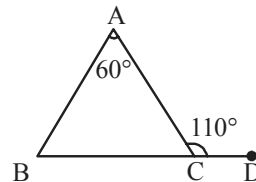
- (a) $\angle 2 = \angle 4$
(b) $\angle 1 = \angle 4$
(c) $\angle 4 = \angle 1 + \angle 2$
(d) $\angle 1 + \angle 2 = \angle 3 + \angle 4$

Assertion & Reason :

DIRECTIONS : Each of these questions contains an Assertion followed by Reason. Read them carefully and answer the question on the basis of following options. You have to select the one that best describes the two statements.

- (a) If both **Assertion** and **Reason** are correct and **Reason** is the correct explanation of **Assertion**.
(b) If both **Assertion** and **Reason** are correct but **Reason** is not the correct explanation of **Assertion**.
(c) If **Assertion** is correct but **Reason** is incorrect.
(d) If **Assertion** is incorrect but **Reason** is correct.

- Assertion :** The triangle is possible to form by three line segments of lengths 10 cm, 12 cm, 13 cm.
Reason : The sum of any two sides of a triangle is less than the third side.
- Assertion :** In $\triangle ABC$, measure of $\angle ABC$ is 50° .



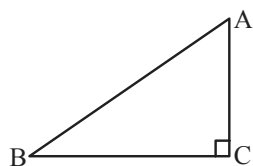
Reason : If a side of a triangle is produced then the exterior angle so formed is equal to the sum of the two interior opposite angles.

Passage Based Questions :

DIRECTIONS : Study the given passage(s) and answer the following questions.

PASSAGE-I

In a $\triangle ABC$, $\angle C = 90^\circ$, then $AB^2 = BC^2 + AC^2$

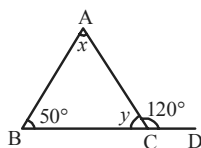


- The hypotenuse of a right angled triangle is 15 cm. If one of the remaining two sides is 9 cm, find the length of the other side.
(a) 10 cm (b) 12 cm
(c) 15 cm (d) 17 cm
- The lengths of the sides of two triangles are given below. Which of the following is a right angled triangle?
(i) $a = 15$ cm, $b = 20$ cm and $c = 25$ cm
(ii) $a = 7$ cm, $b = 6$ cm and $c = 13$ cm
(a) only (i) (b) only (ii)
(c) both (i) and (ii) (d) None of these
- A man goes 12 m due east and then 9 m due north. How far is he from his initial position?
(a) 15 m (b) 17 m
(c) 21 m (d) 23 m

PASSAGE-II

If a side of a triangle is produced then the exterior angle so formed is equal to the sum of the two interior opposite angles.

- In the given figure, find the values of x and y .

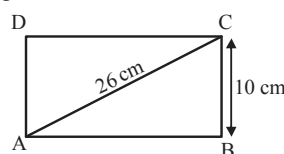


- One side of a triangle is produced and the exterior angle so formed is 140° . If the interior opposite angles are in the ratio. 2 : 5, find the measure of each angle of the triangle.
(a) $45^\circ, 75^\circ, 60^\circ$ (b) $40^\circ, 70^\circ, 70^\circ$
(c) $40^\circ, 100^\circ, 40^\circ$ (d) $40^\circ, 75^\circ, 65^\circ$
- If the sides of a triangles are produced in order, then the sum of the exterior angles so formed is
(a) 180° (b) 270°
(c) 90° (d) 360°

Integer Type Questions :

DIRECTIONS : Answer the following questions. The answer to each of the question is a single digit integer, ranging from 0 to 9.

- If the angles of a $\triangle ABC$ are in the ratio 11 : 13 : 12, then the unit digit of the smallest angle is
- What is the sum of digits of length of the rectangle ABCD?



- If an exterior angle of a triangle is 125° and one of the interior opposite angles is 55° , then other angle is $10 \times k$. Find k.
- In a right angle triangle OAB, $\angle A = 90^\circ$, OA = 24 m and AB = 10 m. Find the sum of digits of length OB.
- How many altitudes does a triangle have?
- In a right-angled triangle ABC, if angle B = 90° , BC = 3 cm and AC = 5 cm, then the length of side AB is

Multiple Matching Question :

DIRECTIONS : Following question has four statements (A, B, C and D) given in Column-I and six statements (p, q, r, s, t, u) in Column-II. Any given statement in Column-I can have correct matching with one or more statement(s) given in Column-II.

- | Column-I | Column-II |
|--|------------------|
| (A) In a $\triangle ABC$, if $\angle A = 30^\circ$, $\angle B = 50^\circ$, then $\angle C = ?$ | (p) 130° |
| (B) In a $\triangle ABC$, BC is extended to D. If $\angle ABC = 90^\circ$, $\angle BAC = 40^\circ$, then $\angle ACD = ?$ | (q) 90° |
| (C) In a right angled triangle, greatest angle must be of measure. | (r) 100° |
| (D) In a $\triangle ABC$, if $\angle A = x$, $\angle B = 2\angle A$, $\angle C = 3\angle A$, then $x =$ | (s) 30° |
| | (t) Right angle |
| | (u) Obtuse angle |

SOLUTIONS

Brief Explanations of Selected Questions

1 EXERCISE

Fill in the Blanks :

1. median
2. equal
3. hypotenuse
4. obtuse
5. 60°
6. a straight angle
7. altitude
8. Scalene
9. equal
10. greater

True / False :

1. False
2. False
3. False
4. False
5. True
6. False
7. False
8. False
9. False
10. False

Match the Columns :

1. (A) $\rightarrow$ (s); (B) $\rightarrow$ (p); (C) $\rightarrow$ (q); (D) $\rightarrow$ (r)

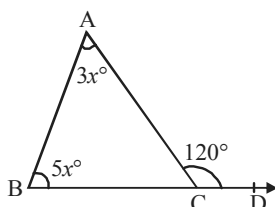
Very Short Answer Questions :

1. Let the measure of each unknown angle be x° .
 $\therefore x + x + 70 = 180$
 $\Rightarrow 2x = (180 - 70)$
 $\Rightarrow 2x = 110 \Rightarrow x = 55$
 Hence, each unknown angle measures 55° .
2. $\angle ACD = \angle A + \angle B = 75^\circ + 45^\circ = 120^\circ$
3. $\angle ABC = \angle ACD - \angle A = 65^\circ - 32^\circ = 33^\circ$
 $\Rightarrow x = 33^\circ$
 Now, $y^\circ + 65^\circ = 180^\circ$ [Linear pair]
 $\Rightarrow y^\circ = 180^\circ - 65^\circ = 115^\circ$
4. (i) $5 \text{ cm} + 7 \text{ cm} = 12 \text{ cm}$, which is greater than third side.
 $\therefore$ The triangle is possible.
 (ii) $2 \text{ cm} + 10 \text{ cm} = 12 \text{ cm}$, which is less than third side.
 $\therefore$ Triangle is not possible.
 (iii) $3 \text{ cm} + 4 \text{ cm} = 7 \text{ cm}$, which is greater than third side.
 $\therefore$ Triangle is possible.
 (iv) $2 \text{ cm} + 5 \text{ cm} = 7 \text{ cm}$, which is equal to the third side.
 $\therefore$ Triangle is not possible.

5. (i) $a^2 + b^2 = 5^2 + 9^2 = 25 + 81 = 106 \neq 11^2$
 $\Rightarrow a^2 + b^2 \neq c^2$
 $\therefore$ The triangle is not a right triangle.
 (ii) $a^2 + b^2 = 15^2 + 20^2 = 225 + 400 = 625$
 $= 25^2 = c^2$
 $\Rightarrow a^2 + b^2 = c^2$
 $\therefore$ The triangle is a right triangle.
6. $\angle A + \angle B + \angle C = 180^\circ$
 $\Rightarrow 2x + 15^\circ + x - 20^\circ + 3x - 55^\circ = 180^\circ$
 $\Rightarrow 6x - 60^\circ = 180^\circ \Rightarrow x = \frac{240^\circ}{6} = 40^\circ$
7. Let the measure of the angles of the triangle be $(2x)^\circ$, $(2x)^\circ$ and $(5x)^\circ$
 $\therefore 2x + 2x + 5x = 180^\circ$
 $\Rightarrow 9x = 180^\circ \Rightarrow x = 20^\circ$
 So, the angles are $(2 \times 20)^\circ$, $(2 \times 20)^\circ$, $(5 \times 20)^\circ$, i.e. 40° , 40° and 100° .
8. (i) Since $PR = RQ$
 $\therefore m = 50^\circ$ [Isosceles triangle property]
 Now, $\angle P + \angle Q + \angle R = 180^\circ$ [Angle sum property]
 $\Rightarrow 50^\circ + 50^\circ + x = 180^\circ$
 $\Rightarrow x = 180^\circ - 100^\circ = 80^\circ$
 (ii). Since $EG = GF$
 $\therefore \angle F = \angle E = x$ [Isosceles triangle property]
 Now, $\angle E + \angle F = 105^\circ$ [Exterior angle property of a Δ]
 $\Rightarrow x + x = 105^\circ$
 $\Rightarrow 2x = 105^\circ$
 $\Rightarrow x = \frac{105^\circ}{2} = 52.5^\circ$
9. (i) $b^2 = 8^2 + 15^2 = 64 + 225 = 289 = 17^2$
 $\Rightarrow b = 17$
 (ii) $c^2 = 5^2 + 12^2 = 25 + 144 = 169 = 13^2$
 $\Rightarrow c = 13$
 (iii) $d^2 = 29^2 - 21^2 = 841 - 441 = 400 = (20)^2$
 $\Rightarrow d = 20$
 (iv) $x^2 = 15^2 - 9^2 = 225 - 81 = 144 = 12^2$
 $\Rightarrow x = 12$

Short Answer Questions :

1. Let the given interior opposite angles be $(3x)^\circ$ and $(5x)^\circ$.



- $\therefore 3x + 5x = 120 \Rightarrow 8x = 120 \Rightarrow x = 15$.
 $\therefore \angle A = (3 \times 15)^\circ = 45^\circ$, $\angle B = (5 \times 15)^\circ = 75^\circ$.
 But, $\angle A + \angle B + \angle C = 180^\circ$
 $\therefore 45^\circ + 75^\circ + \angle C = 180^\circ$
 $\Rightarrow 120^\circ + \angle C = 180^\circ$
 $\Rightarrow \angle C = 180^\circ - 120^\circ = 60^\circ$.
2. (i) Since $DE \parallel BC$ and ADB is the transversal, so
 $\angle ADE = \angle ABC = 55^\circ$
 (ii) Now, In $\triangle ADE$,
 $\angle A + \angle ADE + \angle AED = 180^\circ$
 $\Rightarrow 65^\circ + 55^\circ + \angle AED = 180^\circ$
 $\Rightarrow \angle AED = 180^\circ - 120^\circ = 60^\circ$
 (iii) Since $DE \parallel BC$ and AEC is the transversal.
 $\therefore \angle C = \angle AED = 60^\circ$
3. $\angle PRS$ is an exterior angle to $\triangle PQR$
 $\therefore \angle PRS = \angle QPR + \angle PQR$
 $\Rightarrow y = 30^\circ + 40^\circ = 70^\circ$
 $\angle PTS$ is an exterior angle to $\triangle RST$
 $\therefore \angle PTS = \angle TRS + \angle TSR$
 $\Rightarrow x = y + 35^\circ = 70^\circ + 35^\circ = 105^\circ$
4. Let $\angle A = x$,

$$\therefore \angle B = \text{Three-fourth of } \angle A = \frac{3}{4}x$$

$$\angle C = \text{Two-third of } \angle B = \frac{2}{3} \angle B$$

$$= \frac{2}{3} \times \frac{3}{4}x = \frac{1}{2}x$$

Also, $\angle A + \angle B + \angle C = 180^\circ$ (Angle Sum Property)

$$\Rightarrow x + \frac{3}{4}x + \frac{1}{2}x = 180^\circ$$

$$\Rightarrow \frac{4x + 3x + 2x}{4} = 180^\circ$$

$$\Rightarrow 9x = 4 \times 180$$

$$\Rightarrow x = \frac{4 \times 180}{9} = 80^\circ$$

$$\therefore \angle A = x = 80^\circ$$

$$\angle B = \frac{3}{4}x = \frac{3}{4} \times 80^\circ = 60^\circ$$

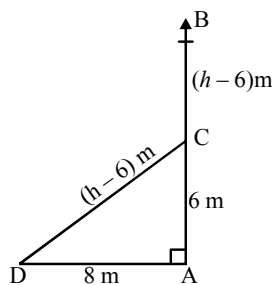
$$\angle C = \frac{1}{2}x = \frac{1}{2} \times 80^\circ = 40^\circ$$

5. (i) Since $PR = QR$
 $\therefore \angle Q = \angle QPR = 59^\circ$
 Now, $\angle Q + \angle QPR + \angle QRP = 180^\circ$
 $\Rightarrow 59^\circ + 59^\circ + x = 180^\circ$
 $\Rightarrow x = 180^\circ - 118^\circ = 62^\circ$
 Now, $x = \angle RPS + \angle RSP$
 $\Rightarrow 62^\circ = 32^\circ + y$
 $\Rightarrow y = 62^\circ - 32^\circ = 30^\circ$
- (ii) Since $AE = AD$
 $\therefore \angle D = \angle E = 63^\circ$
 Now, In $\triangle AED$
 $\Rightarrow \angle AED + \angle ADE + \angle DAE = 180^\circ$
 $\Rightarrow 63^\circ + 63^\circ + \angle DAE = 180^\circ$
 $\Rightarrow \angle DAE = 180^\circ - 126^\circ = 54^\circ$
 Now, $\angle BAC = \angle DAE = 54^\circ$
 [$\therefore$ Vertically opposite $\angle$ s]
 Now, In $\triangle ABC$
 $\angle ABC + \angle ACB + \angle BAC = 180^\circ$
 $\Rightarrow x + x + 54^\circ = 180^\circ$ [$\therefore AB = AC$]
 $\Rightarrow 2x = 180^\circ - 54^\circ = 126^\circ$
 $\Rightarrow x = 63^\circ$
6. We know that in a triangle, the sum of any two sides is greater than the third side.
 In $\triangle ABP$, $AB + BP > AP$... (i)
 In $\triangle APC$, $PC + AC > AP$... (ii)
 Now, adding (i) and (ii), we get
 $AB + BP + PC + AC > AP + AP$
 $\Rightarrow AB + BC + AC > 2AP$
7. In $\triangle ADC$,
 $AD^2 = AC^2 + CD^2$ (Pythagoras Theorem)
 $\Rightarrow (25)^2 = AC^2 + (7)^2$
 $\Rightarrow 625 = AC^2 + 49$
 $\Rightarrow AC^2 = 625 - 49 = 576 = (24)^2$
 $\therefore AC = 24 \text{ cm}$
 In $\triangle ABC$,
 $AB^2 = AC^2 + BC^2$ (Pythagoras Theorem)
 $\Rightarrow (30)^2 = (24)^2 + x^2$
 $\Rightarrow x^2 = 900 - 576 = 324 = (18)^2$
 $\therefore x = 18 \text{ cm}$

8. $(\text{Breadth})^2 = (41)^2 - (40)^2 = 1681 - 1600$
 $= 81 = (9)^2$
 $\Rightarrow \text{Breadth} = 9 \text{ cm}$
 $\text{Perimeter} = 2(l + b) = 2(40 + 9) = 98 \text{ cm}.$

Long Answer Questions :

1. Let $3\angle A = 4\angle B = 6\angle C = x^\circ$. Then,
 $\angle A = \left(\frac{x}{3}\right)^\circ$, $\angle B = \left(\frac{x}{4}\right)^\circ$ and $\angle C = \left(\frac{x}{6}\right)^\circ$.
 Now, $\angle A + \angle B + \angle C = 180^\circ$
 $\Rightarrow \frac{x}{3} + \frac{x}{4} + \frac{x}{6} = 180^\circ$
 $\Rightarrow 4x + 3x + 2x = (180^\circ \times 12)$
 $\Rightarrow 9x = (180 \times 12)$
 $\Rightarrow x = \frac{180 \times 12}{9} = 240^\circ$.
 $\therefore \angle A = \left(\frac{240}{3}\right)^\circ = 80^\circ$,
 $\angle B = \left(\frac{240}{4}\right)^\circ = 60^\circ$ and $\angle C = \left(\frac{240}{6}\right)^\circ = 40^\circ$.
2. Let AB be the tree of height h metres broken at the point C and let CB take the position CD as shown in the figure.



Then, $AC = 6 \text{ m}$, $AD = 8 \text{ m}$ and
 $CD = CB = (h - 6) \text{ m}.$

In $\triangle DAC$ by Pythagoras' theorem, we have:
 $CD^2 = AC^2 + AD^2$

$$\begin{aligned} \Rightarrow (h-6)^2 &= 6^2 + 8^2 = 36 + 64 = 100 = (10)^2 \\ \Rightarrow (h-6) &= 10 \\ \Rightarrow h &= (10 + 6) = 16 \text{ m}. \end{aligned}$$

Hence, the original height of the tree was 16 m.

3. In order to take a shortcut, a person has to travel diagonally to the opposite corner from one corner. Now, $\triangle DCB$ is a right-angled triangle

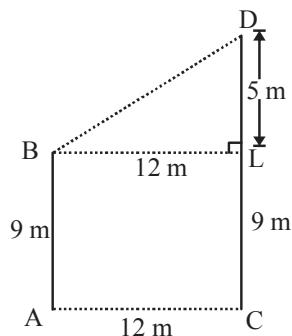
$$\begin{aligned} \therefore DB^2 &= DC^2 + BC^2 \\ &= 60^2 + 25^2 \\ &= 3600 + 625 \\ &= 4225 \end{aligned}$$

$$\therefore DB = \sqrt{4225} = 65 \text{ m}$$

$$\therefore \text{Distance travelled} = 65 \text{ m}$$

4. $AC \parallel DE$
 $\angle BCA = \angle BDE = 40^\circ$
 $\Rightarrow z = 40^\circ$
 Now, $\angle ABD + \angle DBE = 180^\circ$ [Linear Pair]
 $\Rightarrow 70^\circ + y = 180^\circ$
 $\Rightarrow y = 180^\circ - 70^\circ = 110^\circ$
 Now, $\angle ABC = \angle DBE$
 $\Rightarrow u = y = 110^\circ$
 Now, in $\triangle ABC$,
 $\angle BAC + \angle ABC + \angle BCA = 180^\circ$
 $\Rightarrow w + u + z = 180^\circ$
 $\Rightarrow w + 110^\circ + 40^\circ = 180^\circ$
 $\Rightarrow w = 180^\circ - 150^\circ = 30^\circ$
 Now, $AC \parallel DE$
 $\angle BED = \angle BAC$ [Alternate angles]
 $\Rightarrow x = w = 30^\circ$

5. Let AB and CD be the given poles such that $AB = 9 \text{ m}$, $CD = 14 \text{ m}$ and $AC = 12 \text{ m}$. Join BD. From B, draw $BL \perp CD$.



$$\therefore DL = (CD - CL) = (CD - AB) = (14 - 9) \text{ m} = 5 \text{ m}.$$

$$BL = AC = 12 \text{ m}.$$

Now, in right $\triangle BLD$, we have:

$$\begin{aligned} BD^2 &= BL^2 + DL^2 \\ &= \{(12)^2 + (5)^2\} \text{ m}^2 = (144 + 25) \text{ m}^2 = 169 \text{ m}^2 \end{aligned}$$

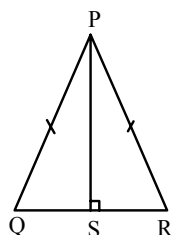
$$\Rightarrow BD = \sqrt{169} \text{ m} = \sqrt{13 \times 13} \text{ m} = 13 \text{ m}.$$

Hence, the distance between their tops is 13 m.

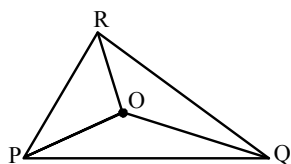
2 EXERCISE

Text-Book Exercise :

1. Draw a line segment PS perpendicular to QR. It can be observed that the length of QS and SR is also same. Therefore, PS is also a median of this triangle.



2. (i) $x = 50^\circ + 70^\circ = 120^\circ$
(ii) $x = 65^\circ + 45^\circ = 110^\circ$
(iii) $x = 40^\circ + 30^\circ = 70^\circ$
(iv) $x = 30^\circ + 60^\circ = 90^\circ$
3. (i) $x + 50^\circ = 115^\circ$
 $\Rightarrow x = 115^\circ - 50^\circ = 65^\circ$
(ii) $x + 90^\circ = 125^\circ$
 $\Rightarrow x = 125^\circ - 90^\circ = 35^\circ$
(iii) $x + 30^\circ = 80^\circ$
 $\Rightarrow x = 80^\circ - 30^\circ = 50^\circ$
(iv) $x + 35^\circ = 75^\circ$
 $\Rightarrow x = 75^\circ - 35^\circ = 40^\circ$
4. If O is a point in the interior of a given triangle, then three triangles $\triangle OPQ$, $\triangle OQR$ and $\triangle ORP$ can be constructed. In a triangle, the sum of the lengths of either two sides is always greater than the third side.



- (i) Yes, $OP + OQ > PQ$
(ii) Yes, $OQ + OR > QR$
(iii) Yes, $OR + OP > PR$
5. In a triangle, the sum of the lengths of either two sides is always greater than the third side.
In $\triangle ABM$,
 $AB + BM > AM$... (i)
Similarly, in $\triangle ACM$,

$$AC + CM > AM \quad \dots (ii)$$

Adding (i) and (ii),

$$AB + BM + MC + AC > AM + AM$$

$$AB + BC + AC > 2AM$$

Yes, the given expression is true.

6. In a triangle, the sum of the lengths of either two sides is always greater than the third side.
In $\triangle ABC$; $AB + BC > CA$... (i)
In $\triangle BCD$; $BC + CD > DB$... (ii)
In $\triangle CDA$; $CD + DA > AC$... (iii)
In $\triangle DAB$; $DA + AB > DB$... (iv)

Adding (i), (ii), (iii) and (iv), we obtain

$$AB + BC + BC + CD + CD + DA + DA + AB > AC + BD + AC + BD$$

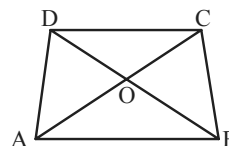
$$\Rightarrow 2AB + 2BC + 2CD + 2DA > 2AC + 2BD$$

$$\Rightarrow 2(AB + BC + CD + DA) > 2(AC + BD)$$

$$\Rightarrow (AB + BC + CD + DA) > (AC + BD)$$

Yes, the given expression is true.

7. In a triangle, the sum of the lengths of either two sides is always greater than the third side.



$$\text{In } \triangle OAB; OA + OB > AB \quad \dots (i)$$

$$\text{In } \triangle OBC; OB + OC > BC \quad \dots (ii)$$

$$\text{In } \triangle OCD; OC + OD > CD \quad \dots (iii)$$

$$\text{In } \triangle ODA; OD + OA > DA \quad \dots (iv)$$

Adding (i), (ii), (iii) and (iv), we obtain

$$OA + OB + OB + OC + OC + OD + OD + OA > AB + BC + CD + DA$$

$$\Rightarrow 2OA + 2OB + 2OC + 2OD > AB + BC + CD + DA$$

$$\Rightarrow 2(OA + OC) + 2(OB + OD) > AB + BC + CD + DA$$

$$\Rightarrow 2(AC) + 2(BD) > AB + BC + CD + DA$$

$$\Rightarrow 2(AC + BD) > AB + BC + CD + DA$$

Yes, the given expression is true.

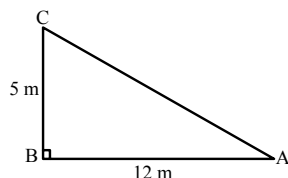
8. Lengths of two sides of a triangle are 12 cm and 15 cm. In a triangle, the sum of the lengths of either two sides is always greater than the third side. Let the side be x .

$$\therefore x + 12 > 15 \Rightarrow x > 3$$

$$\text{and } 12 + 15 > x \Rightarrow 27 > x$$

Hence, third side can measure between 3 and 27.

9. (i) $(2.5)^2 = 6.25$, $(6.5)^2 = 42.25$ and $(6)^2 = 36$
 Here, $36 + 6.25 = 42.25$
 $\Rightarrow (6)^2 + (2.5)^2 = (6.5)^2$
 Hence, these are the sides of a right-angled triangle. Right angle will be opposite of the side of measure 6.5 cm.
- (ii) $(2)^2 = 4$, $(2)^2 = 4$ and $(5)^2 = 25$
 Here, $(2)^2 + (2)^2 \neq (5)^2$
 Hence, these sides are not of a right-angled triangle.
- (iii) $(1.5)^2 = 2.25$, $(2)^2 = 4$ and $(2.5)^2 = 6.25$
 Here, $2.25 + 4 = 6.25$
 $\Rightarrow (1.5)^2 + (2)^2 = (2.5)^2$
 Hence, these are the sides of a right-angled triangle.
 Right angle will be opposite of the side of measure 2.5 cm.
10. In the given figure, BC represents the unbroken part of the tree. Point C represents the point where the tree broke and CA represents the broken part of the tree. Triangle ABC, thus formed, is right-angled at B.



$$\begin{aligned}\text{In } \triangle ABC, \\ AC^2 &= BC^2 + AB^2 \\ &= (5)^2 + (12)^2 \\ &= 25 + 144 = 169 = (13)^2 \\ \therefore AC &= 13\text{m}\end{aligned}$$

$$\text{Original height of the tree} = AC + CB = 13\text{ m} + 5\text{ m} = 18\text{ m}$$

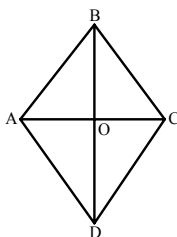
11. Diagonals of a rhombus bisect each other at right angle.
 Let diagonals bisect at O.

$$\therefore AO = OC = \frac{16}{2} = 8\text{ cm}$$

$$\text{and } BO = OD = \frac{30}{2} = 15\text{ cm}$$

Now, In $\triangle AOB$

$$\begin{aligned}\therefore (AO)^2 + (OB)^2 &= (AB)^2 \\ \Rightarrow (AB)^2 &= 8^2 + (15)^2\end{aligned}$$



$$\begin{aligned}&= 64 + 225 \\ &= 289 = (17)^2 \\ \therefore AB &= 17\text{ cm} \\ \text{Perimeter of rhombus} &= 4 \times (17) = 68\text{ cm}\end{aligned}$$

Exemplar Questions :

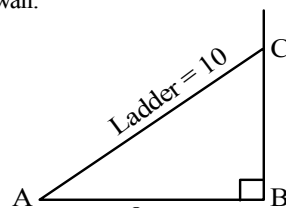
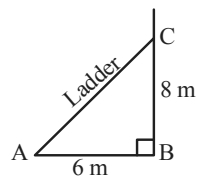
- (i) $a = 20^\circ$, $b = 130^\circ$, $c = 50^\circ$
 (ii) $a = 65^\circ$, $b = 115^\circ$, $c = 25^\circ$
- 40m
- $\angle PON = 90^\circ$, $\angle NPO = 20^\circ$
- Let the measure of third angle be x° .
 Sum of all angles of a triangle is 180° .
 $\therefore 4x + 4x + x = 180^\circ \Rightarrow 9x = 180^\circ$
 $\Rightarrow x = 20^\circ$
 Hence, the measure of angles are 20° , 80° and 80°
- $\angle ACD = \angle A + \angle B$
 $\Rightarrow x + 90^\circ = 80^\circ + 30^\circ$
 $\Rightarrow x = 110^\circ - 90^\circ = 20^\circ$
- $y = 51^\circ$, $x = 129^\circ$
- $z = 160^\circ$
- 49 cm, sum of two sides should be greater than third side.
- (i) By Pythagoras theorem,

$$AC^2 = AB^2 + BC^2$$

$$AC^2 = 6^2 + 8^2$$

$$AC = \sqrt{36 + 64} = \sqrt{100} = 10\text{ m}$$

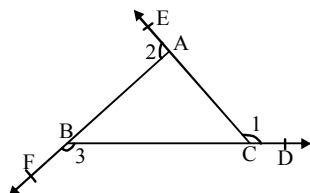
- (ii) If ladder is shifted 8 m away from the foot of wall.



$$\begin{aligned}AC^2 &= AB^2 + BC^2 \\ 10^2 &= 8^2 + BC^2 \\ BC^2 &= 100 - 64 = 36 \\ BC &= 6\text{ m}\end{aligned}$$

HOTS Questions :

- $\angle A = 30^\circ$
- Let the sides BC, CA and AB of $\triangle ABC$ be produced to D, E and F respectively, as shown in the figure.



We know that an exterior angle of a triangle is equal to the sum of its interior opposite angles.

$$\begin{aligned}\therefore \angle 1 &= \angle A + \angle B & \dots (i) \\ \angle 2 &= \angle B + \angle C & \dots (ii) \\ \angle 3 &= \angle C + \angle A & \dots (iii)\end{aligned}$$

On adding (i), (ii) and (iii), we get
 $\angle 1 + \angle 2 + \angle 3 = 2(\angle A + \angle B + \angle C) = 2 \times 180^\circ = 360^\circ$
 Hence, the sum of the exterior angles is 360° .

3. $70^\circ, 60^\circ, 50^\circ$
4. (i) $\angle F = 50^\circ$ (ii) $\angle EOF = 120^\circ$
5. 38m

3 EXERCISE

Single Option Correct :

1. (a) Sum of two unknown angle
 $= 180^\circ - 75^\circ = 105^\circ$
 Second angle $= 2 \times \frac{105^\circ}{5} = 42^\circ$
 Third angle $= 3 \times \frac{105^\circ}{5} = 63^\circ$
2. (b)
3. (b) In $\triangle ABC$, $\angle A + \angle B + \angle ACB = 180^\circ$
 $60^\circ + 45^\circ + \angle ACB = 180^\circ$
 $\angle ACB = 180^\circ - 105^\circ = 75^\circ$
 In $\triangle DCE$, $\angle D + \angle E + \angle DCE = 180^\circ$
 $80^\circ + \angle E + 75^\circ = 180^\circ$
 $\angle E = 180^\circ - 155^\circ = 25^\circ$
4. (d) $AB^2 + BC^2 = AC^2$; $12^2 + 5^2 = AC^2$
 $AC^2 = 144 + 25$
 $AC^2 = 169$, $AC = 13$ cm
5. (b) $a = 75^\circ + 40^\circ = 115^\circ$
 $b = 60^\circ + 40^\circ = 100^\circ$
 $\therefore a + b = 115^\circ + 100^\circ = 215^\circ$
6. (b) Let x, y and z is the angles, where $x = y + z$
 $x + y + z = 180^\circ$ (angle sum property)
 $(y + z) + y + z = 180^\circ$
 $2(y + z) = 180^\circ$
 $y + z = \frac{180^\circ}{2} = 90^\circ$
 $x = 90^\circ$, hence triangle is right angled.
7. (b) $\triangle AQC$ is a right angled triangle
 $(AQ)^2 + (QC)^2 = (AC)^2$
 $(QC)^2 = (AC)^2 - (AQ)^2$
 $(QC)^2 = (37)^2 - (12)^2$
 $1369 - 144 = 1225$
 $(QC)^2 = (35)^2 \Rightarrow QC = 35$
 $BC = BQ + QC = 35 + 35 = 70$
8. (a) In rectangle
 $l^2 + b^2 = d^2$
 $b^2 = d^2 - l^2 = (29)^2 - (20)^2$
 $= 841 - 400 = 441$
 $b = 21$ m
 Perimeter $= 2(20 + 21)$ m $= 82$ m
9. (d) In $\triangle DBA$, $DB = DA$
 $\Rightarrow \angle DBA = \angle BAD = 35^\circ$
 In $\triangle BCD$, $\angle CBD = \angle BCD = 2x$
 In $\triangle ABC$, $\angle A + \angle B + \angle C = 180^\circ$
 $35^\circ + (35^\circ + 2x) + 2x = 180^\circ$
 $4x = 180^\circ - 70^\circ = 110^\circ$
 $x = \frac{110^\circ}{4} = 27.5^\circ$
10. (c) In $\triangle DEC$, $\angle D + \angle C + \angle E = 180^\circ$
 $\angle E = 180^\circ - 61^\circ - 90^\circ = 29^\circ$
 $\angle ABC = \angle BED = 29^\circ$
 $\angle ABC + \angle BCF = 180^\circ$
 $\angle BCF = 180^\circ - 29^\circ = 151^\circ$
11. (b) In $\triangle ABC$, $\angle A + \angle B + \angle ACB = 180^\circ$
 $35^\circ + 110^\circ + \angle ACB = 180^\circ$
 $\angle ACB = 180^\circ - 145^\circ = 35^\circ$
 In $\triangle EFD$, $\angle E + \angle D + \angle EFD = 180^\circ$
 $30^\circ + 100^\circ + \angle EFD = 180^\circ$
 $\angle EFD = 180^\circ - 130^\circ = 50^\circ$

In $\triangle FCG$, $\angle FGC + \angle GCF + \angle CFG = 180^\circ$

$$\angle FGC + 35^\circ + 50^\circ = 180^\circ$$

$$\angle FGC = 180^\circ - 85^\circ = 95^\circ$$

$$\angle FGC = \angle x = 95^\circ \text{ (opposite angles)}$$

12. (a) In $\triangle ABD$, $\angle BAD + \angle b + 90^\circ = 180^\circ$

$$\angle b = 180^\circ - 90^\circ - 48^\circ = 180^\circ - 138^\circ$$

$$\angle b = 42^\circ$$

$$\angle DAC = 360^\circ - (290^\circ + 48^\circ) = 22^\circ$$

In $\triangle ADC$, $\angle DAC + \angle a + 90^\circ = 180^\circ$

$$\angle a = 180^\circ - 90^\circ - 22^\circ$$

$$\angle a = 68^\circ$$

$$\text{difference} = 68^\circ - 42^\circ = 26^\circ$$

More Than One Option Correct :

- (a, b) In isosceles triangle two sides are equal and third side is either 5 cm or 6 cm.
- (b, c) Because in a triangle sum of any two sides must be greater than third side
- (a, b)
- (a, b, c)
- (a, b, c)
- (a, d)
 - Since $AB \parallel CD$
 $\therefore \angle 2 = \angle 4$ [Corresponding angles]
 - Since exterior angle is equal to the sum of interior opposite angles.
 $\therefore \angle 1 + \angle 2 = \angle 3 + \angle 4$

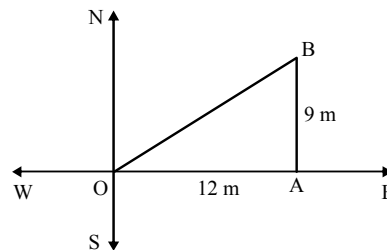
Assertion & Reason :

- (c): As $10 + 12 > 13$, $12 + 13 > 10$, $13 + 10 > 12$
 $\therefore$ Triangle is possible to form by three line segments of lengths 10 cm, 12 cm, 13 cm.
Assertion : correct, **Reason :** incorrect.
- (a): $\angle ACD = \angle ABC + \angle BAC$
 [Exterior angle property]
 $110^\circ = \angle ABC + 60^\circ$
 $\Rightarrow \angle ABC = 110^\circ - 60^\circ = 50^\circ$
Assertion : correct : **Reason :** correct and is the correct explanation of assertion.

Passage Based Questions :

- (b) Let a be the length of the other side
 $\therefore a^2 = (15)^2 - (9)^2$
 $= 225 - 81$
 $= 144 = (12)^2$
 $\Rightarrow a = 12 \text{ cm.}$
 Hence, the length of the other side is 12 cm.

- (a) (i) Here, $a = 15 \text{ cm}$, $b = 20 \text{ cm}$ and $c = 25 \text{ cm}$.
 The largest side is $c = 25 \text{ cm}$.
 Now, $a^2 + b^2 = (15)^2 + (20)^2$
 $= 225 + 400 = 625 = (25)^2$
 $\therefore a^2 + b^2 = c^2$
 Hence, given triangle is a right angled triangle.
 (ii) Here $a = 7 \text{ cm}$, $b = 6 \text{ cm}$ and $c = 13 \text{ cm}$.
 The largest side is $c = 13 \text{ cm}$.
 Now, $a^2 + b^2 = (7)^2 + (6)^2$
 $= 49 + 36 = 85 \neq (13)^2$
 $\Rightarrow a^2 + b^2 \neq c^2$
- (a) Let O be the initial position of the man.
 Finally, he reaches at the point B.



Now, in right $\triangle OAB$, we have

$$OB^2 = (OA^2 + AB^2) = 12^2 + 9^2$$

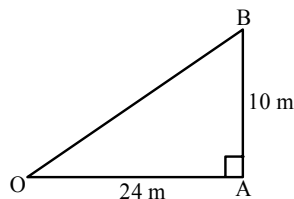
$$= 144 + 81 = 225 = (15)^2$$

Hence, the man is at a distance of 15 m from his initial position.

- (c) $\angle ACD = \angle BAC + \angle ABC$
 $\Rightarrow 120^\circ = x + 50^\circ$
 $\Rightarrow x = (120^\circ - 50^\circ) = 70^\circ$
 In $\triangle ABC$
 $\therefore x + 50^\circ + y = 180^\circ$
 $\Rightarrow 70^\circ + 50^\circ + y = 180^\circ$
 $\Rightarrow y = (180^\circ - 120^\circ) = 60^\circ$
 Hence, $x = 70^\circ$ and $y = 60^\circ$
- (c) Let the given interior opposite angles be $2x$ and $5x$.
 $\therefore 2x + 5x = 140^\circ \Rightarrow 7x = 140^\circ$
 $\Rightarrow x = 20^\circ$
 $\therefore$ One angle $= 2 \times 20 = 40^\circ$
 Other angle $= (5 \times 20^\circ) = 100^\circ$
 Let the third angle be y .
 $\therefore 40^\circ + 100^\circ + y = 180^\circ$
 $\Rightarrow y = (180^\circ - 140^\circ) = 40^\circ$.
 Hence, the angles of the triangle are 40° , 100° and 40° .
- (d)

Integer Type Questions :

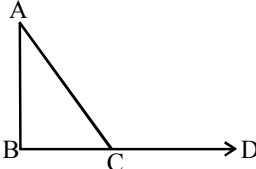
1. (5) Let the angles be $11x$, $13x$, $12x$
 $\therefore$ By angle sum property,
 $11x + 13x + 12x = 180^\circ$
 $36x = 180^\circ \Rightarrow x = \frac{180^\circ}{36} = 5^\circ$
 $\therefore$ Angles are $11 \times 5^\circ = 55^\circ$, $13 \times 5^\circ = 65^\circ$,
 $12 \times 5^\circ = 60^\circ$
 Smallest angle is 55° .
8. (6) Since, ABC is a right angled triangle
 $\therefore AC^2 = AB^2 + BC^2$
 $\Rightarrow AB^2 = AC^2 - BC^2 = (26)^2 - (10)^2$
 $= 676 - 100 = 576 = (24)^2$
 $\Rightarrow AB = 24$ cm
 $\therefore$ The sum of digits of length $= 2 + 4 = 6$
3. (7) Let one of the opposite interior angles be x
 $\therefore 125^\circ = x + 55^\circ$
 $x = 125^\circ - 55^\circ = 70^\circ$
 $\Rightarrow 10 \times k = 70^\circ = 10 \times 7$
 $\Rightarrow k = 7$
4. (8) In right $\triangle OAB$, we have:



$$\begin{aligned}
 OB^2 &= (OA^2 + AB^2) \\
 &= \{(24)^2 + (10)^2\} \text{ m}^2 \\
 &= (576 + 100) \text{ m}^2 = 676 \text{ m}^2 \\
 \Rightarrow OB &= \sqrt{676} \text{ m} = 26 \text{ m.} \\
 \therefore \text{ The sum of digits of length OB} \\
 &= 2 + 6 = 8
 \end{aligned}$$

5. (3) Any triangle have three altitudes.
6. (4) $AC^2 = AB^2 + BC^2$
 $\Rightarrow 5^2 = (AB)^2 + 3^2$
 $\Rightarrow AB^2 = 5^2 - 3^2$
 $= 25 - 9 = 16 = 4^2$
 $\Rightarrow AB = 4$

Multiple Matching Question :

1. (A) $\rightarrow (r, u)$; (B) $\rightarrow (p, u)$; (C) $\rightarrow (q, t)$; (D) $\rightarrow (s)$
 (A) In $\triangle ABC$, by angle sum property,
 $\angle A + \angle B + \angle C = 180^\circ$
 $\angle C = 180^\circ - 30^\circ - 50^\circ = 100^\circ$
 (B) 
 $\angle ACD = \angle ABC + \angle BAC$
 $= 90^\circ + 40^\circ = 130^\circ$
 (C) In a right angled triangle greatest angle is the right angle i.e., 90°
 (D) In $\triangle ABC$, by angle sum property,
 $\angle A + \angle B + \angle C = 180^\circ$
 $\Rightarrow \angle A + 2\angle A + 3\angle A = 180^\circ$
 $\Rightarrow x + 2x + 3x = 180^\circ$
 $\Rightarrow 6x = 180^\circ \Rightarrow x = \frac{180^\circ}{6} = 30^\circ$

Chapter

7

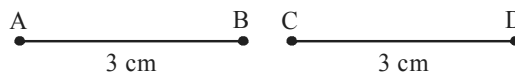
Congruence of Triangles

CONGRUENT FIGURES

Two figures are congruent if they have exactly the same shape and size. In other words, when two figures fit exactly on each other, we call them **congruent figures**. The relation of two congruent figures is called **congruence**. The symbol for congruence is $\cong$.

SOME TYPES OF CONGRUENT FIGURES

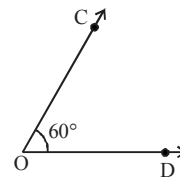
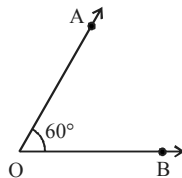
- (i) Two line segments of the same length are congruent.



Since $\overline{AB} = \overline{CD}$ (each 3 cm)

$$\therefore \overline{AB} \cong \overline{CD}$$

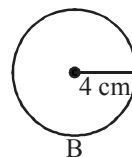
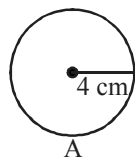
- (ii) Two angles of the same measure are congruent.



Since $\angle AOB = \angle COD$ (each 60°)

$$\therefore \angle AOB \cong \angle COD$$

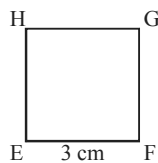
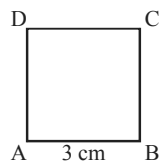
- (iii) Two circles of the same radii are congruent.



Since, radius of circle A = radius of circle B

$$\therefore \text{Circle A} \cong \text{Circle B}$$

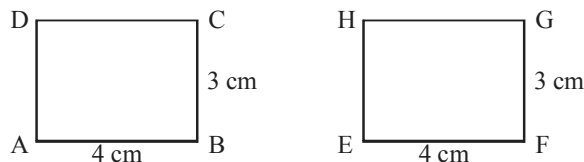
- (iv) Two squares of the same side are congruent.



Since, $AB = EF$ (each 3 cm)

$\therefore$ Sq. ABCD $\cong$ Sq. EFGH

- (v) Two rectangles of the same length and breadth are congruent.



Since, $AB = EF$ (each 4 cm)

$BC = FG$ (each 3 cm)

$\therefore$ Rect. ABCD $\cong$ Rect. EFGH

CONGRUENCE OF TRIANGLES

Two triangles are congruent if pairs of corresponding sides and corresponding angles are equal.

The three angles of a triangle determine its shape and its three sides determine its size. If the three angles and the three sides of a triangle are respectively equal to the corresponding angles and sides of another triangle, then the two triangles are congruent. However, to conclude that the triangles are congruent, it is not necessary that all the six corresponding elements of the two triangles are given to be congruent.

Thus $\triangle PQR \cong \triangle ABC$ if.

$$\angle P = \angle A,$$

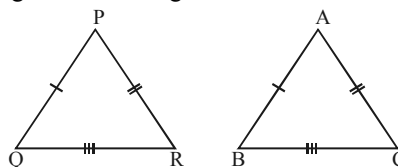
$$\angle Q = \angle B$$

$$\angle R = \angle C$$

$$PQ = AB$$

$$QR = BC$$

$$RP = CA$$



In triangle there are four cases of congruence.

Case 1: (SSS congruence condition) : Two triangles are congruent if the three sides of one triangle are respectively equal to the three sides of the other triangle.

For example:

In $\triangle ABC$ and $\triangle PQR$, $AB = PQ$, $AC = PR$

and $BC = QR$, then $\triangle ABC \cong \triangle PQR$

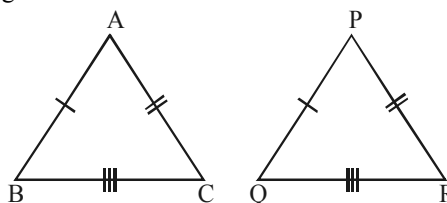


ILLUSTRATION 1 :

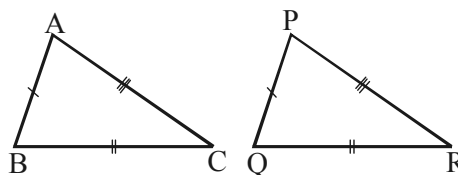
Draw a $\triangle ABC$ with $AB = 5$ cm, $BC = 7$ cm and $CA = 4$ cm. Draw another $\triangle PQR$ with $PQ = 5$ cm, $QR = 7$ cm and $RP = 4$ cm. State about the congruency of triangles.

SOLUTION:

Since, $AB = PQ$, $BC = QR$ and $CA = RP$.

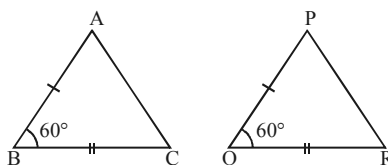
You would observe that the two triangles will cover each other exactly.

$\therefore \triangle ABC \cong \triangle PQR$ (By SSS congruency)



Case 2: (SAS congruence condition): Two triangles are congruent if the two sides and included angle of one triangle are respectively equal to the two sides and the included angle of the other triangle.

For example:



In $\triangle ABC$ and $\triangle PQR$,
 $AB = PQ$
 $\angle ABC = \angle PQR$
 $BC = QR$
 So, $\triangle ABC \cong \triangle PQR$ (by SAS Congruency)

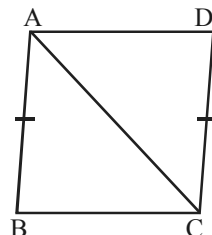
ILLUSTRATION 2 :

In figure, $AB \parallel DC$ and $AB = CD$.

- Is $\angle BAC = \angle DCA$? Why ?
- Is $\triangle ABC \cong \triangle CDA$ by SAS congruence condition ?
- State the three facts you have used to answer (ii).

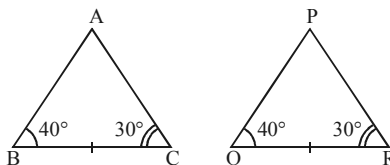
SOLUTION:

- Since, $AB \parallel DC$ and AC is a transversal.
 $\therefore \angle BAC = \angle DCA$ [Alternate angles]
- In $\triangle ABC$ and $\triangle CDA$, we have,
 $AB = CD$ [Given]
 $\angle BAC = \angle DCA$ [Proved in part (i)]
 $AC = CA$ [Common]
 So, by SAS congruence condition, $\triangle ABC \cong \triangle CDA$
- $AB = CD$, $\angle BAC = \angle DCA$ and $AC = CA$.



Case 3: (ASA congruence condition) : Two triangles are congruent if the two angles and the included side of the one are respectively equal to the two angles and the included side of the other.

In $\triangle ABC$ and $\triangle PQR$,
 $BC = QR$
 $\angle ABC = \angle PQR$
 $\angle BCA = \angle QRP$
 So, $\triangle ABC \cong \triangle PQR$



Case 4: (RHS congruence condition) : Two right triangles are congruent if the hypotenuse and one side of the first triangle are respectively equal to the hypotenuse and one side of second triangle.

In right $\triangle ABC$ and $\triangle PQR$,
 $AB = PQ$
 $AC = PR$
 $\angle ABC = \angle PQR$
 $\therefore \triangle ABC \cong \triangle PQR$

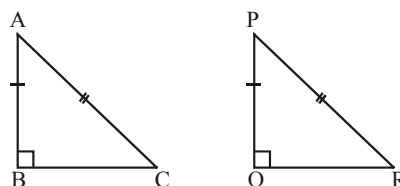


ILLUSTRATION 3 :

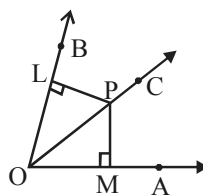
In Fig. $PL \perp OB$ and $PM \perp OA$ such that

$PL = PM$. Is $\triangle PLO \cong \triangle PMO$?

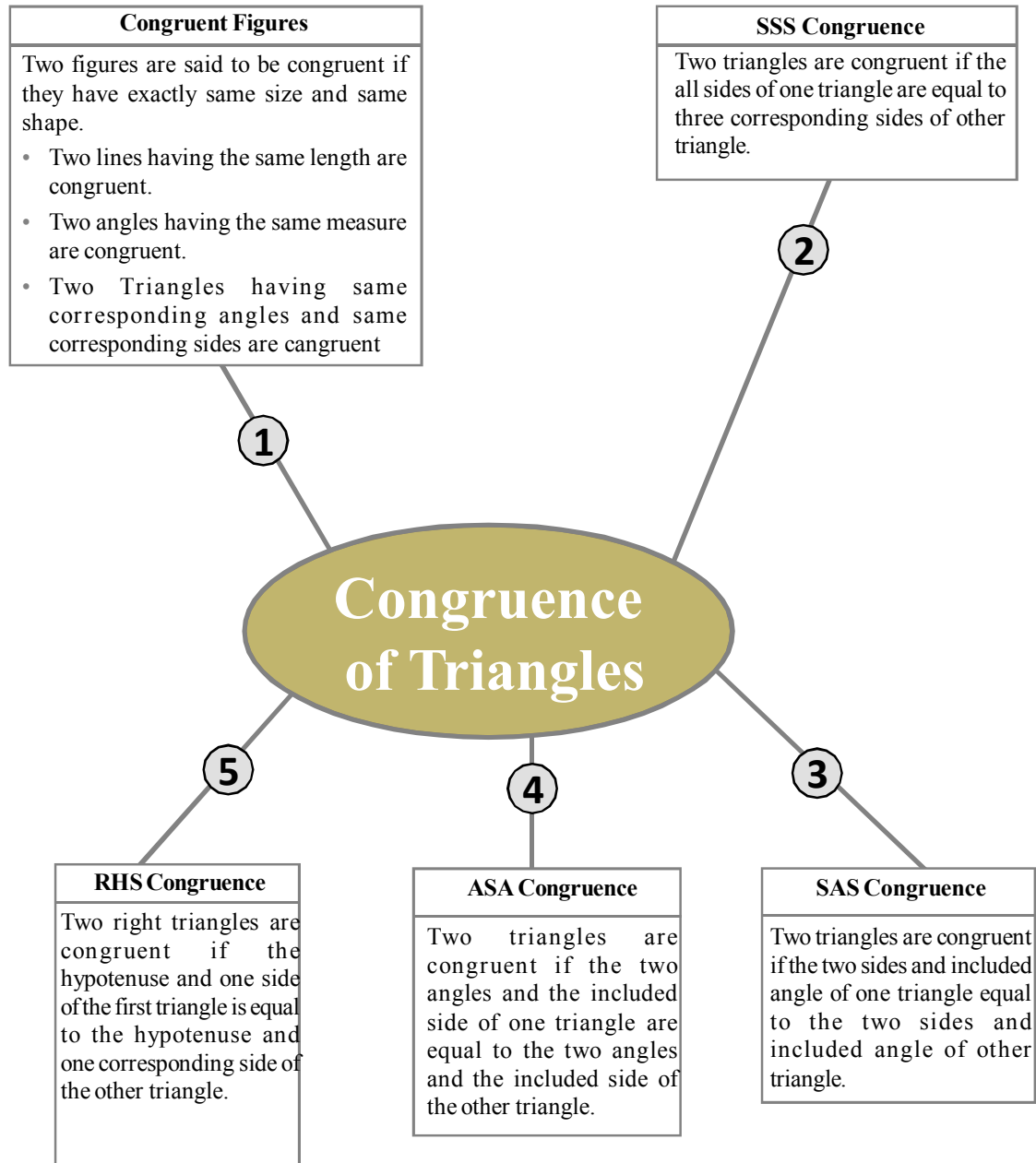
Give reasons in support of your answer.

SOLUTION:

In $\triangle PLO$ and $\triangle PMO$, we have
 $\angle PLO = \angle PMO = 90^\circ$
 $OP = OP$ [Hypotenuse]
 and, $PL = PM$ [Given]
 So, by RHS condition of congruence, we have
 $\triangle PLO \cong \triangle PMO$.



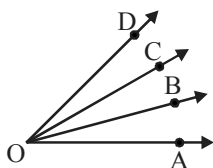
CONCEPT MAP



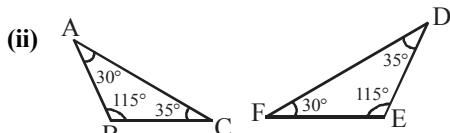
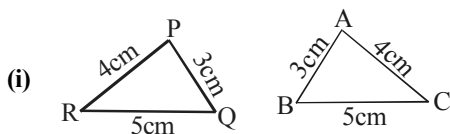
MISCELLANEOUS

SOLVED EXAMPLES

1. In the figure, $\angle AOB = \angle BOC = \angle COD$. Name the angle which is congruent to $\angle AOC$.



- Sol.** It is given that $\angle AOB = \angle COD$
 $\Rightarrow \angle AOB + \angle BOC = \angle COD + \angle BOC$
 $\Rightarrow \angle AOC = \angle BOD$
 $\angle AOC \cong \angle BOD$.
2. Without drawing the triangles, state the correspondence between the sides and the angles of the following pairs of congruent triangles:
- (i) $\triangle ABC \cong \triangle PQR$ (ii) $\triangle ABC \cong \triangle QRP$
- Sol.** (i) We have $\triangle ABC \cong \triangle PQR$
 So, $A \leftrightarrow P$, $B \leftrightarrow Q$ and $C \leftrightarrow R$
 Also, $AB = PQ$, $BC = QR$ and $AC = PR$
 Similarly, $\angle A = \angle P$, $\angle B = \angle Q$ and $\angle C = \angle R$
- (ii) We have $\triangle ABC \cong \triangle QRP$
 So, $A \leftrightarrow Q$, $B \leftrightarrow R$ and $C \leftrightarrow P$
 Also, $AB = QR$, $BC = RP$ and $AC = QP$
 Similarly, $\angle A = \angle Q$, $\angle B = \angle R$ and $\angle C = \angle P$.
3. Are the following pairs of triangles congruent? If yes, then write the criteria of congruency.



- Sol.** (i) Given,
 $PR = AC = 4$ cm,
 $PQ = AB = 3$ cm,
 and $QR = BC = 5$ cm
 $\therefore$ By SSS criterion of congruence,

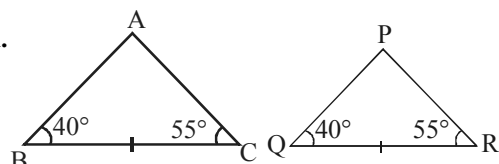
$$\triangle PQR \cong \triangle ABC$$

- (ii) Given, $\angle A = \angle F = 30^\circ$, $\angle B = \angle E = 115^\circ$,
 and $\angle C = \angle D = 35^\circ$

It is not possible to say whether $\triangle ABC$ and $\triangle FED$ are congruent or not, as AAA is not the criterion of congruency and no other criteria such as SSS, SAS, ASA or RHS can be applied in this case.

4. In $\triangle ABC$, $BC = 3.8$ cm, $\angle B = 40^\circ$ and $\angle C = 55^\circ$.
 In $\triangle PQR$, $QR = 3.8$ cm, $\angle Q = 40^\circ$ and $\angle R = 55^\circ$. Are they congruent?

Sol.

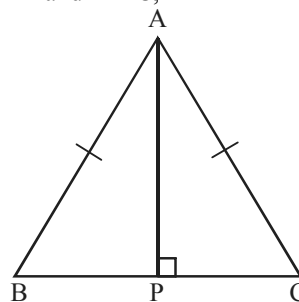


Thus, we have, $BC = QR$
 $\angle B = \angle Q$ and $\angle C = \angle R$.

$\therefore \triangle ABC \cong \triangle PQR$ (By ASA congruency)

5. In the adjoining figure, $\triangle ABC$ is an isosceles triangle with $AB = AC$, $AP \perp BC$. Prove that
 (i) AP bisects $\angle A$ (ii) AP bisects BC

Sol. In $\triangle APB$ and $\triangle APC$,



- $AP \perp BC \Rightarrow \angle APB = \angle APC = 90^\circ$ (i)
 $AB = AC$ (given)(ii)
 and $AP = AP$ (common)(iii)
 $\therefore$ From (i), (ii) and (iii), we have
 $\triangle APB \cong \triangle APC$

[By RHS criterion of congruence]

Hence, $\angle BAP = \angle CAP$ (C.P.C.T)

$\Rightarrow AP$ bisects $\angle A$

also, $BP = PC$ (C.P.C.T)

$\Rightarrow AP$ bisects BC

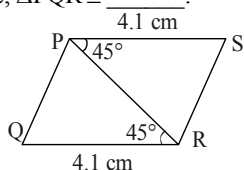
1

EXERCISE

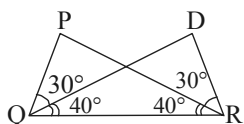
Fill in the Blanks :

DIRECTIONS : Complete the following statements with an appropriate word / term to be filled in the blank space(s).

- Two line segments are congruent, if they are of _____ lengths.
- Two squares are congruent, if they have same _____.
- Two angles are said to be _____, if they have equal measures.
- In figure, $\Delta PQR \cong$ _____.



- Two rectangles are congruent, if they have same _____ and _____.
- In figure, Δ _____ $\cong \Delta$ PQR



- Two circles are congruent if they have _____.
- The congruent figures _____ each other completely.

True / False :

DIRECTIONS : Read the following statements and write your answer as true or false.

- In ΔABC , $AB = 6.5$ cm, $AC = 5$ cm, $BC = 8$ cm and in ΔPQR , $PR = 6.5$, $PQ = 5$ cm, $RQ = 8$ cm. Then $\Delta ABC \cong \Delta PQR$.
- If P is the mid-point of a line segment AB, then we can say that AP and PB are congruent.
- A ten rupee coin is congruent to a five rupee coin.

- Two figures are congruent, if they have the same shape and size.
- If the areas of two circles are the same, they are congruent.
- If the hypotenuse of one right triangle is equal to the hypotenuse of another right triangle, then the triangles are congruent.
- If three angles of two triangles are equal, triangles are congruent.
- If two angles and a side of a triangle are equal to two angles and a side of another triangle, then the triangles are congruent.

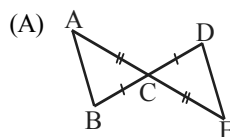
Match the Columns :

DIRECTIONS : Each question contains statements given in two columns which have to be matched. Statements (A, B, C, D) in column-I have to be matched with statements (p, q, r, s) in column-II.

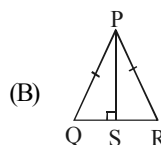
- Match the Columns:

Column-I

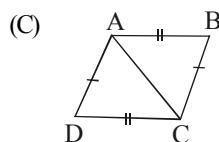
Column-II



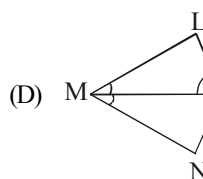
(p) RHS congruency



(q) ASA congruency



(r) SAS congruency

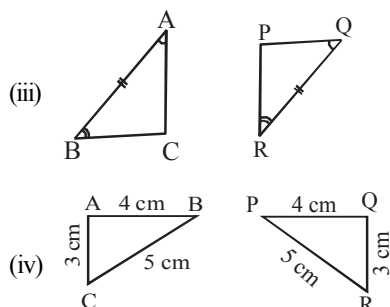
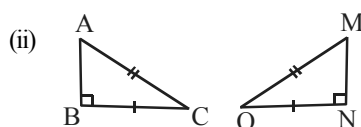
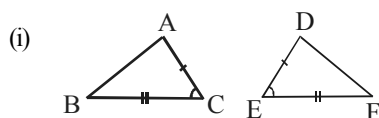


(s) SSS congruency

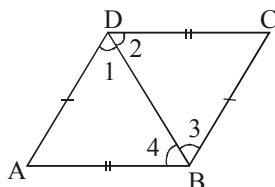
Very Short Answer Questions :

DIRECTIONS : Give answer in one word or one sentence.

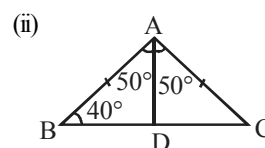
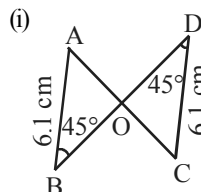
1. State the conditions of congruence satisfied in the following pairs of triangles and name the congruent triangles.



2. State the correspondence between the sides and angles of the following pairs of congruent triangles:
- (i) $\triangle ABC \cong \triangle EFD$ (ii) $\triangle CAB \cong \triangle QRP$
 (iii) $\triangle XZY \cong \triangle QPR$ (iv) $\triangle MPN \cong \triangle SQR$
3. In the given figure ABCD, $AD = BC$, $AB = CD$. Is $\triangle ABD \cong \triangle CDB$. If yes, state the congruent parts.



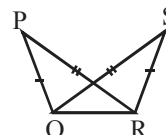
4. State the condition of congruency for the following pairs of triangles and name the congruent triangles.



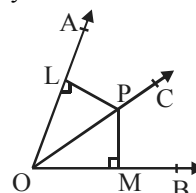
Short Answer Questions :

DIRECTIONS : Give answer in 2-3 sentences.

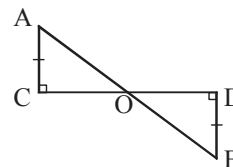
1. Which of the following pairs of triangles are congruent?
- (i) In $\triangle ABC$: $AB = 2$ cm, $AC = 4$ cm, $\angle A = 40^\circ$;
 in $\triangle XYZ$, $XZ = 2$ cm, $YZ = 4$ cm and $\angle Z = 40^\circ$.
- (ii) In $\triangle PQR$: $PQ = 5$ cm, $PR = 6$ cm, $\angle P = 55^\circ$;
 in $\triangle DEF$: $DE = 6$ cm, $EF = 5$ cm, $\angle D = 55^\circ$.
2. In the given figure, we have $PQ = SR$ and $PR = SQ$. Prove that $\triangle PQR \cong \triangle SRQ$



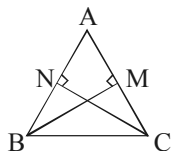
3. In figure $PL \perp OA$ and $PM \perp OB$ such that $PL = PM$. Is $\triangle PLO \cong \triangle PMO$? Give reasons in support of your answer.



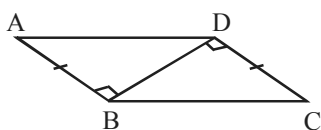
4. Prove that $\triangle AOC \cong \triangle BOD$, where $\angle C = \angle D = 90^\circ$ and $AC = BD$.



5. In the adjoining figure, $\triangle ABC$ is an isosceles triangle in which $AB = AC$. If $BM \perp AC$ and $CN \perp AB$. Prove that $\triangle BMC \cong \triangle CNB$



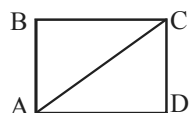
6. In the given figure, $AB \perp BD$, $CD \perp BD$ and $AB = CD$. Prove that $\triangle ABD \cong \triangle CDB$



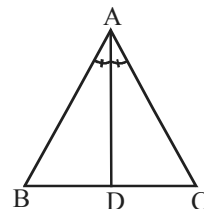
Long Answer Questions :

DIRECTIONS : Give answer in four to five sentences.

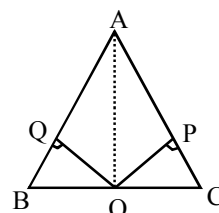
1. In the following figure, $AB \parallel DC$ and $AB = DC$.
- Is $\triangle ACD \cong \triangle CAB$?
 - State the three pairs of matching parts used to answer (i).
 - Which angle is equal to $\angle CAD$?



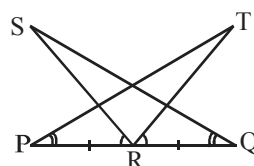
2. In $\triangle ABC$, $AD \perp BC$ and AD bisects $\angle A$. Prove that $\triangle ABD \cong \triangle ACD$. Also prove $\triangle ABC$ is isosceles.



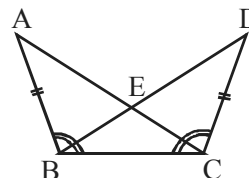
3. In $\triangle ABC$, O is the mid-point of BC . OQ and OP are the perpendiculars from O to AB and AC . If $OP = OQ$, prove that $\triangle ABC$ is an isosceles triangle.



4. Prove that $\triangle SQR$ and $\triangle TPR$ are congruent and $SR = TR$.



5. $AB = DC$ and $\angle ABC = \angle DCB$. Prove that :
- $AC = DB$
 - $BE = CE$

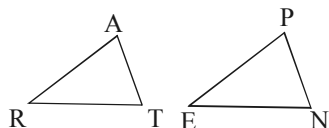


2

EXERCISE

Text-Book Exercise :

1. You want to show that $\triangle ART \cong \triangle PEN$,



- (i) If you have to use SSS criterion, then you need to show

(a) $AR =$ (b) $RT =$

(c) $AT =$

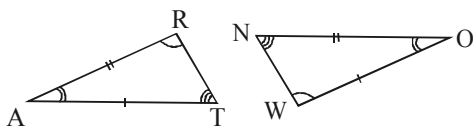
- (ii) If it is given that $\angle T = \angle N$ and you are to use SAS criterion, you need to have

(a) $RT =$ (b) $PN =$

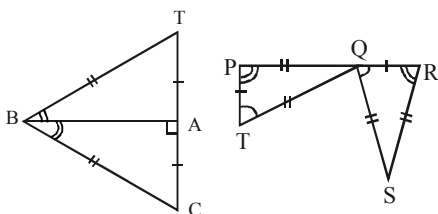
- (iii) If it is given that $AT = PN$ and you are to use ASA criterion, you need to have

(a) $\angle ATR = ?$ (b) $\angle RAT = ?$

2. In $\triangle ABC$, $\angle A = 30^\circ$, $\angle B = 40^\circ$ and $\angle C = 110^\circ$. In $\triangle PQR$, $\angle P = 30^\circ$, $\angle Q = 40^\circ$ and $\angle R = 110^\circ$. A student says that $\triangle ABC \cong \triangle PQR$ by AAA congruence criterion. Is he justified? Why or why not?
3. In the figure, the two triangles are congruent. The corresponding parts are marked. We can write $\triangle RAT \cong ?$

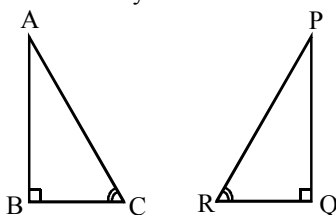


4. Complete the congruence statement:



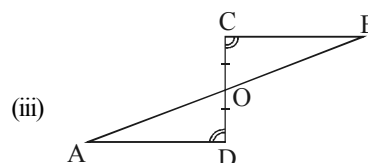
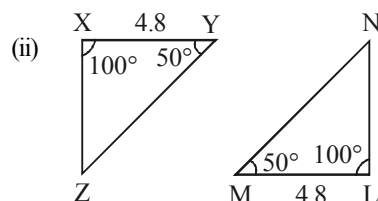
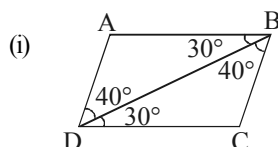
$\triangle BCA \cong ?$ $\triangle QRS \cong ?$

5. Draw a rough sketch of two triangles such that they have five pairs of congruent parts but still the triangles are not congruent.
6. If $\triangle ABC$ and $\triangle PQR$ are to be congruent, name one additional pair of corresponding parts. What criterion did you use?

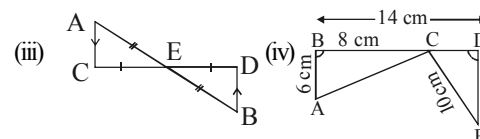
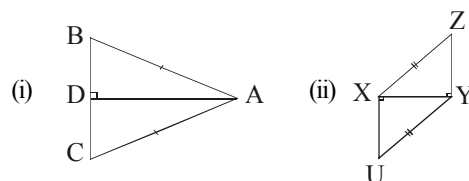


Exemplar Questions :

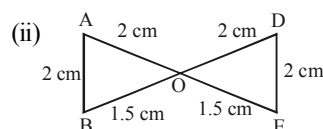
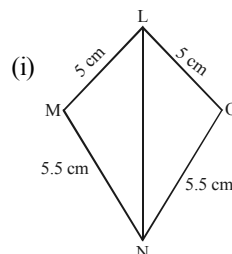
1. In each of the given pairs of triangles applying only ASA congruence criterion, determine which triangles are congruent. Also, write the congruent triangles in symbolic form.

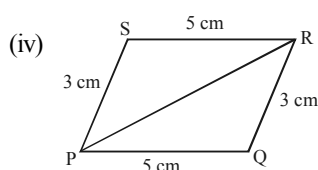
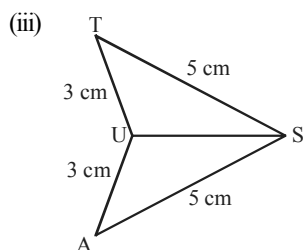


2. In each of the given pairs of triangles using only RHS congruence criterion, determine which pairs of triangles are congruent. In case of congruence, write the result in symbolic form:

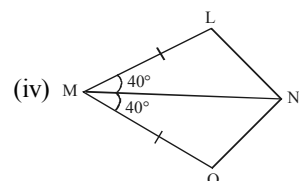
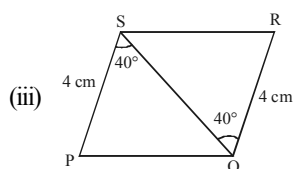
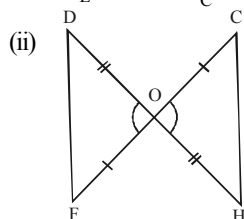
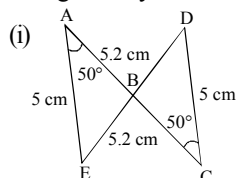


3. In the following pairs of triangles, the lengths of the sides are indicated along the sides. By applying SSS congruence criterion, determine which triangles are congruent. If congruent, write the results in symbolic form.





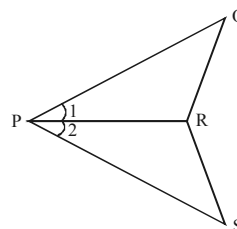
4. Triangles DEF and LMN are both isosceles with $DE = DF$ and $LM = LN$, respectively. If $DE = LM$ and $EF = MN$, then, are the two triangles congruent? Which condition do you use? If $\angle E = 40^\circ$, what is the measure of $\angle N$?
5. In figure, which pairs of triangles are congruent by SAS congruence criterion (condition)? If congruent, write the congruence of the two triangles in symbolic form.



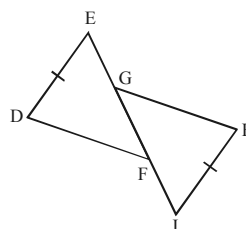
6. State which of the following pairs of triangles are congruent. If yes, write them in symbolic form (you may draw a rough figure).

- (i) $\triangle PQR$: $PQ = 3.5$ cm, $QR = 4.0$ cm, $\angle Q = 60^\circ$
 $\triangle STU$: $ST = 3.5$ cm, $TU = 4$ cm, $\angle T = 60^\circ$
- (ii) $\triangle ABC$: $AB = 4.8$ cm, $\angle A = 90^\circ$, $AC = 6.8$ cm.
 $\triangle XYZ$: $YZ = 6.8$ cm, $\angle X = 90^\circ$, $ZX = 4.8$ cm

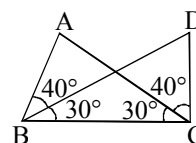
7. In figure, $PQ = PS$ and $\angle 1 = \angle 2$.
- (i) Is $\triangle PQR \cong \triangle PSR$? Give reasons.
- (ii) Is $QR = SR$? Give reasons.



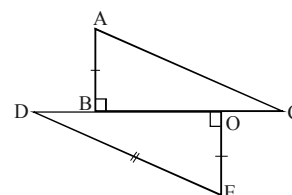
8. In figure, $DE = IH$, $EG = FI$ and $\angle E = \angle I$. Is $\triangle DEF \cong \triangle HIG$? If yes, by which congruence criterion?



9. Observe figure and state the three pairs of equal parts in triangles ABC and DCB.
- (i) Is $\triangle ABC \cong \triangle DCB$? Why?
- (ii) Is $AB = DC$? Why?
- (iii) Is $AC = DB$? Why?

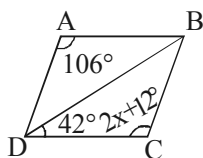


10. In figure, state the three pairs of equal parts in $\triangle ABC$ and $\triangle EOD$. Is $\triangle ABC \cong \triangle EOD$? Why?

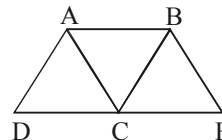


HOTS Questions :

- If $\triangle PQR$ and $\triangle SQR$ are both isosceles triangle on a common base QR such that P and S lie on the same side of QR. Are triangles PSQ and PSR congruent? Which condition do you use?
- In figure $\triangle DAB \cong \triangle BCD$. Find each value.
 (i) x
 (ii) $\angle ADB$



- $\triangle ABC$ is an equilateral triangle. C is the mid-point of DE. $\angle DAC$ and $\angle EBC$ are equal and supplementary. Prove $\triangle DAC \cong \triangle EBC$.



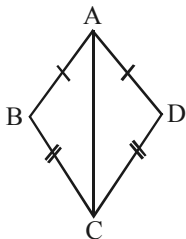
3

EXERCISE

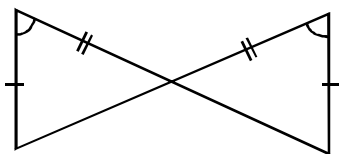
Single Option Correct :

DIRECTIONS : This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONLY ONE is correct.

- The condition that makes the following pair of triangles congruent is

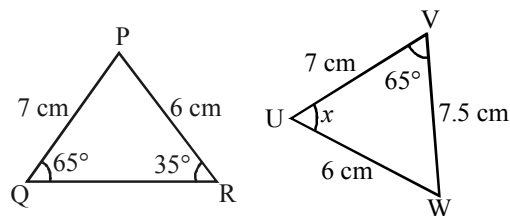


- SSS
 - ASA
 - SAS
 - RHS
- The side included between $\angle A$ and $\angle B$ of $\triangle ABC$ is
 (a) AC (b) CB
 (c) AB (d) CA
 - Which congruence condition, if any can be used to prove the given triangles congruent?



- SSS
- SAS
- AAS
- None of these

- $\triangle ABC$ is an equilateral triangle in which $AD \perp BC$. $\triangle ABD$ is congruent to:
 (a) $\triangle ADC$ (b) $\triangle ACD$
 (c) $\triangle ABC$ (d) $\triangle CDA$
- If triangle PQR is congruent to triangle UVW then value of x is



- 50°
 - 60°
 - 80°
 - 70°
- If ABCD is a square; X is the mid point of AB and Y is the mid point of BC, then considered following statements.
 (i) The triangle ADX and BAY are congruent
 (ii) $\angle DXA = \angle AYB$
 (iii) $DX = AY$
 (a) Only (i) is correct.
 (b) (iv) is correct.
 (c) (i), (ii) & (iii) all are correct.
 (d) Only (ii) is correct.

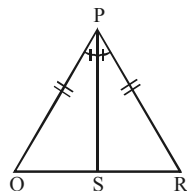
7. Consider the following statements related to the congruency of two right triangles.

- Equality of two sides of one triangle with any two sides of the second makes the triangles congruent.
- Equality of hypotenuse and a side of one triangle with the hypotenuse and a side of the second respectively makes the triangles congruent.
- Equality of the hypotenuse and an acute angle of one triangle with the hypotenuse and an angle of the second respectively makes the triangles congruent.

- (i), (ii) and (iii) are correct
- (i) and (ii) are correct
- (i) and (iii) are correct
- (ii) and (iii) are correct

8. In figure PS is the bisector of $\angle P$ and $PQ = PR$. Then $\triangle PRS$ and $\triangle PQS$ are congruent by the criterion

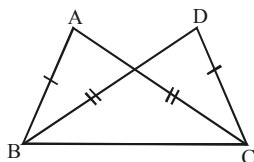
- AAA
- SAS
- SSS
- both (b) and (c)



9. Let ABC and DEF be two triangles in which $AB = DE$, $BC = FD$ and $CA = EF$. The two triangles are congruent under the correspondence $ABC \leftrightarrow \dots$

- DEF
- EDF
- FED
- DFE

10. If $\triangle ABC$ and $\triangle DBC$ are on the same base BC, $AB = DC$ and $AC = DB$, then which of the following gives a congruence relationship?

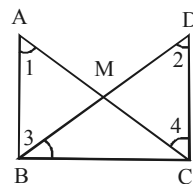


- $\triangle ABC \cong \triangle DBC$
- $\triangle ABC \cong \triangle CBD$
- $\triangle ABC \cong \triangle DCB$
- $\triangle ABC \cong \triangle BCD$

More Than One Option Correct :

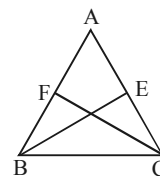
DIRECTIONS : This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONE or MORE may be correct.

- The congruence condition valid for triangles is/are.
 - SSS
 - ASA
 - RHS
 - AAA
- Which of the following are incorrect?
 - If two squares have equal areas, they are congruent.
 - Two equilateral triangles are always
 - If two triangles are equal in area, they are congruent.
 - Two isosceles triangles are congruent.
- If for $\triangle ABC$ and $\triangle DEF$, the correspondence $CAB \leftrightarrow EDF$ gives a congruence, then which of the following is/are true?
 - $AC = DE$
 - $AB = EF$
 - $\angle A = \angle D$
 - $\angle C = \angle E$
- In figure, M is the mid-point of both AC and BD. Then

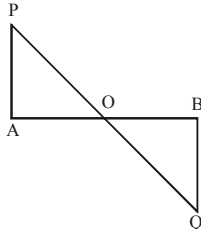


- $\angle 1 = \angle 2$
- $\angle 1 = \angle 4$
- $\angle 2 = \angle 3$
- $\angle 1 = \angle 3$

5. In the adjoining figure, ABC is an isosceles triangle in which $AB = AC$. If E and F be the midpoints of AC and AB respectively, then
- $BE = CF$
 - $\angle BFC = \angle CEB$
 - $BF = CE$
 - $\angle BEC = \angle FBC$



6. In the given figure, if $\triangle OAP \cong \triangle OBQ$, then which of the following is/are true?

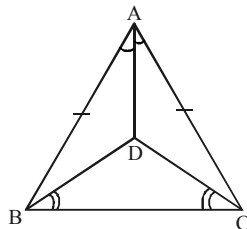


- (a) $AO = BO$ (b) $AP = BQ$
(c) $PO = BO$ (d) $\angle APO = \angle BQO$

Assertion & Reason :

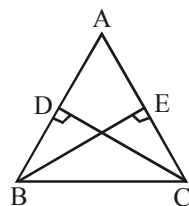
DIRECTIONS : Each of these questions contains an Assertion followed by Reason. Read them carefully and answer the question on the basis of following options. You have to select the one that best describes the two statements.

- (a) If both **Assertion** and **Reason** are correct and **Reason** is the correct explanation of **Assertion**.
(b) If both **Assertion** and **Reason** are correct but **Reason** is not the correct explanation of **Assertion**.
(c) If **Assertion** is correct but **Reason** is incorrect.
(d) If **Assertion** is incorrect but **Reason** is correct.
1. **Assertion:** In the adjoining figure, $\triangle ABD \cong \triangle ACD$,



Reason: If two angles and included side of a triangle are equal to the corresponding two angles and included side of another triangle, then the two triangles are congruent.

2. **Assertion:** In the given figure, $CD \perp AB$, $BE \perp AC$ and $CD = BE$, then $\triangle BEC \cong \triangle BDC$



Reason: In right angled triangles, if the hypotenuse and any one side of a triangle is equal to hypotenuse and corresponding side of the other, then triangles are congruent.

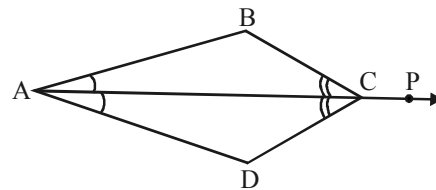
Passage Based Questions :

DIRECTIONS : Study the given passage(s) and answer the following questions.

PASSAGE - I

Two triangles are congruent, if two angles and the included side of one are equal to the corresponding two angles and included side of the other.

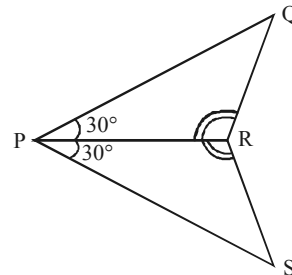
1. In the given figure, $\triangle ABC \cong \dots\dots\dots$



- (a) $\triangle CDA$ (b) $\triangle DAC$
(c) $\triangle ADC$ (d) $\triangle ACD$

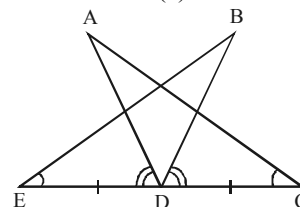
2. In the given figure, $\angle Q = \dots\dots\dots$

- (a) $\angle QPR$ (b) $\angle S$
(c) $\angle PRS$ (d) $\angle PRS$



3. In the given figure, $BD = \dots\dots\dots$

- (a) AD (b) BE
(c) AC (d) None of these

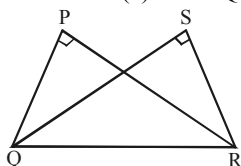


PASSAGE - II

Two right angled triangles are congruent, if the hypotenuse and one side of one triangle are equal to the hypotenuse and corresponding side of the other.

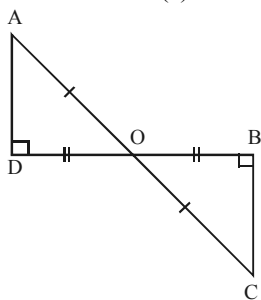
4. In the given figure, triangles PQR and SQR are right angled at P and S respectively and $PQ = SR$, then $\angle PRQ = \dots\dots$

- (a) $\angle SRP$ (b) $\angle SQR$
(c) $\angle QPR$ (d) $\angle SRQ$



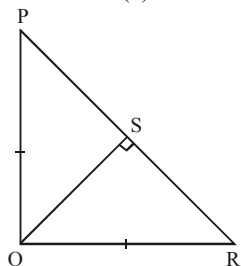
5. In the given figure, $\triangle BOC \cong \dots\dots$

- (a) $\triangle DOA$ (b) $\triangle ADO$
(c) $\triangle AOD$ (d) $\triangle DAO$



6. In the given figure, $PS = \dots\dots$

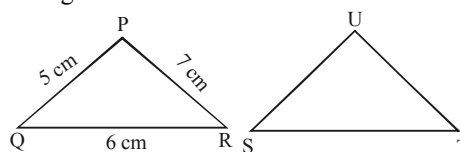
- (a) PQ (b) QS
(c) QR (d) RS



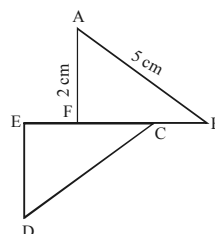
Integer Type Questions :

DIRECTIONS : Answer the following questions. The answer to each of the question is a single digit integer, ranging from 0 to 9.

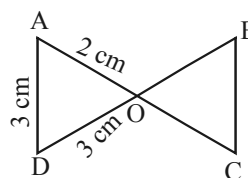
1. If $\triangle PQR$ is congruent to $\triangle STU$, then what is the length of TU?



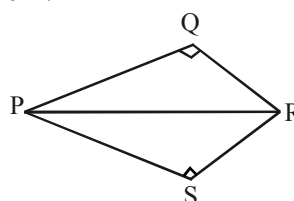
2. In the given figure, $\triangle ABF \cong \triangle DCE$. The value of CD is



3. In the given figure, $\triangle AOD \cong \triangle COB$. The value of OB is



4. In the given figure, $\triangle PQR$ is congruent to $\triangle PSR$ by RHS. The value of $\angle S$ is $(10k)^\circ$. Find the value of k.



SOLUTIONS

Brief Explanations of Selected Questions

1 EXERCISE

Fill in the Blanks :

- equal
- side
- congruent
- ΔRSP
- Length and breadth
- ΔDRQ
- same radius
- superimpose

True / False :

- | | | |
|----------|----------|----------|
| 1. False | 2. True | 3. False |
| 4. True | 5. True | 6. False |
| 7. False | 8. False | |

Match the Columns :

1. (A) $\rightarrow$ (r); (B) $\rightarrow$ (p); (C) $\rightarrow$ (s); (D) $\rightarrow$ (q)

Very Short Answer Questions :

- $\Delta ABC \cong \Delta DFE$ (S.A.S)
 - $\Delta ABC \cong \Delta MNO$ (R.H.S.)
 - $\Delta ABC \cong \Delta QRP$ (A.S.A.)
 - $\Delta ABC \cong \Delta QPR$ (S.S.S.)
- $AB = EF, BC = FD, AC = ED; \angle A = \angle E, \angle B = \angle F$ and $\angle C = \angle D$
 - $CA = QR, AB = RP, BC = PQ; \angle C = \angle Q, \angle A = \angle R$ and $\angle B = \angle P$
 - $XZ = QP, ZY = PR, XY = QR; \angle X = \angle Q, \angle Z = \angle P$ and $\angle Y = \angle R$
 - $MP = SQ, PN = QR, MN = SR; \angle M = \angle S, \angle P = \angle Q$ and $\angle N = \angle R$
- Yes, $\Delta ABD \cong \Delta CDB, \overline{AB} \leftrightarrow \overline{CD}, \overline{CB} \leftrightarrow \overline{AD}, \overline{BD} \leftrightarrow \overline{DB}$ (S.S.S.)
- AAS; $\Delta ABO \cong \Delta CDO$
 - ASA; $\Delta ADB \cong \Delta ADC$

Short Answer Questions :

- $\Delta ABC \cong \Delta ZXY$. by SAS congruence condition.
 - In ΔPQR , the included angle between PQ and PR is $\angle P$.
In ΔDEF , the included angle between DE and EF is $\angle E$.
We have, $PQ = EF = 5$ cm and $PR = DE = 6$ cm, but $\angle P \neq \angle E$.
So, the given triangles are not congruent.
- In ΔPQR and ΔSRQ ,
 $PQ = SR$ (given)
 $PR = SQ$ (given)
 $QR = RQ$ (common)
 $\therefore \Delta PQR \cong \Delta SRQ$ (SSS congruency)
- In ΔPLO and ΔPMO
 $PL = PM$ [given]
 $\angle PLO = \angle PMO = 90^\circ$
[$\because PL \perp OA$ and $PM \perp OB$]
 $PO = PO$ [common]
 $\therefore \Delta PLO \cong \Delta PMO$ (RHS congruency)
- In ΔAOC and ΔBOD ,
 $\angle C = \angle D = 90^\circ$
 $\angle AOC = \angle BOD$ [vertically opposite $\angle$ s]
 $AC = BD$ [given]
 $\therefore \Delta AOC \cong \Delta BOD$ (AAS congruency)
- In ΔBMC and ΔCNB ,
 $\angle BMC = \angle CNB = 90^\circ$ [$\because BM \perp AC$ and $CN \perp AB$]
 $\angle MCB = \angle NBC$ ($\because AB = AC \Rightarrow \angle C = \angle B$)
 $BC = CB$ (common)
 $\therefore \Delta BMC \cong \Delta CNB$ (AAS congruency)
- In ΔABD and ΔCDB ,
 $AB = CD$ (given)
 $\angle ABD = \angle CDB = 90^\circ$
 $BD = DB$ (common)
 $\therefore \Delta ABD \cong \Delta CDB$ (SAS congruency)

Long Answer Questions :

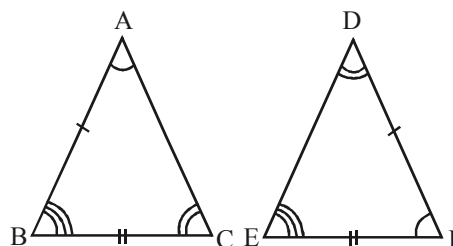
- Yes
 - In $\triangle ACD$ and $\triangle CAB$,
 $DC = AB$ [given]
 $\angle ACD = \angle CAB$ [$\because AB \parallel DC$ and AC is transversal.]
 $AC = AC$ [common]
 $\therefore \triangle ACD \cong \triangle CAB$ (SAS congruency)
 $\angle CAD = \angle ACB$ [C.P.C.T.]
- In $\triangle ABD$ and $\triangle ACD$,
 $\angle BAD = \angle CAD$ [$\because AD$ bisects $\angle A$]
 $AD = AD$ [common]
 $\angle ADB = \angle ADC = 90^\circ$ [$\because AD \perp BC$]
 $\therefore \triangle ABD \cong \triangle ACD$ (ASA congruency)
 $\Rightarrow AB = AC$ [C.P.C.T.]
Hence, $\triangle ABC$ is an isosceles triangle.
- In $\triangle AOQ$ and $\triangle AOP$,
 $\angle AQO = \angle APO = 90^\circ$
 $AO = AO$ [common]
 $OQ = OP$ [given]
 $\therefore \triangle AOQ \cong \triangle AOP$ (RHS congruency)
 $\Rightarrow AQ = AP$ [C.P.C.T.] (i)
and now, in $\triangle BOQ$ and $\triangle COP$,
 $\angle BQO = \angle CPO = 90^\circ$ [$\because OP \perp AC$ and $OQ \perp AB$]
 $OB = OC$ [$\because O$ is the mid-point of BC]
 $OQ = OP$ [given]
 $\therefore \triangle BOQ \cong \triangle COP$ (RHS congruency)
 $\Rightarrow BQ = CP$ [C.P.C.T.] (ii)
From (i) & (ii), we have $AQ + BQ = AP + CP$
 $\Rightarrow AB = AC$
Hence, $\triangle ABC$ is an isosceles triangle.
- $\angle SRP = \angle TRQ$ (given)
 $\Rightarrow \angle SRP + \angle SRT = \angle TRQ + \angle SRT$
 $\Rightarrow \angle TRP = \angle SRQ$ (i)
Now, in $\triangle SQR$ and $\triangle TPR$,
 $\angle SQR = \angle TPR$ (given)
 $QR = PR$ (given)
 $\angle SRQ = \angle TRP$ [from (i)]
 $\therefore \triangle SQR \cong \triangle TPR$ (ASA congruency)
 $\Rightarrow SR = TP$ (CPCT)
- In $\triangle ABC$ and $\triangle DCB$,
 $AB = DC$ [given]
 $\angle ABC = \angle DCB$ [given]
 $BC = CB$ [common]
 $\therefore \triangle ABC \cong \triangle DCB$ [SAS congruency]

- $\therefore AC = DB$ [CPCT]
 $\angle BAC = \angle CDB$ (i) [CPCT]
- In $\triangle ABE$ and $\triangle DCE$,
 $AB = DC$ [given]
 $\angle AEB = \angle DEC$ [vertically opposite angles]
 $\angle BAC = \angle CDB$ [from (i)]
 $\angle ABE = \angle DCE$ [3rd angle of $\triangle ABE$ is equal to 3rd angle of $\triangle DCE$]
 $\therefore \triangle ABE \cong \triangle DCE$ [ASA congruency]
 $\therefore BE = CE$ [CPCT]

2 EXERCISE

Text-Book Exercise :

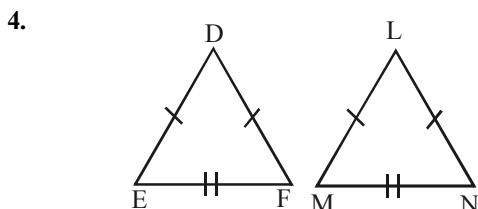
- (a) PE (b) EN (c) PN
 - (a) EN (b) AT
 - (a) $\angle PNE$ (b) $\angle EPN$
- No. This property represents that these triangles have their respective angles of equal measure. However, this gives no information about their sides. Therefore, AAA property does not prove that the two triangles are congruent.
- $\triangle RAT \cong \triangle WON$, by ASA criterion.
- In $\triangle BCA$ and $\triangle BTA$,
 $BC = BT$ (given)
 $CA = TA$ (given)
 $BA = BA$ (common)
 $\therefore \triangle BCA \cong \triangle BTA$ (SSS congruency)
Now, in $\triangle QRS$ and $\triangle TPQ$,
 $\angle PTQ = \angle RQS$ (given)
 $PT = RQ$ (given)
 $\angle QPT = \angle SRQ$ (given)
 $\therefore \triangle QRS \cong \triangle TPQ$ (ASA congruency)
- $\triangle ABC$ and $\triangle DEF$ are not congruent to each other but have five congruent parts i.e.



- $\angle BAC = \angle EFD$, $\angle ABC = \angle DEF$,
 $\angle BCA = \angle EDF$, $AB = DF$ and $BC = EF$
 6. $\angle ABC = \angle PQR = 90^\circ$ and
 $\angle BCA = \angle QRP$ [given]
 For triangles to be congruent $BC = QR$
 $\triangle ABC \cong \triangle PQR$ (ASA criterion)

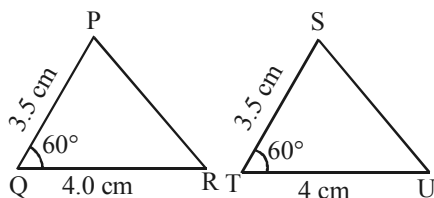
Exemplar Questions :

- (i) $\triangle ABD \cong \triangle CDB$ (ii) $\triangle XYZ \cong \triangle LMN$
 (iii) $\triangle AOD \cong \triangle BOC$
- (i) $\triangle ABD \cong \triangle ACD$ (ii) $\triangle XYZ \cong \triangle YXU$
 (iii) $\triangle ACE \cong \triangle BDE$ (iv) $\triangle ABC \cong \triangle CDE$
- (i) $\triangle LMN \cong \triangle LON$ (ii) $\triangle AOB \cong \triangle DOE$
 (iii) $\triangle STU \cong \triangle SAU$ (iv) $\triangle PSR \cong \triangle RQP$



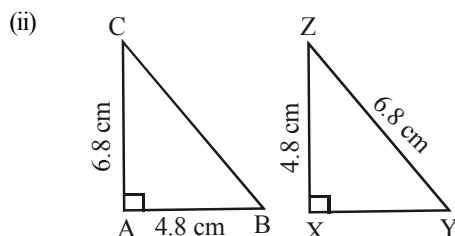
In triangles DEF and LMN,
 $DE = LM$ [given]
 $DF = LN$ [$\because DE = DF, DE = LM \text{ \& } LM = LN$]
 $EF = MN$ [given]
 $\therefore \triangle DEF \cong \triangle LMN$ (SSS congruence criterion)
 In $\triangle DEF$, $\angle E = \angle F = 40^\circ$
 [$\because$ Angles opposite equal sides]
 and $\angle F = \angle N = 40^\circ$ [C.P.C.T.]

- (i) $\triangle ABC \cong \triangle BAE$ (ii) $\triangle DOF \cong \triangle HOC$
 (iii) $\triangle PSQ \cong \triangle RQS$ (iv) $\triangle LMN \cong \triangle OMN$
- (i)



In $\triangle PQR$ and $\triangle STU$,
 $PQ = ST = 3.5 \text{ cm}$ [given]
 $\angle PQR = \angle STU = 60^\circ$ [given]
 $QR = TU = 4 \text{ cm}$ [given]
 $\therefore$ By SAS congruence criterion

$$\triangle PQR \cong \triangle STU$$



In $\triangle ABC$, $\angle A$ is included between two sides
 AB and AC.

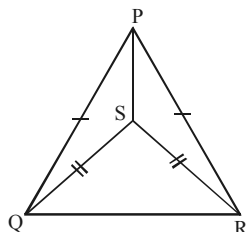
But, In $\triangle XYZ$, $\angle X$ is not included
 between the two sides XZ and YZ.

Therefore, $\triangle ABC$ is not congruent to
 $\triangle XYZ$.

- (i) In $\triangle PQR$ and $\triangle PSR$,
 $PQ = PS$ [given]
 $\angle QPR = \angle SPR$ [$\because \angle 1 = \angle 2$]
 $PR = PR$ [common]
 $\therefore \triangle PQR \cong \triangle PSR$ (SAS congruence criterion)
- Since corresponding parts of congruent
 triangles are equal.
 $\therefore QR = SR$
- In $\triangle DEF$ and $\triangle HIG$,
 $DE = HI$ [given]
 $\angle DEF = \angle HIG$ [given]
 $EG = FI$ [given]
 $\Rightarrow EG + GF = FI + GF$
 $\Rightarrow EF = IG$
 $\therefore$ By SAS congruence criterion
 $\triangle DEF \cong \triangle HIG$
- (i) Yes, In $\triangle ABC$ and $\triangle DCB$,
 $\angle ABC = \angle DCB = 40^\circ + 30^\circ = 70^\circ$ [given]
 $BC = BC$ [common]
 $\angle ACB = \angle DBC = 30^\circ$ [given]
 $\therefore$ By ASA congruence criterion
 $\triangle ABC \cong \triangle DCB$
- (ii) Yes, $AB = DC$ [C.P.C.T.]
 (iii) Yes, $AC = DB$ [C.P.C.T.]
- In $\triangle ABC$ and $\triangle EOD$,
 $AB = EO$ [given]
 $\angle ABC = \angle EOD = 90^\circ$ [given]
 $AC = ED$ [given]
 $\therefore$ By RHS congruence criterion,
 $\triangle ABC \cong \triangle EOD$

HOTS Questions :

1.



In $\triangle PSQ$ and $\triangle PSR$,
 $PQ = PR$ [given]
 $SQ = SR$ [given]
 $PS = PS$ [common]
 $\therefore$ By SSS congruence criterion,
 $\triangle PSQ \cong \triangle PSR$

2. (i) Since $\triangle DAB \cong \triangle BCD$

$$\Rightarrow \angle DAB = \angle BCD \quad [\text{CPCT}]$$

$$\Rightarrow 106^\circ = 2x + 12^\circ$$

$$\Rightarrow 2x = 106^\circ - 12^\circ$$

$$\Rightarrow x = \frac{94^\circ}{2} = 47^\circ$$

(ii) $\angle BCD = \angle DAB = 106^\circ$

In $\triangle BCD$,

$$\angle BCD + \angle BDC + \angle CBD = 180^\circ$$

$$\Rightarrow 106^\circ + 42^\circ + \angle CBD = 180^\circ$$

$$\Rightarrow \angle CBD = 180^\circ - 148^\circ = 32^\circ$$

$$\text{But } \angle ADB = \angle CBD = 32^\circ \quad [\text{C.P.C.T.}]$$

3. Since $\triangle ABC$ is an equilateral triangle

$$\therefore AB = BC = CA \dots (i)$$

Since C is the mid-point of DE.

$$\therefore DC = CE \dots (ii)$$

$$\text{Let } \angle DAC = \angle EBC = x^\circ$$

$$\therefore \angle DAC + \angle EBC = 180^\circ \quad [\text{given}]$$

$$\Rightarrow x^\circ + x^\circ = 180^\circ$$

$$\Rightarrow 2x^\circ = 180^\circ \Rightarrow x^\circ = 90^\circ$$

$$\Rightarrow \angle DAC = \angle EBC = 90^\circ \dots (iii)$$

Now, In $\triangle DAC$ and $\triangle EBC$,

$$AC = BC \quad [\text{from (i)}]$$

$$\angle DAC = \angle EBC = 90^\circ \quad [\text{From (iii)}]$$

$$DC = CE \quad [\text{From (ii)}]$$

$\therefore$ By RHS congruence criterion,

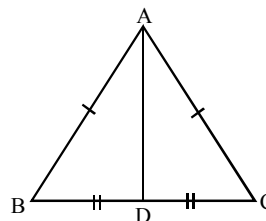
$$\triangle DAC \cong \triangle EBC$$

3 EXERCISE

Single Option Correct :

1. (a) $\triangle ABC \cong \triangle ADC$ (By SSS)
2. (c) AB is included between $\angle A$ and $\angle B$.
3. (b) Because two sides and angle between them is equal in both the triangles.

4. (b)



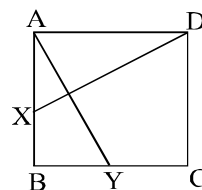
By RHS criteria $\triangle ABD \cong \triangle ACD$

5. (c) In $\triangle PQR$, $\angle P + \angle Q + \angle R = 180^\circ$
 $\angle P = 180^\circ - (65^\circ + 35^\circ) = 180^\circ - 100^\circ$
 $\angle P = 80^\circ$

$$\therefore \triangle PQR \cong \triangle UVW$$

$$\text{Hence, } \angle U = \angle P = x = 80^\circ$$

6. (c)



By RHS, $\triangle ADX \cong \triangle BAY$

Therefore (i), (ii) & (iii) are correct.

7. (d)
8. (b)
9. (b) EDF
10. (c) By SSS congruence criterion
 $\triangle ABC \cong \triangle DCB$

More Than One Option Correct :

1. (a, b, c) AAA is not a congruent criteria.
2. (b, c, d)
3. (a, c, d)
4. (b, c)
5. (a, b, c): $AB = AC$
 $\Rightarrow \angle ABC = \angle ACB$ (angles opposite to)

equal sides)
 $\Rightarrow \angle FBC = \angle ECB$ (i)
 Since, E and F are the mid-points of AC and AB respectively.

$$\therefore 2BF = AB = AC = 2CE$$

$$\Rightarrow BF = CE \quad \text{.....(ii)}$$

Now, in $\triangle BFC$ and $\triangle CEB$,

$$\angle FBC = \angle ECB \quad [\text{from (i)}]$$

$$BF = CE \quad [\text{from (ii)}]$$

$$BC = CB \quad (\text{common})$$

$$\therefore \triangle BFC \cong \triangle CEB \quad (\text{SAS congruency})$$

$$\therefore CF = BE, BF = CE \text{ and } \angle BFC = \angle CEB \quad (\text{CPCT})$$

6. (a, b, d):

Assertion & Reason :

- (b) In $\triangle ABD$ and $\triangle ACD$,
 $AB = AC$ (given)
 $\angle BAD = \angle CAD$ (given)
 $AD = AD$ (common)
 $\triangle ABD \cong \triangle ACD$ (SAS congruency)
Assertion: correct, **Reason:** correct
- (d) In $\triangle CDB$ and $\triangle BEC$,
 $CD = BE$ (given)
 $\angle BDC = \angle CEB = 90^\circ$ ($\because CD \perp AB$ and $BE \perp AC$)
 $BC = CB$ (common hypotenuse)
 $\therefore \triangle BEC \cong \triangle CDB$ (RHS congruency)
Assertion: incorrect, **Reason:** correct

Passage Based Questions :

- (c) In $\triangle ABC$ and $\triangle ADC$,
 $\angle BAC = \angle DAC$ [given]
 $\angle BCA = \angle DCA$ [given]
 $AC = AC$ [common]
 $\therefore \triangle ABC \cong \triangle ADC$ (ASA congruency)
- (b) In $\angle PQR$ and $\angle PSR$,
 $\angle QPR = \angle SPR = 30^\circ$ [given]
 $PR = PR$ [common]
 $\angle QRP = \angle SRP$ [given]
 $\therefore \triangle PQR \cong \triangle PSR$ (ASA congruency)
 $\Rightarrow \angle Q = \angle S$ [C.P.C.T.]

- (a) $\angle ADE = \angle BDC$ [given]
 $\Rightarrow \angle ADE + \angle ADB = \angle BDC + \angle ADB$
 $\Rightarrow \angle BDE = \angle ADC$ (i)

Now, In $\triangle BDE$ and $\triangle ADC$

$$\angle BDE = \angle ADC \quad [\text{From (i)}]$$

$$ED = CD \quad [\text{given}]$$

$$\angle BED = \angle ACD \quad [\text{given}]$$

$$\therefore \triangle BDE \cong \triangle ADC \quad (\text{ASA congruency})$$

$$\Rightarrow BD = AD \quad [\text{C.P.C.T.}]$$

- (b) In $\triangle PQR$ and $\triangle SRQ$,
 $\angle QPR = \angle RSQ = 90^\circ$ [given]
 $PQ = SR$ [given]
 $QR = QR$ [common]
 $\therefore \triangle PQR \cong \triangle SRQ$ (RHS congruency)
 $\Rightarrow \angle PRQ = \angle SQR$ [C.P.C.T.]

- (a) In $\triangle BOC$ and $\triangle DOA$,
 $\angle OBC = \angle ODA = 90^\circ$ [given]
 $OB = OD$ [given]
 $OC = OA$ [given]
 $\therefore$ By RHS congruence criterion,
 $\triangle BOC \cong \triangle DOA$

- (d) In $\triangle PSQ$ and $\triangle RSQ$,
 $\angle PSQ = \angle RSQ = 90^\circ$ [given]
 $PQ = RQ$ [given]
 $QS = QS$ [common]
 $\therefore$ By RHS congruence criterion
 $\triangle PSQ \cong \triangle RSQ$
 $\Rightarrow PS = RS$ [C.P.C.T.]

Integer Type Questions :

- (6) By congruency criteria SSS $\triangle PQR \cong \triangle STU$
 $\Rightarrow PQ = ST, QR = TU$ & $RP = US$
 So, $TU = QR = 6$ cm
- (5) Since $\triangle ABF \cong \triangle DCE$
 $\therefore CD = BA = 5$ cm
- (3) Since $\triangle AOD \cong \triangle COB$
 $\therefore OB = OD = 3$ cm
- (9) Since $\triangle PQR \cong \triangle PSR$
 $\therefore \angle S = \angle Q = 90^\circ$ [given]
 $\Rightarrow (10k)^\circ = (10 \times 9)^\circ$
 $\Rightarrow k = 9$

Chapter

8

Comparing Quantities

RATIO

Ratio is comparison of two or more quantities of the same kind using division. It shown as $a : b$. The first term a is called antecedent and term b is called consequent.

Properties of Ratio

1. In a ratio, the order of terms is very important. i.e., the ratio $2 : 3$ is different from the ratio $3 : 2$.
2. Since ratio is a fraction, the ratio will remain unchanged if each term of the ratio is multiplied or divided by the same non-zero number.

$$\text{e.g. } 4 : 7 = \frac{4}{7} = \frac{4 \times 3}{7 \times 3} = \frac{12}{21} \Rightarrow 4 : 7 = 12 : 21$$

3. (a) Ratio exists between quantities of the same kind, i.e.
 - (i) Ratio cannot exist between height and weight.
 - (ii) We cannot write a ratio between the age of a student and marks obtained by him.(b) Since ratio is a number, it has no units.
(c) To find the ratio of quantities of same kind, quantities should be in same unit.
4. To compare two ratios, we either convert them into equivalent like fractions (fractions with same denominator by finding the LCM of the denominators) or convert them to decimal form.

e.g. To compare $3 : 5$ and $2 : 3$

$$\frac{3}{5} \text{ and } \frac{2}{3}$$

LCM of 5 and 3 = 15

$$\frac{3}{5} = \frac{3}{5} \times \frac{3}{3} = \frac{9}{15} \text{ and } \frac{2}{3} \times \frac{5}{5} = \frac{10}{15}$$

$$\text{Since } \frac{10}{15} > \frac{9}{15} \Rightarrow \frac{2}{3} > \frac{3}{5}$$

i.e. $2 : 3 > 3 : 5$

PROPORTION

A proportion is an equation that states that two ratios are equal.

Four numbers are said to be in proportion if the ratio of the first two is equal to the ratio of the last two i.e. a, b, c and d are said to be in proportion if $a : b = c : d$.

This is expressed as $a : b :: c : d$. The first and fourth terms are called the **extremes** and second and third terms are called the **means**.

Product of extremes = Product of means

ILLUSTRATIONS 1 :

Determine the following numbers are in proportion or not 1.2, 2.7, 0.4, 0.9.

SOLUTION:

Product of extremes = $1.2 \times 0.9 = 1.08$

Product of means = $2.7 \times 0.4 = 1.08$.

Since product of extremes = Product of means

So these numbers are in proportion.

UNITARY METHOD

The method of finding the value of the required number of quantity by first finding the value of the unit quantity is called unitary method.

For example : A scooter consumes 28 litres of petrol in covering 2100 km. Thus, petrol will be needed to cover a distance of 3600 km can be calculated as:

2100 km can be covered in 28 litres of petrol

$\therefore$ 1 km can be covered in $\left(28 \times \frac{1}{2100}\right)$ litres

$\therefore$ 3600 km can be covered in $\left(\frac{28}{2100} \times 3600\right)$ litres = 48 litres.

PERCENTAGE

The word 'per cent' is an abbreviation of the Latin phrase 'per centum' which means per hundred or hundredths.

Thus, the term per cent means out of hundred or for every hundred.

By a certain per cent we mean that many hundredths.

For example, x percent is written as $x\%$ and $x\% = \frac{x}{100}$

Converting Percentage to Fraction or Decimals

To convert a given percentage to a fraction or decimal, divide it by 100 and remove the % sign.

For example : To convert $\frac{4}{5}\%$ into a fraction and decimal.

We divide it by 100 and remove the sign of %.

$$\therefore \left(\frac{4}{5} \div 100\right) = \frac{4}{5} \times \frac{1}{100} = \frac{1}{5 \times 20} = \frac{1}{100} = 0.01$$

Converting Fractions or Decimals to Percentage

To convert a given fraction or decimal into percentage, multiply it by 100 and put the % sign.

$$\frac{a}{b} = \left(\frac{a}{b} \times 100\right)\%$$

ILLUSTRATION 2 :

Convert the following into percent :

(i) $\frac{7}{50}$

(ii) 0.25

SOLUTION:

(i) $\frac{7}{50} = \left(\frac{7}{50} \times 100 \right) \% = (7 \times 2) \% = 14\%$

(ii) First convert the decimal into fraction and then convert that fraction into percent

i.e. $0.25 = \frac{25}{100} \Rightarrow \frac{25}{100} = \left(\frac{25}{100} \times 100 \right) \% = 25\%$

Increase or Decrease as Percent

- Percentage increase = $\left(\frac{\text{Increase in quantity}}{\text{Original quantity}} \times 100 \right) \%$
- Percentage decrease = $\left(\frac{\text{Decrease in quantity}}{\text{Original quantity}} \times 100 \right) \%$

PROFIT AND LOSS

Cost Price: The price at which an article is purchased is known as its.

The cost price is abbreviated as C.P.

Selling Price: The price at which an article is sold is known as its.

The selling price is abbreviated as S.P.

Profit: If the selling price (S.P.) of an article is greater than the cost price (C.P.), then the difference between the selling price and cost price is called *Profit*.

i.e. Gain = S.P. – C.P., if S.P. > C.P.

$$\text{Gain}\% = \frac{\text{Gain} \times 100}{\text{C.P.}}$$

Loss: If the selling price (S.P.) of an article is less than the cost price (C.P.), the difference between the cost price (C.P.) and the selling price (S.P.) is called Loss.

i.e. Loss = C.P. – S.P., if C.P. > S.P.

$$\text{Loss}\% = \frac{\text{Loss} \times 100}{\text{C.P.}}$$

ILLUSTRATION 3 :

By selling a washing machine for ₹ 7200, Rajesh loses 10%. Find the Cost Price of the washing machine.

SOLUTION:

Let cost price of washing machine be '₹ x'

Given, SP = ₹ 7200, % Loss = 10%

As % loss is given, so CP > SP

$$\Rightarrow \text{SP} = \text{CP} - \text{Loss}$$

$$\Rightarrow 7200 = x - 10 \% \text{ of } x$$

$$\Rightarrow x - \frac{10}{100}x = 7200$$

$$\Rightarrow \frac{9x}{10} = 7200$$

$$\Rightarrow x = \frac{7200 \times 10}{9} = 8000$$

$\therefore$ Cost price of washing machine = ₹ 8000

SIMPLE INTEREST

Whenever we borrow money from some lending source such as a bank or a financial institution, we have to pay some extra money to the service provider, which depends upon the sum borrowed by us and the period of time for which we wish to borrow it. This extra money is called **Interest**. The borrowed money is called **Principal**.

On the other hand, when we deposit money in a bank for safekeeping, we earn interest. Interest is calculated according to an agreement which specifies the rate of interest. Generally the rate of interest is taken as “per cent per annum” which means “per ₹100 per year.”

For example, a rate of ‘10% per annum’, means ₹ 10 on ₹ 100 for 1 year.

When interest is calculated simply on the original principal, it is known as **Simple Interest**. When the interest for a specific period is added to the principal, then the sum is called the **Amount**.

$$\therefore \text{S.I.} = \frac{P \times R \times T}{100}, \quad A = P + \text{S.I.}$$

$$P = \frac{100 \times \text{S.I.}}{R \times T}, \quad R = \frac{100 \times \text{S.I.}}{P \times T}, \quad T = \frac{100 \times \text{S.I.}}{R \times P}$$

Where, S.I. = Simple interest, P = Principal, R = Rate of interest, T = Time, A = Amount

ILLUSTRATION :

At what rate percent by simple interest, will a sum of money double itself in 5 years 4 months ?

SOLUTION:

Let $P = ₹ x$

Amount $A = ₹ 2x$

$$\therefore \text{S.I.} = A - P = ₹ 2x - ₹ x = ₹ x$$

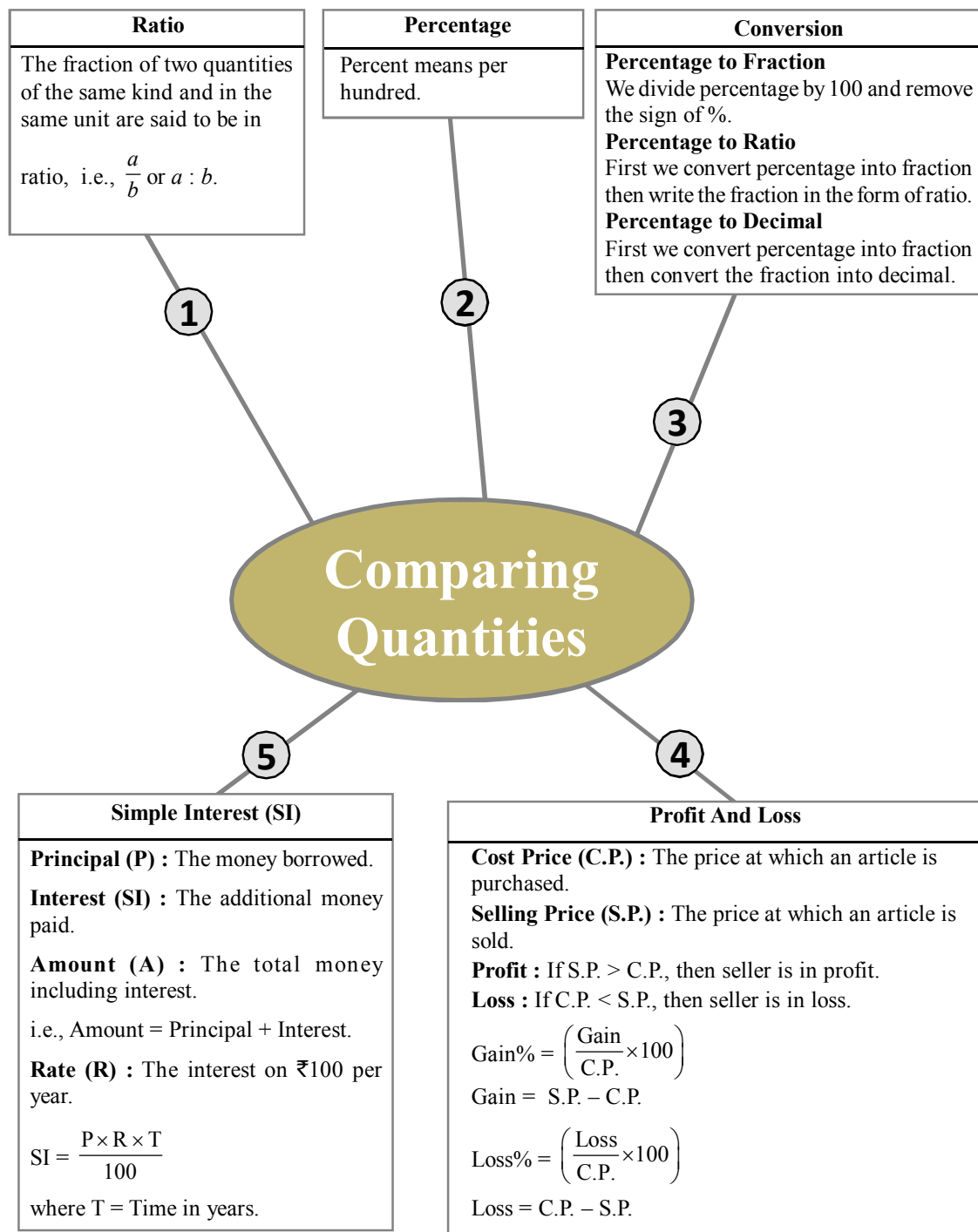
$$T = 5 \text{ years } 4 \text{ months} = 5\frac{4}{12} \text{ years} = 5\frac{1}{3} \text{ years} = \frac{16}{3} \text{ years}$$

Let R be the rate percent per annum.

$$\text{Using } R = \frac{\text{S.I.} \times 100}{P \times T}, \text{ We get } R = \frac{x \times 100}{x \times \frac{16}{3}} = \frac{300}{16} = 18.75.$$

Hence required rate = 18.75% p.a.

CONCEPT MAP



MISCELLANEOUS

SOLVED EXAMPLES

1. Divide ₹ 1350 among A, B and C in the ratio 2 : 3 : 4.

Sol. Sum of ratio terms = $(2 + 3 + 4) = 9$.

$$\text{A's share} = ₹ \left(1350 \times \frac{2}{9} \right) = ₹ 300.$$

$$\text{B's share} = ₹ \left(1350 \times \frac{3}{9} \right) = ₹ 450$$

$$\text{C's share} = ₹ \left(1350 \times \frac{4}{9} \right) = ₹ 600.$$

2. The first, third and fourth terms of a proportion are 12, 8, and 14, respectively. Find the second term.

Sol. Let the second term be x .

Then, 12, x , 8, 14 are in proportion.

Product of the extremes = Product of the means

$$\text{or } 12 \times 14 = x \times 8$$

$$\text{or } 168 = 8x$$

$$\text{or } x = 21$$

Hence, the second term of the proportion is 21.

3. Are 30, 40, 45, 60 in proportion?

Sol. We have,

$$30 : 40 = \frac{30}{40} = \frac{3}{4} \quad \text{and,} \quad 45 : 60 = \frac{45}{60} = \frac{3}{4}$$

$$\therefore 30 : 40 = 45 : 60$$

Hence, 30, 40, 45, 60 are in proportion.

4. The strength of a high school is 2,500. Out of this 600 are in pre-primary section, 800 in primary section, 700 in middle school, and the remaining are in high school. Find the percentage of students in each section.

Sol. Total students in the school = 2500

Percentage of pre-primary students

$$= \frac{600}{2500} \times 100 = 24\%$$

The percentage of students in primary section

$$= \frac{800}{2500} \times 100 = 32\%$$

The percentage of students in middle school

$$= \frac{700}{2500} \times 100 = 28\%$$

The students in high school

$$= 2500 - (600 + 800 + 700)$$

$$= 2500 - 2100 = 400$$

The percentage of students in high school

$$= \frac{400}{2500} \times 100 = 16\%$$

5. 6% of a certain number of oranges in a basket were found to be rotten. If there were 15 rotten oranges, how many fresh oranges were there in the basket?

Sol. Let the total number of oranges be x

$$\therefore 6\% \text{ of } x = 15$$

$$\Rightarrow \frac{6}{100} \times x = 15$$

$$\Rightarrow x = \frac{15 \times 100}{6} = 250$$

$$\therefore \text{Number of fresh oranges} = 250 - 15 = 235$$

6. Express the following as per cent :

(i) 6 : 5

(ii) 30 : 80

(iii) 0.002

(iv) 1.2

Sol. (i) $6 : 5 = \frac{6}{5} = \left(\frac{6}{5} \times 100 \right) \% = 120\%$

$$(ii) \quad 30 : 80 = \frac{30}{80} = \left(\frac{30}{80} \times 100 \right) \% = 37.5\%$$

$$(iii) \quad 0.002 = \frac{2}{1000} = \left(\frac{2}{1000} \times 100 \right) \% = 0.2\%$$

$$(iv) \quad 1.2 = \frac{12}{10} = \left(\frac{12}{10} \times 100 \right) \% = 120\%$$

7. A shopkeeper increased the price of a saree from ₹ 2000 by 15%. He then sold the saree at 10% less of the increased price. Find the

percentage increase or decrease in the price of the saree.

- Sol.** Original cost of the saree = ₹ 2000
 Percentage increase in price = 15%
 $\therefore$ 15% of 2000 = ₹ 300
 $\therefore$ Increased price = 2000 + 300 = ₹ 2300
 Percentage decrease in the increased price = 10%
 $\therefore$ 10% of ₹ 2300 = ₹ 230
 $\therefore$ Final price of the saree = 2300 – 230 = ₹ 2070
 $\therefore$ Change in price = 2070 – 2000 = ₹ 70 (increase)
 $\therefore$ Percentage increase

$$= \left(\frac{70}{2000} \times 100 \right) \% = 3.5\%$$

- 8. If the cost price of 15 pens is equal to the selling price of 20 pens, find the loss per cent.**

- Sol.** Let the CP of each pen be ₹ 1.
 CP of 20 pens = ₹ 20.
 SP of 20 pens = CP of 15 pens = ₹ 15.
 Thus, CP = ₹ 20 and SP = ₹ 15
 $\Rightarrow$ loss = (CP) – (SP) = ₹ (20 – 15) = ₹ 5

$$\Rightarrow \text{loss}\% = \left(\frac{\text{loss}}{\text{CP}} \times 100 \right) \%$$

$$= \left(\frac{5}{20} \times 100 \right) \% = 25\%.$$

Hence, loss % = 25%.

- 9. Anita borrowed ₹ 400 from her friend at the rate of 12% per annum for $2\frac{1}{2}$ years. Find the interest and amount paid by her.**

Sol. We have,

$$P = ₹ 400, R = 12\% \text{ per annum}, T = \frac{5}{2} \text{ years.}$$

Let I be the interest and A be the amount. Then,

$$I = \frac{PRT}{100} = ₹ \left(\frac{400 \times 12 \times \frac{5}{2}}{100} \right) = ₹ 120$$

$$\therefore A = P + I = ₹ 400 + ₹ 120 = ₹ 520$$

- 10. The simple interest on a certain sum of money at the rate of 5% p.a. for 8 years is ₹ 840. At what rate of interest the same amount of interest can be received on the same sum after 5 years?**

Sol. S.I. = ₹ 840, $R = 5\%$, $T = 8$ years.

$$P = \frac{I \times 100}{R \times T}$$

$$\text{Principal} = ₹ \left(\frac{100 \times 840}{5 \times 8} \right) = ₹ 2100.$$

Now, $P = ₹ 2100$, S.I. = ₹ 840, $T = 5$ years.

$$\therefore \text{Rate} = \left(\frac{100 \times 840}{2100 \times 5} \right) \% = 8\%$$

1

EXERCISE

Fill in the Blanks :

DIRECTIONS : Complete the following statements with an appropriate word / term to be filled in the blank space(s).

- $18\frac{3}{4}\% = \underline{\hspace{2cm}} : \underline{\hspace{2cm}}$
- 120% of 50 km = $\underline{\hspace{2cm}}$.
- $\frac{8}{5} = \underline{\hspace{2cm}}\%$
- A $\underline{\hspace{2cm}}$ with its denominator 100 is called a per cent.
- 15 kg is $\underline{\hspace{2cm}}\%$ of 50 kg.
- If the price of sugar is decreased by 20%, then the new price of 3kg sugar originally costing ₹ 120 will be $\underline{\hspace{2cm}}$.
- Sonal bought a bed sheet for ₹ 400 and sold it for ₹ 440. Her $\underline{\hspace{2cm}}\%$ is $\underline{\hspace{2cm}}$.
- If 20 lemons are bought for ₹ 10 and sold at 5 for three rupees, then profit % in the transaction is $\underline{\hspace{2cm}}\%$.
- 25 ml is $\underline{\hspace{2cm}}$ per cent of 5 litres.
- If A is increased by 20%, it equals B. If B is decreased by 50%, it equals C. Then $\underline{\hspace{2cm}}\%$ of A is equal to C.

True / False :

DIRECTIONS : Read the following statements and write your answer as true or false.

- If $8 : x :: 48 : 18$, then $x = 3$.
- $\frac{3}{4}$ as rate per cent is 75%.
- $12\frac{1}{2}\%$ expressed as a fraction is $\frac{1}{8}$.
- $2 : 5 = 25\%$
- 80% of 450 = 360.
- 20% of 1 litre = 200 ml.
- Loss is reckoned on selling price.
- Loss = (CP) – (SP).

$$9. \text{ Rate} = \frac{100 \times \text{SI}}{\text{P} \times \text{T}}$$

- Simple interest on ₹ 1000 at 5% per annum for 73 days is ₹ 10.

Match the Columns :

DIRECTIONS : Each question contains statements given in two columns which have to be matched. Statements (A, B, C, D) in column-I have to be matched with statements (p, q, r, s) in column-II.

- | 1. Column - I | Column - II |
|---|-------------|
| (A) At $x\%$ p.a. rate of interest ₹ 350 yields ₹ 105 in 6 years then x will be | (p) 4 |
| (B) What percent of 8 is 64 ? | (q) 25% |
| (C) What percent of 4.8 litres is 1200 ml? | (r) 5 |
| (D) In y years, ₹ 150 yield an interest of ₹ 36 at 6% p.a. | (s) 800% |

Very Short Answer Questions :

DIRECTIONS : Give answer in one word or one sentence.

- ₹ 2000 is divided among P, Q, R in the ratio 2 : 3 : 5. What share will Q get?
- The sides of a triangle are in ratio 4 : 5 : 9 if its perimeter is 72 cm. Find its longest side.
- $6\frac{1}{4}\%$ of y is 2, then find the value of y .
- What percentage of $2\frac{6}{7}$ is $\frac{2}{5}$?
- If 45% of 90 = $x\%$ of 81 find the value of x
- 270 candidates appeared for an examination, of which 252 passed. What is the pass percentage?
- Find the length whose 30% equals to 24 cm.
- Find the simple interest on ₹ 500 at 6% p.a. for 1 year.

9. What percent of $\frac{2}{7}$ is $\frac{1}{35}$?

10. What is 5% of 10% of ₹ 175 ?

Short Answer Questions :

DIRECTIONS : Give answer in 2-3 sentences.

1. A man buys a radio for ₹ 600 and sells it at a gain of 25%. What is the SP of the radio ?
2. The cost of 6 tins of milk is ₹ 300. Find the cost of 10 such tins of milk.
3. In a school 720 are boys and 40% are girls. Find the number of girls.
4. If the ratio of the cost price and selling price of an article is 5 : 6. Find the gain percent.
5. An item when sold for ₹ 1169 earned 40% profit on the cost price. What is the cost price ?
6. If 40% of a number is 256, then find 25% of that number.
7. If selling price of an article is $\frac{4}{3}$ of its cost price, what is profit%?
8. In an office the office hours are 10:30 a.m. to 5:30 p.m. and in between 30 minutes are spent on lunch. Find the ratio of office hours to the time spent on lunch.
9. The cost of 1 litre milk is ₹ 7.50. What is the cost of 30.5 litre milk ?

10. 35% population of a town are men and 40% are women. If the number of children are 20,000, then find the number of women.

Long Answer Questions :

DIRECTIONS : Give answer in four to five sentences.

1. Gopi borrowed ₹ 1800 at 12% per annum for 2 years and Krishna borrowed ₹ 1200 at 18% per annum for 3 years. What is the ratio of interests paid by them ?
2. A customer saves ₹ 15 when a discount of 20% on the marked price is offered. What did he pay for the article?
3. In an examination there were 640 boys and 360 girls. 60% of the boys and 80% of the girls were successful. What was the percentage of failure ?
4. On selling an article for ₹ 76, there is a profit of 52%. If it is sold for ₹ 75, then find the profit percent.
5. Divide ₹ 8000 into two parts so that the SI on the first part for 5 years at 12% per annum is equal to SI on the second part for 2 years at 18% per annum?
6. A sum of ₹ 450 amounts to ₹ 495 in 2 years. In what time will the sum of ₹ 820 amount to ₹ 943 at the same rate?

2

EXERCISE

Text-Book Exercise :

1. Find the ratio of :
(i) ₹ 5 to 50 paise (ii) 15 kg to 210 g
(iii) 9 m to 27 cm
(iv) 30 days to 36 hours
2. Population of Rajasthan = 570 lakhs and population of UP = 1660 lakhs
Area of Rajasthan = 3 lakhs km² and area of UP = 2 lakhs km²
(i) How many people are there per km² in both these states?
(ii) Which state is less populated?
3. Convert the given decimal fractions of percents.
(i) 0.65 (ii) 2.1
(iii) 0.02 (iv) 12.35
4. Find the whole quantity if
(i) 5% of it is 600
(ii) 12% of it is ₹ 1080
(iii) 40% of it is 500 km
(iv) 70% of it is 14 minutes
(v) 8% of it is 40 litres

5. Out of 15,000 voters in a constituency, 60% voted. Find the percentage of voters who did not vote. Can you now find how many actually did not vote?
6. Tell what is the profit or loss in the following transactions, Also find profit percent or loss percent in each case.
 - (i) Gardening shears bought for ₹ 250 and sold for ₹ 325.
 - (ii) A refrigerator bought for ₹ 12,000 and sold at ₹ 13,500.
 - (iii) A cupboard bought for ₹ 2,500 and sold at ₹ 3,000.
 - (iv) A skirt bought for ₹ 250 and sold at ₹ 150.
7. The population of a city decreased from 25,000 to 24,500. Find the percentage decrease.
8. Juhi sells a washing machine for ₹ 13,500. She loses 20% in the bargain. What was the price at which she bought it?
9.
 - (i) Chalk contains calcium, carbon and oxygen in the ratio 10 : 3 : 12. Find the percentage of carbon in chalk.
 - (ii) If in a stick of chalk, carbon, is 3 g, what is the weight of the chalk stick?
10. Find the amount to be paid at the end of 3 years in each case:
 - (i) Principal = ₹ 1,200 at 12% p.a.
 - (ii) Principal = ₹ 7,500 at 5% p.a.
11. If Meena gives an interest of ₹ 45 for one year at 9% rate p.a. What is the sum she has borrowed?
5. In a furniture shop, 24 tables were bought at the rate of ₹ 450 per table. The shopkeeper sold 16 of them at the rate of ₹ 600 per table and the remaining at the rate of 400 per table. Find her gain or loss percent.
6. Medha deposited 20% of her money in a bank. After spending 20% of the remainder, she has ₹ 4800 left with her. How much did she originally have.?
7. The cost of a flower vase got increased by 12%. If the current cost is ₹ 896, what was its original cost?
8. Radhika borrowed ₹ 12000 from her friends. Out of which ₹ 4000 were borrowed at 18% and the remaining at 15% rate of interest per annum. What is the total interest after 3 years?
9. Dhruvika invested money for a period from May 2006 to April 2008 at rate of 12% per annum. If interest received by her is ₹ 1620, find the money invested.
10. A person wanted to sell a scooter at a loss of 25%. But at the last moment he changed his mind and sold the scooter at a loss of 20%. If the difference in the two SP's is ₹ 4000, then find the CP of the scooter.

HOTS Questions :

Exemplar Questions :

1. Express $16\frac{2}{3}\%$ as a ratio in the lowest form.
2. What is 800% of 800?
3. What percent of ₹ 80 is ₹ 100?
4. 800 kg of mortar consists of 55% sand, 33% cement and rest lime. What is the mass of lime in mortar?
1. What is the difference in SI if a sum of ₹ 4000 is borrowed by a person for 4 years and the rate of interest is increased from 4% to 5%?
2. Out of a sum of ₹ 625, a part was lent at 5% and the remaining part at 10% simple interest. If the interest on the first part after two years is the same as S.I. on the second part after 4 years, then find the second part.
3. If A's salary is 50% more than B's, then by what % B's salary is less than A's salary?
4. A man purchased two articles for ₹ 3000 each. By selling those he gains 20% on the one and loses 20% on the other. Find the gain (or) loss per cent on the whole transaction.
5. What percent of ab is $a\%$ of $b + b\%$ of a ?

3

EXERCISE

Single Option Correct :

DIRECTIONS : This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONLY ONE is correct.

- (180 % of x) $\div 2 = 504$, then x is equal to
(a) 400 (b) 480
(c) 560 (d) 600
- 80% of 50% of 250% of 34 = ?
(a) 34 (b) 40
(c) 42.5 (d) 43
- A man spends 92% of his monthly income if he saves ₹ 220 per month, his monthly income is (in ₹)
(a) 2850 (b) 2650
(c) 2750 (d) 2700
- In an orchard, $12\frac{1}{2}$ % of the trees are apple tree. If the total number of trees in an orchard is 640. The number of trees other than apple trees are :
(a) 720 (b) 560
(c) 660 (d) 640
- In an examination, 94% of the candidates passed and 48 candidates are failed. Total candidates appeared in examination are
(a) 700 (b) 800
(c) 900 (d) 850
- If 10% of m is twice the 20% of n , then $m : n$ is equal to
(a) 2 : 3 (b) 1 : 4
(c) 4 : 1 (d) 5 : 2
- Three numbers A, B and C are in the ratio 12 : 15 : 25. If the sum of these numbers is 312, then the ratio between the difference of A and B and the difference of C and B is
(a) 3 : 10 (b) 7 : 3
(c) 3 : 7 (d) 10 : 3
- Rahul bought a motor cycle for ₹ 45000 and paid ₹ 200 for its cartage. He sold it at a gain of 10%. What was the S.P. of motor cycle?
(a) ₹ 54500 (b) ₹ 44520
(c) ₹ 49720 (d) ₹ 95200
- Mohan sold his watch for ₹ 190, thus bearing a loss of 5%. The cost price of the watch is
(a) ₹ 237.50 (b) ₹ 220
(c) ₹ 210 (d) ₹ 200
- Half of 1% written as a decimal is
(a) 0.2 (b) 0.02
(c) 0.05 (d) 0.005
- The sum of money that will produce ₹ 1770 as simple interest in 15 years at 8% per annum is
(a) ₹ 1475 (b) ₹ 3120
(c) ₹ 2800 (d) ₹ 3200
- Find the cost of one saree, if cost of 12 sarees is ₹ 3600.
(a) ₹ 300 (b) ₹ 350
(c) ₹ 250 (d) ₹ 200
- First, third and fourth terms of a proportion are 6, 12 and 36. Then the second term is
(a) 12 (b) 18
(c) 16 (d) 108
- In what time will ₹ 72 become ₹ 81 at $6\frac{1}{4}$ % p.a. SI?
(a) $1\frac{1}{2}$ year (b) $2\frac{1}{2}$ years
(c) 2 years (d) None of these
- Bhanu borrowed a certain sum of money at 12% per annum for 3 years and Madhuri borrowed the same sum at 24% per annum for 10 years. Find the ratio of their amounts.
(a) 1 : 3 (b) 2 : 1
(c) 2 : 3 (d) 2 : 5

More Than One Option Correct :

DIRECTIONS : This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONE or MORE may be correct.

- Which of the following arrangement of the numbers 75, 4, 3, 100 form a Proportion ?
(a) 100, 3, 75, 4 (b) 3, 4, 75, 100
(c) 3, 100, 4, 75 (d) 3, 75, 4, 100

2. To get 35% of a number, the number should be multiplied by
 (a) $\frac{7}{20}$ (b) 3.5
 (c) $\frac{7}{50}$ (d) 0.35
3. Which of the following ratio is/are equal to 2 : 3?
 (a) 14 : 21 (b) 15 : 30
 (c) 34 : 51 (d) 4 : 9
4. A shirt is made of cotton and polyster only. Then which of the following can be possible composition of shirt?
 (a) $\frac{2}{5}$ part cotton and $\frac{3}{5}$ part polyster
 (b) 70% cotton and 30% polyster
 (c) $\frac{7}{9}$ part Cotton and $\frac{2}{9}$ part Polyster
 (d) 50% Cotton and 20% polyster
5. If 1 litre solution contains 600 ml milk and 400 ml water, then which of the following is/are true for composition of solution?
 (a) 60% milk and 40% water
 (b) 70% milk and 30% water
 (c) Milk : Water :: 3 : 2
 (d) Milk : Water :: 7 : 3

Assertion & Reason :

DIRECTIONS : Each of these questions contains an Assertion followed by Reason. Read them carefully and answer the question on the basis of following options. You have to select the one that best describes the two statements.

- (a) If both **Assertion** and **Reason** are correct and **Reason** is the correct explanation of **Assertion**.
 (b) If both **Assertion** and **Reason** are correct but **Reason** is not the correct explanation of **Assertion**.
 (c) If **Assertion** is correct but **Reason** is incorrect.
 (d) If **Assertion** is incorrect but **Reason** is correct.
1. **Assertion :** Cost Price of an article is ₹ 100, its selling price is ₹ 150, Profit is 33.3%.
Reason : Profit or loss percent is calculated on cost price.
2. **Assertion:** In a class of 90 students 60% are boys it means 54 students are boys.
Reason: In that class 30% students are girls.

3. **Assertion:** An article is sold at ₹ 1425 at a loss of 5%. Its C.P is ₹ 1500.

Reason: If there is loss, cost price is always greater than selling price.

Passage Based Questions :

DIRECTIONS : Study the given passage(s) and answer the following questions.

PASSAGE - I

In a school, there are 180 students out of which 70% are girls. Also, there are 10 teachers out of which 80% teachers are female.

- Ratio of boys to girls is
 (a) 4 : 9 (b) 3 : 7
 (c) 2 : 3 (d) 5 : 7
- Ratio of number of boys to number of male teachers is
 (a) 27 : 1 (b) 29 : 2
 (c) 25 : 2 (d) 31 : 2
- Ratio of number of girls to number of male teacher
 (a) 1 : 63 (b) 2 : 29
 (c) 63 : 1 (d) 29 : 2
- Number of female teachers is what percentage of total number of students.

- (a) $3\frac{4}{7}\%$ (b) $3\frac{1}{8}\%$
 (c) $4\frac{1}{6}\%$ (d) $4\frac{4}{9}\%$

PASSAGE - II

₹ 5000 is invested at a rate of 5% p.a. simple interest for 5 years. After 5 years whole amount is again invested for 2 years at a rate of 3% p.a.

- Find simple interest in 5 years.
 (a) ₹ 1000 (b) ₹ 1200
 (c) ₹ 1250 (d) ₹ 1300
- How much money is invested after 5 years.
 (a) ₹ 6000 (b) ₹ 7000
 (c) ₹ 6500 (d) ₹ 6250
- Find total interest in 7 years.
 (a) ₹ 1625 (b) ₹ 1600
 (c) ₹ 1750 (d) ₹ 1550
- What will be the total amount after 7 years.
 (a) ₹ 8000 (b) ₹ 9000
 (c) ₹ 6625 (d) ₹ 7000

Integer Type Questions :

DIRECTIONS : Answer the following questions. The answer to each of the question is a single digit integer, ranging from 0 to 9.

1. Value of x in Proportion $2 : 3 :: 4 : x$ is
2. The value of 5.5% of $40 + 3\%$ of 60 is
3. If ₹ 250 amounts to ₹ 285 in 2 years, the rate percent per annum is _____ %
4. In how many years will the simple interest on a sum of money be equal to the principal at the rate of $16\frac{2}{3}\%$ per annum?
5. What percent of 6.5 litres is 130 ml?

Multiple Matching Question :

DIRECTIONS : Following question has four statements (A, B, C and D) given in Column-I and six statements (p, q, r, s, t, u) in Column-II. Any given statement in Column-I can have correct matching with one or more statement(s) given in Column-II.

- | 1. | Column-I | Column-II |
|-----|--------------------------|-------------|
| (A) | 21 days of 5 weeks | (p) 1 : 4 |
| (B) | 225% equal | (q) 100 : 7 |
| (C) | 120 seconds to 8 minutes | (r) 27 : 12 |
| (D) | 4 m to 28 cm | (s) 9 : 4 |
| | | (t) 3 : 5 |
| | | (u) 2 : 8 |

SOLUTIONS

Brief Explanations of Selected Questions

1 EXERCISE

Fill in the Blanks :

- | | | |
|---------------|---------|---------|
| 1. 3 : 16 | 2. 60km | 3. 160 |
| 4. fraction | 5. 30 | 6. ₹ 96 |
| 7. Profit, 10 | 8. 20 | 9. 0.5 |
| 10. 60 | | |

True / False :

- | | | | |
|---------|----------|----------|----------|
| 1. True | 2. True | 3. True | 4. False |
| 5. True | 6. True | 7. False | 8. True |
| 9. True | 10. True | | |

Match the Columns :

1. (A) $\rightarrow$ (r); (B) $\rightarrow$ (s); (C) $\rightarrow$ (q); (D) $\rightarrow$ (p)
 (A) P = ₹ 350, S.I = ₹ 105, T = 6 years

$$R = \frac{S.I \times 100}{P \times T} = \frac{105 \times 100}{350 \times 6} = 5\% \text{ p.a.}$$

- (B) Let 64 is x% of 8

$$\Rightarrow \frac{x}{100} \times 8 = 64 \Rightarrow x = \frac{64 \times 100}{8} = 800\%$$

- (C) 1200 ml = 1.2l

$$\frac{1.2l}{4.8l} \times 100 = 25\%$$

- (D) P = ₹ 150, S.I = ₹ 36, R = 6% p.a.

$$T = \frac{S.I \times 100}{P \times R} = \frac{36 \times 100}{150 \times 6} = 4 \text{ years}$$

Very Short Answer Questions :

1. Total parts = 2 + 3 + 5 = 10
 $Q \text{ gets} = 3 \times \frac{2000}{10} = ₹ 600$
2. Total Parts = 4 + 5 + 9 = 18
 {Perimeter of triangle = Sum of all sides
 = 72 cm}

$$\text{longest side} = 9 \times \frac{72}{18} \text{ cm} = 36 \text{ cm}$$

3. $6\frac{1}{4}\%$ of y = 2

$$\frac{25}{4 \times 100} \times y = 2 \Rightarrow y = 32$$

4. $\left(\frac{\frac{2}{5}}{2\frac{6}{7}} \right) \times 100 = \frac{\frac{2}{5}}{\frac{20}{7}} \times 100 = \frac{2}{5} \times \frac{7}{20} \times 100 = 14\%$

5. 45% of 90 = x% of 81

$$\frac{45}{100} \times 90 = \frac{x}{100} \times 81 \Rightarrow x = \frac{45 \times 90}{81} = 50$$

6. $\frac{252}{270} \times 100 = \frac{280}{3} = 93\frac{1}{3}\%$

7. 30% of x = 24

$$\frac{30}{100} \times x = 24 \Rightarrow x = \frac{24 \times 100}{30} = 80$$

8. S.I = $\frac{500 \times 6 \times 1}{100} = ₹ 30$

9. $\frac{x}{100} \times \frac{2}{7} = \frac{1}{35} \Rightarrow x = \frac{7 \times 100}{35 \times 2} = 10$

10. 10% of ₹ 175 = $175 \times \frac{10}{100} = \frac{175}{10}$

$$5\% \text{ of } \frac{175}{10} = \left(\frac{175}{10} \right) \times \frac{5}{100} = \frac{175}{200} = 0.875$$

Short Answer Questions :

1. $SP = C.P. + 25\% \text{ of } CP = \left(1 + \frac{25}{100} \right) CP$
 $= \frac{(100+25)}{100} \times 600 = ₹ 750$

2. Cost of 10 tins of milk = $\frac{300}{6} \times 10 = ₹ 500$

3. % of boys = $(100 - 40)\% = 60\%$
Let total number of students be x
 60% of $x = 720$

$$\frac{60}{100} \times x = 720$$

$$x = \frac{720 \times 100}{60} = 1200$$

$$\text{Number of girls} = \frac{40}{100} \times 1200 = 480$$

4. $\frac{C.P.}{S.P.} = \frac{5}{6} \Rightarrow S.P. = \frac{6}{5} C.P.$

$$\text{gain \%} = \frac{S.P. - C.P.}{C.P.} \times 100$$

$$= \left(\frac{6}{5} - 1 \right) \frac{C.P.}{C.P.} \times 100$$

$$= \frac{1}{5} \times 100 = 20\%$$

5. $S.P. = ₹ 1169$, Profit = 40%
 $S.P. = C.P. + 40\%$ of $C.P.$

$$\Rightarrow 1169 = \left(1 + \frac{40}{100} \right) C.P.$$

$$\Rightarrow C.P. = \frac{1169 \times 100}{(100 + 40)} = \frac{1169 \times 100}{140} = ₹ 835$$

6. $\frac{40}{100}x = 256 \Rightarrow x = 640$,

$$\therefore \frac{25}{100} \times 640 = 160$$

7. $S.P. = \frac{4}{3} C.P.$, $\frac{S.P.}{C.P.} = \frac{4}{3}$

$$\text{Profit\%} = \left(\frac{S.P. - C.P.}{C.P.} \right) \times 100 = \left(\frac{S.P.}{C.P.} - 1 \right) \times 100$$

$$= \left(\frac{4}{3} - 1 \right) \times (100) = \frac{100}{3} = 33\frac{1}{3}\%$$

8. Office hours = $10 : 30$ a.m. to $5 : 30$ p.m.
 $= 7$ hrs = 420 min.

$$\text{Lunch time} = 30 \text{ min.}$$

$$\text{Ratio} = 420 : 30 = 14 : 1$$

9. Cost of 1 litre milk = ₹ 7.50

$$\text{Cost of } 30.5 \text{ litre of milk} = ₹ (7.50 \times 30.5) \\ = ₹ 228.75$$

10. Let total population be x .

$$\therefore 35\% \text{ are men and } 40\% \text{ are women,} \\ 25\% \text{ are children.}$$

$$\Rightarrow \frac{25}{100} \times x = 20,000 \Rightarrow x = 80,000$$

$$\text{Number of women} = \frac{40}{100} \times 80,000 = 32,000$$

Long Answer Questions :

1.

Gopi	Krishna
$P = ₹ 1800$	$P = ₹ 1200$
$R = 12\%$	$R = 18\%$
$T = 2$ years	$T = 3$ years

$$I_1 = \frac{PTR}{100}$$

$$I_2 = \frac{PTR}{100}$$

$$= \frac{1800 \times 2 \times 12}{100}$$

$$= \frac{1200 \times 3 \times 18}{100}$$

$$= ₹ 432$$

$$= ₹ 648$$

$$I_1 : I_2 = 432 : 648 = 2 : 3.$$

2. Let the original price be ₹ x .

$$\therefore 20\% \text{ of } x = ₹ 15$$

$$\Rightarrow \frac{20}{100}x = 15$$

$$\Rightarrow x = \frac{15 \times 100}{20} = ₹ 75$$

$$\therefore \text{He pays} = ₹ (75 - 15) = ₹ 60$$

3. Number of failures = 40% of $640 + 20\%$ of 360

$$= \frac{40}{100} \times 640 + \frac{20}{100} \times 360 = 256 + 72 = 328$$

$$\therefore \% \text{ of failures} = \left(\frac{328}{640 + 360} \right) \times 100$$

$$= \frac{328}{1000} \times 100 = 32.8\%$$

4. S.P of an article = ₹ 76

Profit = 52%

$$SP = CP + 52\% \text{ of } CP = \left(1 + \frac{52}{100}\right) CP$$

$$C.P = \frac{76 \times 100}{(100 + 52)} = \frac{100}{2} = ₹ 50$$

If S.P = ₹ 75

Gain = ₹ (75 - 50) = ₹ 25

$$\text{Gain}\% = \frac{25}{50} \times 100 = 50\%$$

5. Let the first part be x.

Second part = 8000 - x

$$\therefore \frac{x \times 5 \times 12}{100} = \frac{(8000 - x) \times 2 \times 18}{100}$$

$$\Rightarrow 5x = 24000 - 3x$$

$$\Rightarrow 8x = 24000$$

$$\Rightarrow x = 3000$$

$\therefore$ First part = ₹ 3000.

Second part = ₹ 5000.

6. P = ₹ 450; A = ₹ 495; T = 2 years;

$$I = A - P = 495 - 450 = ₹ 45$$

$$R = \frac{100 \times I}{P \times T} = \frac{100 \times 45}{450 \times 2} = 5\%$$

Now take

$$P = ₹ 820; A = ₹ 943; R = 5\%; T = ?$$

$$I = A - P = 943 - 820 = ₹ 123.$$

$$T = \frac{100 \times I}{P \times R} = \frac{100 \times 123}{820 \times 5} = 3 \text{ years.}$$

2 EXERCISE

Text-Book Exercise :

1. (i) Required ratio

$$= \frac{₹ 5}{50 \text{ Paise}} = \frac{500 \text{ Paise}}{50 \text{ Paise}} = \frac{10}{1} \text{ or } 10 : 1.$$

$$(ii) \text{ Required ratio} = \frac{15 \text{ kg}}{210 \text{ g}} = \frac{15000 \text{ g}}{210 \text{ g}}$$

$$= \frac{500}{7} \text{ or } 500 : 7.$$

$$(iii) \text{ Required ratio} = \frac{9\text{m}}{27\text{cm}} = \frac{900\text{cm}}{27\text{cm}} \\ = \frac{100}{3} \text{ or } 100 : 3.$$

$$(iv) \text{ Required ratio} = \frac{30 \text{ days}}{36 \text{ hrs}} = \frac{720 \text{ hrs}}{36 \text{ hrs}} \\ = \frac{20}{1} \text{ or } 20 : 1.$$

2. (i) Population of Rajasthan in 3 km² area = 570 lakhs

Population of Rajasthan in 1 km² area

$$= \frac{570}{3} = 190$$

Population of U.P in 2 km² area = 1660

Population of U.P in 1 km² area

$$= \frac{1660}{2} = 830$$

- (ii) Population per km² area is lesser for Rajasthan. So, Rajasthan is less populated.

3. (i) $0.65 \times 100\% = 65\%$

$$(ii) 2.1 \times 100\% = 210\%$$

$$(iii) 0.02 \times 100\% = 2\%$$

$$(iv) 12.35 \times 100\% = 1235\%$$

4. Let the whole quantity be x.

- (i) 5% of x = 600

$$\Rightarrow \frac{5}{100} \times x = 600 \Rightarrow x = 12000$$

- (ii) 12% of x = ₹ 1080

$$\Rightarrow \frac{12}{100} \times x = ₹ 1080 \Rightarrow x = ₹ 9000$$

- (iii) 40% of x = 500 km

$$\Rightarrow \frac{40}{100} \times x = 500 \text{ km} \Rightarrow x = 1250 \text{ km}$$

- (iv) 70% of x = 14 min

$$\Rightarrow x \times \frac{70}{100} = 14 \text{ min} \Rightarrow x = 20 \text{ min}$$

- (v) 8% of x = 40 l

$$\Rightarrow \frac{8}{100} \times x = 40 \text{ l} \Rightarrow x = 500 \text{ l}$$

5. Percentage of voters who voted = 60%
Percentage of voters who did not vote

- $= 100\% - 60\% = 40\%$
 Number of voters who did not vote
 $= 40\% \text{ of } 15000$
 $= \frac{40}{100} \times 15000 = 6000$
6. (i) Profit = $325 - 250 = ₹ 75$
 Profit % = $\frac{75}{250} \times 100 = 30\%$
 (ii) Profit = $13500 - 12000 = ₹ 1500$
 Profit % = $\frac{1500}{12000} \times 100 = 12.5\%$
 (iii) Profit = $3000 - 2500 = ₹ 500$
 Profit % = $\frac{500}{2500} \times 100 = 20\%$
 (iv) Loss = $250 - 150 = ₹ 100$
 Loss % = $\frac{100}{250} \times 100 = 40\%$
7. Decrease in population = $25000 - 24500 = 500$
 Decrease % = $\frac{500}{25000} \times 100 = 2\%$
8. Let the cost price be ₹ x .
 $\therefore$ Loss = $20\% \text{ of } x$
 Cost price - Loss = Selling price
 $\Rightarrow x - \left(\frac{20}{100} \times x\right) = 13500$
 $\Rightarrow x - \frac{1}{5}x = 13500 \Rightarrow \frac{4}{5}x = 13500$
 $\therefore x = 13500 \times \frac{5}{4} = 16875$
9. (i) Ratio of calcium, carbon and oxygen
 $= 10 : 3 : 12$
 $\Rightarrow$ Percentage of carbon
 $= \frac{3}{25} \times 100\% = 12\%$
 (ii) Let the weight of the chalk stick be x g.
 $12\% \text{ of } x = 3 \Rightarrow \frac{12}{100} \times x = 3$
 $\therefore x = 3 \times \frac{100}{12} = 25\text{g}$
10. (i) S.I. = $\frac{1200 \times 12 \times 3}{100} = ₹ 432$

$$\text{Amount} = ₹ (1200 + 432) = ₹ 1632$$

$$(ii) \text{ S.I.} = \frac{7500 \times 5 \times 3}{100} = ₹ 1125$$

$$\text{Amount} = ₹ (7500 + 1125) = ₹ 8625$$

$$11. \text{ S.I.} = \frac{P \times R \times T}{100}$$

$$\Rightarrow 45 = \frac{P \times 9 \times 1}{100} \Rightarrow P = ₹ 500$$

Exemplar Questions :

1. $16\frac{2}{3}\% = \frac{50}{3 \times 100} = \frac{1}{6} = 1 : 6$
2. $800\% \text{ of } 800 = \frac{800}{100} \times 800 = 6400$
3. $\left(\frac{100}{80} \times 100\right)\% = 125\%$
4. Percentage of lime = $100\% - (55 + 33)\% = 12\%$
 $\therefore$ The mass of lime = $12\% \text{ of } 800 \text{ kg}$
 $= \left(\frac{12}{100} \times 800\right) \text{ kg} = 96 \text{ kg}$
5. C.P. of 24 tables = ₹ $(24 \times 450) = ₹ 10800$
 S.P. of 16 tables = ₹ $(16 \times 600) = ₹ 9600$
 S.P. of 8 tables = ₹ $(8 \times 400) = ₹ 3200$
 Total S.P. = ₹ $(9600 + 3200) = ₹ 12800$
 Gain = ₹ $12800 - ₹ 10800 = ₹ 2000$
 Gain % = $\left(\frac{2000}{10800} \times 100\right)\% = 18.52\%$
6. Let Medha initially have ₹ 100.
 $\therefore$ After depositing ₹ 20 in bank, she left with ₹ 80.
 Money she spends = $20\% \text{ of } 80$
 $= \frac{20 \times 80}{100} = ₹ 16$
 Now, money left with her
 $= ₹ (80 - 16) = ₹ 64$
 When she left with ₹ 64, she initially had
 $= ₹ 100$
 $\therefore$ When she left with ₹ 4800, she initially had
 $= ₹ \frac{100}{64} \times 4800 = ₹ 7,500$

7. Let the original cost be ₹ 100.
 $\therefore$ The cost after 12% increase = ₹ 112
 When current cost is ₹ 112, the original cost is ₹ 100.
 $\therefore$ When current cost is ₹ 896, the original cost

$$= \frac{₹100 \times 896}{112} = ₹800$$

8. S.I. on ₹ 4000 = $\frac{4000 \times 18 \times 3}{100} = ₹2160$

S.I. on ₹ 8000 = $\frac{8000 \times 15 \times 3}{100} = ₹3600$

$\therefore$ Total S.I. = ₹ (2160 + 3600) = ₹ 5760

9. Let the money invested be ₹ P

$$\therefore P = \frac{S.I \times 100}{R \times T} = \frac{1620 \times 100}{12 \times 2} = ₹6750$$

10. Let the C.P of the scooter be ₹ 100.
 When loss is 25%, then S.P. = ₹ 75
 When loss is 20%, the S.P. = ₹ 80
 Difference in the two S.P. = ₹ (80 - 75) = ₹ 5
 When difference is ₹ 5, C.P is ₹ 100.
 $\therefore$ When difference is ₹ 4000, then

$$C.P. = ₹ \left(\frac{100 \times 4000}{5} \right) = ₹80000$$

HOTS Questions :

1. $P = ₹4000$; $T = 4$ years, $R_1 = 4\%$; $R_2 = 5\%$
 Difference in two S.Is.

$$= \frac{4000 \times 5 \times 4}{100} - \frac{4000 \times 4 \times 4}{100}$$

$$= 800 - 640 = ₹160$$
2. Let Ist part be x and IInd part be $(625 - x)$.

$$\therefore \frac{x \times 5 \times 2}{100} = \frac{(625 - x) \times 10 \times 4}{100}$$

$$x = ₹500$$
 $\therefore$ the second part = $625 - 500 = ₹125$
3. Let B's salary be ₹ 100, so A's salary is ₹ 150.
 So, % of B's salary less than A's salary

$$= \frac{50}{150} \times 100 = \frac{100}{3} = 33\frac{1}{3}\%$$

4. Total CP = $3000 + 3000 = ₹6000$
 Gain = 20% of 3000 = ₹ 600
 SP of Ist = ₹ (3000 + 600) = ₹ 3600
 Loss = 20% of 3000 = ₹ 600
 S.P. of IInd = ₹ (3000 - 600) ₹ 2400
 Total of SP = ₹ 3600 + ₹ 2400 = ₹ 6000.
 $\therefore$ No profit, no loss.

5. $\frac{ab}{100} + \frac{ba}{100} = \frac{2ab}{100} = 2\%$ of ab

3 EXERCISE

Single Option Correct :

1. (c) $(180\% \text{ of } x) \times \frac{1}{2} = 504$

$$\Rightarrow \frac{180}{100} \times x \times \frac{1}{2} = 504 \Rightarrow \frac{9x}{10} = 504$$

$$\Rightarrow x = \frac{504 \times 10}{9} = 560$$
2. (a) $\frac{80}{100} \times \frac{50}{100} \times \frac{250}{100} \times 34$

$$= \frac{4}{5} \times \frac{1}{2} \times \frac{5}{2} \times 34 = 34$$
3. (c) Man saves $(100 - 92) = 8\%$ of his salary
 Let his salary be ₹ x
 $\therefore 8\% \text{ of } x = 220$

$$\Rightarrow x = \frac{220 \times 100}{8} = ₹2750$$
4. (b) Number of apple trees

$$= \frac{25}{2 \times 100} \times 640 = 80$$
 Number of trees which are not apple tree

$$= 640 - 80 = 560$$
5. (b) % of candidates failed in the exam

$$= (100 - 94)\% = 6\%$$
 Let total candidates be x

$$\therefore \frac{6}{100} \times x = 48 \Rightarrow x = \frac{48 \times 100}{6} = 800$$

6. (c) $\frac{10}{100}m = \frac{2 \times 20}{100}n$
 $m = 4n \Rightarrow m : n = 4 : 1$
7. (a) A : B : C = 12 : 15 : 25
 Total parts = 52
 One part = $\frac{312}{52} = 6$
 $\therefore A = 12 \times 6 = 72, B = 15 \times 6 = 90,$
 $C = 25 \times 6 = 150$
 $\frac{B - A}{C - B} = \frac{90 - 72}{150 - 90} = \frac{18}{60} = \frac{3}{10} = 3 : 10$
8. (c) C.P of motor cycle = 45000 + 200 = 45200
 gain % = $\frac{\text{gain}}{\text{CP}} \times 100$
 $\Rightarrow \text{gain} = \frac{10}{100} \times 45200 = ₹ 4520$
 S.P = 45200 + 4520 = ₹ 49720
9. (d) S.P of watch = ₹ 190
 Loss = 5%
 SP = CP - 5% of CP
 $\Rightarrow 190 = \left(1 - \frac{5}{100}\right) \text{CP}$
 $\Rightarrow \text{CP} = \frac{190 \times 100}{95} = ₹ 200$
10. (d) Half of 1% = $\frac{1}{200} = 0.005$
11. (a) $P = \frac{\text{S.I} \times 100}{R \times T} = \frac{1770 \times 100}{15 \times 8}$
 $P = ₹ 1475$
12. (a) Cost of 1 saree = ₹ $\frac{3600}{12} = ₹ 300$.
13. (b) $6 : x :: 12 : 36$
 $\Rightarrow x = \frac{6 \times 36}{12} = 18$.

14. (c) $A = P + \text{S.I.} \Rightarrow A = P \left(1 + \frac{TR}{100}\right)$

$$81 = 72 \left(1 + \frac{T \times \frac{25}{4}}{100}\right) \Rightarrow \frac{16 + T}{16} = \frac{81}{72}$$

$$16 + T = 18 \Rightarrow T = 2 \text{ years.}$$

15. (d) Bhanu Madhuri
 $T_1 = 3 \text{ years} \quad T_2 = 10 \text{ years}$
 $R_1 = 12\% \quad R_2 = 24\%$

$$\frac{A_1}{A_2} = \frac{P \left(1 + \frac{T_1 R_1}{100}\right)}{P \left(1 + \frac{T_2 R_2}{100}\right)} = \frac{100 + T_1 R_1}{100 + T_2 R_2}$$

$$= \frac{100 + 3 \times 12}{100 + 10 \times 24} = \frac{136}{340} = \frac{2}{5}$$

$$\therefore A_1 : A_2 = 2 : 5.$$

More Than One Option Correct :

1. (b, d) $\frac{3}{4} = \frac{75}{100}$ and $\frac{3}{75} = \frac{4}{100}$
2. (a, d) 35% of a number = $\frac{35}{100} \times \text{number}$
 $= \frac{7}{20} \times \text{number} = 0.35 \times \text{number}$
3. (a, c) $14 : 21 = \frac{14}{21} = \frac{2}{3} = 2 : 3$
 $34 : 51 = \frac{34}{51} = \frac{2}{3} = 2 : 3$
4. (a, b, c.)
 (a) Since $\frac{2}{5} + \frac{3}{5} = \frac{5}{5} = 1$
 (b) $70\% + 30\% = 100\%$
 (c) $\frac{7}{9} + \frac{2}{9} = \frac{9}{9} = 1$
 (d) $50\% + 20\% = 70\% \neq 100\%$.
 So, it is not possible.
5. (a, c)

- (a) 60% of 1 litre = 600 ml
and 40% of 1 litre = 400 ml
(b) 70% of 1 litre = 700 ml
and 30% of 1 litre = 300 ml
Which is not possible
(c) $600 : 400 = \frac{600}{400} = \frac{3}{2} = 3 : 2$
(d) $600 : 400 = 3 : 2 \neq 7 : 3$

Assertion & Reason :

1. (d) Profit = ₹ 150 – ₹ 100 = ₹ 50
Profit % = $\frac{50}{100} \times 100 = 50\%$
2. (c) Assertion : correct, Reason : Incorrect
3. (b)

Passage Based Questions :

1. (b) Number of girls $\frac{70}{100} \times 180 = 126$
Number of boys = $180 - 126 = 54$
Ratio = $\frac{54}{126} = \frac{3}{7}$ i.e. 3 : 7
2. (a) Male teachers = $\frac{20}{100} \times 10 = 2$
Ratio = $\frac{54}{2} = \frac{27}{1}$ i.e. 27 : 1
3. (c) Ratio = $\frac{126}{2} = 63 : 1$
4. (d) Female teacher = $\frac{80}{100} \times 10 = 8$
Required % = $\frac{8}{180} \times 100 = \frac{40}{9} = 4\frac{4}{9} \%$
5. (c) P = ₹ 5000, R = 5%, T = 5 years.
S.I = $\frac{5000 \times 5 \times 5}{100} = ₹ 1250$
6. (d) Amount = P + S.I

$$= ₹ 5000 + ₹ 1250 = ₹ 6250$$

∴ ₹ 6250 is invested after 5 years.

7. (a) S.I in 2 years

$$S.I = \frac{6250 \times 2 \times 3}{100} = ₹ 375$$

$$\text{Total S.I} = 1250 + 375 = ₹ 1625$$

8. (c) Total amount = $5000 + 1625 = ₹ 6625$

Integer Type Questions :

1. (6) $\frac{2}{3} = \frac{4}{x} \Rightarrow 2x = 12 \Rightarrow x = \frac{12}{2} = 6$
2. (4) 5.5 % of 40 + 3% of 60
$$= \frac{11}{2 \times 100} \times 40 + \frac{3}{100} \times 60$$
$$= \frac{11}{5} + \frac{9}{5} = \frac{20}{5} = 4$$

3. (7) P = ₹ 250, A = ₹ 285, T = 2 years
S.I = ₹ 285 – ₹ 250 = ₹ 35
$$R = \frac{S.I \times 100}{P \times T} = \frac{35 \times 100}{250 \times 2} = 7\% \text{ p.a.}$$

4. (6) Let principal be ₹ x and time be t years.
For simple interest on a sum of money be equal to principal, we have
$$x = \frac{x \times t \times 50}{3 \times 100} \Rightarrow t = 6 \text{ years}$$

5. (2) Let $x\%$ of 6.5 litres is 130 ml
$$\frac{x}{100} \times 6500 = 130 \Rightarrow x = \frac{130}{65} = 2\%$$

Multiple Matching Question :

1. (A) → (t); (B) → (r, s); (C) → (p, u); (D) → (q)

Chapter

9

Rational Numbers

INTRODUCTION

You have already studied natural numbers, whole numbers, fractions and integers previously. In this chapter, number system is extended further. We shall now study the concept of rational numbers alongwith their addition, subtraction, multiplication and division.

RATIONAL NUMBER

A number in form of $\frac{a}{b}$ where a and b are integers and $b \neq 0$ is known as rational number.

For example: $\frac{1}{2}, \frac{-6}{7}, \frac{3}{-2}, \frac{-5}{-8}$ is a rational number.

If both the numerator and denominator are positive or negative then, a rational number is said to be positive rational number.

For example: $\frac{5}{7}, \frac{-18}{-27}, \frac{-16}{-13}, \frac{8}{9}$

If its numerator and denominator are of opposite signs, then a rational number is said to be negative.

For example: $\frac{-3}{4}, \frac{8}{-6}, \frac{-28}{11}$

Whole numbers as rational numbers

Zero is a rational number, since we can write it as $\frac{0}{1}$.

Natural numbers as rational numbers

Every natural number is rational number but a rational number need not to be natural number as $\frac{1}{1}, \frac{2}{1}, \frac{3}{1}, \frac{4}{1}$

etc. are natural numbers and rational numbers but $\frac{2}{3}, \frac{6}{8}, \frac{1}{3}$ etc are rational number which can't be natural numbers.

Integers as rational numbers

Every integer is a rational number but a rational number need not to be an integer.

For example: $1 = \frac{1}{1}, 2 = \frac{2}{1}, 3 = \frac{3}{1} \dots \frac{n}{1}$ are integers but rational number like $\frac{5}{7}, \frac{-8}{9}, \frac{11}{-13}$ are not integers.

Fractions as rational numbers

Every fraction is a rational number but a rational number need not to be a fraction.

Let $\frac{a}{b}$ is a fraction where a and b are natural numbers. Since every natural number is an integer so a and

b are integers so fraction $\frac{a}{b}$ where $b \neq 0$ is a rational number.

For example:

A number like $\frac{5}{-6}$ is a rational number but not a fraction because its denominator -6 is not a natural number.

Equivalent Rational Numbers

If $\frac{a}{b}$ is a rational number and n is a non-zero integer then on multiplying the numerator and denominator of

rational number by n we will get its equivalent rational number as $\frac{a}{b} = \frac{a \times n}{b \times n}$

For example: $-\frac{3}{4} = \frac{(-3) \times 2}{4 \times 2} = \frac{(-3) \times 3}{4 \times 3} = \frac{(-3) \times 4}{4 \times 4} = \dots$

$$-\frac{3}{4} = \frac{-6}{8} = \frac{-9}{12} = \frac{-12}{16} \dots\dots$$

All these rational numbers are equal to one another and are called **equivalent rational numbers**.

Standard form of a Rational Number

A rational number is said to be in its standard form when its denominator is positive and it is in its lowest term.

Rational number can be in its standard form by following steps :

1. Make the denominator of rational number positive.
2. Divide both the numerator and denominator by their HCF.

For example: Let us convert the following in their standard form.

$$\begin{aligned} \text{(i)} \quad 2\frac{4}{9} \\ = \frac{22}{9} \end{aligned}$$

$$\text{(ii)} \quad -\frac{25}{100}$$

HCF of 25 and 100 = 25

$$-\frac{25 \div 25}{100 \div 25} = \frac{-1}{4}$$

$$\begin{aligned} \text{(iii)} \quad \frac{4}{-11} \\ = \frac{4}{-11} \times \frac{-1}{-1} = \frac{-4}{11} \end{aligned}$$

$$\text{(iv)} \quad \frac{9}{15}$$

HCF of 9 and 15 is 3

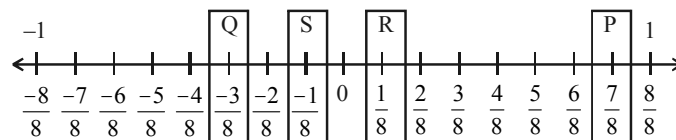
$$\text{so } \frac{9 \div 3}{15 \div 3} = \frac{3}{5}$$

NOTE: If the denominator of rational number is negative then multiply both the numerator and denominator by -1 to make denominator positive.

RATIONAL NUMBER ON NUMBER LINE

To express rational numbers on appropriate number lines, divide the unit length into the number of equal parts as the denominator of the rational number and then mark the numbers on the line.

For example : $P = \frac{7}{8}$, $Q = \frac{-3}{8}$, $R = \frac{1}{8}$, $S = \frac{1}{-8}$



COMPARISON OF RATIONAL NUMBERS

Positive rational numbers are always greater than negative rational numbers. But if we have to compare two positive or two negative rational numbers we will consider the following cases :

Same Denominators

If two rational numbers have same denominators, then

- Compare the numerators of both numbers.
- The rational number with greater numerator is the greater rational number.

Same Numerators

In this case, compare the denominators of both numbers. The rational number with the lesser denominator is the larger rational number.

Different numerator and denominator

In this case, we compare such rational numbers by two methods.

First Method:

- Express each rational number with its positive denominator.
- Find LCM of positive denominators.
- Express each given rational numbers with LCM as the common denominator.
- The number having greater numerator is greater.

ILLUSTRATION 1 :

Compare : $\frac{3}{-14}$ and $-\frac{5}{21}$

SOLUTION:

$$\frac{3}{-14} \text{ and } -\frac{5}{21}$$

$$\text{LCM of 14 and 21} = 2 \times 7 \times 3 = 42$$

$$-\frac{3 \times 3}{14 \times 3} = \frac{-9}{42}, \quad \frac{-5 \times 2}{21 \times 2} = \frac{-10}{42}$$

$$\frac{-9}{42} > \frac{-10}{42}$$

$$\text{So, } \frac{-3}{14} > \frac{-5}{12}$$

Second Method:

To compare two rational numbers $\frac{a}{b}$ and $\frac{c}{d}$, we compare the products $a \times d$ and $b \times c$ and define their inequality accordingly.

$$\begin{array}{l|l} \text{If } a \times d > b \times c & \text{If } a \times d < b \times c \\ \text{then } \frac{a}{b} > \frac{c}{d} & \text{then } \frac{a}{b} < \frac{c}{d} \end{array}$$

ILLUSTRATION 2 :

$$\text{Compare } \frac{-5}{9} \text{ and } \frac{11}{-16}$$

SOLUTION:

$$-\frac{5}{9} \text{ and } -\frac{11}{16}$$

$$\text{Now, } -5 \times 16 \text{ and } 9 \times -11$$

$$\text{So, } -80 > -99$$

$$\text{So, } -\frac{5}{9} > -\frac{11}{16}$$

RATIONAL NUMBERS BETWEEN TWO RATIONAL NUMBERS

There are unlimited rational numbers between any two rational numbers.

ILLUSTRATION 3 :

$$\text{List three rational numbers between } \frac{-4}{5} \text{ and } \frac{-2}{3}.$$

SOLUTION:

$$\text{Write two rational numbers in standard form; } \frac{-4}{5}, \frac{-2}{3}$$

$$\text{Give equivalent fraction of } \frac{-4}{5} \text{ and } \frac{-2}{3} \text{ with same denominators.}$$

$$\frac{-4}{5} = \frac{-4}{5} \times \frac{6}{6} = \frac{-24}{30}; \quad \frac{-2}{3} = \frac{-2}{3} \times \frac{10}{10} = \frac{-20}{30}$$

$$\text{Numbers between } \frac{-24}{30} \text{ and } \frac{-20}{30} \text{ are}$$

$$\frac{-24}{30} < \frac{-23}{30} < \frac{-22}{30} < \frac{-21}{30} < \frac{-20}{30}$$

$$\Rightarrow \frac{-4}{5} < \frac{-23}{30} < \frac{-22}{30} < \frac{-21}{30} < \frac{-2}{3}$$

OPERATIONS OF RATIONAL NUMBERS

Like all other numbers, arithmetic operations are applicable to rational numbers as well. Arithmetic operations on rational numbers include addition, subtraction, multiplication and division.

1. Addition of rational numbers

Case 1: When denominators of given rational numbers are same

Let $\frac{p}{q}$ and $\frac{r}{q}$ are two rational numbers. Then $\frac{p}{q} + \frac{r}{q} = \left(\frac{p+r}{q} \right)$

For example: Add : $\frac{7}{-11}$ and $\frac{3}{11}$

$$\frac{-7}{11} + \frac{3}{11} = \frac{-7+3}{11} = \frac{-4}{11}$$

Case 2: When denominator of given numbers are different.

1. Find the LCM of denominators of the given rational numbers.
2. Express each of the given rational numbers with above LCM as the common denominator.
3. Now add the numerators and divide by the L.C.M

For example: Add : $\frac{7}{-27} + \frac{11}{18}$ $-\frac{7}{27} + \frac{11}{18}$

LCM of 27 and 18 = $3 \times 3 \times 3 \times 2 = 54$

$$-\frac{7 \times 20}{27 \times 2} = -\frac{14}{54} \quad \text{and} \quad \frac{11 \times 3}{18 \times 3} = \frac{33}{54}$$

$$\therefore -\frac{14}{54} + \frac{33}{54} = \frac{-14+33}{54} = \frac{19}{54}$$

2. Subtraction of rational number

Case. I : When denominator of the rational numbers are same

Let $\frac{p}{q}$ and $\frac{r}{q}$ are two rational numbers, then $\frac{p}{q} - \frac{r}{q} = \frac{p-r}{q}$

Case. II : When denominators of given rational numbers are different.

Step. I : Find the LCM of denominators of the given rational numbers

Step. II : Express each of the given numbers with above LCM as the common denominator.

Step. III : Now subtract the numerators and divide by the LCM.

For example: Subtract $\frac{7}{8}$ from $\frac{5}{12}$

$$\frac{5}{12} - \frac{7}{8} = \frac{5}{12} + \frac{-7}{8}$$

LCM of 12 and 8 = $2 \times 2 \times 3 \times 2 = 24$

$$= \frac{5 \times 2 + (-7) \times 3}{24} = \frac{10 + (-21)}{24} = \frac{-11}{24}$$

3. Multiplication of rational numbers:

The product of two rational numbers is the product of their numerators divided by the product of their denominators.

$$\text{Product of two rational numbers} = \frac{\text{Product of numerators}}{\text{Product of denominator}}$$

Thus if $\frac{a}{b}$ and $\frac{c}{d}$ are two rational numbers, then $\frac{a}{b} \times \frac{c}{d} = \frac{a \times c}{b \times d}$

For example: (i) $\frac{-7}{15} \times \frac{5}{-14} = \frac{-7 \times 5}{15 \times -14} = \frac{-35}{-210} = \frac{1}{6}$

(ii) $\frac{-7}{9} \times \frac{-4}{5} = \frac{28}{45}$

Reciprocal or multiplicative inverse of a rational number

The reciprocal of a rational number $\frac{a}{b}$ is $\frac{b}{a}$ or $\left(\frac{a}{b}\right)^{-1} = \frac{b}{a}$

It is also termed as the multiplicative inverse of rational number.

NOTE: (i) Reciprocal of 0 does not exist. (ii) Reciprocal of 1 is 1 (iii) Reciprocal of -1 is -1.

ILLUSTRATION 4 :

Write reciprocal of: (i) $\frac{-7}{9}$ (ii) -5

SOLUTION:

(i) $\frac{-7}{9} = \frac{9}{-7}$ (ii) $-5 = (-5)^{-1} = \frac{1}{-5}$

4. Division of rational numbers

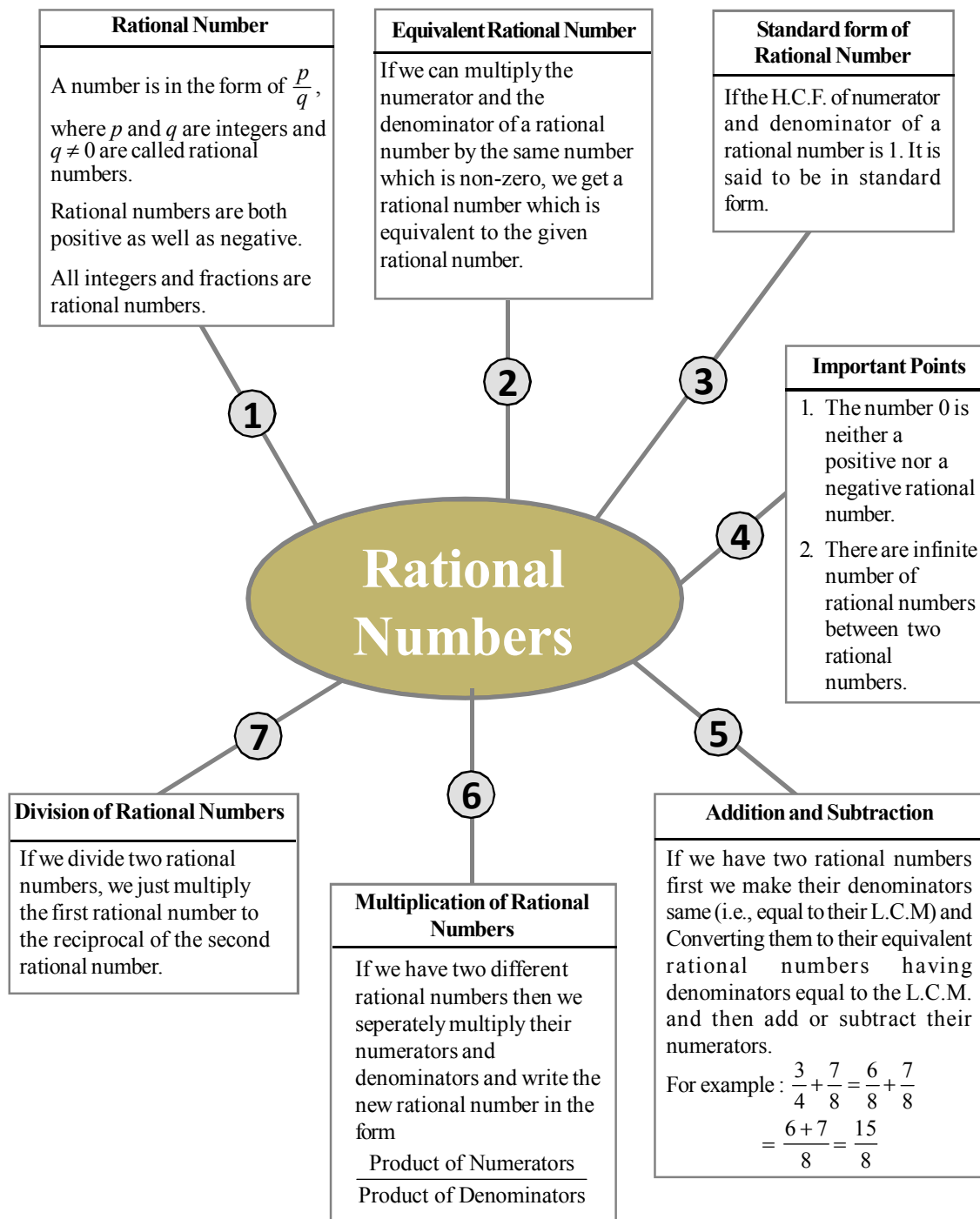
If $\frac{a}{b}$ and $\frac{c}{d}$ are two rational numbers such that $\frac{c}{d} \neq 0$ then,

$$\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \left(\text{reciprocal of } \frac{c}{d}\right) = \frac{a}{b} \times \frac{d}{c}$$

For example:

$$\frac{16}{21} \div \frac{-4}{3} = \frac{16}{21} \times \left(\text{reciprocal of } \frac{-4}{3}\right) = \frac{16}{21} \times \frac{3}{-4} = \frac{-4}{7}$$

CONCEPT MAP



MISCELLANEOUS

SOLVED EXAMPLES

1. Express $\frac{-3}{8}$ as a rational number with

- (i) denominator = 32,
(ii) denominator = -40

Sol. (i) Denominator of $\frac{-3}{8}$ is 8.

By what number should we multiply 8 to get 32?

Clearly, such number is $32 \div 8 = 4$.

So, we multiply its numerator and denominator by 4.

$$\therefore \frac{-3}{8} = \frac{(-3) \times 4}{8 \times 4} = \frac{-12}{32}.$$

$$\text{Hence, } \frac{-3}{8} = \frac{-12}{32}.$$

- (ii) By what number should we multiply 8 to get (-40)?

Clearly, such number is $(-40) \div 8 = -5$

$$\therefore \frac{-3}{8} = \frac{-3 \times (-5)}{8 \times (-5)} = \frac{15}{-40}$$

2. Express $\frac{-247}{228}$ in standard form.

Sol. The given number is $\frac{-247}{228}$.

HCF of 247 and 228 is 19.

So, we divide its numerator and denominator by 19.

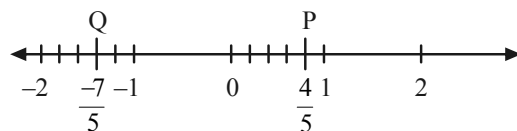
$$\therefore \frac{-247}{228} = \frac{(-247) \div 19}{228 \div 19} = \frac{-13}{12}.$$

$$\text{Hence, } \frac{-247}{228} = \frac{-13}{12} \text{ (in standard form).}$$

3. Represent $\frac{4}{5}$ and $\frac{-7}{5}$ on number line.

Sol. For $\frac{4}{5}$: divide the unit gap between 0 and +1

into five equal parts. Each part represents $\frac{1}{5}$.



Point P represents $\frac{4}{5}$ on the number line.

$$\text{For } \frac{-7}{5} : \frac{-7}{5} = -1\frac{2}{5} \text{ and } -2 < -1\frac{2}{5} < -1$$

We divide the unit gap between -1 and -2 into five equal parts. Each part represents $\frac{-1}{5}$. Point

Q represents $\frac{-7}{5}$ on the number line.

4. Arrange the following rational numbers in

ascending order : $\frac{5}{8}, \frac{1}{-3}, \frac{-3}{5}$

Sol. Rewrite the given rational numbers with positive denominators.

$$\frac{5}{8} = \frac{5}{8}, \frac{1}{-3} = \frac{-1}{3}, \frac{-3}{5} = \frac{-3}{5}$$

Since LCM of denominators 8, 3 and 5 is 120, we write the numbers so that they have the common denominator 120 :

$$\begin{aligned}\frac{5}{8} &= \frac{5 \times 15}{8 \times 15} = \frac{75}{120} \\ \frac{1}{-3} &= \frac{-1}{3} = \frac{-1 \times 40}{3 \times 40} = \frac{-40}{120} \\ \frac{-3}{5} &= \frac{-3 \times 24}{5 \times 24} = \frac{-72}{120}\end{aligned}$$

Now, we compare the numerators :
Since, $-72 < -40 < 75$,

$$\therefore \frac{-72}{120} < \frac{-40}{120} < \frac{75}{120}$$

$$\text{i.e., } \frac{-3}{5} < \frac{-1}{3} < \frac{5}{8}$$

5. Find the sum: $\left(\frac{-5}{9} + \frac{-7}{12} + \frac{11}{18}\right)$

Sol. LCM of 9, 12, 18 = 36.

$$\therefore \frac{-5}{9} = \frac{(-5) \times 4}{9 \times 4} = \frac{-20}{36},$$

$$\frac{-7}{12} = \frac{(-7) \times 3}{12 \times 3} = \frac{-21}{36}$$

$$\text{and } \frac{11}{18} = \frac{11 \times 2}{18 \times 2} = \frac{22}{36}.$$

$$\therefore \frac{-5}{9} + \frac{-7}{12} + \frac{11}{18} = \frac{-20}{36} + \frac{-21}{36} + \frac{22}{36}$$

$$= \frac{(-20) + (-21) + 22}{36} = \frac{(-41) + 22}{36} = \frac{-19}{36}.$$

Hence, the required sum is $\frac{-19}{36}$.

6. What should be added to $\frac{-7}{12}$ so as to get $\frac{9}{16}$?

Sol. Given : Sum of the given numbers = $\frac{9}{16}$

$$\text{The given number} = \frac{-7}{12}$$

$$\begin{array}{r|l} 2 & 16, 12 \\ 2 & 8, 6 \\ \hline & 4, 3 \end{array}$$

$$\text{LCM} = 2 \times 2 \times 4 \times 3 = 48$$

$$\text{Required number} = \text{Sum} - \text{Given number}$$

$$\begin{aligned}&= \frac{9}{16} - \left(-\frac{7}{12}\right) = \frac{9}{16} + \frac{7}{12} = \frac{9 \times 3 + 7 \times 4}{48} \\ &= \frac{27 + 28}{48} = \frac{55}{48}.\end{aligned}$$

7. If a machine can make $23\frac{3}{4}$ kg of potato chips in 1 hour, what quantity of potato chips can it make in $10\frac{1}{3}$ hours?

Sol. Quantity of chips can be made in

$$10\frac{1}{3} \text{ hours} = 23\frac{3}{4} \times 10\frac{1}{3}$$

$$= \frac{95}{4} \times \frac{31}{3} = \frac{2945}{12} = 245\frac{5}{12} \text{ kg}$$

8. List five rational numbers between $\frac{-4}{5}$ and $\frac{-2}{3}$.

Sol. LCM of 5 and 3 is 15.

$$\therefore \frac{-4}{5} = \frac{-4 \times 3}{5 \times 3} = \frac{-12}{15} = \frac{-12 \times 3}{15 \times 3} = \frac{-36}{45}$$

$$\text{and } \frac{-2}{3} = \frac{-2 \times 5}{3 \times 5} = \frac{-10}{15} = \frac{-10 \times 3}{15 \times 3} = \frac{-30}{45}$$

Clearly,

$$\frac{-36}{45} < \frac{-35}{45} < \frac{-34}{45} < \frac{-33}{45} < \frac{-32}{45} < \frac{-31}{45} < \frac{-30}{45}$$

$$\Rightarrow \frac{-4}{5} < \frac{-7}{9} < \frac{-34}{45} < \frac{-11}{15} < \frac{-32}{45} < \frac{-31}{45} < \frac{-2}{3}$$

$\therefore$ Five rational numbers between $\frac{-4}{5}$ and

$$\frac{-2}{3} \text{ are } \frac{-7}{9}, \frac{-34}{45}, \frac{-11}{15}, \frac{-32}{45} \text{ and } \frac{-31}{45}.$$

9. The product of two rational numbers is $\frac{-4}{5}$. If one of them is $\frac{8}{35}$, find the other.

$$\text{Sol. The other number} = \frac{-4}{5} \div \frac{8}{35} = \frac{(-1) \times 7}{1 \times 2} = \frac{-7}{2}$$

Thus, the required other number is $\frac{-7}{2}$.

1

EXERCISE

Fill in the Blanks :

DIRECTIONS : Complete the following statements with an appropriate word / term to be filled in the blank space(s).

- $(\quad) \div \left(\frac{-7}{5}\right) = \frac{-2}{3}$
- $\left(\frac{-65}{14}\right) \div (\quad) = 2\frac{1}{2}$
- $\left(\frac{-3}{8}\right) + (\quad) = \frac{5}{12}$
- Multiplicative inverse of $-1\frac{3}{4}$ is $\quad$.
- $-\frac{3}{8}$ is a $\quad$ rational number.
- $-\frac{1}{2}$ is $\quad$ than $\frac{1}{5}$.
- $\frac{-27}{45}$ and $\frac{-3}{5}$ represent $\quad$ rational numbers.
- The reciprocal of $\quad$ does not exist.
- If p and q are positive integers, then $\frac{p}{q}$ is a $\quad$ rational number and $\frac{p}{-q}$ is a $\quad$ rational number.
- Additive inverse of $\frac{2}{3}$ is $\quad$.

True / False :

DIRECTIONS : Read the following statements and write your answer as true or false.

- Two rationals with different numerators can never be equal.
- If $\frac{p}{q}$ is a rational number and a is a non-zero integer, then $\frac{p \times a}{q \times a}$ is a rational number not equivalent to $\frac{p}{q}$.
- Every integer is a rational number but every rational number need not be an integer.
- The product of an integer and a rational number can never be a natural number.
- The quotient of division of an integer by a rational number can never be an integer.
- Every negative rational number is less than zero.
- 0 is a rational number and it is neither negative nor positive.
- A rational number is said to be positive if its numerator and denominator are either both positive integers or both negative integers.
- On the number line for integers, every number on the right is smaller than all the number on its left.
- Associative property holds both in addition and in subtraction.

Match the Columns :

DIRECTIONS : Each question contains statements given in two columns which have to be matched. Statements (A, B, C, D) in column-I have to be matched with statements (p, q, r, s) in column-II.

1.	Column - I	Column - II
(A)	$\frac{-9}{10} + \frac{22}{15}$	(p) $\frac{-54}{55}$
(B)	$-2\frac{1}{9} - 6$	(q) $\frac{-15}{4}$
(C)	$-\frac{6}{5} \times \frac{9}{11}$	(r) $\frac{17}{30}$
(D)	$\frac{3}{13} \div \frac{-4}{65}$	(s) $\frac{-73}{9}$

Very Short Answer Questions :

DIRECTIONS : Give answer in one word or one sentence.

1. Fill in the blanks with correct symbol using $>$, $<$ or $=$:

(i) $\frac{5}{-7} \square \frac{-5}{8}$ (ii) $\frac{-6}{11} \square \frac{5}{8}$

(iii) $\frac{3}{5} \square \frac{7}{12}$ (iv) $\frac{5}{-8} \square \frac{-6}{13}$

2. Which of the following are positive rational numbers?

(i) $\frac{3}{-5}$ (ii) $\frac{-11}{15}$ (iii) $\frac{-5}{-8}$

(iv) $\frac{37}{53}$ (v) $\frac{0}{3}$ (vi) 8

3. Express each of the following in the standard form.

(i) $\frac{-45}{135}$ (ii) $\frac{66}{-110}$

(iii) $\frac{112}{392}$ (iv) $\frac{-168}{231}$

4. Are the rational numbers in each pair equivalent rational numbers?

(i) $\frac{-2}{7}$ and $\frac{-18}{63}$ (ii) $\frac{-15}{21}$ and $\frac{25}{-35}$

(iii) $\frac{18}{-4}$ and $\frac{27}{-6}$ (iv) $\frac{7}{5}$ and $\frac{847}{605}$

5. Express $\frac{-15}{16}$ as a rational number with :

- (i) numerator = -30
(ii) numerator = 75
(iii) denominator = 48
(iv) denominator = -96

6. Fill in the blanks:

(i) $\frac{-3}{4} = \frac{\dots}{12} = \frac{\dots}{28}$

(ii) $\frac{-5}{-8} = \frac{\dots}{16} = \frac{25}{\dots}$

(iii) $\frac{7}{-9} = \frac{-14}{\dots} = \frac{35}{\dots}$

(iv) $\frac{-8}{\dots} = \frac{4}{13} = \frac{\dots}{-65}$

7. Find the additive inverse of:

(i) $\frac{-11}{15}$ (ii) $\frac{15}{-4}$ (iii) $\frac{-18}{-13}$

(iv) 0 (v) $\frac{1}{-6}$

8. Find the multiplicative inverse or reciprocal of each of the following :

(i) $\frac{13}{25}$ (ii) $\frac{-6}{19}$ (iii) $\frac{-3}{-5}$

(iv) -1 (v) 0

9. Find the value of x in the following.

(i) $\frac{7}{9} = \frac{63}{x}$ (ii) $\frac{x}{14} = \frac{15}{-21}$

(iii) $\frac{24}{x} = \frac{21}{35}$ (iv) $\frac{-x}{3} = \frac{60}{36}$

(v) $\frac{12}{13} = \frac{x}{156}$ (vi) $\frac{2}{x} = \frac{8}{54}$

10. Represent the following rational numbers on the number line:

(i) $-\frac{1}{2}$ (ii) $-\frac{1}{3}$

(iii) $\frac{2}{3}$ (iv) $\frac{-3}{5}$

Short Answer Questions :

DIRECTIONS : Give answer in 2-3 sentences.

1. Add the following

(i) $\frac{3}{4}$ and $\frac{-5}{4}$ (ii) $\frac{-7}{3}$ and $\frac{-13}{3}$

(iii) $\frac{5}{-6}$ and $\frac{2}{3}$ (iv) $\frac{13}{15}$ and $\frac{-3}{10}$

2. Subtract :

(i) $\frac{3}{4}$ from $\frac{2}{3}$ (ii) $\frac{-5}{7}$ from $\frac{-2}{5}$

(iii) $\frac{-9}{7}$ from -1 (iv) 5 from $\frac{-3}{5}$

3. Arrange in order from least to greatest,

(i) $\frac{1}{3}, \frac{3}{7}, \frac{2}{5}$ (ii) $\frac{-9}{6}, \frac{-4}{3}, \frac{-17}{12}$

(iii) $\frac{-3}{5}, \frac{7}{-10}, \frac{-5}{6}$

4. Arrange in descending order.

(i) $\frac{3}{4}, \frac{11}{8}, \frac{9}{16}, \frac{33}{16}$ (ii) $\frac{1}{7}, \frac{-1}{3}, \frac{-2}{8}$

(iii) $\frac{3}{-5}, \frac{-7}{10}, \frac{8}{-10}, \frac{-17}{15}$

5. Multiply:

(i) $\frac{3}{20}$ and $\frac{15}{18}$ (ii) $\frac{-5}{36}$ and $\frac{-3}{20}$

(iii) $\frac{36}{7}$ and $\frac{-14}{9}$ (iv) $\frac{-36}{5}$ and $\frac{25}{-9}$

6. Divide:

(i) $\frac{6}{7}$ by $\frac{-3}{14}$ (ii) $\frac{5}{12}$ by $\frac{4}{5}$

(iii) $\frac{-4}{5}$ by $\frac{-2}{5}$ (iv) $\frac{-8}{9}$ by $\frac{3}{-4}$

7. List six rational numbers between -1 and 0 .

8. Find 3 rational numbers between $\frac{3}{4}$ and $\frac{1}{2}$.

9. The sum of two rational numbers is -5 . If one of the numbers is $\frac{-13}{6}$, find the other.

10. The product of two rational numbers is $\frac{-8}{9}$. If one of the numbers is $\frac{-4}{15}$, find the other.

Long Answer Questions :

DIRECTIONS : Give answer in four to five sentences.

1. Subtract the sum of $\frac{-2}{3}$ and $\frac{1}{3}$ from the sum of $\frac{5}{3}$ and $\frac{-4}{3}$.

2. Solve: $\frac{-5}{6} + \frac{11}{10} + \frac{-9}{32}$

3. The cost of 15 pencils is ₹ $37\frac{1}{2}$. Find the cost of each pencil.

4. Simplify:

$$\left[\frac{5}{23} + \left(-\frac{18}{115} \right) + \left(\frac{-28}{138} \right) \right] \times \left[\left(\frac{23}{14} \right) \div \left(\frac{69}{17} \right) \right]$$

5. Simplify:

(i) $\left(\frac{25}{8} \times \frac{2}{5} \right) - \left(\frac{3}{5} \times \frac{-10}{9} \right)$

(ii) $\left(\frac{-5}{9} \times \frac{72}{-125} \right) - \left(\frac{11}{17} \times \frac{34}{55} \right) + \left(\frac{28}{-13} \times \frac{-52}{21} \right)$

2

EXERCISE

Text-Book Exercise :

1. List five rational numbers between:

(i) -2 and -1 (ii) $\frac{-4}{5}$ and $\frac{-2}{3}$

2. Write four more rational numbers in each of the following patterns:

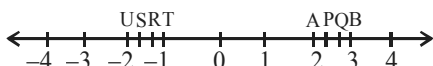
(i) $\frac{-1}{4}, \frac{-2}{8}, \frac{-3}{12}, \dots$

(ii) $\frac{-1}{6}, \frac{2}{-12}, \frac{3}{-18}, \frac{4}{-24}, \dots$

3. Draw the number line and represent the following rational numbers on it:

(i) $\frac{3}{4}$ (ii) $\frac{-5}{8}$ (iii) $\frac{-7}{4}$

4. The points, P, Q, R, S, T, U, A and B on the number line are such that, $TR = RS = SU$ and $AP = PQ = QB$. Name the rational numbers represented by P, Q, R and S.



5. Which of the following pairs represent the same rational number?

(i) $-\frac{7}{21}$ and $\frac{3}{9}$ (ii) $-\frac{16}{20}$ and $\frac{20}{-25}$

(iii) $-\frac{2}{-3}$ and $\frac{2}{3}$ (iv) $-\frac{3}{5}$ and $-\frac{12}{20}$

6. Fill in the boxes with the correct symbol out of $>$, $<$ and $=$.

(i) $-\frac{5}{7} \square \frac{2}{3}$ (ii) $-\frac{4}{5} \square -\frac{5}{7}$

(iii) $-\frac{7}{8} \square \frac{14}{-16}$ (iv) $-\frac{8}{5} \square -\frac{7}{4}$

7. Write the following rational numbers in ascending order:

(i) $-\frac{3}{5}, -\frac{2}{5}, -\frac{1}{5}$ (ii) $-\frac{1}{3}, -\frac{2}{9}, -\frac{4}{3}$

(iii) $-\frac{3}{7}, -\frac{3}{2}, -\frac{3}{4}$

8. Find the sum:

(i) $-\frac{9}{10} + \frac{22}{15}$ (ii) $-\frac{3}{-11} + \frac{5}{9}$

9. Find

(i) $\frac{7}{24} - \frac{17}{36}$ (ii) $\frac{5}{63} - \left(-\frac{6}{21}\right)$

(iii) $-\frac{6}{13} - \left(-\frac{7}{15}\right)$

10. Find the product:

(i) $\frac{9}{2} \times \left(-\frac{7}{4}\right)$ (ii) $\frac{3}{10} \times (-9)$

(iii) $-\frac{6}{5} \times \frac{9}{11}$

11. Find the value of:

(i) $-\frac{1}{8} \div \frac{3}{4}$ (ii) $-\frac{2}{13} \div \frac{1}{7}$

(iii) $-\frac{7}{12} \div \left(-\frac{2}{13}\right)$ (iv) $\frac{3}{13} \div \left(-\frac{4}{65}\right)$

Exemplar Questions :

1. Express $\frac{3}{4}$ as a rational number with denominator:

(i) 36 (ii) -80

2. If $-\frac{5}{7} = \frac{x}{28}$, then find the value of x .

3. Simplify:

(i) $\frac{13}{11} \times \left(-\frac{14}{5}\right) + \frac{13}{11} \times \left(-\frac{7}{5}\right) + \left(-\frac{13}{11}\right) \times \frac{34}{5}$

(ii) $\frac{6}{5} \times \frac{3}{7} - \frac{1}{5} \times \frac{3}{7}$

4. Find the reciprocal of the following :

(i) $\left(\frac{1}{2} \times \frac{1}{4}\right) + \left(\frac{1}{2} \times 6\right)$ (ii) $\frac{20}{51} \times \frac{4}{91}$

(iii) $\frac{3}{13} \div \frac{-4}{65}$

(iv) $\left(-5 \times \frac{12}{15}\right) - \left(-3 \times \frac{2}{9}\right)$

5. Find a rational number exactly halfway between:

(i) $-\frac{5}{13}$ and $-\frac{7}{9}$ (ii) $\frac{1}{15}$ and $\frac{1}{12}$

6. Taking $x = \frac{-4}{9}$, $y = \frac{5}{12}$ and $z = \frac{7}{18}$, find :

(i) $(x \div y) \times z$ (ii) $(x - y) + z$

(iii) $x + (y + z)$ (iv) $x \div (y \div z)$

7. What should be multiplied with $-\frac{5}{8}$ to obtain the nearest integer?

8. From a rope 68 m long, pieces of equal size are cut. If length of one piece is $4\frac{1}{4}$ m, find the number of such pieces.

9. Insert 3 equivalent rational numbers between

- (i) $-\frac{1}{2}$ and $\frac{1}{5}$ (ii) 0 and -10

HOTS Questions :

1. Divide the sum of $-\frac{8}{7}$ and $\frac{5}{14}$ by their product.

2. Simplify $\left(\frac{15}{28} \times \frac{-119}{9}\right) + \left(\frac{-19}{20} \times \frac{-30}{-57}\right)$
 $+ \left(\frac{-39}{3} \times \frac{14}{5} \times \frac{-12}{56}\right)$

3. What should be added to $\left(\frac{1}{-2} - \frac{3}{4} \text{ of } \frac{-8}{15}\right)$ so that the sum is the product of $\frac{-7}{50}$ and $1\frac{1}{14}$?

4. Simplify: $-\frac{8}{15} + \frac{7}{20} - \frac{-11}{35} + \frac{1}{5}$

5. Simplify:
 $\left(\frac{1}{12} + \left(\frac{-3}{4}\right) + \frac{7}{8}\right) \times \left(3\frac{2}{5} - \frac{7}{10} + \left(\frac{-2}{15}\right) - 10\frac{1}{30}\right)$

3

EXERCISE

Single Option Correct :

DIRECTIONS : This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONLY ONE is correct.

1. The multiplicative inverse of $-\frac{1}{8}$ is

- (a) -1 (b) 8
 (c) -8 (d) 1

2. The reciprocal of $\frac{1}{3} \times -\frac{9}{4}$ is

- (a) $\frac{4}{3}$ (b) $-\frac{9}{12}$
 (c) $-\frac{4}{3}$ (d) $-\frac{27}{4}$

3. Which of the following rational numbers show ascending order?

- (a) $\frac{-2}{5}, \frac{-3}{5}, \frac{-1}{5}$ (b) $\frac{-4}{3}, \frac{-1}{3}, \frac{-2}{9}$
 (c) $\frac{-3}{2}, \frac{-3}{7}, \frac{-3}{4}$ (d) $\frac{-2}{9}, \frac{5}{2}, \frac{3}{4}$

4. Arrange the following rational numbers in ascending order.

$$\frac{7}{9}, \frac{5}{6}, \frac{11}{12}, \frac{-1}{3}, \frac{-3}{4}$$

- (a) $\frac{11}{12}, \frac{-3}{4}, \frac{-1}{3}, \frac{7}{9}, \frac{5}{6}$
 (b) $\frac{-3}{4}, \frac{-1}{3}, \frac{7}{9}, \frac{5}{6}, \frac{11}{12}$
 (c) $\frac{7}{9}, \frac{-1}{3}, \frac{5}{6}, \frac{11}{12}, \frac{-3}{4}$
 (d) $\frac{-3}{4}, \frac{-1}{3}, \frac{5}{6}, \frac{7}{9}, \frac{11}{12}$

5. Arrange the following in descending order.

$$\frac{-13}{20}, \frac{6}{15}, \frac{7}{12}, \frac{-9}{10}, \frac{3}{5}$$

- (a) $\frac{7}{12}, \frac{6}{15}, \frac{-13}{20}, \frac{3}{5}, \frac{-9}{10}$
 (b) $\frac{-9}{10}, \frac{-13}{20}, \frac{6}{15}, \frac{7}{12}, \frac{3}{5}$

- (c) $\frac{3}{5}, \frac{7}{12}, \frac{6}{15}, \frac{-13}{20}, \frac{-9}{10}$
- (d) $\frac{3}{5}, \frac{7}{12}, \frac{6}{15}, \frac{-9}{10}, \frac{-13}{20}$
6. Which of the following are rational numbers?
- $\frac{22}{39}, \frac{731}{0}, 288, 1\frac{3}{4}, 8\frac{1}{3}, \frac{17}{3}$
- (a) All are rational numbers.
- (b) All except $\frac{731}{0}$
- (c) Only $1\frac{3}{4}$ and $8\frac{1}{3}$.
- (d) Only $\frac{22}{39}$ and $\frac{17}{3}$
7. The rational number lying between $\frac{5}{6}$ and $\frac{6}{7}$ is
- (a) $\frac{1}{2}$ (b) $\frac{15}{21}$
- (c) $\frac{35}{42}$ (d) $\frac{71}{84}$
8. 3 rational numbers between $\frac{1}{2}$ and $\frac{1}{3}$ are
- (a) $\frac{5}{9}, \frac{11}{18}, \frac{2}{3}$ (b) $\frac{7}{18}, \frac{4}{9}, \frac{13}{27}$
- (c) Both the above (d) None of these
9. Express $\frac{42}{-63}$ as a rational number with denominator 3.
- (a) $-\frac{1}{3}$ (b) $-\frac{4}{3}$
- (c) $-\frac{2}{3}$ (d) $-\frac{8}{3}$
10. The product of two rational numbers is $\frac{-9}{16}$. If one of the numbers is $\frac{-4}{3}$, then the other number is:
- (a) $\frac{36}{48}$ (b) $\frac{25}{64}$
- (c) $\frac{27}{49}$ (d) $\frac{27}{64}$
11. Name the property of multiplication illustrated by
- $$\frac{-4}{3} \times \left(\frac{-6}{5} + \frac{8}{7} \right) = \left(\frac{-4}{3} \times \frac{-6}{5} \right) + \left(\frac{-4}{3} \times \frac{8}{7} \right)$$
- (a) Associative property
- (b) Commutative property
- (c) Distributive property
- (d) None of these
12. Out of the rational numbers $\frac{-5}{11}, \frac{-5}{12}, \frac{-5}{17}$, which is greater?
- (a) $\frac{-5}{11}$ (b) $\frac{-5}{12}$
- (c) $\frac{-5}{17}$ (d) None

More Than One Option Correct :

DIRECTIONS : This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONE or MORE may be correct.

1. Which of the following statements is true?
- (a) $\frac{-5}{8}$ lies to the left of 0 on the number line.
- (b) $\frac{3}{7}$ lies to the right of 0 on the number line.
- (c) The rational numbers $\frac{1}{3}$ and $\frac{-7}{3}$ are on opposite sides of 0 on the number line.
- (d) None of these

2. Choose the rational number which lies between rational numbers $-\frac{2}{5}$ and $-\frac{1}{5}$:

- (a) $-\frac{1}{4}$ (b) $-\frac{3}{10}$
(c) $\frac{3}{10}$ (d) $-\frac{7}{20}$

3. Which of the following statement is/are false?

- (a) $\frac{3}{-8} > \frac{-12}{32}$ (b) $\frac{3}{-8} = \frac{-12}{32}$
(c) $\frac{3}{-8} < \frac{-12}{32}$ (d) $\frac{3}{5} > \frac{4}{3}$

4. Which of the following statements are false about a rational number $-\frac{2}{3}$?

- (a) It lies to the left side of '0' on the number line.
(b) It lies to the right side of '0' on the number line.
(c) It is not possible to represent on the number line.
(d) It cannot be determined on which side the number lies.

5. Read the statement(s) and choose the correct options.

- (a) $-\frac{5}{7} < -\frac{4}{7}$
(b) 1 is only rational number which is its own reciprocal.
(c) The sum of rational numbers $\frac{a}{b}$ and $\frac{c}{d}$ is

$$\frac{a+b}{c+d}$$

- (d) Reciprocal of a negative number is always negative.

Assertion & Reason :

DIRECTIONS : Each of these questions contains an Assertion followed by Reason. Read them carefully and answer the question on the basis of following options. You have to select the one that best describes the two statements.

- (a) If both **Assertion** and **Reason** are correct and **Reason** is the correct explanation of **Assertion**.
(b) If both **Assertion** and **Reason** are correct but **Reason** is not the correct explanation of **Assertion**.
(c) If **Assertion** is correct but **Reason** is incorrect.
(d) If **Assertion** is incorrect but **Reason** is correct.

1. **Assertion:** Zero is a rational number.

Reason: We can write 0 as $\frac{0}{a}$, where $a \neq 0$

2. **Assertion :** Rational numbers are always closed under division.

Reason : Division by zero is not defined.

Passage Based Questions :

DIRECTIONS : Study the given passage(s) and answer the following questions.

PASSAGE - I

If $\frac{a}{b}$ and $\frac{c}{d}$ are two rational numbers, then

$$\frac{a}{b} \times \frac{c}{d} = \frac{a \times c}{b \times d}$$

1. The value of $\frac{-7}{15} \times \frac{45}{56}$ is

- (a) $-\frac{9}{8}$ (b) $\frac{3}{8}$
(c) $-\frac{3}{8}$ (d) $\frac{9}{8}$

2. The product of $-\frac{4}{5}$ and $-\frac{5}{12}$

- (a) $\frac{1}{3}$ (b) $-\frac{48}{25}$
(c) $-\frac{25}{48}$ (d) $-\frac{1}{3}$

3. If $x = \frac{-22}{11}$, and $y = \frac{-3}{8}$, then $x \times y$ is

- (a) $-\frac{3}{4}$ (b) $-\frac{6}{11}$
(c) $\frac{6}{11}$ (d) $\frac{3}{4}$

PASSAGE - II

If $\frac{a}{b}$ and $\frac{c}{d}$ are two rational numbers, then $\frac{a}{b} \div \frac{c}{d}$

$$= \frac{a}{b} \times \left(\text{reciprocal of } \frac{c}{d} \right) = \frac{a}{b} \times \frac{d}{c}$$

4. The value of $\frac{-4}{3} \div \frac{16}{21}$ is

- (a) $\frac{64}{63}$ (b) $\frac{-7}{4}$
 (c) $\frac{-64}{63}$ (d) $\frac{7}{4}$

5. The reciprocal of $\frac{-3}{11} \div \frac{7}{55}$

- (a) $\frac{-21}{55}$ (b) $\frac{7}{15}$
 (c) $\frac{-7}{15}$ (d) $\frac{-15}{7}$

6. If $x = \frac{-4}{9}$ and $y = \frac{7}{18}$, then $x \div y$ is

- (a) $\frac{-8}{7}$ (b) $\frac{-7}{8}$
 (c) $\frac{7}{8}$ (d) $\frac{4}{7}$

Integer Type Questions :

DIRECTIONS : Answer the following questions. The answer to each of the question is a single digit integer, ranging from 0 to 9.

- How many pieces of equal size can be cut from a rope of 30 meters long, each measuring $3\frac{3}{4}$ meters ?
- Find the value of $\frac{-36}{7} \times \frac{-14}{9}$.
- The value of $\frac{-7}{2} \times \frac{7}{6} \times \frac{-12}{7}$ is

4. If $\frac{27}{16} \div \frac{a}{b} = \frac{-15}{8}$. Then $a + b$ is

5. If we divide $\frac{3}{5}$ by $\frac{4}{9}$ and multiply the result by

$$\frac{-2}{9} + \frac{1}{3}, \text{ then we get } \frac{k}{20}. \text{ Find } k.$$

Multiple Matching Question :

DIRECTIONS : Following question has four statements (A, B, C and D) given in Column-I and six statements (p, q, r, s, t, u) in Column-II. Any given statement in Column-I can have correct matching with one or more statement(s) given in Column-II.

- Match the numbers in column-I with equivalent numbers in column-II.

Column - I	Column - II
(A) $\frac{-36}{63}$	(p) $\frac{-14}{49}$
(B) $\frac{2}{-7}$	(q) $\frac{-7}{9}$
(C) $\frac{-98}{126}$	(r) $\frac{-4}{7}$
(D) $\frac{51}{119}$	(s) $\frac{3}{7}$
	(t) $\frac{6}{-21}$
	(u) $\frac{102}{238}$

SOLUTIONS

Brief Explanations
of
Selected Questions

1 EXERCISE

Fill in the Blanks :

- $\frac{14}{15}$
- $\frac{-13}{7}$
- $\frac{19}{24}$
- $\frac{-4}{7}$
- negative
- smaller
- same
- zero
- positive, negative
- $\frac{-2}{3}$

True / False :

- False
- False
- True
- False
- False
- True
- True
- True
- False
- False

Match the Columns :

- (A) $\rightarrow$ (r); (B) $\rightarrow$ (s); (C) $\rightarrow$ (p); (D) $\rightarrow$ (q)

Very Short Answer Questions :

- $<$
 - $<$
 - $>$
 - $<$
- $\frac{-5}{-8}$
 - $\frac{37}{53}$
 - $\frac{8}{1}$
- $\frac{-1}{3}$
 - $\frac{-3}{5}$
 - $\frac{2}{7}$
 - $\frac{-8}{11}$
- yes
 - yes
 - yes
 - yes
- $\frac{-30}{32}$
 - $\frac{75}{-80}$
 - $\frac{-45}{48}$
 - $\frac{90}{-96}$
- $\frac{-3}{4} = \frac{-9}{12} = \frac{-21}{28}$
 - $\frac{-5}{-8} = \frac{10}{16} = \frac{25}{40}$
 - $\frac{7}{-9} = \frac{-14}{18} = \frac{35}{-45}$
 - $\frac{-8}{-26} = \frac{4}{13} = \frac{-20}{-65}$

- $\frac{11}{15}$
 - $\frac{15}{4}$
 - $\frac{-18}{13}$
 - 0
 - $\frac{1}{6}$
- $\frac{25}{13}$
 - $\frac{-19}{6}$
 - $\frac{5}{3}$
 - 1
 - does not exist
- 81
 - 10
 - 40
 - 5
 - 144
 - 13.5

Short Answer Questions :

- $$\frac{3}{4} + \left(\frac{-5}{4} \right) = \frac{3+(-5)}{4}$$

$$= \frac{3-5}{4} = -\frac{2}{4} = -\frac{1}{2}$$
 - $$\frac{-7}{3} + \left(\frac{-13}{3} \right) = \frac{(-7)+(-13)}{3} = -\frac{20}{3}$$
 - $$-\frac{5}{6} + \frac{2}{3} = -\frac{5}{6} + \frac{4}{6} = \frac{-5+4}{6} = -\frac{1}{6}$$
 - $$\frac{13}{15} + \left(\frac{-3}{10} \right) = \frac{13 \times 2 + (-3) \times 3}{30}$$

$$= \frac{26+(-9)}{30} = \frac{17}{30}$$
- $$\left(\frac{2}{3} - \frac{3}{4} \right) = \frac{2}{3} + \frac{-3}{4} = \frac{8+(-9)}{12} = -\frac{1}{12}$$
 - $$\frac{-2}{5} - \left(\frac{-5}{7} \right) = \frac{-2}{5} + \frac{5}{7} = \frac{-14+25}{35} = \frac{11}{35}$$
 - $$-1 - \left(\frac{-9}{7} \right) = -1 + \frac{9}{7} = \frac{-7+9}{7} = \frac{2}{7}$$
 - $$\frac{-3}{5} - 5 = \frac{-3-25}{5} = \frac{-28}{5}$$

3. (i) LCM of 3, 7 and 5 is 105.

$$\therefore \frac{1}{3} = \frac{35}{105}, \frac{3}{7} = \frac{45}{105}, \frac{2}{5} = \frac{42}{105}$$

Since, $35 < 42 < 45$

$$\therefore \frac{1}{3} < \frac{2}{5} < \frac{3}{7}$$

- (ii) LCM of 6, 3 and 12 is 12.

$$\therefore \frac{-9}{6} = \frac{-18}{12}, \frac{-4}{3} = \frac{-16}{12}, \frac{-17}{12}$$

Since, $-18 < -17 < -16$

$$\therefore \frac{-9}{6} < \frac{-17}{12} < \frac{-4}{3}$$

- (iii) LCM of 5, 10 and 6 is 30.

$$\therefore \frac{-3}{5} = \frac{-18}{30}, \frac{-7}{10} = \frac{-21}{30}, \frac{-5}{6} = \frac{-25}{30}$$

Since, $-25 < -21 < -18$

$$\therefore \frac{-5}{6} < \frac{-7}{10} < \frac{-3}{5}$$

4. (i) LCM of 4, 8 and 16 is 16.

$$\therefore \frac{3}{4} = \frac{12}{16}, \frac{11}{8} = \frac{22}{16}, \frac{9}{16}, \frac{33}{16}$$

Since, $33 > 22 > 12 > 9$

$$\therefore \frac{33}{16} > \frac{22}{16} > \frac{12}{16} > \frac{9}{16}$$

- (ii) LCM of 7, 3 and 8 is 168.

$$\therefore \frac{1}{7} = \frac{24}{168}, \frac{-1}{3} = \frac{-56}{168}, \frac{-2}{8} = \frac{-42}{168}$$

Since, $24 > -42 > -56$

$$\therefore \frac{1}{7} > \frac{-2}{8} > \frac{-1}{3}$$

- (iii) LCM of 5, 10 and 15 is 30.

$$\therefore \frac{-3}{5} = \frac{-18}{30}, \frac{-7}{10} = \frac{-21}{30}, \frac{-8}{15} = \frac{-24}{30},$$

$$\frac{-17}{15} = \frac{-34}{30}$$

Since, $-18 > -21 > -24 > -34$

$$\therefore \frac{-3}{5} > \frac{-7}{10} > \frac{-8}{15} > \frac{-17}{15}$$

$$5. (i) \frac{3}{20} \times \frac{15}{18} = \frac{1}{8} \quad (ii) \frac{-5}{36} \times \frac{-3}{20} = \frac{1}{48}$$

$$(iii) \frac{36}{7} \times \frac{-14}{9} = -8 \quad (iv) \frac{-36}{5} \times \frac{25}{-9} = 20$$

$$6. (i) \frac{6}{7} \div \left(\frac{-3}{14} \right) = \frac{6}{7} \times \left(\frac{-14}{3} \right) = -4$$

$$(ii) \frac{5}{12} \div \frac{4}{5} = \frac{5}{12} \times \frac{5}{4} = \frac{25}{48}$$

$$(iii) \frac{-4}{5} \div \left(\frac{-2}{5} \right) = \frac{-4}{5} \times \left(\frac{-5}{2} \right) = 2$$

$$(iv) \frac{-8}{9} \div \left(\frac{3}{-4} \right) = \frac{-8}{9} \times \left(\frac{-4}{3} \right) = \frac{32}{27}$$

$$7. \text{ We may write, } -1 = \frac{-7}{7}.$$

$$\frac{-7}{7} < \frac{-6}{7} < \frac{-5}{7} < \frac{-4}{7} < \frac{-3}{7} < \frac{-2}{7} < \frac{-1}{7} < 0$$

$\Rightarrow$ Six rational numbers between -1 and 0 are

$$\frac{-6}{7}, \frac{-5}{7}, \frac{-4}{7}, \frac{-3}{7}, \frac{-2}{7} \text{ and } \frac{-1}{7}.$$

8. L.C.M of 2 and 4 = 4

$$\frac{3}{4} = \frac{3}{4} \times \frac{4}{4} = \frac{12}{16}, \frac{1}{2} = \frac{1}{2} \times \frac{8}{8} = \frac{8}{16}$$

$$\text{Now, } \frac{8}{16} < \frac{9}{16} < \frac{10}{16} < \frac{11}{16} < \frac{12}{16}$$

So, 3 rational numbers between $\frac{3}{4}$ and $\frac{1}{2}$ are

$$\frac{9}{16}, \frac{10}{16} \text{ and } \frac{11}{16}.$$

9. Required number

$$= -5 - \left(\frac{-13}{6} \right) = \frac{-30 + 13}{6} = \frac{-17}{6}$$

10. Required number

$$= \frac{-8}{9} \div \frac{-4}{15} = \frac{-8}{9} \times \frac{15}{-4} = \frac{10}{3}$$

Long Answer Questions :

1. The sum of $\frac{-2}{3}$ and $\frac{1}{3} = \frac{-2+1}{3} = \frac{-1}{3}$

The sum of $\frac{5}{3}$ and $\frac{-4}{3} = \frac{5-4}{3} = \frac{1}{3}$

Now, subtract $\left(\frac{-1}{3}\right)$ from $\frac{1}{3}$.

$$\therefore \frac{1}{3} - \left(\frac{-1}{3}\right) = \frac{1}{3} + \frac{1}{3} = \frac{2}{3}.$$

2. The LCM of 6, 10 and 32 = 480

$$\begin{aligned} \frac{-5}{6} + \frac{11}{10} + \frac{-9}{32} &= \frac{(-5) \times 80}{6 \times 80} + \frac{11 \times 48}{10 \times 48} + \frac{(-9) \times 15}{32 \times 15} \\ &= \frac{-400}{480} + \frac{528}{480} + \frac{-135}{480} = \frac{-7}{480} \end{aligned}$$

3. Cost of 15 pencils = ₹ $\frac{75}{2}$.

$$\begin{aligned} \text{Cost of 1 pencil} &= ₹ \left(\frac{75}{2} \div 15 \right) = ₹ \left(\frac{75}{2} \times \frac{1}{15} \right) \\ &= ₹ \frac{5}{2} = ₹ 2.50. \end{aligned}$$

$$\begin{aligned} 4. \quad & \left[\frac{5}{23} + \left(\frac{18}{-115} \right) + \left(\frac{-28}{138} \right) \right] \times \left[\left(\frac{23}{14} \right) \div \left(\frac{69}{17} \right) \right] \\ &= \left[\frac{5}{23} - \frac{18}{115} - \frac{28}{138} \right] \times \left(\frac{23}{14} \times \frac{17}{69} \right) \\ &= \left(\frac{150 - 108 - 140}{690} \right) \times \left(\frac{17}{42} \right) \\ &= \left(\frac{-98}{690} \right) \times \frac{17}{42} = \frac{-119}{2070} \end{aligned}$$

$$\begin{aligned} 5. \quad (i) \quad & \left(\frac{25}{8} \times \frac{2}{5} \right) - \left(\frac{3}{5} \times \frac{-10}{9} \right) = \frac{5}{4} - \frac{-2}{3} \\ &= \frac{5 \times 3 - (-2) \times 4}{12} = \frac{15 + 8}{12} = \frac{23}{12} \end{aligned}$$

$$(ii) \quad \left(\frac{-5}{9} \times \frac{72}{-125} \right) - \left(\frac{11}{17} \times \frac{34}{55} \right) + \left(\frac{28}{-13} \times \frac{-52}{21} \right)$$

$$\begin{aligned} &= \frac{1 \times 8}{1 \times 25} - \frac{1 \times 2}{1 \times 5} + \frac{4 \times 4}{1 \times 3} = \frac{8}{25} - \frac{2}{5} + \frac{16}{3} \\ &= \frac{8 \times 3 - 2 \times 15 + 16 \times 25}{75} \\ &= \frac{24 - 30 + 400}{75} = \frac{424 - 30}{75} = \frac{394}{75}. \end{aligned}$$

2 EXERCISE

Text-Book Exercise :

1. (i) $-2 = \frac{-12}{6}$ and $-1 = \frac{-6}{6}$

Five rational numbers between -2 and -1 are

$$\frac{-11}{6}, \frac{-10}{6}, \frac{-9}{6}, \frac{-8}{6}, \frac{-7}{6}.$$

(ii) $\frac{-4}{5} = \frac{-4 \times 9}{5 \times 9} = \frac{-36}{45}$ and

$$\frac{-2}{3} = \frac{-2 \times 15}{3 \times 15} = \frac{-30}{45}$$

Five rational numbers between $\frac{-4}{5}$ and $\frac{-2}{3}$ are

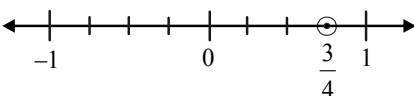
$$\frac{-2}{3} \text{ are } \frac{-35}{45}, \frac{-34}{45}, \frac{-33}{45}, \frac{-32}{45}, \frac{-31}{45}.$$

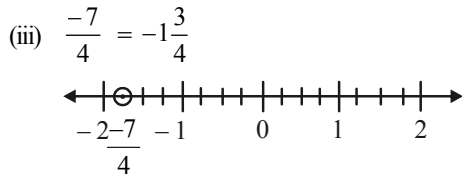
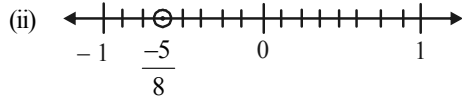
2. (i) $\frac{-1 \times 4}{4 \times 4}, \frac{-1 \times 5}{4 \times 5}, \frac{-1 \times 6}{4 \times 6}, \frac{-1 \times 7}{4 \times 7}$

i.e., $\frac{-4}{16}, \frac{-5}{20}, \frac{-6}{24}, \frac{-7}{28}$

(ii) $\frac{1 \times 5}{-6 \times 5}, \frac{1 \times 6}{-6 \times 6}, \frac{1 \times 7}{-6 \times 7}, \frac{1 \times 8}{-6 \times 8}$

i.e., $\frac{5}{-30}, \frac{6}{-36}, \frac{7}{-42}, \frac{8}{-48}$

3. (i) 



4. P represents $\frac{7}{3}$, Q represents $\frac{8}{3}$
R represents $-\frac{4}{3}$, S represents $-\frac{5}{3}$

5. (i) No, $\frac{-7}{21} = \frac{-1}{3}$ and $\frac{3}{9} = \frac{1}{3}$

- (ii) Yes, $\frac{-16}{20} = \frac{-4}{5}$; $\frac{20}{-25} = \frac{-4}{5}$

- (iii) Yes, $\frac{-2}{-3} \times \frac{(-1)}{(-1)} = \frac{2}{3}$

- (iv) Yes, $\frac{-12}{20} = \frac{-3}{5}$

6. (i) $\frac{-5}{7} = \frac{-15}{21}$; $\frac{2}{3} = \frac{14}{21}$

Therefore, $\frac{-5}{7} \boxed{<} \frac{2}{3}$,

- (ii) $\frac{-4}{5} = \frac{-28}{35}$; $\frac{-5}{7} = \frac{-25}{35}$

Therefore, $\frac{-4}{5} \boxed{>} \frac{-5}{7}$

- (iii) $\frac{14}{-16} = \frac{7}{-8}$

Therefore, $\frac{7}{-8} \boxed{=} \frac{14}{-16}$

- (iv) $\frac{-8}{5} = \frac{-32}{20}$; $\frac{-7}{4} = \frac{-35}{20}$

Therefore, $\frac{-8}{5} \boxed{>} \frac{-7}{4}$

7. (i) As $-3 < -2 < -1$,

$$\therefore \frac{-3}{5} < \frac{-2}{5} < \frac{-1}{5}$$

- (ii) $\frac{-1 \times 3}{3 \times 3}, \frac{-2}{9}, \frac{-4 \times 3}{3 \times 3}$ i.e., $\frac{-3}{9}, \frac{-2}{9}, \frac{-12}{9}$

As $-12 < -3 < -2$,

$$\therefore \frac{-4}{3} < \frac{-1}{3} < \frac{-2}{9}$$

- (iii) $\frac{-3 \times 4}{7 \times 4}, \frac{-3 \times 14}{2 \times 14}, \frac{-3 \times 7}{4 \times 7}$

$$\text{i.e., } \frac{-12}{28}, \frac{-42}{28}, \frac{-21}{28}$$

As $-42 < -21 < -12$,

$$\therefore \frac{-3}{2} < \frac{-3}{4} < \frac{-3}{7}$$

8. (i) L.C.M. of 10 and 15 is 30.

$$\begin{aligned} \therefore \frac{-9}{10} + \frac{22}{15} &= \frac{-9 \times 3}{10 \times 3} + \frac{22 \times 2}{15 \times 2} \\ &= \frac{-27}{30} + \frac{44}{30} = \frac{-27+44}{30} = \frac{17}{30} \end{aligned}$$

- (ii) $\frac{-3}{-11} + \frac{5}{9} = \frac{3}{11} + \frac{5}{9}$

$$\begin{aligned} \therefore \frac{3}{11} + \frac{5}{9} &= \frac{3 \times 9}{11 \times 9} + \frac{5 \times 11}{9 \times 11} \\ &= \frac{27}{99} + \frac{55}{99} = \frac{27+55}{99} = \frac{82}{99} \end{aligned}$$

9. (i) $\therefore \frac{7}{24} - \frac{17}{36} = \frac{7 \times 3}{24 \times 3} - \frac{17 \times 2}{36 \times 2}$

$$= \frac{21}{72} - \frac{34}{72} = \frac{21-34}{72} = \frac{-13}{72}$$

- (ii) $\frac{5}{63} - \left(\frac{-6}{21} \right) = \frac{5}{63} + \frac{2}{7}$

$$= \frac{5}{63} + \frac{2 \times 9}{7 \times 9} = \frac{5}{63} + \frac{18}{63} = \frac{23}{63}$$

$$\begin{aligned} \text{(iii)} \quad \frac{-6}{13} - \left(\frac{-7}{15} \right) &= \frac{-6}{13} + \frac{7}{15} \\ &= \frac{-6 \times 15}{13 \times 15} + \frac{7 \times 13}{15 \times 13} \\ &= \frac{-90}{195} + \frac{91}{195} = \frac{-90+91}{195} = \frac{1}{195} \end{aligned}$$

$$10. \quad \text{(i)} \quad \frac{9}{2} \times \left(\frac{-7}{4} \right) = \frac{9 \times (-7)}{2 \times 4} = \frac{-63}{8}$$

$$\text{(ii)} \quad \frac{3}{10} \times (-9) = \frac{3}{10} \times \frac{(-9)}{1} = \frac{-27}{10}$$

$$\text{(iii)} \quad \frac{-6}{5} \times \frac{9}{11} = \frac{-6 \times 9}{5 \times 11} = \frac{-54}{55}$$

$$11. \quad \text{(i)} \quad \frac{-1}{8} \div \frac{3}{4} = \frac{-1}{8} \times \frac{4}{3} = \frac{-4}{24} = -\frac{1}{6}$$

$$\text{(ii)} \quad \frac{-2}{13} \div \frac{1}{7} = \frac{-2}{13} \times 7 = \frac{-14}{13}$$

$$\text{(iii)} \quad \frac{-7}{12} \div \left(\frac{-2}{13} \right) = \frac{-7}{12} \times \frac{13}{-2} = \frac{91}{24}$$

$$\text{(iv)} \quad \frac{3}{13} \div \left(\frac{-4}{65} \right) = \frac{3}{13} \times \frac{65}{(-4)} = -\frac{15}{4}$$

Exemplar Questions :

$$1. \quad \text{(i)} \quad \frac{3}{4} = \frac{3 \times 9}{4 \times 9} = \frac{27}{36}$$

$$\text{(ii)} \quad \frac{3}{4} = \frac{3 \times (-20)}{4 \times (-20)} = \frac{-60}{-80}$$

$$2. \quad x = \frac{-5 \times 28}{7} = -20$$

$$\begin{aligned} 3. \quad \text{(i)} \quad \frac{13}{11} \times \left(\frac{-14}{5} \right) + \frac{13}{11} \times \left(\frac{-7}{5} \right) + \left(\frac{-13}{11} \right) \times \frac{34}{5} \\ &= \frac{13}{11} \times \left[\frac{-14}{5} - \frac{7}{5} - \frac{34}{5} \right] \\ &= \frac{13}{11} \times \frac{(-55)}{5} = -13 \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad \frac{6}{5} \times \frac{3}{7} - \frac{1}{5} \times \frac{3}{7} \\ &= \left(\frac{6}{5} - \frac{1}{5} \right) \times \frac{3}{7} = \frac{5}{5} \times \frac{3}{7} = \frac{3}{7} \end{aligned}$$

$$4. \quad \text{(i)} \quad \left(\frac{1}{2} \times \frac{1}{4} \right) + \left(\frac{1}{2} \times 6 \right) = \frac{1}{8} + 3 = \frac{25}{8}$$

$$\therefore \text{Reciprocal of } \frac{25}{8} = \frac{8}{25}$$

$$\text{(ii)} \quad \frac{20}{51} \times \frac{4}{91} = \frac{80}{4641}$$

$$\therefore \text{Reciprocal of } \frac{80}{4641} = \frac{4641}{80}$$

$$\text{(iii)} \quad \frac{3}{13} \div \frac{-4}{65} = \frac{3}{13} \times \frac{65}{-4} = \frac{15}{-4}$$

$$\therefore \text{Reciprocal of } \frac{15}{-4} = \frac{-4}{15}$$

$$\begin{aligned} \text{(iv)} \quad \left(-5 \times \frac{12}{15} \right) - \left(-3 \times \frac{2}{9} \right) &= (-4) - \left(\frac{-2}{3} \right) \\ &= -4 + \frac{2}{3} = \frac{-10}{3} \end{aligned}$$

$$\therefore \text{Reciprocal of } \frac{-10}{3} = \frac{-3}{10}$$

$$5. \quad \text{(i)} \quad \text{Required rational number}$$

$$\begin{aligned} &= \frac{1}{2} \left[\left(\frac{5}{-13} \right) + \left(\frac{-7}{9} \right) \right] \\ &= \frac{1}{2} \left[\frac{(-5 \times 9) + (-7 \times 13)}{117} \right] \\ &= \frac{1}{2} \left[\frac{-45 - 91}{117} \right] = \frac{1}{2} \times \frac{-136}{117} = \frac{-68}{117} \end{aligned}$$

$$\text{(ii)} \quad \text{Required rational number}$$

$$\begin{aligned} &= \frac{1}{2} \left[\frac{1}{15} + \frac{1}{12} \right] = \frac{1}{2} \times \left(\frac{4+5}{60} \right) \\ &= \frac{1}{2} \times \frac{9}{60} = \frac{3}{40} \end{aligned}$$

$$6. \quad (i) \quad (x \div y) \times z = \left[\left(\frac{-4}{9} \right) \div \frac{5}{12} \right] \times \frac{7}{18}$$

$$= \left[\frac{-4}{9} \times \frac{12}{5} \right] \times \frac{7}{18} = \frac{-16}{15} \times \frac{7}{18} = \frac{-56}{135}$$

$$(ii) \quad (x - y) + z = \left[\left(\frac{-4}{9} \right) - \frac{5}{12} \right] + \frac{7}{18}$$

$$= \left[\frac{-16 - 15}{36} \right] + \frac{7}{18} = \frac{-31}{36} + \frac{7}{18}$$

$$= \frac{-31 + 14}{36} = \frac{-17}{36}$$

$$(iii) \quad x + (y + z) = \frac{-4}{9} + \left(\frac{5}{12} + \frac{7}{18} \right)$$

$$= \frac{-4}{9} + \left(\frac{15 + 14}{36} \right) = \frac{-4}{9} + \frac{29}{36}$$

$$= \frac{-16 + 29}{36} = \frac{13}{36} = \frac{1}{2}$$

$$(iv) \quad x \div (y \div z) = \frac{-4}{9} \div \left(\frac{5}{12} \div \frac{7}{18} \right)$$

$$= \frac{-4}{9} \div \left(\frac{5}{12} \times \frac{18}{7} \right)$$

$$= \frac{-4}{9} \div \frac{15}{14} = \frac{-4}{9} \times \frac{14}{15} = \frac{-56}{135}$$

$$7. \quad \text{Nearest integer to } \frac{-5}{8} \text{ is } -1.$$

Let the required number be x .

$$\therefore x \times \left(\frac{-5}{8} \right) = -1 \Rightarrow x = \frac{8}{5}$$

$$8. \quad \text{Number of pieces} = 68 \div 4 \frac{1}{4} = 68 \div \frac{17}{4}$$

$$= 68 \times \frac{4}{17} = 16$$

$$9. \quad (i) \quad \frac{-1}{2} = \frac{-1 \times 5}{2 \times 5} = \frac{-5}{10} \text{ and } \frac{1}{5} = \frac{1 \times 2}{5 \times 2} = \frac{2}{10}$$

Rational number between $\frac{-5}{10}$ and $\frac{2}{10}$ is

$$\frac{1}{10}.$$

$\therefore \frac{1}{10}, \frac{2}{20}$ and $\frac{3}{20}$ are required rational numbers.

(ii) Rational number between 0 and -10

$$= \frac{1}{2}[0 + (-10)] = -5$$

$\therefore$ Required rational numbers are -5,

$$\frac{-10}{2} \text{ and } \frac{-15}{3}.$$

HOTS Questions :

$$1. \quad \text{Sum} = \frac{-8}{7} + \frac{5}{14} = \frac{-16 + 5}{14} = \frac{-11}{14}$$

$$\text{Product} = \frac{-8}{7} \times \frac{5}{14} = \frac{-40}{98} = \frac{-20}{49}$$

$$\text{Divide } \frac{-11}{14} \text{ by } \frac{-20}{49} = \frac{-11}{14} \div \left(\frac{-20}{49} \right)$$

$$= \frac{-11}{14} \times \left(\frac{-49}{20} \right) = \frac{539}{280} = \frac{77}{40}$$

$$2. \quad \left(\frac{15}{28} \times \frac{-119}{9} \right) + \left(\frac{-19}{20} \times \frac{-30}{-57} \right)$$

$$+ \left(\frac{-39}{3} \times \frac{14}{5} \times \frac{-12}{56} \right)$$

$$= \left(\frac{5}{4} \times \frac{-17}{3} \right) + \left(-\frac{1}{2} \right) + \left(-13 \times \frac{-3}{5} \right)$$

$$= \frac{-85}{12} - \frac{1}{2} + \frac{39}{5} = \frac{-425 - 30 + 468}{60} = \frac{13}{60}$$

3. Let the number to be added be x .

$$\therefore x + \left(\frac{1}{-2} - \frac{3}{4} \text{ of } \frac{-8}{15} \right) = \frac{-7}{50} \times 1 \frac{1}{14}$$

$$\Rightarrow x + \left[\left(\frac{1}{-2} \right) - \frac{3}{4} \times \left(\frac{-8}{15} \right) \right] = \frac{-7}{50} \times \frac{15}{14}$$

$$\Rightarrow x + \left[\left(\frac{-1}{2} \right) + \frac{2}{5} \right] = \frac{-3}{20}$$

$$\Rightarrow x + \left(\frac{-5+4}{10} \right) = \frac{-3}{20} \Rightarrow x - \frac{1}{10} = \frac{-3}{20}$$

$$\Rightarrow x = \frac{-3}{20} + \frac{1}{10} = \frac{-3+2}{20} = \frac{-1}{20}$$

$$4. \quad \frac{8}{-15} + \frac{7}{20} - \frac{-11}{35} + \frac{1}{5} = -\frac{8}{15} + \frac{7}{20} + \frac{11}{35} + \frac{1}{5}$$

$$\begin{array}{r} 5 \overline{) 15, 20, 35, 5} \\ \underline{3, 4, 7, 1} \\ \therefore \text{LCM} = 5 \times 3 \times 4 \times 7 = 420 \end{array}$$

$$= \frac{-8 \times 28 + 7 \times 21 + 11 \times 12 + 1 \times 84}{420}$$

$$= \frac{-224 + 147 + 132 + 84}{420}$$

$$= \frac{-224 + 363}{420} = \frac{139}{420}$$

$$5. \quad \left(\frac{1}{12} + \left(\frac{-3}{4} \right) + \frac{7}{8} \right) \times \left(3\frac{2}{5} - \frac{7}{10} + \left(\frac{-2}{15} \right) - 10\frac{1}{30} \right)$$

$$= \left(\frac{1}{12} - \frac{3}{4} + \frac{7}{8} \right) \times \left(\frac{17}{5} - \frac{7}{10} - \frac{2}{15} - \frac{301}{30} \right)$$

$$= \left(\frac{2-18+21}{24} \right) \times \left(\frac{102-21-4-301}{30} \right)$$

$$= \left(\frac{5}{24} \right) \times \left(\frac{-224}{30} \right) = \frac{-5}{24} \times \frac{224}{30} = \frac{-14}{9}$$

3 EXERCISE

Single Option Correct :

$$1. \quad (c) \quad -\frac{1}{8} \times -8 = 1$$

$$2. \quad (c) \quad \frac{1}{3} \times -\frac{9}{4} = -\frac{3}{4}$$

Reciprocal of $-\frac{3}{4}$ is $-\frac{4}{3}$

$$3. \quad (b) \quad \frac{-4}{3}, \frac{-1}{3}, \frac{-2}{9}$$

$$4. \quad (b) \quad \frac{-3}{4}, \frac{-1}{3}, \frac{7}{9}, \frac{5}{6}, \frac{11}{12}$$

$$5. \quad (c) \quad \frac{3}{5}, \frac{7}{12}, \frac{6}{15}, \frac{-13}{20}, \frac{-9}{10}$$

$$6. \quad (b) \quad \text{All except } \frac{731}{0}$$

$$7. \quad (d) \quad \frac{71}{84}$$

$$8. \quad (b) \quad \frac{7}{18}, \frac{4}{9}, \frac{13}{27}$$

$$9. \quad (c)$$

$$10. \quad (d) \quad \frac{-9/16}{-4/3} = \frac{-9}{16} \times \frac{-3}{4} = \frac{27}{64}$$

$$11. \quad (c) \quad \text{Since } a \times (b + c) = (a \times b) + (a \times c) \text{ is distributive property.}$$

$$12. \quad (c)$$

More Than One Option Correct :

$$1. \quad (a, b, c) \quad 2. \quad (a, b, d) \quad 3. \quad (a, c, d)$$

$$4. \quad (b, c, d) \quad 5. \quad (a, d)$$

Assertion & Reason :

$$1. \quad (a)$$

$$2. \quad (d) \quad \text{Assertion : incorrect ; Reason : correct}$$

Let us take two rational numbers $\frac{3}{5}$

and $\frac{0}{3}$. Then, $\frac{3}{5} \div \frac{0}{3}$ is not a rational number.

Passage Based Questions :

$$1. \quad (c) \quad \frac{-7}{15} \times \frac{45}{56} = \frac{-7 \times 45}{15 \times 56} = \frac{-345}{840} = \frac{-3}{8}$$

$$2. \quad (a) \quad \left(\frac{-4}{5}\right) \times \left(\frac{-5}{12}\right) = \frac{(-4) \times (-5)}{5 \times 12} = \frac{1}{3}$$

$$3. \quad (d) \quad x \times y = \left(\frac{-22}{11}\right) \times \left(\frac{-3}{8}\right) = \frac{3}{4}$$

$$4. \quad (b) \quad \frac{-4}{3} \div \frac{16}{21} = \frac{-4}{3} \times \frac{21}{16} = \frac{-84}{48} = \frac{-7}{4}$$

$$5. \quad (c) \quad \frac{-3}{11} \div \frac{7}{55} = \frac{-3}{11} \times \frac{55}{7} = \frac{-15}{7}$$

$$\therefore \text{Reciprocal of } \frac{-3}{11} \div \frac{7}{55} = \frac{-7}{15}$$

$$6. \quad (a) \quad x \div y = \frac{-4}{9} \div \frac{7}{18} = \frac{-4}{9} \times \frac{18}{7} = \frac{-8}{7}$$

$$\Rightarrow \frac{27}{16} \times \frac{b}{a} = \frac{-15}{8}$$

$$\Rightarrow \frac{b}{a} = \frac{-15}{8} \div \frac{27}{16} = \frac{-15}{8} \times \frac{16}{27} = \frac{-10}{9}$$

$$\Rightarrow \frac{a}{b} = \frac{9}{-10} = \frac{9 \times (-1)}{(-10) \times (-1)} = \frac{-9}{10}$$

Hence, the required number is $(-9 + 10)$
i.e., 1.

$$5. \quad (3) \quad \left(\frac{3}{5} \div \frac{4}{9}\right) \times \left(\frac{-2}{9} + \frac{1}{3}\right) = \frac{k}{20}$$

$$\Rightarrow \left(\frac{3}{5} \times \frac{9}{4}\right) \times \left(\frac{-2+3}{9}\right) = \frac{k}{20}$$

$$\Rightarrow \left(\frac{27}{20}\right) \times \left(\frac{1}{9}\right) = \frac{k}{20}$$

$$\Rightarrow \frac{3}{20} = \frac{k}{20} \Rightarrow k = 3.$$

Integer Type Questions :

$$1. \quad (8) \quad \frac{30}{3\frac{3}{4}} = \frac{30}{15/4} = 30 \times \frac{4}{15} = 8$$

$$2. \quad (8) \quad \frac{-36}{7} \times \frac{-14}{9} = \frac{(-36) \times (-14)}{7 \times 9} = 8.$$

$$3. \quad (7)$$

$$4. \quad (1) \quad \frac{27}{16} \div \frac{a}{b} = \frac{-15}{8}$$

Multiple Matching Questions :

$$1. \quad (A) \rightarrow (r); (B) \rightarrow (p, t); (C) \rightarrow (q); (D) \rightarrow (s, u)$$

Chapter

10

Practical Geometry

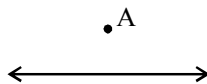
INTRODUCTION

In the previous class, you have learnt to draw a line segment of given length, a line perpendicular to a given line segment, an angle, an angle bisector, a circle, etc.

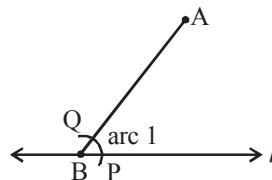
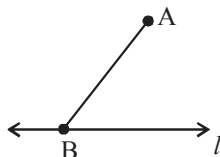
CONSTRUCTION OF A LINE PARALLEL TO A GIVEN LINE THROUGH A POINT NOT ON THE LINE

Steps:

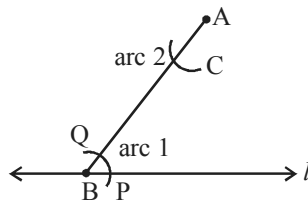
1. Take any line l and a point A outside l



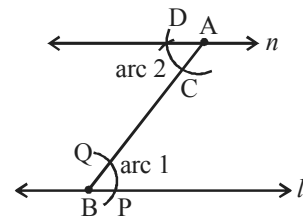
2. Take any point B on the line l and join B to A .
3. With B as centre and a convenient radius, draw an arc 1 cutting l at P and AB at Q .



4. Now, with A as centre and the same radius as in step 3, draw an arc 2 cutting AB at R .



5. Place the point of compasses at P and adjust the opening so the pencil point is at Q .
6. With the same opening as in step 5 and with C as centre, draw an arc at D .
7. Draw a line n by joining AD .



ELEMENTS OF TRIANGLE

A triangle has six elements, three sides and three angles.

A triangle can be constructed if any of the following conditions are fulfilled.

- (i) Three sides of the triangle are given.
- (ii) Two sides and the included angle between them are given.
- (iii) Two angles and the side between them are given.
- (iv) The hypotenuse and a leg in the case of a right-angled triangle are given.

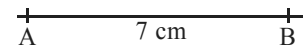
CONSTRUCTION OF TRIANGLES

When Three Sides of the Triangle are Given

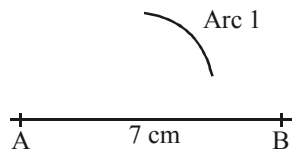
Construct a triangle ABC with $AB = 7$ cm, $BC = 6$ cm and $AC = 5$ cm.

Steps of Construction

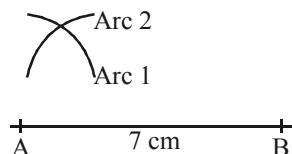
Step 1: Draw a line and mark two points A and B on it such that $AB = 7$ cm.



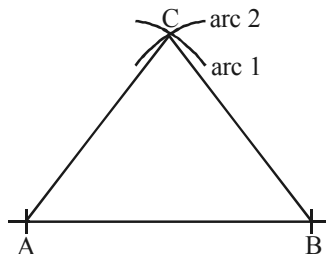
Step 2: Open your compasses to a length of 5 cm draw an arc 1 with A as centre.



Step 3: Now open your compasses to a length of 6 cm draw an arc 2 with B as centre.



Step 4: Label as C the point where the two arcs intersect. Join A to C and B to C.



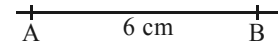
Then the triangle ABC so obtained is the required triangle

When Two Sides and the Included Angle is Given

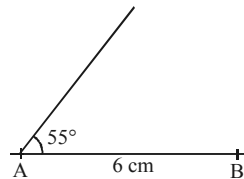
Construct a triangle with $AB = 6$ cm, $AC = 4$ cm and $\angle BAC = 55^\circ$

Steps of Construction

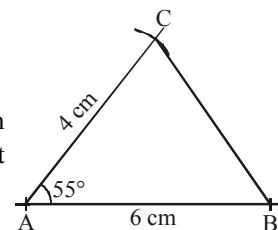
Step 1. Draw $AB = 6$ cm.



Step 2. Using your protractor draw an angle of 55° at A. Make the arm of the angle quite long.



Step 3. Use your compasses to measure the length of AC, i.e. 4 cm, on your ruler and with the point of your compasses A, draw an arc to cut the arm of the angle. This is the point C. Join C and B.

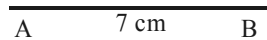


When two Angles and the Side between them are Given

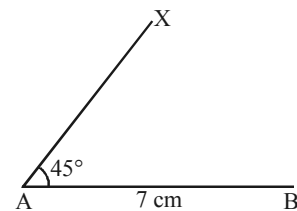
Construct a triangle ABC as $AB = 7$ cm, $\angle A = 45^\circ$, $\angle B = 30^\circ$. Calculate the third angle.

Steps of Construction

Step 1. Draw a line segment $AB = 7$ cm long.

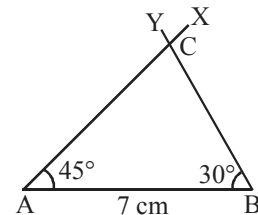


Step 2. Using your protractor, draw an angle BAX of 45° at A.



Step 3. Use your protractor to draw the second angle ABY equal to 30° at point B. AX and BY meet at C.

$$\begin{aligned}\angle A + \angle B + \angle C &= 180^\circ \\ \text{or } 45^\circ + 30^\circ + \angle C &= 180^\circ \\ \Rightarrow 75^\circ + \angle C &= 180^\circ \\ \Rightarrow \angle C &= 180 - 75 \\ \Rightarrow \angle C &= 105^\circ\end{aligned}$$



To Construct a Right Triangle when its Hypotenuse and one side are Given

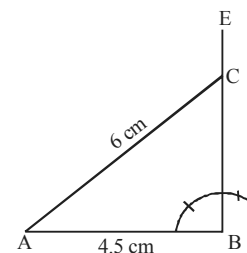
Construct a right $\triangle ABC$, right angled at B, in which $AC = 6$ cm and $AB = 4.5$ cm.

Steps of Construction

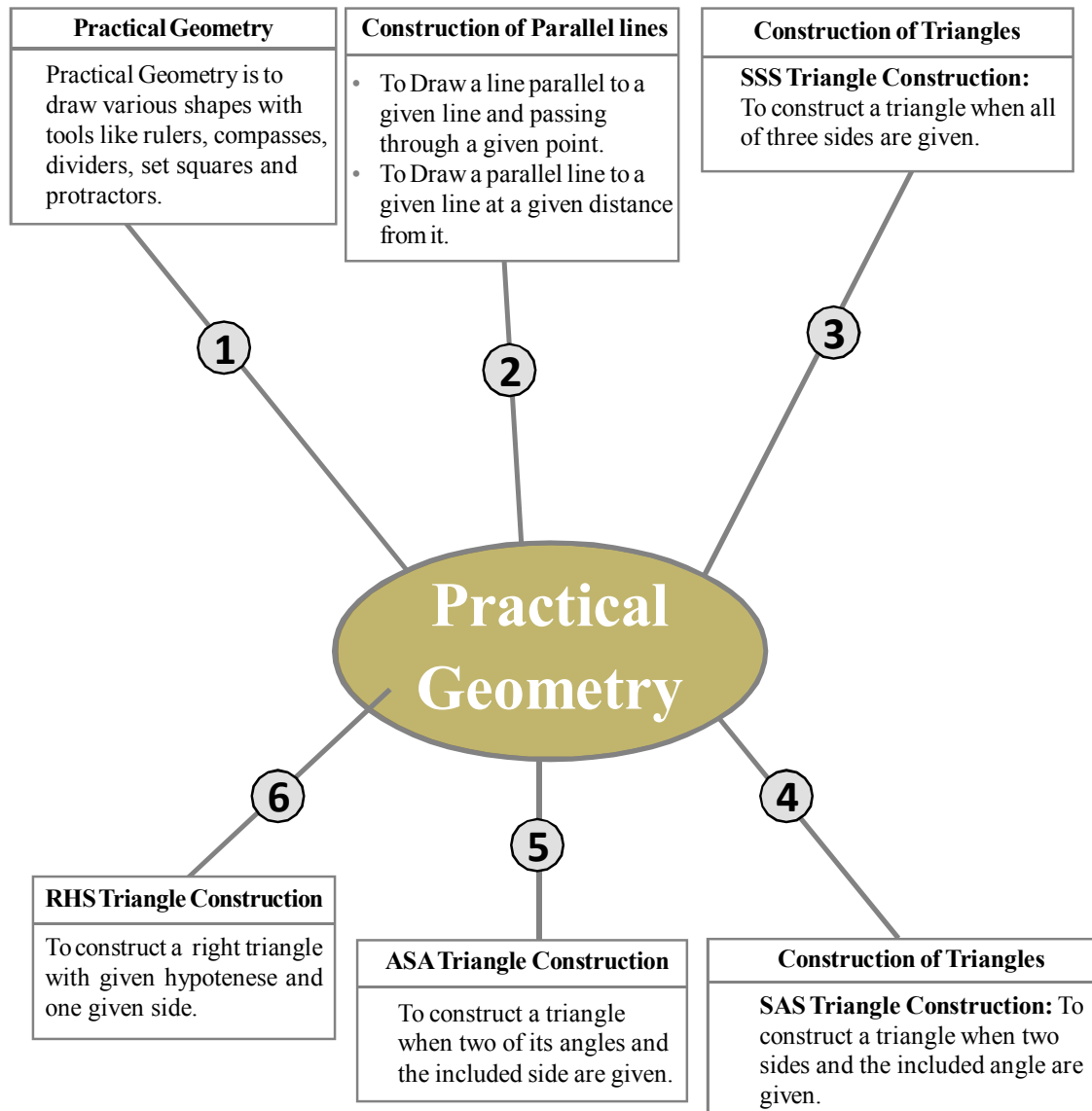
Step 1. Draw a line segment $AB = 4.5$ cm and draw $BE \perp AB$ at B.

Step 2. With A as centre and radius = 6 cm draw an arc to cut BE at C.

Step 3. Join A to C.



CONCEPT MAP

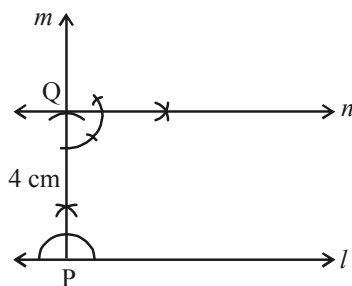


MISCELLANEOUS

SOLVED EXAMPLES

1. Construct a line parallel to a given line at a distance of 4 cm from it.

Sol.



Steps of construction:

Step 1: Let l be the given line.

Step 2: Take a point P on the given line l .

Step 3: At P , construct a line m perpendicular to l .

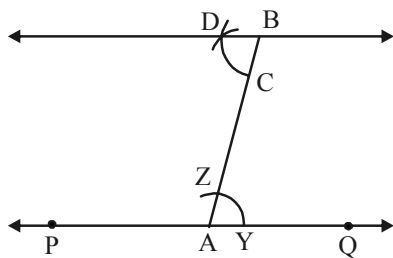
Step 4: With P , as centre and radius 4 cm draw an arc that cuts line m at Q .

Step 5: At Q , construct a line perpendicular to m , say n .

The line n is the required line parallel to the line l and at a distance of 4 cm from it.

2. Draw a line parallel to a given line PQ through a point B outside it.

Sol Steps of construction:



Step 1: Take any point A on PQ .

Step 2: Join AB .

Step 3: With A as the centre, draw an arc cutting PQ and AB at Y and Z , respectively.

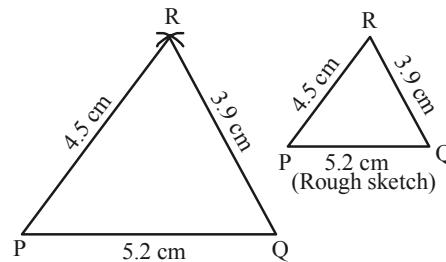
Step 4: With B as the centre and the same radius as in step 3, draw an arc on the opposite side of AB to cut AB at C .

Step 5: With C as the centre and radius equal to YZ , draw an arc cutting the arc drawn in step 4 at D .

Step 6: Join BD and produce it in both directions to obtain the required line.

3. Construct a $\triangle PQR$ in which $PQ = 5.2$ cm, $PR = 4.5$ cm and $QR = 3.9$ cm.

Sol. First draw a rough sketch of $\triangle PQR$,



Steps of construction:

Step 1: Draw a line segment $PQ = 5.2$ cm.

Step 2: With P as centre and radius 4.5 cm, draw an arc.

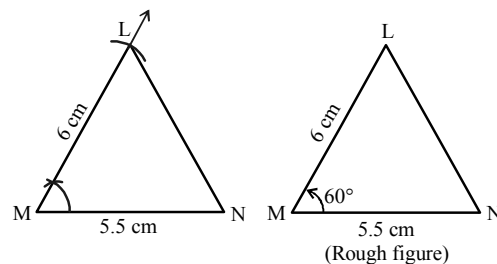
Step 3: With Q as centre and radius 3.9 cm, draw another arc, cutting the previous arc at R .

Step 4: Join PR and QR .

Then, $\triangle PQR$ is the required triangle.

4. Construct a $\triangle LMN$ such that $LM = 6$ cm, $MN = 5.5$ cm and $\angle LMN = 60^\circ$.

Sol. First draw a rough diagram of the figure.



Steps of construction :

Step 1 : Draw the base of the triangle $MN = 5.5$ cm using a ruler.

Step 2 : At M , draw an angle of 60° using a ruler and a compass.

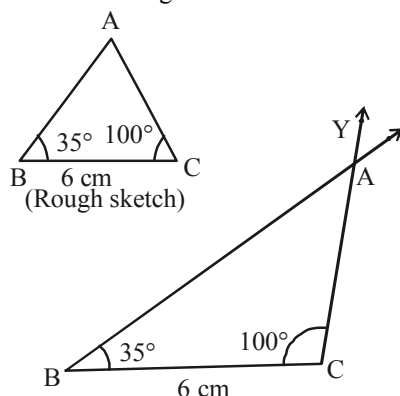
Step 3 : With M as centre, cut off 6 cm on the arm so constructed using the compass at point L .

Step 4 : Join LN .

$\therefore \triangle LMN$ is the required triangle.

5. Draw $\triangle ABC$ in which $BC = 6$ cm, $\angle B = 35^\circ$ and $\angle C = 100^\circ$. Measure $\angle A$.

Sol. First draw a rough sketch of $\triangle ABC$.



Steps of construction :

Step 1 : Draw the line segment $BC = 6$ cm.

Step 2 : Draw $\angle CBX$, such that $\angle CBX = 35^\circ$.

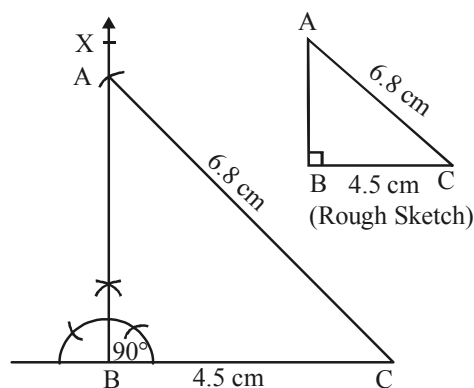
Step 3 : Draw $\angle BCY$ with Y on the same side of BC as X , such that $\angle BCY = 100^\circ$

Step 4 : Let BX and CY intersect at A .

Now, $\triangle ABC$ is the required triangle. By measurement we find that $\angle A = 45^\circ$.

6. Construct a $\triangle ABC$ in which base $BC = 4.5$ cm, $\angle B = 90^\circ$ and hypotenuse $AC = 6.8$ cm.

Sol. First draw a rough sketch of $\triangle ABC$,



Steps of construction :

Step 1 : Draw a line segment $BC = 4.5$ cm

Step 2 : Construct $\angle CBX = 90^\circ$.

Step 3 : With C as centre and radius 6.8 cm, draw an arc, cutting BX at A .

Step 4 : Join AC .

$\therefore \triangle ABC$ is the required triangle.

1

EXERCISE

Fill in the Blanks :

DIRECTIONS : Complete the following statements with an appropriate word / term to be filled in the blank space(s).

1. The sum of the angles of a triangle is ____.
2. The sum of any two sides a triangle is always ____ than the third side.
3. A triangle should have ____ elements to construct it.
4. To construct an equilateral triangle the measure of ____ side should be given.
5. We can construct an ____ triangle when base and two equal base angles are given.
6. We construct a ____ triangle when hypotenuse is 7.5 cm long and one of the legs is 4.5 cm long.
7. To construct a triangle, we should have two ____ and the included angle.

True / False :

DIRECTIONS : Read the following statements and write your answer as true or false.

1. Each acute angle of an isosceles right triangle measures 60° .
2. A right triangle cannot have an obtuse angle.
3. The distance between the two parallel lines is the same everywhere.
4. We can draw exactly one triangle whose angles are 70° , 30° and 80° .
5. Triangle with length of sides as 5 cm, 6 cm and 11 cm can be constructed.
6. The sum of the two acute angles in a right angled triangle is 90° .

Short Answer Questions :

DIRECTIONS : Give answer in 2-3 sentences.

1. Construct a triangle ABC if the lengths of its sides are given by $AB = 6$ cm, $BC = 7$ cm and $AC = 5$ cm.
2. Construct $\triangle ABC$ in which $\angle B = 60^\circ$, $AB = 5$ cm and $BC = 6$ cm.
3. Construct $\triangle PQR$ such that $QR = 7$ cm, $\angle PQR = 60^\circ$, $\angle PRQ = 30^\circ$.
4. Construct a right $\triangle ABC$, right angled at B, in which $AC = 6$ cm and $AB = 4.5$ cm.
5. Draw a line parallel to a given line at a distance of 3 cm from it.

Long Answer Questions :

DIRECTIONS : Give answer in four to five sentences.

1. Construct a line parallel to a given line l and passing through a given point P out side it.
2. Draw $\triangle ABC$ in which $AB = 4.5$ cm, $BC = 5$ cm and $CA = 6$ cm. Also, draw the perpendicular bisector of BC.
3. Construct a $\triangle ABC$ in which $AB = 5$ cm, $AC = 4.3$ cm and $\angle A = 60^\circ$.
4. Draw triangle ABC with $\angle C$ a right angle, $AB = 6.2$ cm and $BC = 4.5$ cm.
5. Construct a right-angled triangle whose hypotenuse measures 5.6 cm and one of whose acute angles measures 30° .

2

EXERCISE

Text-Book Exercise :

1. Draw a line l . Draw a perpendicular to l at any point on l . On this perpendicular choose a point X, 4 cm away from l . Through X, draw a line m parallel to l .
2. Draw ΔPQR with $PQ = 4$ cm, $QR = 3.5$ cm and $PR = 4$ cm. What type of triangle is this?
3. Construct an isosceles triangle in which the lengths of each of its equal sides is 6.5 cm and the angle between them is 110° .
4. Construct ΔPQR if $PQ = 5$ cm, $m\angle PQR = 105^\circ$ and $m\angle QRP = 40^\circ$.
5. Examine whether you can construct ΔDEF such that $EF = 7.2$ cm, $m\angle E = 110^\circ$ and $m\angle F = 80^\circ$. Justify your answer.
6. Construct an isosceles right angled triangle ABC, where $m\angle ACB = 90^\circ$ and $AC = 6$ cm.

Exemplar Questions

1. Given a line l and a point M on it draw a perpendicular MP to l where $MP = 5.2$ cm and a line q parallel to l through P.
2. Construct a right-angled isosceles triangle with one side (other than hypotenuse) of length 4.5 cm.
3. Construct an obtuse angled triangle which has a base of 5.5 cm and base angles of 30° and 120° .
4. Draw two parallel lines at a distance of 2.2 cm apart.
5. Construct an equilateral triangle ABC of side 6 cm.

3

EXERCISE

Single Option Correct :

DIRECTIONS : This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONLY ONE is correct.

1. If two angles in a triangle are 65° and 85° , then the third angle is
(a) 30° (b) 45°
(c) 60° (d) 90°
2. Two sides of an isosceles triangle are 5 cm and 6 cm. Then the length of the third side is
(a) 5 cm (b) 6 cm
(c) 5 cm or 6 cm (d) None of these
3. A triangle always has
(a) exactly one acute angle.
(b) exactly two acute angles
(c) at least two acute angles.
(d) none of these
4. In a ΔABC , if $\angle B$ is an obtuse angle then the longest side is
(a) AB (b) BC
(c) AC (d) none
5. ΔABC is an isosceles triangle with $\angle C = 90^\circ$ and $AC = 5$ cm. Then, $AB = ?$
(a) 2.5 cm (b) 5 cm
(c) 10 cm (d) $5\sqrt{2}$ cm
6. One of the acute angle of a right angled triangle is 62° . The measure of other acute angle is
(a) 50° (b) 28°
(c) 36° (d) 48°
7. A triangle can be constructed by taking two of its angle as:
(a) $110^\circ, 40^\circ$ (b) $70^\circ, 115^\circ$
(c) $135^\circ, 45^\circ$ (d) $90^\circ, 90^\circ$

8. A triangle can be constructed by taking its sides as:
- 1.8 cm, 2.6 cm, 4.4 cm
 - 2 cm, 3 cm, 4 cm
 - 2.4 cm, 2.4 cm, 6.4 cm
 - 3.2 cm, 2.3 cm, 5.5 cm.

More Than One Option Correct :

DIRECTIONS : This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONE or MORE may be correct.

- To construct a triangle ABC, which of the following parts is necessary?
 - Length of AB
 - Length of BC
 - Measure of $\angle B$
 - Only (a) and (b)
- In a triangle PQR, it is given $\angle P = 60^\circ$, $PR = PQ = 5.5$ cm. Which of the following is true about the construction of the triangle?
 - $RQ = 5.5$ cm
 - $\triangle PQR$ is an equilateral triangle
 - $\triangle PQR$ is not an equilateral triangle
 - $RQ = 11$ cm
- If a , b and c are lengths of sides of a triangle, then which of the following is true?
 - $a + b > c$
 - $b - c < a$
 - $b + c > a$
 - $b > a + c$
- Which of the following statements is correct?
 - The difference of any two sides of a triangle is less than the third side.
 - A triangle cannot have two obtuse angles.
 - A triangle cannot have an obtuse angle and a right angle.
 - None of these.
- Two sides of a triangle are 7 cm and 10 cm. Which of the following length can be the length of the third side ?
 - 14 cm
 - 17 cm
 - 13 cm
 - 3 cm

Assertion & Reason :

DIRECTIONS : Each of these questions contains an Assertion followed by Reason. Read them carefully and answer the question on the basis of following options. You have to select the one that best describes the two statements.

- If both **Assertion** and **Reason** are **correct** and Reason is the **correct explanation** of Assertion.
 - If both **Assertion** and **Reason** are correct, but Reason is **not the correct explanation** of Assertion.
 - If **Assertion** is **correct** but **Reason** is **incorrect**.
 - If **Assertion** is **incorrect** but **Reason** is **correct**.
- Assertion :** A triangle can be constructed if at least three of its elements are given. One of the given elements must be a side.
Reason : A unique triangle can be constructed if three of its angles are given.
 - Assertion :** While constructing a triangle, it is convenient and helpful to draw a rough sketch of the triangle and indicate the data on it.
Reason : This suggest the steps of construction.

Passage Based Questions :

DIRECTIONS : Study the given passage(s) and answer the following questions.

PASSAGE

Given a triangle ABC in which $BC = 6$ cm, $\angle ABC = 120^\circ$ and $AB = 7$ cm.

- What is the name of the triangle which can be drawn from the given data?
 - Equilateral triangle
 - Obtuse angled triangle
 - Right angled triangle
 - Isosceles triangle
- If $AB = 6$ cm, then the name of the triangle be
 - Acute angled triangle
 - Isosceles triangle
 - Equilateral triangle
 - Right angled triangle

SOLUTIONS

Brief Explanations of Selected Questions

1 EXERCISE

Fill in the Blanks :

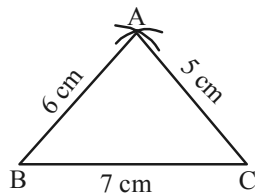
1. 180°
2. greater
3. three
4. one
5. isosceles
6. right angled
7. sides

True / False :

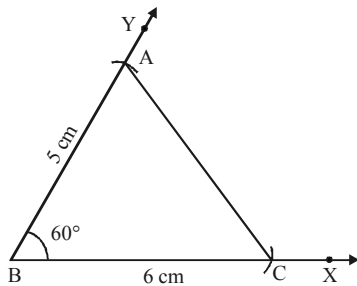
1. False
2. True
3. True
4. False
5. False
6. True

Short Answer Questions

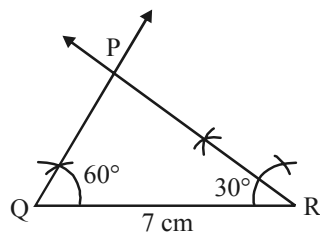
1.



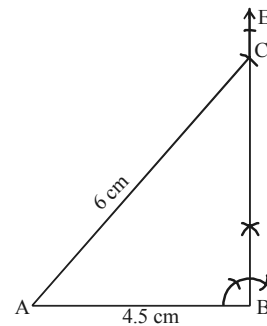
2.



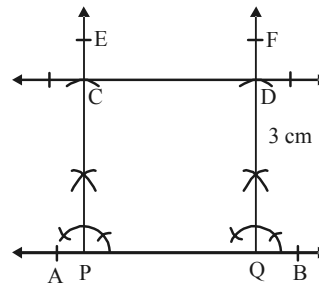
3.



4.

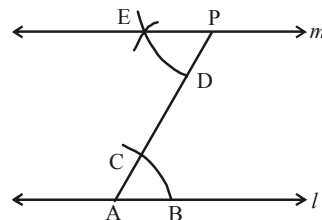


5.



Long Answer Questions :

1.



Steps of construction:

Step 1 : Take a point A on l and join PA.

Step 2 : With A as centre, draw an arc of any radius that cuts l and PA at B and C, respectively.

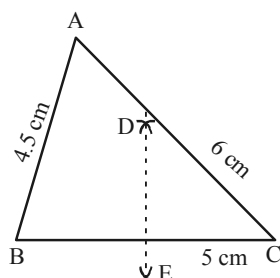
Step 3 : With P as centre, draw an arc of same radius (as in step 2) which cuts PA at D.

Step 4 : With D as centre and radius BC, draw an arc that cuts the previous arc at E.

Step 5 : Join PE and extend it on either side to get line m .

$\therefore$ Line m is the required line parallel to l and passing through point P.

2.



Steps of construction:

Step 1 : Draw a line segment $BC = 5$ cm.

Step 2 : With B as centre and radius $AB = 4.5$ cm, draw an arc.

Step 3 : With C as centre and radius $AC = 6$ cm, draw an arc intersecting the previously drawn arc at A.

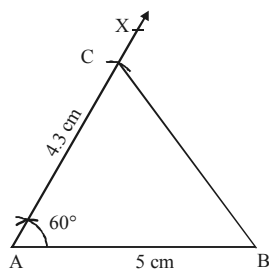
Step 4 : Join AB and AC to obtain the desired triangle.

Step 5 : With B as centre and radius more than half of BC, draw two arcs on both sides of BC.

Step 6 : With centre C and the same radius as in step 5, draw two arcs intersecting the arcs drawn in step 5 at D and E.

Step 7 : Join DE to obtain the required perpendicular bisector of BC.

3.



Steps of construction :

Step 1 : Draw a line segment $AB = 5$ cm.

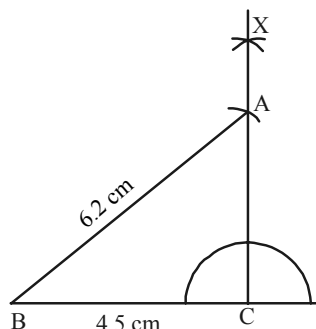
Step 2 : Construct $\angle BAX = 60^\circ$

Step 3 : Along AX, cut off $AC = 4.3$ cm.

Step 4 : Join BC.

Then, $\triangle ABC$ is the required triangle.

4.



Steps of construction:

Step 1 : Draw a line segment BC of length 4.5 cm.

Step 2 : Draw $\angle BCX$ of measure 90° .

Step 3 : With B as centre and radius $AB = 6.2$ cm, draw an arc to intersect ray CX at A.

Step 4 : Join BA to obtain the desired triangle ABC.

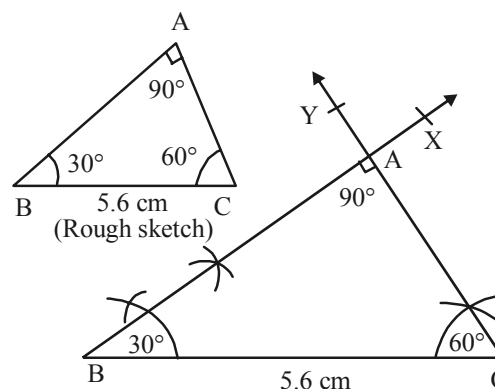
5. Let us consider a right-angled $\triangle ABC$ in which $\angle A = 90^\circ$, hypotenuse $BC = 5.6$ cm and $\angle B = 30^\circ$. We know that the sum of the angles of a triangle is 180° .

$$\Rightarrow \angle A + \angle B + \angle C = 180^\circ$$

$$\Rightarrow 90^\circ + 30^\circ + \angle C = 180^\circ$$

$$\Rightarrow 120^\circ + \angle C = 180^\circ$$

$$\Rightarrow \angle C = 180^\circ - 120^\circ = 60^\circ.$$



Steps of construction:

Step 1 : Draw a line segment $BC = 5.6$ cm.

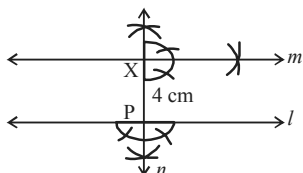
Step 2 : Construct $\angle CBX = 30^\circ$ and $\angle BCY = 60^\circ$

Step 3 : Let BX and CY intersect at the point A. Then, $\triangle ABC$ is the required triangle.

2 EXERCISE

Text-Book Exercise :

1.


Steps of construction:

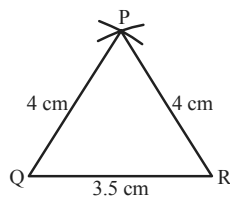
Step 1 : Draw a line l and take a point P on it.

Step 2 : At point P , draw a perpendicular line n .

Step 3 : Take $PX = 4$ cm on line n .

Step 4 : At point X , again draw a perpendicular line m .
Thus, $m \parallel l$

2.


Steps of construction:

Step 1 : Draw a line segment $QR = 3.5$ cm.

Step 2 : Taking Q as centre and radius 4 cm, draw an arc.

Step 3 : Similarly, taking R as centre and radius 4 cm, draw another arc which intersects previously drawn arc at P .

Step 4 : Join PQ and PR .

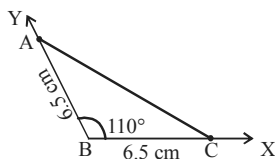
Hence, $\triangle PQR$ is the required isosceles triangle.

3.

Steps of construction:

Step 1 : Draw a line segment $BC = 6.5$ cm.

Step 2 : At point B , draw an angle of 110° with the help of protractor, i.e., $\angle YBC = 110^\circ$.



Step 3 : Taking B as centre, draw an arc with radius 6.5 cm, which cuts BY at point A .

Step 4 : Join AC .

Hence, $\triangle ABC$ is the required isosceles triangle.

4. We know that sum of angles of a triangle is 180° .

$$\therefore m\angle PQR + m\angle QRP + m\angle QPR = 180^\circ$$

$$\Rightarrow 105^\circ + 40^\circ + m\angle QPR = 180^\circ$$

$$\Rightarrow 145^\circ + m\angle QPR = 180^\circ$$

$$\Rightarrow m\angle QPR = 180^\circ - 145^\circ$$

$$\Rightarrow m\angle QPR = 35^\circ$$

Steps of construction:

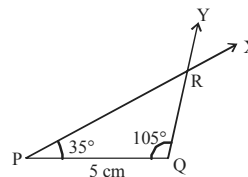
Step 1 : Draw a line segment $PQ = 5$ cm.

Step 2 : At point P , draw $\angle XPQ = 35^\circ$ with the help of protractor.

Step 3 : At point Q , draw $\angle YQP = 105^\circ$ with the help of protractor.

Step 4 : XP and YQ intersect at point R .

Thus, $\triangle PQR$ is the required triangle.



5. Given: In $\triangle DEF$, $m\angle E = 110^\circ$ and $m\angle F = 80^\circ$.

Using angle sum property of triangle

$$\angle D + \angle E + \angle F = 180^\circ$$

$$\Rightarrow \angle D + 110^\circ + 80^\circ = 180^\circ$$

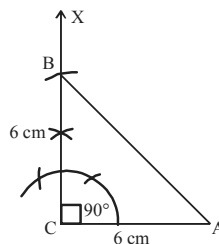
$$\Rightarrow \angle D + 190^\circ = 180^\circ$$

$$\Rightarrow \angle D = 180^\circ - 190^\circ = -10^\circ$$

Which is not possible.

$\therefore \triangle DEF$ with given measurements cannot be constructed.

6.


Steps of construction:

Step 1 : Draw a line segment $AC = 6$ cm.

Step 2 : At point C , draw $XC \perp CA$.

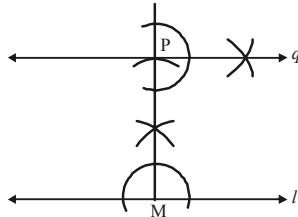
Step 3 : Taking C as centre and radius 6 cm, draw an arc cutting CX at point B .

Step 4 : Join BA .

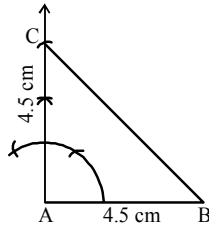
Thus, $\triangle ABC$ is the required isosceles right angled triangle.

Exemplar Questions

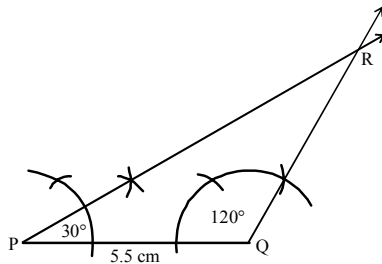
1.

**Steps of construction :****Step 1 :** Draw a line l .**Step 2 :** Take a point M on it.**Step 3 :** Draw an angle of 90° at M with l which is perpendicular to l at M .**Step 4 :** With M as centre and radius 5.2 cm, draw an arc which intersects the perpendicular at point P .**Step 5 :** At P , draw an angle of 90° with PM and produce to make a line q . Line q is the required line parallel to line l .

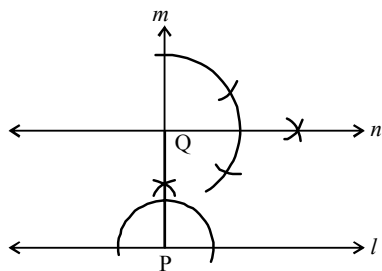
2.



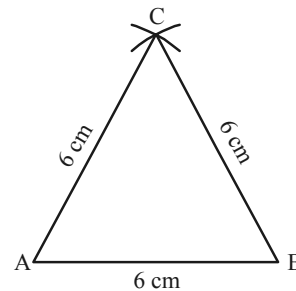
3.



4.

**Steps of construction :****Step 1 :** Draw a line l .**Step 2 :** Take a point P on it.**Step 3 :** At P , draw a line $m \perp l$.**Step 4 :** With P as centre and radius 2.2 cm, draw an arc cutting m at Q .**Step 5 :** At Q , draw a line $n \perp m$. Thus n is the required line parallel to l at a distance of 2.2 cm.

5.



3 EXERCISE

Single Option Correct :

- (a) $\therefore$ Sum of the angles in a triangle $= 180^\circ$
i.e., $65^\circ + 85^\circ + x = 180^\circ \Rightarrow x = 30^\circ$
- (c) $\therefore$ In an isosceles triangle lengths of any two sides must be equal. So the length of the third side may be either 5 cm or 6 cm.
- (c) 4. (c) 5. (d)
- (b) 7. (a) 8. (b)

More Than One Option Correct :

- (a, b, c) 2. (a, b) 3. (a, b, c)
- (a, b, c) 5. (a, c)

Assertion & Reason :

- (c) **Assertion :** correct, **Reason :** incorrect.
- (a) **Assertion :** correct, **Reason :** correct.

Passage Based Questions :

- (b) Obtuse angled triangle as $\angle ABC = 120^\circ$.
- (b) Isosceles triangle as $AB = BC = 6$ cm.

Chapter

11

Perimeter and Area

INTRODUCTION

In the previous class, you have already learnt the concept of area and perimeter; and how to find the area and perimeter of plane figures like rectangle and square. In this class, you will learn about perimeter and area of a few more plane figures.

PERIMETER

Perimeter of a closed figure is the distance around its edges.
or

Perimeter is the sum of the lengths of all the sides

Perimeter of Square

Perimeter of square = $4 \times \text{side}$

Perimeter of Rectangle

Perimeter of Rectangle = $2 \times (l + b)$

where l = length of rectangle

and b = breadth of rectangle

AREA

Amount of plane or region or surface enclosed by a figure is called area.

Area of Square

Area of square = Side $\times$ Side

Area of Rectangle

Area of rectangle = Length $\times$ Breadth

or $= l \times b$

ILLUSTRATION : 1

A garden is 80 m long and 65 m broad. A path 5 m wide is to be fruit outside all around it along its border. Find the area of the path.

SOLUTION :

Let ABCD represents the garden and shaded region is 5m path around it

Area of path = Area of EFGH – Area of ABCD

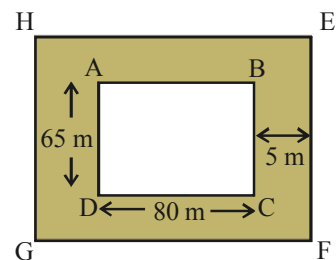
HG = $(65 + 5 + 5)$ m = 75 m

HE = $(80 + 5 + 5)$ m = 90 m

Area of EFGH = $90 \times 75 = 6750 \text{ m}^2$

Area of ABCD = $80 \times 65 = 5200 \text{ m}^2$

Area of path = $6750 - 5200 = 1550 \text{ m}^2$



NOTE: When the path runs inside, twice the width of path should be subtracted from the length and breadth of the outer rectangle.

ILLUSTRATION : 2

A grassy plot is $80\text{m} \times 50\text{m}$. Two cross paths each 4m wide are constructed at right angles through the centre of the field, such that each path is parallel to one of the sides of the rectangle. Find the total area used as path.

SOLUTION :

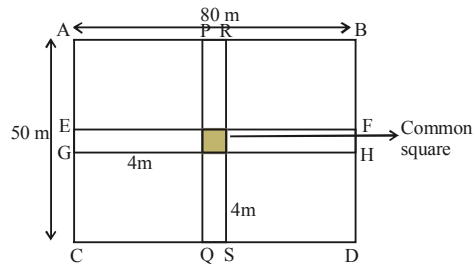
From the figure:

$$\text{Area of EFGH} = (80 \times 4)\text{m}^2 = 320\text{m}^2$$

$$\text{Area of PQRS} = (50 \times 4)\text{m}^2 = 200\text{m}^2$$

$$\text{Area of common square} = (4 \times 4)\text{m}^2 = 16\text{m}^2$$

$$\begin{aligned}\text{Area of cross path} &= (320 + 200 - 16)\text{m}^2 \\ &= 504\text{m}^2\end{aligned}$$



NOTE: From above fig. it is clear that darker region has been considered twice, because it has been included in both parts EFGH and PQRS. So in order to get the total area of the cross paths, we need to subtract this area from the sum of the areas of the two paths.

TRIANGLE AS PARTS OF RECTANGLES

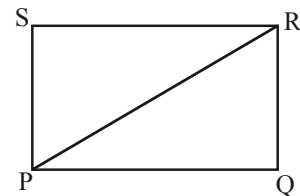
The diagonal of a rectangle divides it into two equal triangles. Thus, the sum of the areas of the two triangles is the same as the area of the rectangle.

For example: PQRS is a rectangle.

PR divides the rectangle into two triangles; $\triangle PQR$ and $\triangle PRS$.

$$\therefore \text{Area of each triangle} = \frac{1}{2} \times (\text{area of rectangle})$$

$$= \frac{1}{2} \times (PQ \times QR) \text{ or } \frac{1}{2} \times (PS \times SR)$$



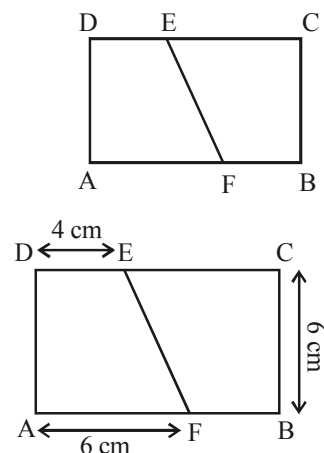
Generalising for other Congruent Parts of Rectangles

When a rectangle is divided into two equal parts, such that one part is congruent or superpose on the other part, then the area of one part is equal to the area of the other part.

$$\text{Area of each congruent part} = \frac{1}{2} \times (\text{Area of rectangle ABCD})$$

ILLUSTRATION : 3

A farmer divides his field into two equal parts. He wants to grow potatoes in one of these equal parts. Find the area of the field used for growing potatoes.



SOLUTION :

$$\begin{aligned}
 \text{Area of one of equal parts} &= \frac{1}{2} \times (\text{Area of rectangle ABCD}) \\
 &= \frac{1}{2} \times (AB \times CB) \\
 &= \frac{1}{2} \times \{(4 + 6) \times 6\} = \frac{1}{2} \times (10 \times 6) = 30 \text{ cm}^2
 \end{aligned}$$

AREA OF PARALLELOGRAM

A parallelogram is a quadrilateral, whose each pair of opposite sides are parallel.

Let ABCD be a parallelogram. DP and CQ are perpendiculars on AB.

$\therefore AB = DC$ and $DP = CQ$

Here, area of the parallelogram = area of the rectangle formed

Also, length of parallelogram ABCD = length of rectangle PQCD

or $AB = PQ$... (i)

and height of parallelogram ABCD = breadth of rectangle PQCD

or $DP = CQ$... (ii)

Now, Area of parallelogram ABCD = Area of rectangle PQCD

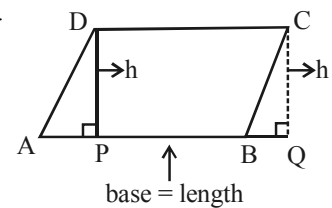
= length $\times$ breadth

= $PQ \times CQ$

= $AB \times DP$ [from (i) and (ii)]

= $b \times h$

$\therefore$ Area of parallelogram = base $\times$ height = $b \times h$



AREA OF TRIANGLE

A diagonal of a parallelogram divides the parallelogram into two equal triangles having same areas. In the adjoining figure, diagonal AC divides parallelogram ABCD into two equal triangles.

CF and AE are the altitudes of the parallelogram.

Since the sum of the areas of both the triangles is equal to the area of the parallelogram.

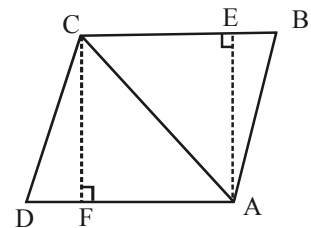
$\therefore$ Area of parallelogram = $2 \times$ Area of $\triangle ADC$

$\Rightarrow$ base $\times$ height = $2 \times$ Area of $\triangle ADC$

$\Rightarrow AD \times CF = 2 \times$ Area of $\triangle ADC$

$\Rightarrow \frac{1}{2} \times (AD \times CF) = \text{Area of } \triangle ADC$

or $\frac{1}{2} \times (\text{base} \times \text{height}) = \text{Area of } \triangle ADC$



NOTE : All the congruent triangles are equal in area but the triangles equal in area need not be congruent.

ILLUSTRATION : 4

A field is in form of parallelogram, has one of its diagonal 40 m long and the perpendicular distance of this diagonal from either of the outlying vertices is 10 m. Find the area of the field.

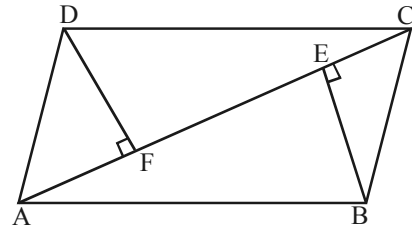
SOLUTION :

Diagonal AC = 42 m.

BE = DF = 10 m.

Area of parallelogram = Area of $\triangle ABC$ + Area of $\triangle ADC$

$$\begin{aligned}
 &= \frac{1}{2} \times AC \times BE + \frac{1}{2} \times AC \times DF \\
 &= \frac{1}{2} \times 42 \times (10 + 10) \\
 &= \frac{1}{2} \times 42 \times 20 = 420 \text{ m}^2
 \end{aligned}$$

**ILLUSTRATION : 5**

Find the area of triangle, whose base is 9.6 cm and altitude is 5 cm.

SOLUTION :

$$\begin{aligned}
 \text{Area of triangle} &= \frac{1}{2} \times \text{base} \times \text{height} \\
 &= \frac{1}{2} \times 9.6 \times 5 \\
 &= 24 \text{ cm}^2
 \end{aligned}$$

CIRCLE**Circumference of a Circle**

The Circumference is the distance around the circle.

Circumference = $2\pi \times \text{radius}$

or $C = 2\pi r$... (i)

As we know, $2 \times \text{radius} = \text{diameter}$

or $2r = d$... (ii)

$\therefore C = 2\pi r = 2r \times \pi = \pi d$

ILLUSTRATION : 6

How many times will the wheel of a car rotate in a journey of 66 km if the diameter of wheel is 50 cm?

SOLUTION :

$$\text{Diameter of wheel (D)} = 50 \text{ cm}, r = \frac{D}{2} = \frac{50}{2} = 25 \text{ cm}$$

$$\therefore \text{Circumference of the wheel} = 2\pi r = 2\pi \times 25$$

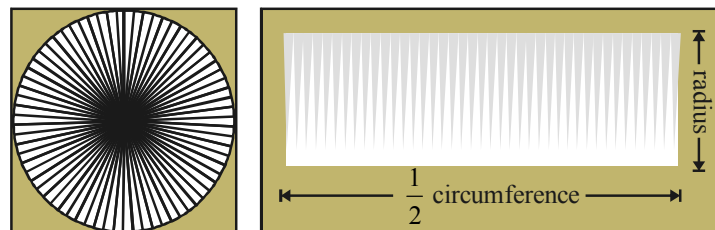
$$= 100\pi = \frac{2200}{7} = 314.28 \text{ cm.}$$

$$\text{Length of journey} = 66 \text{ km} = 66 \times 1000 \times 100 \text{ cm.} = 6600000 \text{ cm.}$$

$$\text{No. of times wheel will rotate to cover the journey} = \frac{\text{Length of journey}}{\text{Circumference of wheel}}$$

$$= \frac{6600000}{314.28} = 21000$$

Area of a Circle



Suppose the given circle is divided into 64 sectors.

When arranging these sectors, then 2 sectors together give nearly a rectangle.

Here, the length of the rectangle is the length of 32 sectors i.e. half of circumference and the breadth of this rectangle is the radius of the circle.

$$\therefore l = \frac{1}{2} \times 2\pi r = \pi r \text{ and } b = r$$

Now, Area of the circle = Area of rectangle formed

$$\begin{aligned} &= l \times b \\ &= \pi r \times r \\ &= \pi r^2 \end{aligned}$$

$$\therefore \text{Area of the circle} = \pi r^2$$

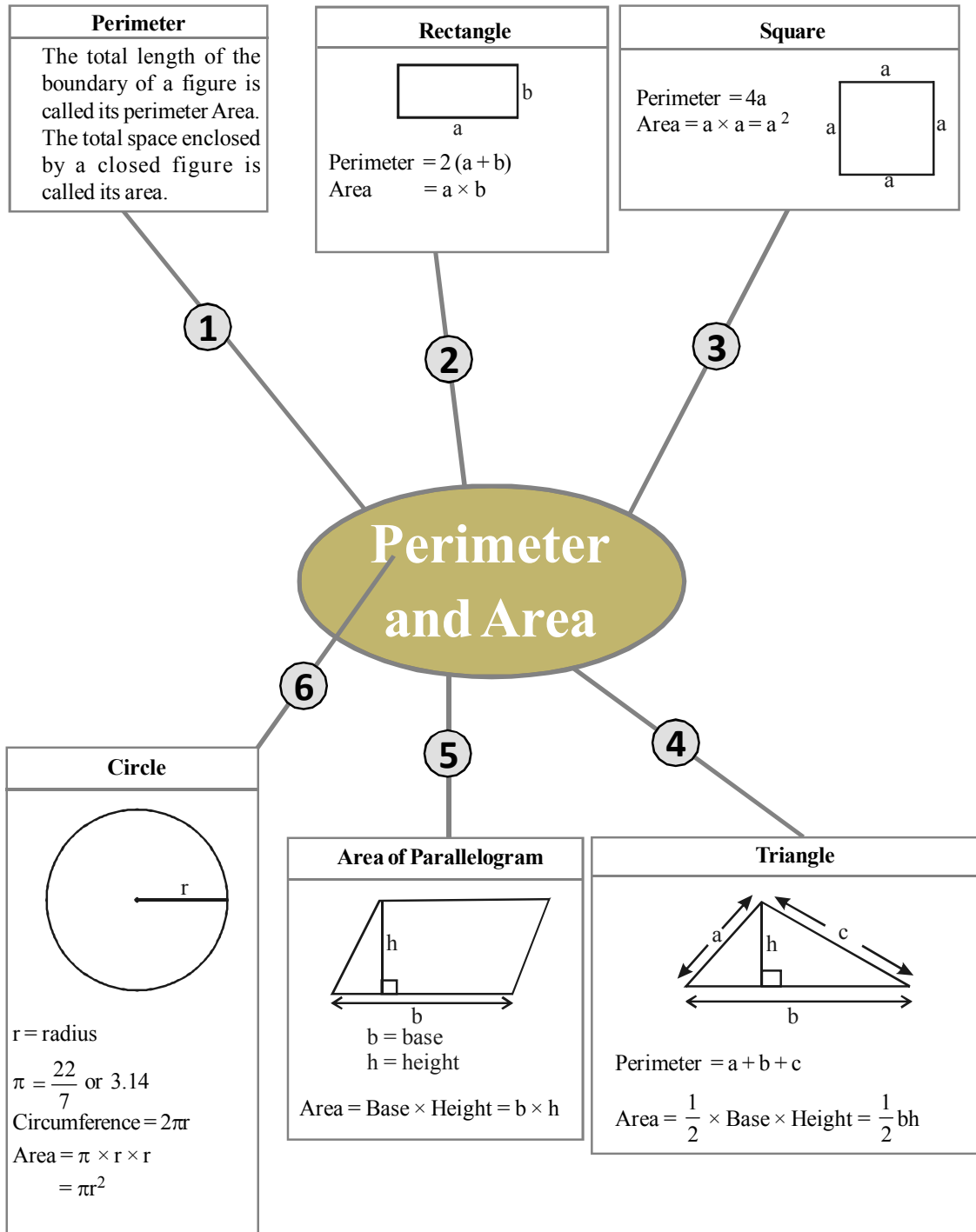
ILLUSTRATION : 7

Find the area of circle when radius is 7 cm.

SOLUTION :

$$\text{Area of circle} = \pi r^2 = \frac{22}{7} \times 7 \times 7 = 154 \text{ cm}^2$$

CONCEPT MAP



MISCELLANEOUS SOLVED EXAMPLES

1. Find the breadth of a rectangular plot of land, if its area is 440 sq. m and length is 22 m. Also, find its perimeter.

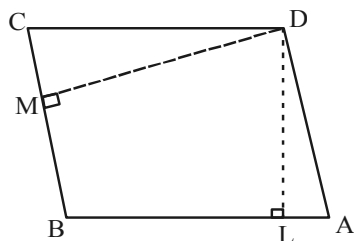
Sol. We have,
 l = Length of the plot = 22 m,
 Area of the plot = 440 sq. metre
 Let the breadth of the plot be b metres. Then,

$$\text{Breadth} = \frac{\text{Area}}{\text{Length}} \Rightarrow b = \frac{440}{22} = 20 \text{ m}$$

$$\text{Perimeter} = 2(l + b) = 2 \times (22 + 20) \text{ m} \\ = 2 \times 42 \text{ m} = 84 \text{ m.}$$

Hence, the breadth of the plot is 20 m and the perimeter is 84 m.

2. In the given fig. ABCD is a parallelogram. $DL \perp AB$ and $DM \perp BC$. If $AB = 18$ cm, $BC = 12$ cm and $DM = 10$ cm. Find DL .



Sol.

Area of parallelogram ABCD are equal when we take BC as base and DM as altitude and AB as base and DL as altitude.

$$\text{i.e., } AB \times DL = BC \times DM$$

$$18 \times DL = 12 \times 10$$

$$DL = \frac{12 \times 10}{18} = \frac{20}{3} = 6\frac{2}{3} \text{ cm}$$

3. Find the height of a triangle whose base is 15 cm and area 120 cm².

Sol. Let the required height be h cm. Then,

$$\frac{1}{2} \times 15 \times h = 120$$

$$\left[\therefore \text{area} = \frac{1}{2} \times \text{base} \times \text{height} \right]$$

$$\Rightarrow h = \frac{120 \times 2}{15} \Rightarrow h = 16 \text{ cm}$$

4. The diameter of a wheel of a car is 63 cm. Find the distance travelled by the car during the period in which it makes 1000 revolutions.

Sol. Diameter of the wheel = 63 cm

$$\Rightarrow \text{radius of the wheel} = \left(\frac{63}{2} \right) \text{ cm}$$

$$\Rightarrow \text{circumference of the wheel} = 2\pi r$$

$$= \left(2 \times \frac{22}{7} \times \frac{63}{2} \right) \text{ cm} = 198 \text{ cm} = 1.98 \text{ m.}$$

Distance covered by the wheel in 1 revolution = 1.98 m.

Distance covered by the wheel in 1000 revolutions = $(1.98 \times 1000) \text{ m} = 1980 \text{ m.}$

Hence, the distance covered by the car = 1980 m.

5. Find the area of a circle whose circumference is the same as the perimeter of a square of side 11 cm.

Sol. Side of the square = 11 cm

$$\therefore \text{Its perimeter} = 4 \times 11 = 44 \text{ cm}$$

Circumference of the circle = perimeter of the square = 44 cm

We know that $2\pi r = C$

$$\therefore 2 \times \frac{22}{7} \times r = 44$$

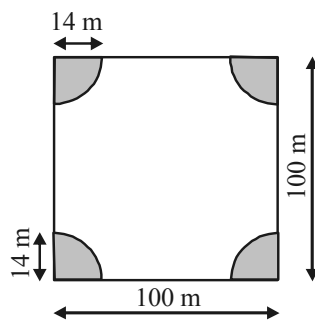
$$\Rightarrow r = 44 \times \frac{1}{2} \times \frac{7}{22} = 7 \text{ cm}$$

$$\therefore \text{Area of the circle} = \pi r^2 = \frac{22}{7} \times (7)^2$$

$$= \frac{22}{7} \times 7 \times 7$$

$$= 154 \text{ cm}^2.$$

6. A square park has each side of 100 m. At each corner of the park, there is a flower bed in the form of a quadrant of radius 14 m as shown in figure. Find the area of the remaining part of the park.



Sol. We have,

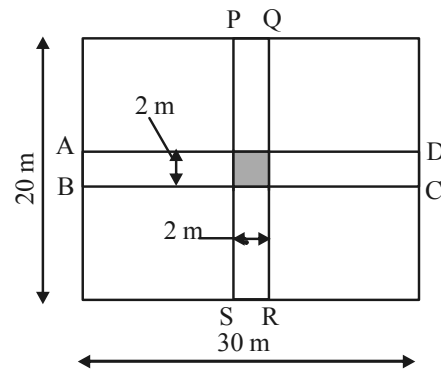
$$\text{Area of each quadrant of radius 14 m} = \frac{1}{4}(\pi r^2)$$

$$= \frac{1}{4} \times \frac{22}{7} \times 14 \times 14 = 154 \text{ m}^2$$

$\therefore$ Area of 4 quadrants $(4 \times 154) \text{ m}^2 = 616 \text{ m}^2$
 Area of square park having side 100 m long
 $= (100 \times 100) \text{ m}^2 = 10000 \text{ m}^2$
 Hence, Area of the remaining part of the park
 $= 10000 - 616 = 9384 \text{ m}^2$.

7. A rectangular lawn is 30 m by 20 m. It has two roads each 2 m wide running in the middle of it, one parallel to the length and the other parallel to the breadth. Find the area of the roads.

Sol.



Area of road ABCD $= 30 \times 2 = 60 \text{ m}^2$
 Area of road PQRS $= 20 \times 2 = 40 \text{ m}^2$
 Area of the shaded common square $= 2 \times 2 = 4 \text{ m}^2$
 Hence, Area of the roads $= 60 + 40 - 4 = 96 \text{ m}^2$.

1

EXERCISE

Fill in the Blanks :

DIRECTIONS : Complete the following statements with an appropriate word / term to be filled in the blank space(s).

- The distance covered by a farmer around a field of 120 m length and 80 m width is _____ m.
- 1 hectare = _____ m^2 .
- All the congruent triangles have _____ area.
- _____ squares each of side 1 m make a square of side 5 km.
- Perimeter of a regular polygon = length of one side $\times$ _____.
- If a wire in the shape of a rectangle is rebent into a square, then the _____ of both shapes remain same, but _____ may vary.
- Perpendicular dropped on the base of a parallelogram from the opposite vertex is known as the corresponding _____ of the base.
- Ratio of the circumference of a circle to its diameter is denoted by symbol _____.
- $1 \text{ m}^2 =$ _____ cm^2 .
- $1 \text{ cm}^2 =$ _____ mm^2 .

True / False :

DIRECTIONS : Read the following statements and write your answer as true or false.

- If perimeter of two parallelograms are equal, then their areas are also equal.
- All congruent triangles are equal in area.
- 1 cm^2 is the area of a square whose side is of length 1 cm.
- One hectare is the area of a square whose side is one hectometre long.
- Area of a triangle = (base $\times$ height).
- Area of a circle = $2\pi r^2$.
- If perimeter of two rectangles are equal, then their areas are also equal.
- $500 \text{ m}^2 = 5$ hectares

Match the Columns :

DIRECTIONS : Each question contains statements given in two columns which have to be matched. Statements (A, B, C, D) in column-I have to be matched with statements (p, q, r, s) in column-II.

1. Column - I (Figure)	Column - II (Area)
(A) Triangle	(p) $(\text{side})^2$
(B) Square	(q) $\pi (\text{radius})^2$
(C) Circle	(r) $\frac{1}{2} \text{ base} \times \text{height}$
(D) Parallelogram	(s) $\text{base} \times \text{height}$

Very Short Answer Questions :

DIRECTIONS : Give answer in one word or one sentence.

- Length and breadth of a rectangle are 3.2 m and 150 cm. Then find the area
- A playground which is 250 m long and 20 m broad is to be fenced with wire. How much wire is needed?
- One side of a parallelogram is 14 cm. Its distance from the opposite side is 16.5 cm. Find the area of the parallelogram.
- Find the area of the circle whose radius is :
(i) 2.1 cm (ii) 4.2 cm
(iii) 6.3 cm (iv) 7.7 cm
- Find the circumference of a circle whose diameter is
(i) 7 cm (ii) 4.2 cm (iii) 11.2 cm
- Find the area in square centimetres of the triangle whose base and altitude are as under:
(i) Base = 15 cm, Altitude = 8 cm
(ii) Base = 1.5 m, Altitude = 0.8 m
- Find the area, in square metres, of a rectangle whose
(i) length = 4.5 m, breadth = 1.6 m
(ii) length = 720 cm, breadth = 25 cm
(iii) length = 3 dm 6 cm, breadth = 2 dm 9 cm.

8. Find the area, in square centimetres, of a square whose side is
 - (i) 2.7 dm
 - (ii) 50 mm
9. Find the area of a rhombus in which
 - (i) each side = 12 cm and height = 7.5 cm,
 - (ii) each side = 2 dm and height = 12.6 cm.
10. Find the area of a quadrant if the radius is 7 cm.

Short Answer Questions :

DIRECTIONS : Give answer in 2-3 sentences.

1. The circumference of a circle is 44 m; then find the area of the circle.
2. The area of a circle is 2464 m^2 , then find the diameter.
3. If the ratio of the areas of two squares is 9 : 1 then find the ratio of their perimeter.
4. The base and height of a triangle are in the ratio 7 : 4. If the area of the triangle is 87.5 cm^2 , then find the base.
5. What is the cost of levelling the field in the form of a parallelogram at the rate of ₹ 50 per 10 m^2 , the base and perpendicular distance from the opposite side being 54 m and 24 m respectively.
6. Find the number of square shaped tin sheets of side 20 cm that can be cut off from a square tin sheet of side 1 m.
7. How many bricks 20 cm by 10 cm will be needed to pave the floor of a room 25 m long and 16 m wide?
8. The length and breadth of a rectangular field are 120 m and 75 m respectively. Find
 - (i) The area of the field and the cost of turfing it at ₹ 15 per m^2 ,
 - (ii) The perimeter of the field and the cost of fencing it at ₹ 40 per m.

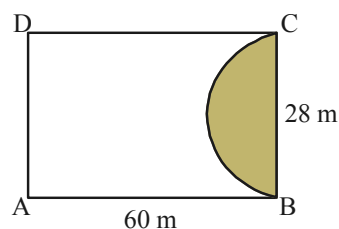
Long Answer Questions :

DIRECTIONS : Give answer in four to five sentences.

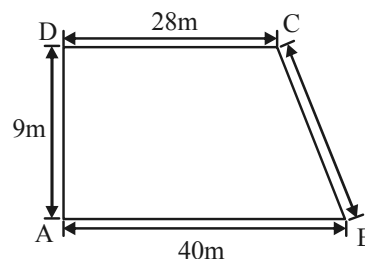
1. Area of four walls of a rectangular room is 77 m^2 . The length and breadth of the room are 7.5 m and 3.5 m respectively. Find the height of the room.
2. Circumference of a circular field is 1188 m. A road 7 m wide is constructed outside the field. Find its area.
3. A wire bent in the form of a circle of radius 42 cm is cut and again bent in the form of a square. Find the ratio of the regions enclosed by the

circle and the square in the two cases.

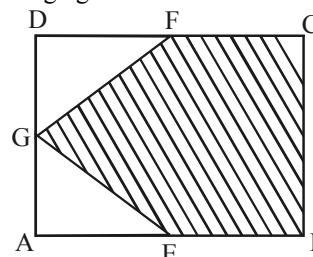
4. A plot is in the form of a rectangle ABCD having semi-circle on BC as shown in figure. The semi-circle portion is grassy while the remaining plot is without grass. Find the area of the plot without grass where $AB = 60 \text{ m}$ and $BC = 28 \text{ m}$.



5. In the given figure, ABCD is a field in the form of a quadrilateral in which $AB \parallel CD$. If $\angle DAB = 90^\circ$. Find the area of field.



6. A table cover of size $4 \text{ m} \times 2 \text{ m}$ is spread on a meeting table so that 0.25 m of the cover is hanging all around the table. Find the cost of polishing the table top at ₹ 2.25 per square metre.
7. Calculate the area of the shaded region of the following figure.



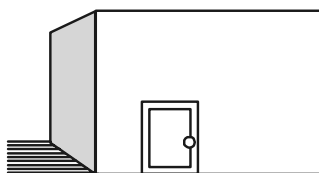
ABCD is a rectangle having length 30 cm, breadth 20 cm. E, F and G are midpoints of AB, CD and AD respectively.

8. A rectangular lawn is 60 m by 40 m and has two roads, each 5 m wide running in the middle of it, one parallel to its length and the other parallel to the breadth. Find the cost of constructing the roads at ₹ 80 per m^2 .

2 EXERCISE

Text-Book Exercise :

1. A wire is in the shape of a rectangle. Its length is 40 cm and breadth is 22 cm. If the same wire is rebent in the shape of a square, what will be the measure of each side. Also find which shape encloses more area?
2. A door of length 2 m and breadth 1 m is fitted in a wall. The length of the wall is 4.5 m and the breadth is 3.6 m (see the given figure). Find the cost of white washing the wall, if the rate of white washing the wall is ₹ 20 per m^2 .



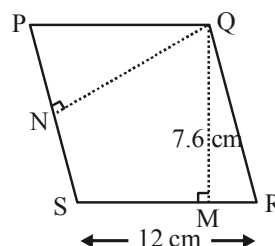
3. Find the missing values:

S.No.	Base	Height	Area of the parallelogram
(i)	20 cm		246 cm^2
(ii)		15 cm	154.5 cm^2
(iii)		8.4 cm	48.72 cm^2
(iv)	15.6 cm		16.38 cm^2

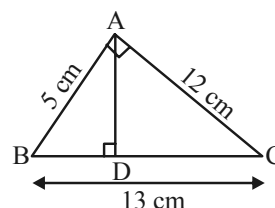
4. Find the missing values:

S. No.	Base	Height	Area of triangle
(i)	15 cm	—	87 cm^2
(ii)	—	31.4 mm	1256 mm^2
(iii)	22 cm	—	170.5 cm^2

5. PQRS is a parallelogram (see the given figure). QM is the height from Q to SR and QN is the height from Q to PS. If $SR = 12 \text{ cm}$ and $QM = 7.6 \text{ cm}$. Find:

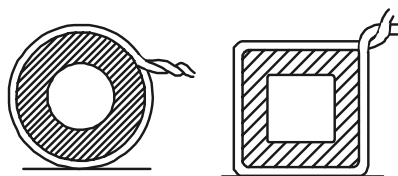


- (i) The area of the parallelogram PQRS
 - (ii) QN, if $PS = 8 \text{ cm}$
6. $\triangle ABC$ is right angled at A (see the given figure). AD is perpendicular to BC. If $AB = 5 \text{ cm}$, $BC = 13 \text{ cm}$ and $AC = 12 \text{ cm}$, find the area of $\triangle ABC$. Also find the length of AD.

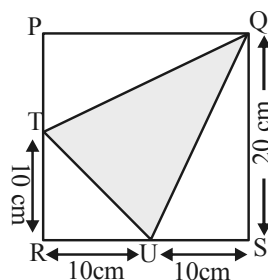


7. A gardener wants to fence a circular garden of diameter 21 m. Find the length of the rope he need to purchase, if he makes 2 rounds of fence, Also find the costs of the rope, if it cost ₹ 4 per meter. (Take $\pi = \frac{22}{7}$)
8. Saima wants to put a lace on the edge of a circular table cover of diameter 1.5 m. Find the length of the lace required and also find its cost if one meter of the lace costs ₹ 15. (Take $\pi = 3.14$)
9. A circle of radius 2 cm is cut out from a square piece of an aluminium sheet of side 6 cm. What is the area of the left over aluminium sheet? (Take $\pi = 3.14$)
10. How many times a wheel of radius 28 cm must rotate to go 352 m? (Take $\pi = \frac{22}{7}$)

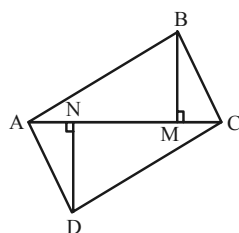
11. A verandah of width 2.25 m is constructed all along outside a room which is 5.5 m long and 4 m wide. Find.
- the area of the verandah.
 - the cost of cementing the floor of the verandah at the rate of ₹ 200 per m^2
12. Pragya wrapped a cord around a circular pipe of radius 4 cm (adjoining figure) and cut off the length required of the cord. Then she wrapped it around a square box of side 4 cm (also shown). Did she have any cord left? ($\pi = 3.14$)



13. Find the area of the shaded portions :



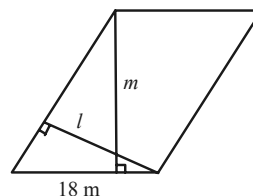
14. Find the area of the quadrilateral ABCD. Here, $AC = 22$ cm, $BM = 3$ cm, $DN = 3$ cm, and $BM \perp AC$, $DN \perp AC$



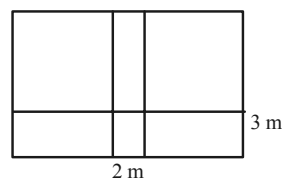
Exemplar Questions :

- Area of an isosceles triangle is 48 cm^2 . If the altitudes corresponding to the base of the triangle is 8 cm, find the perimeter of the triangle.
- Perimeter of a parallelogram shaped land is 96 m and its area is 270 square metres. If one of the

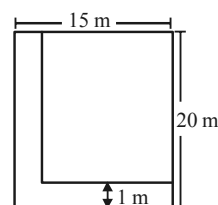
sides of this parallelogram is 18 m, find the length of the other side. Also, find the lengths of altitudes l and m .



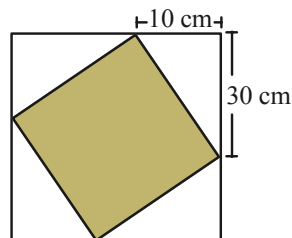
3. A school playground is divided by a 2 m wide path which is parallel to the width of the playground, and a 3 m wide path which is parallel to the length of the ground. If the length and width of the playground are 120 m and 80 m respectively, find the area of the remaining playground.



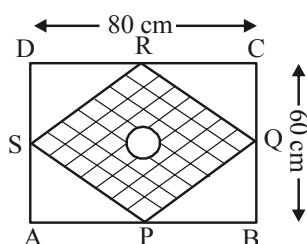
4. In a park of dimensions $20 \text{ m} \times 15 \text{ m}$, there is a L shaped 1 m wide flower bed as shown in figure. Find the total cost of manuring for the flower bed at the rate of ₹ 45 per m^2 .



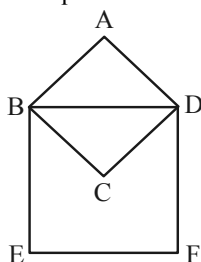
5. A design is made up of four congruent right triangles as shown in figure. Find the area of the shaded portion.



6. A rectangular field is 48 m long and 12 m wide. How many right triangular flower beds can be laid in this field, if sides including the right angle measure 2 m and 4 m, respectively?
7. ABCD is a given rectangle with length as 80 cm and breadth as 60 cm. P, Q, R, S are the mid points of sides AB, BC, CD, DA respectively. A circular rangoli of radius 10 cm is drawn at the centre as shown in figure. Find the area of shaded portion.

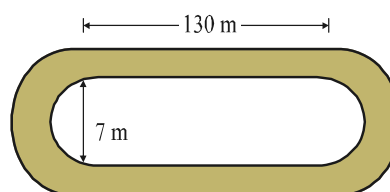


8. 4 squares each of side 10 cm have been cut from each corner of a rectangular sheet of paper of size $100\text{ cm} \times 80\text{ cm}$. From the remaining piece of paper, an isosceles right triangle is removed whose equal sides are each of 10 cm length. Find the area of the remaining part of the paper.
9. The moon is about 384000 km from earth and its path around the earth is nearly circular. Find the length of path described by moon in one complete revolution. (Take $\pi = 3.14$)
10. In figure ABCD is a square with $AB = 15\text{ cm}$ find the area of the square BDFE.

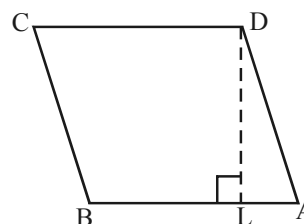


HOTS Questions :

1. The perimeter of a circular plot is equal to that of a square plot. What is the ratio of their respective areas?
2. An athletic track 14 m wide consists of two straight sections 130 m long joining semicircular ends whose inner radius is 3.5 m. Find the area of the shaded region.



3. In the following figure, ABCD is a parallelogram. $DL \perp AB$ and $AB = 13\text{ cm} = AD$. If the area of parallelogram is 156 cm^2 . Find AL.



4. The perimeter of a trapezium is 52 cm and its both non-parallel sides are equal to 10 cm and its altitude is 8 cm. Find its area.
5. One diagonal of a rhombus is 24 cm and its side is 13 cm. Find the area of the rhombus.

3

EXERCISE

Single Option Correct :

DIRECTIONS : This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONLY ONE is correct.

- The area of a square field is 0.5 hectare. The length of its diagonal is
(a) 100m (b) 50m
(c) 250m (d) $50\sqrt{2}$
- The circumference of a circle is 44 cm. Its area is
(a) 308 cm^2 (b) 154 cm^2
(c) 77 cm^2 (d) 616 cm^2
- The length and breadth of a rectangular park are in the ratio 4 : 3 and its perimeter is 56 m. The area of the field is
(a) 192 m^2 (b) 300 m^2
(c) 432 m^2 (d) 228 m^2
- If the length and breadth of a rectangle are doubled then its perimeter is
(a) tripled (b) doubled
(c) made half (d) None of these
- The difference between circumference and radius of a circle is 37 m. The circumference of that circle is
(a) 7m (b) 44m
(c) 154m (d) None of these
- If the circumference of a circle is $\frac{30}{\pi}$, then the diameter of the circle is
(a) 60π (b) $\frac{15}{\pi}$
(c) $\frac{30}{\pi^2}$ (d) 30
- The area of a circle of radius 5 cm is numerically what percent of its circumference?
(a) 200 (b) 225
(c) 240 (d) 250
- The area of a rectangular lawn is $\frac{5}{16}$ hectare. The ratio of the base : height is 1 : 5. The height of the lawn is
(a) 5m (b) 20m
(c) 30m (d) 125m

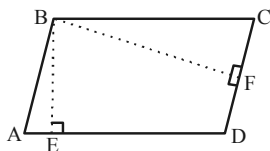
- The circumference of a circle whose area is $36\pi^2$, is
(a) 9π (b) 18π
(c) 12π (d) 81π
- The area of a semicircle of radius $6r$ is
(a) $18\pi^2$ (b) $4\pi^2$
(c) $12\pi^2$ (d) $2\pi^2$
- If the radius of a circle is doubled, the area becomes
(a) 9 times (b) 3 times
(c) 6 times (d) 4 times
- If $1\text{ m}^2 = x\text{ cm}^2$, then the value of x is
(a) 1000 (b) 10000
(c) 100000 (d) 1000000

More Than One Option Correct :

DIRECTIONS : This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONE or MORE may be correct.

- Which of the following statement (s) is/are true about the area of a circle?
(a) Area of a circle is equal to four times the area of a quadrant of same radius.
(b) Area of a circle cannot be divided into two equal parts.
(c) If the measure of diameter of a circle is given we can determine its area.
(d) Area of circle = πr^2 , where r is the diameter of the circle
- In reference to a circle the value of π is equal to
(a) $\frac{\text{area}}{\text{circumference}}$ (b) $\frac{\text{area}}{(\text{radius})^2}$
(c) $\frac{\text{circumference}}{\text{diameter}}$ (d) $\frac{\text{circumference}}{\text{radius}}$
- Circumference of a circle is always
(a) more than three times of its diameter
(d) three times of its diameter
(c) less than three times of its diameter
(d) more than six times of its radius

4. Area of parallelogram ABCD (see figure) is equal to



- (a) $DE \times DC$ (b) $BE \times AD$
 (c) $BF \times DC$ (d) $BE \times BC$
5. 1 km^2 equals to
- (a) 1000 m^2 (b) 100 hm^2
 (c) 1000000 m^2 (d) 1000 hm^2

Assertion & Reason :

DIRECTIONS : Each of these questions contains an Assertion followed by Reason. Read them carefully and answer the question on the basis of following options. You have to select the one that best describes the two statements.

- (a) If both **Assertion** and **Reason** are correct and **Reason** is the correct explanation of **Assertion**.
 (b) If both **Assertion** and **Reason** are correct but **Reason** is not the correct explanation of **Assertion**.
 (c) If **Assertion** is correct but **Reason** is incorrect.
 (d) If **Assertion** is incorrect but **Reason** is correct.
1. **Assertion :** Altitude of a triangle whose area is 45 cm^2 and base 15 cm is 5 cm .

Reason : Area of a triangle = $\frac{1}{2} \times \text{base} \times \text{altitude}$

2. **Assertion :** If the area of a square with side 'a' is equal to the area of a triangle with base 'a', then the altitude of the triangle is 'a'.
Reason : The area of a circle of radius 5 cm is numerically 250 percent of its circumference.

Passage Based Questions :

DIRECTIONS : Study the given passage(s) and answer the following questions.

PASSAGE - I

A piece of wire in the form of a rectangle 12 cm long and 10 cm broad is reshaped and bent into the form of circle.

1. Length of the wire is
 (a) 44 cm (b) 18 cm
 (c) 12 cm (d) 15 cm
2. Radius of the circle is
 (a) 4 cm (b) 8 cm
 (c) 6 cm (d) 7 cm
3. Area of the circle is
 (a) 215 cm^2 (b) 212 cm^2
 (c) 318 cm^2 (d) 154 cm^2

PASSAGE - II

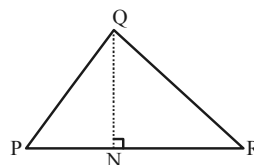
A horse is tied to a pole fixed at one corner of a $30 \text{ m} \times 30 \text{ m}$ square field of grass, by means of 10 m long rope.

4. The area of that part of the field in which the horse can graze is
 (a) 78.5 m^2 (b) 77 m^2
 (c) 76 m^2 (d) 75 m^2
5. The increase in the grazing area if the rope were 20 m long instead of being 10 m long, is
 (a) 236 (b) 237
 (c) 235.5 (d) 238

Integer Type Questions :

DIRECTIONS : Answer the following questions. The answer to each of the question is a single digit integer, ranging from 0 to 9.

1. What is the radius of the largest circle that can be cut out of the rectangle measuring 12 cm in length and 10 cm in breadth?
 2. Area of a right-angled triangle is 6 cm^2 . If its smallest side is 3 cm , then its hypotenuse is
 3. Area of triangle PQR is 28 cm^2 (see figure). If altitude QN is 7 cm , then its base PR is



4. Area of a rectangle and the area of a circle are equal. If the dimensions of the rectangle are $14 \text{ cm} \times 11 \text{ cm}$, then radius of the circle is
 5. A wire is bent to form a square of side 11 cm . If the wire is rebent to form a circle, its radius is

Multiple Matching Question :

DIRECTIONS : Following question has four statements (A, B, C and D) given in Column-I and five statements (p, q, r, s, t) in Column-II. Any given statement in Column-I can have correct matching with one or more statement(s) given in Column-II.

1. Match the following units.

Column-I	Column-II
(A) 1 cm^2	(p) 10000 m^2
(B) 1 m^2	(q) 1000000 m^2
(C) 1 km^2	(r) 10000 cm^2
(D) 1 hectare	(s) 100 mm^2
	(t) 1000000 mm^2

SOLUTIONS

Brief Explanations of Selected Questions

1 EXERCISE

Fill in the Blanks :

- | | |
|--------------------|--------------------|
| 1. 400m | 2. 10000 |
| 3. equal | 4. 2,50,00,000 |
| 5. number of sides | 6. perimeter, area |
| 7. height/altitude | 8. π |
| 9. 10000 | 10. 100 |

True / False :

- | | | |
|----------|----------|----------|
| 1. False | 2. True | 3. True |
| 4. True | 5. False | 6. False |
| 7. False | 8. False | |

Match the Columns :

1. (A) $\rightarrow$ (r); (B) $\rightarrow$ (p); (C) $\rightarrow$ (q); (D) $\rightarrow$ (s)

Very Short Answer Questions :

- $l = 3.2$ m and $b = 150$ cm = 1.5 m.
Area = $l \times b = 3.2 \times 1.5 = 4.80$ sq. m.
- Clearly the playground is of rectangular shape.
 $\therefore$ The required length of the wire
 $= 2$ (length + breadth) = 2 (250 + 20)
 $= 540$ m.
- Area of the parallelogram = base $\times$ height
 $= (14 \times 16.5) \text{ cm}^2 = 231 \text{ cm}^2$.
- (i) 13.86 cm^2 (ii) 55.44 cm^2
(iii) 124.74 cm^2 (iv) 186.34 cm^2
- (i) 22 cm (ii) 13.2 cm
(iii) 35.2 km
- (i) Area of the triangle = $\left(\frac{1}{2} \times 15 \times 8\right) \text{ cm}^2$
 $= 60 \text{ cm}^2$
(ii) Area of the triangle = $\left(\frac{1}{2} \times 150 \times 80\right) \text{ cm}^2$
 $= 6000 \text{ cm}^2$

- (i) Area of the rectangle = $4.5 \times 1.6 \text{ m}^2$
 $= 7.20 \text{ m}^2$
(ii) Area of the rectangle = $7.2 \times 0.25 \text{ m}^2$
 $= 1.8 \text{ m}^2$
(iii) Length = 3 dm 6 cm = 36 cm = 0.36 m
[$\because 1 \text{ dm} = 10 \text{ cm}$]
Breadth = 2 dm 9 cm = 29 cm = 0.29 m
 $\therefore$ Area of the rectangle = $0.36 \times 0.29 \text{ m}^2$
 $= 0.1044 \text{ m}^2$
- (i) Side of the square = 2.7 dm = 27 cm
 $\therefore$ Area of the square = $(27)^2 \text{ cm}^2$
 $= 729 \text{ cm}^2$
(ii) Side of the square = 50 mm = 5 cm
Area of the square = $(5)^2 \text{ cm}^2 = 25 \text{ cm}^2$
- (i) 90 cm^2 (ii) 252 cm^2
- 38.5 cm^2

Short Answer Questions :

- $2\pi r = 44 \Rightarrow r = \frac{44}{2 \times \frac{22}{7}} = 7$ m
Area of a circle = $\pi r^2 = \frac{22}{7} \times 7 \times 7 = 154 \text{ m}^2$
- $\pi r^2 = 2464 \Rightarrow r^2 = \frac{2464}{\frac{22}{7}} = 784$
 $\Rightarrow r = \sqrt{784} = 28$ m
 $\Rightarrow$ Diameter = $2 \times 28 \text{ m} = 56 \text{ m}$
- Let the areas of two square be $9x^2$ and x^2 respectively. Then their sides will be $\sqrt{9x^2}$ and $\sqrt{x^2}$, i.e. $3x$ and x respectively. Hence, the ratio of their perimeter = $\frac{4 \times 3x}{4 \times x} = 3:1$

4. Let base be $7x$ and height be $4x$. Then, the area

$$\text{of the triangle} = \frac{1}{2} \times 7x \times 4x = 14x^2 \text{ cm}^2$$

$$\text{But, } 14x^2 = 87.5 \Rightarrow x = 2.5$$

$$\therefore \text{Base} = 2.5 \times 7 = 17.5 \text{ cm}$$

5. Area of the parallelogram = $54 \text{ m} \times 24 \text{ m} = 1296 \text{ m}^2$

$$\text{Total cost} = ₹ \frac{1296}{10} \times 50 = ₹ 6480$$

6. Required number of sheets = $\frac{100 \times 100}{20 \times 20} = 25$
 ($\because 1 \text{ m} = 100 \text{ cm}$)

7. Area of the room = $25 \text{ m} \times 16 \text{ m} = 400 \text{ m}^2$
 Area of the brick = $20 \text{ cm} \times 10 \text{ cm}$

$$= \frac{1}{5} \text{ m} \times \frac{1}{10} \text{ m} = \frac{1}{50} \text{ m}^2$$

$\therefore$ Number of bricks required

$$= \left(400 \div \frac{1}{50} \right) = 20,000.$$

8. (i) Area of the field = $(120 \times 75) \text{ m}^2 = 9000 \text{ m}^2$.
 Cost of turfing = $₹ (9000 \times 15) = ₹ 135000$.
 (ii) Perimeter of the field = $2(l + b)$ units
 $= 2(120 + 75) \text{ m} = (2 \times 195) \text{ m} = 390 \text{ m}$.
 $\therefore$ The cost of fencing = $₹ (390 \times 40)$
 $= ₹ 15600$.

Long Answer Questions :

1. Let h be the height of the room in metre, then

$$\text{Area of four walls} = 2(\text{length} + \text{breadth}) \times h$$

$$\Rightarrow 77 = 2(7.5 + 3.5) \times h$$

$$\Rightarrow h = \frac{77}{2 \times 11} \Rightarrow h = 3.5 \text{ m}$$

2. $2\pi r = 1188 \Rightarrow r = \frac{1188}{22 \times 2} \times 7 = 189 \text{ m}$

$$\therefore \text{Outer radius} = R = 189 + 7 = 196 \text{ m}$$

$$\therefore \text{Area of road} = \pi(R^2 - r^2)$$

$$= \frac{22}{7} (196^2 - 189^2) \text{ m}^2 = 8470 \text{ m}^2$$

3. Length of wire = $2\pi \times 42 = 84\pi \text{ cm}$.

Let x be the side of the square.

$$\therefore 4x = 84\pi$$

$$\Rightarrow x = 21\pi$$

Area of the circle : Area of the square

$$= \pi(42)^2 : (21\pi)^2$$

$$= \pi \times 42 \times 42 : 21 \times 21 \times \pi \times \pi$$

$$= 4 : \pi = 4 : \frac{22}{7} = 14 : 11$$

4. Area of the plot without grass

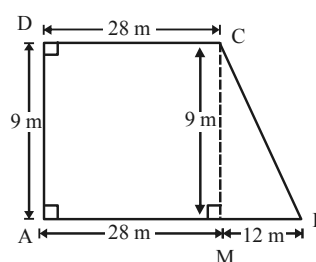
$$= \text{area of the rectangle} - \text{area of the semi-circle}$$

$$= 60 \times 28 - \frac{1}{2} \times \pi \times \left(\frac{28}{2} \right)^2$$

$$= 60 \times 28 - 11 \times 28 = 28(60 - 11)$$

$$= 1372 \text{ m}^2.$$

- 5.



Draw the perpendicular CM on AB from C.

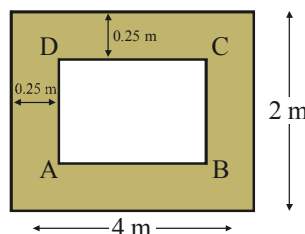
$$\therefore BM = AB - AM = 40 - 28 = 12 \text{ m}$$

Area of the field = area of rectangle AMCD

+ area of right angled triangle MBC

$$= 28 \times 9 + \frac{1}{2} \times 12 \times 9 = 306 \text{ m}^2.$$

- 6.



$$\begin{aligned}
 \text{Area of top ABCD} &= AB \times BC \\
 &= (4 - 0.25 - 0.25) \times (2 - 0.25 - 0.25) \\
 &= 3.5 \times 1.5 = 5.25 \text{ m}^2 \\
 \therefore \text{Cost of polishing ABCD} &= 5.25 \times 2.25 \\
 &= ₹ 11.81.
 \end{aligned}$$

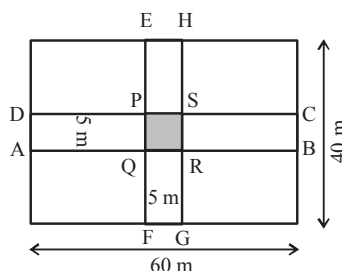
7. Area of rectangle = $l \times b = 30 \times 20 = 600 \text{ cm}^2$

$$\begin{aligned}
 \text{Area of } \triangle DGF &= \frac{1}{2} \times b \times h = \frac{1}{2} \times 10 \times 15 \\
 &= 75 \text{ cm}^2.
 \end{aligned}$$

Similarly, area of $\triangle AGE = 75 \text{ cm}^2$.

$$\begin{aligned}
 \text{Area of shaded region} &= 600 - (75 + 75) \\
 &= 600 - 150 = 450 \text{ cm}^2
 \end{aligned}$$

8.



Let ABCD and EFGH be the two roads.

$$\begin{aligned}
 \text{Area of two roads} &= (\text{area ABCD}) + (\text{area EFGH}) - (\text{area PQRS}) \\
 &= \{(60 \times 5) + (40 \times 5) - (5 \times 5)\} \text{ m}^2 \\
 &= (300 + 200 - 25) \text{ m}^2 = 475 \text{ m}^2.
 \end{aligned}$$

$$\begin{aligned}
 \text{Cost of constructing the roads} &= ₹(475 \times 80) \\
 &= ₹ 38000.
 \end{aligned}$$

2 EXERCISE

Text-Book Exercise :

- Perimeter of rectangle = Perimeter of square
 $\therefore 2(\text{Length} + \text{Breadth}) = 4 \times \text{Side of square}$
 $\Rightarrow 2(40 + 22) = 4 \times \text{Side of square}$
 $\therefore \text{Side of square} = \frac{124}{4} = 31 \text{ cm}$
 Now, area of rectangle = $40 \times 22 = 880 \text{ cm}^2$
 Area of square = $(\text{Side})^2 = 31 \times 31 = 961 \text{ cm}^2$
 $\therefore$ The square encloses more area.
- Area of wall = Length $\times$ Breadth
 $= 4.5 \times 3.6 = 16.2 \text{ m}^2$
 Area of door = $2 \times 1 = 2 \text{ m}^2$

$$\text{Area to be white-washed} = 16.2 - 2 = 14.2 \text{ m}^2$$

$$\begin{aligned}
 \therefore \text{Cost of white-washing } 14.2 \text{ m}^2 \text{ area} \\
 = 14.2 \times 20 = ₹ 284
 \end{aligned}$$

3. Area of parallelogram = Base $\times$ Height

$$(i) \quad h = \frac{246}{20} = 12.3 \text{ cm}$$

$$(ii) \quad b = \frac{154.5}{15} = 10.3 \text{ cm}$$

$$(iii) \quad b = \frac{48.72}{8.4} = 5.8 \text{ cm}$$

$$(iv) \quad h = \frac{16.38}{15.6} = 1.05 \text{ cm}$$

4. Area of triangle = $\frac{1}{2} \times \text{Base} \times \text{Height}$

Let b be the base of triangle and h be the height of triangle.

$$(i) \quad \text{Area} = \frac{1}{2} \times 15 \times h = 87 \text{ cm}^2$$

$$\Rightarrow h = \frac{87 \times 2}{15} = 11.6 \text{ cm}$$

$$(ii) \quad \frac{1}{2} \times b \times 31.4 = 1256$$

$$\Rightarrow b = \frac{1256 \times 2}{31.4} = 80 \text{ mm}$$

$$(iii) \quad \frac{1}{2} \times 22 \times h = 170.5$$

$$\Rightarrow h = \frac{170.2 \times 2}{22} = 15.5 \text{ cm}$$

5. (i) Area of parallelogram = Base $\times$ Height
 $= SR \times QM = 12 \times 7.6 = 91.2 \text{ cm}^2$

(ii) $PS = 8 \text{ cm}$

$$\begin{aligned}
 \text{Area of parallelogram} &= \text{Base} \times \text{Height} \\
 &= PS \times QN = 91.2 \text{ cm}^2
 \end{aligned}$$

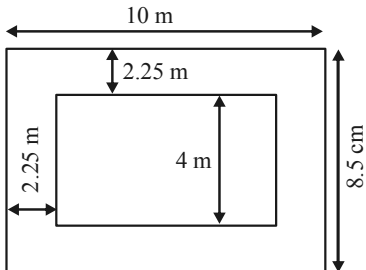
$$\Rightarrow 8 \times QN = 91.2 \Rightarrow QN = 11.4 \text{ cm}$$

6. Area of triangle = $\frac{1}{2} \times \text{Base} \times \text{Height}$

$$= \frac{1}{2} \times AB \times AC = \frac{1}{2} \times 5 \times 12 = 30 \text{ cm}^2$$

$$\text{Also, area of } \triangle ABC = \frac{1}{2} \times AD \times BC$$

$$\Rightarrow 30 = \frac{1}{2} \times AD \times 13 \Rightarrow AD = 4.6 \text{ cm}$$

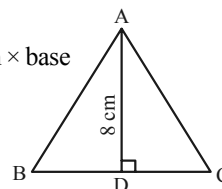
7. Circumference = $2\pi r = 2 \times \frac{22}{7} \times \frac{21}{2} = 66$ m
 Length of rope required = 2×66 m = 132 m
 Cost of 132 m rope = $4 \times 132 = ₹ 528$
8. $2\pi r = 2 \times 3.14 \times \frac{d}{2} = 2 \times 3.14 \times \frac{1.5}{2} = 4.71$ m
 Cost of 4.71 m lace = $4.71 \times 15 = ₹ 70.65$
9. Area of square-shaped sheet = $(6)^2 = 36$ cm²
 Area of circle = $3.14 \times 2 \times 2 = 12.56$ cm²
 Area of remaining sheet = $36 - 12.56 = 23.44$ cm²
10. Circumference = $2\pi r = 2 \times \frac{22}{7} \times 28 = 176$ cm
 Number of rotations = $\frac{352 \text{ m}}{176 \text{ cm}} = \frac{35200}{176} = 200$
11. (i) Area of room = $l \times b = 5.5 \times 4 = 22$ m²
- 
- Area of a room including the verandah
 = $8.5 \times 10 = 85$ m²
 Area of verandah = $85 - 22 = 63$ m²
- (ii) Cost of cementing 63 m² area of the floor of the verandah = $200 \times 63 = ₹ 12600$
12. Perimeter of the square = $4 \times \text{Side of the square}$
 = $4 \times 4 = 16$ cm
 Perimeter of circular pipe = $2\pi r = 2 \times 3.14 \times 4 = 25.12$ cm
 Length of cord left = $25.12 - 16 = 9.12$ cm
13. Area (QTU) = area (PQSR) – area (TRU) – area (SUQ) – area (PQT)
 = $(20 \times 20) - \frac{1}{2}(10 \times 10) - \frac{1}{2}(20 \times 10) - \frac{1}{2}(20 \times 10)$
 = $400 - 50 - 100 - 100 = 150$ cm²
14. Area (ABCD) = Area (ABC) + Area (ADC)
 = $\frac{1}{2}(3 \times 22) + \frac{1}{2}(3 \times 22) = 33 + 33 = 66$ cm²

Exemplar Questions :

1. Area of triangle = $\frac{1}{2} \times \text{base} \times \text{height}$

$$\Rightarrow 48 \text{ cm}^2 = \frac{1}{2} \times 8 \text{ cm} \times \text{base}$$

$$\Rightarrow \text{base} = 12 \text{ cm.}$$



Since, the altitude to the base of an isosceles triangle bisects the base.

$$\therefore BD = DC = \frac{12}{2} \text{ cm} = 6 \text{ cm.}$$

$$AB = \sqrt{AD^2 + BD^2} = \sqrt{8^2 + 6^2}$$

$$= \sqrt{64 + 36} = \sqrt{100} = 10 \text{ cm.}$$

$$\therefore \text{Perimeter of the triangle} = 10 \text{ cm} + 10 \text{ cm} + 12 \text{ cm} = 32 \text{ cm.}$$

2. Let the length of other side of the parallelogram be x metre.

$$\therefore \text{Perimeter of parallelogram} = 2(x + 18) \text{ m}$$

$$\Rightarrow 96 = 2(x + 18)$$

$$\Rightarrow 48 = x + 18 \Rightarrow x = 30 \text{ m.}$$

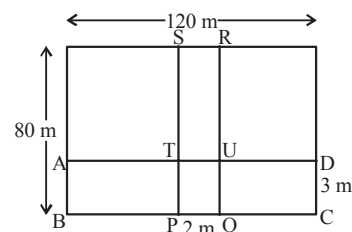
Now, Area of parallelogram = base $\times$ height

$$\Rightarrow 270 = 18 \times m \Rightarrow m = \frac{270}{18} = 15 \text{ metre.}$$

Again, Area of parallelogram = base $\times$ height

$$\Rightarrow 270 = 30 \times l \Rightarrow l = \frac{270}{30} = 9 \text{ metre.}$$

- 3.



$$\text{Area of path ABCD} = 120 \times 3 = 360 \text{ m}^2$$

$$\text{Area of path PQRS} = 80 \times 2 = 160 \text{ m}^2$$

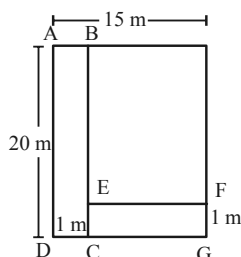
$$\text{Area of common path PQUT} = 3 \times 2 = 6 \text{ m}^2$$

$$\text{Area of play ground} = 120 \times 80 = 9600 \text{ m}^2$$

$$\text{Area used as path} = 360 + 160 - 6 = 514 \text{ m}^2$$

$$\therefore \text{The area of the remaining play ground} = (9600 - 514) \text{ m}^2 = 9086 \text{ m}^2.$$

4.



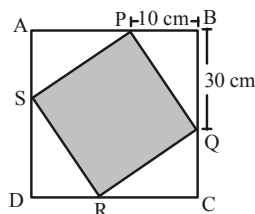
$$\text{Area of } ABCD = 20 \times 1 = 20 \text{ m}^2$$

$$\text{Area of } EFGC = 14 \times 1 = 14 \text{ m}^2$$

$$\therefore \text{Area of the flower bed} = (20 + 14) \text{ m}^2 = 34 \text{ m}^2$$

$$\text{The cost of manuring} = ₹ 45 \times 34 = ₹ 1530.$$

5.



$$\text{Area of } \triangle PBQ = \frac{1}{2} \times PB \times BQ$$

$$= \frac{1}{2} \times 10 \times 30 = 150 \text{ cm}^2$$

$$\text{Area of 4 congruent triangles} = 4 \times 150 \text{ cm}^2 = 600 \text{ cm}^2$$

$$\text{Now, } AB = AP + PB = (30 + 10) \text{ cm} = 40 \text{ cm.}$$

$$\text{and } BC = BQ + QC = (30 + 10) \text{ cm} = 40 \text{ cm.}$$

$$\text{Area of } ABCD = (40 \times 40) \text{ cm}^2 = 1600 \text{ cm}^2$$

$$\text{Area of } PQRS = (1600 - 600) \text{ cm}^2 = 1000 \text{ cm}^2.$$

6. Area of rectangular field = $(48 \times 12) \text{ m}^2 = 576 \text{ m}^2$

$$\text{Area of a triangular flower bed} = \frac{1}{2} \times 2 \times 4 = 4 \text{ m}^2$$

$$\text{Number of triangular beds} = 576 \div 4 = 144.$$

7. Area of $\triangle RCQ = \frac{1}{2} \times RC \times CQ = \frac{1}{2} \times 40 \times 30 = 600 \text{ cm}^2$.Similarly, Area of $\triangle RDS$, $\triangle SAP$ and $\triangle PBQ$ is also 600 cm^2 .

$$\therefore \text{Area of 4 triangles} = (4 \times 600) \text{ cm}^2 = 2400 \text{ cm}^2$$

$$\text{Area of rectangle } ABCD = 80 \times 60 = 4800 \text{ cm}^2$$

Area of PQRS

$$= \text{Area of } ABCD - \text{Area of 4 triangles}$$

$$= 4800 - 2400 = 2400 \text{ cm}^2.$$

$$\text{Area of circular rangoli} = \pi r^2 = 3.14 \times 10 \times 10 = 314 \text{ cm}^2.$$

$$\therefore \text{Area of shaded portion} = (2400 - 314) \text{ cm}^2 = 2086 \text{ cm}^2.$$

8. Area of a square paper = $(10 \text{ cm})^2 = 100 \text{ cm}^2$.

$$\text{Area of 4 squares} = (4 \times 100) \text{ cm}^2 = 400 \text{ cm}^2.$$

Area of an isosceles right triangle

$$= \left(\frac{1}{2} \times 10 \times 10 \right) \text{ cm}^2 = 50 \text{ cm}^2$$

$$\text{Area of rectangular sheet} = (100 \times 80) \text{ cm}^2 = 8000 \text{ cm}^2.$$

Area of remaining part of paper

$$= 800 - (400 + 50) = 350 \text{ cm}^2.$$

9. Distance between Earth and the moon

$$= 384000 \text{ km.}$$

$$\text{The length of path described by moon} = 2\pi r$$

$$= 2 \times 3.14 \times 384000 \text{ km} = 2411520 \text{ km.}$$

$$10. \quad BD = \sqrt{(AB)^2 + (AD)^2} = \sqrt{15^2 + 15^2}$$

$$= \sqrt{225 + 225} = \sqrt{450} = 15\sqrt{2} \text{ cm}$$

$$\text{Area of square } BDFE = (\text{side})^2$$

$$= (15\sqrt{2})^2 \text{ cm}^2 = 450 \text{ cm}^2.$$

HOTS Questions :1. Let $4x$ be the perimeter of the square, so that its area is x^2 .

$$\text{Circumference} = 2\pi r = 4x$$

$$\therefore r = \frac{2x}{\pi}$$

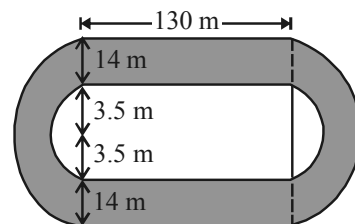
$$\therefore \text{Area} = \pi r^2 = \pi \left(\frac{2x}{\pi} \right)^2 = \pi \frac{4}{\pi^2} x^2$$

 $\therefore$ Ratio of area of the circle and the square

$$= \frac{4}{\pi} x^2 : x^2 = 4 : \pi \text{ or } 4 : \frac{22}{7}$$

$$= 28 : 22 \text{ or } 14 : 11.$$

2.



Area of shaded region = $2 \times (130 \times 14) + 2 \times$

$$\left[\frac{1}{2} \times \frac{22}{7} \times (14 + 3.5)^2 - \frac{1}{2} \times \frac{22}{7} \times (3.5)^2 \right]$$

$$= 3640 + \frac{22}{7} (306.25 - 12.25)$$

$$= 4564 \text{ m}^2$$

3. Area = 156 cm^2

$$AB \times DL = 156$$

$$DL = \frac{156}{AB} = \frac{156}{13} = 12 \text{ cm.}$$

In right $\triangle ADL$,

$$AD^2 = DL^2 + LA^2$$

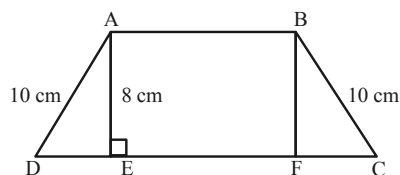
$$AL^2 = AD^2 - DL^2 = 13^2 - 12^2$$

$$= 169 - 144 = 25$$

$$AL = \sqrt{25} = 5 \text{ cm.}$$

4. As shown in figure, ADE is a triangle.

$$\therefore DE^2 = 10^2 - 8^2 = 6^2 \text{ or } DE = 6 \text{ cm.}$$



By symmetry, $FC = 6 \text{ cm.}$

Let $AB = x$

$$\text{Then, } (x + 10 + 6 + x + 6 + 10) = 52$$

$$\text{or } 2x + 32 = 52 \text{ or } x = 10 \text{ cm.}$$

$$\therefore \text{Area of trapezium} = 2 \times \left(\frac{1}{2} \times 6 \times 8 \right) + 10 \times 8$$

$$= 128 \text{ cm}^2.$$

5. Let diagonals be d_1 and d_2 . Since diagonals of rhombus intersect at right angle

$$(\text{Side})^2 = \left[\frac{1}{2} \times d_1 \right]^2 + \left[\frac{1}{2} \times d_2 \right]^2$$

$$(13^2) = \left[\frac{1}{2} \times d_1 \right]^2 + \left(\frac{1}{2} \times 24 \right)^2$$

$$\left(\frac{1}{2} \times d_1 \right)^2 = 169 - 144 = 25$$

$$\therefore d_1 = 10 \text{ cm}$$

$$\therefore \text{Area} = 2 \left[\frac{1}{2} \times d_1 \times \frac{d_2}{2} \right]$$

$$= \frac{1}{2} \times 10 \times 24 = 120 \text{ sq. cm.}$$

3 EXERCISE

Single Option Correct :

1. (a) 2. (b) 3. (a)
4. (b) Let the initial length and breadth be l and b respectively.
So, the initial perimeter = $2(l + b)$
The final perimeter = $2(2l + 2b) = 4(l + b)$

$$\text{Now, } \frac{\text{Final perimeter}}{\text{Initial perimeter}} = \frac{4(l + b)}{2(l + b)} = 2.$$

5. (b) $2\pi r - r = 37$

$$r = \frac{37}{2\pi - 1} = \frac{37}{2 \times \frac{22}{7} - 1} = 7 \text{ m}$$

$$\text{Circumference} = 2\pi r = 2 \times \frac{22}{7} \times 7 = 44 \text{ m.}$$

6. (c) $2\pi r = \frac{30}{\pi}, 2r = \frac{30}{\pi^2}.$

7. (d)

9. (c)

8. (d)

10. (a) Area = $\frac{\pi(6r)^2}{2} = \frac{\pi \times 6r \times 6r}{2} = 18\pi r^2$

11. (d)

12. (b)

More Than One Option Correct :

1. (a, c)

2. (b, c)

3. (a, d)

4. (b, c, d)

5. (b, c)

$$1 \text{ km}^2 = 100 \text{ hm}^2 \text{ and } 1 \text{ km}^2 = 1000000 \text{ m}^2$$

Assertion & Reason :

1. (d) **Assertion** : incorrect; **Reason** : correct

$$\text{Altitude} = \frac{2(\text{Area})}{\text{Base}} = \frac{2 \times 45}{15} = 6 \text{ cm}$$

2. (d) **Assertion** : incorrect; **Reason** : correct

$$a \times a = \frac{1}{2} \times a \times h \Rightarrow h = 2a$$

$$\frac{25\pi}{10\pi} \times 100$$

$$\Rightarrow \text{Required \%} = \frac{25 \times 100}{10} = 250 \%$$

Passage Based Questions :

1. (a) Length of the wire = $2(12 + 10) = 44 \text{ cm}$
2. (d) $2\pi r = 44 \Rightarrow r = \frac{44 \times 7}{22 \times 2} = 7 \text{ cm}$
3. (d) $\text{Area} = \pi r^2 = \frac{22}{7} \times 7 \times 7 = 154 \text{ cm}^2$
4. (a) Required Area = $\frac{1}{4} \times (\pi \times 10^2)$
 $= \frac{1}{4} \times 3.14 \times 100 = 78.5 \text{ cm}^2$
5. (c) Area of the field in which the horse can now graze
 $= \frac{1}{4} \times \pi \times (20)^2 = 314 \text{ m}^2$
 $\therefore \text{Increase in grazing area} = 314 - 78.5$
 $= 235.5 \text{ m}^2$

Integer Type Questions :

1. (5) The largest circle with diameter 10 cm can be cut.
Hence, radius = $\frac{10}{2} \text{ cm} = 5 \text{ cm}$
2. (5) $\text{Area} = \frac{1}{2} \times b \times h \Rightarrow 6 = \frac{1}{2} \times 3 \times h$
 $\Rightarrow h = \frac{6 \times 2}{3} = 4 \text{ cm}$
Hypotenuse = $\sqrt{b^2 + h^2} = \sqrt{3^2 + 4^2}$
 $= \sqrt{9 + 16} = \sqrt{25} = 5 \text{ cm}$
3. (8) $\text{Area} = \frac{1}{2} \times \text{QN} \times \text{PR} \Rightarrow 28 = \frac{1}{2} \times 7 \times \text{PR}$
 $\Rightarrow \text{PR} = \frac{28 \times 2}{7} = 8 \text{ cm}$
4. (7) Area of rectangle = Area of circle
 $\Rightarrow (14 \times 11) \text{ cm}^2 = \pi r^2$
 $\Rightarrow \frac{14 \times 11 \times 7}{22} = r^2$
 $\Rightarrow r = \sqrt{7 \times 7} = 7 \text{ cm}$
5. (7) Length of wire = $4 \times 11 \text{ cm} = 44 \text{ cm}$
Now, Circumference of circle = 44 cm
 $2\pi r = 44 \text{ cm} \Rightarrow r = 7 \text{ cm}$

Multiple Matching Question :

1. (A) $\rightarrow$ (s); (B) $\rightarrow$ (r, t); (C) $\rightarrow$ (q); (D) $\rightarrow$ (p).

Chapter

12

Algebraic Expressions

INTRODUCTION

In this chapter, we will look at the definitions of terms, constant, variable, like and unlike terms. Besides, we shall also discuss the addition and subtraction of algebraic expressions.

ALGEBRAIC EXPRESSION

A combination of constants and variables connected by some or all of the four fundamental operations $+$, $-$, $\times$ and $\div$ is called an *algebraic expression*.

Constant:

A symbol having fixed numerical value is called constant.

For example: 5, 3.6, $\frac{3}{4}$ etc.

Variable:

A symbol which can have various numerical value is called variable.

For example: x, y, z etc.

TERMS OF AN ALGEBRAIC EXPRESSION:

The different parts of algebraic expression separated by the sign $+$ or $-$, are called the terms of algebraic expression.

For example: $3 + 2xy + 5x^2y$ have three terms 3, $2xy$ and $5x^2y$

FACTORS OF A TERM:

Each term of algebraic expression consists of a product of constants and variables. A constant factor is called numerical factor, while variable factor is known as a *literal factor*.

For example: The term $-3xy$ is a product of the factors -3 , x and y.

COEFFICIENT OF TERM

In a term of an algebraic expression any of the factors with the sign of the term is called the *coefficient* of the product of the other factors in that term.

For example: In $3ab$, the coefficient of a is $3b$ and the coefficient of b is $3a$.

LIKE AND UNLIKE TERMS

Like Terms

In an algebraic expression, terms having the same variables and powers are called *like terms*. Their coefficients do not need to match.

Unlike Terms

The terms that are not like i.e. they do not have the same variables or powers are called *unlike terms*.

For example: In the expression $4yx - 2y^2 + 4x^2 + 2xy$; $4yx$ and $2xy$ are like terms.

VARIOUS TYPES OF ALGEBRAIC EXPRESSIONS:

- (i) **Monomials:** An algebraic expression which contain only one term, is called *monomial*.
For examples: $5x$, $2y$, $3xy$, $-3a^2$, $-5a^2b$ etc are *monomials*.
- (ii) **Binomials:** An algebraic expression which contain two terms is called *binomial*.
For examples: $5x + y$, $2y + z$, $-3x^2 + y^2$, $2a^2 + ab$ etc. are *binomials*.
- (iii) **Trinomials:** An algebraic expression contain three terms are called *trinomials*.
For examples: $(a + 2b + 5c)$, $(x^2 + xy + 6y^2)$, $(x^2 - y^3 - z^3)$ etc are *trinomials*.
- (iv) **Quadrinomials:** An algebraic expression containing four terms are called *quadrinomials*.
For example: $(2x + y + 3z + 5)$, $(x^2 + 2y^2 + zz + xz)$, $(x^3 + y^3 + z^3 + 3xy)$ etc are *quadrinomials*.
- (v) **Polynomials:** An expression containing one or more terms is called a *polynomial*.

NOTE : A polynomial can be a monomial, trinomial or binomial.

ILLUSTRATION : 1

State which of the following pairs of terms are of like terms and which are of unlike terms:

- (i) $-a^2b^2$, $2ab^2$ (ii) $9x$, $-3x$ (iii) $-6ab$, $-3ab$, (iv) 4 , $13y$

SOLUTION :

- (i) $-a^2b^2$, $2ab^2$
These are unlike terms
 $\therefore$ They have variables with different powers
- (ii) $9x$, $-3x$
These are like terms
 $\therefore$ They have same variables with same powers.
- (iii) $-6ab$, $-3ab$
These are like terms
 $\therefore$ They have same variables with same powers.
- (iv) 4 , $13y$
These are unlike terms as one of the terms is constant and other contains variable.

ADDITION OF ALGEBRAIC EXPRESSIONS

To add two algebraic expressions, we collect different groups of like terms and find the sum of like terms in each group.

ILLUSTRATION : 1

Add; $7x^2 - 4x + 5$, $-3x^2 + 2x - 1$ and $5x^2 - x + 9$.

$$\begin{aligned} & (7x^2 - 4x + 5) + (-3x^2 + 2x - 1) + (5x^2 - x + 9) \\ &= 7x^2 - 4x + 5 - 3x^2 + 2x - 1 + 5x^2 - x + 9 \\ &= 7x^2 - 3x^2 + 5x^2 - 4x + 2x - x + 5 - 1 + 9 \\ &= (7 - 3 + 5)x^2 + (-4 + 2 - 1)x + (5 - 1 + 9) \\ &= 9x^2 - 3x + 13 \end{aligned}$$

In order to subtract an algebraic expression from another, we change the signs (from $-$ to $+$ or $+$ to $-$) of all the terms of the expression which is to be subtracted and then the two expressions are added.

Subtract $(2x^3 - 4x^2 + 3x + 5)$ from $(4x^3 + x^2 + x + 6)$

$$\begin{aligned} & (4x^3 + x^2 + x + 6) - (2x^3 - 4x^2 + 3x + 5) \\ &= 4x^3 + x^2 + x + 6 - 2x^3 + 4x^2 - 3x - 5 \\ &= (4-2)x^3 + (1+4)x^2 + (1-3)x + (6-5) \\ &= 2x^3 + 5x^2 - 2x + 1 \end{aligned}$$

If the perimeter of a square is a , then its perimeter $= 4a$

(iii) Perimeter of a rectangle = $2 \times (\text{length} + \text{breadth})$

If the length of a rectangle is l and breadth is b , then its perimeter = $2(l + b)$

For example: If $l = 4$ cm and $b = 3$ cm, where l and b are length and breadth of a rectangle respectively, then the perimeter of the rectangle = $2(l + b)$

$$= 2(4 + 3)$$

$$= 14 \text{ cm}$$

Formulae for Area

(i) Area of triangle = $\frac{1}{2} \times \text{base} \times \text{height}$

If b is the base and h is the height of triangle, then its area, $A = \frac{1}{2} \times b \times h$

(ii) Area of square = $(\text{length of its side})^2$

So, the area of a square = a^2 , where a is side of square.

(iii) Area of rectangle = $\text{length} \times \text{Breadth}$

So, the area of a rectangle = lb , where l and b are length and breadth respectively of rectangle.

For example: If side of a square is 5 cm, then the area of square = a^2

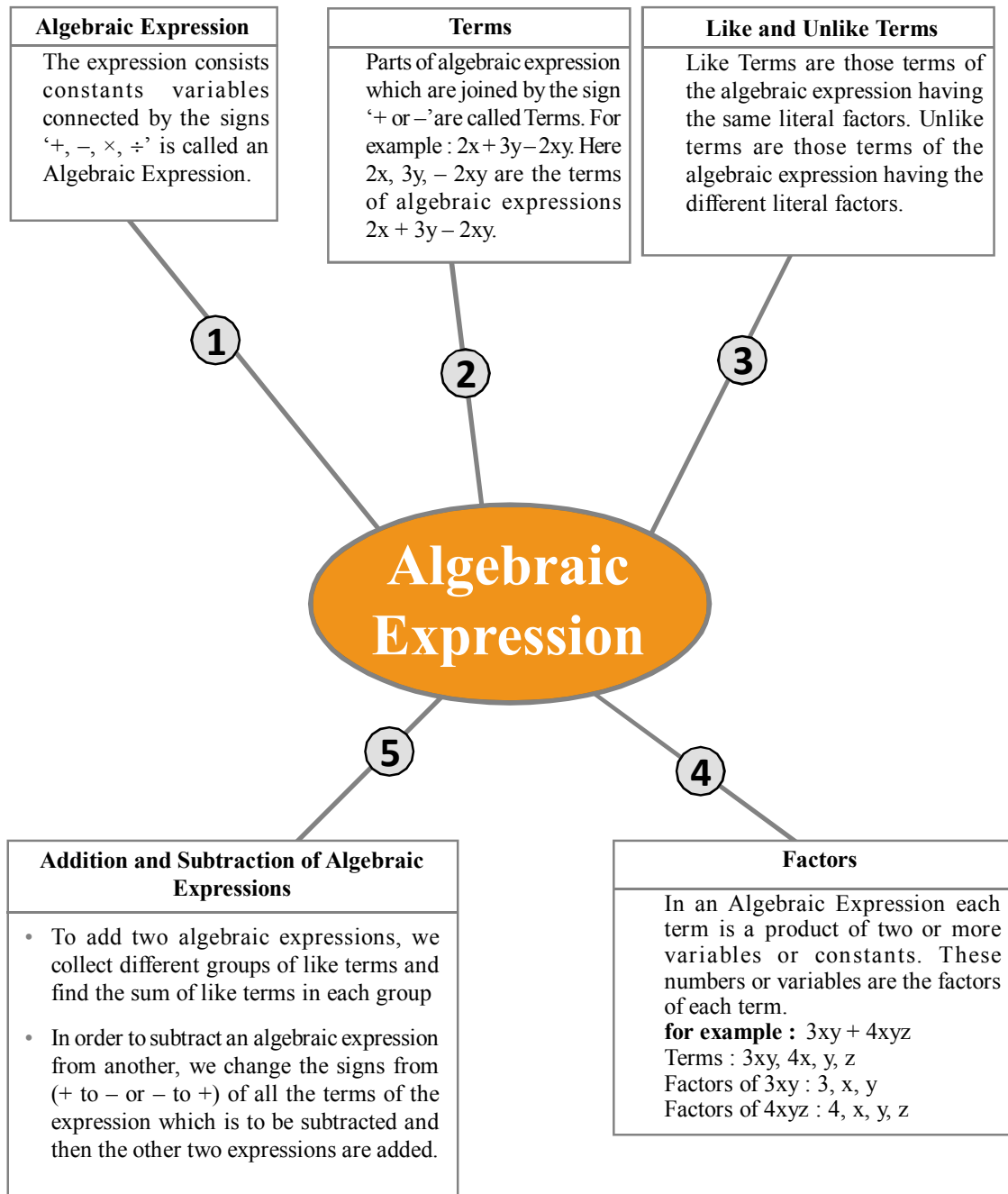
$$= (5)^2$$

$$= 25 \text{ cm}^2$$

Rules for Number Patterns

1. If a natural number is denoted by n , then its successor is $(n + 1)$.
2. If a natural number is denoted by n , $2n$ is an even number and $(2n + 1)$ is an odd number.

CONCEPT MAP



MISCELLANEOUS

SOLVED EXAMPLES

1. Classify into monomials, binomials and trinomials:

(i) $3x - 5y$ (ii) x^2

(iii) $a + b - ab$ (iv) 50

(v) $2x^2 + y^2 - xy$

Sol. Monomials : x^2 , 50

Binomial : $3x - 5y$

Trinomials : $a + b - ab$, $2x^2 + y^2 - xy$

2. Identify the like terms in each of the following algebraic expressions:

(i) $3a + 2b - c + \frac{3}{2}a - 4 + 3b$

(ii) $xy^2 + 3x^2y - 4x^2y^3 - 5y^2x - 2z^2x + 3xz^2$

Sol. (i) In the algebraic expression $3a + 2b - c + \frac{3}{2}a - 4 + 3b$, the groups of like terms are

$3a$, $\frac{3}{2}a$ and $2b$, $3b$.

(ii) In the algebraic expression $xy^2 + 3x^2y - 4x^2y^3 - 5y^2x - 2z^2x + 3xz^2$, the groups of like terms are xy^2 , $-5y^2x$ and $-2z^2x$, $3xz^2$.

3. Simplify: $(a^2 + b^2 + 2ab) - (a^2 + b^2 - 2ab)$

Sol. $(a^2 + b^2 + 2ab) - (a^2 + b^2 - 2ab)$

$= a^2 + b^2 + 2ab - a^2 - b^2 + 2ab$

$= a^2 - a^2 + b^2 - b^2 + 2ab + 2ab$

$= (1 - 1)a^2 + (1 - 1)b^2 + (2 + 2)ab$

$= 0a^2 + 0b^2 + 4ab$

$= 4ab$ [$0 \cdot a^2 = 0$ and $0 \cdot b^2 = 0$]

4. From the sum of $4x^4 - 3x^3 + 6x^2$, $4x^3 + 4x - 3$ and $-3x^4 - 5x^2 + 2x$ subtract $5x^4 - 7x^3 - 3x + 4$.

Sol. The sum of $4x^4 - 3x^3 + 6x^2$, $4x^3 + 4x - 3$ and $-3x^4 - 5x^2 + 2x$ is given by

$(4x^4 - 3x^3 + 6x^2) + (4x^3 + 4x - 3) + (-3x^4 - 5x^2 + 2x)$

$= 4x^4 - 3x^4 - 3x^3 + 4x^3 + 6x^2 - 5x^2 + 4x + 2x - 3$

$= x^4 + x^3 + x^2 + 6x - 3$

Now, we have to subtract $5x^4 - 7x^3 - 3x + 4$ from $x^4 + x^3 + x^2 + 6x - 3$.

$\therefore$ Required expression $= (x^4 + x^3 + x^2 + 6x - 3) - (5x^4 - 7x^3 - 3x + 4)$

$= x^4 + x^3 + x^2 + 6x - 3 - 5x^4 + 7x^3 + 3x - 4$

$= (x^4 - 5x^4) + (x^3 + 7x^3) + x^2 + 6x + 3x - 3 - 4$
 $= -4x^4 + 8x^3 + x^2 + 9x - 7.$

5. How much does $2a^2 - 5a + 4$ exceed $3a^3 - 5a^2 + 7a - 9$?

Sol. Required expression $= (2a^2 - 5a + 4) - (3a^3 - 5a^2 + 7a - 9)$

$= 2a^2 - 5a + 4 - 3a^3 + 5a^2 - 7a + 9$

$= -3a^3 + 2a^2 + 5a^2 - 5a - 7a + 4 + 9$

$= -3a^3 + 7a^2 - 12a + 13.$

6. Evaluate each of the following algebraic expressions for $x=2, y=-3, z=-2, a=2, b=3$:

(i) $4xy^2 - 3yz^2 + 4x^2z$

(ii) $x^3 + y^3 + 3xyz + ab$

(iii) $5 + 4z^3 - 6y + 7a + xy$

Sol. Substituting $x=2, y=-3, z=-2, a=2$ and $b=3$ in the given expressions, we have

(i) $4xy^2 - 3yz^2 + 4x^2z$
 $= 4 \times 2 \times (-3)^2 - 3 \times (-3) \times (-2)^2 + 4 \times (2)^2 \times (-2)$

$= 4 \times 2 \times 9 - 3 \times (-3) \times 4 + 4 \times 4 \times -2$

$= 72 + 36 - 32 = 76$

(ii) $x^3 + y^3 + 3xyz + ab = 2^3 + (-3)^3 + 3 \times 2 \times (-3) \times (-2) + 2 \times 3$
 $= 8 + (-27) + 36 + 6 = 23$

(iii) $5 + 4z^3 - 6y + 7a + xy$
 $= 5 + 4 \times (-2)^3 - 6 \times (-3) + 7 \times 2 + 2 \times -3$
 $= 5 + 4 \times -8 + 6 \times 3 + 7 \times 2 - 2 \times 3$
 $= 5 - 32 + 18 + 14 - 6$
 $= 37 - 38 = -1$

7. Priya borrowed ₹ 2000 for t years at the rate of $r\%$ p.a. If interest is calculated as simple interest, then find the algebraic expression to find amount.

Sol. Principal = ₹ 2000

Time = t years

Rate = $r\%$ p.a.

$\therefore$ Simple interest $= \frac{\text{Principle} \times \text{Rate} \times \text{Time}}{100}$

$= \frac{2000 \times r \times t}{100} = 20rt$

Amount = Simple interest + Principal

$= 20rt + 2000$

1

EXERCISE

Fill in the Blanks :

DIRECTIONS : Complete the following statements with an appropriate word / term to be filled in the blank space(s).

- $3a^2b$ and $-7ba^2$ are _____ terms.
- $-5a^2b$ and $-5b^2a$ are _____ terms.
- The speed of car is 55 km/hrs. The distance covered in y hours is _____.
- In the expression $2\pi r$, the algebraic variable is _____.
- Sum or difference of two like terms is _____.
- If $(x^2y + y^2 + 3)$ is subtracted from $(3x^2y + 2y^2 + 5)$, then coefficient of y in the result is _____.
- If Rohit has $5xy$ toffees and Shantanu has $20yx$ toffees, then Shantanu has _____ more toffees.
- The terms with same algebraic factors are called _____.
- Numerical factor in any term of a polynomial is called _____ of the term.
- The terms with different algebraic factors are called _____.

True / False :

DIRECTIONS : Read the following statements and write your answer as true or false.

- $(3a - b + 3) - (a + b)$ is a binomial.
- A polynomial with more than two terms is a trinomial.
- A binomial has more than two terms.
- $4p$ is the numerical coefficient of q^2 in $-4pq^2$.
- A trinomial can be a polynomial.
- In like terms, variables and their powers are the same.
- $5a$ and $5b$ are unlike terms.
- In like terms, the numerical coefficients should also be the same.

- Sum of $x^2 + x$ and $y + y^2$ is $2x^2 + 2y^2$.
- If we add a monomial and binomial, then answer can never be a monomial.

Match the Columns :

DIRECTIONS : Each question contains statements given in two columns which have to be matched. Statements (A, B, C, D) in column-I have to be matched with statements (p, q, r, s) in column-II.

1.	Column-I (Expression)	Column-II (Coefficient of x)
	(A) $4x - 3y$	(p) y^2
	(B) $8 - x + y$	(q) -1
	(C) $y^2x - y$	(r) $-5z$
	(D) $2z - 5xz$	(s) 4

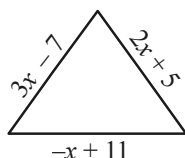
Very Short Answer Questions :

DIRECTIONS : Give answer in one word or one sentence.

- If a and b are respectively the sum and product of coefficients of terms in the expression $x^2 + y^2 + z^2 - xy - yz - zx$, then find $a + 2b$.
- If $P = 3x^3 + 3x^2 + 3x + 3$ and $Q = 3x^2 - 3x + 3$, then find $P - Q$.
- Write down the coefficient of x in each of the following:

(i) $3x$	(ii) $-4ax$
(iii) $5xy^2$	(iv) xyz
(v) $-\frac{3}{2}x + 5$	(vi) $-\frac{5}{2}xyz^2$
- In a class of $(4y^2 + y - 8)$ students, $2y + 16$ play students football and the rest play basketball. How many students play basketball?
- $a^2 - (-a)^2$ is equal to
- What should be added to $-3a + 7b - 16$ to get the sum -5 ?
- Find the coefficient of a^2 in the sum of $-5a^2 + 8ab, -3a^2, 14a^2 - 11ab$

8. Find the perimeter of the given triangle.



9. Write down the algebraic expression whose terms are :

(i) $7x, -5, 8x^2$
 (ii) $-3p^2, 2p^2, -6pq, -5$

(iii) $\frac{2}{3}x^2, -\frac{4}{7}xy, 3y^2$

10. Write all the terms of each of the following algebraic expressions :

(i) $2a + 3b - 7c$ (ii) $a^2 - b^2 - 2ab$
 (iii) $9x + 4y^2 - 6z$ (iv) $a^2 + 3ab + b^2 - 1$

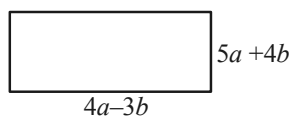
11. Write the coefficient of :

(i) x in $-6xy^2$ (ii) q in qx^2
 (iii) a in $\frac{2}{3}a$ (iv) x^2y in $-\frac{4}{9}ax^2y$
 (v) xy in $-4xyz$ (vi) z in $-z$

Short Answer Questions :

DIRECTIONS : Give answer in 2-3 sentences.

1. Subtract :
 (i) $5a - 3b$ from $2a + b - 2$
 (ii) $-x^2 - 3z$ from $5x^2 - y + z + 7$
 (iii) $-2a + b + 4d$ from $4a - 2b - c$.
 2. Which expression can be used to represent the perimeter of the given rectangle.



3. Add $5x^2 - 7x + 3, -8x^2 + 2x - 5$ and $7x^2 - x - 2$.
 4. Subtract $(2x^2 - 5x + 7)$ from $(3x^2 + 4x - 6)$.
 5. Simplify : $3x(4x - 5) + 4x(3x + 5)$ and find the value for (i) $x = 3$, (ii) $x = \frac{1}{2}$.
 6. What should be the value of a if the value of $2x^2 + x - a$ equals to 5, when $x = 0$?
 7. The length of a rectangle is 6 m more than its breadth. The perimeter of the rectangle is 60 m. Find the length and breadth of the rectangle.
 8. If $A = 3a - 2b, B = -2a + 3b$ and $C = -a - b$, then find $A + B + C$.

Long Answer Questions :

DIRECTIONS : Give answer in four to five sentences.

1. From the sum of $a^2 + 4b^2 - 6ab, a^2 - b^2 + 2ab, b^2 + 6$ and $a^2 - 4ab$, subtract $-2a^2 + b^2 - ab + a$.
 2. Add : $\left(3x^2 - \frac{1}{5}x + \frac{7}{3}\right) + \left(-\frac{1}{4}x^2 + \frac{1}{3}x - \frac{1}{6}\right) + \left(-2x^2 - \frac{1}{2}x + 5\right)$
 3. Take away $\left(\frac{8}{5}x^2 - \frac{2}{3}x^3 + \frac{3}{2}x - 1\right)$ from $\left(\frac{x^3}{5} - \frac{3}{2}x^2 + \frac{2}{3}x + \frac{1}{4}\right)$
 4. Subtract the sum of $10m - 8n + 6p^2$ and $-7m - 4n + 12p^2$ from the sum $2m - 4n + 6p^2$ and $m - 8n + 12p^2$.

2 EXERCISE

Text-Book Exercise :

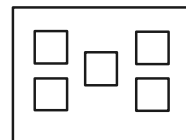
- Identify the terms and their factors in the following expressions. Show the terms and factors by tree diagrams.
 (i) $1 + x + x^2$ (ii) $y - y^3$
 (iii) $5xy^2 + 7x^2y$ (iv) $-ab + 2b^2 - 3a^2$
- Identify terms and factors in the expressions given below:
 (i) $xy + 2x^2y^2$
 (ii) $pq + q$
 (iii) $1.2ab - 2.4b + 3.6a$
 (iv) $\frac{3}{4}x + \frac{1}{4}$
- Identify like terms in the following:
 (i) $-xy^2, -4yx^2, 8x^2, 2xy^2, 7y, -11x^2, -100x, -11yx, 20x^2y, -6x^2, y, 2xy, 3x$
 (ii) $10pq, 7p, 8q, -p^2q^2, -7qp, -100q, -23, -12q^2p^2, -5p^2, 41, 2405p, 78qp, 13p^2q, qp^2, 701p^2$
- Simplify combining like terms:
 (i) $21b - 32 + 7b - 20b$
 (ii) $p - (p - q) - q - (q - p)$
 (iii) $3a - 2b - ab - (a - b + ab) + 3ab + b - a$
 (iv) $5x^2y - 5x^2 + 3yx^2 - 3y^2 + x^2 - y^2 + 8xy^2 - 3y^2$
- Subtract:
 (i) $-m^2 + 5mn$ from $4m^2 - 3mn + 8$
 (ii) $-x^2 + 10x - 5$ from $5x - 10$
 (iii) $5a^2 - 7ab + 5b^2$ from $3ab - 2a^2 - 2b^2$
- What should be subtracted from $2a + 8b + 10$ to get $-3a + 7b + 16$?
- What should be taken away from $3x^2 - 4y^2 + 5xy + 20$ to obtain $-x^2 - y^2 + 6xy + 20$?
- From the sum of $4 + 3x$ and $5 - 4x + 2x^2$, subtract the sum of $3x^2 - 5x$ and $-x^2 + 2x + 5$.
- Simplify these expressions and find their values if $x = 3, a = -1, b = -2$.
 (i) $2 - 8x + 4x + 4$ (ii) $10 - 3b - 4 - 5b$
 (iii) $2a - 2b - 4 - 5 + a$
- If $p = -10$, find the value of $p^2 - 2p - 100$
- What should be the value of a if the value of $2x^2 + x - a$ equals to 5, when $x = 0$?

- Use the given algebraic expression to complete the table of number patterns.

S. No.	Expression	Terms									
		1 st	2 nd	3 rd	4 th	5 th	...	10 th	...	100 th	...
(i)	$2n - 1$	1	3	5	7	9	-	19	-	-	-
(ii)	$7n + 20$	27	34	41	48	-	-	-	-	-	-
(iii)	$n^2 + 1$	2	5	10	17	-	-	-	-	10,001	-

Exemplar Questions :

- If $P = -(x - 2)$, $Q = -2(y + 1)$ and $R = -x + 2y$, find a , when $P + Q + R = ax$.
- From the sum of $x^2 - y^2 - 1$, $y^2 - x^2 - 1$ and $1 - x^2 - y^2$ subtract $-(1 + y^2)$.
- Subtract the sum of $12ab - 10b^2 - 18a^2$ and $9ab + 12b^2 + 14a^2$ from the sum of $ab + 2b^2$ and $3b^2 - a^2$.
- Arjun bought a rectangular plot with length x and breadth y and then sold a triangular part of it whose base is y and height is z . Find the area of the remaining part of the plot.
- Amisha has a square plot of side m and another triangular plot with base and height each equal to m . What is the total area of both plots?
- The rate of planting the grass is ₹ x per square metre. Find the cost of planting the grass on a triangular lawn whose base is y metres and height is z metres.
- In a rectangular plot, 5 square flower beds of side $(x + 2)$ metres each have been laid (see figure given below). Find the total cost of fencing the flower beds at the cost of ₹ 50 per 100 metres:



- A wire is $(7x - 3)$ metres long. A length of $(3x - 4)$ metres is cut for use. Now, answer the following questions:
 (i) How much wire is left?

- (ii) If this left out wire is used for making an equilateral triangle. What is the length of each side of the triangle so formed?
9. Rohan's mother gave him ₹ $3xy^2$ and his father gave him ₹ $5(xy^2 + 2)$. Out of this total money he spent ₹ $(10 - 3xy^2)$ on his birthday party. How much money is left with him?
10. How much does $93p^2 - 55p + 4$ exceed $13p^3 - 5p^2 + 17p - 90$?
2. What values of a and b make the given statement true?
 $(ay^2 + 3xy - 9x^2) - (-4y^2 + 8xy + bx^2)$
 $= 10y^2 - 5xy - 10x^2$
3. Place the last two terms in each of the following expressions in brackets preceded by minus sign.
 (i) $4x + 6y - 7z$
 (ii) $ax - by + cz + 4q$.
4. If $2x - 2 = 4$ and $0.05y = 0.15$, find the value of $y^2 - x^2$.

HOTS Questions :

1. If $||| + \equiv + \square$ represents $4x^2 + 3y^2 + z$, then write the algebraic expression for $\equiv + \square\square\square + |$.

3

EXERCISE

Single Option Correct :

DIRECTIONS : This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONLY ONE is correct.

1. What will be the simplified form of $21b - 32 + 5b - 20b$?
 (a) $21b - 20b - 32 + 7b$
 (b) $21b - 20b + 7b - 32$
 (c) $6b - 32$
 (d) None of the above
2. The algebraic expression of the statement : "number 5 added to three times the product of numbers m and n ".
 (a) $5mn + 3$ (b) $2m + 5n$
 (c) $3mn + 5$ (d) None of the above
3. What should be added to $x^2 + xy + y^2$ to obtain $2x^2 + 3xy$?
 (a) $-x^2 - 2xy + y^2$ (b) $x^2 - 2xy - y^2$
 (c) $x^2 + 2xy - y^2$ (d) $x^2 + 2xy + y^2$
4. Find the values of the expressions
 (i) $4x - 3$ and (ii) $19 - 5x^2$ for $x = 2$
 (a) 5, -1 (b) 5, 4
 (c) -1, 5 (d) 5, 1
5. What must be subtracted from $x^4 + 2x^2 - 3x + 7$ to get $x^3 + x^2 + x - 1$?
 (a) $x^4 - x^3 + x^2 - 4x + 8$
 (b) $x^3 + x^2 - 4x + 8$
 (c) $x^4 - x^3 + x^2 + 4x - 8$
 (d) $x^4 - x^3 - x^2 + 4x - 8$
6. The algebraic expression for "Twice of x added to y squared" is
 (a) $3x + y^2$ (b) $(x + 2)y^2$
 (c) $(x + 2)y^2$ (d) $2x + y^2$
7. What should be added to $-3p + 7q - 16$ to get the sum 8?
 (a) 8 (b) $-3p + 7q + 8$
 (c) $3p - 7q + 8$ (d) $3p - 7q + 24$
8. $(3A + B) - 3(A - B)$ equals
 (a) $4A$ (b) $4B$
 (c) $2A + 2B$ (d) $4A - 2B$
9. The coefficient of xy in $3zy + 7xyz - 2z^2x$ is
 (a) $3z$ (b) -2
 (c) $7yz$ (d) $7z$
10. The factors of the term $-xy^2$ are
 (a) $x \times y \times y$ (b) $-1 \times y \times y$
 (c) $-1 \times x \times y$ (d) $-1 \times x \times y \times y$

More Than One Option Correct :

DIRECTIONS : This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONE or MORE may be correct.

- Which of the following is/are monomial?
(a) $4p^2q - 4pq^2$ (b) $7mn$
(c) $8x^3y^5$ (d) $3x + 2$
- Which of the following expressions give 2 when $x = 5$?
(a) $4x - 5$ (b) $x - 3$
(c) $3x - 7$ (d) $x^2 - x - 18$
- Which of the following expressions give 2 when $x = 0, y = -1$.
(a) $2x + 2y$ (b) $2x^2 + y^2 + 1$
(c) $2x^2y + 2xy^2 + xy$ (d) $x^2 + xy + 2$
- Which of the following pairs have like terms?
(a) $-9yx, 20x^2y$ (b) $5p^2q, 101p^2q$
(c) $-a^2b^2, 4ab$ (d) $3ax^2, 4x^2a$
- Which of the following is/are binomial?
(a) $4ab - 5b^2 - 3a^2$ (b) $5mn - m^2$
(c) $8x^2 - 2$ (d) $5p^2 - 3q^2 + pq$

Assertion & Reason :

DIRECTIONS : Each of these questions contains an Assertion followed by Reason. Read them carefully and answer the question on the basis of following options. You have to select the one that best describes the two statements.

- If both **Assertion** and **Reason** are correct and **Reason** is the correct explanation of **Assertion**.
 - If both **Assertion** and **Reason** are correct but **Reason** is not the correct explanation of **Assertion**.
 - If **Assertion** is correct but **Reason** is incorrect.
 - If **Assertion** is incorrect but **Reason** is correct.
- Assertion :** $2a + 3b + c$ is a trinomial
Reason : An algebraic expression which contains only three terms is called trinomial.
 - Assertion :** An equation changes if the L.H.S. and R.H.S. are interchanged.
Reason : Algebraic expression must contain atleast one variable.

Passage Based Questions :

DIRECTIONS : Study the given passage(s) and answer the following questions.

PASSAGE-I

While adding or subtracting algebraic expressions, we add or subtract numerical co-efficients of like terms. For example: $ax^2y + bx^2y = (a + b)x^2y$, where a and b are numerical co-efficients of x^2y .

- Adding $2b^2 + 3ba$ and $-b^2 - ba$, we get
(a) $3b^2 + 4ba$ (b) $b^2 + 2ba$
(c) $4b^2 + 3ba$ (d) $2b^2 + 3ab$
- Subtract $-a^2 + 10a - 3$ from $5a - 7$.
(a) $a^2 - 5a - 4$ (b) $5a^2 - 4a + 1$
(c) $2a^2 - 3a + 5$ (d) $7a^2 + 5a$
- Add : $3p^2q^2 - 4pq + 5, -10p^2q^2, 15 + 9pq + 7p^2q^2$
(a) $20p^2q^2 + pq + 20$ (b) $5pq + 20$
(c) $5pq + 10$ (d) $p^2q^2 + 5pq + 20$

Integer Type Questions :

DIRECTIONS : Answer the following questions. The answer to each of the question is a single digit integer, ranging from 0 to 9.

- If $m = 2$, the value of $\frac{5m}{2} - 4$ is
- Find the value of the expression $x^2 + 2x + 1$ when $x = -1$
- The value of $25x^2 + 16y^2 + 40xy$ at $x = 1$ and $y = -1$ is
- Coefficient of xy^2 in $4x^2y + 3xy^2$ is
- If $x = 2$ and $y = -2$, then the value of $x^2 + y^2$ is

Multiple Matching Question :

DIRECTIONS : Following question has four statements (A, B, C, D) given in Column-I and six statements (p, q, r, s, t, u) in Column-II. Any given statement in Column-I can have correct matching with one or more statement(s) given in Column-II.

- | 1. | Column-I | Column-II |
|-----|--------------|----------------------------|
| (A) | Monomial | (p) $x^2y + x^3y + x^4y^5$ |
| (B) | Binomial | (q) $4m^2n + 3n^2m + 3$ |
| (C) | Trinomial | (r) $9x^2y^3z^4$ |
| (D) | Quadrinomial | (s) $3x^2 + 5y^2$ |
| | | (t) $z^2 - 3z + y + 8$ |
| | | (u) $9uv + 7$ |

SOLUTIONS

Brief Explanations of Selected Questions

1 EXERCISE

Fill in the Blanks :

- like
- unlike
- $55y$
- r
- a like term
- $2x^2$
- $15xy$
- like terms
- numerical coefficient or coefficient.
- unlike terms

True / False :

- False
- False
- False
- False
- True
- True
- True
- False
- False
- False

Match the Columns :

- (A) $\rightarrow$ (s), (B) $\rightarrow$ (q), (C) $\rightarrow$ (p), (D) $\rightarrow$ (r)

Very Short Answer Questions:

- -2
- $3x^3 + 6x$
- (i) 3
- (ii) $-4a$
- (iii) $5y^2$
- (iv) yz
- (v) $-\frac{3}{2}$
- (vi) $-\frac{5}{2}yz^2$
- $4y^2 - y - 24$
- 0
- $3a - 7b + 11$
- 6
- $4x + 9$
- (i) $7x - 5 + 8x^2$
- (ii) $-3p^2 + 2p^2 - 6pq - 5$
- (iii) $\frac{2}{3}x^2 - \frac{4}{7}xy + 3y^2$
- (i) $2a, 3b, -7c$
- (ii) $a^2, -b^2, -2ab$
- (iii) $9x, 4y^2, -6z$
- (iv) $a^2, 3ab, b^2, -1$
- (i) $-6y^2$
- (ii) x^2
- (iii) $\frac{2}{3}$
- (iv) $-\frac{4}{9}a$
- (v) $-4z$
- (vi) -1

Short Answer Questions :

- $$\begin{array}{r} 2a + b - 2 \\ 5a - 3b \\ \hline -3a + 4b - 2 \end{array}$$
 - $$\begin{array}{r} 5x^2 - y + z + 7 \\ -x^2 - 3z \\ \hline 6x^2 - y + 4z + 7 \end{array}$$
 - $$\begin{array}{r} 4a - 2b - c \\ -2a + b + 4d \\ \hline 6a - 3b - c - 4d \end{array}$$
- $18a + 2b$
- $$\begin{aligned} & (5x^2 - 7x + 3) + (-8x^2 + 2x - 5) + (7x^2 - x - 2) \\ &= 5x^2 - 8x^2 + 7x^2 - 7x + 2x - x + 3 - 5 - 2 \\ &= (5 - 8 + 7)x^2 + (-7 + 2 - 1)x + (3 - 5 - 2) \\ &= 4x^2 - 6x - 4. \end{aligned}$$
- $$\begin{aligned} & (3x^2 + 4x - 6) - (2x^2 - 5x + 7) \\ &= 3x^2 + 4x - 6 - 2x^2 + 5x - 7 \\ &= (3 - 2)x^2 + (4 + 5)x + (-6 - 7) \\ &= x^2 + 9x - 13. \end{aligned}$$
- $$\begin{aligned} & 3x(4x - 5) + 4x(3x + 5) \\ &= 12x^2 - 15x + 12x^2 + 20x \\ &= 24x^2 + 5x \end{aligned}$$
 - For $x = 3$

$$\begin{aligned} 24x^2 + 5x &= 24 \times (3)^2 + 5 \times 3 \\ &= 24 \times 3 \times 3 + 15 \\ &= 216 + 15 = 231 \end{aligned}$$
 - For $x = \frac{1}{2}$

$$\begin{aligned} 24x^2 + 5x &= 24 \times \left(\frac{1}{2}\right)^2 + 5 \times \frac{1}{2} \\ &= 24 \times \frac{1}{2} \times \frac{1}{2} + \frac{5}{2} \\ &= 6 + \frac{5}{2} = \frac{12 + 5}{2} = \frac{17}{2} \end{aligned}$$
- When $x = 0$; $2x^2 + x - a = 5$,

$$\therefore (2 \times 0) + 0 - a = 5$$

$$0 - a = 5 \Rightarrow a = -5$$
- Let breadth of the rectangle be x m

$$\therefore \text{Length of rectangle} = (x + 6) \text{ m}$$

Perimeter = 60 m

$$\therefore 2(x + 6 + x) = 60$$

$$\Rightarrow 2(2x + 6) = 60$$

$$\Rightarrow 4x + 12 = 60$$

$$\Rightarrow 4x = 60 - 12 = 48$$

$$\Rightarrow x = \frac{48}{4} = 12$$

$\therefore$ Breadth of rectangle = 12 m

Length of rectangle = $(12 + 6)$ m = 18 m

$$8. \quad A + B + C = 3a - 2b - 2a + 3b - a - b$$

$$= (3 - 2 - 1)a + (-2 + 3 - 1)b$$

$$= (3 - 3)a + (3 - 3)b = 0$$

Long Answer Questions :

$$1. \quad \begin{array}{r} a^2 + 4b^2 - 6ab \\ a^2 - b^2 + 2ab \\ + \quad b^2 \quad + 6 \\ \hline a^2 \quad - 4ab \end{array}$$

$$\text{Sum} = 3a^2 + 4b^2 - 8ab + 6$$

Now subtract the last expression from this sum

$$\begin{array}{r} 3a^2 + 4b^2 - 8ab + 6 \\ - 2a^2 + b^2 - ab + a \\ + \quad - \quad + \quad - \\ \hline \end{array}$$

$$\text{Difference} = 5a^2 + 3b^2 - 7ab + 6 - a$$

$$2. \quad \left(3x^2 - \frac{1}{5}x + \frac{7}{3}\right) + \left(-\frac{1}{4}x^2 + \frac{1}{3}x - \frac{1}{6}\right) + \left(-2x^2 - \frac{1}{2}x + 5\right)$$

$$= 3x^2 - \frac{1}{4}x^2 - 2x^2 - \frac{1}{5}x + \frac{1}{3}x - \frac{1}{2}x + \frac{7}{3} - \frac{1}{6} + 5$$

$$= \left(3 - \frac{1}{4} - 2\right)x^2 + \left(-\frac{1}{5} + \frac{1}{3} - \frac{1}{2}\right)x + \left(\frac{7}{3} - \frac{1}{6} + 5\right)$$

$$= \left(\frac{12 - 1 - 8}{4}\right)x^2 + \left(\frac{-6 + 10 - 15}{30}\right)x + \left(\frac{14 - 1 + 30}{6}\right)$$

$$= \frac{3}{4}x^2 - \frac{11}{30}x + \frac{43}{6}$$

$$3. \quad \left(\frac{x^3}{5} - \frac{3}{2}x^2 + \frac{2}{3}x + \frac{1}{4}\right) - \left(\frac{8}{5}x^2 - \frac{2}{3}x^3 + \frac{3}{2}x - 1\right)$$

$$= \frac{x^3}{5} - \frac{3}{2}x^2 + \frac{2}{3}x + \frac{1}{4} - \frac{8}{5}x^2 + \frac{2}{3}x^3 - \frac{3}{2}x + 1$$

$$= \left(\frac{1}{5} + \frac{2}{3}\right)x^3 + \left(-\frac{3}{2} - \frac{8}{5}\right)x^2 + \left(\frac{2}{3} - \frac{3}{2}\right)x + \left(\frac{1}{4} + 1\right)$$

$$= \frac{(3+10)}{15}x^3 + \left(\frac{-15-16}{10}\right)x^2 + \frac{(4-9)}{6}x + \frac{(1+4)}{4}$$

$$= \frac{13}{15}x^3 - \frac{31}{10}x^2 - \frac{5}{6}x + \frac{5}{4}$$

$$4. \quad \text{Sum of } 10m - 8n + 6p^2 \text{ and } -7m - 4n + 12p^2$$

$$= 10m - 8n + 6p^2 - 7m - 4n + 12p^2$$

$$= (10 - 7)m + (-8 - 4)n + (6 + 12)p^2$$

$$= 3m - 12n + 18p^2$$

Now, sum of $2m - 4n + 6p^2$ and $m - 8n + 12p^2$

$$= 2m - 4n + 6p^2 + m - 8n + 12p^2$$

$$= (2 + 1)m + (-4 - 8)n + (6 + 12)p^2$$

$$= 3m - 12n + 18p^2$$

$\therefore$ Required difference

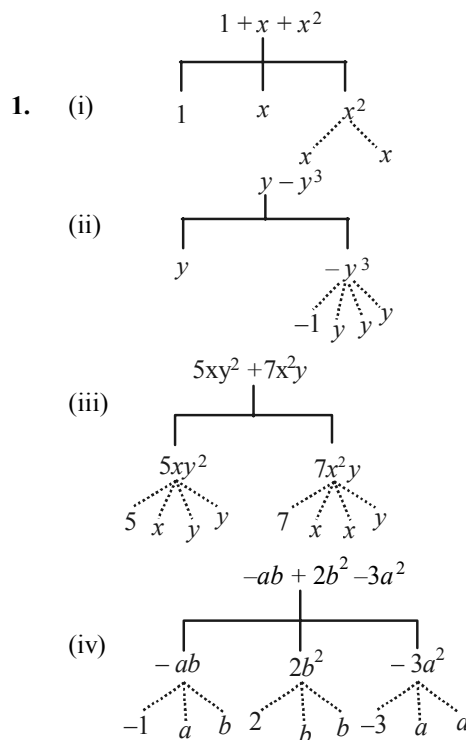
$$= 3m - 12n + 18p^2 - (3m - 12n + 18p^2)$$

$$= 3m - 12n + 18p^2 - 3m + 12n - 18p^2$$

$$= (3 - 3)m + (-12 + 12)n + (18 - 18)p^2 = 0$$

2 EXERCISE

Text-Book Exercise :



2.

	Expression	Terms	Factors
(i)	$xy + 2x^2y^2$	xy	x, y
(ii)	$pq + q$	pq q	p, q q
(iii)	$1.2ab - 24b + 3.6a$	$1.2ab$ $-24b$ $3.6a$	$1.2, a, b$ $-24, b$ $3.6, a$
(iv)	$\frac{3}{4}x + \frac{1}{4}$	$\frac{3}{4}x$ $\frac{1}{4}$	$\frac{3}{4}, x$ $\frac{1}{4}$

3. (i) $-xy^2, 2xy^2, -4yx^2, 20x^2y, 8x^2, -11x^2, -6x^2, 7y, y, -100x, 3x, -11yx, 2xy$.
(ii) $10pq, -7qp, 78qp, 7p, 2405p, 8q, -100q, -p^2q^2, -12q^2p^2, -23, 41, -5p^2, 701p^2, 13p^2q, qp^2$
4. (i) $8b - 32$
(ii) $p - q$
(iii) $a + ab$
(iv) $8x^2y + 8xy^2 - 4x^2 - 7y^2$
5. (i) $(4m^2 - 3mn + 8) - (-m^2 + 5mn)$
 $= 4m^2 - 3mn + 8 + m^2 - 5mn$
 $= 5m^2 - 8mn + 8$
(ii) $(5x - 10) - (-x^2 + 10x - 5)$
 $= 5x - 10 + x^2 - 10x + 5$
 $= x^2 - 5x - 5$
(iii) $(3ab - 2a^2 - 2b^2) - (5a^2 - 7ab + 5b^2)$
 $= 3ab - 2a^2 - 2b^2 - 5a^2 + 7ab - 5b^2$
 $= 10ab - 7a^2 - 7b^2$
6. Let A be the required term.
 $\therefore (2a + 8b + 10) - A = -3a + 7b + 16$
 $\Rightarrow A = 2a + 8b + 10 - (-3a + 7b + 16)$
 $= 2a + 8b + 10 + 3a - 7b - 16$
 $= 5a + b - 6$
7. Required term
 $= (3x^2 - 4y^2 + 5xy + 20) - (-x^2 - y^2 + 6xy + 20)$
 $= 3x^2 - 4y^2 + 5xy + 20 + x^2 + y^2 - 6xy - 20$
 $= 4x^2 - 3y^2 - xy$
8. Sum of $4 + 3x$ and $5 - 4x + 2x^2$
 $= (4 + 3x) + (5 - 4x + 2x^2)$
 $= 4 + 3x + 5 - 4x + 2x^2$

$$= -x + 2x^2 + 9$$

Now, sum of $3x^2 - 5x$ and $-x^2 + 2x + 5$

$$= (3x^2 - 5x) + (-x^2 + 2x + 5)$$

$$= 3x^2 - 5x - x^2 + 2x + 5$$

$$= 2x^2 - 3x + 5$$

$$\text{Required difference} = (-x + 2x^2 + 9) - (2x^2 - 3x + 5)$$

$$= -x + 2x^2 + 9 - 2x^2 + 3x - 5$$

$$= 2x + 4$$

9. (i) $-4x + 6; -6$ (ii) $-8b + 6; 22$

(iii) $3a - 2b - 9; -8$

10. 20

11. -5

12. (i) $2n - 1 \rightarrow 100^{\text{th}} : 199$

(ii) $7n + 20 \rightarrow 5^{\text{th}} : 55; 10^{\text{th}} : 90; 100^{\text{th}} : 720$

(iii) $n^2 + 1 \rightarrow 5^{\text{th}} : 26; 10^{\text{th}} : 101$

Exemplar Questions :

1. $P + Q + R = ax$
 $\Rightarrow -(x - 2) + [-2(y + 1)] + (-x + 2y) = ax$
 $\Rightarrow -x + 2 - 2y - 2 - x + 2y = ax$
 $\Rightarrow -x - x - 2y + 2y + 2 - 2 = ax$
 $\Rightarrow -2x = ax \Rightarrow a = -2$.
2. $[(x^2 - y^2 - 1) + (y^2 - x^2 - 1) + (1 - x^2 - y^2) - \{-(1 + y^2)\}]$
 $= (x^2 - x^2 - x^2 - y^2 + y^2 - y^2 - 1 - 1 + 1) - (-1 - y^2)$
 $= -x^2 - y^2 - 1 + 1 + y^2 = -x^2$.
3. The sum of $12ab - 10b^2 - 18a^2$ and $9ab + 12b^2 + 14a^2$
 $= 12ab - 10b^2 - 18a^2 + 9ab + 12b^2 + 14a^2$
 $= 12ab + 9ab - 10b^2 + 12b^2 - 18a^2 + 14a^2$
 $= 21ab + 2b^2 - 4a^2$.
The sum of $ab + 2b^2$ and $3b^2 - a^2$
 $= ab + 2b^2 + 3b^2 - a^2$
 $= ab + 5b^2 - a^2$.
Now,
 $ab + 5b^2 - a^2 - (21ab + 2b^2 - 4a^2)$
 $= ab + 5b^2 - a^2 - 21ab - 2b^2 + 4a^2$
 $= -20ab + 3b^2 + 3a^2$.
4. Area of rectangular plot $= x \times y = xy$.
Area of triangular part $= \frac{1}{2} \times y \times z = \frac{yz}{2}$
Area of remaining part $= xy - \frac{yz}{2} = y \left(x - \frac{z}{2} \right)$
5. Area of square plot $= (\text{side})^2 = m^2$
Area of triangular plot $= \frac{1}{2} \times m \times m = \frac{m^2}{2}$

$$\text{Total area of both plots} = m^2 + \frac{m^2}{2} = \frac{3}{2}m^2$$

6. Area of triangular lawn $= \frac{1}{2} \times y \times z = \frac{yz}{2}$

$$\text{The cost of planting grass} = ₹x \times \frac{yz}{2} = ₹\frac{xyz}{2}$$

7. Side of a square flower bed $= (x + 2)$ metres
 Perimeter of the square $= 4(x + 2)$ metres
 Perimeter of 5 squares $= 5 \times 4(x + 2)$
 $= (20x + 40)$ metres.
 Cost of fencing 100 metres $= ₹50$.

$$\text{Cost of fencing 1 metre} = ₹\frac{50}{100} = ₹0.50.$$

$$\text{Cost of fencing } (20x + 40) \text{ metres} \\ = ₹0.50 \times (20x + 40) = ₹(10x + 20).$$

8. (i) Length of wire left $= (7x - 3) - (3x - 4)$
 $= 7x - 3 - 3x + 4$
 $= 7x - 3x + 4 - 3$
 $= (4x + 1)$ metres.
 (ii) Perimeter of equilateral triangle $= (4x + 1)$ m
 $\therefore$ Length of each side of triangle

$$= \left(\frac{4x + 1}{3} \right) \text{ m}$$

9. Total money Rohan gets $= ₹(3xy^2) + ₹5(xy^2 + 2)$
 $= ₹(3xy^2 + 5xy^2 + 10)$
 $= ₹(8xy^2 + 10)$

$$\text{Money spent by him} = ₹(10 - 3xy^2) \\ \therefore \text{Money left} = ₹(8xy^2 + 10) - ₹(10 - 3xy^2) \\ = ₹(8xy^2 + 10 - 10 + 3xy^2) = ₹11xy^2.$$

10. $(93p^2 - 55p + 4) - (13p^3 - 5p^2 + 17p - 90)$
 $= 93p^2 - 55p + 4 - 13p^3 + 5p^2 - 17p + 90$
 $= -13p^3 + 98p^2 - 72p + 94.$

Hots Questions :

- $4y^2 + 3z + x^2$
- $a = 6, b = 1$
- (i) $4x - (-6y + 7z)$
 (ii) $ax - by - (-cz - 4q)$
- 0

3 EXERCISE

Single Option Correct :

- (c)
- (c)
- (c)
- (a)
- (a)
- (d)
- (d)
- (b)
- (d)
- (d)

More Than One Option Correct :

- (b, c)
- (b, d)
- (b, d)
- (b, d)
- (b, c)

Assertion & Reason :

- (a)
- (d)

Passage Based Questions :

- (b) $(2b^2 + 3ba) + [(-b^2) + (-ba)]$
 $= (2b^2 - b^2) + (3ba - ba) = b^2 + 2ba.$
- (a) $(5a - 7) - (-a^2 + 10a - 3)$
 $= 5a - 7 + a^2 - 10a + 3 = a^2 - 5a - 4.$
- (b) $(3p^2q^2 - 4pq + 5) + (-10p^2q^2)$
 $+ (15 + 9pq + 7p^2q^2)$
 $= (3p^2q^2 - 10p^2q^2 + 7p^2q^2)$
 $+ (-4pq + 9pq) + 5 + 15$
 $= 5pq + 20.$

Integer Type Questions :

- (1) $\frac{5m}{2} - 4 = \frac{5 \times 2}{2} - 4 = 5 - 4 = 1$
- (0)
- (1) $25x^2 + 16y^2 + 40xy$
 At $x = 1$ and $y = -1$
 $25(1)^2 + 16(-1)^2 + 40(1)(-1)$
 $= 25 + 16 - 40 = 1.$
- (3)
- (8) $x^2 + y^2 = (2)^2 + (-2)^2 = 4 + 4 = 8.$

Multiple Matching Questions :

- (A) $\rightarrow$ (r); (B) $\rightarrow$ (s, u); (C) $\rightarrow$ (p, q); (D) $\rightarrow$ (t).

Chapter

13

Exponents and Powers

INTRODUCTION

Exponents are the mathematical shorthand that tells us to multiply the same number by itself for a specified number of times. The lesson will not only explain how the exponent works but also discuss the six distinct properties or rules, that govern its use.

EXPONENTS

Exponents are used to express large numbers in shorter form to make them easy to read, understand and compare.

For example : $625 = 5 \times 5 \times 5 \times 5$
 $= 5^4$

Here, $(5)^4 \rightarrow$ exponent or power
 Base

It is read as “five to the power 4”.

ILLUSTRATION 1 :

Using exponents, write

(i) $n \times n \times n \times n \times n \times n$

SOLUTION:

(i) $n \times n \times n \times n \times n \times n = n^6$

ILLUSTRATION 2 :

Find the value of :

(i) 4^4

SOLUTION:

(i) $4^4 = 4 \times 4 \times 4 \times 4 = 256$

ILLUSTRATION 3 :

Express in power notation :

(i) -216

SOLUTION:

(i) $-216 = (-6) \times (-6) \times (-6) = (-6)^3$

(ii) $-4 \times -4 \times -4 \times -4$

(ii) $-4 \times -4 \times -4 \times -4 = (-4)^4$

(iii) $\left(\frac{2}{3}\right)^4$

(iii) $\left(\frac{2}{3}\right)^4 = \frac{2}{3} \times \frac{2}{3} \times \frac{2}{3} \times \frac{2}{3} = \frac{2 \times 2 \times 2 \times 2}{3 \times 3 \times 3 \times 3} = \frac{16}{81}$

(ii) $\frac{25}{49}$

(ii) $\frac{25}{49} = \frac{5 \times 5}{7 \times 7} = \frac{5^2}{7^2} = \left(\frac{5}{7}\right)^2$

ILLUSTRATION 4 :

Simplify :

$$(i) \quad (-3)^2 \times (10)^4 \qquad (ii) \quad \left(\frac{3}{4}\right)^2 \times \left(\frac{-1}{2}\right)^3 \times 2^3 \qquad (iii) \quad 3^3 + 2^4$$

SOLUTION :

$$(i) \quad (-3)^2 \times 10^4 = (-3) \times (-3) \times 10 \times 10 \times 10 \times 10 = 9 \times 10000 = 90000$$

$$(ii) \quad \left(\frac{3}{4}\right)^2 \times \left(\frac{-1}{2}\right)^3 \times 2^3 = \frac{3}{4} \times \frac{3}{4} \times \left(\frac{-1}{\cancel{2}_1}\right) \times \left(\frac{-1}{\cancel{2}_1}\right) \times \left(\frac{-1}{\cancel{2}_1}\right) \times \cancel{2}^1 \times \cancel{2}^1 \times \cancel{2}^1 = \frac{9}{16} \times (-1) = \frac{-9}{16}$$

$$(iii) \quad 3^3 + 2^4 = (3 \times 3 \times 3) + (2 \times 2 \times 2 \times 2) = 27 + 16 = 43$$

LAWS OF EXPONENTS**Law 1 : Multiplying powers with the same base**If a is any non-zero rational number and m, n are whole numbers, then $a^m \times a^n = a^{m+n}$

i.e. in case of multiplication, if base are same, simply add the powers.

For example : $3^5 \times 3^2 = 3 \times 3 \times 3 \times 3 \times 3 \times 3 \times 3 = 3^7$
 or $3^5 \times 3^2 = 3^{5+2} = 3^7$

Law 2 : Finding quotients of powers of the same numberIf a is any non-zero rational number and m, n are whole numbers, then $a^m \div a^n = a^{m-n}$

i.e. for division with same bases, subtract the exponents.

For example : $(-2)^7 \div (-2)^5 = \frac{(-2) \times (-2) \times (-2) \times (-2) \times (-2) \times (-2) \times (-2)}{(-2) \times (-2) \times (-2) \times (-2) \times (-2)}$
 $= (-2) \times (-2) = (-2)^2 = (-2)^{7-5} = (-2)^7 \div (-2)^5 = (-2)^{7-5} = (-2)^2$

Law 3 : Finding a power of a power of a rational numberFor any non-zero rational number, if m and n are whole numbers, then $(a^m)^n = a^{mn}$

i.e. to find a power of a power of a number, we multiply the exponents.

For example : $((-3)^5)^3 = (-3)^5 \times (-3)^5 \times (-3)^5$
 $= (-3)^{5+5+5} = (-3)^{15} = (-3)^{5 \times 3}$
 $= ((-3)^5)^3 = (-3)^{5 \times 3} = (-3)^{15}$

Law 4 : Multiplying bases with the same exponentsIf a and b are two non-zero rational numbers and n is a whole number, then $a^n \times b^n = (ab)^n$.

For example : $2^4 \times 3^4 = 2 \times 2 \times 2 \times 2 \times 3 \times 3 \times 3 \times 3$
 $= 6 \times 6 \times 6 \times 6 = 6^4 = (2 \times 3)^4$

or $2^4 \times 3^4 = (2 \times 3)^4 = 6^4$

Law 5 : Finding power of a quotientFor any two rational numbers a and b , and n is a whole number, then $a^n \div b^n = \left(\frac{a}{b}\right)^n$

For example : $5^4 \div 3^4 = \frac{5^4}{3^4} = \frac{5 \times 5 \times 5 \times 5}{3 \times 3 \times 3 \times 3} = \frac{5}{3} \times \frac{5}{3} \times \frac{5}{3} \times \frac{5}{3} = \left(\frac{5}{3}\right)^4$

Law 6 : Zero exponent

If a is any rational number (except 0), then, $a^0 = 1$.

For example : $(-10)^0 = 1$

ILLUSTRATION 5 :

Simplify and write the answer in exponential form :

(i) $\left(\frac{24}{11}\right)^3 \times \left(\frac{11}{8}\right)^3$

(ii) $\left(\frac{3}{4}\right)^{17} \div \left(\frac{3}{4}\right)^4$

(iii) $\left(\frac{7}{3}\right)^5 \times \left(\frac{7}{3}\right)^6 \times \left(\frac{7}{3}\right)^{20}$

(iv) $(2^{20} \div 2^{15}) \times 2^3$

(v) $\left(\frac{3^7}{3^2}\right) \times 3^5$

(vi) $\frac{4^5 \times a^8 b^3}{4^5 \times a^5 b^2}$

SOLUTION :

(i) $\left(\frac{24}{11}\right)^3 \times \left(\frac{11}{8}\right)^3 = \left(\frac{\cancel{24}^3}{\cancel{11}_3} \times \frac{\cancel{11}^3}{\cancel{8}_3}\right) = 3^3$

(ii) $\left(\frac{3}{4}\right)^{17} \div \left(\frac{3}{4}\right)^4 = \left(\frac{3}{4}\right)^{17-4} = \left(\frac{3}{4}\right)^{13}$

(iii) $\left(\frac{7}{3}\right)^5 \times \left(\frac{7}{3}\right)^6 \times \left(\frac{7}{3}\right)^{20} = \left(\frac{7}{3}\right)^{5+6+20} = \left(\frac{7}{3}\right)^{31}$

(iv) $(2^{20} \div 2^{15}) \times 2^3 = (2)^{20-15} \times 2^3 = 2^5 \times 2^3 = 2^{5+3} = 2^8$

(v) $\left(\frac{3^7}{3^2}\right) \times 3^5 = (3)^{7-2} \times 3^5 = 3^5 \times 3^5 = 3^{5+5} = 3^{10}$

(vi) $\frac{4^5 \times a^8 b^3}{4^5 \times a^5 b^2} = \frac{4^5}{4^5} \times \frac{a^8}{a^5} \times \frac{b^3}{b^2} = (4)^{5-5} \times (a)^{8-5} \times (b)^{3-2} = 4^0 \times a^3 \times b = 1 \times a^3 b = a^3 b$

STANDARD FORM OR SCIENTIFIC NOTATION OF LARGE NUMBERS

A given number is said to be in standard form if it is expressed as $m \times 10^n$, where m lies between 1 and 10 and n is a positive integer.

This standard form of a number is also known as Scientific Notation.

ILLUSTRATION 6 :

Express the following numbers in standard form :

(i) 270659

(ii) 40000000

SOLUTION :

(i) $270659 = 2.70659 \times 10^5$

(ii) $40000000 = 4 \times 10^7$

ILLUSTRATION 7 :

Write each of the following in usual form.

(i) 6.28×10^6

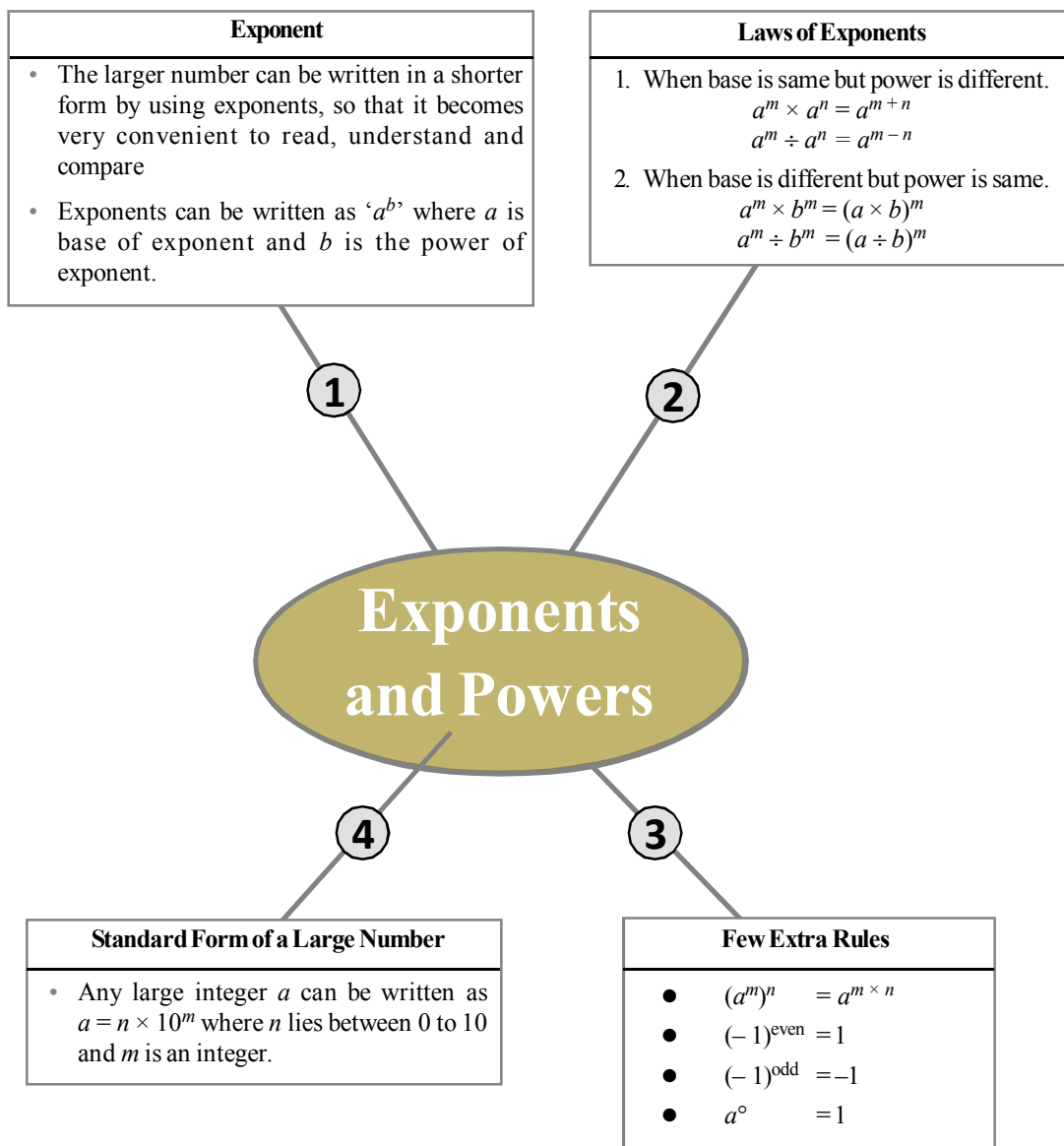
(ii) 9.2×10^3

SOLUTION :

(i) $6.28 \times 10^6 = 6280000$

(ii) $9.2 \times 10^3 = 9200$

CONCEPT MAP



MISCELLANEOUS

SOLVED EXAMPLES

1. Find the product of the cube of $\frac{-2}{3}$ and the square of $\frac{4}{-5}$.

Sol. $\left(\text{Cube of } \frac{-2}{3}\right) \times \left(\text{Square of } \frac{4}{-5}\right)$
 $= \left(\frac{-2}{3}\right)^3 \times \left(\frac{4}{-5}\right)^2$
 $= \frac{(-2) \times (-2) \times (-2)}{3 \times 3 \times 3} \times \frac{4 \times 4}{(-5) \times (-5)}$
 $= \frac{-8}{27} \times \frac{16}{25} = \frac{-128}{675}$

2. If $a = 2$ and $b = 3$, then find the values of each of the following:

(ii) $a^b + b^a$ (iv) $\left(\frac{1}{a} + \frac{1}{b}\right)^a$

Sol. (ii) We have $a^b + b^a = 2^3 + 3^2 = 2 \times 2 \times 2 + 3 \times 3$
 $= 8 + 9 = 17$

(iv) $\left(\frac{1}{a} + \frac{1}{b}\right)^a = \left(\frac{1}{2} + \frac{1}{3}\right)^2 = \left(\frac{3+2}{6}\right)^2$
 $= \left(\frac{5}{6}\right)^2 = \frac{5}{6} \times \frac{5}{6} = \frac{5 \times 5}{6 \times 6} = \frac{25}{36}$

3. Simplify and express each of the following as power of a rational number :

(i) $\left(\frac{2}{3}\right)^3 \times \left(-\frac{6}{7}\right)^2 \times \left(-\frac{7}{4}\right) \times \frac{3}{2}$

(ii) $-\left(\frac{2}{5}\right)^2 \times \left(\frac{5}{7}\right)^2 \times \frac{49}{5} + \left(-\frac{4}{5}\right)^3 \times \frac{5}{4} \times \frac{3}{4}$

Sol. (i) $\left(\frac{2}{3}\right)^3 \times \left(-\frac{6}{7}\right)^2 \times \left(-\frac{7}{4}\right) \times \frac{3}{2}$
 $= \frac{2^3}{3^3} \times \frac{6^2}{7^2} \times -\frac{7}{4} \times \frac{3}{2}$

$$= \frac{8}{27} \times \frac{36}{49} \times \frac{-7}{4} \times \frac{3}{2}$$

$$= \frac{8 \times 36 \times -7 \times 3}{27 \times 49 \times 4 \times 2} = \frac{1 \times 36 \times -1 \times 1}{9 \times 7 \times 1 \times 1}$$

$$= \frac{-36}{9 \times 7} = \frac{-4}{7} = \left(\frac{-4}{7}\right)^1$$

(ii) $-\left(\frac{2}{5}\right)^2 \times \left(\frac{5}{7}\right)^2 \times \frac{49}{5} + \left(-\frac{4}{5}\right)^3 \times \frac{5}{4} \times \frac{3}{4}$
 $= -\frac{4}{25} \times \frac{25}{49} \times \frac{49}{5} + \frac{-64}{125} \times \frac{5}{4} \times \frac{3}{4}$
 $= -\frac{4}{1} \times \frac{1}{1} \times \frac{1}{5} + \frac{-4}{25} \times \frac{1}{1} \times \frac{3}{1}$
 $= -\frac{4}{5} - \frac{12}{25} = \frac{-20-12}{25}$
 $= -\frac{32}{25} = \left(\frac{-32}{25}\right)^1$

4. Express 3125 in exponential form.

Sol.

5	3125
5	625
5	125
5	25
	5

$$3125 = 5 \times 5 \times 5 \times 5 \times 5$$

$$\therefore 3125 = 5^5$$

5. Write exponential form for $8 \times 8 \times 8 \times 8$ taking base as 2.

Sol. We have,
 $8 \times 8 \times 8 \times 8 = 8^4$
 $= (2^3)^4$ [$\because 8 = 2 \times 2 \times 2 = 2^3$]
 $= 2^{3 \times 4} = 2^{12}$ [$\because (a^m)^n = a^{mn}$]

6. Simplify:

(i) $2^{55} \times 2^{60} - 2^{97} \times 2^{18}$

(ii) $2^3 \times a^3 \times 5a^4$

(iii) $\frac{3^n + 3^{n+1}}{3^{n+1} - 3^n}$, where n is a natural number.

Sol. (i) $2^{55} \times 2^{60} - 2^{97} \times 2^{18}$
 $= 2^{55+60} - 2^{97+18} = 2^{115} - 2^{115} = 0$

(ii) $2^3 \times a^3 \times 5a^4 = 2^3 \times 5 \times a^3 \times a^4$
 $= 8 \times 5 \times a^{3+4} = 40 \times a^7 = 40a^7$

(iii) $\frac{3^n + 3^{n+1}}{3^{n+1} - 3^n} = \frac{3^n + 3^n \times 3^1}{3^n \times 3^1 - 3^n}$
 $= \frac{3^n \times 1 + 3^n \times 3}{3^n \times 3 - 3^n \times 1} = \frac{3^n(1+3)}{3^n(3-1)} = \frac{3^n \times 4}{3^n \times 2}$
 $= \frac{3^n \times 2^2}{3^n \times 2} = \frac{3^n}{3^n} \times \frac{2^2}{2} = 3^{n-n} \times 2^{2-1}$
 $= 3^0 \times 2^1 = 1 \times 2 = 2.$

7. Find the value of x in each case:

(i) $(-3)^{x-1} = -243$

(ii) $\left(\frac{4}{5}\right)^8 \times \left(\frac{4}{5}\right)^7 = \left(\frac{4}{5}\right)^{2x+1}$

(iii) $(7)^6 \div (7)^{4-m} = 7^8$

Sol. (i) $(-3)^{x-1} = -243$

$$\Rightarrow (-3)^{x-1} = (-3)^5$$

In an equation, when bases on both sides are equal, their powers must also be equal.

$$\therefore x-1=5 \Rightarrow x=6$$

(ii) $\left(\frac{4}{5}\right)^8 \times \left(\frac{4}{5}\right)^7 = \left(\frac{4}{5}\right)^{2x+1}$

$$\Rightarrow \left(\frac{4}{5}\right)^{8+7} = \left(\frac{4}{5}\right)^{2x+1}$$

$$\Rightarrow \left(\frac{4}{5}\right)^{15} = \left(\frac{4}{5}\right)^{2x+1}$$

$$\Rightarrow 2x+1=15 \Rightarrow 2x=14$$

$$\Rightarrow x=7$$

(iii) $(7)^6 \div (7)^{4-m} = 7^8$

$$\Rightarrow (7)^{6-(4-m)} = 7^8$$

$$\Rightarrow (7)^{6-4+m} = 7^8$$

$$\Rightarrow (7)^{2+m} = 7^8$$

$$\Rightarrow 2+m=8$$

$$\Rightarrow m=8-2$$

$$\Rightarrow m=6$$

1

EXERCISE

Fill in the Blanks :

DIRECTIONS : Complete the following statements with an appropriate word / term to be filled in the blank space(s).

1. Value of $\frac{10^{22} + 10^{20}}{10^{20}}$ is _____.

2. The reciprocal of $\left(-\frac{2}{3}\right)^3$ is _____.

3. $\left(\frac{3}{4}\right)^0$ is equal to _____.

4. $\frac{2}{3} \times \left(\frac{2}{3}\right)^4 =$ _____.

5. $\left(\frac{1}{5}\right)^7 \div \left(\frac{-1}{5}\right)^2$ is equal to _____.

6. If $9 \times 3^n = 3^6$, then $n =$ _____.

7. $\left(\frac{a}{b}\right)^{-16} =$ _____.

8. $\left\{6^{-1} + \left(\frac{3}{2}\right)^{-1}\right\}^{-1} =$ _____.

9. $\left[\left\{\left(\frac{-1}{3}\right)^2\right\}^{-2}\right]^{-1} =$ _____.

10. $(6^{-1} - 8^{-1})^{-1} + (2^{-1} - 3^{-1})^{-1} =$ _____.

True / False :

DIRECTIONS : Read the following statements and write your answer as true or false.

1. $\left[\left(\frac{1}{7}\right)^7\right]^7$ is the reciprocal of 7^{49} .

2. $\left(\frac{-3}{5}\right)^{100} = \frac{-3^{100}}{5^{100}}$.

3. $(30 + 30)^3 = 30^3 + 30^3$.

4. Is the value of $\left(\frac{-5}{9}\right)^{9-3 \times 2-3}$ is 1.

5. Is $\left(\frac{1}{7} \times \frac{1}{7^2}\right) \div \frac{1}{7^3} = 1$.

6. The standard form of 0.000255 is 2.55×10^{-5} .

7. The usual form of 7.2×10^{-4} is 0.00072.

8. The base and exponent of $\left(\frac{1}{9}\right)^{-4}$ is -4 and

$\frac{1}{9}$.

9. If $p = 3$ and $q = 4$, then $p^0 + q^0$ will be 7.

10. This expression $\left(\frac{1}{2}\right)^6 \times \left(\frac{1}{2}\right)^{-4} \times \left(\frac{1}{2}\right)^5$ can be

solved like this $\left(\frac{1}{2}\right)^{6+(-4)+(5)}$.

Match the Columns :

DIRECTIONS : Each question contains statements given in two columns which have to be matched. Statements (A, B, C, D) in column-I have to be matched with statements (p, q, r, s) in column-II.

1.	Column-I (Expression)	Column-II (Exponent form)
	A. $a \times a \times b \times a \times a \times b \times b$	(p) $c^2.a^4$
	B. $a^5 \times a \times a \times b^3 \times b^2$	(q) $a^6b^6c^4$
	C. $c \times c \times a \times a^2 \times a$	(r) $a^4.b^3$
	D. $a^2 \times b^3 \times c^4 \times a^4 \times b^3$	(s) $a^7.b^5$

2.	Column-I (Number)	Column-II (Scientific Notation)
	A. 360	(p) 3.6×10^4
	B. 36000	(q) 3.6×10^6
	C. 3600000	(r) 3.6×10
	D. 36	(s) 3.6×10^2

3. Column-I Column-II

(A) If $a^m \cdot a^n = a^{mn}$

(p) $\frac{7}{48}$

then the value of
 $m(n-2) + n(m-2)$ is

(B) If $2^x = 4^y = 8^z$ and

(q) 0

$$\left(\frac{1}{2x} + \frac{1}{4y} + \frac{1}{6z}\right) = \frac{24}{7},$$

then the value of z is

(C) Value of

(r) $\frac{1}{9}$

$$\frac{3^{12+n} \times 9^{2n-7}}{3^{5n}} \text{ is}$$

Very Short Answer Questions :

DIRECTIONS : Give answer in one word or one sentence.

1. Evaluate:

(i) $\left(\frac{-4}{3}\right)^3$

(ii) $\left(\frac{-1}{2}\right)^4$

2. Express the following in exponential form.

(i) $\frac{2}{5} \times \frac{2}{5} \times \frac{2}{5}$

(ii) $\frac{-7}{3} \times \frac{-7}{3} \times \frac{-7}{3} \times \frac{-7}{3}$

3. Express in power notation:

(i) -343

(ii) $\frac{49}{25}$

(iii) $-\frac{8}{27}$

(iv) $\frac{81}{625}$

4. Find the value of :

(i) $(-5)^3 \times (-1)^7 \times 2^2$ (ii) $\left(\frac{3}{4}\right)^2 \times \left(-\frac{1}{2}\right)^3 \times 2^3$

(iii) $2^2 + 3^3 + 4^3$

5. Is 2^3 equal to 3^2 ?

6. Express the following with a single exponent :

(i) $(2^3)^5 \times (2^7)^2$ (ii) $\left(\left(\frac{-3}{5}\right)^2 \times \left(-\frac{3}{5}\right)^4\right)^3$

7. Evaluate:

(i) x^0

(ii) $7^0 + 8^0 + 9^0$

(iii) $(29^0 - 23^0) \times 16^0$ (iv) $(-2)^{3 \times 5 - 4 - 11}$

8. Express each of the following products in exponential form :

(i) $9^2 \times 9^{18} \times 9^7$

(ii) $a \times a^2 \times a^3$

9. Find the value of $\left(\frac{3}{5}\right)^7 \times \left(\frac{3}{5}\right)^4$.

10. Find the value of $\left(\frac{5}{6}\right)^{12} \div \left(\frac{5}{6}\right)^{12}$.

11. Find the value of $\left(\frac{2}{3}\right)^2 \div \left(\frac{6}{19}\right)^0$.

12. Find the value of $2^3 \times 3^3$.

13. $8 \div \left(\frac{1}{2}\right)^0 = \underline{\hspace{2cm}}$.

14. Value of $3p^2q^2$ when $p = 3$ and $q = 4$ is $\underline{\hspace{2cm}}$.

15. Product of $2^{2a+7} \times 2^{7a-6}$ is equal to $\underline{\hspace{2cm}}$.

16. The value of $\left[(5^2)^3 \times 5^4\right] \div 5^7$ is $\underline{\hspace{2cm}}$.

17. $\frac{a}{b} \times \frac{a}{b} \times \frac{a}{b} \times \dots \times \frac{a}{b}$ n times equals to $\underline{\hspace{2cm}}$.

18. The standard form of 5000000 is $\underline{\hspace{2cm}}$.

Short Answer Questions :

DIRECTIONS : Give answer in 2-3 sentences.

1. Find the value of

(i) $(5x)^3$ when $x = -\frac{2}{5}$

(ii) $(-2ab)^4$ when $a = \frac{3}{5}$, $b = \frac{-1}{2}$

2. Express each of the following with a single exponent:

(i) $(5^5)^3 \times (5^4)^8$ (ii) $\left(\left(\frac{-1}{3}\right)^2\right)^3 \times \left(\left(\frac{-1}{3}\right)^4\right)^5$

(iii) $[(-7)^5 \times (-7)^4]^2$ (iv) $(-4x^3)^4 \times (-x^2)^4$

3. Rewrite 4963 in expanded form using exponents.

4. Evaluate:

(i) $\left(\frac{2}{7}\right)^4 \times \left(\frac{2}{7}\right)^{-5}$

$$(ii) \left(\frac{4}{3}\right)^2 \times \left(\frac{2}{3}\right)^3 \times \frac{1}{4}$$

$$(iii) \left[\left(\frac{1}{2}\right)^{-1} \times (-4)^{-1}\right]^2$$

$$(iv) \left(\frac{3}{2}\right)^{-1} \div \left(\frac{-2}{5}\right)^{-1}$$

$$(v) \left(\frac{-4}{5}\right)^2 \times \frac{15}{2} \times \frac{1}{6} \times \left(\frac{125}{64}\right)$$

$$(vi) \left[\left(\frac{3}{7}\right)^2 \times \left(\frac{3}{7}\right)^3\right] \div \left(\frac{3}{7}\right)^8$$

5. Find the value of :

$$\frac{2}{5} \times \left(\frac{2}{5}\right)^3 \div \left(\frac{3}{5}\right)^4$$

6. Simplify and write the answer with positive exponents.

$$(i) (3+4)^{-1} \times (3^{-1}+4^{-1})$$

$$(ii) \left[\left(\frac{5}{4}\right)^{-1} \times \left(\frac{2}{5}\right)^{-1}\right]^{-1}$$

$$(iii) (5^{-1}-7^{-1})^{-1} + (3^{-1}-5^{-1})^{-1}$$

7. If $\frac{x}{y} = \left(\frac{-1}{3}\right)^{-2} \div \left(\frac{2}{3}\right)^{-3}$, find the value of $\left(\frac{x}{y}\right)^{-3}$.

8. If $5^{2x+1} \div 25 = 125$, find the value of x .

9. Simplify and express the result as a power of 2.

$$(i) (3^4 \times 3^5) \div 3^9$$

$$(ii) \left[\left(\frac{1}{5}\right)^6 \div \left(\frac{1}{5}\right)^5\right] \div \frac{1}{5}$$

10. By what number should we multiply 7^{-5} , so that the product may be equal to 7?

11. (i) By what number should $\left(\frac{-4}{5}\right)^2$ be multiplied, so that the product is $\left(\frac{2}{3}\right)^3$?

- (ii) By what number should 13^{-2} be divided to get 13^5 ?

12. If $\frac{x}{y} = \left(\frac{2}{3}\right)^{-5} \times \left(\frac{2}{3}\right)^{-2}$, find $\left(\frac{x}{y}\right)^{-2}$ and $\left(\frac{x}{y}\right)^2$.

13. If $\frac{x}{y} = \left(\frac{5}{2}\right)^{-1} \times \left(\frac{8}{9}\right)^0$, find the value of $\left(\frac{x}{y}\right)^{-2}$.

14. Find x , so that:

$$(i) \left(\frac{2}{5}\right)^3 \times \left(\frac{2}{5}\right)^4 = \left(\frac{2}{5}\right)^{2x-1}$$

$$(ii) \left(\frac{3}{2}\right)^3 \div \left(\frac{3}{2}\right)^4 = \left(\frac{3}{2}\right)^{2x+1}$$

$$(iii) 3^{-9} \times 3^x = 3$$

$$(iv) 4^x \times 4^{-8} = 4^{-3}$$

$$(v) \left(\frac{-3}{5}\right)^{-4} \times \left(\frac{-3}{5}\right)^6 = \left(\frac{5}{3}\right)^{2x}$$

$$(vi) 5^x \times 5^{-2} \times 5^4 = 5^{-3}$$

15. Verify that $\left(\frac{1}{3}\right)^{-3} + \left(\frac{1}{2}\right)^{-5} - \left(\frac{1}{2}\right)^{-5} = 27$.

16. Find x , so that $\left(\frac{2}{3}\right)^{-2} \times \left(\frac{2}{3}\right)^{-9} = \left(\frac{2}{3}\right)^{2x-1}$.

2

EXERCISE

Text-Book Exercise :

- Which one is greater 8^2 or 2^8 ?
- Compare the following numbers:
 - 2.7×10^{12} ; 1.5×10^8
 - 4×10^{14} ; 3×10^{17}
- Simplify and write the answer in the exponential form.
 - $\left(\frac{3^7}{3^2}\right) \times 3^5$
 - $(6^2 \times 6^4) \div 6^3$
 - $[(2^2)^3 \times 3^6] \times 5^6$
 - $8^2 \div 2^3$
- Simplify:
 - $\frac{12^4 \times 9^3 \times 4}{6^3 \times 8^2 \times 2^7}$
 - $2^3 \times a^3 \times 5a^4$
 - $\frac{2 \times 3^4 \times 2^5}{9 \times 4^2}$
- Express the following numbers in the standard form:
 - 5985.3
 - 70,040,000,000

Exemplar Questions :

- Arrange in ascending order: $2^5, 3^3, 2^3 \times 2^1, (3^3)^2, 3^5, 4^0, 2^3 \times 3^1$
- Arrange in descending order: $2^{2+3}, (2^2)^3, 2 \times 2^2, \frac{3^5}{3^2}, 3^2 \times 3^0, 2^3 \times 5^2$
- By what number should $(-4)^5$ be divided so that the quotient may be equal to $(-4)^3$?
- If $\frac{p}{q} = \left(\frac{3}{2}\right)^2 \div \left(\frac{9}{4}\right)^0$, find the value of $\left(\frac{p}{q}\right)^3$.
- Find the reciprocal of the rational number $\left(\frac{1}{2}\right)^2 \div \left(\frac{2}{3}\right)^3$
- Find the value of:
 - $(-7)^{2 \times 7 - 6 - 8}$
 - $(2^0 + 3^0 + 4^0)(4^0 - 3^0 - 2^0)$
 - $2 \times 3 \times 4 \div 2^0 \times 3^0 \times 4^0$
 - $(8^0 - 2^0) \times (8^0 + 2^0)$
- Find the value of n , where n is an integer and

$$2^{n-5} \times 6^{2n-4} = \frac{1}{12^4 \times 2}$$

- Express the following in usual form:
 - 8.01×10^7
 - 1.75×10^{-3}
- How many times of 30 must be added together to get a sum equal to 30^7 ?
- Simplify and express each of the following in exponential form:

$$(i) \left[\left(\frac{7}{11}\right)^5 \div \left(\frac{7}{11}\right)^2 \right] \times \left(\frac{7}{11}\right)^2$$

$$(ii) \left(\frac{a^6}{a^4}\right) \times a^5 \times a^0$$

$$(iii) \left[\left(\frac{3}{5}\right)^3 \times \left(\frac{3}{5}\right)^8 \right] \div \left[\left(\frac{3}{5}\right)^2 \times \left(\frac{3}{5}\right)^4 \right]$$

$$(iv) (5^{15} \times 5^{10}) \times 5^5$$

- Evaluate

$$(i) \frac{7^8 \times a^{10} b^7 c^{12}}{7^6 \times a^8 b^4 c^{12}} \quad (ii) \frac{5^4 \times 7^4 \times 2^7}{84 \times 49 \times 5^3}$$

$$(iii) \frac{3^4 \times 12^3 \times 36}{2^5 \times 6^3} \quad (iv) \left(\frac{6 \times 10}{2^2 \times 5^3}\right)^2 \times \frac{25}{27}$$

$$(v) \frac{6^4 \times 9^2 \times 25^3}{3^2 \times 4^2 \times 15^6}$$

- If $2^{n+2} - 2^{n+1} + 2^n = c \times 2^n$, find the value of c .

HOTS Questions :

- Simplify, using law of exponents : $[729^3 \div 729] \div 3^8$
- Simplify and express as a power of a rational number $\left(\frac{4}{3}\right)^2 \times \left(\frac{2}{3}\right)^3 \times \frac{1}{4}$.
- If $x^4 = 256$, then find the value of 6^{2x-8} .
- Simplify $(5 \times 10^{12}) \div (8 \times 10^5)$ and write the answer in scientific notation.
- If $x = \frac{1}{6} - \frac{7}{3}$ of $\frac{2}{-7} \div \frac{4}{9}$, find the value of $x^3 - 1$.
- If $27^{2x-1} = (243)^3$, then find the value of 5^{2x-4} .

3

EXERCISE

Single Option Correct :

DIRECTIONS : This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONLY ONE is correct.

- Simplify: $(10)^2 + (10)^{-4} \div (10)^{-2}$.
 (a) $\frac{10001}{100}$ (b) $\frac{10101}{100}$
 (c) $\frac{10100}{100}$ (d) $\frac{1001}{100}$
- Using law of exponents, determine x , such that : $1000 = 2x \times 5x$
 (a) 10 (b) 3
 (c) 2 (d) 1
- The value of $\left(\frac{5}{7}\right)^\circ + \left(\frac{5}{7}\right)^\circ \times \left(\frac{5}{7}\right)^\circ - \left(\frac{5}{7}\right)^\circ$ is
 (a) 10 (b) 0
 (c) 3 (d) 1
- $(4^3)^4 \div (4^2)^3 \times (4^5)^0 =$
 (a) 6^4 (b) 4^6
 (c) 2^6 (d) 6^2
- Which of the following when simplified is not equal to 27 ?
 (a) $[9^{10} \div 9^8] \div 3$
 (b) $3^{12} \div 3^8$
 (c) $(3^8 \div 3^6) \times (4^0 + 5^0 + 6^0)$
 (d) $9^0 \times 3^3$
- Which of the following is equivalent to $\frac{(-5x)^4}{-(5x)^4}$?
 (a) -1 (b) 1
 (c) $-625x^4$ (d) $\frac{1}{625x^4}$
- The missing exponent in the following expression is : $10^5 \times 10^- = 10000000$
 (a) 4 (b) 7
 (c) 0 (d) 2

- 16^2 as a power of 2 is
 (a) 2^6 (b) 2^8
 (c) 2^4 (d) 2^{16}
- A boy was asked to write $2^5 \times 9^2$ but he wrote 2592. The numerical difference between the two is
 (a) 0 (b) 1
 (c) 2 (d) 3
- Which of the following statement is true?
 (a) $4^3 > 3^4$ (b) $2^2 + 3^2 = (2+3)^2$
 (c) $3^5 < 5^3$ (d) $2^8 > 8^2$

More Than One Option Correct :

DIRECTIONS : This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONE or MORE may be correct.

- $[(-3)^2]^3$ is equal to
 (a) 729 (b) $(-3)^6$
 (c) $(-3)^5$ (d) -243
- Value of $\frac{10^{22} + 10^{20}}{10^{20}}$ is
 (a) $10^2 + 1$ (b) 10^{42}
 (c) 101 (d) 10^{22}
- Which of the following is/are equal to 1?
 (a) $\frac{2^3 \times 3^2}{4 \times 18}$ (b) $\left[(-2)^3 \times (-2)^4\right] \div (-2)^7$
 (c) $\frac{2^3 \times 5^3}{5 \times 25}$ (d) $\frac{2^4}{(7^0 \times 3^0)}$
- Which of the following is/are equal to $\left(\frac{-5}{4}\right)^4$?
 (a) $\frac{(-5)^4}{4^4}$ (b) $\frac{5^4}{(-4)^4}$
 (c) $-\frac{5^4}{4^4}$
 (d) $\left(-\frac{5}{4}\right) \times \left(-\frac{5}{4}\right) \times \left(-\frac{5}{4}\right) \times \left(-\frac{5}{4}\right)$

5. Which of the following is/are true?

- (a) $3^2 > 2^3$ (b) $4^3 = 2^6$
(c) $3^3 = 9$ (d) $2^5 > 5^2$

Assertion & Reason :

DIRECTIONS : Each of these questions contains an Assertion followed by Reason. Read them carefully and answer the question on the basis of following options. You have to select the one that best describes the two statements.

- (a) If both **Assertion** and **Reason** are correct and **Reason** is the correct explanation of **Assertion**.
(b) If both **Assertion** and **Reason** are correct but **Reason** is not the correct explanation of **Assertion**.
(c) If **Assertion** is correct but **Reason** is incorrect.
(d) If **Assertion** is incorrect but **Reason** is correct.

1. **Assertion (A):** Value of $\left(\frac{4}{9}\right)^{\frac{3}{2}} \times \left(\frac{4}{9}\right)^{\frac{1}{2}}$ is $\left(\frac{4}{9}\right)^2$.

Reason (R): For any two rational numbers a and b and for any integers m and n, we have $a^m \times a^n = a^{m+n}$

2. **Assertion (A):** Value of $(27)^{\frac{6}{5}} \div (27)^{\frac{1}{5}}$ is 1×3^3

Reason (R): For any two rational numbers a and b and for any integers m and n, we have $a^m \div a^n = a^{m-n}$

Passage Based Questions :

DIRECTIONS : Study the given passage(s) and answer the following questions.

PASSAGE - 1

Let K represents any terminating decimal or a number from 1 to 9.999... (or $1 \leq K < 10$). Then we can express very large and very small numbers in special form such as $K \times 10^n$, where n is any integer.

This form of writing numbers is known as scientific notation. This form of numbers is also called the standard form.

1. Scientific notation 0.23 is

- (a) 2.3×10^{-1} (b) 2.3

- (c) 2.3×10 (d) $\frac{2.3}{10}$

2. Scientific notation of .0000072 is

- (a) 7.2×10^{-7} (b) 7.2×10^7
(c) 7.2×10^{-6} (d) 72×10^{-6}

3. Usual form of 1.05×10^3 is

- (a) 105 (b) 1050
(c) 10500 (d) 105000

PASSAGE-2

For any non-zero integers a and b and whole numbers m and n,

- (a) $a^m \times a^n = a^{m+n}$
(b) $a^m \div a^n = a^{m-n}$, $m > n$
 $= \frac{1}{a^{n-m}}$, $n > m$

- (c) $(a^m)^n = a^{mn}$
(d) $a^m \times b^m = (ab)^m$

4. Simplify: $\frac{25 \times 216 \times 729}{6^6 \times 5^5 \times 9^4}$

- (a) $\frac{1}{(30)^7 (9)^8}$ (b) $\frac{1}{(5)^6 (6)^9 (9)^7}$

- (c) $\frac{1}{(5)^2 (6)^3 (9)^3}$ (d) $\frac{1}{(30)^3 \times 9}$

5. Simplify and express the following in exponential form.

$$\left\{ \left(\frac{6}{5} \right)^2 \times x^8 \times y^7 \right\} \div \left\{ 2^{10} \times \left(\frac{x}{3} \right)^4 \times \left(\frac{y}{3} \right)^3 \right\}$$

- (a) $\frac{3^9}{5^2 \times 2^8} (xy)^4$ (b) $\frac{5^2 \times 2^5}{3^9} (xy)^4$

- (c) $\left(\frac{3}{10} \right)^4 (xy)^4$ (d) $\frac{3^9}{5^2 \times 2^8} x^{12} y^8$

6. Find the value of

$$(2^3 + 3^3 + 4^3) \times \frac{1}{(24)^6}$$

- (a) $\frac{1}{(24)^3}$ (b) $\left(\frac{11}{3^6} \right)$

- (c) $\frac{11}{3^3 \times 8^6}$ (d) $\frac{33}{(24)^6}$

Integer Type Questions :

DIRECTIONS : Answer the following questions. The answer to each of the question is a single digit integer, ranging from 0 to 9.

- Which power of 8 is equal to 2^6 ?
- $(1^0 + 2^0 + 3^0)$ is equal to
- In standard form, the number 42369.2 is written as 4.23692×10^n where n is equal to
- What is the value of $\frac{1}{55^0} \times \frac{1}{555^0}$?

- What is the value of $\left[\frac{1}{(7/5)^0} \right]^4$?

Multiple Matching Question :

DIRECTIONS : Following question has five statements (A, B, C, D and E) given in Column-I and six statements (p, q, r, s, t, u) in Column-II. Any given statement in Column-I can have correct matching with one or more statement(s) given in Column-II.

1.	Column - I	Column - II
A.	$(3^3)^2$	(p) $(0^{18})^{15}$
B.	$(0^6)^{12}$	(q) $(16)^3$
C.	$(644^3)^0$	(r) $(9)^3$
D.	$(2^2)^6$	(s) $(3^0)^{644}$
		(t) 1
		(u) 729

SOLUTIONS

Brief Explanations of Selected Questions

1 EXERCISE

Fill in the Blanks :

1. $\frac{10^{20}(10^2+1)}{10^{20}} = (10^2+1) = 101.$

2. $\left(-\frac{2}{3}\right)^3 = \frac{-8}{27}$

$\Rightarrow \text{Reciprocal} = \left(\frac{-27}{8}\right).$

3. 1

4. $\left(\frac{2}{3}\right)^5$

5. $\left(\frac{1}{5}\right)^5$

6. $9 \times 3^n = 3^6 \Rightarrow 3^2 \times 3^n = 3^6$
 $\Rightarrow 3^{2+n} = 3^6$
 $\Rightarrow 2+n=6 \Rightarrow n=4$

7. $\left(\frac{b}{a}\right)^{16}$

8. $\left\{6^{-1} + \left(\frac{3}{2}\right)^{-1}\right\}^{-1} = \left(\frac{1}{6} + \frac{2}{3}\right)^{-1}$

$= \left(\frac{1+4}{6}\right)^{-1} = \left(\frac{5}{6}\right)^{-1} = \frac{6}{5}$

9. $\left[\left\{\left(\frac{-1}{3}\right)^2\right\}^{-2}\right]^{-1} = \left[\left(\frac{-1}{3}\right)^{2 \times (-2)}\right]^{-1}$
 $= \left[\left(\frac{-1}{3}\right)^{-4}\right]^{-1} = \left(\frac{-1}{3}\right)^{(-4) \times (-1)} = \left(\frac{-1}{3}\right)^4 = \frac{1}{81}$

10. $(6^{-1} - 8^{-1})^{-1} + (2^{-1} - 3^{-1})^{-1}$

$$= \left(\frac{1}{6} - \frac{1}{8}\right)^{-1} + \left(\frac{1}{2} - \frac{1}{3}\right)^{-1}$$

$$= \left(\frac{4-3}{24}\right)^{-1} + \left(\frac{3-2}{6}\right)^{-1} = \left(\frac{1}{24}\right)^{-1} + \left(\frac{1}{6}\right)^{-1}$$

$$= \left(\frac{24}{1}\right) + \left(\frac{6}{1}\right) = (24+6) = 30$$

True / False :

- | | | |
|----------|----------|----------|
| 1. True | 2. True | 3. False |
| 4. True | 5. True | 6. False |
| 7. True | 8. False | 9. False |
| 10. True | | |

Match the Columns :

1. (A) $\rightarrow (r)$; (B) $\rightarrow (s)$; (C) $\rightarrow (p)$; (D) $\rightarrow (t)$

A : $a \times a \times a \times a \times b \times b \times b \times b = a^4.b^3$

B : $a^5 \times a \times a \times b^3 \times b^2 = a^7.b^5$

C : $c \times c \times a \times a^2 \times a = c^2.a^4$

D : $a^2 \times a^4 \times b^3 \times b^3 \times c^4 = a^6.b^6.c^4$

2. (A) $\rightarrow (s)$; (B) $\rightarrow (p)$; (C) $\rightarrow (q)$; (D) $\rightarrow (r)$

3. (A) $\rightarrow (q)$; (B) $\rightarrow (p)$; (C) $\rightarrow (r)$

(A) $a^m.a^n = a^{mn}$

$\Rightarrow a^{m+n} = a^{mn}$

$\Rightarrow m+n=mn$

$m(n-2) + n(m-2) = 2mn - 2(m+n)$

$= 2mn - 2mn = 0$

(B) $2^x = 4^y = 8^z \Rightarrow 2^x = 2^{2y} = 2^{3z}$

$\Rightarrow x=2y=3z$

Consider

$$\frac{1}{2x} + \frac{1}{4y} + \frac{1}{6z} = \frac{1}{2x} + \frac{1}{2x} + \frac{1}{2x}$$

$$= \frac{3}{2x}$$

$$\frac{3}{2x} = \frac{24}{7} \Rightarrow x = \frac{7}{16}$$

$\therefore z = \frac{7}{48}$

$$(C) \frac{3^{12+n} \times 9^{2n-7}}{3^{5n}} = \frac{3^{12} \cdot 3^n \times 3^{4n} \cdot 3^{-14}}{(3^n)^5}$$

$$= 3^{-2} = \frac{1}{9}$$

Very Short Answer Questions:

- We have,

$$\left(\frac{-4}{3}\right)^3 = \frac{-4}{3} \times \frac{-4}{3} \times \frac{-4}{3} = \frac{-64}{27}$$
 - We have,

$$\left(\frac{-1}{2}\right)^4 = \frac{-1}{2} \times \frac{-1}{2} \times \frac{-1}{2} \times \frac{-1}{2} = \frac{1}{16}$$
- $\left(\frac{2}{5}\right)^3$
 - $\left(\frac{-7}{3}\right)^4$
- $(-7)^3$
 - $\left(\frac{7}{5}\right)^2$
- $\left(-\frac{2}{3}\right)^3$
 - $\left(\frac{3}{5}\right)^4$
- 500
 - $-\frac{9}{16}$
 - 95
- $2^3 = 2 \times 2 \times 2 = 8$
 $3^2 = 3 \times 3 = 9$
 $\therefore 9 > 8$
 $\therefore 3^2 > 2^3$
- $(2^3)^5 \times (2^7)^2 = 2^{15} \times 2^{14} = 2^{29}$
 - $$\left[\left(\frac{-3}{5}\right)^2 \times \left(\frac{-3}{5}\right)^4\right]^3 = \left[\left(\frac{-3}{5}\right)^{2+4}\right]^3$$

$$= \left[\left(\frac{-3}{5}\right)^6\right]^3 = \left(\frac{-3}{5}\right)^{18}$$
- $x^0 = 1$
 - $7^0 + 8^0 + 9^0 = 1 + 1 + 1 = 3$
 - $(29^0 - 23^0) \times 16^0 = (1 - 1) \times 1 = 0$
 - $(-2)^{3 \times 5 - 4 - 11} = (-2)^{15 - 4 - 11} = (-2)^0 = 1$
- $9^{2+18+7} = 9^{27}$
 - $a^{1+2+3} = a^6$
- $\left(\frac{3}{5}\right)^{11}$

- 1
- $\frac{4}{9} \div 1 = \frac{4}{9}$
- $8 \times 27 = 216$
- $8 \div \left(\frac{1}{2}\right)^0 = 8 \div 1 = 8$
- $3p^2q^2 = 3 \times (3)^2 \times (4)^2 = 432$
- 2^{9a+1}
- $[5^6 \times 5^4] \div 5^7 = 5^{10-7} = 5^3$
- $\left(\frac{a}{b}\right)^n$
- 5×10^6

Short Answer Questions :

- $(5x)^3 = \left(5 \times \frac{-2}{5}\right)^3 \Rightarrow (-2)^3 \Rightarrow (-8)$
 - $$(-2ab)^4 = \left(-2 \times \frac{3}{5} \times \frac{-1}{2}\right)^4$$

$$= \left(\frac{3}{5}\right)^4 = \frac{81}{625}$$
- $5^{15} \times 5^{32} = 5^{47}$
 - $$\left(\frac{-1}{3}\right)^6 \times \left(\frac{-1}{3}\right)^{20} = \left(\frac{-1}{3}\right)^{26} = \left(\frac{1}{3}\right)^{26}$$
 - $[(-7)^{5+4}]^2 = (-7)^{9 \times 2} = (-7)^{18}$
 - $$(-4x^3)^4 \times (-x)^{2 \times 4} = (-4)^4 \times x^{3 \times 4} \times (-x)^8$$

$$= 256x^{20}$$
- $$4963 = (4 \times 1000) + (9 \times 100) + (6 \times 10) + (3 \times 1)$$

$$= (4 \times 10 \times 10 \times 10) + (9 \times 10 \times 10) + (6 \times 10) + (3 \times 1)$$

$$= 4 \times 10^3 + 9 \times 10^2 + 6 \times 10^1 + (3 \times 1)$$
- $$\left(\frac{2}{7}\right)^{4+(-5)} = \left(\frac{2}{7}\right)^{-1} = \frac{7}{2}$$
 - $$\frac{4^2}{3^2} \times \frac{2^3}{3^3} \times \frac{1}{2^2} = \frac{4^2}{3^{2+3}} \times 2 = \frac{32}{243}$$
 - $$\left[2 \times \left(\frac{-1}{4}\right)\right]^2 = \left(\frac{-1}{2}\right)^2 = \frac{1}{4}$$

$$(iv) \left(\frac{2}{3}\right) \div \left(\frac{-5}{2}\right) = \frac{2}{3} \times \left(\frac{-2}{5}\right) = \left(\frac{-4}{15}\right)$$

$$(v) \frac{16}{25} \times \frac{15}{2} \times \frac{1}{6} \times \frac{125}{64} = \frac{25}{16}$$

$$(vi) \left(\frac{3}{7}\right)^{2+3} \div \left(\frac{3}{7}\right)^8 = \left(\frac{3}{7}\right)^{5-8} = \left(\frac{7}{3}\right)^3$$

$$5. \frac{2}{5} \times \left(\frac{2}{5}\right)^3 \div \left(\frac{3}{5}\right)^4 = \left(\frac{2}{5}\right)^4 \times \left(\frac{5}{3}\right)^4$$

$$= \frac{2^4}{5^4} \times \frac{5^4}{3^4} = \frac{2^4}{3^4} = \frac{16}{81}$$

$$6. (i) \left(\frac{1}{12}\right) \quad (ii) \left(\frac{1}{2}\right) \quad (iii) (25)$$

$$7. \frac{x}{y} = \left(\frac{-3}{1}\right)^2 \div \left(\frac{3}{2}\right)^3 = 3^2 \times \frac{2^3}{3^3} = \frac{8}{3}$$

$$\Rightarrow \left(\frac{x}{y}\right)^{-3} = \left(\frac{8}{3}\right)^{-3} = \frac{3^3}{8^3} = \frac{27}{512}$$

$$8. 5^{2x+1} \div 25 = 125$$

$$\Rightarrow \frac{5^{2x+1}}{25} = 125$$

$$\Rightarrow \frac{5^{2x+1}}{5^2} = 5^3 \Rightarrow 5^{2x+1-2} = 5^3$$

$$\Rightarrow 5^{2x-1} = 5^3 \Rightarrow 2x-1 = 3$$

$$\Rightarrow x = 2$$

$$9. (i) (3^4 \times 3^5) \div 3^9 = 3^{4+5} \div 3^9 = 3^9 \div 3^9 = 3^{9-9} = 3^0 = 1 = 2^0$$

$$(ii) \left[\left(\frac{1}{5}\right)^6 \div \left(\frac{1}{5}\right)^5\right] \div \frac{1}{5} = \left[\left(\frac{1}{5}\right)^{6-5}\right] \div \frac{1}{5}$$

$$= \left(\frac{1}{5}\right)^1 \div \left(\frac{1}{5}\right)^1 = \left(\frac{1}{5}\right)^0 = 1 = 2^0$$

$$10. \text{ Let the number be } x.$$

$$\text{Then } 7^{-5} \times x = 7$$

$$\text{or } x = \frac{7}{7^{-5}} = 7 \times 7^5 = 7^6$$

$$11. (i) \frac{25}{54} \quad (ii) (13)^{-7}$$

$$12. \frac{x}{y} = \left(\frac{2}{3}\right)^{-5+(-2)} = \left(\frac{2}{3}\right)^{-7}$$

$$\therefore \left(\frac{x}{y}\right)^{-2} = \left[\left(\frac{2}{3}\right)^{-7}\right]^{-2} = \left(\frac{2}{3}\right)^{14}$$

$$\text{or } \left(\frac{x}{y}\right)^2 = \left(\frac{2}{3}\right)^{-14}$$

$$13. \frac{25}{4}$$

$$14. (i) 4 \quad (ii) -1 \quad (iii) 10$$

$$(iv) 5 \quad (v) -1 \quad (vi) (-5)$$

$$16. \left(\frac{2}{3}\right)^{(-2)+(-9)} = \left(\frac{2}{3}\right)^{2x-1}$$

$$\Rightarrow (-11) = 2x - 1 \Rightarrow -10 = 2x \Rightarrow -5 = x$$

2 EXERCISE

Text-Book Exercise :

$$1. 8^2 = 8 \times 8 = 64$$

$$2^8 = 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 = 256$$

$$\text{Clearly, } 2^8 > 8^2$$

$$2. (i) 2.7 \times 10^{12} > 1.5 \times 10^8$$

$$(ii) 4 \times 10^{14} < 3 \times 10^{17}$$

$$3. (i) \left(\frac{3^7}{3^2}\right) \times 3^5 = (3^{7-2}) \times 3^5 = 3^5 \times 3^5$$

$$= 3^{5+5} = 3^{10}$$

$$(ii) (6^2 \times 6^4) \div 6^3 = 6^{2+4} \div 6^3 = 6^6 \div 6^3$$

$$= 6^{6-3} = 6^3$$

$$(iii) [(2^2)^3 \times 3^6] \times 5^6 = [2^6 \times 3^6] \times 5^6$$

$$= (2 \times 3)^6 \times 5^6$$

$$= (2 \times 3 \times 5)^6 = 30^6$$

$$(iv) 8 = 2 \times 2 \times 2 = 2^3$$

$$\text{Therefore } 8^2 \div 2^3 = (2^3)^2 \div 2^3$$

$$= 2^6 \div 2^3 = 2^{6-3} = 2^3$$

$$4. (i) \text{ We have } \frac{12^4 \times 9^3 \times 4}{6^3 \times 8^2 \times 27}$$

$$\begin{aligned}
 &= \frac{(2^2 \times 3)^4 \times (3^2)^3 \times 2^2}{(2 \times 3)^3 \times (2^3)^2 \times 3^3} \\
 &= \frac{(2^2)^4 \times (3)^4 \times 3^{2 \times 3} \times 2^2}{2^3 \times 3^3 \times 2^{2 \times 3} \times 3^3} \\
 &= \frac{2^8 \times 2^2 \times 3^4 \times 3^6}{2^3 \times 2^6 \times 3^3 \times 3^3} = \frac{2^{10} \times 3^{10}}{2^9 \times 3^6} \\
 &= 2^{10-9} \times 3^{10-6} = 2^1 \times 3^4 = 162 \\
 \text{(ii)} \quad &2^3 \times a^3 \times 5a^4 = 2^3 \times a^3 \times 5 \times a^4 \\
 &= 8 \times 5 \times a^{3+4} = 40a^7 \\
 \text{(iii)} \quad &\frac{2 \times 3^4 \times 2^5}{9 \times 4^2} = \frac{2 \times 3^4 \times 2^5}{3^2 \times (2^2)^2} = \frac{2^{1+5} \times 3^4}{2^4 \times 3^2} \\
 &= 2^{6-4} \times 3^{4-2} \\
 &= 2^2 \times 3^2 = 4 \times 9 = 36 \\
 5. \quad &\text{(i)} \quad 5985.3 = 5.9853 \times 1000 = 5.9853 \times 10^3 \\
 &\text{(ii)} \quad 70,040,000,000 = 7.004 \times 10,000,000,000 \\
 &= 7.004 \times 10^{10}
 \end{aligned}$$

Exemplar Questions :

- (a) Ascending order : $4^0, 2^3 \times 2, 2^3 \times 3^1, 3^3, 2^5, 3^5, (3^3)^2$
- Descending order : $2^3 \times 5^2, (2^2)^3, 2^{2+3}, \frac{3^5}{3^2}, 3^2 \times 3^0, 2 \times 2^2$
- $(-4)^2$ or 16
- $\frac{729}{64}$
- $\frac{32}{27}$
- (i) 1
(iii) 24
- (ii) -3
(iv) 0
- $n=0$
- (i) 80100000
(ii) 0.00175
- 30^6
- (i) $\left(\frac{7}{11}\right)^5$
(iii) $\left(\frac{3}{5}\right)^5$
- (ii) a^7
(iv) 5^{30}
- (a) $49a^2b^3$
(c) 729
(e) 1
- (b) 3920
(d) $1/75$
- $c=3$

Hots Questions :

- 81
- $\left(\frac{2}{3}\right)^5$
- 3.1
- 6.25×10^6
- $x = \frac{1}{6} - \frac{7}{3}$ of $\frac{2}{-7} \times \frac{9}{4} = \frac{1}{6} - \frac{7}{3} \times \frac{9}{-14}$
 $\Rightarrow x = \frac{1}{6} + \frac{3}{2} = \frac{1+9}{6} = \frac{10}{6} = \frac{5}{3}$
 $\therefore x^3 - 1 = \left(\frac{5}{3}\right)^3 - 1 = \frac{125}{27} - 1 = \frac{98}{27} = 3\frac{17}{27}$
- $27^{2x-1} = (243)^3 \Rightarrow (3)^{6x-3} = (3)^{15}$
 $\Rightarrow 6x-3 = 15 \Rightarrow x=3$
 $\therefore (5)^{2x-4} = (5)^{2(3)-4} = 25$

3 EXERCISE

Single Option Correct :

- (a) $100 + (10)^{(-4)-(-2)} = 100 + (10)^{-2}$
 $= 100 + \frac{1}{100} = \frac{10001}{100}$
- (a) $10 \times 100 = (2 \times 5)x^2$
 $\Rightarrow 100 = x^2$
 $\Rightarrow (10)^2 = x^2 \Rightarrow x=10$
- (d)
- (b)
- (b)
- (a)
- (d)
- (b)
- (a)
- (d)

More Than One Option Correct :

- (a, b) $[(-3)^2]^3 = (-3)^{2 \times 3} = (-3)^6 = 729$
- (a, c) $\frac{10^{22} + 10^{20}}{10^{20}} = \frac{10^{20}(10^2 + 1)}{10^{20}}$
 $= 10^2 + 1 = 101$
- (a, b, c)
- (a, b, d)
- (a, b, d)

Assertion & Reason :

- (a) Assertion and Reason both are correct and Reason is the correct explanation for Assertion.
- (a) Both 'Assertion' and 'Reason' are correct. 'R' is the correct explanation for

‘Assertion’.

$$(27)^{\frac{6}{5}} \div (27)^{\frac{1}{5}} = (27)^{\frac{6}{5} - \frac{1}{5}} = (27)^{\frac{5}{5}} = 27$$

$$= 1 \times 27 = 1 \times 3^3$$

Passage Based Questions :

PASSAGE-1

1. (a) $0.23 = \frac{23}{100} = \frac{23}{10} \times \frac{1}{10} = 2.3 \times 10^{-1}$
2. (c) $0.0000072 = \frac{72}{10000000} = \frac{72}{10 \times 10^6}$
 $= \frac{72}{10} \times 10^{-6} = 7.2 \times 10^{-6}$
3. (b) $1.05 \times 10^3 = \frac{105}{100} \times 10^3$
 $= 105 \times 10^{-2+3} = 105 \times 10 = 1050$

PASSAGE-2

4. (d) $\frac{25 \times 216 \times 729}{6^6 \times 5^5 \times 9^4} = \frac{5^2 \times 6^3 \times 9^3}{6^6 \times 5^5 \times 9^4}$
 $= \frac{1}{6^{6-3} \times 5^{5-2} \times 9^{4-3}} = \frac{1}{6^3 \times 5^3 \times 9}$
 $= \frac{1}{(6 \times 5)^3 \times 9} = \frac{1}{(30)^3 \times 9}$
5. (a) $\frac{\left(\frac{6}{5}\right)^2 \times x^8 \times y^7}{(2)^{10} \times \left(\frac{x}{3}\right)^4 \times \left(\frac{y}{3}\right)^3}$
 $= \frac{(2 \times 3)^2 \times x^8 \times y^7 \times (3)^4 \times (3)^3}{(5)^2 \times (2)^{10} \times x^4 \times y^3}$
 $= \frac{3^{4+3+2} \times x^{8-4} \times y^{7-3}}{(5)^2 \times (2)^{10-2}}$

$$= \frac{3^9 \times x^4 \times y^4}{5^2 \times 2^8} = \frac{3^9}{5^2 \times 2^8} (xy)^4$$

6. (c) $\left[2^3 + 3^3 + 4^3\right] \times \frac{1}{24^6}$
 $= [8 + 27 + 64] \times \frac{1}{24^6}$
 $= \frac{99}{(3 \times 8)^6} = \frac{99}{3^6 \times 8^6} = \frac{11 \times 9}{3^4 \times 3^2 \times 8^6}$
 $= \frac{11 \times 9}{3^4 \times 9 \times 8^6} = \frac{11}{3^4 \times 8^6}$

Integer Type Questions :

1. (2) $2^6 = (2^3)^2 = 8^2$
2. (3)
3. (4) $42369.2 = 4.23692 \times 10^4$
4. (1) $\frac{1}{55^0} \times \frac{1}{55^0} = \frac{1}{1} \times \frac{1}{1} = 1$
5. (1) $\left[\frac{1}{\left(\frac{7}{5}\right)^0}\right]^4 = \left(\frac{1}{1}\right)^4 = 1$

Multiple Matching Questions :

1. (A) $\rightarrow (r, u)$; (B) $\rightarrow (p)$; (C) $\rightarrow (s, t)$; (D) $\rightarrow (q)$
 (A) $(3^3)^2 = (3^2)^3 = (9)^3 = 729$
 (B) $(0^6)^{12} = (0^{18})^{15} = 0$
 (C) $(644^3)^0 = 1$ and $(3^0)^{644} = (1)^{644} = 1$
 (D) $(2^2)^6 = (4)^6 = 4 \times 4 \times 4 \times 4 \times 4 \times 4$
 $= 16 \times 16 \times 16 = (16)^3$

Chapter

14

Symmetry

INTRODUCTION

Symmetry is when a shape looks identical to its original shape after being flipped or turned. It exists everywhere in our life. It also occurs in many areas of mathematics.

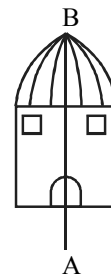
SYMMETRY

A shape if gets folded such that one side fits exactly on the other is known as symmetrical shape. The fold line is known as the line of symmetry, which is also called mirror line.

Rectangle has two lines of symmetry.

For example;

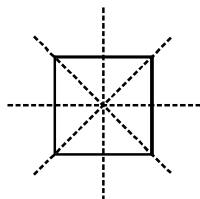
The line AB divides the picture into two identical parts. If the picture is folded along the line AB, one half of the picture will exactly coincide with the other part. In other words one half is the mirror image of the other half. In such cases we say that the picture is symmetrical and AB is called the *line of symmetry*.



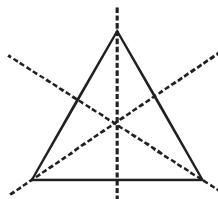
Lines of Symmetry for Regular Polygons

A regular polygon is a polygon whose all sides have the same length and all angles are equal in measure. The regular polygons are symmetrical figures. The number of lines of symmetry in a regular polygon is equal to the number of sides. Thus, they have multiple lines of symmetry.

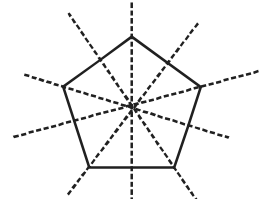
for example,



Square
(4 lines of symmetry)

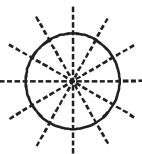


Equilateral triangle
(3 lines of symmetry)



Regular pentagon
(5 lines of symmetry)

NOTE: A circle has infinite lines of symmetry.

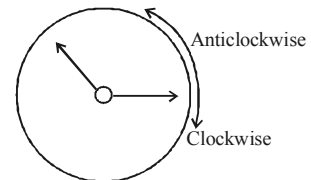


ROTATIONAL SYMMETRY

Rotation

Rotation is the movement of an object around a point through a given number of degrees in a clockwise or anticlockwise direction.

For example: Figure shows rotation of hands of clock through point O.



Other examples are wheel of bicycle, car, motorcycle, truck etc, blades of ceiling fan.

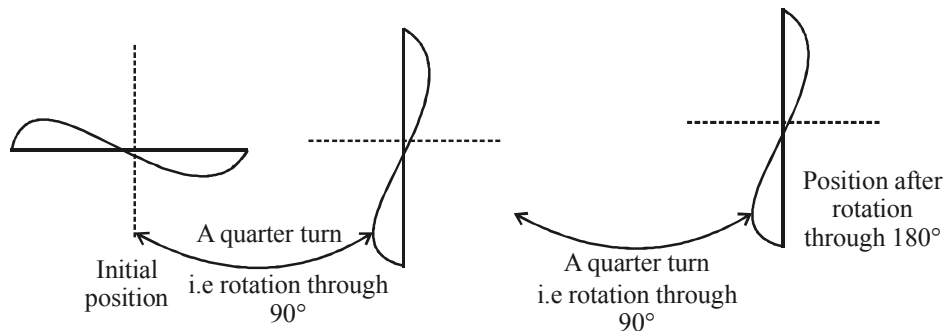
Centre of Rotation

When an object rotates, its shape and size do not change. The rotation turns an object about a fixed point. This point is known as **centre of rotation**.

Angle of Rotation

The angle of turning during rotation is called angle of rotation. A full turn, means 360° , a half turn is 180° and a quarter turn is 90° .

A figure is said to have rotational symmetry if it fits into itself more than once during a full turn, i.e. rotation through 360° . The number of times the figure fits into itself during rotation by 360° , is called the **order of rotational symmetry**.



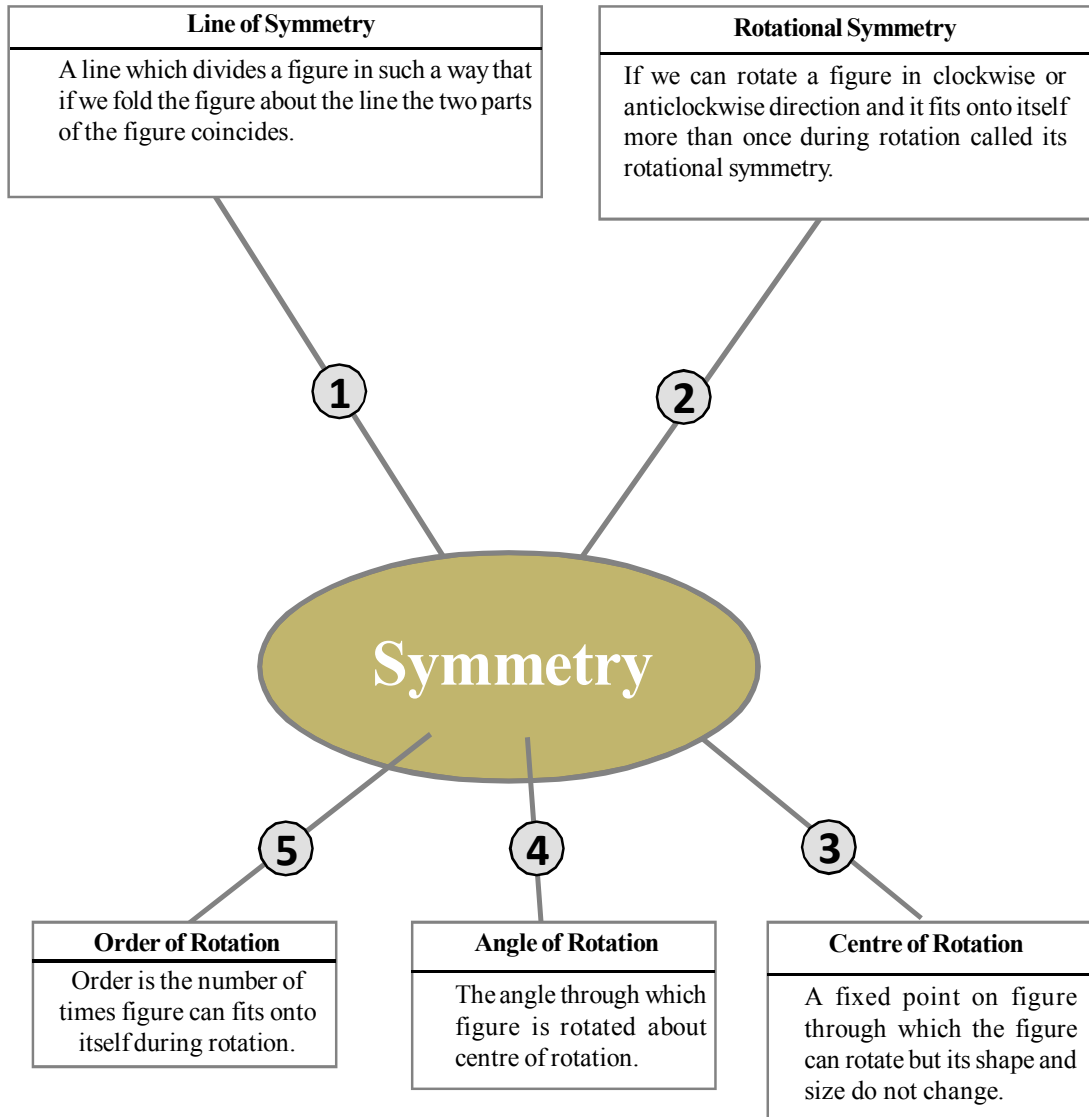
It has a rotational symmetry of order 2

LINE SYMMETRY AND ROTATIONAL SYMMETRY

There are some geometrical shapes which have both line symmetry and rotational symmetry such as:

1. **Square:** It has 4 lines of symmetry and rotational symmetry of order 4.
2. **Rectangle:** It has 2 lines of symmetry and rotational symmetry of order 2.
3. **Equilateral triangle:** It has 3 lines of symmetry and rotational symmetry of order 3.
4. **Letter H:** It has 2 line of symmetry and rotational symmetry of order 2.

CONCEPT MAP

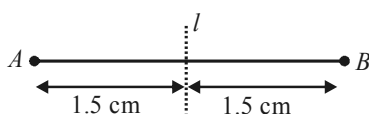


MISCELLANEOUS

SOLVED EXAMPLES

1. Draw a line segment of 3 cm and draw its line of symmetry.

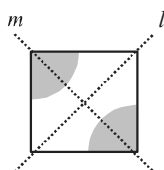
Sol.



2. Which of the following figures have more than one line of symmetry?

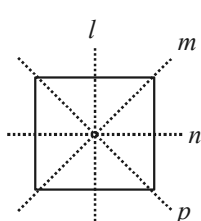


Sol. (i)



It has two lines of symmetry.

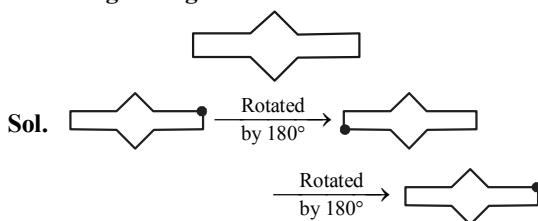
(ii)



It has four lines of symmetry.

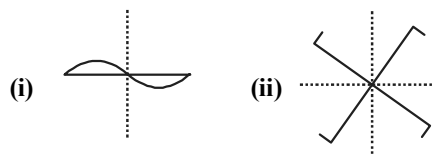
Hence, both figures have more than one line of symmetry.

3. What is the order of rotational symmetry of the given figure?

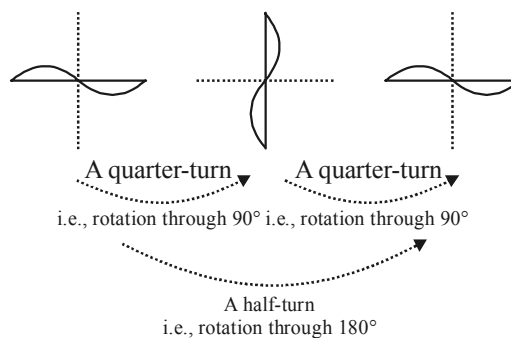


The order of rotational symmetry of figure is 2.

4. Which of the following figures have rotational symmetry? Also, find the order of rotational symmetry.

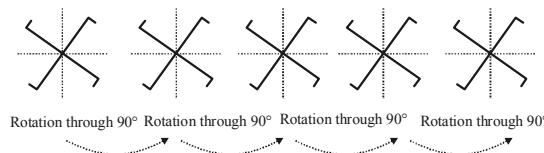


Sol. (i) We observe that figure fits onto itself when it is rotated through 180° .

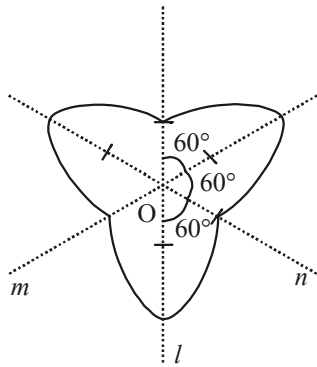


Thus, it has rotational symmetry of order 2.

(ii) We observe that the given figure when rotated through 90° , 180° , 270° and 360° , it coincides with itself. So, it has rotational symmetry of order 4.

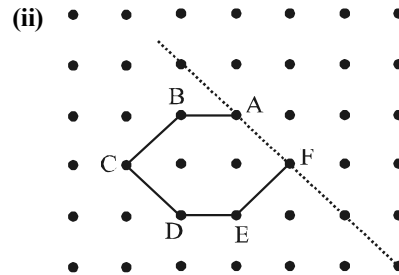
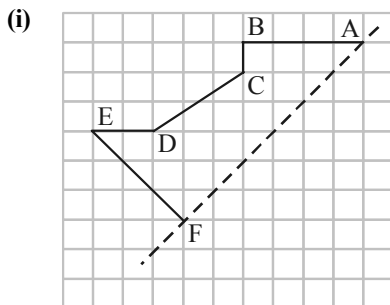


5. Discuss the rotational and line symmetry of the following figure.

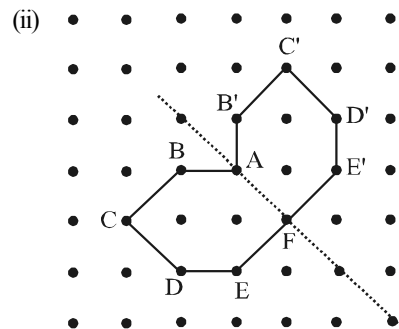
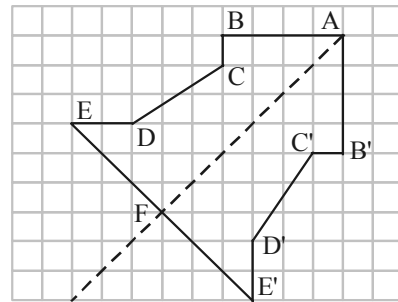


Sol. We observe that the figure fits onto itself when it is rotated through 120° , 240° and 360° .
So, it has rotational symmetry of order 3.
We also observe that the given figure has lines l , m and n as three lines of symmetry.

6. **Copy the diagram and complete each shape to be symmetric about the mirror line.**



Sol. (i)



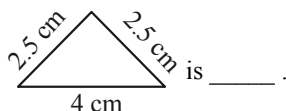
1

EXERCISE

Fill in the Blanks :

DIRECTIONS : Complete the following statements with an appropriate word / term to be filled in the blank space(s).

1. Regular polygons have the same order of rotational symmetry as the number of _____.
2. A line segment is symmetrical about its _____.
3. A triangle having both line of symmetry and rotational symmetry is _____.
4. The line about which the figure is folded to get overlapped is _____ of symmetry.
5. Each of the letters H, N, S and Z has a rotational symmetry of order _____.
6. Line of symmetry for an angle is its _____.
7. Rhombus is a figure that has _____ lines of symmetry and has a rotational symmetry of order _____.
8. In an isosceles right triangle, the number of lines of symmetry is _____.
9. Order of rotational symmetry of



True / False :

DIRECTIONS : Read the following statements and write your answer as true or false.

1. The angle of rotation of a square is 120° .
2. A 2-D shape has rotational symmetry if it takes the appearance of its original shape two or more times during a full turn.
3. Centre of rotation is the point about which the object is turning or rotating.
4. The letter Z has no line of symmetry.
5. A scalene triangle has no line of symmetry.
6. Alphabet E has zero order of rotational symmetry.
7. The centre of rotation of a square is the point of intersection of its diagonals.

8. The order of rotational symmetry of a rectangle is four.

Match the Columns :

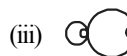
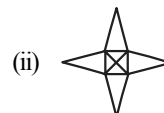
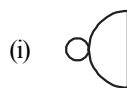
DIRECTIONS : Each question contains statements given in two columns which have to be matched. Statements (A, B, C, D) in column-I have to be matched with statements (p, q, r, s) in column-II.

- | 1. | Column-I | Column-II |
|-----|----------|---|
| (A) | | (p) Infinite order of rotational symmetry |
| (B) | | (q) Only one line of symmetry |
| (C) | | (r) No symmetry |
| (D) | | (s) Only two lines of symmetry |

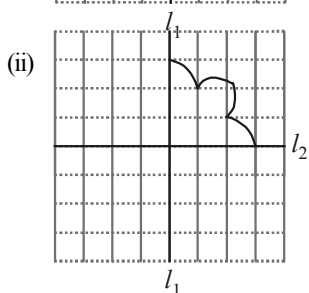
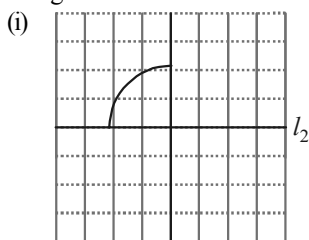
Very Short Answer Questions :

DIRECTIONS : Give answer in one word or one sentence.

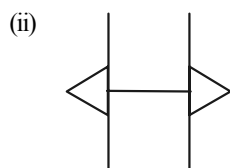
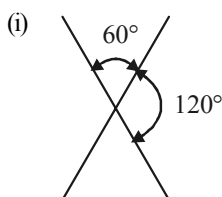
1. Draw the axis of symmetry of the following figures:



2. Which of the letters of the English alphabet have just one line of symmetry? Write the letters and draw their lines of symmetry.
3. Which of the letters of the English alphabet have exactly two lines of symmetry? Write the letters and draw their lines of symmetry.
4. Complete each shape to make it symmetrical along the mirror line:



5. What do you call a rectangle that has four lines of symmetry?
6. Discuss the rotational symmetry of the following figures. Also find the order of symmetry.



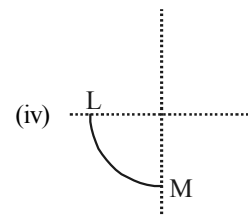
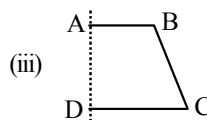
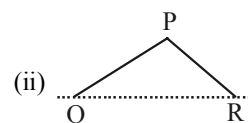
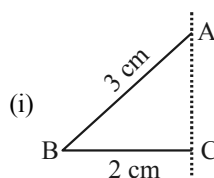
7. Given below is a part of a shape with centre of rotation O. The order of rotational symmetry is 6. Complete the figure.



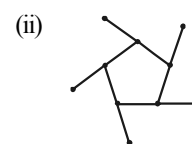
Short Answer Questions :

DIRECTIONS : Give answer in 2-3 sentences.

1. Part of a geometrical figure is given in each of the diagrams below. Complete the figures so that the dotted line in each case is a line of symmetry of the completed figure.



2. Rotate the following figure about its central point at an angle 90° , 120° , 180° .
3. Which kind of symmetry do the following figures show. Also find the value of rotation angle for each.



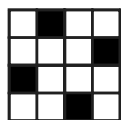
4. Draw two figures which show only reflection symmetry, but not the rotational symmetry. Also draw the axis of symmetry.
5. Draw any 2 figures which neither show the reflection symmetry nor the rotational symmetry.

2

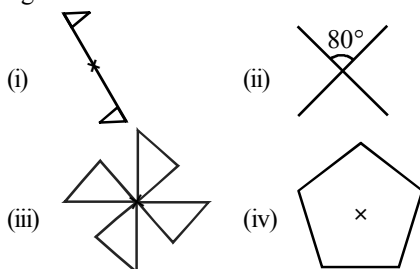
EXERCISE

Text-Book Exercise :

1. Copy the figure given here:
Take any one diagonal as a line of symmetry and shade a few more squares to make the figure symmetric about a diagonal. Is there more than one way to do that? Will the figure be symmetric about both the diagonals?



2. State the number of lines of symmetry for the following figures:
- An equilateral triangle
 - An isosceles triangle
 - A scalene triangle
 - A square
 - A rectangle
 - A rhombus
 - A parallelogram
 - A quadrilateral
 - A regular hexagon
 - A circle
3. What letters of the English alphabet have reflectional symmetry (i.e., symmetry related to mirror reflection) about:
- a vertical mirror
 - a horizontal mirror
 - both horizontal and vertical mirrors
4. What other name can you give to the line of symmetry of:
- an isosceles triangle?
 - a circle?
5. Give the order the rotational symmetry for each figure:



6. Name any two figures that have both line symmetry and rotational symmetry.
7. If a figure has two or more lines of symmetry, should it have rotational symmetry of order more than 1?
8. Fill in the blanks:

Shape	Centre of Rotation	Order of Rotation	Angle of Rotation
Square			
Rectangle			
Rhombus			
Equilateral triangle			
Regular hexagon			
Circle			
Semi-circle			

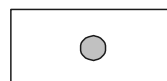
9. Name the quadrilaterals which have both line and rotational symmetry of order more than 1.
10. After rotating by 60° about a centre, a figure looks exactly the same as its original position. At what other angles will this happen for the figure?
11. Can we have a rotational symmetry of order more than 1 whose angle of rotation is:
- 45°
 - 17° ?

Exemplar Questions :

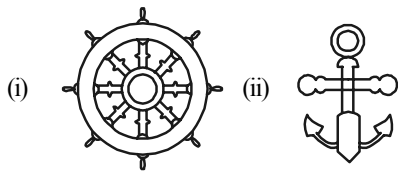
1. State whether the figure shows rotational symmetry. If yes, then what is the order of rotational symmetry?



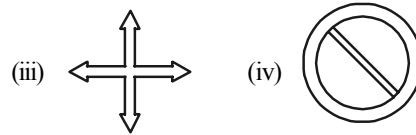
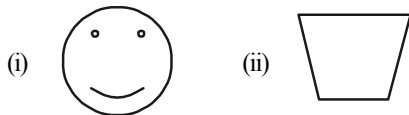
2. The flag of Japan is shown below. How many lines of symmetry does the flag have?



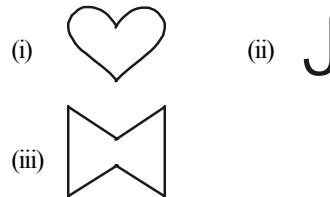
3. Which of the figures given below have both line and rotational symmetry?



4. Tell whether each figure has rotational symmetry or not.



5. Draw all lines of symmetry for each of the following figures as given below:



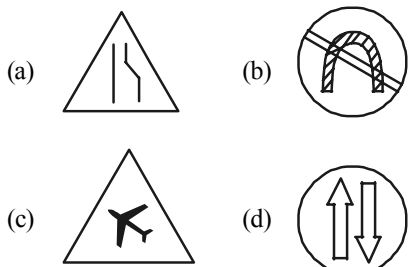
3

EXERCISE

Single Option Correct :

DIRECTIONS : This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONLY ONE is correct.

1. The other name given to the line of symmetry of an isosceles triangle is
(a) angle bisector (b) altitude
(c) median (d) all of these
2. Which of the following road signs has a line symmetry?



3. A triangle having one line of symmetry but showing no rotational symmetry is

- (a) Isosceles triangle
(b) Scalene triangle
(c) Equilateral triangle
(d) None of these

4. After rotation by 60° about the centre, a figure looks exactly the same as its original position. At which of the following angles will this not happen again for the figure.
(a) 180° (b) 240°
(c) 90° (d) 300°

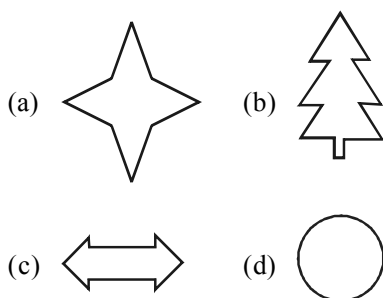
5. The figure  has the order of

rotational symmetry as :

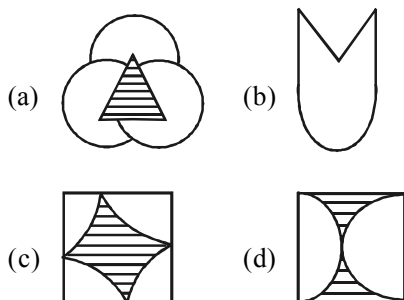
- (a) 4 (b) 6
(c) 5 (d) None

6. Which of the following letters has both line symmetry and rotational symmetry?
(a) A (b) U
(c) H (d) W

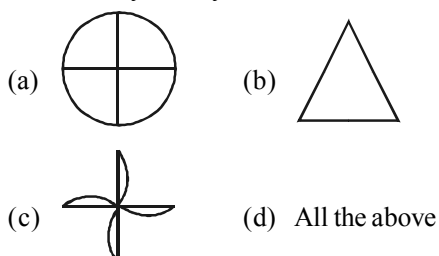
7. A quadrilateral having rotational symmetry, but no line symmetry is
 (a) Square (b) Rectangle
 (c) Parallelogram (d) Rhombus
8. The figure having only two lines of symmetry in the following is



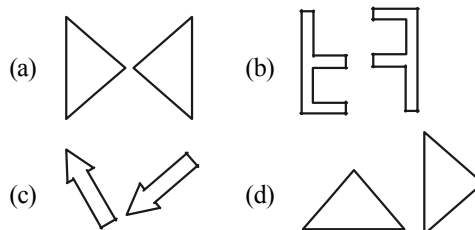
9. Which of the following figures does not have any line of symmetry?



10. A rectangle has _____
 (a) one line of symmetry.
 (b) two lines of symmetry.
 (c) three lines of symmetry.
 (d) None of these.
11. Which of the following figures have rotational symmetry of order more than 1?



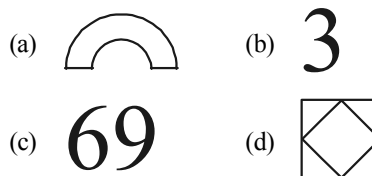
12. Which of the following are reflections of each other?



More Than One Option Correct :

DIRECTIONS : This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONE or MORE may be correct.

1. Which of the following has a line of symmetry?



2. Which of the following triangle has line of symmetry?

- (a) an equilateral triangle.
 (b) an isosceles triangle.
 (c) a scalene triangle.
 (d) all of the above.

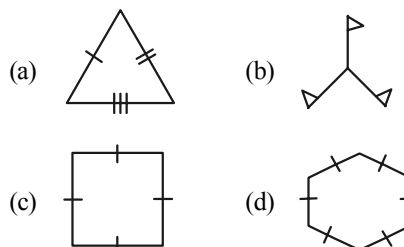
3. Which of the following alphabet has a line of symmetry?

- (a) A (b) B
 (c) Q (d) O

4. Which of the following has a line of symmetry

- (a) equilateral triangle (b) square
 (c) semi-circle (d) parallelogram

5. Which of the following shapes has rotational symmetry?



Passage Based Questions :

DIRECTIONS : Study the given passage and answer the following questions.

PASSAGE

A figure is said to have rotational symmetry if it fits onto itself more than once during a full turn, i.e., rotation through 360° . The number of times the figure fits onto itself during rotation by 360° , is called the order of rotational symmetry.

1. The order of rotational symmetry in the figure given below is



- (a) 4 (b) 8
(c) 6 (d) Infinitely many
2. The order of rotational symmetry in the figure given below is



- (a) 4 (b) 2
(c) 1 (d) Infinitely many

3. Order of rotational symmetry of  is
- (a) 6 (b) 8
(c) 7 (d) 5

Integer Type Questions :

DIRECTIONS : Answer the following questions. The answer to each of the question is a single digit integer, ranging from 0 to 9.

- The number of letters in the word SNAIL that have line symmetry is
- How many lines of symmetry are there in a square?
- How many lines of symmetry are there in a rhombus?
- A regular hexagon has _____ order of rotational symmetry.
- A windmill has a rotational symmetry of the order $2 \times K$. Find K.
- Find the order of rotational symmetry of English alphabet B.

SOLUTIONS

Brief Explanations of Selected Questions

1 EXERCISE

Fill in the Blanks :

1. sides
2. perpendicular bisector
3. equilateral triangle
4. axis
5. 2
6. bisector
7. 2, 2
8. one
9. 1

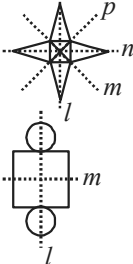
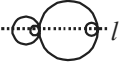
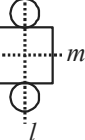
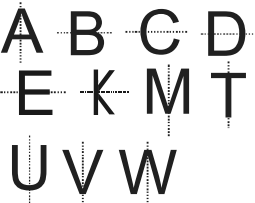
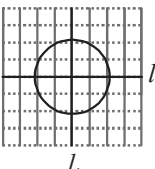
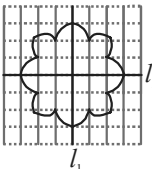
True / False :

1. False
2. True
3. True
4. True
5. True
6. False
7. True
8. False

Match the Column :

1. (A) → (r); (B) → (s); (C) → (q); (D) → (p)

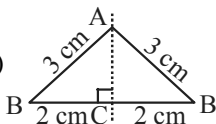
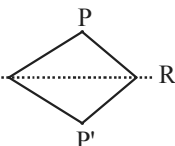
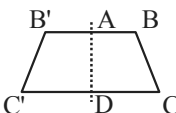
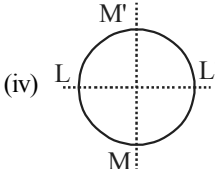
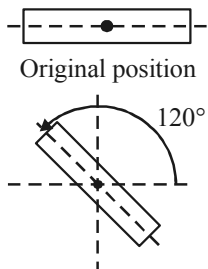
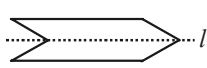
Very Short Answer Questions:

1. (i)  (ii) 
(iii)  (iv) 
2. 
3. 
4. (i)  (ii) 
5. Square
6. (i) Angle of rotational symmetry = 180° ,
Order of rotational symmetry = two

- (ii) Angle of rotational symmetry = 180° ,
Order of rotational symmetry = two



Short Answer Questions :

1. (i)  (ii) 
(iii)  (iv) 
2. 
Original position
120°
180°
3. Both figures show the rotational symmetry but not the reflection symmetry.
(i) Since order of rotation is 3. Hence, rotation angle = $\frac{360^\circ}{3} = 120^\circ$
(ii) Since order of rotation is 5. Hence, rotation angle = $\frac{360^\circ}{5} = 72^\circ$
4. (i)  (ii) 
Answer may vary
5. (i)  (ii) 
Answer may vary.

2 EXERCISE

Text-Book Exercise :

-
- | | | | |
|--------|--------------|---------|----------|
| (i) 3 | (ii) 1 | (iii) 0 | (iv) 4 |
| (v) 2 | (vi) 2 | (vii) 0 | (viii) 0 |
| (ix) 6 | (x) Infinite | | |
- (i) Vertical mirror - A, H, I, M, O, T, U, V, W, X and Y

(ii) Horizontal mirror - B, C, D, E, H, I, O and X

(iii) Both horizontal and vertical mirror - H, I, O and X
- (i) The other name of line of symmetry of an isosceles triangle is median or altitude.

(ii) The other name of line of symmetry of a circle is diameter.
- | | | | |
|-------|--------|---------|--------|
| (i) 2 | (ii) 2 | (iii) 4 | (iv) 5 |
|-------|--------|---------|--------|
- Rectangle and Square
- Yes, For example : A square has 4 lines of symmetry and it has rotational symmetry of order 4.
-

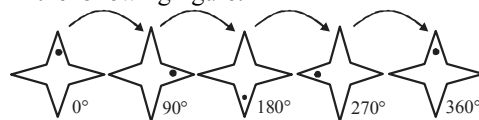
Shape	Centre of Rotation	Order of Rotation	Angle of Rotation
Square	Intersecting point of diagonals.	4	90°
Rectangle	Intersecting point of diagonals.	2	180°
Rhombus	Intersecting point of diagonals.	2	180°
Equilateral triangle	Intersecting point of medians.	3	120°
Regular hexagon	Intersecting point of long diagonals.	6	60°
Circle	Centre	Infinite	At every angle
Semi-circle	Mid-point of diameter	1	360°

- Square, rectangle, rhombus have both line and rotational symmetry of order more than 1.
- Other angles will be 120° , 180° , 240° , 300° , 360° .
- (i) If the angle of rotation is 45° , then order of rotational symmetry would be 8.

(ii) If the angle of rotation is 17° then symmetry of order is not possible because 360° is not completely divided by 17° .

Exemplar Questions :

- The given figure shows rotational symmetry. The order of symmetry = 4, which is clear from the following figure:



- 2 lines of symmetry
- (i) No (ii) No (iii) Yes (iv) Yes
- (i) One line of symmetry

(ii) No line of symmetry

(iii) Two lines of symmetry
- (i) One line of symmetry

(ii) No line of symmetry

(iii) Two lines of symmetry

3 EXERCISE

Single Option Correct :

- (d)
- (c)
- (a)
- (c)
- (c)
- (c)
- (c)
- (c)
- (a)
- (b)
- (d)
- (a)

More Than One Option Correct :

- (a, b, d)
- (a, b)
- (a, b, d)
- (a, b, c)
- (c, d)

Passage Based Questions :

- (c)
- (b)
- (b)

Integer Type Questions :

- (2)
- (4) There are 4 lines of symmetry in a square.
- (2)
- (6)
- (2) The windmill has a rotational symmetry of order 4.

$\therefore 2K = 4 \Rightarrow K = 2$

Rotated by 360°

$\therefore$ Order of rotational symmetry is 1.

Chapter 15

Visualising Solid Shapes

INTRODUCTION

In our daily use we encounter lot of things of different shapes and sizes. For example: mobile phones, dice, pipe, balls, gas cylinder etc.

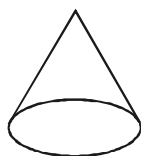
Can you identify the various kinds of shapes from the above list and sort them out. This chapter will enable you to see how the dimensions of different solid shapes can be drastically different. In this chapter we will study about different shapes, their edges, faces and corners; and how to draw their sketches.

PLANE FIGURES AND SOLID SHAPES

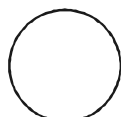
Any closed three dimensional shape is known as **solid**.

It has 3-dimensions i.e, length, width and height and occupies space

Some solids are:



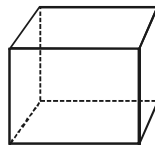
Cone



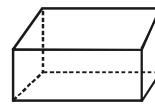
Sphere



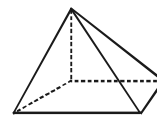
Cylinder



Cube



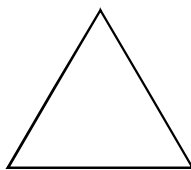
Cuboid



Prism

Shapes having two measurements like length and breadth are called 2-dimensional shapes. These shapes can be easily drawn on paper.

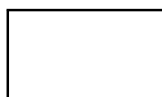
Some Plane Shapes are:



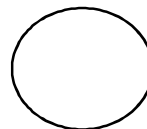
Triangle



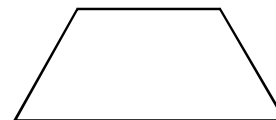
Square



Rectangle



Circle



Quadrilateral

Faces Edges and Vertices

Face: Each flat surface of a solid is called a face.

Vertex: When three or more faces meet at one point, that point is called a vertex.

Edge: The line segment that is the intersection of two faces is called an *edge*.

The table given below, shows the number of faces, vertices and edges of different solids.

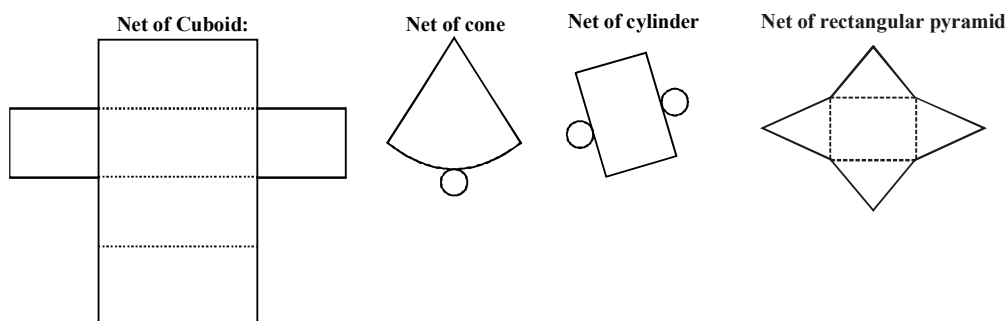
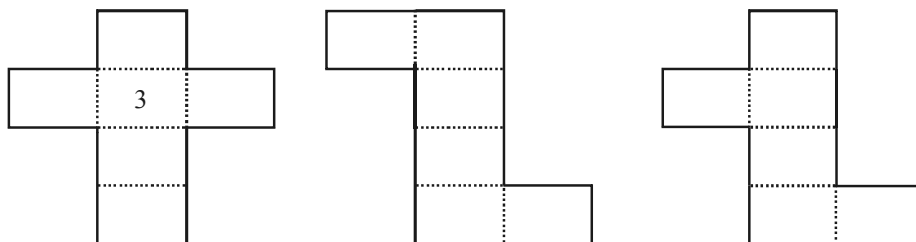
Shape	No. of faces	No. of vertices	No. of edges
1. Cubes	6	8	12
2. Cuboid	6	8	12
3. Cylinder	3	0	2
4. Pyramid (rectangular)	5	5	8
5. Cone	2	1	1

NETS OF SOLID SHAPES

A flat shape that can be folded to make a solid shape is called a net.

It is the outline of its faces joined together from which a model of the solid can be made. Nets of some of the solid figures are shown below.

Net of Cube: It may be following types.



NOTE : A rectangular pyramid has a rectangular base and triangles on the four sides.

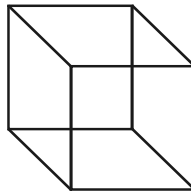
DRAWING SOLIDS ON A FLAT SURFACE

When we draw a solid shape on a paper, the images should be drawn as to make them appear three-dimensional. It is a visual illustration. There are two methods to draw 3-D figures.

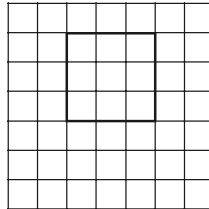
1. Oblique sketch
2. Isometric sketch

Oblique Sketch

Oblique sketches give a clear idea of how the solid looks like, when seen from the front. We do not see certain faces. If we look at figure, just by look we can say that it is a figure of cube although its all edges are not equal. Thus, this type of sketch by which we just get the idea of the given solid is called the **oblique sketch**.

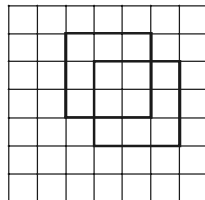


Let us learn the technique for drawing a cube of $(3 \times 3 \times 3)$. We need a squared paper.



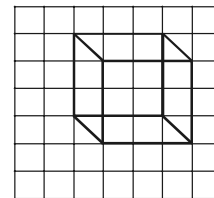
Step 1

Draw the front face



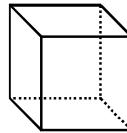
Step 2

Draw the opposite face which is somewhat off-set from step-1. Sizes of the faces have to be same



Step 3

Join the Corresponding vertices of both the faces.



Step 4

Redraw using dotted lines for hidden edges.
(It is a convention)
The sketch is ready now.

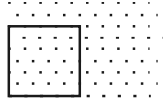
In the oblique sketch above,

- (i) The sizes of the front face and its opposite are same
- (ii) The edges, which are all equal in a cube, appear so in the sketch, though the actual measures of edges are not taken so.

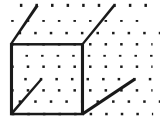
Isometric Sketches

An isometric sheet is a special sheet on which dots are formed on a pattern of equilateral triangle. On isometric sheet, we can draw sketches in which measurements also agree with those of a given solid. Such sketches are known as **isometric sketches**.

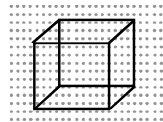
Let us draw an isometric sketch of a cube of dimensions $3 \times 3 \times 3$ (which means each of its edge is 3 units).



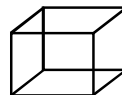
Step 1
Draw a square
to show the front
face



Step 2
Draw four parallel
line segments of length
3 units starting from
the four corners of the
square



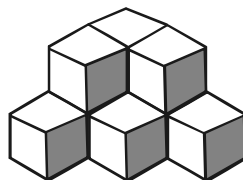
Step 3
Connect the matching
corners with appropriate
line segments.



Step 4
This is an isometric sketch
of the cube

VISUALISING SOLID OBJECTS

Sometimes when we look at solid shapes, some of the faces may be hidden from our view. Let us count the number of cubes present in following figure.

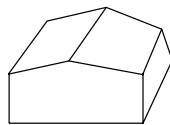


We can say that there are 9 cubes.
3 in the top row and 6 in the bottom row.

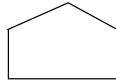
Different Sections of a Solid Can be Viewed in Many Ways:

- (a) One way is to view by cutting or **slicing** the shape, which would result in the cross-section of the solid.
- (b) Another way is by observing a 2-D **shadow** of a 3-shape.
- (c) A third way is to look at the shape from different angles; the **front-view**, the **side-view** and the **top-view** can provide a lot of information about the shape observed.

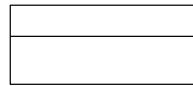
Here is an example of how one gets different views of a given building.



Building



Front view

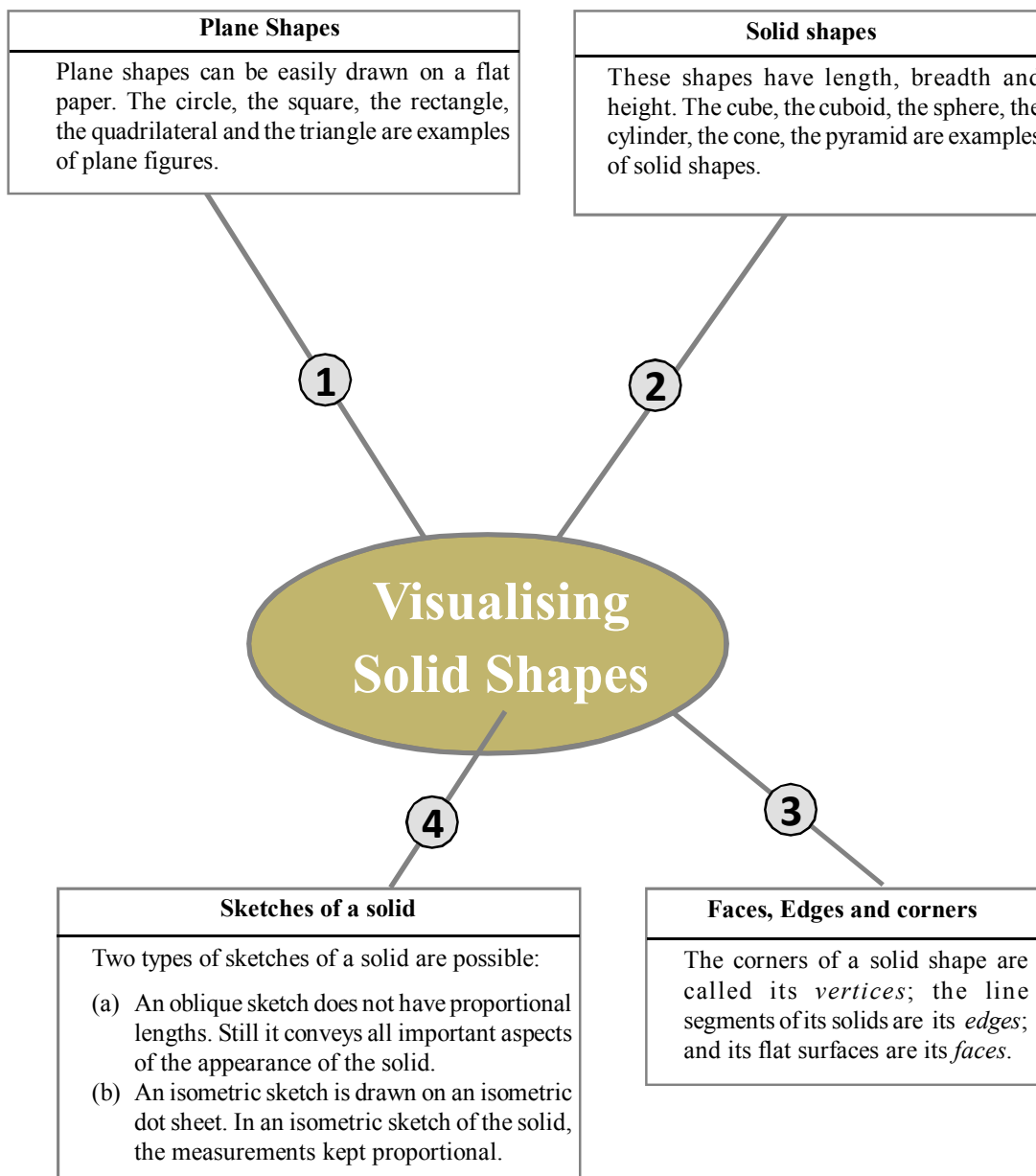


Side view



Top view

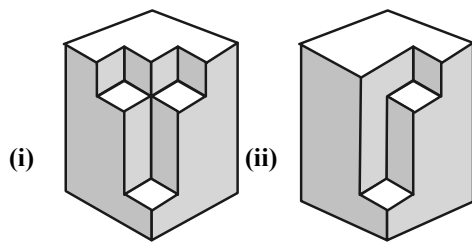
CONCEPT MAP



MISCELLANEOUS

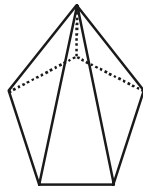
SOLVED EXAMPLES

1. Which of the following solids has the greater number of faces?

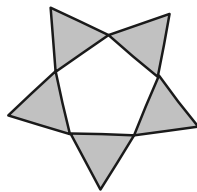


Sol. Figure (i) has 15 faces and figure (ii) has 11 faces. Hence, (i) has greater number of faces.

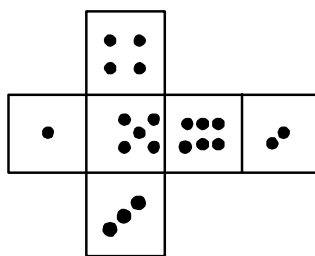
2. Draw the net of the given solid.



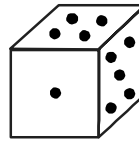
Sol. Net of the solid is given by



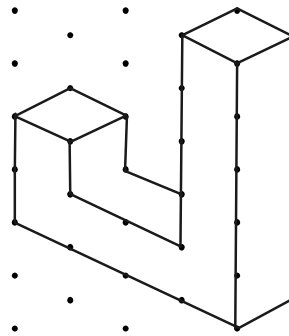
3. The net of a die is shown, which die can be formed by using the net?



Sol.

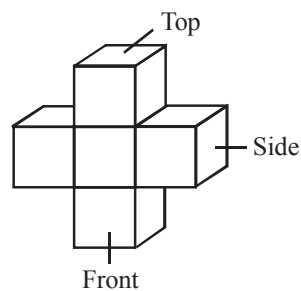


4. How many unit cubes are needed to build the solid?

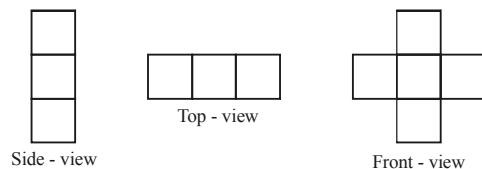


Sol. 9

5. Draw rough sketches of the top, side and front view of the following figure.



Sol.



1 EXERCISE

Fill in the Blanks :

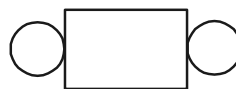
DIRECTIONS : Complete the following statements with an appropriate word / term to be filled in the blank space(s).

1. A cylinder has _____ circular edges.
2. A square pyramid has five faces one of which is a square face and the rest four are _____.
3. Every cube is also a _____.
4. Shape having one curved surface and one plane surface is _____.
5. If a shape is completely bounded by plane faces, _____ is the least number of faces it may have.
6. Brick is an example of _____.
7. A cuboid is also known as _____.
8. Objects having fixed shape and size and occupying fixed space are called _____.
9. Out of _____ faces of a rectangular pyramid _____ are triangles and base is _____.
10. Out of _____ faces of a triangular prism, _____ are rectangles and _____ are triangles.

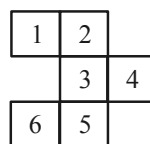
True / False :

DIRECTIONS : Read the following statements and write your answer as true or false.

1. The cross section of a cube is a square.
2. A cross section is the surface made by slicing through a three dimensional object.
3. A flat shape that can be folded to make a solid shape is called a net.
4. The cube can cast a shadow in the shape of a pentagon.
5. When three cubes each with 3 cm edge are placed side by side. The solid so obtained is a cuboid.
6. When a dice is given a horizontal cut we get a square shape as cross-section.
7. This is the net of a cone.



8. This is a net for a die.

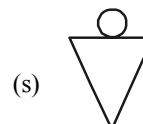
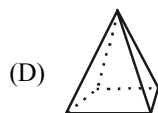
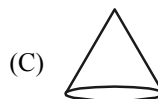
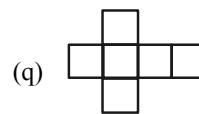
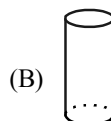
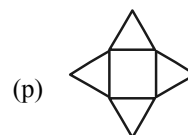
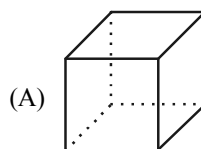


9. The opposite faces of a die always have a total of seven dots on them.

Match the Columns :

DIRECTIONS : Each question contains statements given in two columns which have to be matched. Statements (A, B, C, D) in column-I have to be matched with statements (p, q, r, s) in column-II.

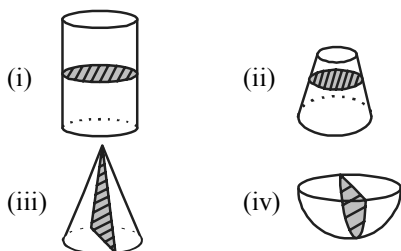
1. Match the following solids with their nets :



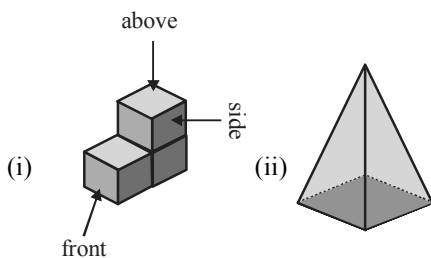
Short Answer Questions :

DIRECTIONS : Give answer in 2-3 sentences.

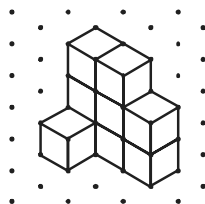
- Give three examples of each : (i) a cone (ii) a sphere (iii) a cuboid (iv) a cylinder
- Name the different plane shapes needed to draw the net of:
 - a cube
 - a triangular prism
 - a triangular pyramid
 - a cylinder
- What cross-section would you get when you give a (a) vertical cut (b) horizontal cut to the following objects?
 - a basket ball
 - a die
 - an unsharpened pencil
- Name the shape of the cross-section in each solid?



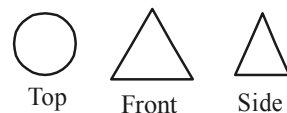
- Draw the top, front and side view of following solids.



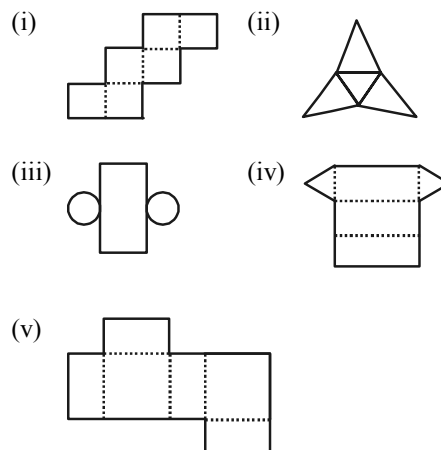
- The diagram shows a solid, made of cubes drawn on a dotted (isometric) paper. Draw its top, front and side view.



- Sketch and name the solid with the given views.



- Name the figures whose nets are given below.



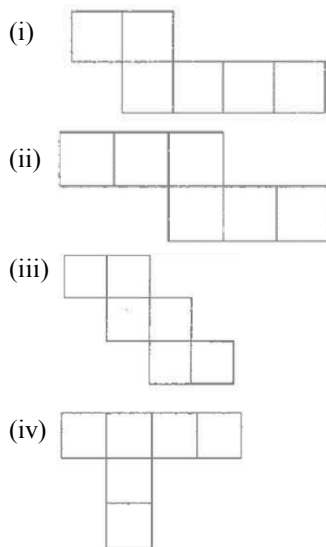
- Complete the following table and verify Euler's formula in each case.

	Faces (F)	Edges (E)	Vertex (V)	$V + F = E + 2$

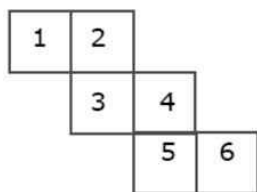
2 EXERCISE

Text-Book Exercise :

1. Identify the nets which can be used to make cubes (cut out copies of the nets and try it):



2. Can this be a net for a die? Explain your answer.

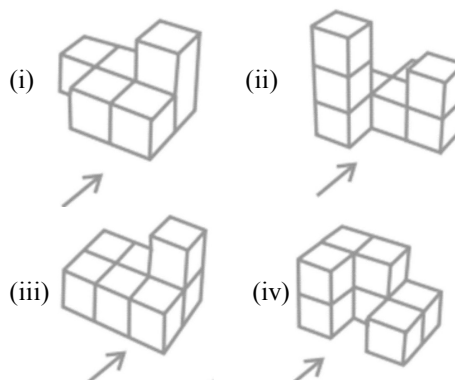


3. The dimensions of a cuboid are 5 cm, 3 cm and 2 cm. Draw three different isometric sketches of this cuboid.
4. Three cubes each with 2 cm edge are placed side by side to form a cuboid. Sketch an oblique or isometric sketch of this cuboid.
5. What cross-sections do you get when you give a:
- vertical cut
 - horizontal cut to the following solids?
 - A brick
 - A round apple
 - A die
 - A circular pipe
 - An ice-cream cone.

6. Examine if the following are true statements:
- The cube can cast a shadow in the shape of a rectangle.
 - The cube can cast a shadow in the shape of a hexagon.

Exemplar Questions :

- Draw a net of a cuboid having same breadth and height, but length double the breadth.
- Draw the nets of the following:
 - Triangular prism
 - Tetrahedron
 - Cuboid
- Draw the net of triangular pyramid with base as equilateral triangle of side 3 cm and slant edges 5 cm.
- Find the number of cubes in each of the following figures and in each case give the top, front, left side and right side view (arrow indicating the front view).



5. How many faces does the figure has?



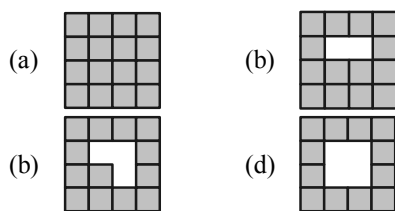
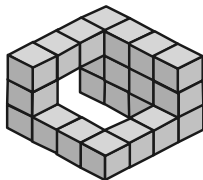
3

EXERCISE

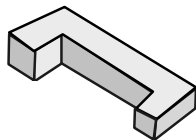
Single Option Correct :

DIRECTIONS : This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONLY ONE is correct.

1. Which of the following figure is correct about the solid as viewed from above?



2. How many corners does the following solid have?

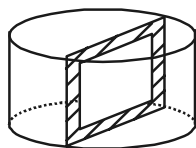


- (a) 12 (b) 16
(c) 14 (d) 18

3. Which of these solids has the maximum number of vertices?

- (a) Cone (b) Cylinder
(c) Cuboid (d) Triangular pyramid

4. A slice of cheese is cut from the cylinder shaped cheese as shown. What name can be given to the cross-section?

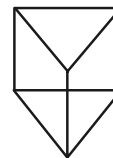


- (a) Circle (b) Square
(c) Rectangle (d) Triangle

5. Name a three-dimensional figure from which a cross-section of a circle can be made.

- (a) Cube (b) Cuboid
(c) Cone (d) Pyramid

6. Which is true statement about the figure?



- (a) The top view of the solid is a rectangle.
(b) The side view is a triangle.
(c) The side view is a rectangle.
(d) The front view is a triangle.

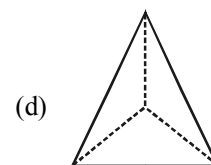
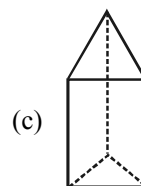
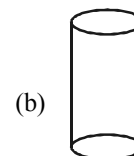
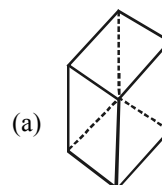
7. A solid object when seen from one side, looks like this.



The same solid, when viewed from top, looks like this.



Which of these shapes could it be?

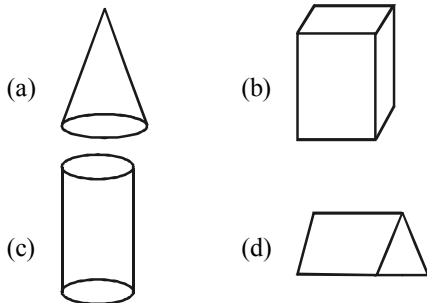


8. Madhavi glued 4 white cubes together as shown below. Then she painted the entire figure red.

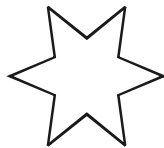


How many faces of the 4 cubes were painted red?

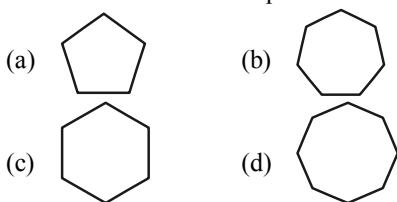
- (a) 4 (b) 9
(c) 18 (d) 24
9. Which of the following figure has six faces?



10. How many corners does this shape below have?



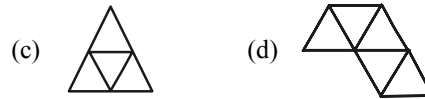
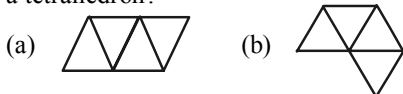
- (a) 6 (b) 10
(c) 12 (d) 13
11. Which shape best represents a hexagonal prism when viewed from the top?



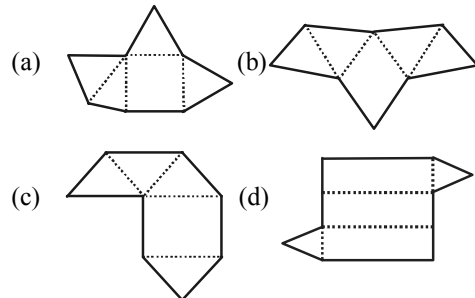
More Than One Option Correct :

DIRECTIONS : This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONE or MORE may be correct.

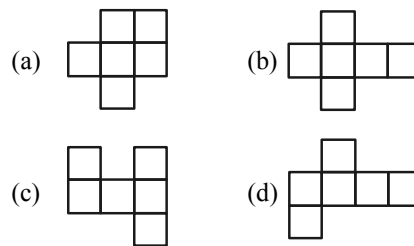
1. Which of the following are correct nets to make a tetrahedron?



2. Which of the following nets could not form a pyramid?



3. Which of these nets will not fold up to give a cube?

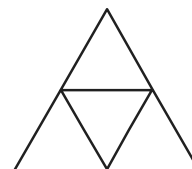


4. The solids combined to form the given solid are



- (a) Cylinder (b) Sphere
(c) Cone (d) Cube

5. The solid whose net is given below is



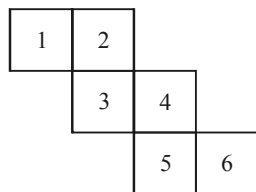
- (a) Tetrahedron
(b) Triangular pyramid
(c) Prism
(d) None of these

Assertion & Reason :

DIRECTIONS : Each of these questions contains an Assertion followed by Reason. Read them carefully and answer the question on the basis of following options. You have to select the one that best describes the two statements.

- If both **Assertion** and **Reason** are correct and **Reason** is the correct explanation of **Assertion**.
- If both **Assertion** and **Reason** are correct but **Reason** is not the correct explanation of **Assertion**.
- If **Assertion** is correct but **Reason** is incorrect.
- If **Assertion** is incorrect but **Reason** is correct.

- Assertion:** The following net is a net for a perfect die.



Reason: The total number of dots on the opposite faces of a perfect die is equal to seven.

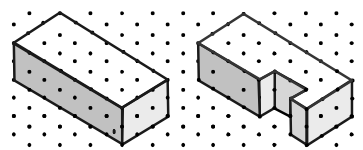
- Assertion:** In the oblique sketches of the solids, the lengths are not alike, as they should be in actual.

Reason: An isometric sheet divides the paper into small equilateral triangles made up of dots or lines.

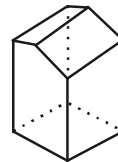
Integer Type Questions :

DIRECTIONS : Answer the following questions. The answer to each of the question is a single digit integer, ranging from 0 to 9.

- How many unit cubes were removed from the solid on the left to obtain the solid on the right?



- How many faces does the solid have?



- How many edges does the following figure have?



- The number of vertices a triangular prism has

SOLUTIONS

Brief Explanations of Selected Questions

1 EXERCISE

Fill in the Blanks :

1. 2 2. triangular faces 3. cuboid
4. cone 5. 4 6. cuboid
7. prism 8. solids
9. 5, 4, rectangle 10. 5, 3, 2

True / False :

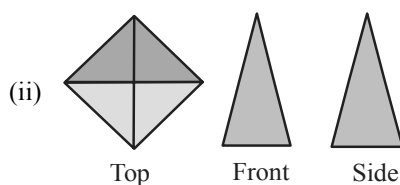
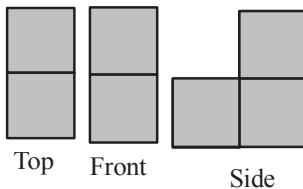
1. True 2. True 3. True
4. False 5. True 6. True
7. False 8. False 9. True

Match the Columns :

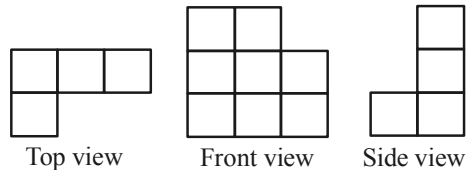
1. (A) → (q), (B) → (r), (C) → (s), (D) → (p)

Short Answer Questions :

1. (i) Ice-cream cone, cap, traffic cone
(ii) A football, a lemon, a round orange
(iii) A book, a brick, a duster
(iv) A circular pipe, a glass, a cold drink can
2. (i) square (ii) triangle, rectangle
(iii) triangle (iv) rectangle, circle
3. (i) (a) circle (b) circle
(ii) (a) square (b) square
(iii) (a) rectangle (b) circle
4. (i) circle (ii) circle
(iii) (isosceles) triangle (iv) semi-circle
5. (i)



6.



7.



8.

- (i) Cube (ii) Triangular pyramid
- (iii) Cylinder (iv) Triangular prism
- (v) Cuboid

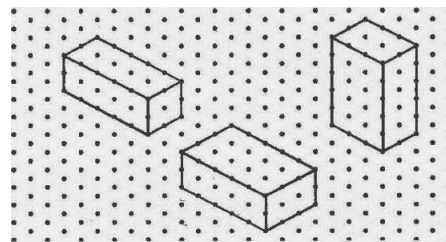
9.

Faces (F)	Edges (E)	Vertex (V)	$V+F=E+2$
6	12	8	$8+6=12+2$
4	6	4	$4+4=6+2$
9	16	9	$9+9=16+2$
7	15	10	$10+7=15+2$

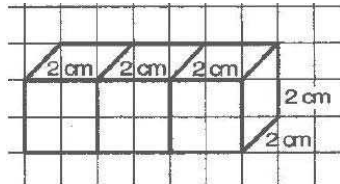
2 EXERCISE

Text-Book Exercise :

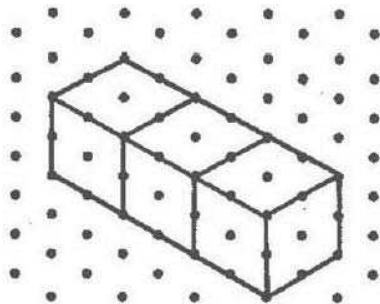
1. Cube's nets are (ii) and (iii).
2. No, this cannot be a net for a die. Because one pair of opposite faces will have 1 and 4 on them and another pair of opposite faces will have 3 and 6 on them whose total is not equal to 7.
3. Three different isometric sketches are given below:



4. Oblique sketch of cuboid:



Isometric sketch of cuboid :



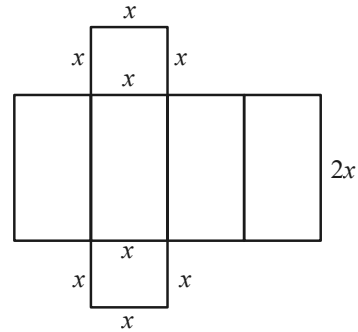
- 5.

S. no.	Name of solid	Vertical cut	Horizontal cut
(i)	A brick		
(ii)	A round apple		
(iii)	A die		
(iv)	A circular pipe		
(v)	An ice cream cone		

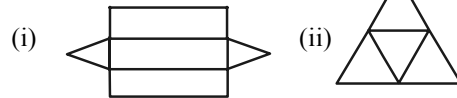
6. (i) True (ii) False

Exemplar Questions :

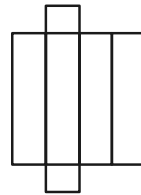
- 1.



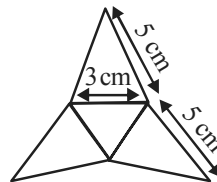
- 2.



- (iii)



- 3.



4. (i) 6 (ii) 8
(iii) 7 (iv) 8
5. 16

3 EXERCISE

Single Option Correct :

1. (d) 2. (b) 3. (c) 4. (c)
5. (c) 6. (c) 7. (c) 8. (c)
9. (b) 10. (c) 11. (c)

More Than One Option Correct :

1. (a, c) 2. (a, c, d) 3. (a, c)
4. (a, c) 5. (a, b)

Assertion & Reason :

1. (d) **Assertion :** incorrect; **Reason :** correct
In a perfect die the sum of dots on opposite faces is seven.
The total numbers of dots on the opposite faces is not equal to seven i.e.,

$$6 + 3 = 9 \neq 7$$

$$4 + 1 = 5 \neq 7$$

Hence, the given net is not a net of a perfect die.

2. (b)

Integer Type Questions :

1. (4) 2. (7) 3. (8) 4. (6)

